GRAND CANYON NATIONAL PARK
GRAND CANYON, ARIZONA

Rehabilitate North Rim Infrastructure

GRCA 318719

SPECIFICATIONS

Volume A (Division 01)

CONSTRUCTION DOCUMENTS

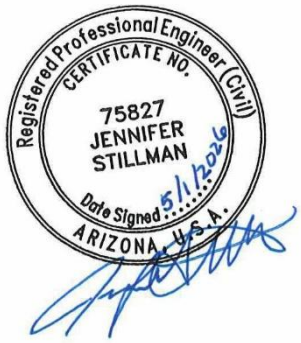
May 2026

NATIONAL PARK SERVICE
DENVER SERVICE CENTER

This page intentionally left blank.

SEALS AND SIGNATURES - Volume A

It is a violation of applicable laws and regulations governing professional licensing and registration for any person, unless acting under the direction of the licensed and registered design professional(s) indicated above, to alter in any way the Specifications in this project manual.

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Ryan Rhoades, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 01 11 00, 01 26 01, 01 27 00, 01 31 00, 01 32 16, 01 32 33, 01 33 23, 01 33 23.A, 01 35 05, 01 35 13.22, 01 35 23, 01 35 73, 01 35 91, 01 40 00, 01 42 00, 01 50 00, 01 57 19.11, 01 57 19.12, 01 57 23, 01 65 50, 01 67 00, 01 71 21, 01 73 00, 01 73 29, 01 73 40, 01 73 50, 01 73 60, 01 74 19, 01 77 00, 01 78 23, 01 78 23.A, 01 79 00, 01 79 00.A, 01 81 13, 01 91 14
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Juan Coronado, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 01 40 00.A, 01 45 25, 01 73 20, 01 81 10
	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Jennifer Stillman, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Section 01 61 03, 01 74 15, 01 75 00

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Josie Brainard, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 01 81 33

END OF SEALS AND SIGNATURES

TABLE OF CONTENTS

DIVISION 01 — GENERAL REQUIREMENTS
01 00 00 - DIVISION 01 DIVIDER
01 11 00 - SUMMARY OF WORK
01 26 01 - CONTRACT MODIFICATION PROCEDURES
01 27 00 - DEFINITION OF CONTRACT LINE ITEMS
01 31 00 - PROJECT MANAGEMENT AND COORDINATION
01 32 16 - CONSTRUCTION SCHEDULE
01 32 33 - PHOTO DOCUMENTATION
01 33 23 - SUBMITTAL PROCEDURES
01 33 23.A - SubmittalList UPDATED
01 35 05 - ENVIRONMENTAL PROTECTION AND SPECIAL CONTROLS
01 35 13.22 - ARCHEOLOGICAL PROTECTION
01 35 23 - SAFETY REQUIREMENTS
01 35 73 - DELEGATED DESIGN PROCEDURES
01 35 91 - HISTORIC PRESERVATION TREATMENT PROCEDURES
01 40 00 - QUALITY REQUIREMENTS
01 40 00.A - Statement of Structural Tests and Special Inspections
01 42 00 - REFERENCE STANDARDS
01 45 25 - TESTING CONCRETE STRUCTURES FOR WATERTIGHTNESS
01 50 00 - TEMPORARY FACILITIES AND CONTROLS
01 57 19.11 - INDOOR AIR QUALITY MANAGEMENT
01 57 19.12 - NOISE AND ACCOUSTICS MANAGEMENT
01 57 23 - TEMPORARY STORM WATER POLLUTION PREVENTION
01 61 03 - EQUIPMENT - BASIC REQUIREMENTS
01 65 50 - MATERIAL TRANSPORT - HELICOPTERS
01 67 00 - PRODUCT REQUIREMENTS
01 71 21 - SPECIALTY ENGINEERING REQUIREMENTS
01 73 00 - WINTER OPERATIONS
01 73 20 - OPENINGS AND PENETRATIONS IN CONSTRUCTION
01 73 29 - CUTTING AND PATCHING
01 73 40 - EXECUTION
01 73 50 - TRAIL ACCESS MANAGEMENT
01 73 60 - CONSTRUCTION CAMPS AND STAFF HOUSING
01 74 15 - DISINFECTION OF FACILITIES
01 74 19 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
01 75 00 - CHECKOUT AND START-UP PROCEDURES FOR WATER TREATMENT FACILITIES
01 77 00 - CLOSEOUT PROCEDURES
01 78 23 - OPERATION AND MAINTENANCE DATA
01 78 23.A - CloseoutOperationMaintenanceRequirements UPDATED
01 79 00 - DEMONSTRATION AND TRAINING
01 79 00.A - Demonstration and Training Requirements Table UPDATED
01 81 10 - WIND AND SEISMIC DESIGN CRITERIA
01 81 13 - SUSTAINABLE DESIGN REQUIREMENTS FOR NON-LEED™ PROJECTS
01 81 33 - CYBER SECURITY REQUIREMENTS
01 91 14 - TOTAL BUILDING COMMISSIONING

This page intentionally left blank.

DIVISION 01

GENERAL REQUIREMENTS

This page intentionally left blank.

SECTION 01 11 00 - SUMMARY OF WORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes the following:

1. Work Covered by Contract Documents
2. Work Phases
3. Work Under Other Contracts
4. Government-Furnished Materials
5. Contractor Use of Site
6. Public Use of Site
7. Occupancy Requirements for Buildings
8. Conduct of Operations
9. Work Restrictions
10. Special Construction Requirements
11. Soils Investigation Report
12. Additional Reports

1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Location: The project is generally located at Grand Canyon National Park on the North Rim and innercanyon between the North Rim and Cottonwood Campground. The project areas and associated project volumes are as follows:

1. Project Area 1 – North Rim
   a. Volume 01: Pipelines – North Rim. A utility/road/trail corridor extending between the North Rim Tank Farm, near the existing Emergency Medical Services (EMS) facility on State Route 67, and the Ballfield administrative site between the concessionaire's housing area and the Grand Canyon North Rim Lodge parking lot.
   b. Volume 02: Pipelines – North Rim to Roaring Springs Pumphouse. A utility corridor (two directionally drilled bores) extending from the existing Ballfield administrative site to the Roaring Springs Pumphouse within the innercanyon.
   c. Volume 03: Pipelines - North Rim to Supai Tunnel. A utility corridor extending along the existing waterline alignment between the North Rim Tank Farm and Supai Tunnel, located near the North Kaibab Trail approximately three (3) miles from the trailhead parking area.
   d. Volume 06: North Rim Water Treatment Plant. The North Rim Water Treatment Plant (NR WTP) site is located adjacent to the North Rim Tank Farm, southeast of the EMS building.
   e. Volume 07: NOT USED.
   f. Volume 08: North Rim Wastewater Collection System. The wastewater collection system improvements are generally located in three areas on the North Rim: NPS Administration (Region 1), Campground (Region 2), and Concessionaire Administration (Region 3).
   g. Volume 09: Power Distribution – North Rim. The power distribution system improvements include localized improvements throughout the developed area of the North Rim peninsula including areas near the Grand Canyon Lodge,

Concessionaire Area, Administration area, Campground/Gas Station area, new NR WTP site, the Power Distribution Center (PDC) and additional sites located throughout the North Rim.

2. Project Area 2 - Innercanyon
    a. Volume 04: Pipelines – Roaring Springs Cave to Roaring Springs Pumphouse. A utility corridor along the existing pipeline alignment between the Roaring Springs Cave to the Roaring Springs Pumphouse.
    b. Volume 05: Roaring Springs Pumphouse. The Roaring Springs Pumphouse is located adjacent to Roaring Creek and Roaring Springs helipad, upstream of the confluence with Bright Angel Creek.
    c. Volume 10: Power Distribution – North Rim to Roaring Springs Pumphouse. A utility corridor along the existing overhead power alignment between the Grand Canyon Lodge parking area to Roaring Springs Pumphouse.
    d. Volume 11: Power Distribution – Roaring Springs Pumphouse to Manzanita. A utility corridor following the existing Transcanyon Waterline (TCWL) alignment along Bright Angel Creek and the North Kaibab Trail, from the Roaring Springs Pumphouse to Manzanita.
    e. Volume 12: Power Distribution – Cottonwood Campground. Located at Cottonwood Campground, adjacent to the Ranger Station, which is approximately 6.8 miles from the North Rim via the North Kaibab Trail.

B. The Work is described within thirteen project volumes as follows:

1. Volume 00: General
    a. The item consists of general sheets, notes, abbreviations, symbology that apply to all volumes and standard details and schematics that are referenced in multiple volumes. Includes areas available for designated construction camps, staging, limits of construction, and helicopter operations and landing zones.
2. Volume 01: Pipelines – North Rim
    a. This item consists of approximately 5,500 LF of 4-IN raw water pipeline, 5,500 LF of 4-IN finished water pipeline, and 8,420 LF of 2 IN conduit and fiber optic cable and appurtenances on the North Rim. Work includes a new 20 FT wide, 3,500 LF gravel access road from the Ballfield to near the intersection of Norton Court and SR67. Work includes rough grading of the Ballfield site for construction staging of work associated with Volume 02. The raw water and finished water pipelines are routed from the Ballfield site to the new North Rim WTP/Tank Farm site along the proposed access road and existing trail and utility corridors. The work includes a new 3-phase underground power feed to the North Rim Wastewater Treatment Plant and new 3-phase underground power feed to the new North Rim Water Treatment Plant in existing utility or roadway corridors. The fiber optic is routed from the new overhead power (Volume 10) at the Grand Canyon Lodge parking along Bridle Path to the new North Rim WTP site (Volume 6).
3. Volume 02: Pipelines – North Rim to Roaring Springs Pumphouse
    a. This item consists of approximately 9,350 LF of 4.5-IN raw water pipeline and 9,350 LF of 4.5-IN finished water pipeline installed in two directionally drilled bores with the following telescoping steel casing (approximately 160 LF of 20-IN conductor casing, approximately 3,000 LF of 13 3/8-IN surface casing, approximately 17,700 FT of 9 5/8-IN intermediate casing, and approximately 18,700 LF of 7-IN protection casing) drilled from the Ballfield site bore entry pit to the Roaring Springs Pumphouse bore exit.
4. Volume 03: Pipelines – North Rim to Supai Tunnel
    a. This item consists of approximately 2,930 LF of 2-IN spoolable composite pipe and 520 LF 1.5 IN steel finished water pipeline and appurtenances from the finished water tanks at the NR WTP site to the Supai Tunnel rest area on the

North Kaibab Trail. A new bottle filling station and drain are included at Supai Tunnel rest area. Work includes rock anchors to secure the 2-IN spoolable composite pipe to the canyon wall every 275 FT.

5. Volume 04: Pipelines – Roaring Springs Cave to Roaring Springs Pumphouse
   a. This item consists of anchors to support the existing pipe between Roaring Springs Cave and Roaring Creek, and replacement of the existing power and communication cables between Roaring Springs Pumphouse and Roaring Springs Cave. Additional work includes removal and replacement of the overflow box near the cave outlet.
6. Volume 05: Roaring Springs Pumphouse
   a. This item consists of a new pumping system to convey water from the Roaring Springs Cave to the new raw water tanks located at the NR WTP. The interior of the existing Roaring Springs Pumphouse (RSPH) will be demolished, including all piping, equipment, electrical, and most interior walls. New automatic backwashing strainers, horizontal multi-stage centrifugal pumps, a new liquid hypochlorite disinfection system, and an outdoor valve vault to house isolation valves will be installed. Two new pipes from the bore exit (Volume 02) to the RSPH will be installed. A finished water storage tank will be installed on the exterior to the building which will be connected to the existing TCWL for supply to Manzanita/Cottonwood. Electrical work consists of a new electrical transformer and motor control center to accommodate the new equipment, and new variable frequency drives (VFDs) to accommodate the larger pumps. Site work includes new transformers, buried underground electrical, new yard piping, a new finished water tank, slope stabilization, and improvements to the exterior deck and supporting crib wall.
7. Volume 06: North Rim Water Treatment Plant
   a. This item consists of a new North Rim WTP that will treat the water source from the Roaring Springs Cave with membrane ultrafiltration and chemical disinfection. The North Rim WTP consists of a 6,200 SF building that includes administration, process, electrical, mechanical, and indoor/outdoor maintenance areas. Finished water pumps will be installed in the new North Rim WTP to replace the functionality of the existing North Rim Pumphouse. Two new welded steel raw water tanks will be installed adjacent to the facility and the two existing finished water tanks will be rehabilitated.
8. Volume 07: NOT USED
9. Volume 08: North Rim Wastewater Collection System
   a. This item consists of replacement of approximately 11,622 LF of sanitary sewer piping and 46 manhole replacements at various areas on the North Rim including the Grand Canyon Lodge area.
      1) Region 1 (NPS Administration): Collection system includes approximately 5,538 LF replacement of sanitary sewer pipe with new PVC sanitary sewer pipe, ranging in size from 6 IN to 10 IN diameter, 15 concrete manhole structures.
      2) Region 2 (Campground): Collection system includes approximately 3,966 LF replacement of sanitary sewer pipe with new PVC sanitary sewer pipe, ranging in size from 4 IN to 8 IN diameter, 22 concrete manhole structures. Replacement of building laterals for 10 cabins and 1 comfort station. The laterals will be replaced from the cleanout to the sewer trunk.
      3) Region 3 (Concessions Administration): Collection system includes approximately 2,118 LF replacement of sanitary sewer pipe with new PVC sanitary sewer pipe, ranging in size from 6 IN to 8 IN diameter, 9 concrete manhole structures.
      4) Lift Station #2: Purchase and replacement of the pumps in existing Lift Station #2.

10. Volume 09: Power Distribution – North Rim
    a. This item consists of the replacement of 8 existing transformers and appurtenances, installation of 1 new transformer, replacement of 5 existing junction enclosures, replacement of 1 existing shunt reactor, and installation of 4 new pad mounted switches at the North Rim Power Distribution Center and select locations throughout the North Rim. A 750kVA standby generator will be installed in the existing Power Distribution Center and connected to the existing generation switching scheme. An above ground 20,000 gallon diesel fuel storage tank will be installed at the site adjacent to the existing generation building with a new 400 gallon day tank replacing the existing day tank inside the generation tank. The work also includes the replacement and installation of a new 4000 Amp, 480V switchgear in the generator building. The work includes a new power feed to the NR WWTP and new NR WTP.
11. Volume 10: Power Distribution – North Rim to Roaring Springs Pumphouse
    a. This item consists of approximately 5,325 LF of three-phase power line, fiber optic, and appurtenances following the existing overhead power utility corridor. Steel power poles will be installed on micropile foundations at four locations in the vicinity of the existing poles: North Rim Bench, Upper Bench, Lower Bench, and Roaring Springs Pumphouse. The overhead conductor will utilize a combination of bare aluminum and insulted aluminum conductor based on the steel pole configuration. New cable marker balls will also be installed along the entire overhead line route to help improve helicopter safety. Fiber optic will be installed in the ground neutral line from the North Rim to Roaring Springs Pumphouse.
12. Volume 11: Power Distribution – Roaring Springs Pumphouse to Manzanita
    a. This item consists of the installation of approximately 3,600 LF of three-phase armored power cable following the existing waterline and trail corridor from Roaring Springs Pumphouse to Manzanita. Installation of the cable will include different methods depending on the terrain, ranging from rock trenching, scree cover, and rock anchor support.
13. Volume 12: Power Distribution – Cottonwood Campground
    a. This item consists of a new photovoltaic solar power installed on the roof of a new shade canopy at Cottonwood Campground. Work includes a new outdoor rated battery enclosure, DC/AC inverter, circuit breaker, and connections to the existing Cottonwood ranger station service entrance. New 1000V rated power cables and conduit will be installed between the shade structure and battery enclosure.

C. Project will be constructed under a single prime contract.

D. All construction staff, materials, and equipment for Innercanyon construction will require transportation into and out via helicopter or foot.

1. Contractor shall provide helicopter operations. Use of Government-owned helicopters by contractor staff will not be permitted. See Section 01 65 50 “Material Transport” for requirements on the use of helicopter transportation.
2. No mechanized ground transportation is permitted on innercanyon trails.

E. The Government has selected the following products and services to be sole-sourced for the Project. No product substitutions will be allowed.

1. Volume 03: Spoolable Composite Pipe – FlexSteel
2. Volume 05: Multi-Stage Centrifugal Pumps - Flowserve
3. Volume 06: Ultrafiltration (UF) Membrane Systems – Veolia/Suez
4. Volume 09: Standby Generator – Caterpillar

5. Various volumes – Programmable Logic Controllers (PLCs) – Automation Direct Do-more
6. Various volumes – Water quality instruments – Hach

1.3 WORK PHASES

A. The Work shall be conducted with the following phasing requirements:

1. Volumes 01, 02, 04 and 05 must be ready for use within four hundred and twenty (420) calendar days after receiving Notice to Proceed, to provide the delivery of water to the existing North Rim Tank Farm.
2. Volume 10 work does not have to be done in the same construction season as the rehabilitation of the Roaring Springs Pumphouse; it can occur during or after Volume 5 work. If the work occurs after the Volume 5 substantial completion, it must be completed when the Roaring Springs Pumphouse is offline from November 1 through May 14.

B. Refer to drawings for additional phasing requirements pertaining to sequence of construction and utility operations. Incorporate phasing requirements on drawings into the construction schedule.

C. Tree removal for all packages shall occur between September 1 and March 31 to avoid bird nesting. Work outside this window may be requested and will require a NPS wildlife bird biologist to perform a nest survey. Surveys shall be conducted no more than five days prior to scheduled activity. Refer to Specification 01 35 05 for additional Wildlife Protection Requirements.

D. The contractor shall prepare a Concrete Phasing Plan that identifies the planned sequences of concrete pours for the Volume 6-North Rim WTP. The Concrete Phasing Plan shall evaluate pour sequences relative to the structural details and specifications, including testing specifications. The Concrete Plan shall be submitted 30 days prior to show drawings submittals for the reinforcement or similar structural submittals. The Concrete Phasing Plan is to be included with the Construction Schedule and defined in 01 32 16 Construction Schedule.

1.4 WORK UNDER OTHER CONTRACTS

A. General: Cooperate and coordinate with concurrent active construction contracts contractors so work on those contracts may be carried out smoothly, without interfering with or delaying work under this Contract. Coordinate Work of this Contract with work performed under separate contracts. Some of the concurrent contracts will have a direct and indirect potential impact to this project schedule; successful bidder is encouraged to communicate and coordinate early in schedule development and finalization of baseline schedule.

B. Existing NPS Operations: Contractor helicopter operations will be on or adjacent to the South Rim helibase, North Rim helipad, Roaring Springs helipad, Manzanita helipad, and Cottonwood Campground helipad used by NPS for operations including transporting personnel and materials, emergencies, and search and rescue missions. Contractor shall yield all work to the needs of helibase operations at any time. Helibase operations will have final authority to stop construction work in all areas including during helicopter operations and emergencies, if required. Grand Canyon National Park Helibase is solely responsible and legally bound to managing the airspace that is required to support the project areas via innercanyon. Aviation operations are subject to conforming to internal policy NPS Reference Manual 60 and GRCA's Internal Aviation Management Plan (IAMP). See Specification Section 01 65 50 – Material Transport-Helicopters for the details associated with use and coordination.

C. Facilities Maintenance Division (FMD) operations: The South Rim contractor support facilities are located adjacent to the South Rim Facilities Maintenance Division (FMD) for Grand Canyon National Park. Contractor shall conduct work to have minimal impact to these facilities or operations.

D. Other NPS Operations: NPS conducts other Park operations for official and visitor purposes. Contractor shall conduct work to have minimal impact on these operations and visitor enjoyment. Temporary safety shutdowns in specific geographic location of contractor work may occur due to emergency park operations at any time during the contract period in and or outside of the limits of construction.

E. Concurrent Work: Government has awarded separate contract(s) for the following construction operations at Project site. Those construction operations potentially will occur simultaneously with portions of work under this Contract.

1. GRCA 190083 TransCanyon Waterline Replacement: A separate contract has been awarded for the replacement of the TransCanyon Waterline (TCWL). The work includes improvements to the raw water, potable water, electrical and communications systems within the Innercanyon and South Rim. Specific areas of TCWL project geographic and utility operations interface includes: 1) Rehabilitation of the TCWL between Roaring Springs Pumphouse and Manzanita, and 2) water distribution improvement within the Cottonwood Campground. In addition, the TCWL Contractor will use the South Rim Contractor Support Facilities for staging and a base for helicopter operations. The Contactor for Rehabilitate North Rim Infrastructure shall coordinate Innercanyon work with the NPS and defer helicopter use until completion of the TCWL Replacement project. Currently, the substantial completion date for TCWL Replacement project is summer 2027.
2. GRCA 257282 South Rim Wastewater Treatment Facilities: This project will replace the South Rim Wastewater Treatment Plant with a new facility. The existing plant services all visitor, resident staff housing, lodging, and support facilities at the South Rim. Work may coincide with deliveries, staging, use of Helibase at the South Rim and South Rim contractor support facilities. Planned construction period of 2027 to 2030.
3. Replacement of Overhead Power from Hopi Point to Havasupai Gardens: Will include helicopter flights from the South Rim Helibase and Contractor Staging. Planned construction period of 2026 to 2029.
4. North Rim Temporary Wastewater Treatment Plant
    a. This project will install a temporary wastewater treatment plant to the north of the permanent wastewater treatment plant that was damaged in the Dragon Bravo Fire. Estimated construction period of performance October 2026 through June 2027. Work will be local to the existing WWTP area but may utilize some of the same construction staging areas.
5. Fire Recovery Operations: Hazard tree and debris removal from fire recovery operations on the North Rim and around the WWTP; estimated period of performance April 2026 through November 2028.
6. Rehabilitation of water, power, wastewater infrastructure damaged in fire; estimated period of performance October 2026 thru December 2026.
7. Replacement of SR67 from Entrance Gate to Grand Canyon Lodge:
    a. The construction approach for the SR67 project has identified the need for a temporary asphalt batch plant with a preliminary location being identified as Marble Flats. This site is available for Contractor use as a temporary batch plant if needed. Road work likely in two phases- stabilization and then complete rehabilitation. Planned construction period of 2026 to 2029. However, work will not begin until this project is complete.

8. Aramark CFIP Projects to be completed prior to the North Rim seasonal opening in the spring of 2027 (projects are tentative):
    a. CFIP Project #1: Rehabilitate/Remodel EDR, Dorm and Staff Housing
    b. CFIP Project #2: Food and Beverage at General Store
    c. CFIP Project #3: Motel Renovation and New Hostel Rooms/Admin Space
    d. CFIP Project #4: ABA Conversion of Two Cabins
    e. CFIP Project #5: Rehabilitate (4) Cabins Near General Store
    f. CFIP Project #6: Rehabilitate the Historic Storage/Linen Cabin
9. North Rim Permanent Wastewater Treatment Plant
    a. This project provides a permanent replacement of the existing NR WWTP that was destroyed during the 2025 Dragon Bravo fire. Once the permanent NR WWTP is operational and online, the North Rim Temporary Wastewater Treatment Plant will be decommissioned and removed from the site. Construction of the North Rim Permanent Wastewater Treatment Plant is planned for Spring 2029 through Fall 2031.

1.5 CONTRACTOR USE OF SITE

A. Limits of Construction (LOC): Contractor shall have limited use of site for construction operations.

1. Limit use of premises to areas within the Limits of Construction indicated on drawings. Do not disturb portions of the site beyond these areas. The Contractor shall clearly delineate the boundary of the Limits of Construction (LOC) prior to commencing work using fencing, flagging or other approved methods. Delineation shall be maintained for the duration of construction and shall correspond to the LOC shown on the drawings. Erosion and sediment control features (silt fence, wattles, etc) may be used to satisfy this requirement where they are installed at the LOC boundary.
2. Contractor may request deviations from the Limits of Construction but approval is not guaranteed. Generally, the approval process once all information has been provided is up to 90 days or longer if outside agency coordination is required.

B. Equipment and Material Staging: Confine storage of materials to staging areas shown on plans.

1. Equipment and material staging areas are intended for deliveries and storage of tools and equipment, material stockpiling, construction equipment parking, construction vehicle parking, construction administration trailers, and similar.
2. Confine stockpiling to equipment and material staging areas. Small stockpiles within limits of construction for use in localized trenching work is acceptable.
3. Areas within the limits of construction that are not designated for equipment and material staging shall not be used for this purpose.
4. Equipment and material staging areas should not be used for construction camps including camping units, trailer complexes for living areas, private vehicle parking, and similar.
5. Prior to work commencing in each geographic area within limits of construction, the contractor will site walk each site with NPS COR, CMR and NPS vegetation specialist to identify vegetation to be salvaged. Any vegetation that can't be salvaged will be disposed of outside of the park boundaries. Additional resources specialists will survey areas pending sensitivity of each location.

C. Parking:

1. Private vehicles for construction personnel shall be parked at designated construction camps or outside the Park.
2. Contractor shall provide a shuttle bus for construction personnel from the designated construction camps and other off-site lodging locations.
3. Parking for marked construction vehicles shall be limited to the following areas between April 15 and November 30 (visitor season):
    a. Volume 01: Ballfield staging area
    b. Volume 06: NRWTP staging area
    c. Volume 09: Power Distribution Center - up to 5 vehicles
    d. Helibase Road trailer, staging and construction camp area, adjacent to the existing North Rim WWTP. This can only be used once the contractor for construction of the Temporary WWTP has completed the work.
4. Marked construction vehicles can be parked at the North Kaibab trail parking lot, Grand Canyon Lodge visitor overflow parking, North Rim Campground, NPS administrative facilities, and Concessionaire administrative facilities for no more than 2 hours and up to 2 vehicles at any time in a single location.

D. Preservation of Natural and Manmade Features:

1. Prevent damage to natural surroundings. Restore damaged areas, repairing or replacing damaged trees and plants with like species, at no additional expense to the Government.
2. Provide temporary barriers to protect existing trees and plants and root zones.
3. Provide temporary construction protection mats for all areas where construction may impact a wetland, saturated conditions, or other sensitive areas. All areas that are disturbed must be restored per Specification 01 35 05.
4. Do not remove, injure, or destroy trees or other plants without prior approval. Consult with Contracting Officer (CO) and remove agreed-on roots and branches that interfere with construction. Contractor shall follow Grand Canyon vegetation guidelines related to pruning and root cutting.
5. Do not fasten ropes, cables, or guys to existing trees.
6. Carefully supervise excavating, grading, filling, and other construction operations near trees to prevent damage.
7. All trees over six (6) IN DBH (diameter breast height) shall be salvaged, transported and stacked in four (4) FT sections at a Native American Tribal location at no cost to tribes within 150-mile radius of removal site. Prior approval of tribal locations for delivery and approval by tribes must be demonstrated prior to transporting outside the NPS boundaries. All slash to be removed by contractor and disposed of outside of the park.
8. Maintain work buffer of at least 5 FT from historic features including rock walls along trails and parking areas. Historic rock curbing and sidewalks in front of General Store will be protected in place.
9. Repair and restore trails and ground to original condition.
10. NPS will provide native seeds for contractor re-seeding of post construction disturbed areas. The seeds need to be collected by the NPS and written request for the seed will need to be provided twelve (12) months prior to scheduled re-seeding.
11. Any proposed use of herbicide or pesticide must be approved by NPS CO as part of existing approved products and licensed applicators.
12. Dispose of unsuitable excavated materials for backfill and spoils at the Ballfield per Volume 01 Drawing 013L101.

E. Driveways and Entrances: Keep driveways, trails, and entrances serving NPS and concessionaire premises clear and available to Government employees, and emergency vehicles at all times. The EMS building driveways and entrances shall not be used by the Contractor. Do not use for parking or storage of materials.

1. Schedule deliveries to minimize use of driveways and entrances.
2. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

F. Construction Camp: Establishment of a camp will be permitted at Lindberg Hill, Dusty Road, near the North Rim Temporary WWTP (once the contractor for construction of the Temporary WWTP has completed the work), Cottonwood Campground (use for all innercanyon work), and South Rim Contractor Support Facilities as shown on drawings and specified in 01 73 60 “Construction Camps”. Provide utilities as specified in Section 01 50 00 "Temporary Facilities and Controls". Adhere to park rules, animal deterrent policies, cleanliness policies, and return site to previous condition at completion of work.

G. Hauling Restrictions: Comply with legal load restrictions on hauling of materials. Load restrictions on park roads are identical to state load restrictions with such additional regulations as may be imposed by the Park Superintendent. Information regarding rules and regulations for vehicular traffic on park roads may be obtained from the Office of the Park Superintendent. A special permit will not relieve Contractor of liability for damage which may result from moving of equipment.

1. State Route 67 (SR67) is heavily used for visitor traffic and park operations. Contractor shall not impede flow of traffic for equipment and material deliveries at any time, unless otherwise approved by the Contracting Officer.
2. Contractor shall use the North Rim park entrance gate and South Rim park entrance gate. Contractor personnel providing identification of employment with the Contractor or in a marked vehicle are not required to pay Park entrance fees. Contractor shall not use other entrance roads into the Park.
3. South Rim Shuttle Bus Road near the Contractor Support Facilities on the South Rim is the primary access route for Facility Maintenance Division (FMD), helibase, and wildland fire staff and associated activities. Contractor shall not use Shuttle Bus Road or associated parking facilities for construction purposes without prior approval from the Contracting Officer.

H. SR67 Restrictions: Contractor shall identify jurisdictions, load restrictions, permit requirements, time and calendar restrictions as required for use of SR67. Contractor shall be responsible for obtaining any permits with Arizona Department of Transportation and Coconino County as required.

1.6 PUBLIC USE OF SITE

A. The contractor will attempt, to the extent possible, to schedule any needed trail, rest area, campground, or campsite closures during lower visitation times (e.g., winter, outside peak rim-to-rim times) to minimize visitor impacts.

B. Road restrictions (one lane must be passable with traffic control) may be permitted within park boundaries, when required, upon specific approval of Contracting Officer for a maximum of 8 hours within a single day.

C. The following areas will be closed to the public for the construction durations identified below. Contractor shall submit a trail, rest area, campground, camp site closure plan as part of the construction schedule for review and approval by the Contracting Officer prior to work commencing. Refer to 01 73 50 for additional requirements. Request for closure is based on accepted baseline schedule six months or more prior to start of planned work.

1. The North Kaibab Trail between the Redwall Bridge and Manzanita will be closed when the directional drilling is exiting the rock face near the Roaring Springs Pumphouse.
2. The North Kaibab Trail between the North Rim and Manzanita will be closed during a portion of overhead power installations when the work is overhead or in the vicinity.
3. A portion of the Supai Tunnel rest area will have a localized closure during construction activities for the water spigot, bottle fill station, and associated piping. The North Kaibab Trail will remain open through this area.
4. The North Kaibab Trail will be closed for the pipe installation from the North Rim to Supai Tunnel.
5. Temporary closure of the North Kaibab Trail will be required during installation of the power from Roaring Springs Pumphouse to Manzanita. It is anticipated this closure will only be required when the alignment follows the trail.
6. Cottonwood Campground will have a localized closure during innercanyon construction activities to accommodate a construction camp. The North Kaibab Trail will remain open through this area.
7. A portion of the North Rim Visitor Center Parking Lot will be closed during construction activities within the limits of construction of the North Rim overhead power bench. Traffic will be routed through other areas of the parking area.
8. The Bridle Path follows SR67 and connects the Grand Canyon Lodge with the North Rim Campground, the NPS Administrative Offices, and the North Kaibab Trailhead. Volume 01: Pipelines – North Rim is located within a portion of the Bridle Path. Bridle Path will be closed during the construction. Traffic will be rerouted to the Transept Trail.
9. The Transept Trail links the northern portion of Bridle Path, the NPS Administrative Offices, and the Grand Canyon Lodge and passes nearby the Ballfield site. The Transept Trail will remain open to the public.

1.7 OCCUPANCY REQUIRMENTS FOR BUILIDINGS

A. Existing Buildings

1. Partial Government Occupancy: Government will occupy premises during entire construction period, with exception of areas under construction. Cooperate with Government during construction operations to minimize conflicts and facilitate Government usage. Perform Work so as not to interfere with Government's operations. Maintain existing exits, unless otherwise indicated.
2. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from Contracting Officer.

1.8 CONDUCT OF OPERATIONS

A. Contractor shall conduct his operations in conformance with rules and regulations promulgated by the Secretary of the Interior for the National Park Service, and applicable park rules and regulations prescribed by Park Superintendent.

B. No signs or advertisements (except those specified herein) shall be displayed on the construction site or within the park unless approved by the Contracting Officer.

1.9 WORK RESTRICTIONS

A. On-Site Work Hours:

1. North Rim: Work on the North Rim shall be generally performed during the hours of 6 a.m. to 6 p.m., Monday through Sunday. Work in the Campground and near the Grand Canyon Lodge during the weekends will not start until 9 a.m.
2. Innercanyon: Work in the Innercanyon shall be generally performed during the hours of 6 a.m. to 6 p.m., Monday through Sunday.
3. Project helicopters will generally fly between 8:00 (power on) a.m. and 5:00 p.m. (shut down) each day with generally no flights between 5 p.m. and 8:00 a.m. In no case will flight occur before 1/2 hour prior to sunrise and 1/2 hour after sunset: Instrument Flight Rules (IFR) are prohibited. Deviations from specified flight times must be approved by the South Rim Helibase Manager and Contracting Officer.
4. Night Work Hours: Work outside established work hours can be requested by contractor.
    a. Volume 02: Pipelines – North Rim bore drilling operations is pre-approved for 24-hour operations.
    b. Innercanyon night work will not be approved due to wildlife and/or visitor services.

B. Work on Federal holidays may not be performed without prior consent from the Contracting Officer. Submit requests 5 business days in advance of the work to the Contracting Officer for approval.

C. Existing Utilities

1. Utility Locating:
    a. Notify Contracting Officer and utility companies of proposed locations and times for excavation. NPS will not be responsible for clearance or locating NPS owned utilities. Contractor shall utilize a third party locator and potholing services as required to clear utilities prior to excavation.
    b. Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs, repair utility at no additional expense to the Government.
    c. If damage occurs to an unknown utility, repair utility. An equitable adjustment will be made in accordance with the Changes clause of the contract.
2. Maintenance of Utility Operations
    a. Roaring Springs Pumphouse
        1) Once Volume 05 work is substantially complete, RSPH must remain operational between May 15 and October 31 each year.
        2) Overhead power to RSPH must remain operational between May 15 and October 31 each year.
    b. North Rim Water Distribution System
        1) North Rim Pump Station and water distribution system must remain operational at all times of year.
        2) Two finished water storage tanks shall remain operational at full capacity between November 1 and May 14 due to winter storage requirements on the North Rim.
        3) One finished water storage tank shall remain operational between May 15 and October 31 while the other finished water tank is off-line for rehabilitation.
        4) Authorized restricted use of water and power for North Rim reference 01 50 00-1 section 1.3.
        5) The existing standby generators at the Power Distribution Center must remain operational at all times of year.
    c. North Rim Wastewater Collections
        1) The main North Rim wastewater collections and temporary treatment shall remain operational between April 1 and October 31 each year.

Sewer manholes, gravity sewer segments and lift station shall remain operational between by bypassing and pumping flows as necessary.

2) The collection system at the Administration area and winter treatment system (septic system) shall be operational from November 1 to March 31 each year.

3. Existing Utility Interruptions:
   a. Do not interrupt utilities serving facilities occupied by Government or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:
   b. Notify Contracting Officer not less than fourteen (14) days in advance of proposed utility interruptions.
   c. Do not proceed with utility interruptions without Contracting Officer's written permission.
   d. Hours and length of Utility Shutdowns: to be approved on case-by-case basis through notification and approval by Contracting Officer.

D. Mule Train Routes: Anticipate approximately 2 mule trains per day from North Rim to Supai Tunnel along the North Kaibab Trail between May 1 and October 31. The Contractor will be required to temporarily stop construction activities to allow mule trains to pass the work area unless the North Kaibab Trail is closed for construction as detailed in Specification Section 01 73 50.

E. Communications equipment proposed by type, purpose and geographic location to include inside the limits of construction to the CO in to ensure that there is no communications conflict with NPS operations. This includes but is not limited to radio repeaters, antennas, handheld radios, aviation radio systems, internet equipment that uses cellular and satellites to communicate.

F. Mechanical equipment will have all industry available noise damping and spark arrestor features to mitigate the decibel output of equipment. Failure to provide such features may result in the NPS CO directing non-use of specific equipment for non-compliance until the requirement is made for the specific equipment.

G. Nonsmoking Building/Tobacco Use/Vaping: Smoking is not permitted within building or within 25 feet of entrances, operable windows, or outdoor air intakes.

1. Smoking is not permitted in buildings.
2. Due to concern of wildland fires, smoking is allowed only in a single designated outside area of no larger than 500 sf within designated construction staging areas. This area shall be clear of exposed vegetation and have adequate means for disposal of trash and flammable materials.

1.10 SPECIAL CONSTRUCTION REQUIREMENTS

A. Project Website: A project website administered by NPS CMR will be used for purposes of managing communication and documents during construction stage.

1. See Section 01 31 00 "Project Management and Coordination" for requirements on using Project Website.

B. The A/E has obtained the Approvals to Construct (ATC) for water system work through Arizona Department Environmental Quality (ADEQ), reference 01 31 00-10 1.9/B/1 for list of specific ATCs. The ATCs issued by ADEQ expire within one (1) year from the issue date for initiation of construction and notification for each individual ATC scope permit. Given the time frame from issue date of ATC to construction contract award and notice to proceed (NTP), the ATCs will be

near expiration. The construction contractor will need to upon issuance of NTP submit an extension request to ADEQ for each individual ATC to include the extension request fee. Failure to request extension prior to expiration will result in the construction contractor having to submit a full new request for the ATC. Generally the time extension is half or less the cost of the original ATC permit and less of an administrative requirement. This can be referenced on the public ADEQ website: **DW Engineering Review | Facilities Review Application | ADEQ**. Contractor will be responsible obtaining the AOC (Approval of Construction) post construction of each ATC, paperwork and fees.

C. The North Rim has a seasonal shutdown of visitor services that varies each year. Typically the shutdown to overnight visitors begins on October 15 and the shutdown to day visitors begins on November 30. The North Rim typically reopens to visitor services on or around May 15. Vehicle traffic is often impassable due to snow accumulation and other weather events on SR67 from Jacob Lake to the North Rim Lodge and on other North Rim roadways from first Monday after the first full weekend in December until May 15. The contractor shall maintain access at all times if working at the North Rim during this period. Snow removal of SR67 and Park roads will not be provided by NPS or the ADOT and will be the responsibility of the contractor. ADOT does not maintain the only road access to North Rim, Grand Canyon National Park (project location), SR67 in the winter from the first Monday of the first weekend in December through May 15; reference 01 73 00 Winter Operations.

D. The Contractor shall not construct permanent structures or install permanent utilities at staging areas or Contractor camps unless specifically shown in the Contract Documents or authorized in writing by the Contracting Officer. Any proposal for permanent structures or utilities at these locations must be submitted to the Contracting Officer for review and acceptance prior to construction.

E. Hazardous Materials:

1. The Roaring Springs Pumphouse was tested for Asbestos Containing Material (ACM) and Lead-Based Paint (LBP). The results and findings are contained in the report, “Asbestos Inspection and Lead Paint Screening at Roaring Springs Pump House and Day-Use Structure, North Rim, Grand Canyon, Arizona” prepared by Landmark Environmental, November 7, 2022. The report does not identify ACM within the project structures. LBP may be present on exterior wood siding within the Roaring Springs Pumphouse.
2. The existing two 2-million gallon water storage tanks were tested for Asbestos Containing Material (ACM) and Lead-Based Paint (LBP). The results and findings are contained in the report, “TM NR Finished Water Tank Coating Lead Tests” prepared by HDR Engineering, May 28, 2025. The report identified LBP within the roof coating.
3. LBP abatement or demolition activities shall conform to OSHA Lead in Construction regulations found in 29 CFR Part 1926.62 which require the employer provide worker protection.

F. The drawings survey data of features and conditions are prior to the 2025 Dragon Bravo Fire and 2026 floods that impacted some areas within this drawing set. Contractors are expected to conduct due diligence during the solicitation phase to identify any changes or conflicts between the drawings and existing conditions due to the fire and provide in writing for technical and price proposals. Contractors are expected to conduct due diligence site visit within 45 days of issuance of notice to proceed (NTP) with Contracting Officers Representative (COR) and onsite CMR to field evaluate locations impacted by the fire within scope of contract documents prior to initiation of work. Notify the Contracting Officer in writing of changes or conflicts between the drawings and existing conditions due to the fire that would impact scope, cost, or schedule.

1.11 ADDITIONAL REPORTS

A. The following reports are provided as an appendix to the project specifications. In case of conflict between report and drawings or specifications, the drawings and specifications govern.

1. Geotechnical Investigation Report (GIR); GRCA 318719 Rehabilitate North Rim Infrastructure; Grand Canyon North Rim, Arizona; prepared by GEI; August 11, 2025.
2. Geotechnical Investigation Report (GIR); GRCA 318719 Rehabilitate North Rim Infrastructure; Grand Canyon Inner Canyon, prepared by GEI; August 11, 2025.
3. Geologic Interpretive Summary Report; GRCA 318719 Rehabilitate North Rim Utility Infrastructure, prepared by GEI, March 31, 2026.

B. The following reports are provided for project reference.

1. Bore B Drilling Plan; GRCA North Rim Rehabilitation (244 pages), prepared by Geostock Sandia, April 2, 2026.
2. Asbestos Inspection and Lead Paint Screening at Roaring Springs Pumphouse and Day-Use Structure, North Rim, Grand Canyon, Arizona (29 pages). prepared by Landmark Environmental, November 7, 2022.
3. Finished Water Storage Tank Inspection - Lead Coating Tests (3 pages), prepared by HDR Engineering, July 14, 2025
4. Representative UF Membrane Skid Process and Instrumentation Diagrams (25 pages), prepared by Suez, March 29, 2021.
5. Upper Bench Power Pole Site Visit Notes (22 pages), HDR Engineering, January 31, 2025.
6. North Rim Power System Analysis Report (442 pages), prepared by HDR Engineering, March 24, 2025.
7. Intergovernmental Agreement (IGA 24-0009595-I/P0012024001153): Snow Removal and Boundary Management between ADOT and NPS (8 pages), August 28, 2024.
8. North Rim Geophysical Studies; GRCA 318719: Rehabilitate North Rim Utility Infrastructure (419 pages), Geostock Sandia, May 19, 2025.
9. Trail Management Standards for Grand Canyon National Park's Class IV Stock Trails (27 pages), dated 2007.
10. South Rim Bore Review Memorandum (5 pages), July 24, 2025 and Attachments.
    a. Drawings-
        1) 1987 Bore Design 2 (1 page)
        2) 1987 Bore Design (1 page)
        3) Transcanyon Water Transmission System Improvements Phase 1 (69 pages)
        4) GRCA_113_40078 (6 pages)
        5) GRCA_113_41098 (20 pages)
        6) GRCA_113_41098A (16 pages)
        7) Transcanyon Water Transmission System Improvements Phase 2 (11 pages)
    b. Field Notes
        1) 103431-Folder 47a (101 pages)
        2) 103431-Folder 47b (140 pages)
        3) 103431-Folder 47c (149 pages)
        4) Fieldnotes-Spring 1986 (1 page)
    c. Reports
        1) D-174 Feasibility Study (214 pages)
        2) D-385 Closeout Report (30 pages)
        3) Directional Drill Water Line (5 pages)
        4) Hydrogeologic Report South Rim Borehole (34 pages)
        5) Spe-16169-South Rim (10 pages)

d. Photos
    1) IMG_3339 (1986 Bore Exit).jpg
    2) IMG_3346 (1986 Bore Exit).jpg
    3) South Rim Bore Section.jpg

11. GRCA 318719 Rehabilitate North Rim Infrastructure REVIT drawings for Volumes 5 and 6. Drawings are for informational use only. Contractor to sign Digital/Electronic Media Release form prior to delivery.
12. GRCA 318719 Photo Logs (all photos are prior to the 2025 Dragon-Bravo Fire):
    a. Volume 01: Pipelines – North Rim, prepared by HDR Engineering.
    b. Volume 02: Pipelines – North Rim to Roaring Springs Pumphouse, prepared by HDR Engineering.
    c. Volume 03: Pipelines – North rim to Supai Tunnel, prepared by HDR Engineering.
    d. Volume 04: Pipelines – Roaring Springs Cave to Roaring Springs Pumphouse, prepared by HDR Engineering.
    e. Volume 05: Roaring Springs Pumphouse, prepared by HDR Engineering.
    f. Volume 06: North Rim Water Treatment, prepared by HDR Engineering.
    g. Volume 07: Not Used.
    h. Volume 08: North Rim Wastewater Collection System, prepared by HDR Engineering.
    i. Volume 09: Power Distribution – North Rim, prepared by HDR Engineering.
    j. Volume 10: Power Distribution – North Rim to Roaring Springs Pumphouse, prepared by HDR Engineering.
    k. Volume 11: Power Distribution - Roaring Springs Pumphouse to Manzanita, prepared by HDR Engineering.
    l. Volume 12: Power Distribution - Cottonwood Campground, prepared by HDR Engineering.
13. Grand Canyon National Park's (Park) Parkwide Spill Response Plan

C. Copies of the reports are available to all plan holders with this package.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 01 11 00

This page intentionally left blank.

SECTION 01 26 01 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of administrative and procedural requirements for contract modifications.

1.2 DEFINITIONS AND ALLOWANCES

A. Home Office Overhead: Costs incurred in support of all of a contractor's projects and not attributable to a specific job. The cost for home office overhead is only allowed as a percentage of all direct work excluding profit. The following items represent allowable home office overhead costs identified in Part 31 of the Federal Acquisition Regulation (FAR):

1. Rent
2. Utilities
3. Furnishings
4. Office equipment
5. Executive and management staff not exclusively assigned to the project
6. Support, accounting, and administrative staff
7. Preparation of cost proposals, estimating, and schedule analyses connected with Modifications
8. Estimating and preconstruction services
9. Mortgage costs
10. Real estate and corporate taxes
11. Automobile maintenance and travel costs for home office personnel
12. Home office insurances i.e. structure, automotive, umbrella, flood, etc.
13. Depreciation of equipment and other assets
14. Home office supplies (paper, staples, etc.)
15. Legal services
16. Accounting and data processing
17. Professional fees/registration

B. General Conditions (Field Office Overhead): Management and administrative costs incurred on site for the designated project. Costs associated with preparation of modifications will not be allowed. Costs for these items are to be included only in the general conditions of the modification estimate. Only in the case of a contract time extension are additional general conditions included in modifications. The following items, if applicable, are considered allowable costs for calculating General Conditions:

1. Project Manager (PM), Assistant Project Manager
2. Superintendent, Assistant Superintendent
3. Quality Control, Safety Officer, Environmental Manager, etc.
4. Helicopter Operations Coordinator, Air Base Radio Operator
5. Construction Camp Coordinator
6. Engineers
7. Travel, lodging, and per diem (as established by Federal Travel Regulations)
8. Scheduling
9. Field Office Trailers and associated temporary utilities
10. Field office supplies

a. Mailing and couriers
b. Reproduction costs
c. Storage
d. Phones
e. Computers
f. Copiers
g. Construction camp supplies.

11. Personal vehicles i.e. Superintendent Pickup trucks

C. General Requirements: Costs directly associated with the project and are necessary to perform the actual work of the modification. These costs shall be shown as direct costs in the estimate. The following items, if applicable, are considered allowable costs for calculating General Requirements:

1. Hoisting
2. Material handling
3. Temporary fencing
4. Port-a-lets
5. Trash removal, dumpsters
6. Barricades
7. Small tools
8. Safety supplies
9. Scaffolding
10. Daily cleaning
11. Traffic control
12. Temporary signage
13. Temporary heating and power

D. Personnel Costs: Costs included in the modification must only be for General Conditions staff and workers actually present and working on project site. Modification costs for salaried workers are only allowed within the structure of a maximum 60-hour week and no overtime or holiday pay will be allowed.

1. Worker Hourly Rates are costs directly associated with the individual worker and consist of the following:
   a. Base Rate: The hourly rate paid directly to the worker
   b. Labor Burden: Employer payments of all applicable burdens; includes insurance and taxes the business must pay on behalf of the worker to government entities and educational forums, such as:
      1) Social Security
      2) Medicare
      3) Workers Compensation – Policy and company calculation to be made available.
      4) Federal Unemployment Tax Act (FUTA) - Cap Rate and percentage to be proportionally allocated over one year.
      5) State Unemployment Tax Act (SUTA) - Cap Rate and percentage to be proportionally allocated over one year.
      6) Union agreement costs - Other costs required under an enforceable collective bargaining agreement.
   c. Fringe Benefits: Various non-wage compensations provided to employees such as:
      1) Health Care Insurance Premiums
      2) Cell Phone
      3) Clothing
      4) 401K and Pensions
      5) Vehicle allowances

6) Gas allowance
7) Life insurance premiums
8) Disability insurance
9) Other Fringe Benefits required under an enforceable collective bargaining agreement

E. Bonuses or Deferred Compensation: No Bonus or Deferred Compensation will be allowed within any components of pricing including Home Office Overhead, General Conditions, General Requirements, Hourly Worker Rates, or the direct costs of work.

F. General Liability Insurance: An insurance policy that protects Contractor from claims resulting from bodily injury or property damage to a third party. Include as a separate line item within all modification proposals and provide a current insurance quote upon request.

G. Performance and Payment Bonds: A performance bond is a surety bond issued by an insurance company or bank to guarantee satisfactory completion of a project. The Payment Bond guarantees the Contractor will pay the labor and material costs incurred. Banks and Insurance companies charge a premium for individual project based on a sliding scale related to the size of the project. Include as a separate line item in modification proposals and provide current company bonding rates upon request.

H. Builder's Risk Insurance: Covers the contractor's loss due to fire, high winds, or other natural forces. Not reimbursed by the National Park Service (NPS) and shall not be included in modification proposals.

1.3 MODIFICATION PROPOSAL PRICING REQUIREMENTS

A. General:

1. Proposal shall be received in the format and within the time frame specified in the Request for Proposal (RFP) letter. Costs or delays resulting from failure of contractor to submit within the time frame specified will not be compensable.
2. Proposal shall be detailed with itemized lists of equipment, materials, labor, production rates, overhead, profit, and bond markup for each item. Labor costs must be itemized by craft and hourly rate, including Fringe Benefits and Labor Burden. If the costs of Fringe Benefits and Labor Burden are not itemized, it is assumed they are included in the hourly rate shown, or contractor is not requesting reimbursement. Contractor may utilize the government provided Contractor Estimate Form, or their own form, provided that it contains the same information and level of detail as the Government's form.
3. Requests for extensions of contract time as a result of change must be justified with a Time Impact Analysis (TIA). Refer to Section 01 32 16 "Construction Schedule", for time impact analysis requirements. TIA and associated costs shall be received with the proposal by the date shown within the Request for Proposal letter. Contractor's failure to submit within the specified time frame will be construed as the Contractor waiving right for additional time and no time extension will be allowed.
4. All supporting documentation used to justify the proposed modification will be made available to the Contracting Officer (CO) upon request.
5. Contractor shall review and approve all subcontractor/supplier pricing in detail for proper format, scope, production rates, and pricing prior to submission to NPS. All delay costs associated with not reviewing and approving subcontractor/supplier pricing will be borne by the Contractor.
6. All pricing and production rates within the estimate must be based on fair and reasonable pricing and cannot include built-in contingency.

B. Labor:

1. Contractor shall estimate cost of labor by itemizing each craft involved, indicating worker hourly rate (base rate + labor burden + fringe benefits) for each and itemizing hours required for each craft directly engaged in modification work. Any work proposed requiring overtime work or premium pay shall be itemized separately. Rates shall be in accordance with the Davis-Bacon Act as incorporated herein. Labor Burden may include payroll taxes, Social Security, unemployment insurances, workers compensation insurance, Federal Insurance Contributions Act (FICA), FUTA, and other direct costs resulting from Federal, State or local laws.
2. Itemize labor costs for equipment operators separate from equipment costs.
3. Labor cost for foremen shall only be costs for related work required for the modification.

C. Materials:

1. No markup shall be applied to any material provided by NPS.

D. Equipment:

1. Equipment used for the project must be appropriately sized for work being performed.
2. Do not include costs for "miscellaneous tools and equipment", in your proposal for a replacement value of $500 or less. Costs shown in excess of $500 shall be broken out separately.
3. Regardless of ownership, rates to be used in determining equipment rental costs shall be the lowest cost from one of the following sources:
   a. United States (U.S.) Army Corps of Engineers, Ownership and Operating Expense Schedule (use latest edition and applicable region)
   b. Construction Blue Book
   c. Local equipment rental rates, documented by actual invoice charges, or itemized vendor quotes.
   d. CALTRANS
4. Estimated equipment rates shall include operating costs of all fuel, oil, lubrication, supplies, small tools, necessary attachments, ground engaging components, tires and tracks, routine repairs and maintenance (cost of major repair and overhaul is not allowed per Federal Acquisition Regulation (FAR) 31.105(d)(2)), depreciation, storage, insurance, and all incidentals. Mobilization, if applicable, may be included for equipment solely used on the modification work but must be listed separately.
5. Estimate full rate for equipment only for duration that equipment will be utilized to accomplish work of the modification.
6. Standby unit rates used in accordance with paragraph 1.3, D, 2, above. If the U.S. Army Corp of Engineers is utilized then their standby rates prevail. If Bluebook or local equipment pricing is accepted, then 1/2 of equipment costs minus any operating costs, major repair and overhaul will be accepted. CALTRANS can also be used.
7. If equipment is in standby mode due solely to a documented NPS delay, established standby rate shall apply from the first day of the delay.
8. Equipment not used and on job site for up to five consecutive days may be classified at standby rates, provided the equipment is or has been used solely to perform work on the modification and will be necessary to complete additional modification work. Equipment still on the jobsite but not in use after five consecutive days will not be considered in the modification pricing.
9. Requests for compensation for equipment stand by time must be justified, documented and itemized separately.
10. The estimated timeframe (daily, weekly, monthly) for use of the equipment must reflect the lowest cost to the Government.

E. Establishment and Application of Overhead and Profit Percentages:

1. Home Office Overhead and Profit (OH&P) shall be applied to direct costs only. Profit shall not be applied to overhead amounts; and overhead shall not be applied to profit. Home office overhead shall contain only allowable, allocable, and reasonable costs per the contract documents and FAR Part 31. Profit percentages are based on risk factors found in FAR Part 31which have been applied to the specific type of work included in this project. Negotiated rates shall not exceed the following percentages for OH&P for contractor self-performed work:

   Overhead……10%
   Profit……8.00%

2. Total aggregate limit of markup (OH&P) for Contractor and Subcontractors on modification work shall not exceed 25%. The NPS will not be responsible for allocation of percentages between contractor and subcontractors at any tier.
3. If Contractors form a partnership, partnership may only receive home office overhead and profit in same amount as an individual Contractor (refer to paragraph 1.3,E,1 above). It is the responsibility of the partners to decide on division of revenue.
4. Combined Increases and Decreases: On proposals involving both increases and decreases in the Contract Price, overhead and profit mark-ups are required on net increases and deducted on net decreases.
5. At no time can profit be calculated on Overhead or itself, it must be calculated on direct costs of work only.

PART 2 - PRODUCTS

PART 3 - EXECUTION

END OF SECTION 01 26 01

This page intentionally left blank.

SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section explains in general, what is and is not included in a contract line item (CLIN), and limits or cut-off points where one item ends and another begins.

B. Limits or cut off points where one CLIN ends and another begins shall be referenced to drawing volume numbers and as defined by the Area Designator in the sheet number conventions.

C. If no contract line item exists for a portion of work, include costs in a related item.

1. Volume 00 - General drawings shall apply to all drawing volumes and costs should be included in related CLIN items.

D. Contract Line Item costs shall be developed such that individual Work Item prices, as defined in Section 3.1, can be determined by the Contracting Officer.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 LIST OF CONTRACT LINE ITEMS

A. Contract Line Item Number 1 (CLIN 1): Pipelines – North Rim.

1. This item consists of work shown in drawings Volume 01, Area Designators 10, 11, 12, and 13. It includes approximately 5,500 LF of 4-IN raw water pipeline, 5,500 LF of 4-IN finished water pipeline, 8,420 LF of 2-IN conduit and fiber optic cable and appurtenances on the North Rim, and 800 LF of 6-IN sanitary sewer from the WTP to the collection system. Work includes a new 20 FT wide, 3,500 LF gravel access road from the Ballfield to near the intersection of Norton Court and SR67. Work includes rough grading of the Ballfield site for construction staging. The raw water and finished water is routed from the Ballfield to the new North Rim WTP site along the proposed ballfield access road and existing Bridle Trail and utility corridors. The fiber optic is routed from the Grand Lodge parking lot at the overhead power lines along the Bridle Trail to the new North Rim WTP site, from the North Rim WTP to the Power Distribution Center, and from the Power Distribution Center to the North Rim WWTP.
2. Measurement for payment will be percent complete based on lineal feet installed and tested.
3. Payment will be made at the contract lump sum price.

B. Contract Line Item Number 2 (CLIN 2): Pipelines – North Rim to Roaring Springs Pumphouse.

1. This item consists of Work shown in drawings Volume 02, Area Designators 20 and 21. Bore construction Work has been further subdivided.

2. This item generally consists of approximately 18,700 FT of 4.5 IN raw water pipeline and 4.5 IN finished water pipeline installed in dual bores and telescoping steel casing (~160 FT of 20 IN, 3,000 FT of 13 3/8 IN, 17,700 FT of 9 5/8 IN, and 18,700 FT of 7 IN) drilled from the Ballfield site bore entry pit to the Roaring Springs Pumphouse bore exit.
3. Measurement for payment will be based on the measurement language for each of the bore construction Work CLINs.
4. For CLIN items 2.1 through 2.5, payment will be made at the contract unit price for the constructed unit quantities, not to exceed the quantities provided in the contract price schedule. For CLIN items 2.6 through 2.11, final quantities may exceed estimated quantities and will be paid at the contract unit prices.
5. Contract Line Item Number 2.1 (CLIN 2.1): Personnel Deployment
    a. This item includes project kickoff and preparation, submittal work, and transporting personnel required to support mobilization and commencement of the Work. No costs included in CLIN 2.4 shall be included in CLIN 2.1.
    b. Measurement for payment will be lump sum.
    c. Payment will be made on a percent complete basis in accordance with an approved Schedule of Values. Payment for this CLIN shall be tied to completion and acceptance of discrete deliverables, including preconstruction submittals, planning activities, and personnel mobilization milestones. No payment will be made in advance of work performed. Allocation of value within the Schedule of Values shall not be front-loaded and must reasonably reflect the timing and effort of the associated work activities. The Contracting Officer reserves the right to review and require revision of the Schedule of Values to prevent unbalanced payment.
    d. Payment will be made at the contract lump sum price for Personnel Deployment, prorated based on approved percent complete.
6. Contract Line Item Number 2.2 (CLIN 2.2): Initial Setup (Ready to Drill)
    a. This item includes transporting equipment and materials, site and equipment preparation work including survey, grading, entry and exit location setup, gravel, lumber, and other site preparation work necessary to achieve readiness for drilling operations.
    b. Measurement for payment will be lump sum.
    c. Payment will be made on a percent complete basis in accordance with an approved Schedule of Values and shall be commensurate with actual progress achieved. Final payment for this CLIN will not be made until drilling operations have commenced, defined as physical penetration of the subsurface for the purpose of bore construction. No payment shall be made in advance of work performed. Allocation of value within the Schedule of Values shall not be front-loaded and must reasonably reflect the timing and effort of the associated work activities. The Contracting Officer reserves the right to require reallocation.
7. Contract Line Item Number 2.3 (CLIN 2.3): Equipment Removal
    a. This item includes all demobilization from the Project Site including transporting equipment, rentals, and personnel, restoration, cleanup and all required site restoration.
    b. Measurement for payment will be lump sum.
    c. The Schedule of Values shall include separate line items for demobilization, cleanup, and site restoration activities. Payment for each component shall be made only upon completion and acceptance of that specific work. Payment will be made on a percent complete basis in accordance with an approved Schedule of Values and shall be commensurate with actual demobilization progress achieved. Payment shall be based only on completed and accepted demobilization, removal, cleanup, and restoration activities. No payment shall be made for demobilization activities not yet performed.
    d. Payment will be made at the contract lump sum price for Equipment Removal prorated based on approved percent complete.
8. Contract Line Item Number 2.4 (CLIN 2.4): Labor/Services/Equipment/Rentals

a. This item includes all Directional Drilling labor, services, equipment, and rentals for execution of the Work specified in Section 31 71 00 Directional Drilling. No costs included in CLIN 2.1 shall be included in CLIN 2.4. This may include but not limited to:

1) All contractor and sub-contractor office labor and support.
2) All contractor and sub-contractor field labor and support.
3) All expenses associated with office and field labor.
4) All construction office setup, rental, management, and other related work which may include, but not limited to: office rental, safety trailer, cleaning, and associated transportation costs.
5) All construction camp facilities including but not limited to: all temporary personnel camp(s), waste management, residential trailers, cooking trailers, portable bathroom and shower facilities, garbage containers, and associated setup, maintenance, teardown, and transportation.
6) All surface rentals and services including, procurement, setup, use, and removal of items including and not limited to: ancillary equipment roll off boxes, liners, berms, frac tanks, cleanout tanks, vacuum trucks, manlifts, flare, cellar pumps, sum pumps, centrifugal pumps, diaphragm pumps, hoses, generators, rig matting, fuel equipment, and associated travel costs.
7) Drilling Rig and Equipment including all equipment and support for operation of the drilling rig and tooling. This may include but not limited to: drilling rig, air package, diesel fuel, loader, shaker screens, consumables, rig monitor, blow out preventer, wear bushings and retrieval tool, test plug, rental spools/adapters, drillstring/collars, stabilizers, shock subs, rotating head, jars, roller reamers, hole openers, fishing tools/services, rental subs, casing scrapers, additional pipe racks, drill bits, welding services, rig cleaning, extra associated equipment, and associated transportation costs.
8) Directional drilling services including but not limited to: mud motor, lost in hole insurance, inspection, repairs, electromagnetic MWD, MWD supervisor, magnetic ranging services including personnel and analysis, stabilizers, reamers, non-magnetic drill collars, ellipse of uncertainty analysis, collision avoidance analysis, rotary steerable system, direction drillers.
9) Fluids and solids control work and services associated with circulating cuttings from the borehole which may include but are not limited to: mud/ foam engineer, drilling mud/ foam additives, mud logging, air/mist drilling package, closed loop solids control/operation/management, fly ash, solids disposal, liquids disposal, mud waste disposal, fresh water, and associated transportation.
10) Casing crew and equipment for installing the casing which may include but not limited to: running casing and tubing – all diameters, hydraulic tongs, slips, elevators, cleaning/inspecting casing and tubing, connection/joint make-up services, and workover inspection.
11) Wellhead crew and equipment for installing the wellhead A, B, and C connections and tree.
12) Cementing services and equipment for cementing casing which may include but not limited to: cementing the conductor casing, cementing the surface casing, cementing the intermediate casing, cementing the final casing, and grouting from the borehole exit location.
13) Logging and wireline work and services required to track and locate borehole excavation and casing installation which may include but not limited to: gyro survey, open hole and cased hole survey of the surface casing, open hole and cased hole survey of the intermediate casing, open hole survey and testing of the final casing, and lost in hole protection and insurance.

14) Miscellaneous labor, services, equipment, and rental expenses for work and services required and not explicitly stated herein that are necessary to complete the Work and are not separately compensated for under other CLINs.

b. This CLIN shall constitute full compensation for all labor, services, equipment, and rentals required to perform the Work, except for those costs expressly and exclusively compensated under CLINs 2.6 through 2.11 when properly authorized. Costs included in this CLIN shall not be invoiced, in whole or in part, under any other CLIN. Conditions encountered during drilling operations, including variability in subsurface conditions, shall be considered inherent to the Work and shall not constitute a basis for additional compensation or time extension, except where such conditions strictly meet the activation criteria of CLINs 2.6 through 2.11 as determined under Section 10.a and 10.k.

c. Measurement for payment will be lump sum.

d. Payment will be made at the contract lump sum price on a percent complete basis in accordance with an approved Schedule of Values and based on accepted progress.

9. Contract Line Item Number 2.5 (CLIN 2.5): Materials

a. This CLIN includes all material procurement, delivery, handling, storage, waste, and incidental losses, except where additional compensation is expressly provided under CLINs 2.6 through 2.11. Materials paid under this CLIN shall not be separately compensated under any other CLIN. Materials include all materials required for execution of the Work specified in Section 31 71 00 Directional Drilling, including but not limited to:

1) 20-inch conductor casing including but not limited to: all conductor casing material costs and inspection services.

2) 13 3/8-inch surface casing including but not limited to: all surface casing material costs and inspection services, DV tools, casing packers, float equipment, plugs, centralizers, API thread locking compound, and API thread dope.

3) 9 5/8-inch intermediate casing including but not limited to: all intermediate casing material costs and inspection services, DV tools, casing packers, float equipment, plugs, centralizers, API thread locking compound, and API thread dope.

4) 7-inch final casing including but not limited to: all final casing material costs and inspection services, DV tools, casing packers, float equipment, plugs, centralizers, API thread locking compound, and API thread dope.

5) 4 1/2-inch tubing including but not limited to all tubing material costs and inspection services.

6) Wellhead including but not limited to: all material costs for the wellhead A, B, and C connections, tree, and ancillary materials.

7) Miscellaneous materials expenses including any other materials required and not explicitly stated herein.

b. Measurement for payment will be lump sum.

c. Payment will be made at the contract lump sum price on a percent complete basis in accordance with an approved Schedule of Values and based on accepted progress.

10. General Requirements for Event-Based CLINs (CLINs 2.6 through 2.11)

a. Applicability

1) CLINs 2.6 through 2.11 apply only to the specific conditions expressly described within each CLIN and are intended to compensate the Contractor for discrete, defined events encountered during execution of the Work.

b. Contracting Officer (CO) Concurrence

1) Payment under these CLINs shall require concurrence by the CO.

2) In emergency field conditions requiring immediate action to protect the Work, personnel, or equipment, the Contractor may proceed prior to receiving formal concurrence, provided that:

a) The Contractor notifies the CO immediately, and
b) Written concurrence is obtained within 24 hours.
c) Delay in Contracting Officer eligibility determination shall not result in retroactive commencement of compensable time.

c. Measurement of time:
1) Where payment is based on hourly work:
a) Measurement shall begin only upon Contracting Officer eligibility determination in accordance with this Section. Under no circumstances shall compensable time begin prior to such determination. Work performed prior to Contracting Officer eligibility determination shall not be compensable under event-based CLINs, regardless of when the qualifying condition is first encountered.
b) Measurement shall end when normal drilling operations resume, as determined by the CO.
c) Measurement shall be supported by documentation in accordance with Section 10.h.
d) Where CLIN-specific measurement endpoints are defined, such endpoints shall control, provided they are consistent with the definition of normal drilling operations in Section 10.j.

d. Non-Overlapping Payment
1) Only one event-based CLIN (CLINs 2.6 through 2.11) may be active at any given time for the same condition or occurrence. Where multiple conditions arise from the same underlying cause, the CO will determine the single applicable CLIN for compensation, based on the predominant causative condition, as determined by the CO.
2) Sequential or successive application of multiple event-based CLINs to conditions arising from the same underlying subsurface feature or continuous event shall not be permitted.
3) The CO's determination of the applicable CLIN shall be final unless demonstrably inconsistent with the documented field conditions. The existence of multiple contributing conditions shall not entitle the Contractor to compensation under multiple CLINs for the same event.
4) Disagreement with the CO determination of the predominant causative condition shall not, in itself, establish entitlement to compensation under an alternative CLIN or to additional compensation. Entitlement shall exist only where the Contractor demonstrates that the determination is demonstrably inconsistent with contemporaneous, objective field data.

e. Exclusivity of Payment
1) Payment under these CLINs shall include only those costs directly attributable to the qualifying event.
2) Costs otherwise compensated under CLINs 2.4 and 2.5 shall not be included in payment under CLINs 2.6 through 2.11

f. Maximum Allowable Hours
1) Maximum hours specified for each CLIN represent the upper limit of compensation. Exceeding these limits does not entitle the Contractor to additional compensation unless modified by contract.
2) Exceeding the maximum allowable hours shall not, by itself, constitute a differing site condition, change in scope, or entitlement to additional compensation.
3) The Contractor acknowledges that the maximum allowable hours are intended to represent the maximum compensable limit for qualifying event conditions under these CLINs absent a separately executed modification. Work performed beyond the specified maximum hours is not separately compensated under CLINs 2.6 through 2.11 and shall only be considered for

additional compensation if the Contractor provides timely notice and the Contracting Officer determines, in accordance with the Contract, that such conditions warrant an adjustment.

4) To the maximum extent practicable, compensation for work performed beyond the specified maximum allowable hours shall be based on the contract unit prices. Adjustment to such unit prices will only be considered where the Contractor demonstrates that the nature of the Work has materially changed from that originally contemplated, subject to approval by the Contracting Officer.

5) CLIN-specific maximum hour limitations shall be interpreted consistently with this Section. In the event of any perceived inconsistency, Section 10.f shall govern.

g. Contractor Responsibility

1) The Contractor shall implement reasonable and industry-standard means and methods to avoid and mitigate the occurrence and duration of qualifying events. Efforts that are inefficient, duplicative, or inconsistent with accepted industry practices for the encountered condition, as supported by contemporaneous documentation and standard industry references, shall not be considered reasonable for purposes of compensation.

2) The Contractor shall sequence and perform the Work in a manner that reasonably minimizes the duration and cost of qualifying events. The Contracting Officer may disallow time determined to result from avoidable inefficiency, delay, or suboptimal sequencing of operations

h. Documentation Requirements.

1) The Contractor shall provide daily records prepared at the time the Work is performed to substantiate the qualifying condition, as described in 31 71 00.

2) The CO will determine the adequacy and acceptance of such documentation. Documentation shall be recorded in real time or at regular intervals consistent with standard drilling practices.

3) Records lacking contemporaneous, time-stamped supporting data or not consistent with observed field conditions may be rejected by the CO. Post-event reconstruction without supporting real-time records may be rejected by the CO.

4) Documentation shall include, at a minimum:

a) Drilling parameters

b) Time logs

c) Activities performed

d) Conditions encountered.

i. Definition of Normal Drilling Operations

1) “Normal drilling operations” shall mean drilling has resumed along the intended bore path with stable returns, controlled pressures, and rates of penetration consistent with pre-event conditions and no longer requiring atypical intervention measures, as determined by the CO based on drilling data and field conditions.

2) Short-duration stabilization, partial returns, or temporary improvements that do not demonstrate sustained and repeatable drilling performance consistent with pre-event conditions shall not constitute resumption of normal drilling operations.

j. Order of Precedence

1) In the event of any inconsistency between CLIN 2.4 and CLIN 2.5 and CLINs 2.6 through 2.11, the express terms of CLINs 2.6 through 2.11 shall govern solely for the duration and scope of the qualifying event.

2) All other work conditions not meeting the express activation criteria shall remain subject to CLIN 2.4 and CLIN 2.5 and shall not be separately compensable.

k. No Constructive Activation
   1) No event-based CLIN (CLINs 2.6 through 2.11) shall be deemed activated, initiated, or eligible for compensation based on Contractor interpretation, field conditions, partial performance, or notification alone.
   2) Activation occurs only upon CO eligibility determination.
   3) The absence of immediate CO determination shall not constitute approval, deemed approval, constructive activation, or entitlement to compensation.
   4) Contractor continuation of work prior to CO eligibility determination shall be considered part of the Contractor's means and methods and shall not establish entitlement under any event-based CLIN.

l. Definition of a Discrete Event
   1) A "discrete event" for purposes of CLINs 2.6 through 2.11 shall mean a continuous occurrence of a qualifying condition without restoration of normal drilling operations.
   2) An event shall:
      a) Begin when the qualifying condition first occurs; and
      b) End when normal drilling operations resume, as defined in Section 10.j.
   3) A new event shall not be considered to have occurred unless:
      a) Normal drilling operations have resumed and
      b) Such operations have been maintained for a continuous period of not less than two (2) hours, unless otherwise approved by the CO.
   4) The identification or occurrence of a qualifying condition does not, in itself, establish entitlement to compensation. Entitlement to compensation is strictly limited to the period following CO eligibility determination in accordance with this Section. This applies regardless of any emergency action taken under Section 10.b.
   5) Temporary, partial, or unstable restoration of drilling conditions shall not constitute the end of an event. Interruptions shorter than two (2) hours duration shall be considered part of the same event.
   6) The minimum duration specified herein for defining a new discrete event (two (2) hours of sustained normal drilling operations) is independent of, and shall not modify, any time-based eligibility thresholds specified within individual CLINs.

m. Non-Qualification / Non-Activation
   1) If the CO determines that conditions as described within an event based CLIN (CLINs 2.6 through 2.11) have not been met, then:
      a) The CLIN shall not be activated
      b) Work shall be included in CLIN 2.4 and CLIN 2.5
      c) No additional compensation shall be due.

11. Contract Line Item Number 2.6 (CLIN 2.6): Intersecting a Void While Drilling
   a. General Description:
      1) Intersecting a void while drilling includes all labor, equipment, rentals, services, materials, and other costs necessary for execution of the Work specified in Section 31 71 00 Directional Drilling.
   b. Eligibility Conditions:
      1) CLIN 2.6 shall not be eligible for compensation unless all of the following preconditions are met:
         a) The Contractor demonstrates, using field data, that a void or loss zone has been encountered and materially affected drill path control or bore stability.
         b) The Contractor implements reasonable and documented means and methods consistent with industry practice to continue drilling or regain control of the bore path.

c) The condition results in deviation from normal drilling operations as defined in Section 3.1.B.10(j).

d) CLIN 2.6 includes situations where a drill pipe enters a void, becomes unsupported, bends, and deviates the bore path from design line and grade such that continued drilling and a return to the design bore path after exiting the void is impractical or ill advised.

2) Documentation and Eligibility Determination:

a) The Contractor shall submit contemporaneous field logs, drilling parameters, and operational data demonstrating borehole conditions, loss indicators, and deviation from planned trajectory.

b) Further, CLIN 2.6 is only activated should the following conditions be realized:

- Means and methods response for continuing to execute the Work specified in Section 31 71 00 requires a mechanical intervention in the form of a mechanical whip stock, a cement kickoff plug, or other mechanical interventions to allow for drill pipe to be retracted and advanced at a trajectory revised from the design to advance around the void.
- The Contractor must measure and document a change in inclination or azimuth greater than 7 degrees per 100 feet from the design bore path due to encountering the void and an attempt to drill back into the formation.
- The Contractor must verify the presence of the void. Data that may indicate presence of a void includes, but may not be limited to: increased hook load, weight on bit of zero and no resistance to drilling three or more feet, decrease in bottom hole pressure, loss of circulation, no cuttings returning to surface. The Contractor must provide data and explanation of the data that indicates the presence of a void and provide documentation to the CO in accordance with Section 3.1.B.10 and receive approval from the CO for measurement to be activated.
- Encountering a void while drilling that does not meet the conditions outlined herein will not activate measurement of CLIN 2.6. The CO must approve activation of CLIN 2.6. Activation of CLIN 2.6 shall be in accordance with Section 3.1.B.10, General Requirements for Event-Based CLINs. The qualifying event shall be deemed to begin upon occurrence of the qualifying condition as supported by field documentation; however, compensable time shall begin only upon CO eligibility determination in accordance with Section 10.

3) The CO will determine whether the documented conditions satisfy CLIN 2.6 eligibility based on confirmation of a qualifying condition and associated operational impact.

4) The CO's determination shall constitute the eligibility determination for CLIN 2.6.

5) All conditions described herein shall occur within a single discrete event as defined in Section 10.m

c. Commencement of Compensable Work:

1) Compensable time shall begin no earlier than the time the CO determines that eligibility criteria have been satisfied.

2) Time expended during initial drilling, diagnostic evaluation, and non-mechanical response efforts shall not be included in compensable time under this CLIN.

d. Measurement for Payment:

1) Measurement for payment will be per hour of qualifying work performed after Contracting Officer eligibility determination.
2) The contract unit price shall incorporate all items included in Specification 31 71 00.
3) Measurement for payment will be per hour worked from the time established in Section 3.1.B.10, General Requirements for Event-Based CLINs for conditions as described in Specification 31 71 00 until the excavated borehole has advanced past the location of the void and resumed typical drilling activity. Measurement shall be supported by the daily records required under Section 3.1.B.10.h. Documentation of conditions described in Specification 31 71 00 and the number of hours spent advancing past the void shall be submitted by the Contractor to the CO. The CO will then evaluate whether documentation is acceptable to activate CLIN 2.6.
4) Time used to prepare documentation, submit notifications, or await CO determination shall not be compensable.
5) A total of one hundred and eighty (180) hours shall be the maximum allowable compensation under this CLIN.

e. Payment:
1) Payment will be made at the contract unit price for intersecting a void while drilling, inclusive of all labor, equipment, services, and materials necessary to complete the Work after eligibility determination.
2) The contract unit price shall be full compensation for all work performed under this CLIN and shall not be separately compensated under any other CLIN.

12. Contract Line Item Number 2.7 (CLIN 2.7): Intersecting a Void While Advancing Casing
a. General Description:
1) Intersecting a void while advancing casing includes all labor, equipment, rentals, services, materials, and other costs necessary for execution of the Work specified in Section 31 71 00 Directional Drilling where casing advancement is impeded due to a void or loss zone, resulting in inability to locate or engage the drilled bore path during casing installation.

b. Eligibility Conditions:
1) CLIN 2.7 shall not be eligible for compensation unless all of the following preconditions are met:
a) The Contractor demonstrates that casing advancement is obstructed due to a previously documented void or loss zone encountered during drilling.
b) The Contractor makes reasonable and documented efforts consistent with industry practice to advance casing through the affected zone.
c) The condition prevents continued casing advancement using normal installation methods.
d) CLIN 2.7 includes situations where casing pipe enters a void, fails to line up with the borehole created during drilling, and is unable to enter or find the exit from the void.

c. Documentation of Eligibility Determination:
a) The Contractor must make a reasonable and documented effort consistent with industry standard practices and supported by contemporaneous field documentation to advance the casing through the void along the drilled bore path for a minimum of one hour. The Contractor shall implement the contingency plan required in Section 31 71 00.
b) The Contractor must have previously documented the void at the same location when drilling the borehole. Documentation must include relevant data and an explanation of that data indicating the void.

- Data that may indicate presence of a void includes but may not be limited to the conditions described in Specification 31 71 00. The Contractor must provide data and explanation of the data that indicates the presence of a void and receive approval from the CO for the void to be recognized.

2) Multiple casing advancement attempts, including repositioning, re-engagement, or repeated attempts to locate the bore path, shall be considered part of the same continuous event and shall not reset or extend compensable time.
3) The CO will determine whether the documented conditions satisfy CLIN 2.7 eligibility based on confirmation of a qualifying condition and associated operational impact.
4) The CO's determination shall constitute the eligibility determination for CLIN 2.7.
5) All conditions described herein shall occur within a single discrete event as defined in Section 10.m.

d. Commencement of Compensable Work:
1) Compensable time shall begin no earlier than the time the CO determines that eligibility criteria have been satisfied.
2) Compensable time shall be limited to continuous, reasonable casing advancement efforts. Intermittent, standby, or non-continuous casing operations shall not restart or extend compensable time.

e. Measurement for Payment:
1) Measurement for payment will be per hour of qualifying work performed after CO eligibility determination.
2) Hourly measurement worked from the time established in Section 3.1.B.10, General Requirements for Event-Based CLINs for conditions as described in Specification 31 71 00 until the casing pipe has advanced past the location of the void. Documentation of conditions described in Specification 31 71 00 and the number of hours spent advancing casing past the void shall be submitted by the Contractor to the CO. The CO will then evaluate whether documentation is acceptable to activate CLIN 2.7.
3) Time used to prepare documentation, submit notifications, or await Contracting Officer determination shall not be compensable.
4) A total of one hundred and fifty (150) hours shall be the maximum allowable compensation under this CLIN.

f. Payment:
1) Payment will be made at the applicable contract unit rate structure established herein for the qualifying event, subject to the applicable maximum hourly limitations and activation criteria. The contract unit price shall incorporate all items included in Specification 31 71 00.
2) The contract unit price shall be full compensation for all work performed under this CLIN and shall not be separately compensated under any other CLIN.

13. Contract Line Item Number 2.8 (CLIN 2.8): Excessive Loss of Drilling Medium to Formation

a. General Description:
1) Excessive loss of drilling medium to formation includes all labor, equipment, rentals, services, materials, and other costs necessary for execution of the Work in Specification 31 71 00 where drilling mud is lost to the formation resulting in sustained loss of circulation and inability to maintain return flow under normal drilling operations.

b. Eligibility Conditions:
1) CLIN 2.8 shall not be eligible unless:

a) Sustained loss of circulation occurs for a minimum of twelve (12) continuous hours within a single discrete event, as defined in Section 10.m. Intermittent or cumulative loss of circulation occurring across multiple discrete events shall not be combined to satisfy this requirement.
b) Reasonable efforts using industry-standard loss circulation materials are implemented.
c) Loss condition materially interrupts normal drilling operations.
d) The only drilling mediums applicable for qualification of CLIN 2.8 are drilling mud. Loss of air, foam, or mist drilling mediums do not qualify under CLIN 2.8.

c. Documentation of Eligibility Determination:
1) Sustained loss of circulation with no measurable returns for a continuous minimum period of twelve (12) continuous hours within a single discrete event, as documented by drilling logs and instrumentation, must exist. Intermittent or cumulative loss of circulation occurring across multiple discrete events shall not be combined to satisfy this requirement.
2) Means and methods response for continuing to execute the Work in Specification 31 71 00 must require a reasonable and documented effort consistent with industry standard practices to regain circulation using loss circulation material (LCMs). Should LCMs be insufficient to regain circulation, the suspected loss zone shall be plugged with cement or by other means submitted in writing by the Contractor and approved by the CO. The Contractor shall demonstrate continuous, active, and reasonable efforts to regain circulation throughout the duration of the qualifying loss condition. Periods of inactivity, standby, or non-continuous response shall not be counted toward the required duration or compensable time.
3) The Contractor must document and submit for approval evidence of excessive loss of drilling medium to the CO. Documentation must include relevant data and an explanation of that data indicating the excessive loss of drilling medium.
4) The CO will determine whether the documented conditions satisfy CLIN 2.8 eligibility based on confirmation of a qualifying condition and associated operational impact.
5) The CO's determination shall constitute the eligibility determination for CLIN 2.8.
6) All conditions described herein shall occur within a single discrete event as defined in Section 10.m.

d. Measurement for Payment:
1) Measurement for payment will be per hour worked from the time established in Section 3.1.B.10, General Requirements for Event-Based CLINs for conditions as described in Specification 31 71 00 until circulation and drilling medium (drilling mud only) returns have been re-established. Measurement shall be supported by the daily records required under Section 3.1.B.10.h. Documentation of conditions described in Specification 31 71 00 and the number of hours spent attempting to re-establish circulation and returns shall be submitted by the contractor to the CO. The CO will then evaluate whether documentation is acceptable to activate CLIN 2.8.
2) Measurement for payment will be per hour of qualifying work performed after CLIN 2.8 activation.
3) Time used to prepare documentation, submit notifications, or await CO determination shall not be compensable.
4) A total of one hundred and twenty (120) hours shall be the maximum allowable compensation under this CLIN.

e. Payment:

1) Payment will be made at the applicable contract unit rate structure established herein for the qualifying event, subject to the applicable maximum hourly limitations and activation criteria. The contract unit price shall incorporate all items included in Specification 31 71 00.
2) The contract unit price shall be full compensation for all work performed under this CLIN and shall not be separately compensated under any other CLIN.

14. Contract Line Item Number 2.9 (CLIN 2.9): Excessive Loss of Cement to Formation
    a. General Description:
        1) Excessive loss of cement to formation includes all labor, equipment, rentals, services, materials, and other costs necessary for execution of the Work specified in Section 31 71 00 Directional Drilling where cement placed during casing operations is lost to formation resulting in failure to achieve planned cement volume or annular fill.
    b. Eligibility Conditions:
        1) CLIN 2.9 shall not be eligible for compensation unless all of the following preconditions are met should cement be lost to the formation and the following conditions are realized:
            a) Only cement lost to the formation during cementing of a casing will be considered as applicable to CLIN 2.9. No loss of cement for other purposes will be considered applicable to CLIN 2.9.
            b) Contractor must demonstrate the zone of lost circulation along the annular volume to be cemented by attempting to fill the annular volume with water throughout the extent of the planned cemented bore length as indicated in the Contractor's approved drilling plan. Other methods to demonstrate a lost circulation zone may be considered if approved by the CO.
            c) Cement mix volumes greater than 150 percent above the calculated theoretical required volume (i.e., greater than 2.5 times the theoretical volume). Theoretical cement mix volume shall be based on approved submittals and bore geometry accepted by the CO.
                - Theoretical cement mix volume for each casing pipe shall be calculated per submittal requirements in Specification 31 71 00. Minimum cementing requirements are also described in Specification 31 71 00.
            d) The Contractor must document and submit for approval evidence of excessive loss of cement to the formation to the CO. Documentation must include relevant data and an explanation of that data indicating the excessive loss of cement and presence of a loss circulation zone.
    c. Documentation and Eligibility Determination:
        1) All documentation and cement volume calculations shall comply with the requirements of Section 10 and the specific eligibility and measurement provisions of this CLIN. In the event of a conflict, the more stringent requirement, as determined by the CO, shall apply.
        2) The CO will determine whether the documented conditions satisfy CLIN 2.9 eligibility based on confirmation of a qualifying condition and associated operational impact.
        3) The CO's determination shall constitute the eligibility determination for CLIN 2.9.
        4) All conditions described herein shall occur within a single discrete event as defined in Section 10.m.
    d. Measurement for Payment:
        1) Measurement for payment will be per hour worked from the time established in Section 3.1.B.10, after the CO has determined and documented that the threshold theoretical cement volume specified in Section 2.9.b has been

exceeded based on field measurements and supporting documentation in accordance with Specification 31 71 00, until the loss of cement to formation ceases or the upper limit of allotted hours is reached, whichever comes first.

2) Measurement shall be supported by the daily records required under Section 3.1.B.10.h. Documentation of conditions described in Specification 31 71 00 and the number of hours spent attempting to cement the casing shall be submitted by the contractor to the CO. The CO will then evaluate whether documentation is acceptable to activate CLIN 2.9.

3) Only time reasonably required to address the qualifying loss condition, as determined by the CO based on documented field conditions and response effectiveness, shall be compensable.

4) Only cement volumes and associated time reasonably required to address verified loss conditions, as determined by the CO, shall be considered for compensation. Excessive, inefficient, or non-targeted cement placement shall not be considered in establishing entitlement or duration.

5) Time used to prepare documentation, submit notifications, or await CO determination shall not be compensable.

6) A total of one hundred and twenty (120) hours shall be the maximum allowable compensation under this CLIN.

e. Payment:

1) Payment will be made at the applicable contract unit rate structure established herein for the qualifying event, subject to the applicable maximum hourly limitations and activation criteria. The contract unit price shall incorporate all items included in Specification 31 71 00.

2) The contract unit price shall be full compensation for all work performed under this CLIN and shall not be separately compensated under any other CLIN.

15. Contract Line Item Number 2.10 (CLIN 2.10): Subsurface Fluid Influx

a. General Description:

1) Subsurface fluid influx includes conditions where water or other formation fluids enter the borehole at rates sufficient to interfere with drilling operations and require mitigation or sealing measures and includes all labor, equipment, rentals, services, materials, and other costs necessary for execution of the Work specified in 31 71 00 Directional Drilling should subsurface fluid flow into the borehole at a sufficient rate and volume requiring the Contractor to plug off the location of water influx with cement to squeeze off the water.

b. Eligibility Conditions:

1) CLIN 2.10 shall not be eligible for compensation unless all of the following preconditions are met:

a) Subsurface fluid influx results in measurable interference with drilling operations.

b) The Contractor must make a reasonable and documented effort consistent with industry standard practices to lift cuttings and water inflow with current drilling medium for a minimum period of two (2) hours to attempt to deplete the water inflow and allow for drilling to continue.

c) All conditions described herein shall occur within a single discrete event as defined in Section 10.m. Intermittent or recurring influx conditions separated by less than two (2) hours shall be considered part of the same event.

c. Documentation and Eligibility Determination:

1) The Contractor must document and submit for approval data indicating subsurface fluid influx using real-time drilling records, mud logs and pressure monitoring instrumentation, along with an explanation of the data.

2) Data may include but not limited to: an increase in bottom hole pressure, significant returns of extra water above the water volume or rate injected into the drill string, diluted returns, a change in drilling medium density due to fluid influx.
3) The CO will determine whether the documented conditions satisfy CLIN 2.10 eligibility based on confirmation of a qualifying condition and associated operational impact.
4) The CO's determination shall constitute the eligibility determination for CLIN 2.10.
5) All conditions described herein shall occur within a single discrete event as defined in Section 10.m.

d. Commencement of Compensable Work:
1) Compensable time shall begin no earlier than the time the CO determines that eligibility criteria have been satisfied.
2) Time expended during the initial two (2) hours shall not be included in compensable time under this CLIN.

e. Measurement for Payment:
1) Measurement for payment will be per hour worked from the time established in Section 3.1.B.10, General Requirements for Event-Based CLINs for conditions as described in Specification 31 71 00 until the influx of water has been fully controlled such that normal drilling operations, as defined in Section 10.i, can be sustained or the upper limit of allotted hours is reached, whichever comes first. Measurement shall be supported by the daily records required under Section 3.1.B.10.h. Documentation of conditions described in Specification 31 71 00 and the number of hours spent attempting to re-establish circulation and returns shall be submitted by the contractor to the CO. The CO will then evaluate whether documentation is acceptable to activate CLIN 2.10.
2) Only time during which the Contractor is actively and continuously performing reasonable mitigation efforts shall be considered compensable. Periods of inactivity, standby, or non-continuous operations shall not be included.
3) Measurement for payment will be per hour of qualifying work performed after CLIN 2.10 activation.
4) Time used to prepare documentation, submit notifications, or await CO determination shall not be compensable.
5) A total of ninety (90) hours will be included in the CLIN shall be the maximum allowable compensation under this CLIN.

f. Payment:
1) Payment will be made at the applicable contract unit rate structure established herein for the qualifying event, subject to the applicable maximum hourly limitations and activation criteria. The contract unit price shall incorporate all items included in Specification 31 71 00.
2) The contract unit price shall be full compensation for all work performed under this CLIN and shall not be separately compensated under any other CLIN.

16. Contract Line Item Number 2.11 (CLIN 2.11): Stuck Drill Pipe Following Exit from Canyon Wall
a. General Description:
1) Stuck drill pipe following exit from canyon wall includes all labor, equipment, rentals, services, materials, and other costs necessary for execution of the Work specified in Section 31 71 00 Directional Drilling should partial or full collapse of the borehole occur upon breaking through the canyon wall and/or loss of formation integrity and losing positive pressure creating challenges pulling out of the hole, resulting in the drill

string becoming stuck or unable to be safely removed using routine drilling methods.

b. Eligibility Conditions:
   1) CLIN 2.11 shall not be eligible for compensation unless all of the following preconditions are met:
      a) The Contractor shall perform a reasonable and documented effort consistent with industry standard practices, to recover and/or free the drill pipe using the drilling rig and available tools for a minimum continuous period of six (6) hours.
      b) The minimum six (6) hour recovery effort described herein is a contractor means-and-methods obligation and shall not be compensable under CLIN 2.11 or any other CLIN.
      c) Upon completion of the required recovery effort, the Contractor shall immediately document conditions and provide supporting data to the CO in accordance with Section 3.1.B.10, including contemporaneous field logs, rig reports and daily documentation under Section 10.h.

c. Documentation and Eligibility Determination:
   1) The Contractor shall submit daily records and supporting drilling data demonstrating, borehole conditions at time of incident, recovery attempts performed, observed mechanical or formation-related impediments to pipe removal, and transition from active recovery efforts to event documentation.
   2) The CO will determine whether the documented conditions satisfy the criteria for CLIN 2.11 eligibility based on confirmation of qualifying subsurface conditions, and determination that continued pipe recovery is no longer feasible using standard drilling means and methods.
   3) The CO's determination shall constitute the eligibility determination for activation of CLIN 2.11.
   4) All conditions described herein shall occur within a single discrete event as defined in Section 10.m.

d. Commencement of Compensable Work
   1) Compensable time shall begin no earlier than the time the CO determines that eligibility criteria have been satisfied.
   2) Time expended during: initial drilling operations, borehole cleaning efforts, and the required six (6) hour recovery attempt shall not be included in compensable time under this CLIN.

e. Measurement for Payment:
   1) Measurement for payment will be per hour of qualifying work performed after CLIN 2.11 activation.
   2) Time used to prepare documentation, submit notifications, or await CO determination shall not be compensable.
   3) The event shall be considered concluded when the drill string is recovered, abandoned in place with CO approval, or when an alternative method of completion is directed by the CO.
   4) Periods of inactivity, standby, non-continuous recovery efforts, or inefficient or non-productive operations shall not be considered compensable time under this CLIN.
   5) A total of sixty (60) hours shall be the maximum allowable compensation under this CLIN.

f. Payment:
   1) Payment will be made at the contract unit price for stuck drill pipe following exit from canyon wall, inclusive of all labor, equipment, services, and materials necessary to complete the Work once CLIN 2.11 has been activated.

2) The contract unit price shall be full compensation for all work performed under this CLIN and shall not be separately compensated under any other CLIN.

C. Contract Line Item Number 3 (CLIN 3): Pipelines – North Rim to Supai Tunnel.

1. This item consists of Work shown in drawings Volume 03. It includes approximately 2,930 LF of 2-IN spoolable composite pipe and 520 LF 1.5-IN steel finished water pipeline and appurtenances from the finished water tanks at the new North Rim WTP site to the Supai Tunnel rest area on the North Kaibab Trail. Rehab of the existing bottle filling station, a new bottle filling station, hose bib, pressure reducing valve and vault, and drain are included at Supai Tunnel rest area.
2. Measurement for payment will be percent complete based on lineal feet installed and tested.
3. Payment will be made at the contract lump sum price.

D. Contract Line Item Number 4 (CLIN 4): Pipelines – Roaring Springs Cave to Roaring Springs Pumphouse.

1. This item consists of Work shown in drawings Volume 04. It includes reinforcing the existing 8 IN raw water pipeline from the Roaring Springs Cave to the Roaring Springs Pumphouse.
2. Measurement for payment will be based on percent complete.
3. Payment will be made at the contract lump sum price.

E. Contract Line Item Number 5 (CLIN 5): Roaring Springs Pumphouse.

1. This item consists of Work shown in drawings Volume 05. It includes a new high pressure pumping system to convey water from the Roaring Springs Cave to the new raw water tanks located at the North Rim Water Treatment Plant site. The interior of the existing Roaring Springs Pumphouse (RSPH) will be demolished, including all piping, equipment, electrical, and most interior walls. New automatic backwashing strainers, horizontal multi-stage centrifugal pumps, a new liquid hypochlorite disinfection system, and drainage valves will be installed. Additional piping and valves will be installed within the RSPH for pressure reduction on the return finished water pipeline. A finished water storage tank will be installed on the exterior of the building for finished water storage and will be connected to the to the existing TransCanyon Waterline for supply to Manzanita/Cottonwood. Electrical work consists of new electrical transformer and motor control center to accommodate the new equipment. Site work includes buried underground electrical, new yard piping, a new finished water tank, slope reinforcement, and improvements to the exterior deck and supporting crib wall. The building will include a new exterior roofing materials and HVAC system and the building's chlorine room will be converted into an operator space.
2. Measurement for payment will be based on percent complete based on a schedule of values.
3. Payment will be made at the contract lump sum price.

F. Contract Line Item Number 6 (CLIN 6): North Rim Water Treatment Plant.

1. This item consists of Work shown in drawings Volume 06. It includes a new 100 gpm North Rim Water Treatment Plant (WTP) located adjacent to the existing Emergency Medical Services (EMS) Facility. Two new welded steel 150,000 gallon raw water tanks will be constructed adjacent to the facility. The new WTP will treat the raw water source from the Roaring Springs Cave with a new ultrafiltration membrane system, and chemical feed systems. The new WTP consists of an approximately 6,200 SF building that includes administration, process, electrical, mechanical, and indoor/outdoor maintenance

areas. Two existing 2 million gallon finished water tanks will be rehabilitated including new coating and cathodic protection and new perimeter fencing will be installed. New finished water pumps will be installed in the new WTP to replace the functionality of the existing North Rim Pumphouse. Work includes a temporary access road and paving around the NR WTP.

2. Measurement for payment will be based on percent complete based on a schedule of values.
3. Payment will be made at the contract lump sum price.

G. Contract Line Item Number 7 (CLIN 7): NOT USED.

H. Contract Line Item Number 8 (CLIN 8): North Rim Wastewater Collection System – Sanitary Sewer.

1. This item consists of Work shown in drawings Volume 08. It includes replacement of approximately 5,538 LF of sanitary sewer piping and 15 manholes in Region 1-Administration; approximately 3,966 LF of sanitary sewer piping and 22 manholes in Region 2-Campground; approximately 2,118 LF of sanitary sewer piping and 9 manholes in Region 3-Concessions/Administration Housing; and pumps in Lift Station #2.
2. Measurement for payment will be based on percent complete by Region based on a schedule of values.
3. Payment will be made at the contract lump sum price.

I. Contract Line Item Number 9 (CLIN 9): Power Distribution – North Rim.

1. This item consists of Work shown in drawings Volume 09. It includes the replacement of 8 existing transformers and appurtenances, installation of 1 new transformer, replacement of 5 existing junction enclosures, replacement of 1 existing shunt reactor, and installation of 4 new pad mounted switches at the North Rim Power Distribution Center and select locations throughout the North Rim. A 750kVA standby generator will be installed in the existing Power Distribution Center and connected to the existing generation switching scheme. An above ground 20,000 gallon diesel fuel storage tank will be installed at the site adjacent to the existing generation building with a new 400 gallon day tank replacing the existing day tank inside the generation tank. The work also includes the replacement and installation of a new 4000 Amp, 480V switchgear in the generator building. The work includes a new power feed to the NR WWTP and new NR WTP.
2. Measurement for payment will be based on percent complete based on a schedule of values.
3. Payment will be made at the contract lump sum price.

J. Contract Line Item Number 10 (CLIN 10): Power Distribution – North Rim to Roaring Springs Pumphouse.

1. This item consists of Work shown in drawings Volume 10. It includes approximately 5,325 LF of three-phase power line and appurtenances following the existing overhead power utility corridor. Steel power poles will be installed at four locations in the vicinity of the existing poles: North Rim Bench, Upper Bench, Lower Bench, and Roaring Springs Bench. Existing power poles and lines will be removed. The overhead conductor will utilize a combination of bare aluminum and insulted aluminum conductor based on the steel pole configuration. New cable marker balls will also be installed along the entire overhead line route to help improve helicopter safety.
2. A fourth neutral conductor would also be installed to help detect potential phase to ground faults. Additionally, fiber optic communications would be supplied inside the fourth neutral conductor to provide communications between the RSPH and the new NR WTP.
3. Measurement for payment will be based on percent complete based on a schedule of values.
4. Payment will be made at the contract lump sum price.

K. Contract Line Item Number 11 (CLIN 11): Power Distribution – Roaring Springs Pumphouse to Manzanita.

1. This item consists of Work shown in drawings Volume 11. It includes installation of approximately 3,600 LF of three-phase underground armored power cable following the existing waterline and trail corridor from Roaring Springs Pumphouse to Manzanita. Installation of the cable will include different methods depending on the terrain, ranging from rock trenching, scree cover, rock anchor support, and being supported by the existing waterline.
2. Measurement for payment will be based on percent complete based lineal foot installed.
3. Payment will be made at the contract lump sum price.

L. Contract Line Item Number 12 (CLIN 12): Power Distribution – Cottonwood Campground.

1. This item consists of Work shown in drawings Volume 12. It includes a new 5.16 kWp (kilowatt peak) photovoltaic solar power installed on the roof of a new shade canopy at Cottonwood Campground. Work includes a new 20 kWh (kilowatt hour) battery system, outdoor rated battery enclosure, storage inverter, AC disconnect, DC disconnect, and connection to the existing Cottonwood ranger station.
2. Measurement for payment will be based on percent complete based on a schedule of values.
3. Payment will be made at the contract lump sum price.

END OF SECTION 01 27 00

This page intentionally left blank.

SECTION 01 31 00 - PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:

1. Definitions
2. Construction Coordination
3. Submittals
4. Coordination Drawings
5. Requests for Information (RFIs)
6. Project Website
7. Project Meetings
8. Environmental Coordination
9. Permits

B. Related Requirements:

1. Section 01 32 16 "Construction Schedule" for preparing and submitting Contractor's construction schedule.
2. Section 01 35 05 "Environmental Protection and Special Controls" for coordinating environmental requirements of the Contract.
3. Section 01 73 40 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
4. Section 01 75 00 "Checkout and Start-up Procedures for Water Treatment Facilities" for coordinating start-up requires at the North Rim WTP and related facilities.
5. Section 01 77 00 "Closeout Procedures" for coordinating closeout of the Contract.
6. Section 01 79 00 "Demonstration and Training" for coordinating training requirements for Government staff on new equipment.
7. Section 01 91 14 "Total Building Commissioning" for coordinating the work with Owner's Commissioning Authority.

1.2 DEFINITIONS

A. Agency with Jurisdiction: An Agency that has been granted legal authority over a location, action, or resource. This authority includes the ability to issue a permit or other legal permission document.

B. Construction Permits – Contractor Provided: Permits obtained by the Contractor based on the means and methods to conduct the work. Construction Permits are typically issued by the Agency to the Contractor and not to the National Park Service. Examples of typical Contractor Provided Permits are: National Pollutant Discharge Elimination System (NPDES) Construction Permit (ground disturbance), dewatering permit, permit for work in a roadway (with traffic management plan), temporary utility tap permit, etc.

C. Government Furnished Permits: Obtained by NPS, usually through the Architect/Engineer (A/E), during the design process and are usually concerned with impacts to resources protected by federal law. However, GF permits can include other permits that are obtained for the convenience of the

Government during design. GF permits are "furnished" to the Contractor in the Request for Proposal (RFP) and as part of the final contract documents. The Contractor must comply with GF permit provisions.

1.3 CONSTRUCTION COORDINATION

A. Coordination: Coordinate construction operations included in different Sections of Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, which depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
2. Coordinate installation of different components with other Contractors to ensure maximum accessibility for required maintenance, service, and repair.
3. Make provisions to accommodate items scheduled for later installation.
4. Where availability of space is limited, coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair of components, including mechanical and electrical.
5. Properly plan construction operations to include permit requirements. Allow enough time to execute permit provisions to maintain work schedule, site visits, inspections, and reporting deadlines.

B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.

C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to:

1. Preparation of Contractor's Construction Schedule
2. Preparation of the Schedule of Values
3. Installation and removal of temporary facilities and controls
4. Delivery and processing of submittals
5. Progress meetings
6. Permit requirements
7. Pre-installation conferences
8. Project closeout activities
9. Commissioning activities

1.4 SUBMITTALS

A. Coordination Drawings: Prepare Coordination Drawings if limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.

1. Content: Project-specific information, drawn accurately to scale. Do not base Coordination Drawings on reproductions of Contract Documents or standard printed data. Include following information, as applicable:

a. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems.
b. Indicate dimensions shown on Contract Drawings and make specific note of dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternate sketches to Contracting Officer (CO) for resolution of such conflicts.
c. Indicate space requirements for routine maintenance and anticipated replacement of components during the life of the installation.
d. Show location and size of access doors required for access to concealed dampers, valves, and controls.
e. Indicate required installation sequences.

2. Refer to individual Sections for Coordination Drawing requirements for Work in those Sections.
3. Follow NPS DSC Electronic Drawing Standards. See CAD/BIM (Computer Aided Design/Building Information Modeling) & Drafting Standards page > DSC Requirements section.

B. Coordination Drawing Organization: Organize coordination drawings as follows:

1. Floor Plans and Reflected Ceiling Plans: Show architectural, structural, mechanical, plumbing, fire-protection, fire-alarm, and electrical elements. Show locations of visible ceiling-mounted devices relative to acoustical ceiling grid. Supplement plan drawings with section drawings where required to adequately represent the Work.
2. Plenum Space: Indicate sub-framing for support of ceiling and wall systems, mechanical and electrical equipment, and related Work. Locate components within ceiling plenum to accommodate layout of light fixtures indicated on Drawings. Indicate areas of conflict between light fixtures and other components.
3. Mechanical Rooms: Provide coordination drawings for mechanical rooms showing plans and elevations of mechanical, plumbing, fire-protection, fire-alarm, and electrical equipment.
4. Structural Penetrations: Indicate penetrations and openings required for all disciplines.
5. Slab Edge and Embedded Items: Indicate slab edge locations and sizes and locations of embedded items for metal fabrications, sleeves, anchor bolts, bearing plates, angles, door floor closers, slab depressions for floor finishes, curbs and housekeeping pads, and similar items.
6. Mechanical and Plumbing Work:
   a. Sizes and bottom elevations of ductwork, piping, and conduit runs, including insulation, bracing, flanges, and support systems.
   b. Dimensions of major components, such as dampers, valves, diffusers, access doors, cleanouts and electrical distribution equipment.
   c. Fire-rated enclosures around ductwork.
7. Electrical Work:
   a. Runs of vertical and horizontal conduit 1-1/4 inches in diameter and larger.
   b. Light fixture, exit light, emergency battery pack, smoke detector, and other fire-alarm locations.
   c. Panel board, switch board, switchgear, transformer, busway, generator, and motor control center locations.
   d. Location of pull boxes and junction boxes, dimensioned from column center lines.
8. Fire-Protection System:
   a. Locations of standpipes, mains piping, branch lines, pipe drops, and sprinkler heads.
9. Review: Contracting Officer will review coordination drawings to confirm Work is being coordinated; details of coordination are Contractor's responsibility. If Contracting Officer determines coordination drawings are not prepared in scope or detail, or are otherwise deficient, Contracting Officer will inform Contractor, who shall make changes and resubmit.

C. Coordination Digital Data Files: Prepare coordination digital data files according to:

1. File Submittal Format:
   a. AutoCad.dwg, Version 2018, operating in Microsoft Windows operating system
   b. Submit or post coordination drawing files using Portable Document Format (PDF) file format.
2. Contracting Officer will furnish Contractor one set of digital data files (AutoCad.dwg) of Drawings for use in preparing coordination digital data files.
   a. Contracting Officer makes no representations as to accuracy or completeness of digital data files as they relate to Drawings.
   b. Digital Data Software Program: Drawings are available in AutoCad.dwg.

D. Division 1 documents:

1. Letter designating Project Superintendent
2. Construction Schedule
3. A Comprehensive Schedule of Values
4. Accident Prevention Plan
5. A List of Subcontractors for this project
6. Written statements from Subcontractors certifying compliance with applicable labor standard clauses.
7. Certificates of Insurance and Standard Form SF1413 for Contactor and all Subcontractors
8. Waste Management Plan
9. Quality Control Plan
10. Temporary Storm Water Pollution Prevention Plan (SWPP or UPPP)
11. Indoor Air Quality (IAQ) Management Plan
12. Contractors Commissioning Plan
13. Historic Preservation Treatment Plan (HPTP)
14. List of Required Construction Permits. Include the following information for each permit:
    a. Name of Permit
    b. Agency(ies) with Jurisdiction issuing the permit
    c. Information required from Government to complete permit application

E. Provide items listed to Contracting Officer at least one month prior to Pre-Construction Conference, or as directed by Contracting Officer.

1. Failure to submit items may result in:
   a. Construction conference may be cancelled, Notice to Proceed may not be issued, and Contracting Officer will consider other contractual remedies.
   b. Work shall not commence until written Notice to Proceed has been issued and Certificates of Insurance and SF1413 documents are received for Contractor and all Subcontractors.

1.5 REQUESTS FOR INFORMATION (RFIs)

A. General: Immediately on discovery of the need for additional information or interpretation of Contract Documents, Contractor shall prepare and submit an RFI utilizing form created on NPS/DSC management software-website.

1. Contracting Officer will not respond to RFIs submitted by other entities controlled by Contractor.
2. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.

B. Content of RFI: Include detailed, legible description of item needing information or interpretation and the following:

1. RFI number, numbered sequentially
2. Date
3. RFI subject
4. Specification Section number and title and related paragraphs, as appropriate.
5. Drawing number and detail references, as appropriate.
6. Field dimensions and conditions, as appropriate.
7. Contractor's suggested resolution: If Contractor's suggested resolution impacts Contract Time or Contract Sum, Contractor shall state impact in RFI.
8. Contractor's signature
9. Requested date for response
10. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
    a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.

C. RFI Form: Complete and submit on NPS project management and communication software.

D. Contracting Officer’s Action: Contracting Officer will review each RFI, determine action required, and respond. Contracting Officer will determine critical nature of each RFI and issue response accordingly.

1. The following are not considered to be RFIs and will receive no action:
    a. Requests for approval of submittals.
    b. Requests for approval of substitutions.
    c. Requests for approval of Contractor's means and methods.
    d. Requests for coordination information already indicated in Contract Documents.
    e. Requests for adjustments in Contract Time or Contract Sum.
    f. Requests for interpretation of Architect's actions on submittals.
    g. Incomplete RFIs or inaccurately prepared RFIs.
2. Contracting Officer's action may include a request for additional information; time for response will date from time of receipt of additional information.
3. Contracting Officer's action on RFIs may result in need for a change to Contract Time or Contract Sum. All contract changes will be processed following terms and conditions of contract.

1.6 PROJECT WEB SITE

A. Use website based construction management software, provided by the NPS or their Construction Management Representative (CMR) designee, for communication throughout contract period on:

1. Project directory
2. Project correspondence
3. Meeting agendas and minutes
4. Contract modifications forms and logs
5. RFI form and processing
6. Task and issue management
7. Photo documentation
8. Baseline schedule, schedule updates and calendar management
9. Submittal form and processing
10. Payment coordination documentation

11. Drawing and specification document hosting, viewing, and updating
12. Online document collaboration
13. Reminder and tracking functions
14. Archiving functions
15. Notification of submittal and RFI statuses and current responsible party
16. Permits and addendums

B. Some documents are not suitable to be shared using the construction management software website. Documents containing Personal Identifying Information (PII) (i.e. certified payrolls), cybersecurity related access, or similar shall not be shared using construction management software website and shall be coordinated with the Project team as appropriate.

1.7 PROJECT MEETINGS

A. Preconstruction Conference: Before start of construction, Contracting Officer will arrange an on-site meeting with Contractor. Meeting agenda will include the following as a minimum:

1. Roles & Responsibilities / Lines of Authority
2. Park rules and regulations
3. Jobsite Safety
4. Resolution of comments on required Division 1 documents
5. Coordination of Subcontractors
6. Labor law application
7. Modifications
8. Payments to Contractor
9. Payroll reports
10. Contract time
11. Liquidated damages
12. Contractor Performance Evaluation
13. Display of Hotline posters
14. Notice to proceed
15. Correspondence procedures
16. NPS/DSC Project website
17. Acceptance/rejection of work
18. Progress meetings
19. Submittal procedures
20. NPS Final Accessibility Inspection
21. Environmental requirements
22. Permit requirements
23. As-constructed drawings/operation and maintenance (O&M) manuals.
24. Saturday, Sunday, holiday and night work.
25. Reference materials
26. Value engineering
27. Schedule of Values

B. Progress Meetings: Contracting Officer will schedule weekly meetings with Contractor.

1. Attendees: In addition to Government Representatives, each Contractor, Subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented. Participants at meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Meeting agenda will include:

a. Approval of minutes of previous meetings
b. Submittal status
c. Review of off-site fabrication and delivery schedules.
d. Requests for information (RFI) and other issues.
e. Modifications
f. Work in progress and projected.
    1) Status of required inspections (Special Inspections, Accessibility, etc.)
g. Inspections of work in progress and projected (Special inspections, Accessibility, etc.)
h. Construction Schedule update (provide updated Critical Path Method (CPM)).
i. Status of Project Record Drawings and O&M manuals.
j. Other business relating to work.
k. Permit requirements

C. Preinstallation Conferences: Conduct at Project site before each construction activity that requires coordination with other construction.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend meeting. Advise Contracting Officer of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for particular activity under consideration, including requirements for:
    a. Contract Documents
    b. Options
    c. Related RFIs
    d. Related Change Orders
    e. Purchases
    f. Deliveries
    g. Submittals
    h. Review of mockups
    i. Possible conflicts
    j. Compatibility requirements
    k. Time schedules
    l. Weather limitations
    m. Manufacturer's written instructions
    n. Warranty requirements
    o. Compatibility of materials
    p. Acceptability of substrates
    q. Temporary facilities and controls
    r. Space and access limitations
    s. Regulations of agency(ies) with jurisdiction
    t. Testing and inspecting requirements
    u. Installation procedures
    v. Coordination with other work
    w. Required performance results
    x. Protection of adjacent work
    y. Protection of construction and personnel
3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
4. Reporting: Distribute minutes of meeting to each party present and to other parties requiring information.
5. Do not proceed with installation if conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of Work and reconvene conference at earliest feasible date.

1.8 ENVIRONMENTAL COORDINATION

A. Contractor's Environmental Manager: Designate on-site party responsible for overseeing Contractor's conformance to environmental goals for project and implementing procedures for environmental protection.

1. Qualifications: Minimum 3 years Construction experience on projects of similar size and scope; with environmental procedures similar to this project; familiar with environmental regulations applicable to construction operations.
2. Responsibilities: Responsibilities shall include:
    a. Compliance with applicable Federal, State, and local environmental regulations, including maintaining required documentation.
    b. Implementation of Waste Management Plan (WMP).
    c. Implementation of Indoor Air Quality (IAQ) Management Plan.
    d. Implementation of Storm Water Pollution Prevention Plan (SWPPP).
    e. Present overview of environmental issues and summarize site specific procedures relating to management plans at Preconstruction conference.
    f. Training for Contractor personnel in accordance with position requirements.
    g. Monitoring and documentation of environmental procedures.

B. Perform project quality control in accordance with requirements specified in Related Sections, including:

1. Quality Requirements
2. Regulatory Requirements
3. Indoor Air Quality (IAQ) Management
4. Noise and Acoustics Management
5. Temporary Storm Water Pollution Prevention Environmental Management
6. Construction Waste Management

C. Contractor's Environmental Training Program: Contractor shall provide environmental training for workers performing work on project site. Training shall include:

1. Overview of environmental issues related to building industry.
2. Overview of environmental issues related to Project.
3. Review of site-specific procedures and management plans:
    a. Construction Waste Management
    b. Indoor Air Quality (IAQ) Management
    c. Noise and Acoustics Management
    d. Temporary Storm Water Pollution Prevention
4. Pollution Prevention (P2) practices: Submit evidence of participation in P2 programs illustrating familiarity with P2 practices.
5. Compliance with environmental regulations: As specified in Regulatory Requirements. Submit Contractor 40 CFR (Code of Federal Regulations) employee training records upon request of Contracting Officer.

D. Provide documentation for environmental procedures as specified herein and in accordance with approved Waste Management Plan, IAQ Management Plan, and Storm Water Pollution Prevention Plan.

1.9 PERMITS

A. General:

1. Permits and Responsibilities: Contractor shall, without additional expense to the Government, be responsible for obtaining necessary licenses and permits, and for complying with Federal, State and municipal laws, codes, and regulations applicable to the performance of the work. Contractor shall also be responsible for damages to persons or property that occur as a result of Contractor's fault or negligence; and for materials delivered and work performed until completion and acceptance of the work.
2. For the purpose of this contract, Contractor will not be considered an agent of the Government. Contractor shall comply with appropriate Federal, State and local laws.

B. Government Furnished Permits: During development of the project's design, permits listed below were negotiated and agreed to by the Government. Terms and provisions of these permits shall be adhered to for the duration specified in each permit.

1. Arizona Department of Environmental Quality (ADEQ)
    a. Approvals to Construct (ATC)
        1) North Rim Water Treatment Plant
            a) Water Treatment (Volume 6)
            b) Upper Zone Pumps (Volume 6)
        2) Innercanyon Potable Water Backfeed System
            a) Potable Water Backfeed Pipeline (Volume 1,2,5)
            b) RSPH Storage Tank (Volume 5)
            c) RSPH Chlorine System (Volume 5)
        3) Supai Tunnel Potable Water Backfeed
            a) Pipelines – North Rim to Supai Tunnel (Volume 3)
    b. Air Quality Permit
        1) Diesel Generator (Volume 9)
2. Arizona Department of Transportation
    a. Overall Encroachment Permit for routine maintenance of SR67

C. Potential Permits: Permits listed below were identified during the design process as likely to be required based on typical means and methods of construction. The list is provided to assist Contractor in determining which permits will be required for contract's chosen means and methods. The list shall not be considered complete; it is the Contractors' responsibility to determine means and methods and obtain required permits. Contractor shall obtain all permits required to legally conduct work.

1. Arizona Department of Environmental Quality
    a. AZPDES Construction Activity General Permit (CGP) for Stormwater with accompanying SWPPP
    b. Approvals of Construction (AOC)
        1) North Rim Water Treatment Plant
            a) Water Treatment (Volume 6)
            b) Upper Zone Pumps (Volume 6)
        2) Innercanyon Potable Water Backfeed System
            a) Potable Water Backfeed Pipeline (Volume 1,2,5)
            b) RSPH Storage Tank (Volume 5)
            c) RSPH Chlorine System (Volume 5)
        3) Supai Tunnel Potable Water Backfeed
            a) Pipelines – North Rim to Supai Tunnel (Volume 3)
2. Arizona Department of Transportation
    a. Class A or C Oversize/Overweight Permit
    b. Encroachment Permit for Snow Clearing/Maintenance of SR67 during construction
3. Arizona Department of Water Resources
    a. Notice of Intent
        1) Pipelines – North Rim to RS Pumphouse – Directional Drill (Volume 2)

4. Garkane Energy
    a. Notice of Intent for temporary utilities

D. Coordination with Agency(ies) with Jurisdiction Issuing Permits

1. Coordination: Contact the Agency(ies) with Jurisdiction as needed and sufficiently in advance to avoid delaying work: Coordinate meetings, reporting requirements, inspections, and other requirements.

E. Administrative Procedures:

1. Coordinate scheduling and timing of required administrative provisions of project permits with Agency(ies) with Jurisdiction, Construction Manager, and Park to avoid conflicts.
2. Supply needed information to Agency(ies) with Jurisdiction issuing permits, pay fees required and provide material needed to comply with permit's conditions and provisions.
3. Upload permits to construction management software website when permits are obtained.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 31 00

This page intentionally left blank.

SECTION 01 32 16 – CONSTRUCTION SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of Construction Schedule requirements including:

1. Schedule of Values
2. Construction Schedule Requirements.
3. Construction Schedule Updates.
4. Time Impact Analysis.

B. Purpose: The Construction Schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the work by the Contractor. It shall assist the Contractor and Contracting Officer (CO) in monitoring the progress of the work, evaluating proposed changes, and processing Contractor's monthly progress payments. It shall include the dates in the contract, phases, milestones, occupancies, holidays, weather consideration, a critical path, and the requirements of this section.

1.2 DEFINITIONS

A. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.

1. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times.
2. Predecessor Activity: An activity that precedes another activity in the network.
3. Successor Activity: An activity that follows another activity in the network.

B. Cost Loading: Allocation of the Schedule of Values for completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by the Contracting Officer.

C. Critical Path Method (CPM): Method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.

D. Critical Path: Longest connected chain of interdependent activities through the network schedule that establishes minimum overall Project duration and contains no float.

E. Float: Measure of leeway in starting and completing an activity.

1. Float: Not for the exclusive use or benefit of the Government or Contractor but is jointly owned.
2. Free Float: Amount of time an activity can be delayed without adversely affecting the early start of the successor activity.
3. Total Float: Measure of leeway in starting or completing an activity without adversely affecting planned Project completion date.

F. Resource Loading: Allocation of manpower and equipment necessary for completion of an activity as scheduled.

G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

1.3 SUBMITTALS

A. Electronic Copies: Schedules and reports submitted shall be posted on the NPS/DSC management software website in native electronic file formats. The intent of the Government is to limit the number of printed reports to those determined by the project team as essential.

B. Schedule of Values: After contract award and before Pre-Construction conference, submit schedule of dollar values based on Contract Price Schedule.

C. Construction Baseline Schedule: After contract award and before Pre-Construction conference, submit pdf electronic copy of baseline schedule, showing activities across the entire schedule for entire construction period. Submit a second pdf electronic schedule with four week look ahead starting at mobilization to the site. Utilize Schedule of Values in preparation of Construction Baseline Schedule.

D. Critical Path Method (CPM) Reports: Concurrent with CPM schedule, submit 3 paper copies of the following computer-generated reports. For each activity, include activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.

1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
2. Logic Report: List of predecessor and successor tasks for activities sorted in ascending order by activity number and then early start date, or actual start date if known.
3. Total Float Report: List of activities sorted in ascending order of total float.

E. Construction Schedule Updates: On or before 7th day preceding progress payment request date, submit estimates of percent completion of each schedule activity and necessary supporting data.

F. Construction Schedule Revisions and Time Impact Analysis: For each Construction Schedule revision, submit 2 paper copies of a Time Impact Analysis. Incorporate a Fragmentary Network (Fragnet) into currently accepted Construction Schedule that demonstrating how Contractor proposes to incorporate a modification, change, delay, or Contractor request.

1.4 QUALITY ASSURANCE

A. Contractor shall meet with Contracting Officer on day of the preconstruction conference to go over:

1. Review software limitations, content, and format for reports.
2. Verify availability of qualified personnel needed to develop and update schedule.
3. Discuss constraints including work stages, area separations, interim milestones, and partial Government occupancy/substantial completions.
4. Review delivery dates for Government-furnished products.
5. Review schedule for work of separate Government contracts.
6. Review time required for review of submittals and re-submittals.

7. Review requirements for tests and inspections by independent testing and inspecting agencies.
8. Review time required for completion and startup procedures.
9. Review time required for obtaining and activating permits.
10. Review and finalize list of construction activities to be included in schedule.
11. Review baseline schedule comments, resolve issues and progress on incorporating them.
12. Review procedures for updating schedule.
13. Discuss reporting requirements and establish protocol for naming and transmitting electronic schedules.

B. Contractor's Schedule Representative: Before the preconstruction conference, designate an authorized representative to be responsible for preparing and maintaining the Construction Schedule. Submit resume outlining qualifications of Scheduler to Contracting Officer for acceptance. Scheduler shall have prepared and maintained at least 5 previous schedules of similar size and complexity similar to this Contract, demonstrating proficiency of using scheduling software. Authorized representative will be responsible for preparing the Baseline Schedule, required updates, revisions, Time Impact Analyses, and reports.

1.5 COORDINATION

A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate Contractors.

B. Coordinate Construction Baseline Schedule with Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.

1. In developing Construction Baseline Schedule, ensure Subcontractor’s work at all tiers, and prime Contractor’s work, is included and coordinated.
2. Secure time commitments for performing critical elements of work from parties involved.
3. Coordinate each construction activity in network with other activities and schedule in proper sequence.

PART 2 - PRODUCTS

2.1 SCHEDULE OF VALUES

A. Breakdown each lump-sum item into component work activities used in the schedule for which progress payments may be requested. Work activities broken out within schedule of values shall be integrated into and made a logical part of the construction baseline schedule. Total costs for the component work activities shall equal contract price for that lump-sum item. Contracting Officer may request data to verify accuracy of dollar values. Include mobilization, general condition costs, overhead and profit in the total dollar value of unit price items and in the component work activities for each lump-sum item. Do not include mobilization, general condition costs, overhead or profit as a separate item.

B. Do not break down unit price items. Use only the contract price for unit price items.

C. Total cost of all items shall equal the contract price. The Schedule of Values will form the basis for progress payments and the Construction Schedule.

2.2 CONSTRUCTION SCHEDULE REQUIREMENTS

A. Construction Baseline Schedule: Prepare Construction Baseline Schedule using a computerized, cost and resource-based, time-scaled Critical Path Method network analysis diagram for the Work.

1. Develop a Draft Construction Baseline Schedule with a focused view of the first 120-days of construction within 15 days after date established for the Notice of Award.
2. Develop and finalize Construction Baseline Schedule so it can be accepted for use no later than 45 days after date established for the Notice of Award.
    a. Failure to include any work item required for performance of this Contract shall not excuse Contractor from completing work within applicable completion dates, regardless of Governments acceptance of schedule.
3. Establish procedures for monitoring and updating Construction Baseline Schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.

B. Construction Baseline Schedule Preparation: Prepare a list of all activities required to complete the Work. Using preliminary Critical Path Method network diagram, prepare a skeleton network to identify probable critical paths.

1. Activities: Indicate estimated duration, sequence requirements, and relationship of each activity in relation to other activities.
2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates shall be consistent with Contract milestone dates.
3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the Critical Path Method schedule within the limitations of the Contract Time.
4. Show sequence and interdependence of activities required for completion of work. Ensure work sequences are logical and Construction Baseline Schedule shows a coordinated plan of the work.
5. Resource loading of each activity shall include personnel by labor category and equipment type and capacity proposed to complete the activity in duration shown.
6. Consider seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time.
7. Time Frame: Proposed duration assigned to each activity shall be Contractor's best estimate of time required to complete activity considering the scope and resources planned for activity.
    a. An early finish date may be shown but the late finish date shall be same date as last day of contract period. An early completion schedule shall contain:
        1) Insert an activity titled "Project Float" as a successor to last activity in early project completion schedule network.
        2) Add a milestone titled "Contract End Date" as a successor to the activity "Project Float".
        3) Add duration to the activity "Project Float" as required so the milestone "Contract End Date" equals the last day of Contract Period.
    b. Contract completion date shall not be changed by submission of a schedule that shows an early completion date.
    c. Contractor shall limit use of lead or lag duration’s between schedule activities.
    d. Project Calendars: Develop and incorporate the following calendars:
        1) Administrative Calendar: Include calendar based on a 7-day week to be used on activities based on calendar days. Apply this calendar to administrative tasks or other tasks not affected by non-working days (Federal Holidays, weather, etc.).

2) Project Calendar: Include calendar based on planned work week for the project. Include Federal Holidays, weekends, and non-workdays indicated in contract documents. Apply this calendar to activities not anticipated to be affected by weather. Be clear when identifying number of work days in work week.
3) Weather Calendar: Utilize Project Calendar and show anticipated normal downtime related to weather as non-working time. Weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer. Apply this calendar to activities anticipated to be affected by weather.

e. Activity Duration: Define so no activity is longer than 14 days, except for non-construction activities including mobilization, shop drawings and submittals, fabrication and delivery of materials and equipment.

f. Procurement Activities: Include procurement process activities for long lead items requiring a cycle of more than 60 calendar days and including major items listed below as separate activities in the schedule. Procurement cycle activities can include submittals, approvals, purchasing, fabrication, and delivery.
1) Volume 3 – Pipelines – North Rim to Supai Tunnel
a) Flexsteel
2) Volume 5 – Roaring Springs Pumphouse
a) Raw water pumps
b) Backwash strainers
c) Network servers
d) Motor Control Centers
e) Transformer
3) Volume 6 – North Rim Water Treatment
a) Ultrafiltration membranes
b) Reaction Tanks
c) Motor Control Centers
d) Raw Water Tanks
e) Network servers
4) Volume 9 -Power Distribution - North Rim
a) Standby generator
b) Transformers
c) Diesel fuel storage tank
d) Electrical switchgear
5) Volume 10 – Power Distribution – North Rim to Roaring Springs Pumphouse
a) Micropile foundations
b) Steel power poles
c) Cable

g. Submittal Review Time: Include review and re-submittal times indicated. Coordinate submittal review times in Construction Baseline Schedule.

h. Startup and Testing Time: Include not less than 30 days for startup and testing and commissioning activities.

i. Substantial Completion: Allow time for Government administrative procedures necessary for certification of Substantial Completion. (For more information, refer to Specification 01 77 00 "Closeout Procedures.")

8. Constraints: Include constraints and work restrictions indicated in Contract Documents and as follows in schedule and show how the sequence of Work is affected.
a. Phasing: Arrange list of activities on schedule by Volume.
b. Work Restrictions: Show effect of the following on the schedule:
1) Coordination with existing operations

2) Limitations of continued occupancies
3) Uninterruptible visitor services
4) Partial occupancy before Substantial Completion
5) Use of premises restrictions
6) Provisions for future construction
7) Seasonal variations and associated weather related access to the site
8) Environmental control
9) Permit provisions
10) Trail, rest area, campground and campsite closures

c. Other Constraints:
1) Endangered Species, migration, and nesting activities identified in the NEPA documents
2) Reference 01 11 00 Section 1.3 for other constraints.

d. Work Stages: Indicate important stages of construction for each major portion of the Work.
1) Subcontract awards
2) Submittals
3) Purchases
4) Mockups
5) Fabrication
6) Sample testing
7) Deliveries
8) Installation
9) Tests and inspections
10) Adjusting
11) Curing
12) Building flush-out.
13) Building commissioning activities.

9. Milestones: Include milestones indicated in Contract Documents in schedule, including, but not limited to Notice to Proceed and Substantial Completion,.

C. Joint Review, Revision, and Acceptance:

1. Within seven calendar days of receiving Contractor's proposed Construction Baseline Schedule, Contracting Officer shall review initial Construction Baseline Schedule.
2. Within seven calendar days after review, Contractor shall revise and resubmit Construction Baseline Schedule in accordance with comments presented from review.
3. In the event the Contractor fails to define any element of work, activity, or logic, and the Contracting Officer review does not detect this omission or error, such omission or error, when discovered by Contractor or Contracting Officer, shall be corrected by Contractor within seven calendar days and shall not affect contract period.
4. Upon acceptance of the Construction Baseline Schedule, Contracting Officer saves schedule as a baseline and updates on a monthly basis. Construction schedule update will be used to evaluate Contractor's monthly applications for payment based upon information developed at monthly Construction Schedule update meeting.

D. Cost Correlation/Cost Loaded Schedule: Within 45-days of acceptance of the Construction Baseline Schedule, submit a Cost Loaded Schedule. In the heading of the Construction Baseline Schedule, provide a cost correlation line, indicating planned and actual costs. On the line, show dollar volume of Work performed as of dates used to prepare payment requests.

1. Contractor shall assign cost to construction activities on Construction Baseline Schedule. Costs shall not be assigned to submittal activities unless specified otherwise but may, with Contracting Officer’s approval, be assigned to fabrication and delivery activities. Costs shall be included for testing and commissioning activities, operation and maintenance

manuals, punch list activities, Project Record Documents, and demonstration and training (if applicable).
2. Each activity cost shall reflect an accurate value based on the Contract Price Schedule.
3. Total cost assigned to activities shall equal total Contract Price.

E. Recovery Schedule: When periodic schedule update indicates Work is 14 or more calendar days behind current accepted schedule, a separate recovery schedule indicating means by which Contractor intends to regain compliance with the schedule shall also be submitted. Indicate changes to working hours, working days, crew sizes, and equipment required to achieve compliance, and date by which recovery shall be accomplished.

F. Computer Software: Prepare schedules using a program developed specifically to manage construction schedules.

1. Use Microsoft Project or Primavera, for Windows 7 and newer.

PART 3 - EXECUTION

3.1 CONSTRUCTION SCHEDULE UPDATES

A. Progress Meeting Updates: Provide a 4 week look-ahead schedule, derived from the currently accepted schedule, before each weekly progress meeting. Utilize look-ahead schedule to facilitate and take notes on discussions held during progress meeting. The Contractor shall include a column in the 4-week look ahead schedule to flag any activities requiring resource monitoring (e.g., archaeological, biological, vehicle inspections, etc.).

B. Monthly Schedule Updates:

1. General: Update Construction Schedule on monthly basis to reflect construction progress and activities throughout entire contract period and until project substantial completion. The status date of each schedule update shall be the 7th day preceding the progress payment request date.
2. Procedure: Contractor shall meet with Contracting Officer each month at Construction Schedule update meeting to review progress made through the status date of the Construction Schedule update, including dates activities were started or completed and percentage of work completed on each activity started or completed.
3. Reports: Concurrent schedule revisions, prepare tabulated reports showing:
    a. Identification of activities that have changed
    b. Changes in early and late start dates
    c. Changes in early and late finish dates
    d. Changes in activity durations in workdays
    e. Changes in the critical path
    f. Changes in total float or slack time
    g. Changes in the Contract Time
4. Narrative: Report shall include a brief description of actual progress made during update period; actual and potential delaying activities; impediments to progress; issues related to inclement weather; progress toward established milestones and project float. Report shall include a brief description of work anticipated to be performed in the next month. Minor revisions to the schedule should be identified for evaluation and acceptance or rejection.
5. As Work progresses, indicate Actual Completion percentage for each activity.
6. If schedule update shows a late finish date after contract completion date, include:

    a. Known delays

b. Actions to get back on schedule
c. Pending modifications
d. Impediments or constraints affecting progress

7. Progress Payments: Monthly updating of the currently accepted Construction Schedule shall be an integral part of the process upon which progress payments will be made. If Contractor fails to provide schedule updates or revisions, a portion of the monthly payment may be retained until corrections have been made.

C. Distribution: Distribute copies of accepted schedule to Contracting Officer, Contracting Officers Representative, Construction Management Representative, Subcontractors, testing and inspecting agencies, and other parties identified by Contractor with a need-to-know schedule responsibility.

1. Post copies in Project meeting rooms and temporary field offices.
2. When revisions are made, distribute updated schedules to same parties and post in same locations. Delete parties from distribution when they have completed their assigned portion of the Work.

D. Construction Schedule Revisions:

1. Required Revisions: If, as a result of the monthly schedule update, it appears the currently accepted Construction Schedule no longer represents actual prosecution and progress of the work, Contracting Officer will request, and Contractor shall submit, a revision to the Construction Schedule. Contractor may also request reasonable revisions to currently accepted Construction Schedule in event the Contractor's planning for the work is revised. If Contractor desires to make changes, Contractor shall notify Contracting Officer in writing, stating reason for proposed revision. Accepted revisions shall be incorporated into currently accepted Construction Schedule for next monthly schedule update.
2. Procedure: If revision to currently accepted Construction Schedule is contemplated, Contractor or Contracting Officer shall advise the other in writing at least seven calendar days prior to next monthly schedule update meeting, describing revision and reasons for the revision. Government-requested revisions will be presented in writing to the Contractor, who shall respond in writing within seven calendar days.
3. Reports: Concurrent with making revisions to schedule, prepare tabulated reports showing:
   a. Identification of activities changed
   b. Changes in early and late start dates
   c. Changes in early and late finish dates
   d. Changes in activity durations in workdays
   e. Changes in critical path
   f. Changes in total float or slack time

3.2 TIME IMPACT ANALYSIS FOR CONTRACT MODIFICATIONS CHANGES DELAYS AND CONTRACTOR REQUESTS:

1. Requirements: When contract modifications or changes are initiated, delays experienced, or Contractor desires to revise currently accepted Construction Schedule, Contractor shall submit to Contracting Officer a written time impact analysis illustrating the influence of modification, change, delay, or Contractor request on contract time.
2. Time Extensions: Activity delays, resulting in a late completion date projection, shall not automatically mean an extension of contract time is warranted or due to Contractor. It is possible a modification, change, or delay will not affect existing critical path activities or cause non-critical activities to become critical. A modification, change, or delay may result in absorbing a part of available total float that may exist within an activity chain of the Schedule, not causing any effect on contract time. Time extensions will be granted in accordance with terms of contract.

3. Extension of contract time will be granted only to the extent the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.
4. Procedure: Each time impact analysis shall be submitted within time period stated in a request for proposal, or time period designated under the clauses entitled Changes or Default. In cases where Contractor does not submit a written request for extension of time and a time impact analysis within the designated time, it is mutually agreed that the particular modification, change, delay, or Contractor request does not require an extension of the contract time. Upon acceptance, time impact analysis shall be incorporated into currently accepted Construction Schedule at next monthly schedule update.
5. Contract Modifications: Prepare time-impact analysis using fragnets to demonstrate effect of proposed change on overall Construction Schedule for each proposed contract modification concurrent with submission.

END OF SECTION 01 32 16

This page intentionally left blank.

SECTION 01 32 33 – PHOTO DOCUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for:

1. Existing condition images
2. Periodic construction images

B. See Section 01 77 00 "Closeout Procedures" for a complete listing of closeout documents.

C. See Section 01 79 00 "Demonstration and Training" for submitting videotapes of demonstration of equipment and training of National Park Service (NPS) personnel.

1.2 SUBMITTALS

A. Construction Images: Submit images electronically within 7 days of taking the image. Include:

1. Date, time and number (sequentially number all images) in filename.
2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.

B. Closeout: Submit complete set of digital image electronic files as a Project Record Document**.** Submit on Digital Video Disc (DVD).

1. Provide index as separate file on Disc. List each image as a file name with number, date, and time. Include description and or vantage point image was taken.

PART 2 - PRODUCTS

2.1 FORMAT REQUIREMENTS

A. JPG file format submitted to the CMR hosted project website.

B. All Photo images are to be organized by Volume with a photo contact sheet for image referencing included in each Volume.

PART 3 - EXECUTION

3.1 CONSTRUCTION IMAGES

A. General: Take digital images using the maximum range of depth of field, in-focus, to clearly show the Work. No blurry or out-of-focus areas accepted.

1. Maintain index with each set of Construction images and identify the number, date, time, and description for each.
2. Maintain one set of images accessible in field office at Project site available for reference.

B. Existing Condition Images: Before starting construction, take color digital images of Project site and surrounding properties, including existing items to remain during construction, from different vantage points.

1. Flag construction limits before recording construction images.
2. Take twenty separate images for each Volume reference to show existing conditions before starting Work.
    a. Volume 0 – General
        1) Lindberg Hill construction camp (20)
        2) Lindberg Hill access (10)
        3) Dusty Road construction camp (20)
        4) North Rim WWTP construction camp (20)
        5) WTP staging (10)
        6) Cottonwood Campground construction camp (10)
        7) Cottonwood Campground resthouse (10)
        8) Cottonwood Campground staging area (10)
    b. Volume 01: Pipelines – North Rim
        1) Every 50 FT along North Rim pipe alignment
        2) Every 50 FT along Ballfield access road alignment
        3) Each roadway crossing (5)
        4) Ballfield limits of construction (10)
        5) Ballfield existing site conditions (10)
    c. Volume 02: Pipelines – North Rim to Roaring Springs Pumphouse
        1) Ballfield limits of construction after Volume 01 prep (10)
        2) Ballfield existing site conditions after Volume 01 prep (10)
        3) Bore exit (20)
    d. Volume 03: Pipelines – North Rim to Supai Tunnel
        1) Every 200 FT along Supai tunnel alignment
        2) Supai resthouse (10)
        3) Supai rest area site features (20)
    e. Volume 04: Pipelines – Roaring Springs Cave to Roaring Springs Pumphouse
        1) Every 50 FT along existing pipeline
        2) Existing bridge crossing (5)
        3) Every 50 FT along new pipeline alignment
    f. Volume 05: Roaring Springs Pumphouse
        1) Roaring Springs Pumphouse LOC (20)
        2) Roaring Springs Pumphouse Interior (20)
        3) Roaring Springs Pumphouse Exterior (20)
        4) Roaring Springs Pumphouse Helibad Exterior (10)
        5) Transformer and power alignment (10)
        6) Pipeline alignment from bore exit to Roaring Springs Pumphouse (10)
    g. Volume 06: North Rim Water Treatment Plant
        1) Water Treatment Plant LOC (20)

2) Finished Water Tanks interior (20 each)
3) Finished Water Tanks exterior (20 each)
4) North Rim Pumphouse interior (10)
5) North Rim Pumphouse exterior (10)
6) Existing historical fire pump station (5)
7) Existing EMS driveway entrance and parking (10)

h. Volume 07: NOT USED

i. Volume 08: North Rim Wastewater Collection System
1) Each manhole in project scope interior and exterior
2) Every 50 FT along pipeline replacement
3) Buildings within 50 FT of pipeline replacement (10 each)
4) Exterior hardscape features (curbs, medians, boulders, walls, etc.) and roads within 50 FT of pipeline replacement

j. Volume 09: Power Distribution – North Rim
1) Power Distribution Center LOC (20)
2) Power Distribution Center exterior (20)
3) Power Distribution Center interior (20)
4) Power Distribution Center transformer yard (20)
5) Power Distribution Center site features and roads (20)
6) Each transformer and electrical equipment replacement (10)

k. Volume 10: Power Distribution – North Rim to Roaring Springs Pumphouse
1) North Rim Grand Lodge parking area LOC (20)
2) North Rim bench (20)
3) Upper bench (20)
4) Lower bench (20)
5) RSPH bench (20)

l. Volume 11: Power Distribution – Roaring Springs Pumphouse to Manzanita
1) Every 50 FT along alignment

m. Volume 12: Power Distribution – Cottonwood Campground
1) Solar facility LOCs (20)
2) Cottonwood Ranger Station exterior (10)
3) Cottonwood Storage Building exterior (10)

C. Periodic Construction Images: Take 25 color, digital images for each volume weekly, with timing each month adjusted to coincide with cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last images were taken.

D. Additional Images: Contracting Officer may issue requests for additional images.

1. Three days advance, where feasible.
2. In emergency situations, take additional images within 24 hours of request.
3. Additional images include, but are not limited to:
   a. Immediate follow-up when on-site events result in construction damage or losses.
   b. Fabrication locations away from Project site.
   c. Substantial Completion of a major phase or component of Work.
   d. Extra record images at time of final acceptance.

E. Use of Unmanned Aerial Vehicles (UAVs) / drones for imagery or other purposes is prohibited.

END OF SECTION 01 32 33

This page intentionally left blank.

SECTION 01 33 23 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.2 DEFINITIONS

A. Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.

B. Informational Submittals: Written information that does not require Government's responsive action. Submittals may be rejected for not complying with requirements.

C. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access files.

D. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

1.3 GENERAL SUBMITTAL PROCEDURES

A. General: Prepare and submit submittals required by individual Specification Sections and in some cases as requested in drawings. Types of submittals are indicated in individual specific sections.

1. Contracting Officer (CO) reserves right to require submittals in addition to those called for in individual sections.
2. The contractor will digitally upload submissions into the project management software system. The system will be provided and managed by the Governments on-site Construction Management Representative (CMR). The automated system will receive, review, reject, approve/approve with notations, and/or accept/reject and be the administrative project record. The software platform will allow access to contractor, Engineer of Record, CMR and NPS CO (owner); at award of contract the contractor is to submit list of team members that will need access to the software platform to the Lead CMR. The Lead CMR will assign roles and communication flows for the review and processing of submittals. The system will provide a work-flow status of individual submittals in real time. Weekly Construction Meetings will include discussion section to get clarification on status of a submittal in review or comments on a reviewed submittal transmitted back to contractor. The same system will be used for submission of Requests for Information (RFI) and managed through similar work-flow but do not require a CM-16 cover page. Contractor must provide a cover letter that provides the same basic information as a submittal.
3. Individual submittals and RFI's will be organized and transmitted by Volume with a General Submission that apply to all Volumes. Each CM-16 or RFI Cover Page will be

titled with Volume followed by brief description of the submission. Example(s): Example 1) **Volume 1: Potable Water Pipe;** 2) **General All Volumes: Trench Backfill Material.** Label high priority and long lead items on the CM-16/RFI Cover Letter, clearly mark submissions to allow the project team to prioritize the review the process.

B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Review for legibility, accuracy, completeness, and compliance with Contract Documents.

1. Coordinate submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
2. Coordinate transmittal of different types of submittals for related parts of Work so processing will not be delayed because of need for concurrent review coordination.
    a. Contracting Officer reserves right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

C. Submittal List: Submittal list is attached to the end of this Specification Section as information reference material. The intent is to provide an overall summary of submittal requirements. The requirements of individual Specification Sections and terms and conditions of the Contract still apply regardless of what is shown on submittal list. The contractor is responsible to submit and maintain a complete submittal list. Contractor submittal list will be provided within thirty (30) days from issuance of Notice to Proceed to the Contracting Officer.

D. Processing Time: Allow time for submittal review, including time for re-submittals, as follows. Time for review shall commence when e-mail notification is received by Contracting Officer (or designee) indicating submittal has been posted on NPS management software website and is ready for review. When Contracting Officer has completed review, e-mail notification will be sent to Contractor indicating submittal has been processed. No extension of Contract Time will be authorized because of failure to transmit submittals in advance of Work to permit processing, including re-submittals.

1. Action Submittals
    a. Initial Review: Allow 21 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
    b. Re-submittal Review: Allow 14 days for each review of each re-submittal, time restarts once a resubmittal is required, each with its own 14 days
2. Informational submittals
    a. Review: Allow 10 days for review of each submittal.

E. Approved Equals:

1. For each item proposed as an "approved equal," submit supporting data, including:
    a. Drawings and samples as appropriate.
    b. Comparison of the characteristics of the proposed item with that specified.
    c. Changes required in other elements of the work because of the substitution.
    d. Name, address, and telephone number of vendor.
    e. Manufacturer’s literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service and state source of replacement materials.
2. A request for approval constitutes a representation that Contractor:
    a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
    b. Will provide the same warranties for the proposed item as for the item specified.
    c. Has determined that the proposed item is compatible with interfacing items.

d. Will coordinate installation of an approved item and make changes required in other elements of the work because of the substitution.
e. Waives claims for additional expenses that may be incurred as a result of the substitution; additional expenses such as redesign costs as a result of the substitution from the basis of design listed in specifications will be responsibly of contractor and CO will issue contract cost impact in deduction as to approving the equal, any delay to baseline schedule will be burdened by contractor in cost and schedule

F. Electronic Submittals: Identify and incorporate information in each electronic submittal file as follows:

1. Transmittal Form (CM-16): All submittals shall be transmitted using National Park Service Transmittal Form (CM-16). The form can be downloaded from the DSC Workflows website's Submittal Review . No action will be taken on a submittal item unless accompanied by this Transmittal Form.
   a. Complete the general information at the top of form.
   b. Provide all required information based on submittal type
   c. Attach all related documents.
   d. Sign the Contractor section at bottom of the Transmittal Form (CM-16).
2. Physical samples: Prior submitting physical samples, send informational transmission with the proposed physical samples finishes. Once CO has determined that the proposed samples meet the contract document requirements then the CO will accept the submission for contractor to then send the physical samples. Any submission of physical samples without prior accepted informational submittal may incur additional costs for the contract for a second physical sample submission at no cost to the Government. Complete Transmittal Form (CM-16) on the NPS/DSC management software website as described above. Deliver physical sample to the Contracting Officer (or designee) on site for processing. All comments and actions will be documented on the Transmittal Form (CM-16) on the NPS/DSC management software website.

G. Identification: Submittal number or other unique identifier, including revision identifier.

1. Submittal number shall use a sequential number (e.g. .001). Re-submittals shall include alphabetic suffix after another decimal point (e.g. .001.A).

H. Re-submittals: Make re-submittals using same process used with initial submittal.

1. Note date and content of previous submittal.
2. Note date and content of revision in the title block on the Transmittal Form (CM-16) and clearly indicate extent of revision.
3. Re-submit submittals until they are marked "Approved" or "Approved with notations".

I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, and others as necessary for performance of construction activities.

J. Use for Construction: Use only final submittals with mark indicating "Approved" or "Approved with notations". Ensure notations have been incorporated and, at a minimum, keep one copy of final approved submittal on site for use during construction.

1.4 CONTRACTOR'S USE OF CAD/BIM FILES

A. General: At Contractor's written request, copies of CAD (Computer Aided Design)/BIM (Building Information Modeling) files will be provided to Contractor for Contractor's use in connection with Project as reference material only, subject to:

1. Files provided as is; no format or other changes to files or changes to objects in the drawing will be done by the Government

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
2. Mark each submittal to show which products and options are applicable.
3. As applicable, include:
    a. Manufacturer's product specifications.
    b. Manufacturer's installation instructions: When Contract Documents require compliance with manufacturer's printed instructions, provide one complete set of instructions to Contracting Officer and keep another complete set of instructions at the project site until substantial completion.
    c. Manufacturer's catalog cuts: Submit only pertinent pages; mark each page of standard printed data to identify specific products proposed for use.
    d. Wiring diagrams showing factory-installed wiring.
    e. Printed performance curves.
    f. Operational range diagrams.
    g. Compliance with specified referenced standards.
    h. Testing by recognized testing agency.
4. Submit product data in PDF (portable document format) file format before or concurrent with samples.

B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data , unless submittal of CAD/BIM Drawings is otherwise permitted.

1. Preparation: Fully illustrate requirements in Contract Documents. As applicable, include:
    a. Dimensions
    b. Identification of products
    c. Fabrication and installation drawings
    d. Roughing-in and setting diagrams
    e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring
    f. Shopwork manufacturing instructions
    g. Templates and patterns
    h. Schedules
    i. Notation of coordination requirements
    j. Notation of dimensions established by field measurement
    k. Relationship to adjoining construction clearly indicated
    l. Seal and signature of professional engineer if specified
    m. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring
2. Submit shop drawings as PDF electronic file

C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal

and actual component as delivered and installed. Reference informational submittal process in 01 33 23-3 - 1.3/E/2/e (Physical Samples - above).

1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
2. Complete and post the Transmittal Form (CM-16) on the NPS/DSC management software website for processing and documentation of action on submitted samples.
3. Identification: Attach label on unexposed side of Samples that includes:
    a. Generic description of Sample
    b. Product name and name of manufacturer
    c. Sample source
    d. Submittal Number and title of appropriate Specification Section
4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing full range of colors, textures, and patterns available.
    a. Number of Samples: Submit two full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Contracting Officer will return with options selected.
6. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
    a. Number of Samples: Submit four sets of Samples. Contracting Officer will retain three Sample sets; remainder will be returned.

D. Construction Materials: Contractor is encouraged to submit products made out of recycled or environmentally responsible material. Every effort will be made by National Park Service to approve these materials.

2.2 INFORMATIONAL SUBMITTALS

A. General: Prepare and submit Informational Submittals required by individual Specification Sections.

1. Post informational submittals as PDF electronic files directly to the NPS management software website.
2. Certificates and Certifications: Provide a notarized statement with signature of entity responsible for preparing certification. Certificates and certifications shall be signed by officer or other individual authorized to sign documents on behalf of that entity.
3. Informational submittals that do not comply with requirements specified in Contract Documents will be rejected and one copy will be returned.

B. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."

C. Contractors Construction Schedule: Comply with requirements specified in Section 01 32 16 "Construction Schedule."

D. Accident Prevention Plan: Comply with requirements specified in Section 01 35 23 "Safety Requirements."

E. Schedule of Values: Comply with requirements specified in Section 01 32 16 "Construction Schedule."

F. Waste Recycling Plan: Comply with requirements specified in Section 01 74 19 "Construction Waste Management and Disposal."

G. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."

H. Storm Water Pollution Prevention Plan: Comply with requirements specified in Section 01 57 23 "Temporary Storm Water Pollution Prevention" and storm water permit requirements identified in Section 01 31 00 "Project Management and Coordination."

I. Indoor Air Quality Management Plan: Comply with requirements specified in Section 01 57 19.11 "Indoor Air Quality Management."

J. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

K. Welding Certificates: Prepare written certification that welding procedures and personnel comply with Contract Documents. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on American Welding Society (AWS) forms. Include names of firms and personnel certified.

L. Installer Certificates: Prepare written statements on manufacturer’s letterhead certifying Installer complies with Contract Documents and, where required, is authorized by manufacturer for this specific Project.

M. Manufacturer Certificates: Prepare written statements on manufacturer’s letterhead certifying manufacturer complies with Contract Documents. Include evidence of manufacturing experience where required.

N. Product Certificates: Prepare written statements on manufacturer’s letterhead certifying product complies with Contract Documents.

O. Material Certificates: Prepare written statements on manufacturer’s letterhead certifying material complies with Contract Documents.

P. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency’s standard form, indicating and interpreting test results of material for compliance with requirements in Contract Documents.

Q. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.

R. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.

S. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in Contract Documents.

T. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.

U. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in Contract Documents.

V. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements specified in Section 01 78 23 "Operation and Maintenance Data."

W. Design Data: Prepare written and graphic information, including: performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.

X. Manufacturer's Instructions: Prepare written or published information documenting manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.

Y. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. As applicable, include:

1. Statement on condition of substrates and their acceptability for installation of product.
2. Summary of installation procedures being followed, compliance with requirements and, if not, what corrective action was taken.
3. Results of operational and other tests and a statement of whether observed performance complies with the requirements.

Z. Permit Compliance Products: Prepare required information for compliance with permit provisions. Products include written notification of project startup, suspension, and completion of work; photo documentation of site conditions; reports; and drawings.

2.3 COLOR SUBMITTALS

A. No Designated Color in the Project Specifications

1. Preliminary Color Selection:
   a. Submit a picture, scan or download of the entire color chart(s).
   b. Mail a hard copy of color chart to the project architect at the address provided by the COR.
      i. Project architect will select three colors (maximum).
      ii. COR will review the color selections with the Government.
   c. Pending Government approval, procure two physical samples for each of the selected colors.
2. Final Color Selection:
   a. Submit a picture of the physical samples.

b. Mail one set of physical samples to the Government at the address provided by the COR.
c. Mail the second set of physical samples to the project architect at the address provided by the COR.
    i. Project architect will review the physical samples and make a recommendation for the final color selection.
    ii. COR will review the recommended color selection with the Government.
d. Government will either approve the color recommendation, make a different color selection, or start the color selection process over.

B. Designated Color in the Project Specifications

1. Final Color Selection:
    a. Procure two physical samples for the color listed in the Project Specifications.
    b. Mail one physical sample to the Government at the address provided by the COR.
    c. Mail the second physical sample to the project architect at the address provided by the COR.
        i. Project architect will review the physical sample and approve or disapprove.
        ii. COR will coordinate the review of the physical sample with the Government.
    d. Government will either approve the physical sample or start the color selection process over using the "No Designated Color in the Project Specifications".

PART 3 - EXECUTION

3.1 CONTRACTOR'S REVIEW

A. Review each submittal and check for coordination with other Work of Contract and for compliance with Contract Documents. Note corrections and field dimensions.

3.2 CONTRACTING OFFICER'S ACTION

A. General: Submittals will be disapproved without technical review if identification information is missing, not filled in, or if placed on back of submittal; an incorrect format of submittals is provided; transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of Contractor's approval.

1. Any work done or orders for materials or services placed before approval shall be at Contractor's own risk.

B. Action Submittals: Contracting Officer will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on the Transmittal Form (CM-16). Submittal will be marked as defined below:

1. APPROVED: Acceptable with no corrections.
2. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review is required. Contractor shall address review comments when proceeding with the work.
3. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. Contracting Officer will identify reasons for disapproval. Contractor shall revise and resubmit with changes clearly identified.

C. Informational Submittals: Contracting Officer will review each submittal and will either accept or reject it.

D. Partial submittals are not acceptable, will be considered non-responsive, and will be returned without review.

END OF SECTION 01 33 23

This page intentionally left blank.

Submittal List

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

SUBMITTAL LIST

Project Title: **Rehabilitate North Rim Infrastructure** **March 2026**

ACTION SUBMITTAL REVIEW ESTIMATE

ARCHITECT / ENGINEER REVIEW TIME

SUBMITTAL				REQUIREMENTS																					
				INFORMATIONAL				ACTION																	
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	01 11 00	1.8	SUMMARY OF WORK																					
ALL	01 11 00	1.8	Communications Equipment Plan		X																			
ALL	01 26 01	1.8	CONTRACT MODIFICATION PROCEDURES																					
ALL	01 27 00	1.8	DEFINITION OF CONTRACT LINE ITEMS																					
ALL	01 31 00	1.4	COORDINATION DRAWINGS				X																	
ALL	01 31 00	1.4	DIVISION 01 DOCUMENTS																					
ALL	01 31 00	1.4	Letter designating Project Superintendent				X																	
ALL	01 31 00	1.4	Construction Schedule				X												2				2	
ALL	01 31 00	1.4	Schedule of Values				X												2				2	
ALL	01 31 00	1.4	Accident Prevention Plan				X												2				2	
ALL	01 31 00	1.4	List of Subcontractors				X																	
ALL	01 31 00	1.4	Written Statement certifying compliance with labor standards	X																				
ALL	01 31 00	1.4	Evidence of liability insurance	X																				
ALL	01 31 00	1.4	Waste Management Plan				X																	
ALL	01 31 00	1.4	Quality Control Plan				X																	
ALL	01 31 00	1.4	Temporary Storm Water Pollution Prevention Plan (SWPP or UPPP)				X						4						1				1	
ALL	01 31 00	1.4	Indoor Air Quality (IAQ) Management Plan				X																	
ALL	01 31 00	1.4	Contractors Commissioning Plan				X						1		1	1	1	1	1		1		1	
ALL	01 31 00	1.4	Historic Preservation Treatment Plan				X																	
ALL	01 31 00	1.4	List of required Construction Permits				X																	2
ALL	01 31 00	1.5	REQUEST FOR INFORMATION (RFI)				X					131	131	131	131	131	131	131	131	131	131		130	
ALL	01 31 00	1.6	PROJECT WEB SITE				X																	
ALL	01 31 00	1.7	PROJECT MEETINGS				X					100	200	100	100	100	100	100	1200	100			1200	
ALL	01 31 00	1.7	Look Ahead Work Schedule Format				X																	
ALL	01 31 00	1.7	Pre-Installation Conferences List				X												2				2	
ALL	01 32 16	1.3	CONSTRUCTION SCHEDULE				X												4				4	
ALL	01 32 33	1.3	PHOTO DOCUMENTATION FOR HISTORIC PRESERVATION PROJECTS																					
ALL	01 32 33	1.3	Construction and Close Out Images								X													
ALL	01 33 23	1.3	SUBMITTAL PROCEDURES								X								4				4	
ALL	01 35 05	1.3	ENVIRONMENTAL PROTECTION AND SPECIAL																					
ALL	01 35 05	1.3	Spill Prevention Plan		X																			
ALL	01 35 05	1.3	Rehabiliation Plan		X		X																	
ALL	01 35 13.22	1.3	ARCHEOLOGICAL PROTECTION																					
ALL	01 35 13.22	1.3	Daily Work Schedule		X																			
ALL	01 35 23	1.2	ACCIDENT PREVENTION PLAN (APP)		X																		2	
ALL	01 35 73	1.3	DELEGATED DESIGN PROCEDURES				X																	
ALL	01 35 91	1.3	HISTORIC PRESERVATION TREATMENT PROCEDURES																					
ALL	01 35 91	1.3	Historic Preservation Treatment Plan		X																			
ALL	01 35 91	1.3	Photographs or Videotape				X				X													
ALL	01 40 00	1.4	QUALITY CONTROL PLAN		X														2				2	
ALL	01 40 00	1.4	Qualification Data	X	X																			
ALL	01 40 00	1.4	Daily Reports		X																			
ALL	01 40 00	1.4	Test Reports	X	X																			
ALL	01 40 00	1.4	Accessibility Inspection Report		X																			
ALL	01 40 00	1.4	Off-Site Inspection Reports	X	X																			
ALL	01 40 00	1.4	Permits, Licences, Certificates	X			X																	
ALL	01 40 00.A-1	1.4	STATEMENT OF STRUCTURAL TESTS AND SPECIAL	X										2									2	
5	01 45 25	1.3	TESTING CONCRETE STRUCTURES FOR WATERTIGHTNESS																					
5	01 45 25	1.3	Volume 5: RSPH																					
5	01 45 25	1.3	Watertightness Testing Plan	X	X									2									1	
5	01 45 25	1.3	Repair and Patching Plan		X									2									1	
5	01 45 25	1.3	Shop Drawings		X				X					4					2				2	
6	01 45 25	1.3	TESTING CONCRETE STRUCTURES FOR WATERTIGHTNESS																					
6	01 45 25	1.3	Volume 6: NR WTP																					
6	01 45 25	1.3	Watertightness Testing Plan	X	X									2									1	
6	01 45 25	1.3	Repair and Patching Plan		X									2									1	
6	01 45 25	1.3	Shop Drawings		X				X					4					2				2	
ALL	01 50 00	1.3	TEMPORARY FACILITIES AND CONTROLS																					
ALL	01 50 00	2.3	Office Trailer Plan		X														2				2	

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

SUBMITTAL				REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME										
				INFORMATIONAL				ACTION														
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	01 50 00	3.3	Traffic Control Plan & Signage Plan		X								1						1			1
ALL	01 50 00	3.3	Dewatering Plan		X								2						1			1
5	01 57 19.11	3.3	INDOOR AIR QUALITY MANAGEMENT																			
5	01 57 19.11	1.4	Volume 5: RSPH																			
5	01 57 19.11	1.4	Indoor Air Quality (IAQ) Management Plan		X														2			1
5	01 57 19.11	1.4	Product Data			X																
5	01 57 19.11	1.4	Inspection and Test Reports	X	X																	
6	01 57 19.11	1.4	INDOOR AIR QUALITY MANAGEMENT																			
6	01 57 19.11	1.4	Volume 6: NR WTP																			
6	01 57 19.11	1.4	Indoor Air Quality (IAQ) Management Plan		X														2			1
6	01 57 19.11	1.4	Product Data			X																
6	01 57 19.11	1.4	Inspection and Test Reports	X	X																	
ALL	01 57 23	3.2	SWPPP NPDES Permit (Incluedes NOI: Notice of Intent, NOT: Notice of Termination)		X								4						2			1
1	01 57 23	3.3	Volume 1:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
2	01 57 23	3.3	Volume 2:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
3	01 57 23	3.3	Volume 3:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
6	01 57 23	3.3	Volume 6:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
8	01 57 23	3.3	Volume 8:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
9	01 57 23	3.3	Volume 9:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
10	01 57 23	3.3	Volume 10:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
11	01 57 23	3.3	Volume 11:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
12	01 57 23	3.3	Volume 12:Temporary Storm Water Pollution Prevention Plan (Inspection Schedule, Products)		X								4						2			1
ALL	01 61 03	1.4	EQUIPMENT - EXHIBIT A - Per Applicable Equipment		X		X															
ALL	01 65 50	1.2	MATERIAL TRANSPORT - HELICOPTERS																			
ALL	01 65 50	1.2	Pilot Certification	X																		1
ALL	01 65 50	1.2	Rigger Certification	X																		1
ALL	01 65 50	1.2	ABRO Certification	X																		1
ALL	01 65 50	1.3	Aviation Plan (Includes Independent Communications Equipment Compatability with NPS System)	X	X		X												2			1
ALL	01 65 50	1.3	A-219 Training Certificates	X																		
ALL	01 65 50	3.1	Aviation Schedule (Weekly and Monthly)		X		X												40			20
ALL	01 67 00	1.3	PRODUCT REQUIREMENTS																			
ALL	01 67 00	1.3	Affirmative Procurement Reporting Form				X															
ALL	01 67 00	1.3	Environmental Data (1.3 / A / 2 a thru z)				X												2			
ALL	01 67 00	1.3	Material Safety Data Sheets (a thru p)				X															
ALL	01 67 00	1.3	Chain of Custody Documentation				X						1		1							
ALL	01 71 21	3.2	SPECIALTY ENGINEERING REQUIREMENTS																			
ALL	01 71 21	3.2	SSE Design Documentation Calculations		X		X							4					2			1
ALL	01 73 00	1.5	SNOW REMOVAL																			
ALL	01 73 00	1.5	Annual Winter Work Plan: Season 1 (Submit Per/Season)		X		X												2			1
ALL	01 73 00	1.6	Encroachment Permit: Season 1 (Submit Per/Season)				X												2			1
ALL	01 73 00	1.5	Annual Winter Work Plan: Season 2 (Submit Per/Season)		X		X												2			1
ALL	01 73 00	1.6	Encroachment Permit: Season 2 (Submit Per/Season)				X												2			1
ALL	01 73 00	1.5	Annual Winter Work Plan: Season 3 (Submit Per/Season)		X		X												2			1
ALL	01 73 00	1.6	Encroachment Permit: Season 3 (Submit Per/Season)				X												2			1
5	01 73 20		OPENINGS AND PENETRATIONS IN CONSTRUCTION																			
5	01 73 20	1.4	Shop Drawings - Volume 5: Roaring Springs Pump House				X		X			1		1	1	1	1		1			1
6	01 73 20		OPENINGS AND PENETRATIONS IN CONSTRUCTION																			
6	01 73 20	1.4	Shop Drawings - Volume 6: North Rim WTP				X		X			1		1	1	1	1		1			1
9	01 73 20	1.4	OPENINGS AND PENETRATIONS IN CONSTRUCTION																			
9	01 73 20	1.4	Shop Drawings - Volume 9: Power Distrobution Center - Fuel System				X		X			1		1	1	1	1		1			1
5	01 73 29		CUTTING AND PATCHING PLAN		X																	
5	01 73 29	1.2	Shop Drawings - Volume 5: Roaring Springs Pump House				X		X			1		1	1	1	1		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
9	01 73 29	1.2	CUTTING AND PATCHING PLAN		X																		
9	01 73 29	1.2	Shop Drawings - Volume 9: Power Distrobution Center - Fuel System				X		X			1		1	1	1	1		1			1
ALL	01 73 40	1.2	EXECUTION																				
ALL	01 73 40	1.2	Surveyor Certificates (A,C)				X																
ALL	01 73 40	1.2	Landfill Receipts (B)				X																
ALL	01 73 40	1.2	Quantity Surveys (D)				X																
ALL	01 73 40	1.2	Utility Management Plan (G)																2			1
1	01 73 50		TRAIL ACCESS MANAGEMENT PLAN																				
1	01 73 50	1.3	Volume 1		X								1						2			2
2	01 73 50	1.3	TRAIL ACCESS MANAGEMENT PLAN																				
2	01 73 50	1.3	Volume 2		X								1						2			2
3	01 73 50	1.3	TRAIL ACCESS MANAGEMENT PLAN																				
3	01 73 50	1.3	Volume 3		X								1						2			2
4	01 73 50	1.3	TRAIL ACCESS MANAGEMENT PLAN																				
4	01 73 50	1.3	Volume 4		X								1						2			2
5	01 73 50		TRAIL ACCESS MANAGEMENT PLAN																				
5	01 73 50	1.3	Volume 5		X								1						2			2
6	01 73 50		TRAIL ACCESS MANAGEMENT PLAN																				
6	01 73 50	1.3	Volume 6		X								1						2			2
8	01 73 50		TRAIL ACCESS MANAGEMENT PLAN																				
8	01 73 50	1.3	Volume 8		X								1						2			2
9	01 73 50	1.3	TRAIL ACCESS MANAGEMENT PLAN																				
9	01 73 50	1.3	Volume 9		X								1						2			2
10	01 73 50	1.3	TRAIL ACCESS MANAGEMENT PLAN																				
10	01 73 50	1.3	Volume 10		X								1						2			2
11	01 73 50	1.3	TRAIL ACCESS MANAGEMENT PLAN																				
11	01 73 50	1.3	Volume 11		X								1						2			2
12	01 73 50	1.5	TRAIL ACCESS MANAGEMENT PLAN																				
12	01 73 50	1.5	Volume 12		X								1						2			2
ALL	01 73 60		CONSTRUCTION CAMPS																				
ALL	01 73 60	1.4	NORTH RIM: Contractor Construction Camp Development Plan		X								1						2			2
ALL	01 73 60	1.4	INNERCANYON: Contractor Construction Camp Development Plan (Cottonwood Camp Ground)		X								1						2			2
ALL	01 73 60	1.4	SOUTH RIM: Contractor Construction Camp Development Plan (Contractor Support Facilities)		X								1						1			1
ALL	01 74 15	1.3	DISINFECTION OF FACILITIES																				
ALL	01 74 15	1.3	Product Technical Data			X	X		X				1									
ALL	01 74 15	1.3	Qualifications for use of liquid chlorine				X						1									
ALL	01 74 15	1.3	Volume 6: North Rim WTP Disinfection Plan		X								1						1			1
ALL	01 74 15	1.3	Volume 5: Roaring Springs Pump House		X								2						1			1
ALL	01 74 15	1.3	Pipe by Volume 1, 2, 3, 4		X								2						2			1
ALL	01 74 19	1.4	CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL																				
ALL	01 74 19	1.4	Waste Management Plan		X																		2
ALL	01 74 19	1.4	Progress Documentation				X																
ALL	01 74 19	1.4	Waste Reduction Calculations				X																
ALL	01 74 19	1.4	Records of Donations				X																
ALL	01 74 19	1.4	Records of Sales				X																
ALL	01 74 19	1.4	Recycling and Processing Facility Records				X																
ALL	01 74 19	1.4	Landfill and Incinerator Disposal Records				X																
ALL	01 74 19	1.4	Qualification Data				X																
ALL	01 74 19	1.4	Statement of Refrigerant Recovery				X																
ALL	01 74 19	1.4	Closeout Submittals				X																
5	01 75 00	1.4	CHECKOUT AND START-UP PROCEDURES FOR WATER TREATMENT FACILITIES																				
5	01 75 00	1.5	Demonstration Plan Volume 5: Roaring Springs Pump Station	X	X	X	X						2		2	2	2	2	2			2
6	01 75 00	1.5	CHECKOUT AND START-UP PROCEDURES FOR WATER TREATMENT FACILITIES																				
6	01 75 00	1.5	Demonstration Plan Volume 6: North Rim WTP and Tanks	X	X	X	X						2		2	2	2	2	2			2
9	01 75 00	1.5	CHECKOUT AND START-UP PROCEDURES FOR WATER TREATMENT FACILITIES																				
9	01 75 00	1.5	Demonstration Plan Volume 9: Power Distrobution Center (Generator, ATS, Fuel Tank)	X	X	X	X										4	2	2			2
12	01 75 00	1.5	CHECKOUT AND START-UP PROCEDURES FOR WATER TREATMENT FACILITIES																				
12	01 75 00	1.5	Demonstration Plan Volume 12: Cottonwood Power System	X	X	X	X										4	1	2			2
0	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 0				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
1	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 1				X		X	X	X		4	2			2		2			4

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

ARCHITECT / ENGINEER REVIEW TIME

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	INFORMATIONAL: OTHER	ACTION: SAMPLES	ACTION: SHOP DRAWINGS	ACTION: MANUFACTURER DATA AND INSTRUCTIONS	ACTION: OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
2	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 2				X		X	X	X		4	2			2		2			4
3	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 3				X		X	X	X		4	2			2		2			4
4	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 4				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
5	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 5				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
6	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 6				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
8	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 8				X		X	X	X		24	4	2	4	4	4	4			4
9	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 9				X		X	X	X	1	2	2		12	24	4	4		1	4
10	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 10				X		X	X	X			16			24		4			4
11	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 11				X		X	X	X		4	4			24		4			4
12	01 77 00	1.3	CLOSEOUT PROCEDURES - Record drawing - Volume 12				X		X	X	X		4	4			24		4		1	4
0	01 77 00	1.3	Permit AOCs/ATC - Volume 0				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
1	01 77 00	1.3	Permit AOCs/ATC - Volume 1				X		X	X	X		4	2			2		2			4
2	01 77 00	1.3	Permit AOCs/ATC - Volume 2				X		X	X	X		4	2			2		2			4
3	01 77 00	1.3	Permit AOCs/ATC - Volume 3				X		X	X	X		4	2			2		2			4
4	01 77 00	1.3	Permit AOCs/ATC - Volume 4				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
5	01 77 00	1.3	Permit AOCs/ATC - Volume 5				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
6	01 77 00	1.3	Permit AOCs/ATC - Volume 6				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
8	01 77 00	1.3	Permit AOCs/ATC - Volume 8				X		X	X	X		24	4	2	4	4	4	4			4
9	01 77 00	1.3	Permit AOCs/ATC - Volume 9				X		X	X	X	1	2	2		12	24	4	4		1	4
10	01 77 00	1.3	Permit AOCs/ATC - Volume 10				X		X	X	X			16			24		4			4
11	01 77 00	1.3	Permit AOCs/ATC - Volume 11				X		X	X	X		4	4			24		4			4
12	01 77 00	1.3	Permit AOCs/ATC - Volume 12				X		X	X	X		4	4			24		4		1	4
0	01 77 00	1.3	Warranties - Volume 0				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
1	01 77 00	1.3	Warranties - Volume 1				X		X	X	X		4	2			2		2			4
2	01 77 00	1.3	Warranties - Volume 2				X		X	X	X		4	2			2		2			4
3	01 77 00	1.3	Warranties - Volume 3				X		X	X	X		4	2			2		2			4
4	01 77 00	1.3	Warranties - Volume 4				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
5	01 77 00	1.3	Warranties - Volume 5				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
6	01 77 00	1.3	Warranties - Volume 6				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
8	01 77 00	1.3	Warranties - Volume 8				X		X	X	X		24	4	2	4	4	4	4			4
9	01 77 00	1.3	Warranties - Volume 9				X		X	X	X	1	2	2		12	24	4	4		1	4
10	01 77 00	1.3	Warranties - Volume 10				X		X	X	X			16			24		4			4
11	01 77 00	1.3	Warranties - Volume 11				X		X	X	X		4	4			24		4			4
12	01 77 00	1.3	Warranties - Volume 12				X		X	X	X		4	4			24		4		1	4
0	01 77 00	1.3	Substantial Completion - Volume 0				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
1	01 77 00	1.3	Substantial Completion - Volume 1				X		X	X	X		4	2			2		2			4
2	01 77 00	1.3	Substantial Completion - Volume 2				X		X	X	X		4	2			2		2			4
3	01 77 00	1.3	Substantial Completion - Volume 3				X		X	X	X		4	2			2		2			4
4	01 77 00	1.3	Substantial Completion - Volume 4				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
5	01 77 00	1.3	Substantial Completion - Volume 5				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
6	01 77 00	1.3	Substantial Completion - Volume 6				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
8	01 77 00	1.3	Substantial Completion - Volume 8				X		X	X	X		24	4	2	4	4	4	4			4
9	01 77 00	1.3	Substantial Completion - Volume 9				X		X	X	X	1	2	2		12	24	4	4		1	4
10	01 77 00	1.3	Substantial Completion - Volume 10				X		X	X	X			16			24		4			4
11	01 77 00	1.3	Substantial Completion - Volume 11				X		X	X	X		4	4			24		4			4
12	01 77 00	1.3	Substantial Completion - Volume 12				X		X	X	X		4	4			24		4		1	4
0	01 77 00	1.3	Testing and Balencing - Volume 0				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
1	01 77 00	1.3	Testing and Balencing - Volume 1				X		X	X	X		4	2			2		2			4
2	01 77 00	1.3	Testing and Balencing - Volume 2				X		X	X	X		4	2			2		2			4
3	01 77 00	1.3	Testing and Balencing - Volume 3				X		X	X	X		4	2			2		2			4
4	01 77 00	1.3	Testing and Balencing - Volume 4				X		X	X	X	2	2	2	2	2	2	2	2	2	1	4
5	01 77 00	1.3	Testing and Balencing - Volume 5				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
6	01 77 00	1.3	Testing and Balencing - Volume 6				X		X	X	X	8	8	8	8	8	8	8	8	8	2	8
8	01 77 00	1.3	Testing and Balencing - Volume 8				X		X	X	X		24	4	2	4	4	4	4			4
9	01 77 00	1.3	Testing and Balencing - Volume 9				X		X	X	X	1	2	2		12	24	4	4		1	4
10	01 77 00	1.3	Testing and Balencing - Volume 10				X		X	X	X			16			24		4			4
11	01 77 00	1.3	Testing and Balencing - Volume 11				X		X	X	X		4	4			24		4			4
12	01 77 00	1.3	Testing and Balencing - Volume 12				X		X	X	X		4	4			24		4		1	4
0	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 0		X	X			X	X	X		2	2	2	2	2	2	2			2
1	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 2		X	X			X	X	X		2	2	2	2	2	2	2			2
2	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 3		X	X			X	X	X		2	2	2	2	2	2	2			2
3	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 4		X	X			X	X	X		2	2	2	2	2	2	2			2

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
4	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 5		X	X			X	X	X		2	2	2	2	2	2	2			2
5	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 6		X	X			X	X	X		2	2	2	2	2	2	2			2
8	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 9		X	X			X	X	X		2	2	2	2	2	2	2			2
9	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 10		X	X			X	X	X		2	2	2	2	2	2	2			2
10	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 11		X	X			X	X	X		2	2	2	2	2	2	2			2
11	01 78 23	1.2	OPERATION AND MAINTENANCE DATA - Volume 12		X	X			X	X	X		2	2	2	2	2	2	2			2
12	01 79 00	1.2	DEMONSTRATION AND TESTING																			
5	01 79 00	1.2	DEMONSTRATION AND TESTING -Instruction Program & Video - Volume 5: Roaring Springs Pumphouse		X		X															
6	01 79 00	1.3	DEMONSTRATION AND TESTING - Instruction Program & Video - Volume 6: North Rim WTP		X		X															
8	01 79 00	1.2	DEMONSTRATION AND TESTING Instruction Program & Video- Volume 8: Sanitary Lift Station		X		X												1			1
9	01 79 00	1.2	DEMONSTRATION AND TESTING Instruction Program & Video - Volume 9: Generator/ATS		X		X												1			1
9	01 79 00	1.2	DEMONSTRATION AND TESTING Instruction Program & Video - Volume 9: Fuel Tank System		X		X												1			1
12	01 79 00	1.2	DEMONSTRATION AND TESTING Instruction Program & Video - Volume 12: Cottonwood Power System		X		X												1			1
5	01 81 10	1.7	WIND AND SEISMIC DESIGN CRITERIA Structural Calculations - Volume 5: Roaring Springs Pump House		X		X				X			2			2		1			1
6	01 81 10	1.7	WIND AND SEISMIC DESIGN CRITERIA Structural Calculations - Volume 6: North Rim WTP		X		X				X			2			2		1			1
9	01 81 10	1.7	WIND AND SEISMIC DESIGN CRITERIA Structural Calculations - Volume 9: Fuel Tank System		X		X				X			2			2		1			1
10	01 81 10	1.7	WIND AND SEISMIC DESIGN CRITERIA Structural Calculations - Volume 10: Overhead Power North Rim to Roaring Springs Pump Station		X		X				X			2			2		1			1
ALL	01 81 13	1.3	FEDERAL SUSTAINABILITY DOCUMENTATION				X												1			1
ALL	01 81 33	1.2	CYBERSECURITY REQUIREMENTS																			
ALL	01 81 33	1.2	Comprehensive Asset Inventory				X				X							8	1			1
ALL	01 81 33	1.2	Network Diagrams				X				X							8	1			1
ALL	01 81 33	1.2	Software Source Code				X				X							2	1			1
ALL	01 81 33	1.2	Sensitve Project Information for the Full Project Scope				X				X							2	1			1
5	01 91 14	1.6	TOTAL BUILDING COMMISSIONING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	01 91 14	1.6	Two-week look ahead schedules				X										1	1	1	1		1
5	01 91 14	1.6	Certificates of Readiness	X													1	1	1	1		1
5	01 91 14	1.6	Contractor's Commissioning Representatative Qualifications				X										1	1	1	1		1
5	01 91 14	1.6	Commissioning Plan		X												1	1	1	1		1
5	01 91 14	1.6	Pre Functional Checklists				X										1	1	1	1		1
5	01 91 14	1.6	Owner's project requirements				X										1	1	1	1		1
5	01 91 14	1.6	Edited functional performance test forms				X										1	1	1	1		1
5	01 91 14	1.6	List of test instrumentation, equipment, and monitoring devices				X										1	1	1	1		1
5	01 91 14	1.6	Deficiency report and resolution record				X										1	1	1	1		1
5	01 91 14	1.6	Closeout documentation				X										1	1	1	1		1
6	01 91 14	1.6	TOTAL BUILDING COMMISSIONING VOLUME 6: NORTH RIM WTP																			
6	01 91 14	1.6	Two-week look ahead schedules				X										1	1	1	1		1
6	01 91 14	1.6	Certificates of Readiness	X													1	1	1	1		1
6	01 91 14	1.6	Contractor's Commissioning Representatative Qualifications				X										1	1	1	1		1
6	01 91 14	1.6	Commissioning Plan		X												1	1	1	1		1
6	01 91 14	1.6	Pre Functional Checklists				X										1	1	1	1		1
6	01 91 14	1.6	Owner's project requirements				X										1	1	1	1		1
6	01 91 14	1.6	Edited functional performance test forms				X										1	1	1	1		1
6	01 91 14	1.6	List of test instrumentation, equipment, and monitoring devices				X										1	1	1	1		1
6	01 91 14	1.6	Deficiency report and resolution record				X										1	1	1	1		1
6	01 91 14	1.6	Closeout documentation				X										1	1	1	1		1
5	02 41 00	1.4	DEMOLITION VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	02 41 00	1.4	Demolition and Removal Plan			X						2		2	2	2	2	2	2			2
5	02 41 00	1.4	Notification of Intended Demolition Start																			1
5	02 41 00	1.4	Field Quality Control Test Results		X																	1
5	02 41 00	1.4	Qualification Statements	X																		1
8	02 41 00	1.4	DEMOLITION VOLUME 8: NORTH RIM WW Collection System - Lift Station																			
8	02 41 00	1.4	Demolition and Removal Plan			X						2		2	2	2	2	2	2			2
8	02 41 00	1.4	Notification of Intended Demolition Start				X															

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
8	02 41 00	1.4	Field Quality Control Test Results		X																	
8	02 41 00	1.4	Qualification Statements	X																		
9	02 41 00	1.4	DEMOLITION VOLUME 9: NORTH RIM PDC																			
9	02 41 00	1.4	Demolition and Removal Plan			X						2		2	2	2	2	2	2			2
9	02 41 00	1.4	Notification of Intended Demolition Start				X															
9	02 41 00	1.4	Field Quality Control Test Results		X																	
9	02 41 00	1.4	Qualification Statements	X																		
10	02 41 00	1.4	DEMOLITION VOLUME 10: OH Power																			
10	02 41 00	1.4	Demolition and Removal Plan			X						2		2	2	2	2	2	2			4
10	02 41 00	1.4	Notification of Intended Demolition Start				X															
10	02 41 00	1.4	Field Quality Control Test Results		X																	
10	02 41 00	1.4	Qualification Statements	X																		
11	02 41 00	1.4	DEMOLITION VOLUME 11: Power RSPH to Manzanita																			
11	02 41 00	1.4	Demolition and Removal Plan			X						2		2	2	2	2	2	2			2
11	02 41 00	1.4	Notification of Intended Demolition Start				X															
11	02 41 00	1.4	Field Quality Control Test Results		X																	
11	02 41 00	1.4	Qualification Statements	X																		
12	02 41 00	1.4	DEMOLITIONVOLUME 12: Power System Cottonwood Campground																			
12	02 41 00	1.4	Demolition and Removal Plan			X						2		2	2	2	2	2	2			2
12	02 41 00	1.4	Notification of Intended Demolition Start				X															
12	02 41 00	1.4	Field Quality Control Test Results		X																	
12	02 41 00	1.4	Qualification Statements	X																		
5	03 01 30	1.5	REPAIR AND REHABILITATION OF EXISTING CONSTRUCTION																			
5	03 01 30	1.5	REPAIR AND REHABILITATION OF EXISTING CONSTRUCTION VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	03 01 30	1.5	Shop Drawings						X					1						1		1
5	03 01 30	1.5	Product Data							X				1						1		1
5	03 01 30	1.5	Samples					X						1						1		1
5	03 01 30	1.5	Test Proceedures								X			1						1		1
5	03 01 30	1.5	Certifications	X										1						1		1
5	03 01 30	1.5	Test and Evaluation Reports		X									1						1		1
5	03 01 30	1.5	Manufactures Instructions			X								1						1		1
5	03 01 30	1.5	Field Quality Control				X							1		1				1		1
5	03 01 30	1.5	Qualifications Statement				X							1						1		1
5	03 01 30	1.5	Closeout				X							1						1		1
10	03 01 30	1.5	REPAIR AND REHABILITATION OF EXISTING CONSTRUCTION VOLUME 10: NORTH RIM PDC																			
10	03 01 30	1.5	Shop Drawings						X				1	1						1		1
10	03 01 30	1.5	Product Data							X			1	1						1		1
10	03 01 30	1.5	Samples					X					1	1						1		1
10	03 01 30	1.5	Test Proceedures								X		1	1						1		1
10	03 01 30	1.5	Certifications	X									1	1						1		1
10	03 01 30	1.5	Test and Evaluation Reports		X								1	1						1		1
10	03 01 30	1.5	Manufactures Instructions			X							1	1						1		1
10	03 01 30	1.5	Field Quality Control				X						1	1						1		1
10	03 01 30	1.5	Qualifications Statement				X						1	1						1		1
10	03 01 30	1.5	Closeout				X						1	1						1		1
11	03 01 30	1.5	REPAIR AND REHABILITATION OF EXISTING CONSTRUCTION VOLUME 11: Power RSPH to Manzanita																			
11	03 01 30	1.5	Shop Drawings						X				1	1						1		1
11	03 01 30	1.5	Product Data							X			1	1						1		1
11	03 01 30	1.5	Samples					X					1	1						1		1
11	03 01 30	1.5	Test Proceedures								X		1	1						1		1
11	03 01 30	1.5	Certifications	X									1	1						1		1
11	03 01 30	1.5	Test and Evaluation Reports		X								1	1						1		1
11	03 01 30	1.5	Manufactures Instructions			X							1	1						1		1
11	03 01 30	1.5	Field Quality Control				X						1	1						1		1
11	03 01 30	1.5	Qualifications Statement				X						1	1						1		1
11	03 01 30	1.5	Closeout				X						1	1						1		1
12	03 01 30	1.5	REPAIR AND REHABILITATION OF EXISTING CONSTRUCTION VOLUME 12: Power System Cottonwood Campground																			
12	03 01 30	1.5	Shop Drawings						X					1						1		1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
12	03 01 30	1.5	Product Data							X				1						1			1
12	03 01 30	1.5	Samples					X						1						1			1
12	03 01 30	1.5	Test Proceedures								X			1						1			1
12	03 01 30	1.5	Certifications	X										1						1			1
12	03 01 30	1.5	Test and Evaluation Reports		X									1						1			1
12	03 01 30	1.5	Manufactures Instructions			X								1						1			1
12	03 01 30	1.5	Field Quality Control				X							1						1			1
12	03 01 30	1.5	Qualifications Statement				X							1						1			1
12	03 01 30	1.5	Closeout				X							1						1			1
6	03 05 05	1.5	CONCRETE TESTING AND INSPECTION Volume 6: North Rim WTP D																				
6	03 05 05	1.5	Shop Drawings						X					1						1			1
6	03 05 05	1.5	Certifications	X										1						1			1
6	03 05 05	1.5	Product Technical Data - Mix Design							X				1						1			1
6	03 05 05	1.5	Pre-Inspection Report							X				1						1			1
6	03 05 05	1.5	Post-Inspection/Break Report: 7 day							X				1						1			1
6	03 05 05	1.5	Post-Inspection/Break Report: 14 day							X				1						1			1
6	03 05 05	1.5	Post-Inspection/Break Report: 30 day							X				1						1			1
5	03 05 05	1.5	Volume 5: Roaring Springs Pump House																				
5	03 05 05	1.5	Shop Drawings						X					1						1			1
5	03 05 05	1.5	Certifications	X										1						1			1
5	03 05 05	1.5	Product Technical Data - Mix Design							X				1						1			1
5	03 05 05	1.5	Pre-Inspection Report							X				1						1			1
5	03 05 05	1.5	Post-Inspection/Break Report: 7 day							X				1						1			1
5	03 05 05	1.5	Post-Inspection/Break Report: 14 day							X				1						1			1
5	03 05 05	1.5	Post-Inspection/Break Report: 30 day							X				1						1			1
1	03 05 05	1.5	CONCRETE TESTING AND INSPECTION Volume 1 Pipeline																				
1	03 05 05	1.5	Shop Drawings						X					1						1			1
1	03 05 05	1.5	Certifications	X										1						1			1
1	03 05 05	1.5	Product Technical Data - Mix Design							X				1						1			1
1	03 05 05	1.5	Pre-Inspection Report							X				1						1			1
1	03 05 05	1.5	Post-Inspection/Break Report: 7 day							X				1						1			1
1	03 05 05	1.5	Post-Inspection/Break Report: 14 day							X				1						1			1
1	03 05 05	1.5	Post-Inspection/Break Report: 30 day							X				1						1			1
2	03 05 05	1.5	CONCRETE TESTING AND INSPECTION Volume 2 Pipeline																				
2	03 05 05	1.5	Shop Drawings						X					1						1			1
2	03 05 05	1.5	Certifications	X										1						1			1
2	03 05 05	1.5	Product Technical Data - Mix Design							X				1						1			1
2	03 05 05	1.5	Pre-Inspection Report							X				1						1			1
2	03 05 05	1.5	Post-Inspection/Break Report: 7 day							X				1						1			1
2	03 05 05	1.5	Post-Inspection/Break Report: 14 day							X				1						1			1
2	03 05 05	1.5	Post-Inspection/Break Report: 30 day							X				1						1			1
3	03 05 05	1.5	CONCRETE TESTING AND INSPECTION Volume 3 Pipeline																				
3	03 05 05	1.5	Shop Drawings						X					1						1			1
3	03 05 05	1.5	Certifications	X										1						1			1
3	03 05 05	1.5	Product Technical Data - Mix Design							X				1						1			1
3	03 05 05	1.5	Pre-Inspection Report							X				1						1			1
3	03 05 05	1.5	Post-Inspection/Break Report: 7 day							X				1						1			1
3	03 05 05	1.5	Post-Inspection/Break Report: 14 day							X				1						1			1
3	03 05 05	1.5	Post-Inspection/Break Report: 30 day							X				1						1			1
4	03 05 05	1.5	CONCRETE TESTING AND INSPECTION Volume 4: Pipeline																				
4	03 05 05	1.5	Shop Drawings						X					1						1			1
4	03 05 05	1.5	Certifications	X										1						1			1
4	03 05 05	1.5	Product Technical Data - Mix Design							X				1						1			1
4	03 05 05	1.5	Pre-Inspection Report							X				1						1			1
4	03 05 05	1.5	Post-Inspection/Break Report: 7 day							X				1						1			1
4	03 05 05	1.5	Post-Inspection/Break Report: 14 day							X				1						1			1
4	03 05 05	1.5	Post-Inspection/Break Report: 30 day							X				1						1			1
ALL	03 11 13	1.4	FORMWORK COMMON TO ALL VOLUMES																				
ALL	03 11 13	1.4	Sample - Wood Form for Surfaces Exposed to View					X						1		1				1			1
ALL	03 11 13	1.4	Sample - Metal Form for Surfaces Exposed to View					X						1		1				1			1
ALL	03 11 13	1.4	Sample - Void Form					X						1		1				1			1
ALL	03 11 13	1.4	Sample - Metal/Wood Form for Surfaces Not Exposed to View					X						1		1				1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME											
				INFORMATIONAL				ACTION															
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	03 11 13	1.4	Product Technical Data - Ties			X	X			X		1		1					1			1
ALL	03 11 13	1.4	Product Technical Data - Form Release Material			X	X			X		1		1					1			1
5	03 11 13	1.4	FORMWORK VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	03 11 13	1.4	Formwork		X						X	1		1					1			1
5	03 11 13	1.4	Sample - Wood Form for Surfaces Exposed to View					X				1		1					1			1
5	03 11 13	1.4	Sample - Metal Form for Surfaces Exposed to View					X				1		1					1			1
5	03 11 13	1.4	Sample - Void Form					X				1		1					1			1
5	03 11 13	1.4	Sample - Metal/Wood Form for Surfaces Not Exposed to View					X				1		1					1			1
5	03 11 13	1.4	Product Technical Data - Ties			X	X			X		1		1					1			1
5	03 11 13	1.4	Product Technical Data - Form Release Material			X	X			X		1		1					1			1
6	03 11 13	1.4	FORMWORK VOLUME 6: NORTH RIM WTP																			
6	03 11 13	1.4	Formwork		X						X	1		1					1			1
6	03 11 13	1.4	Sample - Wood Form for Surfaces Exposed to View					X				1		1					1			1
6	03 11 13	1.4	Sample - Metal Form for Surfaces Exposed to View					X				1		1					1			1
6	03 11 13	1.4	Sample - Void Form					X				1		1					1			1
6	03 11 13	1.4	Sample - Metal/Wood Form for Surfaces Not Exposed to View					X				1		1					1			1
6	03 11 13	1.4	Product Technical Data - Ties			X	X			X		1		1					1			1
6	03 11 13	1.4	Product Technical Data - Form Release Material			X	X			X		1		1					1			1
ALL	03 15 19	1.5	COMMON TO ALL VOLUMES																			
ALL	03 15 19	1.5	Shop Drawings						X					1					1			1
ALL	03 15 19	1.5	Product Data - Cast in Place Anchor		X					X				1					1			1
ALL	03 15 19	1.5	Product Data - Anchor Rods		X					X				1					1			1
ALL	03 15 19	1.5	Product Data - Reinforcing Anchorage Adhesive		X					X				1					1			1
ALL	03 15 19	1.5	Product Data - Post Installed Concrete Anchors		X					X				1					1			1
ALL	03 15 19	1.5	Delegated Design Professional's Instruments of Service Submittals		X						X			1					1			1
ALL	03 15 19	1.5	Calculations by Delegated Design Professional								X			1					1			1
ALL	03 15 19	1.5	Field Quality Control Submittals	X										1					1			1
ALL	03 15 19	1.5	Reports of Supplier's Visits to Site				X							1					1			1
3	03 15 19	1.5	VOLUME 3: NR TANK FARM TO SUPAI TUNNEL																			
3	03 15 19	1.5	Shop Drawings						X					1					1			1
3	03 15 19	1.5	Product Data - Cast in Place Anchor		X					X				1					1			1
3	03 15 19	1.5	Product Data - Anchor Rods		X					X				1					1			1
3	03 15 19	1.5	Product Data - Reinforcing Anchorage Adhesive		X					X				1					1			1
3	03 15 19	1.5	Product Data - Post Installed Concrete Anchors		X					X				1					1			1
3	03 15 19	1.5	Delegated Design Professional's Instruments of Service Submittals		X						X			1					1			1
3	03 15 19	1.5	Calculations by Delegated Design Professional								X			1					1			1
3	03 15 19	1.5	Field Quality Control Submittals	X										1					1			1
3	03 15 19	1.5	Reports of Supplier's Visits to Site				X							1					1			1
5	03 15 19	1.5	VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	03 15 19	1.5	Shop Drawings						X					1					1			1
5	03 15 19	1.5	Product Data - Cast in Place Anchor		X					X				1					1			1
5	03 15 19	1.5	Product Data - Anchor Rods		X					X				1					1			1
5	03 15 19	1.5	Product Data - Reinforcing Anchorage Adhesive		X					X				1					1			1
5	03 15 19	1.5	Product Data - Post Installed Concrete Anchors		X					X				1					1			1
5	03 15 19	1.5	Delegated Design Professional's Instruments of Service Submittals		X						X			1					1			1
5	03 15 19	1.5	Calculations by Delegated Design Professional								X			1					1			1
5	03 15 19	1.5	Field Quality Control Submittals	X										1					1			1
5	03 15 19	1.5	Reports of Supplier's Visits to Site				X							1					1			1
6	03 15 19	1.5	VOLUME 6: NORTH RIM WTP																			
6	03 15 19	1.5	Shop Drawings						X					1					1			1
6	03 15 19	1.5	Product Data - Cast in Place Anchor		X					X				1					1			1
6	03 15 19	1.5	Product Data - Anchor Rods		X					X				1					1			1
6	03 15 19	1.5	Product Data - Reinforcing Anchorage Adhesive		X					X				1					1			1
6	03 15 19	1.5	Product Data - Post Installed Concrete Anchors		X					X				1					1			1
6	03 15 19	1.5	Delegated Design Professional's Instruments of Service Submittals		X						X			1					1			1
6	03 15 19	1.5	Calculations by Delegated Design Professional								X			1					1			1
6	03 15 19	1.5	Field Quality Control Submittals	X										1					1			1
6	03 15 19	1.5	Reports of Supplier's Visits to Site				X							1					1			1
ALL	03 21 00	1.3	REINFORCEMENT SHOP DRAWINGS AND PRODUCT DATA						X	X				4					1			1
ALL	03 31 30	1.3	CONCRETE, MATERIALS AND PROPORTIONING																			
ALL	03 31 30	1.3	CONCRETE, MATERIALS AND PROPORTIONING COMMON TO ALL VOLUMES																			
ALL	03 31 30	1.3	Schedule of Concrete Materials	X	X	X								2					1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

ARCHITECT / ENGINEER REVIEW TIME

	SUBMITTAL			REQUIREMENTS																						
				INFORMATIONAL				ACTION																		
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	03 31 30	1.3	Mix Designs		X									2					1			1
ALL	03 31 30	1.3	Reinforcing Bars: Cast, Welded, Smooth Dowels, Proprietary Reinforcing Mechanical Splices, Headed Deformed Bars			X			X					2					1			1
ALL	03 31 30	1.3	Welded Wire, Accessories: Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports			X			X					2					1			1
ALL	03 31 30	1.3	Accessories: Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports			X			X					2					1			1
ALL	03 31 30	1.3	Mill Certificates	X					X					2					1			1
ALL	03 31 30	1.3	Welding Certification	X					X					2					1			1
ALL	03 31 30	1.3	Shop Drawings						X					2					1			1
5	03 31 30	1.3	CONCRETE, MATERIALS AND PROPORTIONING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	03 31 30	1.3	Schedule of Concrete Materials	X	X	X								2					1			1
5	03 31 30	1.3	Mix Designs		X									2					1			1
5	03 31 30	1.3	Reinforcing Bars: Cast, Welded, Smooth Dowels, Proprietary Reinforcing Mechanical Splices, Headed Deformed Bars			X			X					2					1			1
5	03 31 30	1.3	Welded Wire, Accessories: Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports			X			X					2					1			1
5	03 31 30	1.3	Accessories: Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports			X			X					2					1			1
5	03 31 30	1.3	Mill Certificates	X					X					2					1			1
5	03 31 30	1.3	Welding Certification	X					X					2					1			1
5	03 31 30	1.3	Shop Drawings						X					2					1			1
6	03 31 30	1.3	CONCRETE, MATERIALS AND PROPORTIONING VOLUME 6: NORTH RIM WTP																			
6	03 31 30	1.3	Schedule of Concrete Materials	X	X	X								2					1			1
6	03 31 30	1.3	Mix Designs		X									2					1			1
6	03 31 30	1.3	Reinforcing Bars: Cast, Welded, Smooth Dowels, Proprietary Reinforcing Mechanical Splices, Headed Deformed Bars			X			X					2					1			1
6	03 31 30	1.3	Welded Wire, Accessories: Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports			X			X					2					1			1
6	03 31 30	1.3	Accessories: Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports			X			X					2					1			1
6	03 31 30	1.3	Mill Certificates	X					X					2					1			1
6	03 31 30	1.3	Welding Certification	X					X					2					1			1
6	03 31 30	1.3	Shop Drawings						X					2					1			1
ALL	03 31 31	1.5	CONCRETE MIXING, PLACING, JOINTING, AND CURING COMMON TO ALL VOLUMES																			
ALL	03 31 31	1.5	Shop Drawings						X					2					1			1
ALL	03 31 31	1.5	Project Data - Joint Fillers			X					X			2					1			1
ALL	03 31 31	1.5	Project Data - Curing Agents			X					X			2					1			1
ALL	03 31 31	1.5	Project Data - Contruction Joint Bonding Adhesive			X					X			2					1			1
ALL	03 31 31	1.5	Project Data - Prefabricated waterstops			X					X			2					1			1
ALL	03 31 31	1.5	Supplier Site Visit Reports				X							2					1			1
ALL	03 31 31	1.5	Delivery Tickets				X							2					1			1
ALL	03 31 31	1.5	Special Procedure Submittals - Curing				X							2					1			1
5	03 31 31	1.5	CONCRETE MIXING, PLACING, JOINTING, AND CURING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	03 31 31	1.5	Shop Drawings						X					2					1			1
5	03 31 31	1.5	Project Data - Joint Fillers			X					X			2					1			1
5	03 31 31	1.5	Project Data - Curing Agents			X					X			2					1			1
5	03 31 31	1.5	Project Data - Contruction Joint Bonding Adhesive			X					X			2					1			1
5	03 31 31	1.5	Project Data - Prefabricated waterstops			X					X			2					1			1
5	03 31 31	1.5	Supplier Site Visit Reports				X							2					1			1
5	03 31 31	1.5	Delivery Tickets				X							2					1			1
5	03 31 31	1.5	Special Procedure Submittals - Curing				X							2					1			1
6	03 31 31	1.5	CONCRETE MIXING, PLACING, JOINTING, AND CURING VOLUME 6: NORTH RIM WTP																			
6	03 31 31	1.5	Shop Drawings						X					2					1			1
6	03 31 31	1.5	Project Data - Joint Fillers			X					X			2					1			1
6	03 31 31	1.5	Project Data - Curing Agents			X					X			2					1			1
6	03 31 31	1.5	Project Data - Contruction Joint Bonding Adhesive			X					X			2					1			1
6	03 31 31	1.5	Project Data - Prefabricated waterstops			X					X			2					1			1
6	03 31 31	1.5	Supplier Site Visit Reports				X							2					1			1
6	03 31 31	1.5	Delivery Tickets				X							2					1			1
6	03 31 31	1.5	Special Procedure Submittals - Curing				X							2					1			1
2	03 31 31	1.5	CONCRETE MIXING, PLACING, JOINTING, AND CURING VOLUME 2: BALLFIELD TO RSPH																			

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME											
				INFORMATIONAL				ACTION															
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
2	03 31 31	1.5	Shop Drawings						X					2					1			1
2	03 31 31	1.5	Project Data - Joint Fillers			X					X			2					1			1
2	03 31 31	1.5	Project Data - Curing Agents			X					X			2					1			1
2	03 31 31	1.5	Project Data - Contruction Joint Bonding Adhesive			X					X			2					1			1
2	03 31 31	1.5	Project Data - Prefabricated waterstops			X					X			2					1			1
2	03 31 31	1.5	Supplier Site Visit Reports				X							2					1			1
2	03 31 31	1.5	Delivery Tickets				X							2					1			1
2	03 31 31	1.5	Special Procedure Submittals - Curing				X							2					1			1
8	03 31 31	1.5	CONCRETE MIXING, PLACING, JOINTING, AND CURING VOLUME 8: NR COLLECTION SYSTEM																			
8	03 31 31	1.5	Shop Drawings						X					2					1			1
8	03 31 31	1.5	Project Data - Joint Fillers			X					X			2					1			1
8	03 31 31	1.5	Project Data - Curing Agents			X					X			2					1			1
8	03 31 31	1.5	Project Data - Contruction Joint Bonding Adhesive			X					X			2					1			1
8	03 31 31	1.5	Project Data - Prefabricated waterstops			X					X			2					1			1
8	03 31 31	1.5	Supplier Site Visit Reports				X							2					1			1
8	03 31 31	1.5	Delivery Tickets				X							2					1			1
8	03 31 31	1.5	Special Procedure Submittals - Curing				X							2					1			1
ALL	04 01 20	1.3	MASONRY CLEANING	X						X												
ALL	04 01 20	1.3	MASONRY CLEANING COMMON TO ALL VOLUMES																			
ALL	04 01 20	1.3	Shop Drawings																1			1
ALL	04 01 20	1.3	Project Data - Detergent-Type Cleaning Solution							X	X	1							1			1
ALL	04 01 20	1.3	Project Data - Manganese or Vanadium-Stained Monsonary							X	X	1							1			1
ALL	04 01 20	1.3	Project Data - Neutralizing Rinse							X	X	1							1			1
ALL	04 01 20	1.3	Certifications	X								1							1			1
6	04 05 13	1.4	MASONRY MORTAR AND GROUT VOLUME 6: NORTH RIM WTP																			
6	04 05 13	1.4	Shop Drawings						X					2					1			1
6	04 05 13	1.4	Liquid Polymeric Admixture								X			2					1			1
6	04 05 13	1.4	Mix Design Motar								X			2					1			1
6	04 05 13	1.4	Mix Design Grout								X			2					1			1
6	04 05 13	1.4	Product Data - Cement							X	X			2					1			1
6	04 05 13	1.4	Product Data - Hydrated Lime							X	X			2					1			1
6	04 05 13	1.4	Product Data - Masonary Mortar							X	X			2					1			1
6	04 05 13	1.4	Product Data - Source/Producer Aggregates and Gradation							X	X			2					1			1
6	04 05 13	1.4	Product Data - Integrated Water Repellent Dose Rate							X	X			2					1			1
6	04 05 13	1.4	Product Data - Masonary Grout							X	X			2					1			1
6	04 05 13	1.4	Product Data - Cementitious Material							X	X			2					1			1
6	04 05 13	1.4	Mock Up								X			2					1			1
6	04 05 13	1.4	Certifications Lab	X										2					1			1
6	04 05 13	1.4	Test Results - Pre-Construction Mortar Test Results				X							2					1			1
6	04 05 13	1.4	Test Results - Pre-Construction Grout Test Results				X							2					1			1
6	04 05 13	1.4	Samples Color 1					X						2					1			1
6	04 05 13	1.4	Samples Color 2					X						2					1			1
6	04 05 13	1.4	Samples Color 3					X						2					1			1
6	04 05 23	1.3	MASONRY ACCESSORIES VOLUME 6: NORTH RIM WTP																			
6	04 05 23	1.3	Shop Drawings						X					2					1			1
6	04 05 23	1.3	Project Data - Brick Veneer Anchoring System							X	X			2					1			1
6	04 05 23	1.3	Project Data - Through Wall Flashing and Stainless-Steel Drip							X	X			2					1			1
6	04 05 23	1.3	Project Data - Mortar Net Protection System							X	X			2					1			1
6	04 05 23	1.3	Project Data - Weep Vents - Cavity Wall							X	X			2					1			1
6	04 05 23	1.3	Project Data - Rebar Positioner Masonary Anchors							X	X			2					1			1
6	04 05 23	1.3	Project Data - Masonary Horizontal Joint Reinforcing Anchor							X	X			2					1			1
6	04 05 23	1.3	Project Data - Mesh Wall Ties							X	X			2					1			1
6	04 05 23	1.3	Project Data - Preformed Rubber Conrol Joint Inserts							X	X			2					1			1
6	04 05 23	1.3	Project Data - Reglets							X	X			2					1			1
6	04 05 23	1.3	Project Data - Grout Screen							X	X			2					1			1
6	04 22 00	1.5	CONCRETE MASONRYVOLUME 6: NORTH RIM WTP																			
6	04 22 00	1.5	Shop Drawings						X					2					1			1
6	04 22 00	1.5	Product Data - Concrete Masonary - CMU							X	X			2					1			1
6	04 22 00	1.5	Product Data - Factory Shot Blast-Face Masonary (Sandblast Face CMU Veneer)							X	X			2					1			1
6	04 22 00	1.5	Product Data - Precast Concrete Lintels							X	X			2					1			1
6	04 22 00	1.5	Product Data - Integral Water Repellent Admixture							X	X			2					1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE											
				INFORMATIONAL				ACTION				ARCHITECT / ENGINEER REVIEW TIME											
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	04 22 00	1.5	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
6	04 22 00	1.5	Test Results				X							2					1			1
6	04 22 00	1.5	Mock Up - Pre-Colored Masonary					X			X			2					1			1
6	04 22 00	1.5	Mock Up - Sound Absorbing Masonary					X			X			2					1			1
6	04 22 00	1.5	Mock Up - Pre-Cast Lintel					X			X			2					1			1
6	04 22 00	1.5	Mock Up - Cast Stone					X			X			2					1			1
5	05 12 00	1.5	STRUCTURAL STEEL - Welding, Bolting, Building Framing, MISC Framing																			
ALL	05 12 00	1.4	STRUCTURAL STEEL - Welding, Bolting, Building Framing, MISC FramingCOMMON TO ALL VOLUMES																			
ALL	05 12 00	1.4	Shop Drawings						X					2					1			1
ALL	05 12 00	1.4	Product Data - Load Indicator Washers							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Load Indicator Washers for High Strenght Bolts							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Alternate Design High-Strenght Bolts							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Head Studs and Deformer Bar Anchors							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Mechanical Anchor Bolts							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Anchor Bolts Sleeves							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Steel - Structural Shapes/Plate							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Pipe							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - HSS - Hollow Structural Sections							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Threaded Rod and Washers							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Welding Electrodes							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Anchor Rods and Bolts							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Non-Shrink Grout							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Crane Rails							X	X			2					1			1
ALL	05 12 00	1.4	Product Data - Mechanical and Adhesive Anchor Bolts for Fastening to Concrete							X	X			2					1			1
ALL	05 12 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
ALL	05 12 00	1.4	Test Results				X							2					1			1
5	05 12 00	1.4	STRUCTURAL STEEL - Welding, Bolting, Building Framing, MISC Framing VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	05 12 00	1.4	Shop Drawings						X					2					1			1
5	05 12 00	1.4	Product Data - Load Indicator Washers							X	X			2					1			1
5	05 12 00	1.4	Product Data - Load Indicator Washers for High Strenght Bolts							X	X			2					1			1
5	05 12 00	1.4	Product Data - Alternate Design High-Strenght Bolts							X	X			2					1			1
5	05 12 00	1.4	Product Data - Head Studs and Deformer Bar Anchors							X	X			2					1			1
5	05 12 00	1.4	Product Data - Mechanical Anchor Bolts							X	X			2					1			1
5	05 12 00	1.4	Product Data - Anchor Bolts Sleeves							X	X			2					1			1
5	05 12 00	1.4	Product Data - Steel - Structural Shapes/Plate							X	X			2					1			1
5	05 12 00	1.4	Product Data - Pipe							X	X			2					1			1
5	05 12 00	1.4	Product Data - HSS - Hollow Structural Sections							X	X			2					1			1
5	05 12 00	1.4	Product Data - Threaded Rod and Washers							X	X			2					1			1
5	05 12 00	1.4	Product Data - Welding Electrodes							X	X			2					1			1
5	05 12 00	1.4	Product Data - Anchor Rods and Bolts							X	X			2					1			1
5	05 12 00	1.4	Product Data - Non-Shrink Grout							X	X			2					1			1
5	05 12 00	1.4	Product Data - Crane Rails							X	X			2					1			1
5	05 12 00	1.4	Product Data - Mechanical and Adhesive Anchor Bolts for Fastening to Concrete							X	X			2					1			1
5	05 12 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
5	05 12 00	1.4	Test Results				X							2					1			1
6	05 12 00	1.4	STRUCTURAL STEEL - Welding, Bolting, Building Framing, MISC Framing VOLUME 6: NORTH RIM WTP																			
6	05 12 00	1.4	Shop Drawings						X					2					1			1
6	05 12 00	1.4	Product Data - Load Indicator Washers							X	X			2					1			1
6	05 12 00	1.4	Product Data - Load Indicator Washers for High Strenght Bolts							X	X			2					1			1
6	05 12 00	1.4	Product Data - Alternate Design High-Strenght Bolts							X	X			2					1			1
6	05 12 00	1.4	Product Data - Head Studs and Deformer Bar Anchors							X	X			2					1			1
6	05 12 00	1.4	Product Data - Mechanical Anchor Bolts							X	X			2					1			1
6	05 12 00	1.4	Product Data - Anchor Bolts Sleeves							X	X			2					1			1
6	05 12 00	1.4	Product Data - Steel - Structural Shapes/Plate							X	X			2					1			1
6	05 12 00	1.4	Product Data - Pipe							X	X			2					1			1
6	05 12 00	1.4	Product Data - HSS - Hollow Structural Sections							X	X			2					1			1
6	05 12 00	1.4	Product Data - Threaded Rod and Washers							X	X			2					1			1
6	05 12 00	1.4	Product Data - Welding Electrodes							X	X			2					1			1
6	05 12 00	1.4	Product Data - Anchor Rods and Bolts							X	X			2					1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE											
				INFORMATIONAL				ACTION				ARCHITECT / ENGINEER REVIEW TIME											
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	05 12 00	1.4	Product Data - Non-Shrink Grout							X	X			2					1			1
6	05 12 00	1.4	Product Data - Crane Rails							X	X			2					1			1
6	05 12 00	1.4	Product Data - Mechanical and Adhesive Anchor Bolts for Fastening to Concrete							X	X			2					1			1
6	05 12 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
6	05 12 00	1.4	Test Results				X							2					1			1
ALL	05 14 00	1.3	STRUCTURAL ALUMINUM - Welding, Bolting, Framing																			
ALL	05 14 00	1.3	STRUCTURAL ALUMINUM - Welding, Bolting, Framing COMMON TO ALL VOLUMES																			
ALL	05 14 00	1.3	Shop Drawings & Erection Plans						X					2					1			1
ALL	05 14 00	1.3	Product Data - Alloy Sheet Metal							X	X			2					1			1
ALL	05 14 00	1.3	Product Data - Nonferrous Bolts and Nuts							X	X			2					1			1
ALL	05 14 00	1.3	Product Data - Stainless Steel Bolts and Nuts							X	X			2					1			1
ALL	05 14 00	1.3	Product Data - Washers							X	X			2					1			1
ALL	05 14 00	1.3	Product Data - Electrodes for Welding Aluminum							X	X			2					1			1
ALL	05 14 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
ALL	05 14 00	1.3	Test Results				X							2					1			1
5	05 14 00	1.3	STRUCTURAL ALUMINUM - Welding, Bolting, Framing VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	05 14 00	1.3	Shop Drawings & Erection Plans						X					2					1			1
5	05 14 00	1.3	Product Data - Alloy Sheet Metal							X	X			2					1			1
5	05 14 00	1.3	Product Data - Nonferrous Bolts and Nuts							X	X			2					1			1
5	05 14 00	1.3	Product Data - Stainless Steel Bolts and Nuts							X	X			2					1			1
5	05 14 00	1.3	Product Data - Washers							X	X			2					1			1
5	05 14 00	1.3	Product Data - Electrodes for Welding Aluminum							X	X			2					1			1
5	05 14 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
5	05 14 00	1.3	Test Results				X							2					1			1
6	05 14 00	1.3	VOLUME 6: NORTH RIM WTP																			
6	05 14 00	1.3	STRUCTURAL ALUMINUM - Welding, Bolting, Framing Shop Drawings & Erection Plans						X					2					1			1
6	05 14 00	1.3	Product Data - Alloy Sheet Metal							X	X			2					1			1
6	05 14 00	1.3	Product Data - Nonferrous Bolts and Nuts							X	X			2					1			1
6	05 14 00	1.3	Product Data - Stainless Steel Bolts and Nuts							X	X			2					1			1
6	05 14 00	1.3	Product Data - Washers							X	X			2					1			1
6	05 14 00	1.3	Product Data - Electrodes for Welding Aluminum							X	X			2					1			1
6	05 14 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
6	05 14 00	1.3	Test Results				X							2					1			1
6	05 21 00	1.3	STEEL JOISTS VOLUME 6: NORTH RIM WTP																			
6	05 21 00	1.3	Shop Drawings						X					2					1			1
6	05 21 00	1.3	Product Data - Manufactured Open-Web Steel Joists							X	X			2					1			1
6	05 21 00	1.3	Product Data - Manufactured Open-Web Steel Joists Accessories							X	X			2					1			1
6	05 21 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
6	05 21 00	1.3	Test Results				X							2					1			1
5	05 30 00	1.4	METAL DECKING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	05 30 00	1.4	Shop Drawings & Erection Plans						X					2					1			1
5	05 30 00	1.4	Product Data - Roof Decking							X	X			2					1			1
5	05 30 00	1.4	Product Data - Fasteners							X	X			2					1			1
5	05 30 00	1.4	Product Data - Manufactures Load Data Tables							X	X			2					1			1
5	05 30 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
5	05 30 00	1.4	Test Results				X							2					1			1
5	05 30 00	1.4	Samples					X						2					1			1
6	05 30 00	1.4	METAL DECKING VOLUME 6: NORTH RIM WTP																			
6	05 30 00	1.4	Shop Drawings & Erection Plans						X					2					1			1
6	05 30 00	1.4	Product Data - Roof Decking							X	X			2					1			1
6	05 30 00	1.4	Product Data - Fasteners							X	X			2					1			1
6	05 30 00	1.4	Product Data - Manufactures Load Data Tables							X	X			2					1			1
6	05 30 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2					1			1
6	05 30 00	1.4	Test Results				X							2					1			1
6	05 30 00	1.4	Samples					X						2					1			1
6	05 50 00	1.3	METAL FABRICATIONS VOLUME 6: NORTH RIM WTP																			
6	05 50 00	1.3	Heavy-Duty Castings, Trench Covers, and Accessories	X	X	X			X	X				2					1			1
6	05 50 00	1.3	Loose Lintels	X	X	X			X	X		1		2					1			1
6	05 50 00	1.3	Modular framing systems	X	X	X			X	X		1		2					1			1
6	05 50 00	1.3	Access Cover	X	X	X			X	X				2					1			1
6	05 50 00	1.3	Bollards	X	X	X			X	X		1		2					1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	05 52 46	1.4	MECHANICALLY FASTENED ALUMINUM RAILINGS VOLUME 6: NORTH RIM WTP																			
6	05 52 46	1.4	Shop Drawings & Erection Plans						X					2						1			1
6	05 52 46	1.4	Product Data - Mechanically Fastened Component Railing System							X	X			2						1			1
6	05 52 46	1.4	Product Data - Guard and Hand Rails							X	X			2						1			1
6	05 52 46	1.4	Product Data - Mounting and Fasteners							X	X			2						1			1
6	05 52 46	1.4	Product Data - Toeboards							X	X			2						1			1
6	05 52 46	1.4	Product Data - Guardrail Gates							X	X			2						1			1
6	05 52 46	1.4	Product Data - Manufactures Load Data Tables							X	X			2						1			1
6	05 52 46	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X				X			2						1			1
6	05 52 46	1.4	Cleaning and Maintenance Data							X	X			2						1			1
6	05 52 46	1.4	Calculations								X			2						1			1
6	05 52 46	1.4	Test Results								X			2						1			1
6	05 52 46	1.4	Samples - Finishes					X						2						1			1
5	06 10 00	1.3	ROUGH CARPENTRY VOLUME 5: ROARING SPRINGS PUMP STATION											2						1			1
5	06 10 00	1.3	Shop Drawings						X					2						1			1
5	06 10 00	1.3	Product Data - Framing Lumber							X	X			2						1			1
5	06 10 00	1.3	Product Data - Non-Structural Plywood							X	X			2						1			1
5	06 10 00	1.3	Product Data - Treated Wood							X	X			2						1			1
5	06 10 00	1.3	Product Data - Fasteners							X	X			2						1			1
5	06 10 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2						1			1
6	06 10 00	1.3	ROUGH CARPENTRY VOLUME 6: NORTH RIM WTP																				
6	06 10 00	1.3	Shop Drawings						X					2						1			1
6	06 10 00	1.3	Product Data - Framing Lumber							X	X			2						1			1
6	06 10 00	1.3	Product Data - Structural Plywood							X	X			2						1			1
6	06 10 00	1.3	Product Data - Exterior Sheathing							X	X			2						1			1
6	06 10 00	1.3	Product Data - Non-Structural Plywood							X	X			2						1			1
6	06 10 00	1.3	Product Data - Treated Wood							X	X			2						1			1
6	06 10 00	1.3	Product Data - Fire-Retardant Treated Material (FRTM)							X	X			2						1			1
6	06 10 00	1.3	Product Data - Fasteners							X	X			2						1			1
6	06 10 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2						1			1
6	06 41 00	1.5	ARCHITECTURAL WOOD VOLUME 6: NORTH RIM WTP																				
6	06 41 00	1.5	Product Data - Architectural Cabinetwork - Millwork							X	X			4						1			1
6	06 41 00	1.5	Product Data - Solid Surface Material - Window Stools and Countertops							X	X			2						1			1
6	06 41 00	1.5	Shop Drawings						X	X	X			2						1			1
6	06 41 00	1.5	Samples: Plastic Laminent Color and Finishes					X			X			2						1			1
6	06 41 00	1.5	Samples: Plastic Hardware (Drawer Pull, Hinges, Shelf Support)					X			X			2						1			1
6	06 41 00	1.5	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2						1			1
6	06 82 00	1.5	FIBERGLASS REINFORCED PLASTIC FABRICATIONS VOLUME 6: NORTH RIM WTP																				
6	06 82 00	1.5	Shop Drawings and Fabrication						X					2						1			1
6	06 82 00	1.5	Product Data - FRP Fasteners							X	X			2						1			1
6	06 82 00	1.5	Certification of Engineers Qualifications	X			X							2						1			1
6	06 82 00	1.5	Color Samples Minimum 3 per Feature					X						2						1			1
6	07 14 00	1.4	FLUID APPLIED WATERPROOFING VOLUME 6: NORTH RIM WTP																				
6	07 14 00	1.4	Shop Drawings - Flashing of penetrations, terminations, expansion joints, protection course attachment and other special conditions.						X					2						1			1
6	07 14 00	1.4	Product Data - Fluid Applied Waterproofing							X	X			2						1			1
6	07 14 00	1.4	Product Data - Fluid Applied Waterproofing Primer							X	X			2						1			1
6	07 14 00	1.4	Product Data - Moisture Curing Polyuretheane Elastomer							X	X			2						1			1
6	07 14 00	1.4	Product Data - Manufacture's Adhesive & Sealent							X	X			2						1			1
6	07 14 00	1.4	Product Data - Flashing Reinforcment - Woven Uncoated Fiberglass Mesh							X	X			2						1			1
6	07 14 00	1.4	Product Data - Fluid Applied Waterproofing							X	X			2						1			1
6	07 14 00	1.4	Product Data - Backer Rod							X	X			2						1			1
6	07 14 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2						1			1
6	07 21 00	1.3	BUILDING INSULATION VOLUME 6: NORTH RIM WTP																				
6	07 21 00	1.3	Product Data - Building Insulation Rigid Extruded Polystyrene Board Insulation							X	X			2						1			1
6	07 21 00	1.3	Product Data - Building Insulation Rigid Polyiscocyanurate Foam Board Insulation							X	X			2						1			1
6	07 21 00	1.3	Product Data - Blanket or Batt Thermal Insulation							X	X			2						1			1
6	07 21 00	1.3	Product Data - Sound Control and Fire Safing Insulation							X	X			2						1			1
6	07 21 00	1.3	Product Data - Vapor Retarder							X	X			2						1			1
6	07 21 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X							2						1			1
6	07 26 00	1.3	UNDER SLAB VAPOR RETARDER VOLUME 6: NORTH RIM WTP																				

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	07 26 00	1.3	Product Data - Vapor Retarder							X	X	2							1			1
6	07 26 00	1.3	Product Data - Vapor Retarder Tape							X	X	2							1			1
6	07 26 00	1.3	Product Data - Pipe Boot							X	X	2							1			1
6	07 26 00	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	07 26 00	1.3	Color Samples					X				2							1			1
6	07 46 10	1.5	CEMENT BOARD SIDING VOLUME 6: NORTH RIM WTP																			
6	07 46 10	1.5	Product Data - Pre-Finished Fiber-Cement Board Lap Siding							X	X	2							1			1
6	07 46 10	1.5	Product Data - Nail Board Polyiso Ridgid Sheathing							X	X	2							1			1
6	07 46 10	1.5	Product Data - Sheet Weather Barier							X	X	2							1			1
6	07 46 10	1.5	Product Data - Fasteners							X	X	2							1			1
6	07 46 10	1.5	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	07 46 10	1.5	Color Sample Mock-Up					X			X	2							1			1
5	07 61 13	1.6	METAL ROOFING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	07 61 13	1.6	Product Data - Roof Sheet Weather Barrier Membrane							X	X	2							1			1
5	07 61 13	1.6	Product Data - Standing Seam Metal Roofing							X	X	2							1			1
5	07 61 13	1.6	Product Data - Standing Seam Metal Facial System							X	X	2							1			1
5	07 61 13	1.6	Product Data - Ancillary Sheet Metal							X	X	2							1			1
5	07 61 13	1.6	Product Data - Accessories and Fasteners							X	X	2							1			1
5	07 61 13	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
5	07 61 13	1.6	Color Sample Mock-Up					X			X	2							1			1
6	07 61 13	1.6	METAL ROOFING VOLUME 6: NORTH RIM WTP																			
6	07 61 13	1.6	Product Data - Vapor Barrier							X	X	2							1			1
6	07 61 13	1.6	Product Data - Thermal Barrier Sheathing							X	X	2							1			1
6	07 61 13	1.6	Product Data - Roof Insulation							X	X	2							1			1
6	07 61 13	1.6	Product Data - Roof Sheet Weather Barrier Membrane							X	X	2							1			1
6	07 61 13	1.6	Product Data - Standing Seam Metal Roofing							X	X	2							1			1
6	07 61 13	1.6	Product Data - Standing Seam Metal Facial System							X	X	2							1			1
6	07 61 13	1.6	Product Data - Soffit Panels							X	X	2							1			1
6	07 61 13	1.6	Product Data - Lightening Protection System Clamps							X	X	2							1			1
6	07 61 13	1.6	Product Data - Ancillary Sheet Metal							X	X	2							1			1
6	07 61 13	1.6	Product Data - Accessories and Fasteners							X	X	2							1			1
6	07 61 13	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	07 61 13	1.6	Color Sample Mock-Up					X			X	2							1			1
5	07 62 00	1.4	SHEET METAL FLASHING AND TRIM VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	07 62 00	1.4	Product Data - Architectural Flashing and Sheet Metal							X	X	2							1			1
5	07 62 00	1.4	Color Sample Mock-Up					X			X	2							1			1
6	07 62 00	1.4	SHEET METAL FLASHING AND TRIM VOLUME 6: NORTH RIM WTP																			
6	07 62 00	1.4	Product Data - Architectural Flashing and Sheet Metal							X	X	2							1			1
6	07 62 00	1.4	Product Data - Stainless Steel Clamping Rings							X	X	2							1			1
6	07 62 00	1.4	Product Data - Metal Flashing and Counter Flashing							X	X	2							1			1
6	07 62 00	1.4	Product Data - Termination Bars							X	X	2							1			1
6	07 62 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	07 62 00	1.4	Color Sample Mock-Up					X			X	2							1			1
6	07 84 00	1.6	FIRESTOPPING VOLUME 6: NORTH RIM WTP																			
6	07 84 00	1.6	Project Data - Firestopping							X	X	2							1			1
6	07 84 00	1.6	Firestopping Schedule with Shop Drawings						X		X	2							1			1
6	07 84 00	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	07 84 00	1.6	Color Sample Mock-Up					X			X	2							1			1
5	07 92 00	1.6	JOINT SEALANTS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	07 92 00	1.6	Product Data - Compressible Sealent							X	X	2							1			1
5	07 92 00	1.6	Product Data - Expandable Foam Sealent							X	X	2							1			1
5	07 92 00	1.6	Product Data - Polyether Foam Sealent							X	X	2							1			1
5	07 92 00	1.6	Product Data - Polysulfide Rubber Sealent							X	X	2							1			1
5	07 92 00	1.6	Product Data - Polyurea Joint Filler							X	X	2							1			1
5	07 92 00	1.6	Product Data - Polyuretheane Sealents							X	X	2							1			1
5	07 92 00	1.6	Product Data - Silicone Sealents							X	X	2							1			1
5	07 92 00	1.6	Product Data - Accessories: Back Rod, Cmpressible Filler, Primer, Joint Cleaner, Bond Breaker							X	X	2							1			1
5	07 92 00	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
5	07 92 00	1.6	Color Samples					X			X	2							1			1
6	07 92 00	1.6	JOINT SEALANTS VOLUME 6: NORTH RIM WTP																			
6	07 92 00	1.6	Product Data - Compressible Sealent							X	X	2							1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE											
				INFORMATIONAL				ACTION				ARCHITECT / ENGINEER REVIEW TIME											
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	07 92 00	1.6	Product Data - Expandable Foam Sealent							X	X	2							1			1
6	07 92 00	1.6	Product Data - Polyether Foam Sealent							X	X	2							1			1
6	07 92 00	1.6	Product Data - Polysulfide Rubber Sealent							X	X	2							1			1
6	07 92 00	1.6	Product Data - Polyurea Joint Filler							X	X	2							1			1
6	07 92 00	1.6	Product Data - Polyuretheane Sealents							X	X	2							1			1
6	07 92 00	1.6	Product Data - Silicone Sealents							X	X	2							1			1
6	07 92 00	1.6	Product Data - Fire Resistant Sealent							X	X	2							1			1
6	07 92 00	1.6	Product Data - Accessories: Back Rod, Cmpressible Filler, Primer, Joint Cleaner, Bond Breaker							X	X	2							1			1
6	07 92 00	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	07 92 00	1.6	Color Samples					X			X	2							1			1
9	07 92 00	1.6	JOINT SEALANTS VOLUME 9: POWER DISTROBUTION CENTER																			
9	07 92 00	1.6	Product Data - Compressible Sealent							X	X	2							1			1
9	07 92 00	1.6	Product Data - Expandable Foam Sealent							X	X	2							1			1
9	07 92 00	1.6	Product Data - Polyether Foam Sealent							X	X	2							1			1
9	07 92 00	1.6	Product Data - Polysulfide Rubber Sealent							X	X	2							1			1
9	07 92 00	1.6	Product Data - Polyurea Joint Filler							X	X	2							1			1
9	07 92 00	1.6	Product Data - Polyuretheane Sealents							X	X	2							1			1
9	07 92 00	1.6	Product Data - Silicone Sealents							X	X	2							1			1
9	07 92 00	1.6	Product Data - Fire Resistant Sealent							X	X	2							1			1
9	07 92 00	1.6	Product Data - Accessories: Back Rod, Cmpressible Filler, Primer, Joint Cleaner, Bond Breaker							X	X	2							1			1
9	07 92 00	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
9	07 92 00	1.6	Color Samples					X			X	2							1			1
6	08 11 00	1.6	HOLLOW METAL DOORS AND FRAMES VOLUME 6: NORTH RIM WTP																			
6	08 11 00	1.6	Product Data - Metal Doors and Frames							X	X	4							1			1
6	08 11 00	1.6	Product Data - Metal Borrowed Lite Window Frames							X	X	4							1			1
6	08 11 00	1.6	Product Data - Jamb Anchors							X	X	4							1			1
6	08 11 00	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	08 11 00	1.6	Schedule of Doors and Frames with Drawings			X					X	4							1			1
6	08 11 00	1.6	Color Samples					X			X	2							1			1
5	08 11 16	1.6	ALUMINUM DOORS AND FRAMES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	08 11 16	1.6	Product Data: Doors and Frames							X	X	4							1			1
5	08 11 16	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
5	08 11 16	1.6	Schedule of Doors and Frames			X					X	4							1			1
5	08 11 16	1.6	Color Samples					X			X	2							1			1
6	08 11 16	1.6	ALUMINUM DOORS AND FRAMES VOLUME 6: NORTH RIM WTP																			
6	08 11 16	1.6	Product Data: Doors and Frames							X	X	4							1			1
6	08 11 16	1.6	Product Data: Windows and Frames							X	X	4							1			1
6	08 11 16	1.6	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	08 11 16	1.6	Schedule of Doors and Frames			X					X	4							1			1
6	08 11 16	1.6	Color Samples					X			X	2							1			1
6	08 14 16	1.3	FLUSH WOOD DOORS VOLUME 6: NORTH RIM WTP																			
6	08 14 16	1.3	Product Data - Flush Interior Wood Doors							X	X	4							1			1
6	08 14 16	1.3	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	08 14 16	1.3	Schedule of Doors and Frames			X					X	4							1			1
6	08 31 00	1.4	ACCESS DOORS VOLUME 6: NORTH RIM WTP																			
6	08 31 00	1.4	Product Data - Ceiling Access Doors							X	X	2							1			1
6	08 31 00	1.4	Product Data - Floor Access Doors							X	X	2							1			1
6	08 31 00	1.4	Product Data - Wall Access Doors							X	X	2							1			1
6	08 31 00	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	08 31 00	1.4	Schedule of Doors			X					X	4							1			1
6	08 33 22	1.4	ALUMINUM ROLLING OVERHEAD DOORS VOLUME 6: NORTH RIM WTP																			
6	08 33 22	1.4	Product Data - Aluminum Rolling Overhead Doors							X	X	2							1			1
6	08 33 22	1.4	Product Data - Motor Operators							X	X	2							1			1
6	08 33 22	1.4	Product Data - Control Systems							X	X	2							1			1
6	08 33 22	1.4	Quality Control - Special Inspections, Quality Control Tests, Certification of Testing Quals	X			X					2							1			1
6	08 33 22	1.4	Schedule of Doors			X					X	4							1			1
6	08 33 22	1.4	Samples					X				2							1			1
5	08 70 00	1.4	FINISH HARDWARE VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	08 70 00	1.4	Product Data - Finish Hardware - Mechanical Door							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Electromechanical Door Hardware							X	X	4							1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

	SUBMITTAL			REQUIREMENTS								ARCHITECT / ENGINEER REVIEW TIME										
				INFORMATIONAL				ACTION														
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
5	08 70 00	1.4	Product Data - Finish Hardware - Butt Hinges							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Continous Hinges							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Flush Bolts							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Door Push Bar and Pulls							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Cylinders and Keying							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Door Protective Trim							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Door Stops and Holders							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Silencers							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Key Control							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Weatherstripping							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Closer							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Exit Devices							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Mortise							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Coordinators							X	X	2							1			1
5	08 70 00	1.4	Product Data - Power Transfer Hinges							X	X	2							1			1
5	08 70 00	1.4	Product Data - Electric Door Wire Harness							X	X	2							1			1
5	08 70 00	1.4	Product Data - Finish Hardware - Cylinders Specified for Doors in Other Sections							X	X	2							1			1
5	08 70 00	1.4	Keying Schedule			X					X	2							1			1
5	08 70 00	1.4	Samples					X				2							1			1
5	08 70 00	1.4	Shop Drawings to Include Wiring Diagram						X			2					2		2			2
5	08 70 00	1.4	Testing Report								X	2					2		1			1
6	08 70 00	1.4	FINISH HARDWARE VOLUME 6: NORTH RIM WTP																			
6	08 70 00	1.4	Product Data - Finish Hardware - Mechanical Door							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Electromechanical Door Hardware							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Butt Hinges							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Continous Hinges							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Flush Bolts							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Door Push Bar and Pulls							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Cylinders and Keying							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Door Protective Trim							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Door Stops and Holders							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Silencers							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Key Control							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Weatherstripping							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Closer							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Exit Devices							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Mortise							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Coordinators							X	X	2							1			1
6	08 70 00	1.4	Product Data - Power Transfer Hinges							X	X	2							1			1
6	08 70 00	1.4	Product Data - Electric Door Wire Harness							X	X	2							1			1
6	08 70 00	1.4	Product Data - Finish Hardware - Cylinders Specified for Doors in Other Sections							X	X	2							1			1
6	08 70 00	1.4	Keying Schedule			X					X	4							1			1
6	08 70 00	1.4	Samples					X				2							1			1
6	08 70 00	1.4	Shop Drawings to Include Wiring Diagram						X			2					2		1			1
6	08 70 00	1.4	Testing Report								X	2					2		1			1
6	08 81 00	1.4	GLASS AND GLAZING VOLUME 6: NORTH RIM WTP																			
6	08 81 00	1.4	Product Data - Glass Glazing							X	X	2							1			1
6	08 81 00	1.4	Product Data - Glass Glazing Sealent							X	X	2							1			1
6	08 81 00	1.4	Glazing Schedule			X					X	2							1			1
6	08 81 00	1.4	Samples					X				2							1			1
6	08 81 00	1.4	Shop Drawings						X			2							1			1
5	08 90 00	1.4	LOUVERS AND VENTS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	08 90 00	1.4	Product Data - Louvers							X	X	2							1			1
5	08 90 00	1.4	Product Data - Vents							X	X	2							1			1
5	08 90 00	1.4	Louver and Vent Schedule			X					X	2							1			1
5	08 90 00	1.4	Samples					X				2							1			1
5	08 90 00	1.4	Shop Drawings						X			2							1			1
6	08 90 00	1.4	LOUVERS AND VENTSVOLUME 6: NORTH RIM WTP																			
6	08 90 00	1.4	Product Data - Louvers							X	X	2							1			1
6	08 90 00	1.4	Product Data - Vents							X	X	2							1			1
6	08 90 00	1.4	Louver and Vent Schedule			X					X	2							1			1
6	08 90 00	1.4	Samples					X				2							1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	08 90 00	1.4	Shop Drawings						X			2							1			1
5	09 06 00	1.2	COLOR SCHEDULE VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	09 06 00	1.2	Samples Interior					X				2							1			1
5	09 06 00	1.2	Samples Exterior					X				2							1			1
5	09 06 00	1.2	Interior Finish Schedule				X				X	2							1			1
5	09 06 00	1.2	Exterior Finish Schedule				X				X	2							1			1
6	09 06 00	1.2	COLOR SCHEDULE VOLUME 6: NORTH RIM WTP																			
6	09 06 00	1.2	Samples Interior					X				2							1			1
6	09 06 00	1.2	Samples Exterior					X				2							1			1
6	09 06 00	1.2	Interior Finish Schedule				X				X	2							1			1
6	09 06 00	1.2	Exterior Finish Schedule				X				X	2							1			1
9	09 06 00	1.2	COLOR SCHEDULE VOLUME 9: POWER DISTROBUTION CENTER																			
9	09 06 00	1.2	Samples Interior					X				2							1			1
9	09 06 00	1.2	Samples Exterior					X				2							1			1
9	09 06 00	1.2	Interior Finish Schedule				X				X	2							1			1
9	09 06 00	1.2	Exterior Finish Schedule				X				X	2							1			1
9	09 06 00	1.2	COLOR SCHEDULE VOLUME 12: COTTONWOOD POWER SYSTEM																			
9	09 06 00	1.2	Samples Interior					X				1							1			1
9	09 06 00	1.2	Samples Exterior					X				1							1			1
9	09 06 00	1.2	Interior Finish Schedule				X				X	1							1			1
9	09 06 00	1.2	Exterior Finish Schedule				X				X	1							1			1
6	09 22 16	1.5	NON-STRUCTURAL METAL FRAMING VOLUME 6: NORTH RIM WTP																			
6	09 22 16	1.5	Shop Drawings						X					1					1			1
6	09 22 16	2.3	Product Data - Cast in Place Anchor							X	X			2					1			1
6	09 22 16	2.3	Product Data - Metal Studs and Floor Tracks							X	X			2					1			1
6	09 22 16	2.3	Product Data - Flexible Track							X	X			2					1			1
6	09 22 16	2.3	Product Data - Head of Wall Accessories							X	X			2					1			1
6	09 22 16	2.3	Product Data - Furring Channels							X	X			2					1			1
6	09 22 16	2.3	Product Data - Metal Backing							X	X			2					1			1
6	09 22 16	2.3	Product Data - Stud-Framed Ceiling/Soffit System							X	X			2					1			1
6	09 22 16	2.3	Product Data - Tie Wire & Wire Hangers							X	X			2					1			1
6	09 22 16	1.5	Calculations by Delegated Design Professional				X							1					1			1
6	09 22 16	1.5	Field Quality Control Submittals	X							X			1					1			1
6	09 29 00	1.6	GYPSUM BOARD VOLUME 6: NORTH RIM WTP																			
6	09 29 00	1.6	Shop Drawings - Joint Control Layout						X					2					1			1
6	09 29 00	1.6	Product Data - Installation Instructions - Manufactures							X	X			2					1			1
6	09 29 00	1.6	Product Data - Trims, Adhesive, Joint Treatment Compound and Tape							X	X			2					1			1
6	09 29 00	1.6	Product Data - Gypsum Board							X	X			2					1			1
6	09 29 00	1.6	Product Data - Moisture Resistant Gypsum Wall Board							X	X			2					1			1
6	09 29 00	1.6	Product Data - Coated Tile Backer Gypsum Board							X	X			2					1			1
6	09 30 13	1.6	CERAMIC TILE VOLUME 6: NORTH RIM WTP																			
6	09 30 13	1.6	Mockup Sample Tile and Grout								X	2							1			1
6	09 30 13	1.6	Product Certification - Master Grade Certificates							X	X	2							1			1
6	09 30 13	1.6	Product Data - Tile							X	X	2							1			1
6	09 30 13	1.6	Product Data - Chemical Resistant Mortar and Grout							X	X	2							1			1
6	09 30 13	1.6	Product Data - Dry Set Portland Cement Mortar and Grout							X	X	2							1			1
6	09 30 13	1.6	Product Data - Divider and Transition Strips and Cap Trim							X	X	2							1			1
6	09 30 13	1.6	Product Data - Leveling Compound							X	X	2							1			1
6	09 30 13	1.6	Product Data - Latex Portland Cement Mortar and Grout							X	X	2							1			1
6	09 30 13	1.6	Product Data - Commercial Portland Cement Grout							X	X	2							1			1
6	09 51 00	1.6	ACCOUSTICAL CEILINGS VOLUME 6: NORTH RIM WTP																			
6	09 51 00	1.6	Shop Drawings - Reflective Ceiling Plan						X			2							1			1
6	09 51 00	1.6	Samples					X			X	2							1			1
6	09 51 00	1.6	Product Data - Panel Type 1							X	X	2							1			1
6	09 51 00	1.6	Product Data - Panel Type 2							X	X	2							1			1
6	09 51 00	1.6	Product Data - Panel Type 3							X	X	2							1			1
6	09 65 36	1.6	VINYL COMPOSITION TILE FLOORING AND RESILIENT BASE VOLUME 6: NORTH RIM WTP																			
6	09 65 36	1.6	Samples					X			X	2							1			1
6	09 65 36	1.6	Product Data - Panel Type 1							X	X	2							1			1
6	09 65 36	1.6	Product Data - Panel Type 2							X	X	2							1			1
6	09 65 36	1.6	Product Data - Panel Type 3							X	X	2							1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	09 65 36	1.6	Product Data - Adhesive							X	X	2							1			1
6	09 67 26	1.6	SEAMLESS EPOXY FLOORING VOLUME 6: NORTH RIM WTP																			
6	09 67 26	1.6	Samples					X			X	2							1			1
6	09 67 26	1.6	Product Data - MSDS							X	X	2							1			1
6	09 67 26	1.6	Product Data - Manufacture Installation Instructions							X	X	2							1			1
6	09 67 26	1.6	Product Data - Epoxy Product							X	X	2							1			1
6	09 67 26	1.6	Certificate of Installer	X								2							1			1
6	09 67 26	1.6	Product Data - Maintenance Requirements							X	X	2							1			1
5	09 84 00	1.6	ACOUSTICAL PANEL SYSTEM VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	09 84 00	1.6	Shop Drawings - Layout and Elevations Plan						X			2							1			1
5	09 84 00	1.6	Samples					X			X	2							1			1
5	09 84 00	1.6	Product Data - Panel System Type 1							X	X	2							1			1
5	09 84 00	1.6	Product Data - Panel System Type 2							X	X	2							1			1
5	09 84 00	1.6	Product Data - Panel System Type 3							X	X	2							1			1
5	09 84 00	1.6	Certification	X						X	X	2							1			1
5	09 91 00	1.6	ARCHITECTUAL PAINTING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	09 91 00	1.6	Samples					X			X	2							1			1
5	09 91 00	1.6	Product Data - Paint Type 1							X	X	2							1			1
5	09 91 00	1.6	Product Data - Paint Type 2							X	X	2							1			1
5	09 91 00	1.6	Product Data - Paint Type 3							X	X	2							1			1
5	09 91 00	1.6	Certification	X						X	X	2							1			1
5	09 91 00	1.6	Test Results	X								2							1			1
6	09 91 00	1.6	ARCHITECTUAL PAINTING VOLUME 6: NORTH RIM WTP																			
6	09 91 00	1.6	Samples					X			X	2							1			1
6	09 91 00	1.6	Product Data - Paint Type 1							X	X	2							1			1
6	09 91 00	1.6	Product Data - Paint Type 2							X	X	2							1			1
6	09 91 00	1.6	Product Data - Paint Type 3							X	X	2							1			1
6	09 91 00	1.6	Certification	X						X	X	2							1			1
6	09 91 00	1.6	Test Results	X								2							1			1
9	09 91 00	1.6	ARCHITECTUAL PAINTING VOLUME 9: POWER DISTROBUTION CENTER																			
9	09 91 00	1.6	Samples					X			X	2							1			1
9	09 91 00	1.6	Product Data - Paint Type 1							X	X	2							1			1
9	09 91 00	1.6	Product Data - Paint Type 2							X	X	2							1			1
9	09 91 00	1.6	Product Data - Paint Type 3							X	X	2							1			1
9	09 91 00	1.6	Certification	X						X	X	2							1			1
9	09 91 00	1.6	Test Results	X								2							1			1
12	09 91 00	1.6	ARCHITECTUAL PAINTING VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	09 91 00	1.6	Samples					X			X	2							1			1
12	09 91 00	1.6	Product Data - Paint Type 1							X	X	2							1			1
12	09 91 00	1.6	Product Data - Paint Type 2							X	X	2							1			1
12	09 91 00	1.6	Product Data - Paint Type 3							X	X	2							1			1
12	09 91 00	1.6	Certification	X						X	X	2							1			1
12	09 91 00	1.6	Test Results	X								2							1			1
5	09 96 00	1.6	HIGH PERFORMANCE INDUSTRIAL COATINGS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	09 96 00	1.6	Samples					X			X	2							1			1
5	09 96 00	1.6	Product Data - Coating Type 1 (Include Manufactures Preparation Requirements)							X	X	2							1			1
5	09 96 00	1.6	Product Data - Coating Type 2 (Include Manufactures Preparation Requirements)							X	X	2							1			1
5	09 96 00	1.6	Product Data - Coating Type 3 (Include Manufactures Preparation Requirements)							X	X	2							1			1
5	09 96 00	1.6	Certification	X						X	X	2							1			1
5	09 96 00	1.6	Test Results	X								2							1			1
6	09 96 00	1.6	HIGH PERFORMANCE INDUSTRIAL COATINGS VOLUME 6: NORTH RIM WTP																			
6	09 96 00	1.6	Samples					X			X	2							1			1
6	09 96 00	1.6	Product Data - Coating Type 1 (Include Manufactures Preparation Requirements)							X	X	2							1			1
6	09 96 00	1.6	Product Data - Coating Type 2 (Include Manufactures Preparation Requirements)							X	X	2							1			1
6	09 96 00	1.6	Product Data - Coating Type 3 (Include Manufactures Preparation Requirements)							X	X	2							1			1
6	09 96 00	1.6	Certification	X						X	X	2							1			1
6	09 96 00	1.6	Test Results	X								2							1			1
8	09 96 00	1.6	HIGH PERFORMANCE INDUSTRIAL COATINGS VOLUME 8: WW COLLECTION AND LS																			
8	09 96 00	1.6	Samples					X			X	2							1			1
8	09 96 00	1.6	Product Data - Coating Type 1 (Include Manufactures Preparation Requirements)							X	X	2							1			1
8	09 96 00	1.6	Product Data - Coating Type 2 (Include Manufactures Preparation Requirements)							X	X	2							1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
8	09 96 00	1.6	Product Data - Coating Type 3 (Include Manufactures Preparation Requirements)							X	X	2							1			1
8	09 96 00	1.6	Certification	X						X	X	2							1			1
8	09 96 00	1.6	Test Results	X								2							1			1
9	09 96 00	1.6	HIGH PERFORMANCE INDUSTRIAL COATINGS VOLUME 9: POWER DISTROBUTION CENTER																			
9	09 96 00	1.6	Samples					X			X	2							1			1
9	09 96 00	1.6	Product Data - Coating Type 1 (Include Manufactures Preparation Requirements)							X	X	2							1			1
9	09 96 00	1.6	Product Data - Coating Type 2 (Include Manufactures Preparation Requirements)							X	X	2							1			1
9	09 96 00	1.6	Product Data - Coating Type 3 (Include Manufactures Preparation Requirements)							X	X	2							1			1
9	09 96 00	1.6	Certification	X						X	X	2							1			1
9	09 96 00	1.6	Test Results	X								2							1			1
10	09 96 00	1.6	HIGH PERFORMANCE INDUSTRIAL COATINGS VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	09 96 00	1.6	Samples					X			X	2							1			1
10	09 96 00	1.6	Product Data - Coating Type 1 (Include Manufactures Preparation Requirements)							X	X	2							1			1
10	09 96 00	1.6	Product Data - Coating Type 2 (Include Manufactures Preparation Requirements)							X	X	2							1			1
10	09 96 00	1.6	Product Data - Coating Type 3 (Include Manufactures Preparation Requirements)							X	X	2							1			1
10	09 96 00	1.6	Certification	X						X	X	2							1			1
10	09 96 00	1.6	Test Results	X								2							1			1
12	09 96 00	1.6	HIGH PERFORMANCE INDUSTRIAL COATINGS VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	09 96 00	1.6	Samples					X			X	2							1			1
12	09 96 00	1.6	Product Data - Coating Type 1 (Include Manufactures Preparation Requirements)							X	X	2							1			1
12	09 96 00	1.6	Product Data - Coating Type 2 (Include Manufactures Preparation Requirements)							X	X	2							1			1
12	09 96 00	1.6	Product Data - Coating Type 3 (Include Manufactures Preparation Requirements)							X	X	2							1			1
12	09 96 00	1.6	Certification	X						X	X	2							1			1
12	09 96 00	1.6	Test Results	X								2							1			1
6	10 11 00	1.6	LIQUID CHALKBOARD AND TACKBOARDS VOLUME 6: NORTH RIM WTP																			
6	10 11 00	1.6	Product Data - Boards and Fastener System							X	X	1										1
1	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 1					X			X		2		2	2	2	2	4		2	4
2	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 2					X			X		2		2	2	2	2	4		2	4
3	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 3					X			X		2		2	2	2	2	4		2	4
4	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 4					X			X		2		2	2	2	2	4		2	4
5	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 5					X			X		2		2	2	2	2	4		2	4
6	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 6					X			X		2		2	2	2	2	4		2	4
8	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 8					X			X		2		2	2	2	2	4		2	4
9	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 9					X			X		2		2	2	2	2	4		2	4
10	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 10					X			X		2		2	2	2	2	4		2	4
11	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 11					X			X		2		2	2	2	2	4		2	4
12	10 14 00	1.6	IDENTIFICATION DEVICES - Tag Schedule & Samples Volume 12					X			X		2		2	2	2	2	4		2	4
5	10 14 23	1.6	SIGNAGE Sign Schedule & Samples Volume 5					X			X		2		2	2	2	2	4		2	4
6	10 14 23	1.6	SIGNAGE Sign Schedule & Samples Volume 6					X			X		2		2	2	2	2	4		2	4
8	10 14 23	1.6	SIGNAGE Sign Schedule & Samples Volume 8					X			X		2		2	2	2	2	4		2	4
9	10 14 23	1.6	SIGNAGE Sign Schedule & Samples Volume 9					X			X		2		2	2	2	2	4		2	4
12	10 14 23	1.6	SIGNAGE Sign Schedule & Samples Volume 12					X			X		2		2	2	2	2	4		2	4
6	10 26 00	1.6	WALL AND DOOR PROTECTION VOLUME 6: NORTH RIM WTP																			
6	10 26 00	1.6	Shop Drawings - Location by Type						X			2							1			1
6	10 26 00	1.6	Product Data - Wall Guard							X	X	1							1			1
6	10 26 00	1.6	Product Data - Corner Guard							X	X	1							1			1
6	10 28 13	1.6	TOILET ACCESSORIES VOLUME 6: NORTH RIM WTP																			
6	10 28 13	1.6	Product Data - Toilet Accessories							X	X	4							1			1
5	10 44 33	1.6	FIRE PROTECTION SPECIALTIES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	10 44 33	1.6	Shop Drawings - Location by Type						X			1							1		2	1
5	10 44 33	1.6	Product Data - Portable Fire Extinguishers & Brackets							X	X	1							1		2	1
5	10 44 33	1.6	Product Data - Fire Extinguisher Cabinets							X	X	1							1		2	1
6	10 44 33	1.6	FIRE PROTECTION SPECIALTIES VOLUME 6: NORTH RIM WTP																			
6	10 44 33	1.6	Shop Drawings - Location by Type						X			1							1		2	1
6	10 44 33	1.6	Product Data - Portable Fire Extinguishers & Brackets							X	X	1							1		2	1
6	10 44 33	1.6	Product Data - Fire Extinguisher Cabinets							X	X	1							1		2	1
9	10 44 33	1.6	VOLUME 9: POWER DISTROBUTION CENTER																			
9	10 44 33	1.6	Shop Drawings - Location by Type						X			1							1		2	1
9	10 44 33	1.6	Product Data - Portable Fire Extinguishers & Brackets							X	X	1							1		2	1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
9	10 44 33	1.6	Product Data - Fire Extinguisher Cabinets							X	X	1							1		2	1
12	10 44 33	1.6	VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	10 44 33	1.6	Shop Drawings - Location by Type						X			1							1		2	1
12	10 44 33	1.6	Product Data - Portable Fire Extinguishers & Brackets							X	X	1							1		2	1
12	10 44 33	1.6	Product Data - Fire Extinguisher Cabinets							X	X	1							1		2	1
6	10 51 13	1.6	METAL LOCKERS AND LOCKER BENCHES VOLUME 6: NORTH RIM WTP																			
6	10 51 13	1.6	Shop Drawings - Location by Type & Numbering System						X			4							1			1
6	10 51 13	1.6	Product Data - Metal Locker & Anchoring							X	X	2							1			1
12	10 73 16	1.6	ALUMINUM CANOPIES VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	10 73 16	1.6	Shop Drawings						X			4		4					1			1
12	10 73 16	1.6	Product Data - Aluminum Canopy System					X		X	X	2		2					1			1
6	11 24 26	1.6	SHOP AND STORAGE EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	11 24 26	1.6	Shop Drawings						X			4							1			1
6	11 24 26	1.6	Product Data - Shelving							X	X	2							1			1
6	11 24 26	1.6	Product Data - Work Bench							X	X	2							1			1
6	11 24 26	1.6	Product Data - Shop Stools							X	X	2							1			1
6	11 24 26	1.6	Product Data - Gas Cylinder Hand Trucks							X	X	2							1			1
5	11 24 26	1.6	Product Data - Industrial Curtain							X	X	2							1			1
6	11 24 26	1.6	Samples					X				2							1			1
6	11 30 13	1.6	APPLIANCES VOLUME 6: NORTH RIM WTP																			
6	11 30 13	1.6	Product Data - Microwave							X	X	2							1			1
6	11 30 13	1.6	Product Data - Refrigerator							X	X	2							1			1
6	11 30 13	1.6	Product Data - Washer and Dryer							X	X	2							1			1
6	11 30 13	1.6	Product Data - Dishwasher							X	X	2							1			1
6	11 30 13	1.6	Product Data - Undercounter Refrigerator							X	X	2							1			1
6	11 30 13	1.6	Product Data - Commercial Ice Maker							X	X	2							1			1
6	11 30 13	1.6	Product Data - Storage Bin							X	X	2							1			1
6	11 30 13	1.6	Samples					X				2							1			1
6	11 53 00	1.6	LABORATORY EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	11 53 00	1.6	Shop Drawings						X					1		4	1		1			1
6	11 53 00	1.6	Product Data -							X	X	2							1			1
6	11 53 13	1.6	LABORATORY FUME HOODS AND EXHAUST DEVICES VOLUME 6: NORTH RIM WTP																			
6	11 53 13	1.6	Shop Drawings						X					1		4	1		1			1
6	11 53 13	1.6	Product Data - Chemical Fume Hood (include bench mounted)							X	X			1		2	1		1			1
6	11 53 13	1.6	Product Data - Fume Extractor Arms (snorkels)							X	X			1		2	1		1			1
6	11 53 13	1.6	Samples					X						1		2	1		1			1
6	11 53 43	1.6	LABORATORY SERVICE FITTINGS AND FIXTURES VOLUME 6: NORTH RIM WTP																			
6	11 53 43	1.6	Shop Drawings						X			2				2			1			1
6	11 53 43	1.6	Product Data - Laboratory Service Fittings							X	X	1				1			1			1
6	11 53 43	1.6	Product Data - Laboratory Valves							X	X	1				1			1			1
6	11 53 43	1.6	Product Data - Laboratory Electrical Pedestal Boxes							X	X	1					1		1			1
6	11 53 43	1.6	Product Data - Laboratory Emergency Plumbing Fixtures							X	X	1				1			1			1
6	11 53 43	1.6	Product Data - Laboratory Emergency Shower Tempering System							X	X	1				2			1			1
6	11 53 43	1.6	Product Data - Laboratory Sink							X	X	1				1			1			1
6	11 53 43	1.6	Product Data - Laboratory Recessed Valve Box							X	X	1				1			1			1
6	11 53 43	1.6	Samples					X								2			1			1
6	12 24 13	1.6	ROLLER WINDOW SHADES VOLUME 6: NORTH RIM WTP																			
6	12 24 13	1.6	Shop Drawings						X			4							1			1
6	12 24 13	1.6	Product Data - Window Coverings							X	X	2							1			1
6	12 24 13	1.6	Samples					X				2							1			1
6	12 35 53	1.6	LABORATORY CASEWORK AND OTHER FURNISHINGS VOLUME 6: NORTH RIM WTP																			
6	12 35 53	1.6	Shop Drawings						X			4		4					1			1
6	12 35 53	1.6	Product Data - Metal Casework							X	X	4		2					1			1
6	12 35 53	1.6	Product Data - Adaptive Laboratory Furniture Systems							X	X	4							1			1
6	12 35 53	1.6	Product Data - Stainless Steel Laboratory Furniture Systems							X	X	4							1			1
6	12 35 53	1.6	Product Data - Cabinet Hardware							X	X	2							1			1
6	12 35 53	1.6	Product Data - Special Purpose Cabinet - Corrosive Storage Cabinet							X	X	2							1			1
6	12 35 53	1.6	Product Data - Special Purpose Cabinet - Flamable Liquid Storage							X	X	2							1			1
6	12 35 53	1.6	Product Data - Laboratory Work Sufaces							X	X	2							1			1
6	12 35 53	1.6	Product Data - Shelving Assemblies							X	X	2		2					1			1
6	12 35 53	1.6	Product Data - Cylinder Restraint Assembly							X	X	2		1					1			1
6	12 35 53	1.6	Product Data - Metal Pipe Drop Enclosure							X	X	2							1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	12 35 53	1.6	Product Data - Drying Rack							X	X	2							1			1
6	12 35 53	1.6	Product Data - Metal Fabrication and Finishes							X	X	4		2					1			1
6	12 35 53	1.6	Product Data - Stainless Steel Fabrications - Work Surfaces, Sinks and Cup Sinks, Canopy Hood							X	X	4		2					1			1
6	12 35 53	1.6	Product Data - Sealent							X	X	4							1			1
6	12 35 53	1.6	Samples					X				2							1			1
6	13 47 13.17	1.6	CATHODIC PROTECTION OF STEEL WATER TANKS VOLUME 6: NORTH RIM WTP																			
6	13 47 13.17	1.6	Shop Drawings						X							4			1			1
6	13 47 13.17	1.6	Product Data - Anodes							X	X					2			1			1
6	13 47 13.17	1.6	Product Data - Reference Electrodes							X	X					2			1			1
6	13 47 13.17	1.6	Product Data - Anode Support Rope							X	X					2			1			1
6	13 47 13.17	1.6	Product Data - Anode Support Clips							X	X					2			1			1
6	13 47 13.17	1.6	Product Data - Wire and Splicing Materials							X	X					2			1			1
6	13 47 13.17	1.6	Product Data - Anode Junction Box, Shunts, Electronic Components							X	X					2			1			1
6	13 47 13.17	1.6	Product Data - Pressure Entrance Fittings							X	X					2			1			1
6	13 47 13.17	1.6	Certifications	X															1			1
ALL	22 05 03	1.3	COMMON TO ALL VOLUMES																			
ALL	22 05 03	1.3	Product Data - Pipe							X	X					1			1			1
ALL	22 05 03	1.3	Product Data - Pipe Fittings							X	X					1			1			1
ALL	22 05 03	1.3	Product Data - Flex Connections, Pressure Gauges, Wye Strainers							X	X					1			1			1
ALL	22 05 03	1.3	Product Data - Accessories							X	X					1			1			1
ALL	22 05 03	1.3	Testing Report								X					1			1			1
5	22 05 29	1.3	HANGERS AND SUPPORTS FOR PLUMBING PIPE AND EQUIPMENT Plumbing Hangers: Pipe Hangers and Supports, Seismic Restraints, Hanger Rods, Inserts, Sleeves, Mechanical Sleeve Seals, Formed Steel Channel, Firestopping Volume 5							X	X			2			1			1
6	22 05 29	1.3	HANGERS AND SUPPORTS FOR PLUMBING PIPE AND EQUIPMENT Plumbing Hangers: Pipe Hangers and Supports, Seismic Restraints, Hanger Rods, Inserts, Sleeves, Mechanical Sleeve Seals, Formed Steel Channel, Firestopping Volume 6							X	X			4			1			1
6	22 11 26	2.2	PACKAGED DOMESTIC WATER PRESSURE BOOSTER SYSTEM																			
6	22 11 26	2.2	Product Data - Booster System	X						X	X		40	1	4	4	4	4	1			1
6	22 11 26	2.2	Shop Drawings - Submersible Sump Pump System						X		X		40	1	4	4	4	4	1			1
5	22 20 00	1.3	PLUMBING FIXTURES AND EQUIPMENT VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	22 20 00	1.3	Shop Drawings						X			2				2			1			1
5	22 20 00	1.3	Product Data - Drains, Roof Drains, Carriers, and Shock Absorbers							X	X					1			1			1
5	22 20 00	1.3	Product Data - Trap Primer							X	X					1			1			1
5	22 20 00	1.3	Samples					X				2				2			1			1
6	22 20 00	1.3	PLUMBING FIXTURES AND EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	22 20 00	1.3	Shop Drawings						X			2				2			1			1
6	22 20 00	1.3	Product Data - Water Closet and Lavatories							X	X	1				2			1			1
6	22 20 00	1.3	Product Data - Stainless Steel Sinks							X	X	1				2			1			1
6	22 20 00	1.3	Product Data - Premolded Mop Sink							X	X					2			1			1
6	22 20 00	1.3	Product Data - Water Closet Seats							X	X	1				1			1			1
6	22 20 00	1.3	Product Data - Lavatory Fittings							X	X					1						1
6	22 20 00	1.3	Product Data - Mop Sink Fittings							X	X					1						1
6	22 20 00	1.3	Product Data - Sink Fittings							X	X					1						1
6	22 20 00	1.3	Product Data - Laboratory Sink Fittings							X	X					1						1
6	22 20 00	1.3	Product Data - Flush Valves							X	X					1						1
6	22 20 00	1.3	Product Data - Emergency Shower and Eye Wash							X	X	1				1			1			1
6	22 20 00	1.3	Product Data - Drains, Roof Drains, Carriers, and Shock Absorbers							X	X					1			1			1
6	22 20 00	1.3	Product Data - Trap Primer							X	X					1			1			1
6	22 20 00	1.3	Product Data - Domestic Water Heater							X	X					1			1			1
6	22 20 00	1.3	Product Data - Reduced Pressure Backflow Preventor							X	X					1			1			1
6	22 20 00	1.3	Product Data - Hot Water Circulation Pump							X	X					1	1		1			1
6	22 20 00	1.3	Product Data - Mixing Valves - Emergency Fixtures, Sinks, Lavatories							X	X					1			1			1
6	22 20 00	1.3	Product Data - Mixing Valves - Expansion Tanks							X	X					1			1			1
6	22 20 00	1.3	Samples					X				1				1			1			1
5	23 05 48	1.3	VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	23 05 48	1.3	Shop Drawings - Vibration Isolators by Type and Location						X					4		2	2	1	1			1
5	23 05 48	1.3	Shop Drawings - Manufactures Drawings						X					4		2	2	1	1			1
5	23 05 48	1.3	Shop Drawings - Piping and Ductwork						X					1		4			1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
5	23 05 48	1.3	Product Data - Vibration Isolators							X	X			2	4	4	4	2	1			1
5	23 05 48	1.3	Product Data - Vibration Isolator Connected Piping and Ductwork							X	X			2	2	2	2	2	1			1
6	23 05 48	1.3	VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	23 05 48	1.3	Shop Drawings - Vibration Isolators by Type and Location						X					4		2	2	2	1			1
6	23 05 48	1.3	Shop Drawings - Manufactures Drawings						X					4		2	2	2	1			1
6	23 05 48	1.3	Shop Drawings - Piping and Ductwork						X					1		4			1			1
6	23 05 48	1.3	Product Data - Vibration Isolators							X	X			2	4	4	4	2	1			1
6	23 05 48	1.3	Product Data - Vibration Isolator Connected Piping and Ductwork							X	X			2	2	2	2	2	1			1
6	23 05 93	1.3	HVAC SYSTEMS - BALANCING AND TESTING VOLUME 6: NORTH RIM WTP																			
6	23 05 93	1.3	Product Data - Air Handling Units							X	X			1		4	2		1			1
6	23 05 93	1.3	Product Data - Fan Coil Units							X	X			1		4	2		1			1
6	23 05 93	1.3	Testing and Balencing Reports		X		X									4			1			1
5	23 09 00	1.3	INSTRUMENTATION AND CONTROL FOR HVAC SYSTEMS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	23 09 00	1.3	Shop Drawings - Wiring and Control Diagrams (I/O)						X							4	2		1			1
5	23 09 00	1.3	Shop Drawings - Basis Of Operations and Sequence Narratives						X							2	1		1			1
5	23 09 00	1.3	Shop Drawings - Actuators						X							4	2		1			1
5	23 09 00	1.3	Product Data - Instrament and Controls Equipment							X	X					4	2		1			1
5	23 09 00	1.3	Product Data - Electric Damper Actuators							X	X			1		4	4	1	1			1
5	23 09 00	1.3	Product Data - Thermostats							X	X	1				4	2		1			1
5	23 09 00	1.3	Product Data - Temperature Sensors							X	X					2	1		1			1
5	23 09 00	1.3	Product Data - Differential Pressure Switch							X	X					2	1		1			1
5	23 09 00	1.3	Product Data - Current Switch							X	X					2	1		1			1
5	23 09 00	1.3	Product Data - Transformer							X	X					2	4		1			1
5	23 09 00	1.3	Product Data - Local Temperature Control Panels							X	X					2	4		1			1
6	23 09 00	1.3	INSTRUMENTATION AND CONTROL FOR HVAC SYSTEMS VOLUME 6: NORTH RIM WTP																			
6	23 09 00	1.3	Shop Drawings - Wiring and Control Diagrams (I/O)						X							4	2		1			1
6	23 09 00	1.3	Shop Drawings - Basis Of Operations and Sequence Narratives						X							2	1		1			1
6	23 09 00	1.3	Shop Drawings - Actuators						X							4	2		1			1
6	23 09 00	1.3	Product Data - Instrament and Controls Equipment							X	X					4	2		1			1
6	23 09 00	1.3	Product Data - Electric Damper Actuators							X	X			1		4	4	1	1			1
6	23 09 00	1.3	Product Data - Thermostats							X	X	1				4	2		1			1
6	23 09 00	1.3	Product Data - Temperature Sensors							X	X					2	1		1			1
6	23 09 00	1.3	Product Data - Differential Pressure Switch							X	X					2	1		1			1
6	23 09 00	1.3	Product Data - Current Switch							X	X					2	1		1			1
6	23 09 00	1.3	Product Data - Transformer							X	X					2	4		1			1
6	23 09 00	1.3	Product Data - Local Temperature Control Panels							X	X					2	4		1			1
5	23 31 00	1.3	HVAC DUCTWORK VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	23 31 00	1.3	Shop Drawings - Scaled Ductwork						X							1		4			1			1
5	23 31 00	1.3	Product Data - Transverse Joints & Connections							X	X					2			1			1
5	23 31 00	1.3	Product Data - Flexible Ducts							X	X					2			1			1
5	23 31 00	1.3	Product Data - Turning Vanes							X	X					2			1			1
5	23 31 00	1.3	Product Data - Access Doors - Ductwork							X	X					2			1			1
5	23 31 00	1.3	Product Data - Backdraft Damper							X	X					2	1		1			1
5	23 31 00	1.3	Product Data - Grills and Registers							X	X	2				2			1			1
5	23 31 00	1.3	Product Data - Air Filters							X	X					1			1			1
5	23 31 00	1.3	Product Data - Manual Dampers							X	X	2				1			1			1
5	23 31 00	1.3	Product Data - Duct Sealer							X	X					1			1			1
5	23 31 00	1.3	Product Data - Supports and Anchors							X	X			4		2			1			1
5	23 31 00	1.3	Samples - Dampers							X	X	2				1			1			1
5	23 31 00	1.3	Samples - Grills							X	X	2				1			1			1
5	23 31 00	1.3	Samples - Ceiling Diffuser							X	X	2				1			1			1
6	23 31 00	1.3	HVAC DUCTWORK VOLUME 6: NORTH RIM WTP																			
6	23 31 00	1.3	Shop Drawings - Scaled Ductwork						X							1		4			1			1
6	23 31 00	1.3	Product Data - Transverse Joints & Connections							X	X					2			1			1
6	23 31 00	1.3	Product Data - Flexible Ducts							X	X					2			1			1
6	23 31 00	1.3	Product Data - Turning Vanes							X	X					2			1			1
6	23 31 00	1.3	Product Data - Access Doors - Ductwork							X	X					2			1			1
6	23 31 00	1.3	Product Data - Backdraft Damper							X	X					2	1		1			1
6	23 31 00	1.3	Product Data - Exhaust Hood							X	X					4	2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	23 31 00	1.3	Product Data - Fire Damper							X	X					2	2		1		1	1
6	23 31 00	1.3	Product Data - Ceiling Diffuser							X	X	2				2			1			1
6	23 31 00	1.3	Product Data - Grills and Registers							X	X	2				2			1			1
6	23 31 00	1.3	Product Data - Air Filters							X	X					1			1			1
6	23 31 00	1.3	Product Data - Manual Dampers							X	X	2				1			1			1
6	23 31 00	1.3	Product Data - Duct Sealer							X	X					1			1			1
6	23 31 00	1.3	Product Data - Supports and Anchors							X	X			4		2			1			1
6	23 31 00	1.3	Samples - Dampers							X	X	2				1			1			1
6	23 31 00	1.3	Samples - Grills							X	X	2				1			1			1
6	23 31 00	1.3	Samples - Ceiling Diffuser							X	X	2				1			1			1
6	23 31 00	1.3	Samples - Exhaust Hood							X	X	2				1			1			1
5	23 34 00	1.3	HVAC FANS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	23 34 00	1.3	Product Data - Vibration Isolation Assemblies							X	X			2		2			1			1
5	23 34 00	1.3	Product Data - Corrosion Protective Coating							X	X			1		1			1			1
5	23 34 00	1.3	Product Data - Wall Mounted Centrifugal Exhaust Fans							X	X					2	1		1			1
5	23 34 00	1.3	Product Data - Wall Mounted Proeller Type Fans							X	X					2	1		1			1
6	23 34 00	1.3	HVAC FANS VOLUME 6: NORTH RIM WTP																			
6	23 34 00	1.3	Product Data - Vibration Isolation Assemblies							X	X			2		2			1			1
6	23 34 00	1.3	Product Data - Corrosion Protective Coating							X	X			1		1			1			1
6	23 34 00	1.3	Product Data - Centrifugal Fans							X	X					2	1		1			1
6	23 34 00	1.3	Product Data - Toilet Room Exhaust Fans							X	X	1				2	1		1			1
5	23 34 00	1.3	REFRIGERANT PIPING SYSTEM VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	23 34 00	1.3	Shop Drawings - Split System Heat Pump and Air Conditioning System						X			1		1		4	2		1			1
5	23 34 00	1.3	Product Data - Refrigerant Piping and Fittings							X	X					2			1			1
5	23 34 00	1.3	Product Data - Hangers and Anchors							X	X			2		2			1			1
5	23 34 00	1.3	Testing and Balencing Reports		X		X									4			1			1
6	23 34 00	1.3	REFRIGERANT PIPING SYSTEM VOLUME 6: NORTH RIM WTP																			
6	23 34 00	1.3	Shop Drawings - Split System Heat Pump and Air Conditioning System						X			1		1		4	2		1			1
6	23 34 00	1.3	Product Data - Refrigerant Piping and Fittings							X	X					2			1			1
6	23 34 00	1.3	Product Data - Hangers and Anchors							X	X			2		2			1			1
6	23 34 00	1.3	Testing and Balencing Reports		X		X									4			1			1
6	23 81 29	1.3	VARIABLE REFRIGERANT FLOW HEAT PUMP SYSTEMS VOLUME 6: NORTH RIM WTP																			
6	23 81 29	1.3	Shop Drawings - Split System Heat Pump						X			1		1		4	2		1			1
6	23 81 29	1.3	Shop Drawings - Duct						X							4			1			1
6	23 81 29	1.3	Product Data - Outdoor Heat Pump Unit (heat/cool)							X	X			1		4	2		1			1
6	23 81 29	1.3	Product Data - Indoor Ceiling Cassette Unit							X	X			1		2	1		1			1
6	23 81 29	1.3	Product Data - Indoor Ducted Air Handling Unit							X	X			2		2	1		1			1
6	23 81 29	1.3	Testing and Balencing Reports		X		X									4			1			1
ALL	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL COMMON TO ALL VOLUMES																			
ALL	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
ALL	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
ALL	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
ALL	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
4	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 4: RS CAVE TO RSPH																			
4	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
4	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
4	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
5	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
5	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
5	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
5	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
6	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 6: NORTH RIM WTP																			
6	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
6	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
6	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
6	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
8	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 8: WW COLLECTION AND LS																			
8	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
8	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
8	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

VOLUME	SUBMITTAL: SPECIFICATION SECTION	SUBMITTAL: PARAGRAPH NUMBER	SUBMITTAL: DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
8	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
9	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
9	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
9	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
9	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
10	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
10	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
10	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
10	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
11	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 11: POWER FROM RSPH TO MANZANITA																			
11	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
11	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
11	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
11	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
12	26 05 00	1.4	COMMON WORK RESULTS FOR ELECTRICAL VOLUME 12: POWER SYSTEM COTTONWOOD CAMPGROUND																			
12	26 05 00	1.4	Product Data - Electrical Equipment Support Pedestals / Racks							X	X						2		1			1
12	26 05 00	1.4	Product Data - Field Tough Up Zinc Rich Primer							X	X						2		1			1
12	26 05 00	1.4	Product Data - Modular Strut & Structural Members / Hardware							X	X						2		1			1
12	26 05 00	1.4	Product Data - Equipment Pads							X	X			2			2		1			1
5	26 05 13	1.4	MEDIUM VOLTAGE CABLE VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 05 13	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
5	26 05 13	1.4	Certification	X			X										1		1			1
5	26 05 13	1.4	Cable Test Reports				X				X						2		1			1
5	26 05 13	1.4	Product Data - Medium Voltage Cable (Armored Wired Cable)							X	X						4		1			1
5	26 05 13	1.4	Product Data - Medium Voltage Cable (Non-Armored Wired Cable)							X	X						4		1			1
5	26 05 13	1.4	Product Data - Cable Terminations, Connections, Terminations and Splices							X	X						2		1			1
5	26 05 13	1.4	Product Data - Fault Indicators							X	X						2		1			1
5	26 05 13	1.4	Product Data - Multi-Point Cable Junctions							X	X						2		1			1
5	26 05 13	1.4	Product Data - Cable Shield Grounding Adapters							X	X						2		1			1
5	26 05 13	1.4	Product Data - Conduit							X	X						2		1			1
6	26 05 13	1.4	MEDIUM VOLTAGE CABLE VOLUME 6: NORTH RIM WTP																			
6	26 05 13	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
6	26 05 13	1.4	Certification	X			X										1		1			1
6	26 05 13	1.4	Cable Test Reports				X				X						2		1			1
6	26 05 13	1.4	Product Data - Medium Voltage Cable (Armored Wired Cable)							X	X						4		1			1
6	26 05 13	1.4	Product Data - Medium Voltage Cable (Non-Armored Wired Cable)							X	X						4		1			1
6	26 05 13	1.4	Product Data - Cable Terminations, Connections, Terminations and Splices							X	X						2		1			1
6	26 05 13	1.4	Product Data - Fault Indicators							X	X						2		1			1
6	26 05 13	1.4	Product Data - Multi-Point Cable Junctions							X	X						2		1			1
6	26 05 13	1.4	Product Data - Cable Shield Grounding Adapters							X	X						2		1			1
6	26 05 13	1.4	Product Data - Conduit							X	X						2		1			1
9	26 05 13	1.4	MEDIUM VOLTAGE CABLE VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 05 13	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
9	26 05 13	1.4	Certification	X			X										1		1			1
9	26 05 13	1.4	Cable Test Reports				X				X						2		1			1
9	26 05 13	1.4	Product Data - Medium Voltage Cable (Armored Wired Cable)							X	X						4		1			1
9	26 05 13	1.4	Product Data - Medium Voltage Cable (Non-Armored Wired Cable)							X	X						4		1			1
9	26 05 13	1.4	Product Data - Cable Terminations, Connections, Terminations and Splices							X	X						2		1			1
9	26 05 13	1.4	Product Data - Fault Indicators							X	X						2		1			1
9	26 05 13	1.4	Product Data - Multi-Point Cable Junctions							X	X						2		1			1
9	26 05 13	1.4	Product Data - Cable Shield Grounding Adapters							X	X						2		1			1
9	26 05 13	1.4	Product Data - Conduit							X	X						2		1			1
10	26 05 13	1.4	MEDIUM VOLTAGE CABLE VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	26 05 13	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
10	26 05 13	1.4	Certification	X			X										1		1			1
10	26 05 13	1.4	Cable Test Reports				X				X						2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
10	26 05 13	1.4	Product Data - Medium Voltage Cable (Armored Wired Cable)							X	X						4		1			1
10	26 05 13	1.4	Product Data - Medium Voltage Cable (Non-Armored Wired Cable)							X	X						4		1			1
10	26 05 13	1.4	Product Data - Cable Terminations, Connections, Terminations and Splices							X	X						2		1			1
10	26 05 13	1.4	Product Data - Fault Indicators							X	X						2		1			1
10	26 05 13	1.4	Product Data - Rock Wall Cable Support Clamps and Cable Support Grips							X	X						2		1			1
10	26 05 13	1.4	Product Data - Multi-Point Cable Junctions							X	X						2		1			1
10	26 05 13	1.4	Product Data - Cable Shield Grounding Adapters							X	X						2		1			1
10	26 05 13	1.4	Product Data - Conduit							X	X						2		1			1
11	26 05 13	1.4	MEDIUM VOLTAGE CABLE VOLUME 11: POWER FROM RSPH TO MANZANITA																			
11	26 05 13	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
11	26 05 13	1.4	Certification	X			X										1		1			1
11	26 05 13	1.4	Cable Test Reports				X				X						2		1			1
11	26 05 13	1.4	Product Data - Medium Voltage Cable (Armored Wired Cable)							X	X						4		1			1
11	26 05 13	1.4	Product Data - Medium Voltage Cable (Non-Armored Wired Cable)							X	X						4		1			1
11	26 05 13	1.4	Product Data - Cable Terminations, Connections, Terminations and Splices							X	X						2		1			1
11	26 05 13	1.4	Product Data - Fault Indicators							X	X						2		1			1
11	26 05 13	1.4	Product Data - Rock Wall Cable Support Clamps and Cable Support Grips							X	X						2		1			1
11	26 05 13	1.4	Product Data - Multi-Point Cable Junctions							X	X						2		1			1
11	26 05 13	1.4	Product Data - Cable Shield Grounding Adapters							X	X						2		1			1
11	26 05 13	1.4	Product Data - Conduit							X	X						2		1			1
12	26 05 13	1.4	MEDIUM VOLTAGE CABLE VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 05 13	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
12	26 05 13	1.4	Certification	X			X										1		1			1
12	26 05 13	1.4	Cable Test Reports				X				X						2		1			1
12	26 05 13	1.4	Product Data - Medium Voltage Cable (Armored Wired Cable)							X	X						4		1			1
12	26 05 13	1.4	Product Data - Medium Voltage Cable (Non-Armored Wired Cable)							X	X						4		1			1
12	26 05 13	1.4	Product Data - Cable Terminations, Connections, Terminations and Splices							X	X						2		1			1
12	26 05 13	1.4	Product Data - Fault Indicators							X	X						2		1			1
12	26 05 13	1.4	Product Data - Rock Wall Cable Support Clamps and Cable Support Grips							X	X						2		1			1
12	26 05 13	1.4	Product Data - Multi-Point Cable Junctions							X	X						2		1			1
12	26 05 13	1.4	Product Data - Cable Shield Grounding Adapters							X	X						2		1			1
12	26 05 13	1.4	Product Data - Conduit							X	X						2		1			1
5	26 05 19	1.4	LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 05 19	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
5	26 05 19	1.4	Certification	X			X										1		1			1
5	26 05 19	1.4	Cable Test Reports				X				X						2		1			1
5	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						4		1			1
5	26 05 19	1.4	Product Data - Building Wire							X	X						4		1			1
5	26 05 19	1.4	Product Data - Power Cable							X	X						4		1			1
5	26 05 19	1.4	Product Data - Control Cables							X	X						2		1			1
5	26 05 19	1.4	Product Data - Shielded VFD Cable							X	X						2		1			1
5	26 05 19	1.4	Product Data - Instrumentation Cable							X	X						2		1			1
5	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						2		1			1
5	26 05 19	1.4	Product Data - Wire Connections							X	X						2		1			1
5	26 05 19	1.4	Product Data - Insulating Tape							X	X						2		1			1
5	26 05 19	1.4	Product Data - Pulling Lubricant							X	X						2		1			1
6	26 05 19	1.4	LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES VOLUME 6: NORTH RIM WTP																			
6	26 05 19	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
6	26 05 19	1.4	Certification	X			X										1		1			1
6	26 05 19	1.4	Cable Test Reports				X				X						2		1			1
6	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						4		1			1
6	26 05 19	1.4	Product Data - Building Wire							X	X						4		1			1
6	26 05 19	1.4	Product Data - Power Cable							X	X						4		1			1
6	26 05 19	1.4	Product Data - Control Cables							X	X						2		1			1
6	26 05 19	1.4	Product Data - Shielded VFD Cable							X	X						2		1			1
6	26 05 19	1.4	Product Data - Instrumentation Cable							X	X						2		1			1
6	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						2		1			1
6	26 05 19	1.4	Product Data - Wire Connections							X	X						2		1			1
6	26 05 19	1.4	Product Data - Insulating Tape							X	X						2		1			1
6	26 05 19	1.4	Product Data - Pulling Lubricant							X	X						2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME										
				INFORMATIONAL				ACTION														
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
9	26 05 19	1.4	LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 05 19	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
9	26 05 19	1.4	Certification	X			X										1		1			1
9	26 05 19	1.4	Cable Test Reports				X				X						2		1			1
9	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						4		1			1
9	26 05 19	1.4	Product Data - Building Wire							X	X						4		1			1
9	26 05 19	1.4	Product Data - Power Cable							X	X						4		1			1
9	26 05 19	1.4	Product Data - Control Cables							X	X						2		1			1
9	26 05 19	1.4	Product Data - Shielded VFD Cable							X	X						2		1			1
9	26 05 19	1.4	Product Data - Instrumentation Cable							X	X						2		1			1
9	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						2		1			1
9	26 05 19	1.4	Product Data - Wire Connections							X	X						2		1			1
9	26 05 19	1.4	Product Data - Insulating Tape							X	X						2		1			1
9	26 05 19	1.4	Product Data - Pulling Lubricant							X	X						2		1			1
10	26 05 19	1.4	LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	26 05 19	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
10	26 05 19	1.4	Certification	X			X										1		1			1
10	26 05 19	1.4	Cable Test Reports				X				X						2		1			1
10	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						4		1			1
10	26 05 19	1.4	Product Data - Building Wire							X	X						4		1			1
10	26 05 19	1.4	Product Data - Power Cable							X	X						4		1			1
10	26 05 19	1.4	Product Data - Control Cables							X	X						2		1			1
10	26 05 19	1.4	Product Data - Shielded VFD Cable							X	X						2		1			1
10	26 05 19	1.4	Product Data - Instrumentation Cable							X	X						2		1			1
10	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						2		1			1
10	26 05 19	1.4	Product Data - Wire Connections							X	X						2		1			1
10	26 05 19	1.4	Product Data - Insulating Tape							X	X						2		1			1
10	26 05 19	1.4	Product Data - Pulling Lubricant							X	X						2		1			1
12	26 05 19	1.4	LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 05 19	1.4	Pull Plan - Tensioning, Breakaway Swivel		X					X	X						4		1			1
12	26 05 19	1.4	Certification	X			X										1		1			1
12	26 05 19	1.4	Cable Test Reports				X				X						2		1			1
12	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						4		1			1
12	26 05 19	1.4	Product Data - Building Wire							X	X						4		1			1
12	26 05 19	1.4	Product Data - Power Cable							X	X						4		1			1
12	26 05 19	1.4	Product Data - Control Cables							X	X						2		1			1
12	26 05 19	1.4	Product Data - Shielded VFD Cable							X	X						2		1			1
12	26 05 19	1.4	Product Data - Instrumentation Cable							X	X						2		1			1
12	26 05 19	1.4	Product Data - Fiber Optic Cable							X	X						2		1			1
12	26 05 19	1.4	Product Data - Wire Connections							X	X						2		1			1
12	26 05 19	1.4	Product Data - Insulating Tape							X	X						2		1			1
12	26 05 19	1.4	Product Data - Pulling Lubricant							X	X						2		1			1
5	26 05 26	1.3	GROUNDING AND BONDING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 05 26	1.3	Product Data - Building/Tank Grounding System							X	X	1		1			4		1			1
5	26 05 26	1.3	Product Data - Grounding System							X	X						2		1			1
5	26 05 26	1.3	Product Data - Exothermic Weld Connections							X	X						2		1			1
5	26 05 26	1.3	Product Data - Prefabricated Composite Test Station							X	X						2		1			1
5	26 05 26	1.3	Product Data - Transformer Grounding System							X	X						2		1			1
6	26 05 26	1.3	GROUNDING AND BONDING VOLUME 6: NORTH RIM WTP																			
6	26 05 26	1.3	Product Data - Building/Tank Grounding System							X	X	1		1			4		1			1
6	26 05 26	1.3	Product Data - Grounding System							X	X						2		1			1
6	26 05 26	1.3	Product Data - Exothermic Weld Connections							X	X						2		1			1
6	26 05 26	1.3	Product Data - Prefabricated Composite Test Station							X	X						2		1			1
6	26 05 26	1.3	Product Data - Transformer Grounding System							X	X						2		1			1
8	26 05 26	1.3	GROUNDING AND BONDING VOLUME 8: WW COLLECTION AND LS																			
8	26 05 26	1.3	Product Data - Building/Tank Grounding System/Manhole							X	X	1		1			4		1			1
8	26 05 26	1.3	Product Data - Grounding System							X	X						2		1			1
8	26 05 26	1.3	Product Data - Exothermic Weld Connections							X	X						2		1			1
8	26 05 26	1.3	Product Data - Prefabricated Composite Test Station							X	X						2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
8	26 05 26	1.3	Product Data - Transformer Grounding System							X	X						2		1			1
9	26 05 26	1.3	GROUNDING AND BONDING VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 05 26	1.3	Product Data - Building/Tank Grounding System							X	X	1		1			4		1			1
9	26 05 26	1.3	Product Data - Grounding System							X	X						2		1			1
9	26 05 26	1.3	Product Data - Grounding Clamps, Connectors and Terminals / Exothermic Weld Connections							X	X						2		1			1
9	26 05 26	1.3	Product Data - Prefabricated Composite Test Station							X	X						2		1			1
9	26 05 26	1.3	Product Data - Transformer/ Swtichgear Grounding System							X	X						2		1			1
10	26 05 26	1.3	GROUNDING AND BONDING VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	26 05 26	1.3	Product Data - Building/Tank Grounding System/Pole							X	X	1		1			4		1			1
10	26 05 26	1.3	Product Data - Grounding System							X	X						3		1			1
10	26 05 26	1.3	Product Data - Grounding Clamps, Connectors and Terminals / Exothermic Weld Connections							X	X						3		1			1
10	26 05 26	1.3	Product Data - Prefabricated Composite Test Station							X	X						2		1			1
11	26 05 26	1.3	GROUNDING AND BONDING VOLUME 11: POWER FROM RSPH TO MANZANITA																			
11	26 05 26	1.3	Product Data - Grounding System							X	X						3		1			1
11	26 05 26	1.3	Product Data - Grounding Clamps, Connectors and Terminals / Exothermic Weld Connections							X	X						3		1			1
12	26 05 26	1.3	GROUNDING AND BONDING VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 05 26	1.3	Product Data - Shade Structure Grounding System							X	X	1		1			4		1			1
12	26 05 26	1.3	Product Data - Grounding System							X	X						2		1			1
12	26 05 26	1.3	Product Data - Grounding Clamps, Connectors and Terminals / Exothermic Weld Connections							X	X						2		1			1
12	26 05 26	1.3	Product Data - Prefabricated Composite Test Station							X	X						2		1			1
12	26 05 26	1.3	Product Data - Battery/Inverter Grounding System							X	X						2		1			1
ALL	26 05 33	1.3	RACEWAYS AND BOXES COMMON TO ALL VOLUMES																			
ALL	26 05 33	1.3	Shop Drawing - Box Layout by Type and Location						X	X	X	4					4		1			1
ALL	26 05 33	1.3	Product Data - Conduit - Pipe, Fittings, Supports							X	X						2		1			1
ALL	26 05 33	1.3	Product Data - Wireways							X	X						2		1			1
ALL	26 05 33	1.3	Product Data - Box: Outlet, Pull and Junction							X	X						2		1			1
5	26 05 33	1.3	RACEWAYS AND BOXES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 05 33	1.3	Shop Drawing - Box Layout by Type and Location						X	X	X	4					4		1			1
5	26 05 33	1.3	Product Data - Conduit - Pipe, Fittings, Supports							X	X						2		1			1
5	26 05 33	1.3	Product Data - Wireways							X	X						2		1			1
5	26 05 33	1.3	Product Data - Box: Outlet, Pull and Junction							X	X						2		1			1
6	26 05 33	1.3	RACEWAYS AND BOXES VOLUME 6: NORTH RIM WTP																			
6	26 05 33	1.3	Shop Drawing - Box Layout by Type and Location						X	X	X	4					4		1			1
6	26 05 33	1.3	Product Data - Conduit - Pipe, Fittings, Supports							X	X						2		1			1
6	26 05 33	1.3	Product Data - Wireways							X	X						2		1			1
6	26 05 33	1.3	Product Data - Box: Outlet, Pull and Junction							X	X						2		1			1
9	26 05 33	1.3	RACEWAYS AND BOXES VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 05 33	1.3	Shop Drawing - Box Layout by Type and Location						X	X	X	4					4		1			1
9	26 05 33	1.3	Product Data - Conduit - Pipe, Fittings, Supports							X	X						2		1			1
9	26 05 33	1.3	Product Data - Wireways							X	X						2		1			1
9	26 05 33	1.3	Product Data - Box: Outlet, Pull and Junction							X	X						2		1			1
6	26 05 43	1.4	ELECTRICAL - EXTERIOR UNDERGROUND VOLUME 6: NORTH RIM WTP																			
6	26 05 43	1.4	Shop Drawing - Manholes and Handholes						X	X	X						4		1			1
6	26 05 43	1.4	Shop Drawing - Underground Conduit and Ductbank						X	X	X						4		1			1
6	26 05 43	1.4	Product Data - Manholes							X	X						2		1			1
6	26 05 43	1.4	Product Data - Handholes							X	X						2		1			1
6	26 05 43	1.4	Product Data - Underground Conduit and Ductbanks							X	X						2		1			1
10	26 05 46	1.3	OVERHEAD POWER LINES VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	26 05 46	1.3	Shop Drawing - Power Pole Installation Per Site - Micropiles		X				X	X	X		4	4			4		1			1
10	26 05 46	1.3	Test Reports - Independent Agency								X			2			2		1			1
10	26 05 46	1.3	Report: Record of Measurements Made During Sagging Operations		X		X				X						4		1			1
10	26 05 46	1.3	Product Data - Bare Conductor							X	X						2		1			1
10	26 05 46	1.3	Product Data - Insulated Conductor - Low Voltage Insulated Conductor (Single)							X	X						2		1			1
10	26 05 46	1.3	Product Data - Messanger Supported Medium Voltage Cables							X	X						2		1			1
10	26 05 46	1.3	Product Data - Optical Ground Wire							X	X						2		1			1
10	26 05 46	1.3	Product Data - Marker Balls							X	X						2		1			1
10	26 05 46	1.3	Product Data - Insulators Porcelain							X	X						2		1			1
10	26 05 46	1.3	Product Data - Surge Arrestors							X	X						2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
10	26 05 46	1.3	Product Data - Anchors							X	X							2		1			1
10	26 05 46	1.3	Product Data - Connectors							X	X							2		1			1
10	26 05 46	1.3	Product Data - Misc Components - OPFW Bonding Ground Wire, Bolted Dead End for OPFW, Coil Bracket for OPGW and ADSS, Bonding Clamps for OPGW, Cable Grip Closed Mesh, Gable Grip Open Mesh							X	X							4		1			1
10	26 05 46	1.3	Product Data - Conduit - Galvanized Ridgid and PVC							X	X							2		1			1
10	26 05 46	1.3	Product Data - Cable Terminations							X	X							2		1			1
10	26 05 46	1.3	Product Data - Insulators Glass Suspension							X	X							2		1			1
ALL	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS COMMON TO ALL VOLUMES																			
ALL	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					4		2		1			1
ALL	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					2		2		1			1
ALL	26 05 48	1.3	Product Data - Bracing							X	X					2		2		1			1
5	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					2		1		1			1
5	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					1		1		1			1
5	26 05 48	1.3	Product Data - Bracing							X	X					1		1		1			1
6	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS VOLUME 6: NORTH RIM WTP																			
6	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					4		2		1			1
6	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					2		2		1			1
6	26 05 48	1.3	Product Data - Bracing							X	X					2		2		1			1
9	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					2		1		1			1
9	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					1		1		1			1
9	26 05 48	1.3	Product Data - Bracing							X	X					1		1		1			1
10	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					2		1		1			1
10	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					1		1		1			1
10	26 05 48	1.3	Product Data - Bracing							X	X					1		1		1			1
11	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS VOLUME 11: POWER FROM RSPH TO MANZANITA																			
11	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					2		1		1			1
11	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					1		1		1			1
11	26 05 48	1.3	Product Data - Bracing							X	X					1		1		1			1
12	26 05 48	1.3	ELECTRICAL SEISMIC RESTRAINT SYSTEMS VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 05 48	1.3	Shop Drawing - Location by Type of Seismic Bracing						X	X	X					2		1		1			1
12	26 05 48	1.3	Calculations Report: Structural Calculations		X		X				X					1		1		1			1
12	26 05 48	1.3	Product Data - Bracing							X	X					1		1		1			1
6	26 05 73	1.2	UNDERGROUND DUCTS AND RACEWAYS FOR ELECTRICAL SYSTEMS VOLUME 6: NORTH RIM WTP																			
6	26 05 73	1.2	Shop Drawing - Manhole						X	X	X					2		2		1			1
6	26 05 73	1.2	Shop Drawing - Handhole						X	X	X					2		2		1			1
6	26 05 73	1.2	Product Data - Duct Bank Conduit Seperation							X	X							2		1			1
1	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 1						X	X	X							2		1			1
2	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 2						X	X	X							2		1			1
3	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 3						X	X	X							2		1			1
4	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 4						X	X	X							2		1			1
5	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 5						X	X	X							2		1			1
6	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 6						X	X	X							2		1			1
8	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 8						X	X	X							2		1			1
9	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 9						X	X	X							2		1			1
10	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 10						X	X	X							2		1			1
11	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 11						X	X	X							2		1			1
12	26 08 13	1.3	ACCEPTANCE TESTING TESTING AND ENERGIZING PLAN Volume 12						X	X	X							2		1			1
5	26 09 13	1.3	ELECTRICAL METERING DEVICES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 09 13	1.3	Shop Drawing - Fabrication and Each Meter Layout						X	X	X					4		1		1			1
5	26 09 13	1.3	Product Data - Digital Metering Equipment: Type A,B,C, Data Logging							X	X					2		1		1			1
6	26 09 13	1.3	ELECTRICAL METERING DEVICES VOLUME 6: NORTH RIM WTP																			
6	26 09 13	1.3	Shop Drawing - Fabrication and Each Meter Layout						X	X	X					4		1		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	26 09 13	1.3	Product Data - Digital Metering Equipment: Type A,B,C, Data Logging							X	X						2		1			1
8	26 09 13	1.3	ELECTRICAL METERING DEVICES VOLUME 8: WW COLLECTION AND LS																			
8	26 09 13	1.3	Shop Drawing - Fabrication and Each Meter Layout						X	X	X						4		1			1
8	26 09 13	1.3	Product Data - Digital Metering Equipment: Type A,B,C, Data Logging							X	X						2		1			1
9	26 09 13	1.3	ELECTRICAL METERING DEVICES VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 09 13	1.3	Shop Drawing - Fabrication and Each Meter Layout						X	X	X						4		1			1
9	26 09 13	1.3	Product Data - Digital Metering Equipment: Type A,B,C, Data Logging							X	X						2		1			1
10	26 09 13	1.3	ELECTRICAL METERING DEVICES VOLUME 10: OVERHEAD POWER NR TO RSPH																			
12	26 09 13	1.3	Shop Drawing - Fabrication and Each Meter Layout						X	X	X						4		1			1
12	26 09 13	1.3	Product Data - Digital Metering Equipment: Type A,B,C, Data Logging							X	X						2		1			1
5	26 09 16	1.4	CONTROL EQUIPMENT ACCESSORIES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 09 16	1.4	Plan: Functional Test Plan		X					X	X						4		1			1
5	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Device						X	X	X						4		1			1
5	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Control Devices						X	X	X						4		1			1
5	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Industrical Control Panel						X	X	X						4		1			1
5	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Station						X	X	X						4		1			1
5	26 09 16	1.4	Product Data - Relays							X	X						2		1			1
5	26 09 16	1.4	Product Data - Contactors							X	X						2		1			1
5	26 09 16	1.4	Product Data - Photosensors and Time Clocks							X	X						2		1			1
5	26 09 16	1.4	Product Data - Alarm Devices - Audio and Visual							X	X						2		1			1
5	26 09 16	1.4	Product Data - Run Time Meter							X	X						2		1			1
5	26 09 16	1.4	Product Data - Control Power Transformer							X	X						2		1			1
5	26 09 16	1.4	Product Data - Actuated Sensor Interface Asi							X	X						4		1			1
5	26 09 16	1.4	Product Data - Termination Equipment: Standard Type Block, Bladed Type Disconnect Block, Grounded Type Block, Fuse Holder							X	X						4		1			1
5	26 09 16	1.4	Product Data - Enclosures: Industrial Control Panel, NEMA 4X Rated, NEMA 7 and 9 Rated, NEMA 12 Enclosure, Control Panel Misc							X	X						4		1			1
5	26 09 16	1.4	Product Data - Operator Control Devices: Selector Switch, Pushbuttons, Indicator Lights							X	X						2		1			1
5	26 09 16	1.4	Product Data - Control Devices: Timers, Relays, Contactors							X	X						2		1			1
5	26 09 16	1.4	Product Data - Industrial Control Panel							X	X						4		1			1
5	26 09 16	1.4	Product Data - Operator Control Station							X	X						4		1			1
6	26 09 16	1.4	CONTROL EQUIPMENT ACCESSORIES VOLUME 6: NORTH RIM WTP																			
6	26 09 16	1.4	Plan: Functional Test Plan		X					X	X						4		1			1
6	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Device						X	X	X						4		1			1
6	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Control Devices						X	X	X						4		1			1
6	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Industrical Control Panel						X	X	X						4		1			1
6	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Station						X	X	X						4		1			1
6	26 09 16	1.4	Product Data - Relays							X	X						2		1			1
6	26 09 16	1.4	Product Data - Contactors							X	X						2		1			1
6	26 09 16	1.4	Product Data - Photosensors and Time Clocks							X	X						2		1			1
6	26 09 16	1.4	Product Data - Alarm Devices - Audio and Visual							X	X						2		1			1
6	26 09 16	1.4	Product Data - Run Time Meter							X	X						2		1			1
6	26 09 16	1.4	Product Data - Control Power Transformer							X	X						2		1			1
6	26 09 16	1.4	Product Data - Actuated Sensor Interface Asi							X	X						4		1			1
6	26 09 16	1.4	Product Data - Termination Equipment: Standard Type Block, Bladed Type Disconnect Block, Grounded Type Block, Fuse Holder							X	X						4		1			1
6	26 09 16	1.4	Product Data - Enclosures: Industrial Control Panel, NEMA 4X Rated, NEMA 7 and 9 Rated, NEMA 12 Enclosure, Control Panel Misc							X	X						4		1			1
6	26 09 16	1.4	Product Data - Operator Control Devices: Selector Switch, Pushbuttons, Indicator Lights							X	X						2		1			1
6	26 09 16	1.4	Product Data - Control Devices: Timers, Relays, Contactors							X	X						2		1			1
6	26 09 16	1.4	Product Data - Industrial Control Panel							X	X						4		1			1
6	26 09 16	1.4	Product Data - Operator Control Station							X	X						4		1			1
8	26 09 16	1.4	CONTROL EQUIPMENT ACCESSORIES VOLUME 8: WW COLLECTION AND LS																			
8	26 09 16	1.4	Plan: Functional Test Plan		X					X	X						1		1			1
8	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Device						X	X	X						1		1			1
8	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Control Devices						X	X	X						1		1			1
8	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Industrical Control Panel						X	X	X						1		1			1
8	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Station						X	X	X						1		1			1
8	26 09 16	1.4	Product Data - Relays							X	X						1		1			1
8	26 09 16	1.4	Product Data - Contactors							X	X						1		1			1
8	26 09 16	1.4	Product Data - Photosensors and Time Clocks							X	X						1		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

	SUBMITTAL			REQUIREMENTS								ARCHITECT / ENGINEER REVIEW TIME											
				INFORMATIONAL				ACTION															
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
8	26 09 16	1.4	Product Data - Alarm Devices - Audio and Visual							X	X						1		1			1
8	26 09 16	1.4	Product Data - Run Time Meter							X	X						1		1			1
8	26 09 16	1.4	Product Data - Control Power Transformer							X	X						1		1			1
8	26 09 16	1.4	Product Data - Actuated Sensor Interface Asi							X	X						1		1			1
8	26 09 16	1.4	Product Data - Termination Equipment: Standard Type Block, Bladed Type Disconnect Block, Grounded Type Block, Fuse Holder							X	X						1		1			1
8	26 09 16	1.4	Product Data - Enclosures: Industrial Control Panel, NEMA 4X Rated, NEMA 7 and 9 Rated, NEMA 12 Enclosure, Control Panel Misc							X	X						1		1			1
8	26 09 16	1.4	Product Data - Operator Control Devices: Selector Switch, Pushbuttons, Indicator Lights							X	X						1		1			1
8	26 09 16	1.4	Product Data - Control Devices: Timers, Relays, Contactors							X	X						1		1			1
8	26 09 16	1.4	Product Data - Industrial Control Panel							X	X						1		1			1
8	26 09 16	1.4	Product Data - Operator Control Station							X	X						1		1			1
9	26 09 16	1.4	CONTROL EQUIPMENT ACCESSORIES VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 09 16	1.4	Plan: Functional Test Plan		X					X	X						4		1			1
9	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Device						X	X	X						4		1			1
9	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Control Devices						X	X	X						4		1			1
9	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Industrical Control Panel						X	X	X						4		1			1
9	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Station						X	X	X						4		1			1
9	26 09 16	1.4	Product Data - Relays							X	X						2		1			1
9	26 09 16	1.4	Product Data - Contactors							X	X						2		1			1
9	26 09 16	1.4	Product Data - Photosensors and Time Clocks							X	X						2		1			1
9	26 09 16	1.4	Product Data - Alarm Devices - Audio and Visual							X	X						2		1			1
9	26 09 16	1.4	Product Data - Run Time Meter							X	X						2		1			1
9	26 09 16	1.4	Product Data - Control Power Transformer							X	X						2		1			1
9	26 09 16	1.4	Product Data - Actuated Sensor Interface Asi							X	X						4		1			1
9	26 09 16	1.4	Product Data - Termination Equipment: Standard Type Block, Bladed Type Disconnect Block, Grounded Type Block, Fuse Holder							X	X						4		1			1
9	26 09 16	1.4	Product Data - Enclosures: Industrial Control Panel, NEMA 4X Rated, NEMA 7 and 9 Rated, NEMA 12 Enclosure, Control Panel Misc							X	X						4		1			1
9	26 09 16	1.4	Product Data - Operator Control Devices: Selector Switch, Pushbuttons, Indicator Lights							X	X						2		1			1
9	26 09 16	1.4	Product Data - Control Devices: Timers, Relays, Contactors							X	X						2		1			1
9	26 09 16	1.4	Product Data - Industrial Control Panel							X	X						4		1			1
9	26 09 16	1.4	Product Data - Operator Control Station							X	X						4		1			1
12	26 09 16	1.4	CONTROL EQUIPMENT ACCESSORIES VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 09 16	1.4	Plan: Functional Test Plan		X					X	X						2		1			1
12	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Device						X	X	X						2		1			1
12	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Control Devices						X	X	X						2		1			1
12	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Industrical Control Panel						X	X	X						2		1			1
12	26 09 16	1.4	Shop Drawing - Fabrication and Layout Drawings: Operator Control Station						X	X	X						2		1			1
12	26 09 16	1.4	Product Data - Relays							X	X						1		1			1
12	26 09 16	1.4	Product Data - Contactors							X	X						1		1			1
12	26 09 16	1.4	Product Data - Photosensors and Time Clocks							X	X						1		1			1
12	26 09 16	1.4	Product Data - Alarm Devices - Audio and Visual							X	X						1		1			1
12	26 09 16	1.4	Product Data - Run Time Meter							X	X						1		1			1
12	26 09 16	1.4	Product Data - Control Power Transformer							X	X						1		1			1
12	26 09 16	1.4	Product Data - Actuated Sensor Interface Asi							X	X						2		1			1
12	26 09 16	1.4	Product Data - Termination Equipment: Standard Type Block, Bladed Type Disconnect Block, Grounded Type Block, Fuse Holder							X	X						2		1			1
12	26 09 16	1.4	Product Data - Enclosures: Industrial Control Panel, NEMA 4X Rated, NEMA 7 and 9 Rated, NEMA 12 Enclosure, Control Panel Misc							X	X						2		1			1
12	26 09 16	1.4	Product Data - Operator Control Devices: Selector Switch, Pushbuttons, Indicator Lights							X	X						1		1			1
12	26 09 16	1.4	Product Data - Control Devices: Timers, Relays, Contactors							X	X						1		1			1
12	26 09 16	1.4	Product Data - Industrial Control Panel							X	X						2		1			1
12	26 09 16	1.4	Product Data - Operator Control Station							X	X						2		1			1
5	26 09 43	1.4	NETWORKED LIGHTING CONTROL SYSTEM VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 09 43	1.4	Shop Drawing - Networked Lighting Control System						X	X	X						4		1			1
5	26 09 43	1.4	Plan- Reflected Ceiling Plan, Sequence of Operation		X		X			X	X						4		1			1
5	26 09 43	1.4	Product Data - System Equipment: Wall Station, Tough Screen Panel, Dimmers, Switches, Occupant Sensor, Photo Sensor, Daylight Harvesting Light Controls, Time Clock							X	X						4		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

ARCHITECT / ENGINEER REVIEW TIME

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
5	26 09 43	1.4	Product Data - System Contact Closures (input/ouput devices), Scene Setting and Selection Controller, Distributed Relay Lighting Controllers, Distributed Relay Lighting Controllers, Relay Based Lighting and Recepticle Control Panels, Energy Control Units for Load Shedding							X	X						4		1			1
5	26 09 43	1.4	Product Data - System Server, Lighting Control System Software, Communication Wire							X	X						4		1			1
5	26 09 43	1.4	Product Data - Interface to: Alarm System, Security System, Building Management System (BMS)							X	X						4		1			1
6	26 09 43	1.4	NETWORKED LIGHTING CONTROL SYSTEM VOLUME 6: NORTH RIM WTP																			
6	26 09 43	1.4	Shop Drawing - Networked Lighting Control System						X	X	X						4		1			1
6	26 09 43	1.4	Plan- Reflected Ceiling Plan, Sequence of Operation		X		X			X	X						4		1			1
6	26 09 43	1.4	Product Data - System Equipment: Wall Station, Tough Screen Panel, Dimmers, Switches, Occupant Sensor, Photo Sensor, Daylight Harvesting Light Controls, Time Clock							X	X						4		1			1
6	26 09 43	1.4	Product Data - System Contact Closures (input/ouput devices), Scene Setting and Selection Controller, Distributed Relay Lighting Controllers, Distributed Relay Lighting Controllers, Relay Based Lighting and Recepticle Control Panels, Energy Control Units for Load Shedding							X	X						4		1			1
6	26 09 43	1.4	Product Data - System Server, Lighting Control System Software, Communication Wire							X	X						4		1			1
6	26 09 43	1.4	Product Data - Interface to: Alarm System, Security System, Building Management System (BMS)							X	X						4		1			1
9	26 12 19	1.4	DISTRIBUTION TRANSFORMERS NETWORKED LIGHTING CONTROL SYSTEM VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 12 19	1.4	Shop Drawing - Transformer: Three Phase						X	X	X						4		1			1
9	26 12 19	1.4	Shop Drawing - Transformer: Single Phase						X	X	X						4		1			1
9	26 12 19	1.4	Shop Drawing - Single Phase Shunt Reactor						X	X	X						4		1			1
9	26 12 19	1.4	Shop Drawing - Insulating Secondary Cover						X	X	X						4		1			1
9	26 12 19	1.4	Shop Drawing - Surge Arrestor						X	X	X						4		1			1
9	26 12 19	1.4	Product Data - Three Phase Transformer							X	X						2		1			1
9	26 12 19	1.4	Product Data - Single Phase Transformer							X	X						2		1			1
9	26 12 19	1.4	Product Data - Single Phase Shunt Reactor							X	X						2		1			1
9	26 12 19	1.4	Product Data - Insulating Secondary Cover							X	X						2		1			1
9	26 12 19	1.4	Product Data - Surge Arrestors							X	X						2		1			1
9	26 12 19	1.4	Sample - Color					X			X						2		1			1
5	26 13 16	1.3	MEDIUM VOLTAGE METAL ENCLOSED LOAD INTERRUPTER SWITCHGEARVOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 13 16	1.3	Testing Data	X		X											1		1			1
5	26 13 16	1.3	Shop Drawing - Load Interrupter Switchgear						X	X	X						4		1			1
5	26 13 16	1.3	Shop Drawing - Fault Interrupter Switchgear						X	X	X						4		1			1
5	26 13 16	1.3	Product Data - Load Interrupter Switchgear							X	X						4		1			1
5	26 13 16	1.3	Product Data - Fault Interrupter Switchgear							X	X						4		1			1
6	26 13 16	1.3	MEDIUM VOLTAGE METAL ENCLOSED LOAD INTERRUPTER SWITCHGEAR VOLUME 6: NORTH RIM WTP																			
6	26 13 16	1.3	Testing Data	X		X											1		1			1
6	26 13 16	1.3	Shop Drawing - Load Interrupter Switchgear						X	X	X						4		1			1
6	26 13 16	1.3	Shop Drawing - Fault Interrupter Switchgear						X	X	X						4		1			1
6	26 13 16	1.3	Product Data - Load Interrupter Switchgear							X	X						4		1			1
6	26 13 16	1.3	Product Data - Fault Interrupter Switchgear							X	X						4		1			1
9	26 13 16	1.3	MEDIUM VOLTAGE METAL ENCLOSED LOAD INTERRUPTER SWITCHGEAR VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 13 16	1.3	Testing Data	X		X											1		1			1
9	26 13 16	1.3	Shop Drawing - Load Interrupter Switchgear						X	X	X						4		1			1
9	26 13 16	1.3	Shop Drawing - Fault Interrupter Switchgear						X	X	X						4		1			1
9	26 13 16	1.3	Product Data - Load Interrupter Switchgear							X	X						4		1			1
9	26 13 16	1.3	Product Data - Fault Interrupter Switchgear							X	X						4		1			1
9	26 13 23	1.3	SECTIONALIZING EQUIPMENT VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 13 23	1.3	Testing Data	X		X											1		1			1
9	26 13 23	1.3	Shop Drawing - Pad Mounted Type Sectionalizing Junction Enclosure						X	X	X						2		1			1
9	26 13 23	1.3	Product Data - Pad Mounted Type Sectionalizing Junction Enclosure							X	X						2		1			1
11	26 13 23	1.3	SECTIONALIZING EQUIPMENT VOLUME 11: POWER FROM RSPH TO MANZANITA																			
11	26 13 23	1.3	Testing Data	X		X											1		1			1
11	26 13 23	1.3	Shop Drawing - Pad Mounted Type Sectionalizing Junction Enclosure						X	X	X						2		1			1
11	26 13 23	1.3	Product Data - Pad Mounted Type Sectionalizing Junction Enclosure							X	X						2		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
5	26 22 13	1.3	LOW-VOLTAGE DISTRIBUTION TRANSFORMERS VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	26 22 13	1.3	Testing Data	X		X												1		1			1
5	26 22 13	1.3	Shop Drawing - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less						X	X	X							4		1			1
5	26 22 13	1.3	Product Data - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less							X	X							4		1			1
6	26 22 13	1.3	LOW-VOLTAGE DISTRIBUTION TRANSFORMERS VOLUME 6: NORTH RIM WTP																				
6	26 22 13	1.3	Testing Data	X		X												1		1			1
6	26 22 13	1.3	Shop Drawing - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less						X	X	X							4		1			1
6	26 22 13	1.3	Product Data - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less							X	X							4		1			1
9	26 22 13	1.3	LOW-VOLTAGE DISTRIBUTION TRANSFORMERS VOLUME 9: POWER DISTROBUTION CENTER																				
9	26 22 13	1.3	Testing Data	X		X												1		1			1
9	26 22 13	1.3	Shop Drawing - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less						X	X	X							4		1			1
9	26 22 13	1.3	Product Data - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less							X	X							4		1			1
12	26 22 13	1.3	LOW-VOLTAGE DISTRIBUTION TRANSFORMERS VOLUME 12: COTTONWOOD POWER SYSTEM																				
12	26 22 13	1.3	Testing Data	X		X												1		1			1
12	26 22 13	1.3	Shop Drawing - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less						X	X	X							4		1			1
12	26 22 13	1.3	Product Data - Low Voltage Distrobution Transformers - Dry-Type 1,000kVA or Less							X	X							4		1			1
5	26 23 00	1.3	SWITCHGEAR VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	26 23 00	1.3	Testing Data	X		X												1		1			1
5	26 23 00	1.3	Shop Drawing - Low Voltage Metal Enclosed Switchgear						X	X	X							4		1			1
5	26 23 00	1.3	Product Data - Low Voltage Metal Enclosed Switchgear							X	X							4		1			1
6	26 23 00	1.3	SWITCHGEAR VOLUME 6: NORTH RIM WTP																				
6	26 23 00	1.3	Testing Data	X		X												1		1			1
6	26 23 00	1.3	Shop Drawing - Low Voltage Metal Enclosed Switchgear						X	X	X							4		1			1
6	26 23 00	1.3	Product Data - Low Voltage Metal Enclosed Switchgear							X	X							4		1			1
9	26 23 00	1.3	SWITCHGEAR VOLUME 9: POWER DISTROBUTION CENTER																				
9	26 23 00	1.3	Testing Data	X		X												1		1			1
9	26 23 00	1.3	Shop Drawing - Generator Paralleleing Switchgear						X	X	X							4		1			1
9	26 23 00	1.3	Product Data - Generator Paralleleing Switchgear							X	X							4		1			1
~~12~~	~~26 23 00~~	~~1.3~~	~~SWITCHGEAR VOLUME 12: COTTONWOOD POWER SYSTEM~~				-								-	-	-	-	-	-		-	-
~~12~~	~~26 23 00~~	~~1.3~~	~~Testing Data~~	~~X~~	-	~~X~~			-	-	-	-		-		-	~~1~~	-	~~1~~			~~1~~
~~12~~	~~26 23 00~~	~~1.3~~	~~Shop Drawing - Low Voltage Metal Enclosed Switchgear~~						~~X~~	~~X~~	~~X~~	-		-		-	~~4~~	-	~~1~~			~~1~~
~~12~~	~~26 23 00~~	~~1.3~~	~~Product Data - Low Voltage Metal Enclosed Switchgear~~						-	~~X~~	~~X~~	-		-		-	~~4~~	-	~~1~~			~~1~~
5	26 24 13	1.3	SWITCHBOARDS VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	26 24 13	1.3	Testing Data	X		X												1		1			1
5	26 24 13	1.3	Shop Drawing - Low Voltage Switchboards						X	X	X							4		1			1
5	26 24 13	1.3	Product Data - Low Voltage Switchboards							X	X							4		1			1
6	26 24 13	1.3	SWITCHBOARDS VOLUME 6: NORTH RIM WTP																				
6	26 24 13	1.3	Testing Data	X		X												1		1			1
6	26 24 13	1.3	Shop Drawing - Low Voltage Switchboards						X	X	X							4		1			1
6	26 24 13	1.3	Product Data - Low Voltage Switchboards							X	X							4		1			1
9	26 24 13	1.3	SWITCHBOARDS VOLUME 9: POWER DISTROBUTION CENTER																				
9	26 24 13	1.3	Testing Data	X		X												1		1			1
9	26 24 13	1.3	Shop Drawing - Low Voltage Switchboards						X	X	X							4		1			1
9	26 24 13	1.3	Product Data - Low Voltage Switchboards							X	X							4		1			1
12	26 24 13	1.3	SWITCHBOARDS VOLUME 12: COTTONWOOD POWER SYSTEM																				
12	26 24 13	1.3	Testing Data	X		X												1		1			1
12	26 24 13	1.3	Shop Drawing - Low Voltage Switchboards						X	X	X							4		1			1
12	26 24 13	1.3	Product Data - Low Voltage Switchboards							X	X							4		1			1
4	26 24 16	1.4	PANELBOARDS VOLUME 4: RS CAVE TO RSPH																				
4	26 24 16	1.4	Testing Data	X		X												1		1			1
4	26 24 16	1.4	Shop Drawing - Branch Circuit Panelboard 400A or Less						X	X	X							4		1			1
4	26 24 16	1.4	Product Data - Branch Circuit Panelboard 400A or Less							X	X							4		1			1
5	26 24 16	1.4	PANELBOARDS VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	26 24 16	1.4	Testing Data	X		X												1		1			1
5	26 24 16	1.4	Shop Drawing - Branch Circuit Panelboard 400A or Less						X	X	X							4		1			1
5	26 24 16	1.4	Product Data - Branch Circuit Panelboard 400A or Less							X	X							4		1			1
6	26 24 16	1.4	PANELBOARDS VOLUME 6: NORTH RIM WTP																				
6	26 24 16	1.4	Testing Data	X		X												1		1			1
6	26 24 16	1.4	Shop Drawing - Branch Circuit Panelboard 400A or Less						X	X	X							4		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	26 24 16	1.4	Product Data - Branch Circuit Panelboard 400A or Less							X	X						4		1			1
8	26 24 16	1.4	PANELBOARDS VOLUME 8: WW COLLECTION AND LS																			
8	26 24 16	1.4	Testing Data	X		X											1		1			1
8	26 24 16	1.4	Shop Drawing - Branch Circuit Panelboard 400A or Less						X	X	X						4		1			1
8	26 24 16	1.4	Product Data - Branch Circuit Panelboard 400A or Less							X	X						4		1			1
9	26 24 16	1.4	PANELBOARDS VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 24 16	1.4	Testing Data	X		X											1		1			1
9	26 24 16	1.4	Shop Drawing - Branch Circuit Panelboard 400A or Less						X	X	X						4		1			1
9	26 24 16	1.4	Product Data - Branch Circuit Panelboard 400A or Less							X	X						4		1			1
12	26 24 16	1.4	PANELBOARDS VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 24 16	1.4	Testing Data	X		X											1		1			1
12	26 24 16	1.4	Shop Drawing - Branch Circuit Panelboard 400A or Less						X	X	X						4		1			1
12	26 24 16	1.4	Product Data - Branch Circuit Panelboard 400A or Less							X	X						4		1			1
5	26 24 19	1.3	MOTOR CONTROL EQUIPMENT VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 24 19	1.3	Testing Data	X		X											1		1			1
5	26 24 19	1.3	Shop Drawing - Motor Control Center						X	X	X						4		1			1
5	26 24 19	1.3	Shop Drawing - Seperately Mounted Motor Starter						X	X	X						4		1			1
5	26 24 19	1.3	Shop Drawing - Manual Motor Starters						X	X	X						4		1			1
5	26 24 19	1.3	Product Data - Motor Control Center							X	X						4		1			1
5	26 24 19	1.3	Product Data - Seperately Mounted Motor Starter							X	X						4		1			1
5	26 24 19	1.3	Product Data - Manual Motor Starters							X	X						4		1			1
6	26 24 19	1.3	MOTOR CONTROL EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	26 24 19	1.3	Testing Data	X		X											1		1			1
6	26 24 19	1.3	Shop Drawing - Motor Control Center						X	X	X						4		1			1
6	26 24 19	1.3	Shop Drawing - Seperately Mounted Motor Starter						X	X	X						4		1			1
6	26 24 19	1.3	Shop Drawing - Manual Motor Starters						X	X	X						4		1			1
6	26 24 19	1.3	Product Data - Motor Control Center							X	X						4		1			1
6	26 24 19	1.3	Product Data - Seperately Mounted Motor Starter							X	X						4		1			1
6	26 24 19	1.3	Product Data - Manual Motor Starters							X	X						4		1			1
8	26 24 19	1.3	MOTOR CONTROL EQUIPMENT VOLUME 8: WW COLLECTION AND LS																			
8	26 24 19	1.3	Testing Data	X		X											1		1			1
8	26 24 19	1.3	Shop Drawing - Motor Control Center						X	X	X						4		1			1
8	26 24 19	1.3	Shop Drawing - Seperately Mounted Motor Starter						X	X	X						4		1			1
8	26 24 19	1.3	Shop Drawing - Manual Motor Starters						X	X	X						4		1			1
8	26 24 19	1.3	Product Data - Motor Control Center							X	X						4		1			1
8	26 24 19	1.3	Product Data - Seperately Mounted Motor Starter							X	X						4		1			1
8	26 24 19	1.3	Product Data - Manual Motor Starters							X	X						4		1			1
9	26 24 19	1.3	MOTOR CONTROL EQUIPMENT VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 24 19	1.3	Testing Data	X		X											1		1			1
9	26 24 19	1.3	Shop Drawing - Motor Control Center						X	X	X						4		1			1
9	26 24 19	1.3	Shop Drawing - Seperately Mounted Motor Starter						X	X	X						4		1			1
9	26 24 19	1.3	Shop Drawing - Manual Motor Starters						X	X	X						4		1			1
9	26 24 19	1.3	Product Data - Motor Control Center							X	X						4		1			1
9	26 24 19	1.3	Product Data - Seperately Mounted Motor Starter							X	X						4		1			1
9	26 24 19	1.3	Product Data - Manual Motor Starters							X	X						4		1			1
9	26 26 13	1.3	PACKAGE POWER SUPPLY VOLUME 9: POWER DISTRIBUTION CENTER																			
9	26 26 13	1.3	Testing Data	X		X											1		1			1
9	26 26 13	1.3	Shop Drawing - Package Power Supply (Include Transformer and Panelboard)						X	X	X						4		1			1
9	26 26 13	1.3	Product Data - Package Power Supply (Include Transformer and Panelboard)							X	X						4		1			1
12	26 26 13	1.3	PACKAGE POWER SUPPLY VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 26 13	1.3	Testing Data	X		X											1		1			1
12	26 26 13	1.3	Shop Drawing - Package Power Supply (Include Transformer and Panelboard)						X	X	X						4		1			1
12	26 26 13	1.3	Product Data - Package Power Supply (Include Transformer and Panelboard)							X	X						4		1			1
5	26 27 26	1.3	WIRING DEVICES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 27 26	1.3	Shop Drawing - Type by Location						X	X	X	1					4		1			1
5	26 27 26	1.3	Product Data - Wall Switch							X	X						1		1			1
5	26 27 26	1.3	Product Data - Recepticles							X	X						1		1			1
5	26 27 26	1.3	Product Data - Device Wall Plate and Coverplates							X	X	1					1		1			1
6	26 27 26	1.3	WIRING DEVICES VOLUME 6: NORTH RIM WTP																			
6	26 27 26	1.3	Shop Drawing - Type by Location						X	X	X	1					4		1			1
6	26 27 26	1.3	Product Data - Wall Switch							X	X						1		1			1
6	26 27 26	1.3	Product Data - Recepticles							X	X						1		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	26 27 26	1.3	Product Data - Device Wall Plate and Coverplates							X	X	1					1		1			1
5	26 28 00	1.3	OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 28 00	1.3	Report - Ground Fault Protection System Test Field Reports				X		X	X	X						1		1			1
5	26 28 00	1.3	Product Data - Low Voltage Circuit Breakers							X	X						1		1			1
5	26 28 00	1.3	Product Data - Low Voltage Fuses							X	X						1		1			1
6	26 28 00	1.3	OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES VOLUME 6: NORTH RIM WTP																			
6	26 28 00	1.3	Report - Ground Fault Protection System Test Field Reports				X		X	X	X						1		1			1
6	26 28 00	1.3	Product Data - Low Voltage Circuit Breakers							X	X						1		1			1
6	26 28 00	1.3	Product Data - Low Voltage Fuses							X	X						1		1			1
8	26 28 00	1.3	OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES VOLUME 8: WW COLLECTION AND LS																			
8	26 28 00	1.3	Report - Ground Fault Protection System Test Field Reports				X		X	X	X						1		1			1
8	26 28 00	1.3	Product Data - Low Voltage Circuit Breakers							X	X						1		1			1
8	26 28 00	1.3	Product Data - Low Voltage Fuses							X	X						1		1			1
9	26 28 00	1.3	OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 28 00	1.3	Report - Ground Fault Protection System Test Field Reports				X		X	X	X						1		1			1
9	26 28 00	1.3	Product Data - Low Voltage Circuit Breakers							X	X						1		1			1
9	26 28 00	1.3	Product Data - Low Voltage Fuses							X	X						1		1			1
12	26 28 00	1.3	OVERCURRENT AND SHORT CIRCUIT PROTECTIVE DEVICES VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 28 00	1.3	Report - Ground Fault Protection System Test Field Reports				X		X	X	X						1		1			1
12	26 28 00	1.3	Product Data - Low Voltage Circuit Breakers							X	X						1		1			1
12	26 28 00	1.3	Product Data - Low Voltage Fuses							X	X						1		1			1
5	26 28 02	1.4	COORDINATED POWER SYSTEM PROTECTION VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 28 02	1.4	Report - Coordinated System Protection Study: Short Circuit, Protection, Arc-Flash, Circuit Breaker Protective Device Settings				X		X	X	X						4		1			1
6	26 28 02	1.4	COORDINATED POWER SYSTEM PROTECTION VOLUME 6: NORTH RIM WTP																			
6	26 28 02	1.4	Report - Coordinated System Protection Study: Short Circuit, Protection, Arc-Flash, Circuit Breaker Protective Device Settings				X		X	X	X						4		1			1
9	26 28 02	1.4	COORDINATED POWER SYSTEM PROTECTION VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 28 02	1.4	Report - Coordinated System Protection Study: Short Circuit, Protection, Arc-Flash, Circuit Breaker Protective Device Settings				X		X	X	X						4		1			1
5	26 28 16	1.3	SAFETY SWITCHES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 28 16	1.3	Report - Summary Table				X		X	X	X						4		1			1
5	26 28 16	1.3	Product Data - Safety Switchs							X	X						4		1			1
6	26 28 16	1.3	SAFETY SWITCHES VOLUME 6: NORTH RIM WTP																			
6	26 28 16	1.3	Report - Summary Table				X		X	X	X						4		1			1
6	26 28 16	1.3	Product Data - Safety Switchs							X	X						4		1			1
9	26 28 16	1.3	SAFETY SWITCHES VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 28 16	1.3	Report - Summary Table				X		X	X	X						4		1			1
9	26 28 16	1.3	Product Data - Safety Switchs							X	X						4		1			1
12	26 28 16	1.3	SAFETY SWITCHES VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 28 16	1.3	Report - Summary Table				X		X	X	X						4		1			1
12	26 28 16	1.3	Product Data - Safety Switchs							X	X						4		1			1
5	26 29 23	1.3	VARIABLE FREQUENCY DRIVES - LOW VOLTAGE VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 29 23	1.3	Report - Summary Table Schedule				X			X	X						4	1	1			1
5	26 29 23	1.3	Shop Drawings with Certifications and Factory Tests Results				X		X	X	X						4	1	1			1
5	26 29 23	1.3	Product Data - Variable Frequency Drive (VFD)							X	X						4	1	1			1
6	26 29 23	1.3	VARIABLE FREQUENCY DRIVES - LOW VOLTAGE VOLUME 6: NORTH RIM WTP																			
6	26 29 23	1.3	Report - Summary Table Schedule				X			X	X						4	1	1			1
6	26 29 23	1.3	Shop Drawings with Certifications and Factory Tests Results				X		X	X	X						4	1	1			1
6	26 29 23	1.3	Product Data - Variable Frequency Drive (VFD)							X	X						4	1	1			1
12	26 31 00	1.3	PHOTOVOLTAIC POWER SYSTEM VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 31 00	1.3	Report - Performance Criteria Meets Requirements				X			X	X						4		1			1
12	26 31 00	1.3	Shop Drawings Contorls, Inverters, Batteries Modules, Disconnects				X		X	X	X						4		1			1
12	26 31 00	1.3	Shop Drawings Wiring Diagram				X		X	X	X						4		1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	INFORMATIONAL: OTHER	ACTION: SAMPLES	ACTION: SHOP DRAWINGS	ACTION: MANUFACTURER DATA AND INSTRUCTIONS	ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
12	26 31 00	1.3	Shop Drawings Panels				X		X	X	X						4		1			1
12	26 31 00	1.3	Shop Drawings Panels / PV Modules							X	X						4		1			1
12	26 31 00	1.3	Product Data - Wiring							X	X						4		1			1
12	26 31 00	1.3	Product Data - Panels / PV Modules							X	X						4		1			1
9	26 32 14	1.3	ENGINE GENERATOR - DIESEL VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 32 14	1.3	Certifications				X			X	X						4		1			1
9	26 32 14	1.3	Shop Drawings - Generator				X		X	X	X			1			4		1			1
9	26 32 14	1.3	Shop Drawings - Fuel Storage Tanks				X		X	X	X			2		2	2	2	1			1
9	26 32 14	1.3	Shop Drawings - Enclosure and Accessories				X		X	X	X						2		1			1
9	26 32 14	1.3	Shop Drawings - Access Stairs and Platform				X		X	X	X			4			1		1			1
9	26 32 14	1.3	Product Data - Generator							X	X			2			8	2	1			1
9	26 32 14	1.3	Product Data - Fuel Storage Tanks							X	X			8		4	8	4	1			1
9	26 32 14	1.3	Product Data - Enclosure and Accessories							X	X						4		1			1
9	26 32 14	1.3	Product Data - Access Stairs and Platform							X	X			8		1	2	1	1			1
12	26 33 00	1.4	BATTERY AND BATTERY CHARGING SYSTEMS VOLUME 12: COTTONWOOD POWER SYSTEM																			
12	26 33 00	1.4	Report - Performance Criteria Meets Requirements				X			X	X						2		1			1
12	26 33 00	1.4	Shop Drawings - Charging and Battery Storage System (includes ventilation/cooling system)				X		X	X	X						4		1			1
12	26 33 00	1.4	Product Data - Batteries							X	X						2		1			1
12	26 33 00	1.4	Product Data - Inverter							X	X						2		1			1
12	26 33 00	1.4	Product Data - Enclosure and Ventilation System							X	X						2		1			1
5	26 35 26	1.4	HARMONIC CORRECTION EQUIPMENT VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 35 26	1.4	Shop Drawings						X	X	X						1	4	1	1		1
5	26 35 26	1.4	Certifications	X						X	X						1	4	1	1		1
5	26 35 26	1.4	Product Data - Harmonic Correction Equipment (includes Active Harmonic Filters)							X	X						1	4	1	1		1
6	26 35 26	1.4	HARMONIC CORRECTION EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	26 35 26	1.4	Shop Drawings						X	X	X						1	4	1	1		1
6	26 35 26	1.4	Certifications	X						X	X						1	4	1	1		1
6	26 35 26	1.4	Product Data - Harmonic Correction Equipment (includes Active Harmonic Filters)							X	X						1	4	1	1		1
5	26 41 13	1.5	LIGHTNING PROTECTION SYSTEM VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 41 13	1.5	Shop Drawings - Building System						X	X	X			1				4		1			1
5	26 41 13	1.5	Certifications	X						X	X							1		1			1
5	26 41 13	1.5	Product Data - Grounding System							X	X							2		1			1
5	26 41 13	1.5	Product Data - Rods, Wire and Mounting Hardare							X	X			1				4		1			1
6	26 41 13	1.5	LIGHTNING PROTECTION SYSTEM VOLUME 6: NORTH RIM WTP																			
6	26 41 13	1.5	Shop Drawings - Building System						X	X	X			1				4		1			1
6	26 41 13	1.5	Shop Drawings - Tank System						X	X	X			1				4		1			1
6	26 41 13	1.5	Certifications	X						X	X							1		1			1
6	26 41 13	1.5	Product Data - Grounding System							X	X							2		1			1
6	26 41 13	1.5	Product Data - Rods, Wire and Mounting Hardare							X	X			1				4		1			1
5	26 43 13	1.5	SURGE PROTECTION DEVICES FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 43 13	1.5	Shop Drawings - Surge Protection System						X	X	X							4		1			1
5	26 43 13	1.5	Certifications	X						X	X							1		1			1
5	26 43 13	1.5	Product Data - Surge Protection Equipment							X	X							4		1			1
6	26 43 13	1.5	SURGE PROTECTION DEVICES FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS VOLUME 6: NORTH RIM WTP																			
6	26 43 13	1.5	Shop Drawings - Surge Protection System						X	X	X							4		1			1
6	26 43 13	1.5	Certifications	X						X	X							1		1			1
6	26 43 13	1.5	Product Data - Surge Protection Equipment							X	X							4		1			1
9	26 43 13	1.5	SURGE PROTECTION DEVICES FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS VOLUME 9: POWER DISTROBUTION CENTER																			
9	26 43 13	1.5	Shop Drawings - Surge Protection System						X	X	X							4		1			1
9	26 43 13	1.5	Certifications	X						X	X							1		1			1
9	26 43 13	1.5	Product Data - Surge Protection Equipment							X	X							4		1			1
5	26 50 00	1.4	INTERIOR AND EXTERIOR LIGHTING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	26 50 00	1.4	Shop Drawings - Lighting Type by Location						X	X	X			2				2		1			1
5	26 50 00	1.4	Certifications	X						X	X							1		1			1
5	26 50 00	1.4	Product Data - Interior Lighting							X	X			4				4		1			1
5	26 50 00	1.4	Product Data - Exterior Lighting							X	X			4				4		1			1
6	26 50 00	1.4	INTERIOR AND EXTERIOR LIGHTING VOLUME 6: NORTH RIM WTP																			

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	26 50 00	1.4	Shop Drawings - Lighting Type by Location						X	X	X	4					4		1			1
6	26 50 00	1.4	Certifications	X						X	X						1		1			1
6	26 50 00	1.4	Product Data - Interior Lighting							X	X	8					8		1			1
6	26 50 00	1.4	Product Data - Exterior Lighting							X	X	8					8		1			1
5	27 10 00	1.4	STRUCTURED CABLING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	27 10 00	1.4	Shop Drawings - By Floor						X	X	X				1	1	4	1	1			1
5	27 10 00	1.4	Plan - Labeling and Color Coding		X						X						4	4	1			1
5	27 10 00	1.4	Certifications and Test Reports	X						X	X						1		1			1
5	27 10 00	1.4	Product Data - Equipment Racks							X	X			1			4		1			1
5	27 10 00	1.4	Product Data - Cable Management Systems							X	X			1			4		1			1
5	27 10 00	1.4	Product Data - Relay Rack Grounding Strip							X	X			1			4		1			1
5	27 10 00	1.4	Product Data - Telephone Telecommunications Outlet Boxes Terminal Cabinets							X	X			1			4		1			1
5	27 10 00	1.4	Product Data - Cable Runway							X	X			1			4		1			1
5	27 10 00	1.4	Product Data - Equipment Mounting Backboard							X	X			2			2		1			1
5	27 10 00	1.4	Product Data - Patch Pannel							X	X			1			1	4	1			1
5	27 10 00	1.4	Product Data - Fiber Optic Enclosure - Rack Mount							X	X			1			1	4	1			1
5	27 10 00	1.4	Product Data - Fiber Optic Enclosure - Wall Mount							X	X			1			1	4	1			1
5	27 10 00	1.4	Product Data - Termination Block							X	X						2	2	1			1
5	27 10 00	1.4	Product Data - Backbone Cabling System - Fiber Optic Outside Plant Cable							X	X						2	2	1			1
5	27 10 00	1.4	Product Data - Backbone Cabling System - Fiber Optic Inside Plant Cable							X	X						2	2	1			1
5	27 10 00	1.4	Product Data - Backbone Cabling System - Fiber Optic Connections							X	X						2	2	1			1
5	27 10 00	1.4	Product Data - Horizontal Cableing System							X	X						4	4	1			1
5	27 10 00	1.4	Product Data - Telecom Outlets - Jacks and Faceplates							X	X	1					1	1	1			1
5	27 10 00	1.4	Product Data - Patch Cords							X	X						1	1	1			1
5	27 10 00	1.4	Product Data - Device Cables							X	X						1	1	1			1
5	27 10 00	1.4	Product Data - Fiber Optic Microduct							X	X						2	2	1			1
6	27 10 00	1.4	STRUCTURED CABLING VOLUME 6: NORTH RIM WTP																			
6	27 10 00	1.4	Shop Drawings - By Floor						X	X	X				1	1	4	1	1			1
6	27 10 00	1.4	Plan - Labeling and Color Coding		X						X						4	4	1			1
6	27 10 00	1.4	Certifications and Test Reports	X						X	X						1		1			1
6	27 10 00	1.4	Product Data - Equipment Racks							X	X			1			4		1			1
6	27 10 00	1.4	Product Data - Cable Management Systems							X	X			1			4		1			1
6	27 10 00	1.4	Product Data - Relay Rack Grounding Strip							X	X			1			4		1			1
6	27 10 00	1.4	Product Data - Telephone Telecommunications Outlet Boxes Terminal Cabinets							X	X			1			4		1			1
6	27 10 00	1.4	Product Data - Cable Runway							X	X			1			4		1			1
6	27 10 00	1.4	Product Data - Equipment Mounting Backboard							X	X			2			2		1			1
6	27 10 00	1.4	Product Data - Patch Pannel							X	X			1			1	4	1			1
6	27 10 00	1.4	Product Data - Fiber Optic Enclosure - Rack Mount							X	X			1			1	4	1			1
6	27 10 00	1.4	Product Data - Fiber Optic Enclosure - Wall Mount							X	X			1			1	4	1			1
6	27 10 00	1.4	Product Data - Termination Block							X	X						2	2	1			1
6	27 10 00	1.4	Product Data - Backbone Cabling System - Fiber Optic Outside Plant Cable							X	X						2	2	1			1
6	27 10 00	1.4	Product Data - Backbone Cabling System - Fiber Optic Inside Plant Cable							X	X						2	2	1			1
6	27 10 00	1.4	Product Data - Backbone Cabling System - Fiber Optic Connections							X	X						2	2	1			1
6	27 10 00	1.4	Product Data - Horizontal Cableing System							X	X						4	4	1			1
6	27 10 00	1.4	Product Data - Telecom Outlets - Jacks and Faceplates							X	X	1					1	1	1			1
6	27 10 00	1.4	Product Data - Patch Cords							X	X						1	1	1			1
6	27 10 00	1.4	Product Data - Device Cables							X	X						1	1	1			1
6	27 10 00	1.4	Product Data - Fiber Optic Microduct							X	X						2	2	1			1
10	27 10 00	1.4	STRUCTURED CABLING VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	27 10 00	1.4	Plan - Labeling and Color Coding		X						X						4	4	1			1
10	27 10 00	1.4	Certifications and Test Reports	X						X	X						1		1			1
10	27 10 00	1.4	Product Data - Cable Management Systems							X	X			1			4		1			1
10	27 10 00	1.4	Product Data - Relay Rack Grounding Strip							X	X			1			4		1			1
10	27 10 00	1.4	Product Data - Telephone Telecommunications Outlet Boxes Terminal Cabinets							X	X			1			4		1			1
10	27 10 00	1.4	Product Data - Cable Runway							X	X			1			4		1			1
10	27 10 00	1.4	Product Data - Equipment Mounting Backboard							X	X			2			2		1			1
10	27 10 00	1.4	Product Data - Patch Pannel							X	X			1			1	4	1			1
10	27 10 00	1.4	Product Data - Fiber Optic Enclosure - Rack Mount							X	X			1			1	4	1			1
10	27 10 00	1.4	Product Data - Termination Block							X	X						2	2	1			1
10	27 10 00	1.4	Product Data - Horizontal Cableing System							X	X						4	4	1			1
10	27 10 00	1.4	Product Data - Patch Cords							X	X						1	1	1			1
10	27 10 00	1.4	Product Data - Device Cables							X	X						1	1	1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	31 10 00	1.4	SITE CLEARINGCOMMON TO ALL VOLUMES																			
ALL	31 10 00	1.4	Product Data - Filter Fabric and Burlap							X	X		2						1			1
ALL	31 10 00	1.4	Product Data - Drainage Fill							X	X		2						1			1
1	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 1		X						X		1						2			2
2	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 2		X						X		1						2			2
3	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 3		X						X		1						2			2
4	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 4		X						X		1						2			2
5	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 5		X						X		1						2			2
6	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 6		X						X		1						2			2
8	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 8		X						X		1						2			2
9	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 9		X						X		1						2			2
10	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 10		X						X		1						2			2
11	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 11		X						X		1						2			2
12	31 10 00	1.4	SITE CLEARING Geographic Clearence Mobilization Plan Volume 12		X						X		1						2			2
ALL	31 22 00	1.4	SITE GRADING	X				X		X												
ALL	31 22 00	1.4	Testing and Certifications	X									8						4			4
ALL	31 22 00	1.4	Import Material Clearence	X	X						X		8						4			4
ALL	31 23 00	1.4	EARTHWORK	X			X	X	X	X												
ALL	31 23 00	1.4	Structural Fill Compliance Submission	X							X		2	2					1			1
ALL	31 23 00	1.4	Non-Structural Fill Compliance Submission	X							X		2	2					1			1
ALL	31 23 00	1.4	Granular Fill under Building Slabs, Electrical Equipment Pads, Manholes and Handholes	X							X		2	2					1			1
ALL	31 23 00	1.4	Geotex Filter Fabric	X							X		2						1			1
ALL	31 23 00	1.4	Expansive Demolition Grout	X							X		2	2					1			1
ALL	31 23 00	1.4	Testing and Certifications	X									8						1			1
ALL	31 23 33	1.4	UTILITIES - TRENCHING AND BACKFILLING						X	X	X											
ALL	31 23 33	1.5	Shop Drawings						X	X					1							
ALL	31 23 33	1.5	Product Data							X	X		1									
ALL	31 23 33	1.5	Qualification Statements	X																		
ALL	31 23 33	1.5	Excavation and Compaction Plan - North Rim		X								4	4					1			1
ALL	31 23 33	1.5	Excavation and Compaction Plan - Inner-Canyon		X								4	4					1			1
ALL	31 23 33	1.4	Third Party Testing Agency	X			X						1	1					1			1
ALL	31 25 00	1.4	SOIL EROSION AND SEDIMENT CONTROL						X	X												
ALL	31 25 00	1.4	Product Data - Erosion Control Logs, Wattles, Matts			X							2						1			1
ALL	31 25 00	1.4	Product Data - Silt Fence			X							2						1			1
ALL	31 25 00	1.4	Product Data - Synthetic Filter Fabric			X							2						1			1
ALL	31 25 00	1.4	Product Data - Wood/Steel Fence Posts			X							2						1			1
ALL	31 25 00	1.4	Product Data - Wire Fencing Support			X							2						1			1
ALL	31 32 19	1.4	SLOPE ARMORING SYSTEM						X		X											
ALL	31 32 19	1.4	Shop Drawings						X				2	2					1			1
ALL	31 32 19	1.4	Product Data - Slope Stability System			X				X	X		4	4					1			1
ALL	31 68 13	1.4	ROCK FOUNDATION ANCHORS	X		X			X	X												
ALL	31 68 13	1.4	Anchor Report		X						X		8	8					4			4
ALL	31 68 13	1.4	Load Test and Certifications for Anchors - (includes Calculations)	X									4	4					2			2
ALL	31 68 13	1.4	Anchor Plan																			
ALL	31 68 13	1.4	Product Data - Anchors			X				X	X		4	4					2			2
ALL	31 68 13	1.4	Product Data - Anchor Washers and Base Plates			X				X	X		4	4					2			2
ALL	31 68 13	1.4	Product Data - Fast Setting Leveling Grout			X				X	X		1	1					1			1
ALL	31 68 13	1.4	Product Data - Resin Anchor Grout - Polyester / Epoxy Resin			X				X	X		1	1					1			1
ALL	32 12 00	1.6	FLEXIBLE PAVING COMMON TO ALL NORTH RIM VOLUMES																			
ALL	32 12 00	1.6	Plan - Pavement Marking		X				X				4						1			1
ALL	32 12 00	1.6	Product Data - Apshalt Concrere Mix Design Road	X		X	X			X	X		2						1			1
ALL	32 12 00	1.6	Product Data - Apshalt Concrere Mix Design Parking Lot	X		X	X			X	X		2						1			1
ALL	32 12 00	1.6	Delivery Tickets & Third Part Testing Field Reports								X		8						1			1
ALL	32 12 00	1.6	Product Data - Joint Adhesive	X		X	X			X	X		1						1			1
ALL	32 12 00	1.6	Product Data - Crack Sealent	X		X	X			X	X		1						1			1
ALL	32 12 00	1.6	Product Data - Pavement Marking Paint and Glass Beads	X		X	X			X	X		2						1			1
ALL	32 12 00	1.6	Product Data - Admix	X		X	X			X	X		1						1			1
ALL	32 12 00	1.6	Certifications / Qualifications	X		X	X						1						1			1
ALL	32 12 00	1.6	Subgrade and Subbase Compaction Results	X		X	X						8						1			1
ALL	32 13 13	1.6	CONCRETE PAVING COMMON TO ALL VOLUMES																			
ALL	32 13 13	1.6	Product Data - Ridgid Concrere Mix Design Structural Slab / Foundation	X		X	X			X	X		2						1			1
ALL	32 13 13	1.6	Product Data - Ridgid Concrere Mix Design Non-Structural	X		X	X			X	X		2						1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	32 13 13	1.6	Delivery Tickets & Third Part Testing Field Reports								X		8						1			1
ALL	32 13 13	1.6	Product Data - Admix	X		X	X			X	X		1						1			1
ALL	32 13 13	1.6	Product Data - Reinforcing Steel (Rebar)	X		X	X			X	X		1	4					1			1
ALL	32 13 13	1.6	Product Data - Water Stops	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Membrane-Forming Curing Compound	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Temporary Cover Material for Curing Concrete	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Preformed Joint Filler	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Hot Poured Joint Sealing Material	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Paper Subgrade Cover	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Concrete Treatment	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Product Data - Pavement Marking Paint and Glass Beads	X		X	X			X	X		2						1			1
ALL	32 13 13	1.6	Product Data - Forms	X		X	X			X	X		2	2					1			1
ALL	32 13 13	1.6	Certifications / Qualifications	X		X	X						1						1			1
ALL	32 13 13	1.6	Subgrade and Subbase Compaction Results	X		X	X						8						1			1
5	32 13 13	1.6	CONCRETE PAVING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	32 13 13	1.6	Shop Drawing - Rebar, Joints			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	CONCRETE PAVING VOLUME 6: NORTH RIM WTP																			
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 1			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 2			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 3			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 4			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 5			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 6			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 7			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 8			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 9			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 10			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 11			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 12			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 13			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 14			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Shop Drawing - Rebar, Joints - Pour 15			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 13 13	1.6	Strength Break Reports								X			8					1			1
ALL	32 15 40	1.6	CRUSHED STONE SURFACING			X		X		X												
ALL	32 15 40	1.6	Product Data - Stone Source Information - Sieve Analysis	X		X	X			X	X		4						1			1
ALL	32 15 40	1.6	Sample 1QT								X		2						1			1
ALL	32 15 40	1.6	Product Data - Geotext Filter Fabric	X		X	X			X	X		1						1			1
ALL	32 15 40	1.6	Subgrade and Subbase Compaction Results	X		X	X						4						1			1
ALL	32 16 23	1.6	CONCRETE SIDEWALK AND STEPS COMMON TO ALL VOLUMES																			
ALL	32 16 23	1.6	Product Data - Ridgid Concrere Mix Design Structural Slab / Foundation	X		X	X			X	X		2						1			1
ALL	32 16 23	1.6	Product Data - Ridgid Concrere Mix Design Non-Structural	X		X	X			X	X		2						1			1
ALL	32 16 23	1.6	Delivery Tickets & Third Part Testing Field Reports								X		8						1			1
ALL	32 16 23	1.6	Product Data - Admix	X		X	X			X	X		1						1			1
ALL	32 16 23	1.6	Product Data - Reinforcing Steel (Rebar)	X		X	X			X	X		1	4					1			1
ALL	32 16 23	1.6	Product Data - Temporary Cover Material for Curing Concrete	X		X	X			X	X		2	2					1			1
ALL	32 16 23	1.6	Product Data - Preformed Joint Filler	X		X	X			X	X		2	2					1			1
ALL	32 16 23	1.6	Product Data - Hot Poured Joint Sealing Material	X		X	X			X	X		2	2					1			1
ALL	32 16 23	1.6	Product Data - Paper Subgrade Cover	X		X	X			X	X		2	2					1			1
ALL	32 16 23	1.6	Product Data - Concrete Treatment	X		X	X			X	X		2	2					1			1
ALL	32 16 23	1.6	Product Data - Pavement Marking Paint and Glass Beads	X		X	X			X	X		2						1			1
ALL	32 16 23	1.6	Product Data - Forms	X		X	X			X	X		2	2					1			1
ALL	32 16 23	1.6	Certifications / Qualifications	X		X	X						1						1			1
ALL	32 16 23	1.6	Subgrade and Subbase Compaction Results	X		X	X						8						1			1
ALL	32 16 23	1.6	Shop Drawing - Rebar, Joints - Pour 1			X	X		X		X	1	1	4	1	1	1	1	1			1
ALL	32 16 23	1.6	Shop Drawing - Rebar, Joints - Pour 2			X	X		X		X	1	1	4	1	1	1	1	1			1
ALL	32 16 23	1.6	Shop Drawing - Rebar, Joints - Pour 3			X	X		X		X	1	1	4	1	1	1	1	1			1
ALL	32 16 23	1.6	Shop Drawing - Rebar, Joints - Pour 4			X	X		X		X	1	1	4	1	1	1	1	1			1
ALL	32 16 23	1.6	Shop Drawing - Rebar, Joints - Pour 5			X	X		X		X	1	1	4	1	1	1	1	1			1
6	32 31 13	1.4	CHAIN LINK FENCE AND GATES VOLUME 6: NORTH RIM WTP																			
6	32 31 13	1.4	Shop Drawings						X	X	X		4						2	1		1
6	32 31 13	1.4	Sample		X						X		1						1	1		1
6	32 31 13	1.4	Product Data - Fence Mesh Panel							X	X		1						1	1		1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
6	32 31 13	1.4	Product Data - Posts - Line, Corner, Gate, Braces and Rails							X	X		1						1	1		1
6	32 31 13	1.4	Product Data - Hardware							X	X		1						1	1		1
6	32 31 13	1.4	Product Data - Automatic Gate System							X	X		2	1		2			2	1		1
6	32 31 13	1.4	Product Data - Manual Gate System							X	X		2	1					1	1		1
6	32 31 13	1.4	Product Data - Concrete for Posts							X	X		2	1					1	1		1
1	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 1		X						X		2						1			1
2	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 2		X						X		2						1			1
4	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 4		X						X		2						1			1
6	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 6		X						X		2						1			1
8	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 8		X						X		2						1			1
9	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 9		X						X		2						1			1
10	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 10		X						X		2						1			1
11	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 11		X						X		2						1			1
12	32 91 13	1.5	FINISH GRADING USDA NRCS Web Soil Survey Report Volume 12		X						X		2						1			1
1	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 1		X						X								1	2		1
2	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 2		X						X								1	2		1
3	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 3		X						X								1	2		1
4	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 4		X						X								1	2		1
5	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 5		X						X								1	2		1
6	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 6		X						X								1	2		1
8	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 8		X						X								1	2		1
9	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 9		X						X								1	2		1
10	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 10		X						X								1	2		1
11	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 11		X						X								1	2		1
12	32 92 21	1.2	NATIVE GRASS SEEDING Product Data - Mulch Plan - Seeding by Geographic Area Volume 12		X						X								1	2		1
ALL	33 05 15	1.5	PRECAST CONCRETE UTILITY STRUCTURES COMMON TO ALL VOLUMES																			
ALL	33 05 15	1.5	Shop Drawings Typical Structures - Valve and Meter Vaults						X	X	X		2	4					1			1
ALL	33 05 15	1.5	Mix Design per Structure		X						X		4	8					1			1
ALL	33 05 15	1.5	Certifications, Calculations and Test Reports	X						X	X		1	4					1			1
ALL	33 05 15	1.5	Product Data - Access Doors							X	X		2	2					1			1
ALL	33 05 15	1.5	Product Data - Manhold Systems - Rings, Frames, Covers							X	X		4	1					1			1
ALL	33 05 15	1.5	Product Data - Premolded Joint Sealent							X	X		1	1					1			1
ALL	33 05 15	1.5	Product Data - Elastomeric Joint Seal							X	X		1	1					1			1
ALL	33 05 15	1.5	Product Data - External Joint Wrap							X	X		1	1					1			1
ALL	33 05 15	1.5	Product Data - Emulsified Fibrated Asphalt Compound							X	X		1	1					1			1
ALL	33 05 15	1.5	Product Data - Verticle Wall Surface Coating							X	X		1	1					1			1
2	33 05 15	1.5	PRECAST CONCRETE UTILITY STRUCTURES VOLUME 2: NR BOREHOLE TO RSPH																			
2	33 05 15	1.5	Shop Drawings - Borehole Vault						X	X	X		4	2					2			2
8	33 05 15	1.5	PRECAST CONCRETE UTILITY STRUCTURES VOLUME 8: WW COLLECTION AND LS																			
8	33 05 15	1.5	Shop Drawings - Structure by Location on Drawings Cross Referenced to Pre-Cast Product Submittal						X	X	X		16						2			2
ALL	33 05 24	1.3	STEEL UTILITY PIPE - COMMON TO ALL VOLUMES	X		X			X	X												
ALL	33 05 24	1.3	Certifications, Statements of Qualification, Calculations	X						X	X		4						1			1
ALL	33 05 24	1.3	Shop Drawing - Pipe Layout						X				4						1			1
ALL	33 05 24	1.3	Fabrication Report		X				X				4						1			1
ALL	33 05 24	1.3	Shop Drawing - Joint Details		X				X				4						1			1
ALL	33 05 24	1.3	Welding Data (Shop and Field)		X				X				2						1			1
ALL	33 05 24	1.3	Product Data - Pipe							X	X		2						1			1
ALL	33 05 24	1.3	Product Data - Fittings (Couplings, Flanges)							X	X		2						1			1
ALL	33 05 24	1.3	Product Data - Fabricate Outlets and Finishes							X	X		2						1			1
ALL	33 05 24	1.3	Product Data - Pipe Handling Equipment to Include Supports for Storage							X	X		2						1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	33 05 24	1.3	Product Data - Wall Sleeves							X	X		2						1			1
ALL	33 05 24	1.3	Product Data - Flanged Joints							X	X		2						1			1
ALL	33 05 24	1.3	Product Data - Coating and Linings - Include Color Chart							X	X		2						1			1
1	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 1		X						X		4						1			1
2	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 2		X						X		4						1			1
3	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 3		X						X		4						1			1
4	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 4		X						X		4						1			1
5	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 5		X						X		4						1			1
6	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 6		X						X		4						1			1
8	33 05 24	1.3	STEEL UTILITY PIPE - Plan - Field Testing Plan Volume 8		X						X		4						1			1
3	33 05 28	1.4	PIPE - SPOOLABLE COMPOSITE VOLUME 3: NR TANK FARM TO SUPAI TUNNEL																			
3	33 05 28	1.4	Shop Drawings						X				1									
3	33 05 28	1.4	Certifications	X																		
3	33 05 28	1.4	Testing Data		X						X		4						1			1
3	33 05 28	1.4	Product Data - Pipe							X	X		4						1			1
3	33 05 28	1.4	Product Data - Fittings							X	X		4						1			1
ALL	33 05 31.11	1.5	POLYVINYL CHLORIDE GRAVITY SEWER PIPE - COMMON TO ALL VOLUMES						X	X												
ALL	33 05 31.11	1.5	Certifications, Statements of Qualification, Calculations	X						X	X		4						1			1
ALL	33 05 31.11	1.5	Shop Drawing - Pipe Layout						X				4						1			1
ALL	33 05 31.11	1.5	Fabrication Report		X				X				4						1			1
ALL	33 05 31.11	1.5	Product Data - Pipe							X	X		2						1			1
ALL	33 05 31.11	1.5	Product Data - Fittings (Couplings, Flanges)							X	X		2						1			1
ALL	33 05 31.11	1.5	Product Data - Wall Sleeves							X	X		2						1			1
ALL	33 05 31.11	1.5	Field Reports - Testing Results							X	X		16						1			1
ALL	33 05 31.11	1.5	Plan - Field Testing Plan																1			1
5	33 05 31.11	1.5	POLYVINYL CHLORIDE GRAVITY SEWER PIPE Plan - Field Testing Plan Volume 5		X						X		4						1			1
6	33 05 31.11	1.5	POLYVINYL CHLORIDE GRAVITY SEWER PIPE Plan - Field Testing Plan Volume 6		X						X		4						1			1
8	33 05 31.11	1.5	POLYVINYL CHLORIDE GRAVITY SEWER PIPE Plan - Field Testing Plan Volume 8		X						X		4						1			1
ALL	33 05 33.23	1.5	PIPE - HDPE		X	X			X	X												
ALL	33 05 33.23	1.5	Certifications, Statements of Qualification, Calculations	X						X	X		4						1			1
ALL	33 05 33.23	1.5	Shop Drawing - Pipe Layout						X				4						1			1
ALL	33 05 33.23	1.5	Fabrication Report		X				X				2						1			1
ALL	33 05 33.23	1.5	Product Data - Pipe							X	X		2						1			1
ALL	33 05 33.23	1.5	Product Data - Fittings: Molded Fused							X	X		2						1			1
ALL	33 05 33.23	1.5	Product Data - Fittings: Adaptor Fittings							X	X		2						1			1
ALL	33 05 33.23	1.5	Product Data - Fittings: Backup Flanges (include nuts, bolts, and gasgets)							X	X		2						1			1
ALL	33 05 33.23	1.5	Product Data - Wall Sleeves							X	X		2						1			1
ALL	33 05 33.23	1.5	Product Data - Butt Fusion Equipment							X	X		2						1			1
ALL	33 05 33.23	1.5	Field Reports - Testing Results							X	X		16						1			1
ALL	33 05 33.23	1.5	Plan - Field Testing Plan		X								10						1			1
ALL	33 05 33.23	1.4	POLYETHYLENE PRESSURE PIPE AND TUBING (AWWA C901 AND AWWA C906)			X			X	X												
ALL	33 05 33.23	1.4	Certifications, Statements of Qualification	X						X	X		4						1			1
ALL	33 05 33.23	1.4	Shop Drawing - Pipe Layout						X				4						1			1
ALL	33 05 33.23	1.4	Fabrication Report		X				X				2						1			1
ALL	33 05 33.23	1.4	Product Data - Pipe							X	X		2						1			1
ALL	33 05 33.23	1.4	Product Data - Fittings: Adaptor Fittings							X	X		2						1			1
ALL	33 05 33.23	1.4	Product Data - Fittings: Backup Flanges (include nuts, bolts, and gasgets)							X	X		2						1			1
ALL	33 05 33.23	1.4	Product Data - Wall Sleeves							X	X		2						1			1
ALL	33 05 33.23	1.4	Field Reports - Testing Results							X	X		16						1			1
ALL	33 05 33.23	1.4	Plan - Field Testing Plan		X								10						1			1
8	33 05 61	1.3	CONCRETE MANHOLES VOLUME 8: WW COLLECTION AND LS																			
8	33 05 61	1.3	Shop Drawings						X	X	X		4						1			1
8	33 05 61	1.3	Boyance Calculations		X						X		1						1			1
8	33 05 61	1.3	Product Data - Sanitary Sewer Manholes and Appurtenances							X	X		2						1			1
8	33 05 61	1.3	Product Data - Drain Manholes and Appurtenances							X	X		2						1			1
8	33 05 61	1.3	Product Data - Sealents / Compounds							X	X		1						1			1
1	33 11 13	1.3	WATER MAIN CONSTRUCTION Disinfection and Testing Plan Volume 1		X						X		2						1			1
2	33 11 13	1.3	WATER MAIN CONSTRUCTION Disinfection and Testing Plan Volume 2		X						X		2						1			1
3	33 11 13	1.3	WATER MAIN CONSTRUCTION Disinfection and Testing Plan Volume 3		X						X		2						1			1
4	33 11 13	1.3	WATER MAIN CONSTRUCTION Disinfection and Testing Plan Volume 4		X						X		2						1			1
ALL	33 12 19	1.3	FIRE HYDRANT		X	X			X	X												
ALL	33 12 19	1.3	Shop Drawing - Hydrant by Location by Product Data by Color Finish					X	X				2						1		2	1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME										
				INFORMATIONAL				ACTION														
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	33 12 19	1.3	Product Data - Fire Hydrant Assembly			X							2						1		2	1
ALL	33 12 19	1.3	Product Data - Control Valve			X							1						1			1
5	33 16 33	1.3	HYDROPNEUMATIC TANKS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	33 16 33	1.3	Shop Drawings - Hydromatic Tank						X	X	X		2						1			1
5	33 16 33	1.3	Product Data - Hydromatic Tank							X	X		2						1			1
5	33 16 33	1.3	Product Data - Instraments and Valves							X	X		2						1			1
5	33 16 33	1.3	Product Data - Automatic Level Control							X	X		2						1			1
6	33 16 33	1.3	HYDROPNEUMATIC TANKS VOLUME 6: NORTH RIM WTP																			
6	33 16 33	1.3	Shop Drawings - Hydromatic Tank						X	X	X		2						1			1
6	33 16 33	1.3	Product Data - Hydromatic Tank							X	X		2						1			1
6	33 16 33	1.3	Product Data - Instraments and Valves							X	X		2						1			1
6	33 16 33	1.3	Product Data - Automatic Level Control							X	X		2						1			1
8	33 31 11	1.3	GRAVITY SEWER PIPELINE AND MANHOLE CONSTRUCTION		X	X				X												
8	33 31 11	1.3	VOLUME 8: WW COLLECTION AND LS																			
8	33 31 11	1.3	Testing Plan		X				X	X	X		1						1			1
8	33 31 11	1.3	Testing Reports		X				X	X	X		1						1			1
9	33 52 10	1.3	SERVICE PIPING - FUEL SYSTEMS		X				X	X												
9	33 52 10	1.3	VOLUME 9: POWER DISTROBUTION CENTER																			
9	33 52 10	1.3	Testing Plan		X					X	X		1			1			1			1
9	33 52 10	1.3	Testing Report							X	X		1			1			1			1
9	33 52 10	1.3	Shop Drawings - Fuel Storage Tank and Appurtenances						X	X	X		2			2			1			1
9	33 52 10	1.3	Product Data - Fuel Storage Tank and Appurtenances		X					X	X		2			2			1			1
9	33 56 13	1.3	ABOVEGROUND FUEL STORAGE TANKS		X	X			X	X												
9	33 56 13	1.3	VOLUME 9: POWER DISTROBUTION CENTER																			
9	33 56 13	1.3	Testing Plan		X					X	X		1			1			1			1
9	33 56 13	1.3	Testing Report							X	X		1			1			1			1
9	33 56 13	1.3	Shop Drawings - Fuel Storage Piping and Appurtenances						X	X	X		2			2			1			1
9	33 56 13	1.3	Product Data - Fuel Storage Piping and Appurtenances		X					X	X		2			2			1			1
9	33 56 16	1.4	FUEL DAY TANK SYSTEMS		X	X			X													
9	33 56 16	1.4	VOLUME 9: POWER DISTROBUTION CENTER																			
9	33 56 16	1.4	Testing Certification							X	X		1			1			1			1
9	33 56 16	1.4	Testing Report							X	X		1			1			1			1
9	33 56 16	1.4	Shop Drawings - Fuel Storage Piping and Appurtenances						X	X	X		2			2			1			1
9	33 56 16	1.4	Product Data - Fuel Storage Piping and Appurtenances		X					X	X		2			2			1			1
10	33 71 16.23	1.6	TUBULAR STEEL POLES			X			X	X												
10	33 71 16.23	1.4	Calculations		X					X	X		3	3					1			1
10	33 71 16.23	1.4	Testing Report							X	X		1	1					1			1
10	33 71 16.23	1.4	Shop Drawings						X	X	X		3	3					1			1
10	33 71 16.23	1.4	Product Data		X					X	X		2	2					1			1
10	33 71 23.17	1.4	MICROPILE FOUNDATIONS VOLUME 10: OVERHEAD POWER NR TO RSPH																			
10	33 71 23.17	1.4	Calculations		X					X	X		2	2					1			1
10	33 71 23.17	1.4	Testing Plan and Report (Mill Test Report)							X	X		2	2					1			1
10	33 71 23.17	1.4	Shop Drawings						X	X	X		4	4					1			1
10	33 71 23.17	1.4	Product Data - Micorpile		X					X	X		2	2					1			1
10	33 71 23.17	1.4	Product Data - Steel Pile Cap		X					X	X		2	2					1			1
5	40 05 00	1.4	PIPE AND PIPE FITTINGS - BASIC REQUIREMENTS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 05 00	1.4	Testing and Disinfection Plan							X	X		2						1			1
5	40 05 00	1.4	Shop Drawings - Process Piping System						X	X	X		4						1			1
5	40 05 00	1.4	Product Data - Process Piping System		X					X	X		2						1			1
6	40 05 00	1.4	PIPE AND PIPE FITTINGS - BASIC REQUIREMENTS VOLUME 6: NORTH RIM WTP																			
6	40 05 00	1.4	Testing and Disinfection Plan							X	X		2						1			1
6	40 05 00	1.4	Shop Drawings - Process Piping System						X	X	X		4						1			1
6	40 05 00	1.4	Product Data - Process Piping System		X					X	X		2						1			1
ALL	40 05 07	1.4	PIPE SUPPORT SYSTEMS COMMON TO ALL VOLUMES																			
ALL	40 05 07	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 07	1.4	Product Data - Hot Dipped Galvanized Fittings		X					X	X		2						1			1
ALL	40 05 07	1.4	Product Data - Dissimilar Fittings		X					X	X		2						1			1
ALL	40 05 07	1.4	Product Data - FRP: Channels, Pipe Clamps, Clevis Hangers, Beam Clamps, Column Base		X					X	X		2						1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
ALL	40 05 07	1.4	Product Data - Concrete Inserts for Hanger Rods, Hanger Rods, MSS SP-58, Trapeze Hangers, Verticle Pipe Supports, Lateral Retraints, Expanding Pipe Supports, Pipe Saddles, Riser Stanchions, Base Plates, Wall Brackets, Pipe Anchors, Seismic Restraints, Pipe Guilds		X					X	X		2						1			1
5	40 05 19	1.4	PIPE - DUCTILE IRON PROCESS PIPE VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 05 19	1.4	Shop Drawings						X	X	X		2						1			1
5	40 05 19	1.4	Product Data - Flanged Adaptors, Restrained Flanged Adaptors, Compression Sleeve Couplings, Insulating Couplings, Reducing Couplings, Transition Coupling, Ceramic Epoxy Lining, Polyethylene Encased Tape, Restrained Joints, Expansion Joints, Dismantling Joints, Ductile Iron Pipe, Fittings/Flanges, Nuts/Bolts, Gaskets, Mechanical Couplings		X					X	X		2						1			1
ALL	40 05 19	1.4	PIPE - DUCTILE IRON PROCESS PIPE COMMON TO ALL VOLUMES																			
ALL	40 05 19	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 19	1.4	Product Data - Flanged Adaptors, Restrained Flanged Adaptors, Compression Sleeve Couplings, Insulating Couplings, Reducing Couplings, Transition Coupling, Ceramic Epoxy Lining, Polyethylene Encased Tape, Restrained Joints, Expansion Joints, Dismantling Joints, Ductile Iron Pipe, Fittings/Flanges, Nuts/Bolts, Gaskets, Mechanical Couplings		X					X	X		2						1			1
ALL	40 05 23	1.4	PIPE -STAINLESS STEEL PIPE AND TUBING VOLUME 5: ROARING SPRINGS PUMP STATION																			
ALL	40 05 23	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 23	1.4	Product Data - Tubing, Pipe, Fittings, Flanges, Nuts/Bolts/Washers, Expansion Joints, Flex Metal Hose, Gaskets		X					X	X		2						1			1
5	40 05 24	1.3	PIPE - STEEL VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 05 24	1.3	Shop Drawings						X	X	X		2						1			1
5	40 05 24	1.3	Product Data - Flanged Adaptors, Braided Flex Connectors, Mechanical Couplings/Fittings, Steel Pipe Fabricated and Milled with Fittings, Flanges, Nuts/Bolts, Gaskets, Insulating Components, Dieletric Unions		X					X	X		2						1			1
ALL	40 05 24	1.3	PIPE - STEEL COMMON to ALL VOLUMES																			
ALL	40 05 24	1.3	Shop Drawings						X	X	X		2						1			1
ALL	40 05 24	1.3	Product Data - Flanged Adaptors, Braided Flex Connectors, Mechanical Couplings/Fittings, Steel Pipe Fabricated and Milled with Fittings, Flanges, Nuts/Bolts, Gaskets, Insulating Components, Dieletric Unions		X					X	X		2						1			1
ALL	40 05 24	1.3	Welder Certifications	X																		
ALL	40 05 24	1.3	Coating Manufacturer's Qualifications	X																		
ALL	40 05 25	1.4	PIPE - UNDERGROUND, PREFABRICATED, INSULATED AND JACKETED COMMON TO ALL VOLUMES																			
ALL	40 05 25	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 25	1.4	Product Data - Insulation, Jacket	X	X					X	X		2						1			1
ALL	40 05 31	1.4	PIPE - PLASTIC VOLUME COMMON TO ALL VOLUMES																			
ALL	40 05 31	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 31	1.4	Product Data - PVC/CPVC Pipe and Fittings, Tubing, Unions, Flanges, Flex Couplings, Seals/O-Rings		X					X	X		2						1			1
ALL	40 05 51	1.4	COMMON REQUIREMENTS FOR PROCESS AND UTILITY VALVES COMMON TO ALL VOLUMES																			
ALL	40 05 51	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 51	1.4	Product Data - Valves		X					X	X		2						1			1
ALL	40 05 52	1.4	MISCELLANEOUS VALVES				X		X													
ALL	40 05 52	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 05 52	1.4	Product Data		X					X	X		2						1			1
1	40 05 61	1.4	GATE VALVES Volume 1							X	X		2						1			1
2	40 05 61	1.4	GATE VALVES Volume 2							X	X		2						1			1
3	40 05 61	1.4	GATE VALVES Volume 3							X	X		2						1			1
4	40 05 61	1.4	GATE VALVES Volume 4							X	X		2						1			1
5	40 05 61	1.4	GATE VALVES Volume 5							X	X		2						1			1
6	40 05 61	1.4	GATE VALVES Volume 6							X	X		2						1			1
1	40 05 63	1.4	BALL VALVES Volume 1							X	X		2						1			1
2	40 05 63	1.4	BALL VALVES Volume 2							X	X		2						1			1
3	40 05 63	1.4	BALL VALVES Volume 3							X	X		2						1			1
4	40 05 63	1.4	BALL VALVES Volume 4							X	X		2						1			1
5	40 05 63	1.4	BALL VALVES Volume 5							X	X		2						1			1
6	40 05 63	1.4	BALL VALVES Volume 6							X	X		2						1			1
ALL	40 05 63	1.4	Shop Drawings						X				1									
1	40 05 64	1.4	BUTTERFLY VALVES Volume 1							X	X		2						1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	REQUIREMENTS: INFORMATIONAL: CERTIFICATIONS OR LABORATORY TESTS	REQUIREMENTS: INFORMATIONAL: REPORTS OR CALCULATIONS OR PLAN	REQUIREMENTS: INFORMATIONAL: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: INFORMATIONAL: OTHER	REQUIREMENTS: ACTION: SAMPLES	REQUIREMENTS: ACTION: SHOP DRAWINGS	REQUIREMENTS: ACTION: MANUFACTURER DATA AND INSTRUCTIONS	REQUIREMENTS: ACTION: OTHER	ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME: ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
2	40 05 64	1.4	BUTTERFLY VALVES Volume 2							X	X		2						1			1
3	40 05 64	1.4	BUTTERFLY VALVES Volume 3							X	X		2						1			1
4	40 05 64	1.4	BUTTERFLY VALVES Volume 4							X	X		2						1			1
5	40 05 64	1.4	BUTTERFLY VALVES Volume 5							X	X		2						1			1
6	40 05 64	1.4	BUTTERFLY VALVES Volume 6							X	X		2						1			1
1	40 05 65	1.3	GLOBE VALVES Volume 1							X	X		2						1			1
2	40 05 65	1.3	GLOBE VALVES Volume 2							X	X		2						1			1
3	40 05 65	1.3	GLOBE VALVES Volume 3							X	X		2						1			1
4	40 05 65	1.3	GLOBE VALVES Volume 4							X	X		2						1			1
5	40 05 65	1.3	GLOBE VALVES Volume 5							X	X		2						1			1
6	40 05 65	1.3	GLOBE VALVES Volume 6							X	X		2						1			1
1	40 05 66	1.4	CHECK VALVES Volume 1							X	X		2						1			1
2	40 05 66	1.4	CHECK VALVES Volume 2							X	X		2						1			1
3	40 05 66	1.4	CHECK VALVES Volume 3							X	X		2						1			1
4	40 05 66	1.4	CHECK VALVES Volume 4							X	X		2						1			1
5	40 05 66	1.4	CHECK VALVES Volume 5							X	X		2						1			1
6	40 05 66	1.4	CHECK VALVES Volume 6							X	X		2						1			1
ALL	40 41 13	1.4	HEAT TRACING CABLE	X			X		X		X											
ALL	40 41 13	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 41 13	1.4	Product Data		X					X	X		2						1			1
ALL	40 42 00	1.4	PIPE, DUCT AND EQUIPMENT INSTALLATION	X			X		X	X												
ALL	40 42 00	1.4	Shop Drawings						X	X	X		2						1			1
ALL	40 42 00	1.4	Product Data		X					X	X		2						1			1
5	40 61 13	1.4	PROCESS CONTROL SYSTEM GENERAL REQUIREMENTS VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 61 13	1.4	Shop Drawings						X	X	X		2						1			1
5	40 61 13	1.4	Product Data		X					X	X		2						1			1
6	40 61 13	1.4	PROCESS CONTROL SYSTEM GENERAL REQUIREMENTS VOLUME 6: NORTH RIM WTP																			
6	40 61 13	1.4	Shop Drawings						X	X	X		2						1			1
6	40 61 13	1.4	Product Data		X					X	X		2						1			1
5	40 61 21	1.4	PROCESS CONTROL SYSTEM TESTING VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 61 21	1.4	Shop Drawings						X	X	X		2						1			1
5	40 61 21	1.4	Product Data		X					X	X		2						1			1
6	40 61 21	1.4	PROCESS CONTROL SYSTEM TESTING VOLUME 6: NORTH RIM WTP																			
6	40 61 21	1.4	Shop Drawings						X	X	X		2						1			1
6	40 61 21	1.4	Product Data		X					X	X		2						1			1
5	40 61 43	1.4	SURGE PROTECTION DEVICES FOR INSTRUMENTATION AND CONTROL EQUIPMENT VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 61 43	1.4	Shop Drawings						X	X	X		2						1			1
5	40 61 43	1.4	Product Data		X					X	X		2						1			1
6	40 61 43	1.4	SURGE PROTECTION DEVICES FOR INSTRUMENTATION AND CONTROL EQUIPMENT VOLUME 6: NORTH RIM WTP																			
6	40 61 43	1.4	Shop Drawings						X	X	X		2						1			1
6	40 61 43	1.4	Product Data		X					X	X		2						1			1
5	40 61 93	1.4	PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 61 93	1.4	Shop Drawings						X	X	X		2						1			1
5	40 61 93	1.4	Product Data		X					X	X		2						1			1
6	40 61 93	1.4	PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST VOLUME 6: NORTH RIM WTP																			
6	40 61 93	1.4	Shop Drawings						X	X	X		2						1			1
6	40 61 93	1.4	Product Data		X					X	X		2						1			1
5	40 62 05	1.4	COMPUTER SYSTEM HARDWARE AND ANCILLARIES VOLUME 5: ROARING SPRINGS PUMP STATION																			
5	40 62 05	1.4	Shop Drawings						X	X	X		2						1			1
5	40 62 05	1.4	Product Data		X					X	X		2						1			1
6	40 62 05	1.4	COMPUTER SYSTEM HARDWARE AND ANCILLARIES VOLUME 6: NORTH RIM WTP																			
6	40 62 05	1.4	Shop Drawings						X	X	X		2						1			1
6	40 62 05	1.4	Product Data		X					X	X		2						1			1
5	40 62 16	1.4	COMPUTER NETWORK AND HUMAN MACHINE INTERFACE HARDWARE VOLUME 5: ROARING SPRINGS PUMP STATION																			

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

ACTION SUBMITTAL REVIEW ESTIMATE

ARCHITECT / ENGINEER REVIEW TIME

SUBMITTAL				REQUIREMENTS																						
				INFORMATIONAL				ACTION																		
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER		ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
5	40 62 16	1.4	Shop Drawings						X	X	X			2						1			1
5	40 62 16	1.4	Product Data		X					X	X			2						1			1
6	40 62 16	1.4	COMPUTER NETWORK AND HUMAN MACHINE INTERFACE HARDWARE VOLUME 6: NORTH RIM WTP																					
6	40 62 16	1.4	Shop Drawings						X	X	X			2						1			1
6	40 62 16	1.4	Product Data		X					X	X			2						1			1
5	40 63 43	1.4	PROGRAMMABLE LOGIC CONTROLLERS VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 63 43	1.4	Shop Drawings						X	X	X			2						1			1
5	40 63 43	1.4	Product Data - PLC		X					X	X			2						1			1
6	40 63 43	1.4	PROGRAMMABLE LOGIC CONTROLLERS VOLUME 6: NORTH RIM WTP																					
6	40 63 43	1.4	Shop Drawings						X	X	X			2						1			1
6	40 63 43	1.4	Product Data - PLC		X					X	X			2						1			1
5	40 66 05	1.4	NETWORK EQUIPMENTVOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 66 05	1.4	Shop Drawings						X	X	X			2						1			1
5	40 66 05	1.4	Product Data		X					X	X			2						1			1
6	40 66 05	1.4	NETWORK EQUIPMENTVOLUME 6: NORTH RIM WTP																					
6	40 66 05	1.4	Shop Drawings						X	X	X			2						1			1
6	40 66 05	1.4	Product Data		X					X	X			2						1			1
5	40 66 40	1.4	TELEMETRY SYSTEMSVOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 66 40	1.4	Shop Drawings						X	X	X			2						1			1
5	40 66 40	1.4	Product Data		X					X	X			2						1			1
6	40 66 40	1.4	TELEMETRY SYSTEMSVOLUME 6: NORTH RIM WTP																					
6	40 66 40	1.4	Shop Drawings						X	X	X			2						1			1
6	40 66 40	1.4	Product Data		X					X	X			2						1			1
5	40 67 00	1.4	CONTROL SYSTEM EQUIPMENT PANELS AND RACKS VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 67 00	1.4	Shop Drawings						X	X	X			2						1			1
5	40 67 00	1.4	Product Data		X					X	X			2						1			1
6	40 67 00	1.4	CONTROL SYSTEM EQUIPMENT PANELS AND RACKS VOLUME 6: NORTH RIM WTP																					
6	40 67 00	1.4	Shop Drawings						X	X	X			2						1			1
6	40 67 00	1.4	Product Data		X					X	X			2						1			1
5	40 67 63	1.4	UNINTERRUPTABLE POWER SUPPLY VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 67 63	1.4	Shop Drawings						X	X	X			2						1			1
5	40 67 63	1.4	Product Data		X					X	X			2						1			1
6	40 67 63	1.4	UNINTERRUPTABLE POWER SUPPLY VOLUME 6: NORTH RIM WTP																					
6	40 67 63	1.4	Shop Drawings						X	X	X			2						1			1
6	40 67 63	1.4	Product Data		X					X	X			2						1			1
5	40 71 00	1.4	FLOW INSTRUMENTATION VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 71 00	1.4	Shop Drawings						X	X	X			2						1			1
5	40 71 00	1.4	Product Data		X					X	X			2						1			1
6	40 71 00	1.4	FLOW INSTRUMENTATIONVOLUME 6: NORTH RIM WTP																					
6	40 71 00	1.4	Shop Drawings						X	X	X			2						1			1
6	40 71 00	1.4	Product Data		X					X	X			2						1			1
6	40 72 00	1.4	LEVEL INSTRUMENTATIONVOLUME 6: NORTH RIM WTP																					
6	40 72 00	1.4	Shop Drawings						X	X	X			2						1			1
6	40 72 00	1.4	Product Data		X					X	X			2						1			1
5	40 73 00	1.4	PRESSURE INSTRUMENTATION VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 73 00	1.4	Shop Drawings						X	X	X			2						1			1
5	40 73 00	1.4	Product Data		X					X	X			2						1			1
6	40 73 00	1.4	PRESSURE INSTRUMENTATION VOLUME 6: NORTH RIM WTP																					
6	40 73 00	1.4	Shop Drawings						X	X	X			2						1			1
6	40 73 00	1.4	Product Data		X					X	X			2						1			1
5	40 74 00	1.4	TEMPERATURE INSTRUMENTATION VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 74 00	1.4	Shop Drawings						X	X	X			2						1			1
5	40 74 00	1.4	Product Data		X					X	X			2						1			1
6	40 74 00	1.4	TEMPERATURE INSTRUMENTATION VOLUME 6: NORTH RIM WTP																					
6	40 74 00	1.4	Shop Drawings						X	X	X			2						1			1
6	40 74 00	1.4	Product Data		X					X	X			2						1			1
5	40 75 00	1.4	PROCESS LIQUID ANALYTICAL MEASUREMENT VOLUME 5: ROARING SPRINGS PUMP STATION																					
5	40 75 00	1.4	Shop Drawings						X	X	X			2						1			1

SUBMITTAL LIST

Project Title: Rehabilitate North Rim Infrastructure — March 2026

	SUBMITTAL			REQUIREMENTS								ACTION SUBMITTAL REVIEW ESTIMATE — ARCHITECT / ENGINEER REVIEW TIME											
				INFORMATIONAL				ACTION															
VOLUME	SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER		ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	PROCESS ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	CONTROLS ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT	FIRE PROTECTION ENGINEER	HDR LEAD CONSTRUCTION
5	40 75 00	1.4	Product Data		X					X	X			2						1			1
6	40 75 00	1.4	PROCESS LIQUID ANALYTICAL MEASUREMENT VOLUME 6: NORTH RIM WTP																				
6	40 75 00	1.4	Shop Drawings						X	X	X			2						1			1
6	40 75 00	1.4	Product Data		X					X	X			2						1			1
5	40 75 93	1.4	PROCESS LIQUID SAMPLING AND ACCESSORIES VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	40 75 93	1.4	Shop Drawings						X	X	X			2						1			1
5	40 75 93	1.4	Product Data		X					X	X			2						1			1
6	40 75 93	1.4	PROCESS LIQUID SAMPLING AND ACCESSORIES VOLUME 6: NORTH RIM WTP																				
6	40 75 93	1.4	Shop Drawings						X	X	X			2						1			1
6	40 75 93	1.4	Product Data		X					X	X			2						1			1
9	40 76 00	1.4	PROCESS GAS ANALYTICAL MEASUREMENT VOLUME 9: POWER DISTROBUTION CENTER																				
9	40 76 00	1.4	Shop Drawings						X	X	X			2						1			1
9	40 76 00	1.4	Product Data		X					X	X			2						1			1
9	40 78 00	1.4	PANEL MOUNTED INSTRUMENTS VOLUME 9: POWER DISTROBUTION CENTER																				
9	40 78 00	1.4	Shop Drawings						X	X	X			2						1			1
9	40 78 00	1.4	Product Data		X					X	X			2						1			1
5	41 22 23	1.4	HOISTS VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	41 22 23	1.4	Shop Drawings						X	X	X			2						1			1
5	41 22 23	1.4	Product Data		X					X	X			2						1			1
6	41 22 23	1.4	HOISTS VOLUME 6: NORTH RIM WTP																				
6	41 22 23	1.4	Shop Drawings						X	X	X			2						1			1
6	41 22 23	1.4	Product Data		X					X	X			2						1			1
6	43 41 15	1.4	FIELD ERECTED WELDED STEEL TANKS VOLUME 6: NORTH RIM WTP																				
6	43 41 15	1.4	Shop Drawings						X	X	X			2						1			1
6	43 41 15	1.4	Product Data		X					X	X			2						1			1
6	43 41 16	1.4	CARBON STEEL TANKS VOLUME 6: NORTH RIM WTP																				
6	43 41 16	1.4	Shop Drawings						X	X	X			2						1			1
6	43 41 16	1.4	Product Data		X					X	X			2						1			1
6	43 41 45	1.4	POLYETHYLENE CHEMICAL TANKS VOLUME 6: NORTH RIM WTP																				
6	43 41 45	1.4	Shop Drawings						X	X	X			2						1			1
6	43 41 45	1.4	Product Data		X					X	X			2						1			1
5	43 41 45	1.4	FIBERGLASS REINFORCED PLASTIC TANKS VOLUME 5: ROARING SPRINGS PUMP STATION																				
5	43 41 45	1.4	Shop Drawings						X	X	X			2						1			1
5	43 41 45	1.4	Product Data	x	X					X	X			2						1			1
6	46 61 33	1.4	UF MEMBRANE EQUIPMENT VOLUME 6: NORTH RIM WTP																				
6	46 61 33	1.4	Shop Drawings						X	X	X			8	2	2	2	2	2	2			2
6	46 61 33	1.4	Product Data		X					X	X			8	2	2	2	2	2	2			2
6	46 61 73	1.4	AUTOMATIC SELF-CLEANING FILTER VOLUME 6: NORTH RIM WTP																				
6	46 61 73	1.4	Shop Drawings						X	X	X			8	2	2	2	2	2	2			2
6	46 61 73	1.4	Product Data		X					X	X			8	2	2	2	2	2	2			2
												Total:	1,255	1,816	1,654	488	880	2,812	645	3,797	361	233	3,826

This page intentionally left blank.

SECTION 01 35 05 ENVIRONMENTAL PROTECTION AND SPECIAL CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

A. This section includes:

1. Wildlife protection.
2. Other environmental protection measures.

1.2 DEFINITIONS

A. Contracting Officer: Includes the Government Contracting Officer (CO).

B. Sensitive Resource Areas: Including, but not limited to, archaeological, paleontological sites, threatened and endangered species areas, and special-status plant habitats.

1.3 SUBMITTALS

A. Spill Prevention and Response Plan: After contract award and before the Pre-Construction conference, submit for informational review, a Spill Prevention and Response Plan. The CO will review the proposed Plan. If the plan requires any revisions or corrections, the Contractor shall resubmit the Plan within 10 days. No progress payments will be made until the Plan is accepted.

B. Rehabilitation Plan: Prior to the start of construction, Contractor shall submit a rehabilitation plan for all disturbed areas. The CO will review the proposed Plan. If the plan requires any revisions or corrections, the Contractor shall resubmit the Plan within 10 days. No progress payments will be made until the Plan is accepted.

PART 2 - PRODUCTS

2.1 SPILL PREVENTION AND RESPONSE PLAN

A. The Spill Prevention and Response Plan (Plan) shall be written to comply with Occupational Safety and Health Administration (OSHA), Environmental Protection Agency (EPA), other federal and local requirements, and project requirements.

PART 3 - EXECUTION

3.1 RESOURCE MONITORING AND PROTECTION

A. Resource Monitors for Sensitive Resources: When work is to occur in or near sensitive resource areas, the Contractor shall not proceed with work in those areas unless the required resource monitor(s) are present, unless otherwise authorized by the Contracting Officer. The Contractor is

to notify the Contracting Officer at least two weeks in advance of proposed work that will occur in or near sensitive resource areas.

B. Resource Protection: Provide and maintain a minimum 16-foot (5-meter) buffer around all identified sensitive resource areas unless another buffer distance is established by GRCA staff. Fencing, flagging, or taping of site boundaries shall correspond to the approved buffer. The NPS will mark avoidance boundaries in the field by flagging, fencing, or taping for Contractor avoidance.

3.2 UNANTICIPATED DISCOVERY PROTOCOL

A. General Requirement: If previously undocumented sensitive resources are encountered at any time during construction, including but not limited to cultural resources, paleontological resources, biological resources (including nests or roosts), or other protected environmental resources, the Contractor shall immediately halt work in the vicinity of the discovery.

B. Initial Actions
1. Secure the discovery area to prevent further disturbance.
2. Do not collect, move, photograph, or otherwise handle materials.
3. Notify the Contracting Officer (CO) immediately. The CO will contact the appropriate Park resource program manager(s).

C. Direction & Authorization
1. Follow CO direction for evaluation, protection, and any required coordination (Cultural Resource Program Manager for archeology/human remains; Park Wildlife Program for biological resources).
2. Work shall not resume in the affected area until authorized by the CO.

D. Cross-References: Implement this protocol in conjunction with:
1. Section 01 35 13.22 “Archeological Protection” (stop-work for human remains and concealed paleontological resources).
2. Section 01 35 05 “Wildlife Protection Requirements” and “Vegetation Clearing Timing – Wildlife Driven Avoidance Requirements”.

3.3 WILDLIFE PROTECTION REQUIREMENTS

A. Wildlife protections are subject to all applicable federal laws, including Migratory Bird Treaty Act and the Endangered Species Act.

B. Critical species include Humpback Chub, Razorback Sucker, California Condor, Mexican Spotted Owl, Desert Bighorn Sheep, and Special Status Bat Species.

C. All areas to be disturbed shall be within the Limits of Construction (LOC) as noted on the drawings and have boundaries flagged by the Contractor before beginning the activity, and all disturbance and project activities shall be confined to the flagged areas. Disturbance outside flagged areas is prohibited.

D. All project vehicles shall be driven at speeds within posted speed limits on existing roads and shall not exceed 20 mph within project boundaries.

E. Project personnel shall exercise caution when commuting to the project area and obey speed limits to minimize any chance for the inadvertent injury or mortality of species encountered on roads leading to and from the project site.

F. Light and heavy construction equipment shall not be used within 0.5 mile from owl Protected Activity Center boundary or known nest/roost sites during the owl breeding season (March 1 to August 31).

G. Contractor shall develop a Spill Prevention and Response Plan based on the Grand Canyon National Park's (Park) Parkwide Spill Response Plan to prevent potential poisoning of wildlife and soil and water contamination.

H. Project sites shall be cleaned up at the end of each day the work is being conducted (e.g., trash disposed of/secured appropriately and scrap materials picked up) to minimize the likelihood of California condors visiting the site. Park wildlife program staff may conduct periodic spot checks to ensure adequate project clean-up measures are being appropriately undertaken.

I. Trash and recycling receptacles and all dumpsters shall be wildlife-proof certified and will be tightly covered to avoid wildlife access.

J. Contractor shall not feed or approach Wildlife.

K. Contractor shall avoid any interaction with condors and immediately contact park wildlife program staff or park dispatch if condors visit a project site. If California condor nesting activity is known within 0.5 miles of a project area, then light and heavy construction equipment (as defined in 01 35 05, Attachment A) in the project area will be restricted during the active nesting season from February 1 to September 30. These dates may be modified based on the most current information, in consultation with the park biologist, the park Section 7 Coordinator, and the U.S. Fish and Wildlife Service.

L. If condors visit a project site, construction project activities shall cease until the condors leave on their own or until techniques, such as hazing, are employed by permitted park personnel that result in the condors leaving the area.

M. Contractor pilots shall minimize aircraft use along the rim and cliffs to the greatest extent possible.

N. Avoid operating helicopters and fixed-wing aircraft within 1,000 FT of eagle nests during the breeding season. Potentially disruptive activities will be minimized in the eagles' direct flight path between their nest and roost sites and important foraging areas. Regardless of season, aircraft corridors shall be located no closer than 1,000 FT vertical or horizontal distance from eagle roost sites.

O. Helicopters shall stay at least 1 mile away from active condor nest locations and vicinities except when human safety will be compromised. The active nesting season is February 1 to September 30. These dates may be modified based on the most current information regarding condor nesting activities (e.g., roosting and fledging) and coordination with the U.S. Fish and Wildlife Service (USFWS). If condors establish a nest within 0.5 mile of helicopter flight paths and/or landing zones, contractors will record helicopter flight numbers by helicopter type and report this information to GRCA's wildlife department, the GRCA section 7 coordinator, and FWS. NPS is adding the following conservation measure: In instances when condors are actively nesting within 0.5 mile of helicopter flight paths and/or landing zones, GRCA will assign a wildlife monitor to observe the area and condor behavior when frequent and high numbers of helicopter flights are expected

P. Helicopters shall stay at least 1,200 FT away from condors in the air or on the ground or cliffs unless human safety concerns override this restriction.

Q. To minimize noise disturbance within Mexican spotted owl Protected Activity Centers (PACs), light helicopters shall remain at least 1,200 FT from the boundary of any designated PAC during the Mexican spotted owl breeding season (March 1 through August 31).

R. To reduce noise impacts on bighorn sheep, helicopters shall remain at least 1,500 FT above ground level and maintain that distance while in the canyon, except during takeoff, landing, or dropping a sling load.

S. Contractor shall comply with the Migratory Bird Treaty Act (16 United States Code 703). Any active bird nest shall be left in place and undisturbed until the young hatch and depart. Tree removal shall occur between September 1 and March 31.

S. Notify Contracting Officer not less than fourteen (14) days in advance of tree removals that will require a survey.

T. If active nests or breeding behavior are detected during these surveys, no vegetation removal activities should be conducted until nestlings have fledged or the nest fails or breeding behaviors are no longer observed. Continued bird monitoring will occur for work approved April 1 through August 31 and work will be stopped and or relocated if nests are found at any time.

U. If using erosion netting, biodegradable matting with a large-diameter natural fiber shall be used to prevent entrapment of wildlife.

V. Park wildlife biologists will train contractor staff, at the preconstruction meeting, to avoid disturbance to any wildlife species (reptiles, migratory birds, raptors, and bats) found nesting, hibernating, estivating, or otherwise living in, or immediately nearby, work sites.

W. Work involving trenching or digging holes, provisions (generally in the form of ramps, with a slope less than 45°) must be made every 50 FT to allow for the escape of animals that may fall into these recesses, or they must be covered in such a way as to prevent animals (vertebrates) from falling into them.

X. If bats are encountered on buildings, please cease all activity and contact the wildlife department for assistance and/or advice in removing them safely. Notify park dispatch and wildlife staff for any physical encounter with bats. If Park wildlife biologists determine that bat specific surveys or roost protection requirements apply, the Contractor shall implement those measures in addition to the requirements of Section 3.5 VEGETATION CLEARING TIMING AND WILDLIFE-DRIVEN AVOIDANCE MEASURES.

Y. Care shall be taken not to disturb any wildlife species (reptiles, migratory birds, raptors, or bats) found nesting, hibernating, estivating, or otherwise living in, or immediately nearby, worksites. Work shall cease if wildlife is encountered in a manner that may result in harm, until Park wildlife biologists provide direction.

Z. Wildlife Department personnel shall be notified/consulted when any wildlife must be disturbed or handled. They will be available to assist with moving/relocating wildlife, when necessary; and/or to make recommendations for relocating any disturbed animals.

AA. All water shall be contained inside buildings or enclosed/covered tanks. No open standing water is permitted.

BB. Vegetation-disturbance limitations implemented for wildlife protection, including seasonal clearing windows, nest-survey requirements, avoidance buffers, and coordination with USFWS, are specified in Section 3.5 VEGETATION CLEARING TIMING AND WILDLIFE-DRIVEN

AVOIDANCE MEASURES. The Contractor shall implement those requirements as wildlife-protection measures for this Project.

3.4 SOUNDSCAPES

A. To reduce noise impacts on sensitive wildlife and areas with natural or wilderness characteristics when flying to and from the work area, helicopters will maintain a minimum 2,000-FT altitude where possible, per Federal Aviation Administration (FAA) Advisory Circular 91-36D Visual Flight Rules (VFR) Flight near Noise-Sensitive Areas.

3.5 VEGETATION CLEARING TIMING AND WILDLIFE-DRIVEN AVOIDANCE MEASURES

A. Seasonal Avoidance Requirement
   1. To protect migratory birds, roosting bat species, and other animals using trees and shrubs for forage, nesting, and shelter, the Contractor shall avoid vegetation clearing to the greatest extent possible between April 1 and August 31, the primary bird nesting season. Protection measures for roosting bat species shall be implemented when directed by Park wildlife staff, consistent with the species-specific guidance provided by the National Park Service or USFWS.
   2. Scheduling shall prioritize vegetation removal outside this window unless infeasible due to construction constraints.

B. Requirements When Clearing Cannot Avoid the Nesting Window
   1. If vegetation removal during April 1–August 31 is unavoidable, surveys are required by a qualified biologist. The survey shall be conducted within five (5) days prior to the scheduled vegetation removal.
   2. If the Park is unable to provide a qualified biologist, the Contractor shall retain a qualified biologist. The biologist shall document the survey(s) in a written report and provide the report to the Park for review and acceptance prior to any vegetation removal.

C. Restrictions Based on Survey Findings
   1. If active nests or breeding behavior are detected during surveys, no vegetation removal or other construction activities shall occur within 150 feet of the nest or observed breeding activity.
   2. The 150-foot restriction remains in place until nestlings have fledged, the nest fails, or breeding behaviors are no longer observed.

D. Proceeding Within the 150-Foot Radius
   1. If vegetation removal is to occur while nests are active or breeding behavior is observed, the Contractor must coordinate with USFWS to determine if a Migratory Bird Treaty Act (MBTA) permit or additional avoidance and/or minimization measures are required.
   2. The Contractor shall implement the U.S. Fish & Wildlife Service Nationwide Avoidance & Minimization Measures for Birds. No work may begin until USFWS guidance or permitting (if required) is complete.

E. Responsibility for Delay Costs
   1. The Contractor shall record all nest surveys, buffers, restrictions, and USFWS coordination in the Vegetation Clearing Log and submit updates weekly to the Contracting Officer (CO) and Park Environmental staff.

F. Documentation

1. The Contractor shall record all nest surveys, buffers, restrictions, and USFWS coordination in the Vegetation Clearing Log and submit updates weekly to the Contracting Officer (CO) and Park Environmental staff.

3.6 VEGETATION AND SOIL MANAGEMENT

A. If erosion-control fencing is used, soil will be piled in front of the fence to avoid creating bare soil and potential for invasive plant species encroachment outside the project area.

B. Trenching and construction equipment transported to the site from outside the park will first be pressure washed to minimize the potential to import invasive plant seed and material to the site.

C. Pruning necessary for the project will adhere to the park's pruning guidelines with the goal of retaining the health and integrity of trees and shrubs treated. Damage to trees or roots in or adjacent to project sites during construction will be avoided as much as possible and, if avoidance is not possible, root pruning guidelines provided by the park's Vegetation Management Program will be followed.

D. Prior to construction the NPS will perform an invasive plant survey within the Limits of Construction shown on the Drawings. The Contractor shall separate the removal of native plants from invasive plants and dispose of invasive plant material outside the limits of the Park. Invasive non-native vegetation shall be treated and/or removed prior to construction activities. 60 days before the start of non-native vegetation treatment or removal actions, the contractor will meet with the Contracting Officer (CO) and GRCA Vegetation Program Manager to discuss planned actions.

E. Imported fill materials required for the project will be obtained from a park-approved source in adherence to Standard Operating Procedure 8213-007 (01 35 05, Attachment B) Invasive Plant Free Forage and Construction Materials.

1. In areas where ground disturbing activities are planned, all topsoil will be properly salvaged, stored, and redistributed. Topsoil is a term used for A (or A and part of B) horizons or the uppermost layer of soil capable of growing and supporting vegetation. Topsoil contains the essential microorganisms, nutrients, organic matter, and physical characteristics necessary to grow and sustain vegetation. Stripping, stockpiling, and reusing topsoil on construction projects is essential for proper revegetation of disturbed areas. To help guide topsoil management actions, the contractor will access the USDA NRCA Web Soil Survey website (https://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm), delineate project areas, and produce customized soil reports specific to each project area (e.g., North Rim WTP, Collection System Improvements, Ballfield, Supai Tunnel, Roaring Springs Pumphouse, etc.) where soil disturbance will occur. The USDA Web Soil Survey generated reports will include soil profile descriptions, soil mapping units, characteristics, topsoil depth, soil textures, rock content, soil depth, available water holding capacity, permeability rates, and drainage classes. 120 days before the start of topsoil management actions, the contractor will meet with the Contracting Officer (CO) and GRCA Vegetation Program Manager to discuss the USDA NRCS Web Soil Survey reports and finalize topsoil management actions per project area.
2. The Contractor will remove all topsoil prior to all other soil disturbing operations. The Contractor will remove the topsoil in one pass, will not mix topsoil with subsoil, and will not compact or drive upon topsoil during removal. Topsoil salvage depth by location may vary, but as a general rule a minimum of 6 inches of material shall be conserved in all cases, and additional topsoil may be salvageable in some locations.

3. Live vegetation less than 3 FT in height and limbs less than 1 inch in diameter shall be incorporated as topsoil in the stockpiles. The contractor will properly dispose or bury all topsoil contaminated with invasive non-native plant species as directed by the Contract Officer (CO) and GRCA Vegetation Program Manager.
4. The salvaged topsoil will be stockpiled in windrowed piles along the outer edge of the limits of construction. If possible, the topsoil will be stockpiled in a disturbed area to minimize the impact to adjacent vegetation. If it is impractical to stockpile topsoil adjacent to the disturbed area or objectionable to the Contracting Officer (CO) and GRCA Vegetation Program Manager due to environmental or other concerns within an area, the contractor will remove the topsoil and stockpile at an approved location.
5. Salvaged topsoil will be separated from the sub-soil and stockpiled in windrowed piles no higher than three FT, and no wider than three FT. The contractor will identify removal areas and stockpiles so that topsoil may be replaced in the same locations. If salvaged topsoil must be stored longer than 3 weeks, topsoil stockpiles will be hydroseeded and hyrdo-mulched by the contractor with locally collected seed provided by the GRCA Vegetation Program.
6. Prior to reapplying the topsoil, the contractor will disk or scarify all disturbed areas to a depth of 6" in a direction perpendicular to the natural flow of water. Prior to topsoil placement, if the Contracting Officer (CO) and GRCA Vegetation Program Manager determines that subsoil is excessively compacted by the NPS or contractors' operations, the contractor will scarify the subsoil to a depth of 24" to de-compact and break the interface between particles.
7. When reapplying topsoil, the contractor will avoid mixing the topsoil with subsoil material. The contractor will spread conserved topsoil to a depth of 4 inches or greater (if additional topsoil is available) over all disturbed soil areas. Topsoil will be spread using the lightest weight equipment applicable. If the Contracting Officer (CO) and GRCA Vegetation Program Manager determine that topsoil is excessively compacted by the NPS or contractors' operations, the contractor will scarify the topsoil to a depth of 8" to de-compact and break the interface between particles.

F. Water will be applied for dust abatement at the following project sites: pipeline excavations, Ballfield, North Rim Water Treatment Plant, collection system pipe replacement.

3.7 OTHER ENVIRONMENTAL PROTECTION MEASURES

A. The contractor shall be responsible for obtaining and complying with all permits necessary to complete the work based on contractor means and methods. Contractor shall pay all permit fees required and provide all materials, labor, and equipment necessary to comply with permit conditions and provisions at no additional expense to the government. Refer to Section 01 31 00 – Project Management and Coordination.

B. Contractor shall provide all storm water control, erosion control, dust control, air quality control, water quality control, and other pollution control measures necessary to comply with all Federal, State, and municipal laws, codes, rules, regulations, and permits applicable to performance of the work at no additional expense to the government. Contractor shall ensure all construction personnel are aware of all applicable requirements.

C. Contractor shall develop and adhere to a Spill Prevention and Response Plan to ensure that fuel, oil, lubricants, and other hazardous material spills do not occur. Plan will be in accordance with the Grand Canyon National Park's (Park) Parkwide Spill Response Plan. If spills occur, cleanup shall be completed by the contractor at no additional expense to the government.

1. When gasoline, diesel fuel, antifreeze, hydraulic fluid or any other chemical contained within the vehicle is released to the pavement or ground, proper corrective, clean-up, and

safety actions specified in the Spill Prevention and Response Plan must be immediately implemented. All vehicles with load rating of 2 tons or greater should carry, at minimum, enough absorbent materials to effectively immobilize the total volume of fluids contained within the vehicle. Repair oil leaks immediately on discovery. Do not use equipment that is leaking. Have oil pans and absorbent material in place prior to beginning repair work. Have the "on scene" capability of catching and absorbing leaks or spillages of petroleum products including antifreeze from breakdowns or repair actions with approved absorbent materials. Keep a supply of acceptable absorbent materials at the job site in the event of spills. Sand or soil are not approved absorbent materials. Use oil pans and absorbent materials to prevent leaks, spills and draining petroleum fluids from falling onto bare ground and paved surfaces during servicing of equipment. Dig up soils contaminated with such fluids, place in appropriate safety containers, and dispose of according to state and/or federal regulations.

2. Equipment found operating on the project that has oil leaks will be shut down and subject to fines.

D. Equipment Washing and Inspections

1. Vehicle and equipment washing is required to preclude entry of seeds, other plant materials (e.g., stems, rhizomes), and soil capable of non-native plant species establishment and spread. The contractor will make arrangements for the Contracting Officer (CO) or GRCA Vegetation Program staff to inspect each vehicle and piece of equipment before entering the project area. All equipment shall be thoroughly pressure washed and cleaned outside of the Park Area before entering. Equipment leaving the park for any reason must be re-washed before returning to the job site.
2. The Contractor shall clean equipment that will be used off surfaced roadways, or approved roadways, prior to the equipment entering the Park Area. It is recommended that cleaning be accomplished by pressure washing, using water only. All equipment surfaces shall be cleaned, with special emphasis given to the undercarriage. This includes, but is not limited to: wheel wells, bumpers, radiator grills, springs, shock absorbers, wire harnesses, skid plates, oil pans, axles, frame ledges, buckets, tracks, belly pans, etc. Cleaning shall also include all dirt and debris inside the cab, or operator area. A Shop-Vac or commercial car wash vacuum system may be used for operator compartments, provided the method used effectively removes all visible debris. Vehicle and equipment cleaning is for the purpose of removing seeds, other plant materials (e.g., stems, rhizomes) and soil accumulations to prevent the potential spread of non-native plants from outside the Park to locations inside the Park. The contractor will make arrangements for the Contracting Officer (CO) or GRCA Vegetation Program staff to inspect each vehicle and piece of equipment before entering the project area. Cleaning shall be performed in a manner that will reasonably remove all visible soil, plant and other foreign material. All equipment that requires washing shall be inspected by government personnel, or contractors/partners assigned to this duty, to ensure compliance.
3. Equipment that requires washing is generally defined as track or rubber-tired machinery intended for use primarily off-road. Examples include backhoes, loaders, earthmovers, graders, rollers, bulldozers, water trucks, etc.; or four wheel drive or passenger vehicles that are driven in off road areas. Equipment may also be stationary, such as a material screening plant, conex storage containers, or a mobile office (mobile offices may be cleaned in the same manner as operator compartments as outlined in 3.3.D.2). The equipment is generally under the direct control of the contractor or sub-contractors, is likely to remain on the construction site for an extended period of time, and may contain adhered soil and seed from other project sites.
4. Equipment/vehicles that do not require washing are defined as those normally used for highway driving only and travel primarily on paved surfaces and well maintained service roads and parking areas. Examples include semi-tractor trailers, dump trucks, concrete trucks, pickup trucks, fuel trucks, service trucks, etc. These delivery and passenger vehicles are generally not under the direct control of the contractor, have no earth-moving function,

and do not remain on the project site. These vehicles have a low probability of carrying adhered soil and seed from other projects. However, every vehicle and piece of equipment is subject to rejection from entering the Park if there are obvious signs (covered in mud, plant material adhered to it, etc.) that it could be transporting non-native species into the Park.

5. The Contractor shall make arrangements for the COR and CMR to inspect each piece of equipment before entering the project. The COR and CMR will maintain records of inspections. Equipment not passing initial inspection must be rewashed outside the Park at no additional expense to the government.
6. Equipment found operating on the project that has not been inspected, will be shut down and subject to fines

E. Upon completion of work, leave area in a clean, natural looking condition as specified in Section 01 57 23 – Temporary Stormwater Pollution Prevention and Section 32 92 21 – Native Grass Seeding. Remove all signs of temporary construction and activities incidental to construction of required permanent work.

END OF SECTION 01 35 05

Appendix A

Equipment Types

Construction projects are multi-faceted and involve the use of several different equipment types that can result in various effects on wildlife. The most common impacts result from ground disturbance, vegetation removal, and noise generation.

For this consultation, equipment types are broken down into the following categories: heavy construction, light construction, and cyclic maintenance. These guidelines and definitions are consistent with previous section 7 consultations and analyses of project impacts throughout Grand Canyon National Park.

Heavy Construction

Heavy construction will require use of large equipment for actions such as earthmoving, rock excavation, and construction of facilities. A description of each activity associated with heavy construction is provided below.

Earthmoving

Earthmoving is typical for construction of facilities and roads where large areas of ground disturbance are necessary. Equipment such as large bulldozers, large dump trucks, scrapers, and excavators for transporting several cubic tons of soil and rock are required for such work.

Rock excavation and trenching

This type of activity will be used in installation or repair of utilities or construction of building foundations. The operation of equipment necessary to excavate rock can be very loud (as can the excavation itself) and can result in vibration. Work involving excavation of rock and trenching will typically require use of equipment such as trenchers, hoe-rams, rock saws, hammer hoes, rippers on bulldozers, and large track-hoes with hydraulic hammers.

Construction of facilities

Work involving construction of tall facilities may require use of a large crane over 100 feet high. This has an effect on the surrounding landscape while in use.

Light Construction

Light construction activities require smaller pieces of equipment than those described for heavy construction, and do not generate as much noise.

Examples of light construction includes such activities as road rehabilitation and maintenance, concrete work, utility installation and repair, trail construction, revegetation and grounds maintenance, sign installation, hazardous vegetation removal, and construction of facilities two stories or less. These activities typically require the use of equipment identified as Light Construction (listed below).

Cyclic Maintenance

Cyclic maintenance equipment primarily includes hand-held mechanized tools needed for activities such as typical building repairs and replacements (roof, siding, etc.), painting, grounds work and maintenance, etc.

Construction Category Equipment List

Heavy Construction	• Large bulldozers • Large dump trucks • Scrapers • Excavators • Rock saws • Large track-hoes with hydraulic hammers • Large cranes • Trenchers • Asphalt re-claimers • Loaders
Light Construction	• Backhoes • Small dump trucks • Chainsaws • Jackhammers • Small bulldozers • Bobcats • Large boom truck • Asphalt pavers • Asphalt distributors • Soil compactors • Pionjars • Graders • Cement mixers • Welders and torches • Woodchippers • Stump grinders • Augers • Forklifts • Mechanized mowers
Cyclic Maintenance	• Air compressors • Scissor lifts • Small boom truck • Polaris 4X4s • Chainsaws • Farm tractors • Mechanized hand tools • Snowcat • Snowmobiles • Street sweepers • Pick-up trucks

	• Snowplows • Snow blowers • Mechanized stripers • Concrete saws • Mechanized sewer rods • Mobile generators • Non-mechanized hand tools

Appendix A is extracted from:
Rehabilitate the North Rim and Roaring Springs Utility Systems
Grand Canyon National Park
Biological Assessment
February 2026
National Park Service – U.S. Department of the Interior

REVISED VERSION
May 17, 2012

United States Department of the Interior

NATIONAL PARK SERVICE
Grand Canyon National Park
P. O. Box 129
Grand Canyon, Arizona 86023-1238

IN REPLY REFER TO:
A56 (GRCA 8211)

Invasive Plant Free Forage, Materials, Vehicles, and Equipment

Standard Operating Procedure

Purpose: This Standard Operating Procedure (SOP) establishes policies, procedures, and requirements for park staff, volunteers, contractors, concessioners, other agency personnel, partners, and visitors transporting into the park and using livestock forage (including hay, straw, mulch, cubes, feed grain, pellets), materials (including construction, erosion control, site restoration), equipment, and vehicles that could potentially contain invasive plant materials.

Scope: This SOP applies to all Grand Canyon National Park (GRCA) employees, volunteers, contractors, concessioners, other agency personnel, partners, and visitors with livestock. All parties listed are responsible for understanding and adhering to this policy, but ultimate enforcement is the responsibility of GRCA staff.

Procedure: Grand Canyon National Park has established a policy to minimize or prevent the spread of invasive exotic plants into the park through contaminated forage or materials, and from vehicles travelling on the park's non-paved roads.

The park has adopted the following regulations concerning invasive plant free forage and plant containing materials:

> Possessing, storing, or transporting any hay, straw, or mulch that has not been certified as free of noxious invasive plants and seeds by a certified officer through an official state program, is prohibited within the boundaries of Grand Canyon National Park.

Forage:
Only certified invasive plant free forage (hay, straw, mulch) can be used in the park. Official certification from outside of Arizona is acceptable. Baled or loose hay, straw and mulch cannot be taken beyond a trailhead into the

backcountry or into the park's proposed wilderness. Only pelletized feed, compressed hay cubes, and feed grain products can be used in the backcountry. **Grazing of stock is not permitted in any portion of the park.**

Materials:
Plant containing materials such as wattles, erosion matting, and mulch must be made of approved materials that are inherently weed free or certified weed free. Top soil, cinders, aggregate, and other construction material often contain plant material; these materials must be surveyed and approved by Vegetation Program staff to contain no plant species on Arizona's noxious weed list or other highly aggressive invasive plant species. Routine bi-annual surveys of all known pit locations will be complete by Vegetation Program staff; any potential new material sources will be surveyed by Vegetation Program staff, upon request, within a three week time period. Contracts and project specifications will include language to ensure this condition is met.

Exclusions:
Persons with a permit specifically authorizing the otherwise prohibited act.

Unlike many western states which have passed legislation requiring invasive plant free forage, Arizona and Utah have voluntary programs. These programs use the North American Weed Management Association (NAWMA) standards for certification. The North America Weed Free Forage Standards are benchmark standards used across the United States.

Invasive Plant Free Purchasing Policy
This policy was created for the purchase and transport of invasive plant free forage and construction, and site restoration materials into the park:

- Purchasers may include government purchase card holders, concessioner employees, contractors, and private livestock owners visiting the park.
- These parties are responsible for obtaining proof of invasive plant free certification of any plant based materials such as hay, straw, mulch, and grain prior to shipment or delivery to the park.
- If more than one producer is used per year or per project, the purchaser must obtain a certificate from each producer.
- If the materials are not able to be certified at the state or regional level, these materials are prohibited from the park unless they have been approved in consultation with the Vegetation Program Manager.
- Each fiscal, purchasers of plant based materials must provide certificates of invasive plant free certification from each producer utilized.
- Purchasers of top soil, cinders, aggregate, or other construction material must verify that the park's Vegetation Program has surveyed and approved the location prior to purchase.
- Purchasers of erosion control and site restoration materials must verify that they are comprised of inherently weed-seed free materials.

REVISED VERSION
May 17, 2012

Invasive Plant Free Contracting Policy

This policy was established to ensure the use of invasive plant free construction and site restoration materials, and to provide public notification of the park's invasive plant free forage and materials policy.

- All project managers, procurement, and contracting staff must ensure that invasive plant free plant based material language is included in the construction and site restoration specifications for all in-park contracts that may use such materials. This includes roadwork, buildings, plant salvage, site restoration, and other contracts which may involve ground disturbance and subsequent erosion control or restoration measures, and any projects that require the import of top soil, cinders, aggregate or any other similar material.
- Wattles and erosion matting should contain materials such as aspen coir or coconut fiber coir that are inherently seed and propagule free and not hay or straw. If bales are required (wattles are preferred), they must be certified invasive plant free straw bales, not hay.
- Mulch material must either be inherently invasive free, such as wood chips or wood straw, or must contain only certified invasive plant free material.
- Hydroseeding and hydromulching will use wood and/or paper fibers.
- These requirements will be clearly stated in the contract specifications.

Vehicle Cleaning Policy

This policy was established to minimize the vehicular and equipment spread of invasive plant materials into the park, and to provide public notification of the park's vehicle cleaning policy. This policy recognizes that it is impractical to expect every business-related vehicle to be washed prior to entering the park; however, the intent is to control the import of invasive materials whenever possible. The emphasis is on those vehicles that travel on non-paved roads while in the park and those that have travelled on non-paved roads prior to entry.

- Prior to entry, contractors and partners must pressure wash all vehicles that could potentially leave paved surfaces in the park.
- Prior to entry, all government vehicles and equipment that have been on non-paved roads outside of the park must be pressure washed. Examples include visiting NPS employees from other park units, visiting or assigned fire crews, leased bulldozers, etc.
- Exceptions to this policy include GRCA river shuttles and GRCA backcountry travel that does not have a vehicle wash station en route; however, these government vehicles should be washed as soon as possible at one of the vehicle wash areas within the park.

Planning and Compliance

Procedures and policies outlined in this document will be referred to and included in park compliance documents in the form of mitigation measures for all applicable projects.

REVISED VERSION
May 17, 2012

Definitions

North American Weed Management Association (NAWMA): A network of public and private professional invasive plant managers who are involved in implementing any phase of a county, municipal, district, state, provincial or federal noxious invasive plant law.

North America Weed Free Forage Standards: These standards are those established by NAWMA and adopted by the ACIA and used for inspection of Arizona forage. The North America Weed Free Forage Standards are designed:

- To provide some assurance to all participants that forage certified through this program meets a minimum acceptable standard.
- To provide continuity between the various provinces and states in the program.
- To limit the spread of noxious invasive plants through forage and mulch.

Forage: includes hay, straw, mulch, and feed grain.

Plant containing construction materials: includes hay, straw, mulch, and seeds. Also includes such erosion control and construction materials as wattles, erosion control blankets, mulch used in hydromulching and hydroseeding, straw or hay bales, and natural fiber mats.

Invasive plant free certification: in Arizona this means certified under the state voluntary program to be free from propagative plant parts and free from invasive plant seed from plants set forth on the Arizona designated noxious invasive plant list and the North American designated noxious invasive plant list. Official certifications from other states will be accepted.

Invasive plant free: means free from propagative parts and free from invasive plant seed from plants identified in state, regional, and national lists of invasive plants published by NAWMA. Sources for invasive plant free forage in Arizona and Utah are listed at the end of this document.

Specific Requirements Established by the Park

1. Responsibilities:

A. Vegetation Program
Contact Invasive Plant Biologist or Vegetation Program Manager

1. Work with Backcountry Information Center to review the private stock use policy and handout given to visitors annually.

2. Coordinate training and provide assistance to work units on technical aspects of invasive plant free forage and materials.

3. Coordinate training and guidance to work units regarding the vehicle washing component of this SOP.

4. Monitor and document findings, including producer certificates, concerning implementation of this SOP each fiscal year.

5. Provide all parties mentioned in this SOP a list of Arizona, California, Colorado, Nevada, and Utah Certified Noxious Weed Free Forage and Mulch providers or direct parties to locations where this information can be found.

6. Survey all proposed borrow pit locations outside of park boundaries for invasive plants in May or June and September or October each year and provide a list of approved pits to all division chiefs for dispersal to their employees.

7. Notify borrow pits of the results of the bi-annual surveys and provide standard recommendations.

8. Provide standard mitigation measures to the park's Office of Planning and Compliance for inclusion in park compliance documents and consistently insert these standard mitigations into projects in the Planning and Environmental Compliance (PEPC) database.

9. Provide guidance and training for Resource Advisors.

10. For large-scale projects or as requested by Project Managers, Vegetation Program staff will attend pre-construction meetings to discuss this policy with contractors.

B. Division Chiefs

1. Ensure all aspects of this SOP are communicated and understood by supervisors and employees. At least annually monitor and document findings concerning implementation of this SOP.

2. Provide feedback regarding implementation of this SOP to the Vegetation Program Manager or Science and Resource Management Division Chief as needed.

3. Ensure that the appropriate staff members provide the Vegetation Program's primary contact with updated certificates of inspection from preferred providers upon request.

C. Supervisors

1. Ensure all employees are aware of the contents of this SOP. Answer any questions regarding this SOP or refer employees to the Vegetation Program's primary contact.

2. Ensure all employees are following the guidelines of this SOP.

3. Communicate any difficulties or failure to follow this SOP to the Division Chief and the Vegetation Program Manager.

D. Employees

1. Read this SOP and ask your Supervisor or Vegetation Program Invasive Plant Biologist or Program Manager to clarify any questions.

2. Follow invasive plant free purchasing policy outlined in this SOP.

3. Report any deviation of this SOP by any park employee, contractor, concessioner, or visitor to the Vegetation Program.

4. Project managers, contracting personnel, and government purchase card holders must provide certificates of inspection to the Vegetation Program for each project bringing materials including hay, straw, mulch, or other plant containing materials into the park.

5. Project managers, Contracting Officer's Representatives (COR), and any employee involved in a project that requires the import of top soil, cinders, aggregate, or other construction material must ensure that the pit location has been approved by the park's Vegetation Program prior to purchase.

6. If multiple shipments of these materials are entering the park over the course of the project from multiple sub-contractors, Project managers, contracting personnel, and government purchase card holders are responsible for ensuring that each shipment utilizes only approved materials. These personnel must provide certificates of inspection to the Vegetation Program upon request.

7. Project managers are responsible for ensuring and documenting that contractors and project partners comply with these contract specifications, and must be able to verify the specifications have been met.

8. Contracting personnel and government purchase card holders who purchase large amounts of forage for the trail crew and law enforcement livestock, must ensure that the vendors always use certified forage in their shipments. These personnel must provide certificates of inspection to the Vegetation Program. For example: if the trail crew purchases their forage

REVISED VERSION
May 17, 2012

from a feed store in Flagstaff, each shipment may come from a different producer. Therefore a certificate of inspection must be provided with each shipment. These certificates of inspection must be provided to the Vegetation Program.

- If multiple producers are involved with one shipment, then multiple certificates must also be provided.
- It is the responsibility of contracting personnel and government purchase card holders to request the specifics on providers from the vendors.

E. Visitors with livestock

1. Ensure all livestock forage transported into the park is certified invasive plant free.

2. Be prepared to prove this certification either with certified colored string, certification tags, or a written certificate. Visitors should also have certified vendor information on hand for verification.

/s/ Lori J. Makarick
Prepared By
Lori J. Makarick
Vegetation Program

June 1, 2011
Date

/s/
Approved
Palma Wilson
Deputy Superintendent

Date

/s/
Approved
Barclay Trimble
Acting Superintendent

Date

REVISED VERSION
May 17, 2012

Certified Vendors of Invasive Plant Free Forage and Mulch

**Vegetation Program staff will update and distribute list annually.*

Arizona
Available from the Arizona Crop Improvement Association website:
http://www.arizonacrop.org/NWFF&M/Grower's%20list.html

Anrac Farms, 7745 N 5th St. Coolidge, AZ 85228. Tel: 480-296-9592. Contact: Scott R. Wilding. Crops Certified: Wheat and Barley Straw.

A.N.S. Inc., 2946 E. Robin Gilbert, AZ 85296. Tel: 480-241-1082. Contact: Brandon Fish. Crop Certified: Wheat Straw.

Desert Ranch Enterprises, 30151 W. Old US 80 Palo Verde, AZ 85343. Tel: 602-680-8320. Contact: Jason Hardison. Crop Certified: Wheat Straw.

Southwest Farm Services, 11926 W. Southern Ave. Tolleson, AZ 85353. Tel: 623-936-4339. Contact: Mike Perez Sr. Crop Certified: Wheat Straw.

Triangle Y, PO Box 2136 Camp Verde, AZ 86322. Tel: 928-699-8628. Contact: Rick Anderson. Crop Certified: Wheat Straw.

Utah
Available from the Utah Department of Agriculture and Food
http://ag.utah.gov/news/publications/documents/Weed-FreeList.pdf

California
Available from the California Department of Food and Agriculture
www.cdfa.ca.gov

Colorado
Available from the Colorado Department of Agriculture
www.colorado.gov/ag/dpi

Nevada
Available from the Nevada Department of Agriculture,
http://agri.nv.gov/nwac/PLANT_WFHProducers.htm

REVISED VERSION
May 17, 2012

List of Borrow Pits Outside of Park Boundaries

**Vegetation Program staff will update and distribute list annually*

BABBITT'S RANCH, TIN HOUSE PIT
Contact: 928-607-4606, Les Wilderman
Mailing Address: P.O. Box 815, Williams, AZ 86046

Site Location:
The survey was conducted at the pit location north of Coconino County Road 347. Access the site from Highway 64 approximately 16 miles south of Grand Canyon National Park's South Entrance. Turn west on CCR 347 (unmarked road with a right hand turn lane) and follow it out past the railroad tracks. The Willaha Pit is the second right past the railroad tracks (the first right is directly past the tracks and runs along them). From the Willaha site, continue west on Coconino County Road 347, 2-4 miles after the railroad tracks, take the left fork and the road will follow the large powerline. Cross creek drainage, stay on the heavily travelled road and the pit will be on the right. (Tin House ranch home and other buildings are past pit at the end of the road). The Tin House Pit is about 17-18 Miles past the Willaha Pit.

CEMEX MATERIALS
Contact: 928-606-0055, Darrell Wendel
928-526-5250, Mike Scharfencamp
Mailing Address: 5200 East Railhead, Flagstaff, AZ 86004

Site Locations:

1. Gray Mountain Pit, Mile marker 455 on North 89 between Flagstaff and Cameron
2. McCormick Pit, Sunset Crater Road
3. 58 Dodge Avenue, Flagstaff

~~**COCONINO CINDERS AND MATERIALS**~~
~~Contact: 928-526-9509~~
~~Mailing Address: 7000 East Route 66, Flagstaff, AZ 86004~~
REMOVED FROM LIST – NOT APPROVED FOR USE

DYNA ROCK AND SAND INC.
Contact: 928-289-9349
Mailing Address: 206 W First St., Winslow, AZ 86047

Site Location:
The survey was conducted at two locations in Winslow, AZ. To access the first site, go east on I-40 from Flagstaff to exit 257. Take a left off the highway onto Hwy 87 and continue over the bridge. Turn left into the site, which is visible from the highway and has a large sign and tower that say "Winslow Redi Mix". To access the rock quarry site, turn right out of the main site and follow Hwy 87 over I-40 and continue on Hwy 87

through Winslow following the signs to Payson (there will be a left turn at a stoplight). Once out of town turn left on Hwy 99 and follow to mile marker 29. Turn left and follow road over two cattle guards—the road will then curve to the right. After a couple miles turn into the site on the right side of the road.

~~FLAGSTAFF CONCRETE PRODUCTS~~
~~Contact: 928-773-9139~~
~~Mailing Address: 2150 East Huntington Drive, Flagstaff, AZ 86004~~
REMOVED FROM LIST – NOT APPROVED FOR USE

FLAGSTAFF LANDSCAPE PRODUCTS
Contact: 928-526-0801
Mailing Address: P.O. Box 30326, Flagstaff, AZ 86003

Site Location:
6500 Old Highway 66. Take I-40 east to Page / Country Club exit. Take exit, then turn left at light. At T intersection, turn left onto 89. Take first left onto Old Route 66. Follow road past mall, new mall extension, and waste water treatment plant. About ½ mile past the mall, you will arrive at the location.

JDM SAND & ROCK
**Note – This is the only North Rim pit available as of 2011.*
Contact: 928-643-7113, Steve or Brian
Mailing Address: P.O. Box 113, Fredonia, AZ 86022

Site Locations:
1. Main Site: 1446 North Highway 89A. From Jacob Lake, go all the way through Fredonia heading north. Go past ADOT and Red Hills Manufacturing on the right. This location will be right after all the large oil tanks on the right.
2. Limestone or "RipRap" Pit: Turn south onto FSR 22 at Dennison Mines Building (Paved Road). Turn west at Mile 8.5 (before Cedar Knoll). Follow dirt road approximately 1.0 miles to limestone quarry on left.
3. Cedar Knoll Pit: Like with Limestone Pit, turn south onto FSR 22. Turn west at Mile 11.6 (past utility poles) onto dirt road just past sign: "trucks entering highway". Drive west approximately 4 miles – stay right and follow windy road around knoll to site.

LP'S EXCAVATING CINDER PIT
Contact: 928-635-4107, Jim Cross
Mailing Address: Williams, AZ 86046

Site Location:
Take I-40 to the westernmost Williams exit (Exit 161). Head south of the highway on the frontage road past Denny's to the east about ¼ mile. The site will be on the south (right) side of the road directly before the Williams Cemetery.

REVISED VERSION
May 17, 2012

MCNALLEY'S RANCH, WILLAHA PIT
Contact: 928-607-4604, Les Wilderman
Mailing Address: P.O. Box 815, Williams, AZ 86046

Site Location:
Highway 64 16 miles south of park's South Entrance gate. Turn west and go to the Coconino County Road 347 sign. The site is 8.1 miles past the sign. Cross the railroad tracks and cattle guard, then turn right. Take the left fork and drive down to the pit.

MILLER MINE-BLACK CINDER PIT MINE
Contact: 928-526-5700, Earnest Caldwell (millermining@gmail.com)
Mailing Address: 46 Leupp Road, Leupp, AZ 86035

Site Location:
The survey was conducted at the location off Leupp Road. To access the site take I-40 east from Flagstaff and take exit 211 for Winona. Continue northwest on Townsend Winona Road to County Road 419/Leupp Road and turn right (north). Follow Leupp Road until mile marker 441 where the pit access road is on the left. Once on the pit access road you will see a black cinder mine on the hill to the right—this is not the site. Continue down the road and after the first dip then crest in the road there will be a road on the left. Follow this road to the pit which is below the access road.

Q-MAX COMPANY
Contact: 928-635-9255
Mailing Address: P.O. Box 877, Williams, AZ 86046

Site Location:
Highway 64, eight miles south of Valle between mile markers 201 and 202 on the west side of the road. A dedicated turn lane exists for the facility and a large sign is on the access road a few hundred yards in from the highway. Piles are visible from highway.

This page intentionally left blank.

SECTION 01 35 13.22 – ARCHEOLOGICAL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of protecting archeological and paleontological resources on the ground surface and contained in soil deposits.

1.2 DEFINITIONS

A. Archeological Resources: Archeological resources are physical evidences of past human activity, including evidences of effects of that activity on the environment. Archeological resources represent both prehistoric and historic time periods, found above and below ground and under water.

B. Paleontological Resources: Paleontological resources means any fossilized remains, traces, or imprints of organisms, preserved in or on the earth's crust, that are of paleontological interest and that provide information about the history of life on earth, except that the term does not include—

1. any materials associated with an archaeological resource
2. any cultural item

C. Archeologically Sensitive Areas: Areas having potential to contain significant (National Register eligible) archeological resources. If archaeological resources cannot be avoided, park archaeologists will develop an appropriate mitigation strategy in consultation with the State Historic Preservation Office and as appropriate with the park's associated Tribes.

D. Paleontological Sensitive Areas: Areas having a potential for producing scientifically significant fossils.

E. Non-sensitive Areas: Areas with little potential of containing significant (National Register eligible) archeological resources.

F. Archeological/Paleontological Monitor: Representative of Government designated to oversee construction activities that could disturb archeological resources.

G. Archeological Resources Protection Act (ARPA) of 1979 (Public Law (P.L.) 96-95; 93 United States Statues at Large (Stat.) 712): defines archeological resources as any material remains of past human life or activities that are of archeological interest and at least 100 years old; Section 4 of the statute describes the requirements that must be met before Federal authorities can issue a permit to excavate or remove any archeological resource on Federal or Indian lands; the curatorial requirements of artifacts, and other materials excavated or removed.

H. National Historic Preservation Act and Statute: The National Historic Preservation Act (NHPA) of 1966 is a federal law that protects the United States' historic and archaeological sites with the goal to preserve cultural heritage, protect sites from federal development, and encourage preservation.

1.3 SUBMITTALS

A. Daily Work Schedule: Submit a Daily work Schedule detailing ground disturbance construction work. Submit to Contracting Officer (CO) 30 days before start of ground disturbing site work.

1.4 QUALITY ASSURANCE

A. At least one week before on-site work begins, and prior to submittal of the Daily Work Schedule, the Contractor shall meet with Contracting Officer and Archeological Monitor to discuss Daily Work Schedule, equipment, and special methods used in archeologically sensitive areas.

B. Contractor shall ensure approved Daily Work Schedule is followed throughout construction. If schedule changes occur, the Contractor will provide an update to the approved submittal and coordinate schedule changes with the CO and NPS Archeological Monitor a minimum of 48 hours in advance of work.

PART 2 - PRODUCTS

2.1 DAILY WORK SCHEDULE

A. Daily Work Schedule is required for work occurring within archeologically sensitive areas. Include all work that is to occur within the area and key the schedule to the drawings to include:

1. Starting and ending dates of ground-disturbing construction.
2. Locations of temporary facilities, such as barriers, field offices, staging areas, sanitary facilities, borrow pits, and haul and access roads.
3. Types of construction, such as clearing, topsoil stripping, structure or trench excavation, landscaping, and post construction clean-up.
4. Methods and equipment used for each type of construction.
5. Plan for relocating work in the event of temporary work stoppages at each archeologically sensitive area.

PART 3 - EXECUTION

3.1 ARCHEOLOGICAL AND PALEONTOLOGICAL PROTECTION

A. Provide and maintain a minimum 16-foot (5-meter) buffer around all identified archeological sites unless another buffer distance is established by GRCA Cultural Resources staff. Fencing, flagging, or taping of site boundaries shall correspond to the approved buffer. The NPS will mark avoidance boundaries in the field by flagging, fencing, or taping for Contractor avoidance.

B. Archeological Monitor will observe all ground-disturbing site work, including construction of temporary facilities, at known archeologically sensitive areas, from a safe location mutually agreed on by Contractor and Monitor. As new ground is broken, Monitor will examine excavated materials, using construction layout centerline, perimeter staking, and GPS as a reference point to record locations of findings.

C. The Secretary of the Interior's Standards for the Treatment of Historic Properties with Guidelines for the Treatment of Cultural Landscapes and Director's Order 28: Cultural Resources Management will be followed for this project.

D. If previously unknown cultural (archeological) resources are encountered during construction, the Contractor shall immediately stop all work in the immediate vicinity of the discovery prior to any further disturbance and notify the Cultural Resource Program Manager, followed by notification to the COR. Work shall not resume in the affected area until the Cultural Resource Program Manager has had sufficient time to investigate, document, and evaluate the discovery, determine appropriate treatment measures, complete any required consultation, and approve resumption of work.

E. Contractor personnel, subcontractor personnel, and all other persons not qualified and expressly authorized by the Cultural Resource Program Manager shall be prohibited from entering or disturbing any archeological site and from collecting, handling, moving, or removing artifacts or other cultural materials. Prohibited materials include, but are not limited to, historic bottles, cans, projectile point, and other artifact types.

F. If human remains are encountered at any time, all work in the immediate vicinity of the discovery shall cease immediately, prior to any further disturbance. The Contractor shall immediately notify the Cultural Resource Program Manager. If an archeologist is not present, the Contractor shall report the location and nature of the discovery to the Cultural Resource Program Manager. Access to the discovery area shall be limited to personnel authorized by the Cultural Resource Program Manager, and all other construction personnel shall leave the area. Human remains shall be treated with respect and dignity at all times. No photographs, video, drawings, sketches, or other visual documentation. Discussion of the discovery except as necessary to notify appropriate staff and implement required procedures, shall be limited. Work shall not resume in the affected area until the Cultural Resource Program Manager authorizes resumption of work.

G. Activities will be conducted in a way that will not damage or move inventoried archeological or paleontological resources. If concealed paleontological resources are encountered during project activities, all necessary steps will be taken to protect them.

H. Although there is no surface evidence of archeological or paleontological resources, if concealed archeological or paleontological resources are encountered during project activities, all necessary steps will be taken to protect them, and an appropriate mitigation strategy will be developed.

I. Buildings and Structures: Preserve and protect all historic buildings and structures within the project area. The Contractor shall prepare a protection plan for any Work that may affect historic buildings or structures. The protection plan shall be reviewed by the GRCA Historical Architect, the GRCA Cultural Resource Program Manager, and other appropriate staff before such Work begins.

3.2 BARRICADES

A. Comply with requirements specified in Section 01 50 00 "Temporary Facilities and Controls."

3.3 ARCHEOLOGICAL OR PALEONTOLOGICAL INVESTIGATION BY NON-NPS PERSONNEL

A. Archeologic or paleontological investigations by non-NPS or unqualified personnel (NPS or non-NPS) is prohibited. All access to investigate, document, access, archeological resources must be approved by the Cultural Resource Program Manager.

3.4 OBSERVATION

A. An Archeological/Paleontological Monitor shall be present during all ground-disturbing activities unless monitoring is determined to be unnecessary by the Cultural Resource Program Manager. The Archeological/Paleontological Monitor, in coordination with the Contractor's field crew, shall establish a safe means of observing active ground-disturbing work while maintaining adequate visibility of the area being disturbed.

3.5 DISCOVERY OF RESOURCES

A. If Archeological/Paleontological Monitor discovers resources, immediate relocation of work to a non-sensitive area may be required to allow the Monitor time to identify, consult, and document resources and, if necessary, develop, and implement, an appropriate mitigation plan. Contractor shall relocate work to non-sensitive areas where monitoring is not normally sensitive required.

B. If resources are discovered while Archeological/Paleontological Monitor is absent, stop work immediately and report the discovery to the Contracting Officer.

3.6 WORK STOPPAGE

A. Contractor shall plan, schedule, and execute work to prevent stoppages at one area from stopping all work at construction site.

END OF SECTION 01 35 13.22

SECTION 01 35 23 - SAFETY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes establishing an effective accident prevention program and providing a safe working environment for personnel and visitors.

1.2 SUBMITTALS

A. Accident Prevention Plan (APP): Submit APP after contract award and before Pre-Construction conference. Contracting Officer (CO) will review proposed APP. If APP requires any revisions or corrections, Contractor shall resubmit Plan within 10 days. No progress payments will be made until the APP is accepted.

1.3 QUALITY ASSURANCE

A. Comply with contract clauses "Accident Prevention" and "Permits and Responsibilities." In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Onsite equipment shall meet 29 CFR 1926 (Code of Federal Regulations) (Occupational Safety and Health Administration (OSHA)) requirements. Failure to comply with requirements of this section and related sections may result in suspension of work.

B. Site Safety Supervisors:

1. Designate authorized onsite representatives for preparation and maintenance of the APP.
    a. Designate Lead Site Safety Supervisor
    b. Designate Innercanyon Site Safety Supervisor in support of the Lead Site Safety Supervisor
    c. Lead and Innercanyon Site Safety Supervisors shall have received Wilderness Safety Training and Wilderness First Aid Training.
    d. Aviation Safety Manager
2. Shall be responsible for:
    a. Implementation and enforcement of the APP
    b. Daily safety inspections
    c. Conducting and documenting weekly and monthly safety meetings
    d. Review of safety requirements at progress meetings
    e. Compilation and maintenance of Safety Data Sheets (SDS) and safety reference materials
    f. Tracking and resolution of safety violations
    g. Site personnel and visitor compliance with site safety and health requirements and APP
    h. Investigation and reporting of accidents and injuries

C. Qualifications of Employees:

1. Physically and able to perform their assigned duties in a safe manner.

2. Do not allow employees whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.
3. Provide operating instructions for equipment. Operators of vehicles, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.
4. All employees for Innercanyon work must be able to hike from Cottonwood to the North Rim on foot. There may be instances where helicopter flights are not available or not feasible due to weather. This might be necessary during summer conditions where extreme heat is common in the Innercanyon.
5. All employees for inner canyon work must be able to stay in construction camps for multiple nights.

1.4 ACCIDENT REPORTING

A. Reportable Accidents: Defined as: death, occupational disease, and/or traumatic injury to employees or the public; fires; and/or property damage by accident in excess of $100.

1. Notify Contracting Officer immediately in the event of a reportable accident.
2. Fill out and forward an Accident/Property Damage Report Form (CM-22) to Contracting Officer within 7 days of a reportable accident. Obtain form from Contracting Officer.

B. Near Miss: Contractor to report all near-miss incidents (including minor incidents).

1. An employee who witnesses a near miss incident must complete the Near Miss Reporting form and submit to a Manager or Human Resources. The reporting system will not result in disciplinary action of the reporter and, if desired by the person reporting, may be anonymous;
2. All near miss incidents will be reviewed by company management to identify the root cause and the weaknesses in the system contributing to the incident. The reporting individual may be asked to participate in the incident investigation; and
3. Investigation results will be used to improve safety systems, hazard control, risk reduction, and to educate employees. All of these represent opportunities for training, feedback on performance, and a commitment to continuous improvement.

PART 2 - PRODUCTS

2.1 ACCIDENT PREVENTION PLAN (APP)

A. APP shall be written to comply with OSHA and project requirements (generic plan is not acceptable) including but not limited to:

1. Name and qualifications of supervisor responsible to carry out program.
2. Weekly and monthly safety meetings shall be documented with topics and attendees.
3. First aid and rescue procedures.
4. Job Hazard Analysis (JHA) for each major phase. List of hazards associated and methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Include initial and continuing training.
5. Planning for possible emergency situations, as detailed in Article 1.2. Such planning shall take nature of construction, site conditions, and degree of exposure of persons and property into consideration.

6. Heat stress management plan.
7. Infectious Disease Preparedness:
    a. Contractors are responsible for their employees' safety and the safety of job site visitors during the performance of this contract. We encourage Contractors to follow guidance from the Department of Labor (DOL), Occupational Safety and Health Administration (OSHA), the Centers for Disease Control and Prevention (CDC), and all other applicable local, city, and state mandates. We encourage Contractors to develop policies for infection prevention and an Infectious Disease Preparedness and Response Plan.
    b. To the extent appropriate, Contractors should include the protective health and safety measures they intend to implement in any accident prevention or safety submittals required under this contract. These plans should contain preventive measures the Contractor intends to follow while performing work on government property as well as responsive and corrective actions to be taken if an employee exhibits symptoms or tests positive for contagion.
    c. Upon contract award, Contractors should communicate with Contracting Officer regarding Contractor decisions and actions to protect the health and safety of workers for the duration of contract performance under which pandemic conditions exist.

2.2 FIRST AID FACILITIES

A. Provide adequate facilities for number of employees and appropriate to construction hazards.

B. Contractor first aid facilities shall be established in locked contractor provided job box at Manzanita Ranger Station, Cottonwood Campground, Roaring Springs Pumphouse, and Supai Tunnel.

2.3 PERSONNEL PROTECTIVE EQUIPMENT (PPE)

A. Selection shall conform to OSHA Subpart E.

PART 3 - EXECUTION

3.1 DAILY SAFETY INSPECTIONS

A. Conduct daily safety inspections and maintain daily safety reports which include:

1. Area/operation inspected
2. Date of inspection
3. Identified hazards
4. Corrective actions taken

3.2 EMERGENCY INSTRUCTIONS

A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.

3.3 FIRE AND LIFE SAFETY

A. Comply with requirements of 2024 International Fire Code Chapter 33.

3.4 HAZARDOUS MATERIALS

A. Hazardous materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.

B. Store hazardous materials in accordance with manufacturer’s and OSHA Subpart D requirements. Maintain Safety Data Sheets (SDS) for each chemical readily available on site.

1. Immediately report spills of hazardous materials to the Park.
2. Maintain a spill emergency response kit.
3. Train employees how to respond to a spill and use emergency response kit.

3.5 PROTECTIVE EQUIPMENT

A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing to another individual.

3.6 SAFETY MEETINGS

A. As a minimum, conduct one weekly 15-minute "toolbox" safety meeting conducted by a foreman or supervisor and attended by construction personnel at worksite. Topics shall coincide with work scheduled for following week. Document and submit meeting minutes to Contracting Officer within one day after meeting.

B. Conduct monthly safety meetings for personnel, contractors, and subcontractors performing work on the site. Notify Contracting Officer of meeting dates and times. Meetings shall be used to: review effectiveness of Contractor's safety effort; resolve current health and safety problems; provide a forum for planning safe construction activities, and for updating Accident Prevention Plan. Contracting Officers Representative will attend meetings and enter results of meetings into the daily log.

3.7 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

A. A hard hat use area shall be designated by Contractor. Hard hat area shall be posted by Contractor in a manner satisfactory to Contracting Officer.

B. It is Contractor's responsibility to require persons working on or visiting site to wear hard hats and PPE in good repair at all times. As a minimum, maintain six hard hats and other APP required equipment.

3.8 TRAINING

A. First Aid: Provide and document training to personnel to ensure prompt and efficient first aid and CPR.

B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage.

C. Contractor staff shall be briefed by Park SAR Supervisor on heat/cold injuries prior to starting innercanyon work.

D. Contractor staff shall be briefed by Park Helibase Supervisor on airspace, helibase (south and north rim), landing zone, aviation corridor safety protocols.

END OF SECTION 01 35 23

This page intentionally left blank.

SECTION 01 35 73 - DELEGATED DESIGN PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. General provisions for delegated design services.
2. Coordination of delegated designs with other Work.
3. Qualifications requirements for delegated design professionals.
4. Limitations on Contracting Officer's review of delegated design Submittals.
5. Responsibilities of delegated design professionals.

B. Scope:

1. Where delegated design is specifically Contractor's responsibility in accordance with the Contract Documents, Contractor shall provide labor, services, other effort, and pay all costs necessary and required to perform delegated design services for Work that will be part of the completed Project as a functioning whole.
2. Perform delegated design Work in accordance with the Contract Documents, delegated design Action Submittals approved by Contracting Officers Representative, and Shop Drawings, product data Submittals, and Samples approved by the associated delegated design professional.
3. Contractor's correction period, general warranty and guarantee, and obligations for safety and protection apply to delegated design Work to the same extent such provisions apply to all other Work under the Contract.
4. Specifications requiring delegated design services include, but are not necessarily limited to, the following:
   a. Section 03 15 19 – Anchorage to Concrete.
   b. Section 03 41 33 – Precast and Prestressed Concrete.
   c. Section 05 21 00 – Steel Joists.
   d. Section 05 50 00 – Metal Fabrications.
   e. Section 05 52 46 – Mechanically Fastened Aluminum Railings.
   f. Section 07 84 00 – Firestopping.
   g. Section 09 22 16 – Non-Structural Metal Framing.
   h. Section 26 41 13 – Lightning Protection System.
   i. Section 32 32 16 – Precast Modular Block Retaining Walls.
   j. Section 40 05 07 – Pipe Support Systems.
   k. Section 40 41 13 – Heat Tracing Cable
   l. Section 43 41 15 – Field Erected Welded Steel Tanks.
   m. Section 46 13 15 – Aluminum Covers – Formed Panel Type
5. Not Delegated Design: The following are not delegated design and are not covered by this Section:
   a. Contractor's use of design professionals for: (1) temporary construction or temporary facilities not part of the completed Project as a functioning whole, or (2) Contractor's means, methods, procedures, techniques, and sequences of construction and safety and protection measures incident thereto.
   b. Certain final designs that, in accordance with commonly accepted practice, are typically prepared by unlicensed, unregistered individuals, including for manufactured or fabricated systems, components or assemblies, not acting under the supervisory control of the design professional in responsible charge,

but who commonly possess appropriate certification from a relevant industry organization, together with appropriate training and experience.

C. Related Requirements:

1. Sections of Divisions 02-46 where delegated design Work is required.

1.2 REFERENCES

A. Terminology:

1. Terminology indicated below are not defined terms and are not indicated with initial capital letters but, when used in this Section and Specifications of Division 02-46 where delegated design Work is required, have the meaning indicated below:
   a. "Delegated design" means preparing the final design of part of the completed, permanent Work by one or more delegated design professionals, in accordance with the Contract Documents. The terms "delegated design", "delegated design services", "delegation of design responsibility", and similar or derivative terms have the same meaning.
   b. "Delegated design professional" means the licensed and registered engineer, architect, geologist, or other design professional retained by or employed by Contractor, Subcontractor, or Supplier to perform delegated design services for delegated design Work and possessing appropriate experience and qualifications for such delegated design services.
   c. "Delegated design Work" means delegated design services, associated construction, and related Work.
   d. "Instruments of service", relative to delegated designs, means delegated design professional's: (1) certifications (including delegated design professional's certification of compliance, as required in this Section, and other certifications required of delegated design professional), (2) reports (where required), (3) design drawings, (4) design specifications, (5) other documents specifically indicated as delegated design professional's "instruments of service" in the Contract Documents, and (6) documents modifying a delegated design (after Contracting Officer's Representative approval of the original delegated design Submittals). "Instruments of service" are to be sealed, signed, and dated by delegated design professional and expressly required as Submittals. Shop Drawings sealed and signed by delegated design professional are delegated design professional's "instruments of service".

1.3 GENERAL PROVISIONS CONCERNING DELEGATED DESIGN SERVICES

A. Delegated Designs - General:

1. This Section augments the requirements of the General Conditions, as may be amended by the Supplementary Conditions, and other provisions of the Contract Documents regarding Contractor's responsibilities for delegated design Work.
2. Delegated design professionals or their employer shall furnish professional liability insurance. Provisions on professional liability insurance are set forth in the Division 1 Specifications.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordination:

1. Coordination - General:
    a. Contractor shall coordinate the services of delegated design professionals with all other elements of the Work.
    b. Contractor has full responsibility for scheduling delegated designs and all related Work.
    c. Allow sufficient time in Progress Schedule for performance of delegated design services, including requests for interpretation or clarification between delegated design professional and Contractor and between Contractor and Contracting Officer.
2. Coordination of Delegated Design Work's Connections to Other Work:
    a. Where delegated design Work connects to other Work designed by Contracting Officer's Representative, existing construction, or both, the delegated design Work shall be consistent with the other Work and existing construction to which delegated design Work connects, and adjacent construction.
    b. Submit details, loading, anchorage, and other coordinating information necessary for the delegated design Work to properly interface with Work designed by Contracting Officer's Representative.
    c. Changes in the Work, whether designed by Contracting Officer's Representative, designed by delegated design professional, or existing construction, necessary as a result of the delegated design are ineligible for increase in Contract Price or Contract Times, unless: (1) otherwise agreed by Contracting Officer, or (2) expressly indicated otherwise elsewhere in the Contract Documents for the associated delegated design Work.
    d. Changes requiring extra compensation, time, or both arising from delegated design aspects needed for convenience of Contractor, Subcontractor, or Supplier, are not grounds for increase in Contract Price or Contract Times.
3. Coordination of Submittals, Fabrication, Production, and Shipment:
    a. Do not release for raw materials procurement, fabrication, production, and shipment to the Site materials, equipment, or systems designed by delegated design professional until the associated delegated design professional has reviewed and approved all associated Shop Drawings, product data, Samples, and (relative to shipment) source quality control Submittals, and such Submittals have been delivered to and accepted by the Contracting Officer.
        1) For delegated design systems that required reactions to be submitted to the Contracting Officer. These submittals shall be submitted and approved first before approval is given for the delegated design submittal.
    b. Allow sufficient time in the Progress Schedule for required Submittals and required actions by delegated design professionals and Contracting Officer's Representative.

1.5 QUALITY ASSURANCE

A. Qualifications:

1. Delegated Design Professionals:
    a. Each delegated design professional shall possess not less than the minimum qualifications set forth in this provision. Where the Specifications for the associated delegated design Work establish more-stringent qualifications requirements, comply with the more-stringent requirements.

b. Each delegated design professional shall comply with all of the following:
   1) Legally qualified, as both an individual and as a business entity, to practice the associated design discipline(s) in the jurisdiction where the Site is located, including possessing current, valid license and registration for the design discipline(s) for which the delegated design professional will render its services on the Project.
   2) Possess not less than five years of experience in the subject design discipline(s).
   3) Served as design professional in responsible charge on not less than five other designs similar in scope and complexity to the Work for which delegated design professional is retained on the Project; construction of such prior projects shall be complete by the start of the Project's construction.

c. Summary of Qualifications: Submit to Contracting Officer summary of delegated design professional's experience and qualifications, including:
   1) Evidence of coverage under appropriate professional liability insurance in accordance with the Contract Documents.
   2) Evidence of delegated design professional's ability to legally conduct business as a design professional in the same jurisdiction as the Site, as a business entity.
   3) Copy of delegated design professional's current, valid personal design professional license and registration for the same jurisdiction as the Site. Such documents shall indicate the individual's name, license or registration number, and dates for which the license or registration is valid.
   4) Other information reasonably requested by Contracting Officer.

1.6 GENERAL PROVISIONS FOR DELEGATED DESIGN SUBMITTALS

A. Under the Division 02-46 Specifications section(s) where delegated design Work is required, furnish to Contracting Officer Submittals such as:

1. Action Submittals:
   a. Delegated design professional's instruments of service Submittals.
2. Informational Submittals:
   a. When delivered to Contracting Officer, the following must bear delegated design professional's Submittal approval stamp:
      1) Shop Drawings, product data Submittals, Samples, testing plans.
      2) Results of source quality control and field quality control activities.
   b. Delegated design professional's calculations.
   c. Reports of delegated design professional's visits to the Site, bearing delegated design professional's dated signature.
   d. Other Informational Submittals required for the subject delegated design Work.

B. Limitations of Contracting Officer's Review of Delegated Design Submittals:

1. Delegated Design Professional's Instruments of Service Submittals:
   a. Contracting Officer's review of delegated design instruments of service Submittals is for the limited purposes indicated herein.
   b. The following disclaimer applies to Contracting Officer's responses to delegated design professional's instruments of service Submittals:
      1) Contracting Officer's review and approval of delegated design instruments of service is only for the limited purpose of verifying that

performance and design criteria given in the Contract were used in the delegated design, and checking for compliance with the Contracting Officer's Representative design concept expressed in the Contract Documents.

2) Contractor is solely responsible for complying with the Contract Documents, Subcontractor and Supplier instructions consistent with the Contract Documents, Owner's directions, and Laws and Regulations.
3) Contractor is solely responsible for obtaining, correlating, confirming, and correcting dimensions at the Site; quantities; information and choices pertaining to fabrication processes; means, methods, sequences, procedures, and techniques of construction; safety precautions and programs incident thereto; and for coordinating the Work of all trades.
4) The Contracting Officer is not responsible for the effects of resubmittals or tracking progress of resubmittals.

2. Delegated Design Informational Submittals:
   a. Other provisions of the Contract Documents notwithstanding, Contracting Officer's review of delegated design Informational Submittals is limited to only:
      1) Verifying the Submittal was furnished as required; and
      2) Submittal generally appears complete (except for calculations); and
      3) Submittal bears delegated design professional's approval stamp; or, for calculations prepared by or for delegated design professional, that such calculations bear delegated design professional's seal, signature, and date; or, for delegated design professional's reports of visits to the Site, that such report is legible, and bears delegated design professional's signature with date.
   b. Contracting Officer receives such Submittals, including delegated design professional's calculations, on behalf of Owner, for Owner's records.
   c. Contracting Officer and others involved in the Project have the right to rely on delegated design professional's approval stamp as meaning that the delegated design professional has performed and appropriate review of the Submittal and determined it to be complete, in accordance with delegated design professional's instruments of service approved by Contracting Officer's Representative, in accordance with delegated design professional's design intent, and in accordance with the Contract Documents.
3. Contracting Officer Representative's Other Comments on Delegated Design Submittals:
   a. Despite the limitations of Contracting Officer's review of Submittals for delegated design Work, should Contracting Officer become aware of, or reasonably suspect existence of, potential of associated delegated design Work to adversely affect health, safety, or welfare of persons, or pose reasonable potential for damage to the Work, work of other contractors, or adjacent property, Contracting Officer's will advise Contractor in writing of general nature of Contracting Officer's concern.
   b. Such advisory by Contracting Officer, if issued, is rendered in good faith and does not in any way constitute:
      1) Contracting Officer's review of all aspects of the delegated design.
      2) Any sharing by Contracting Officer of any of delegated design professional's responsibilities or professional liability.
      3) Any responsibility imposed, in any way, on Contracting Officer for any aspect of the delegated design professional's services or design, beyond the limited purposes of Contracting Officer's review as set forth in the Contract Documents.

c. Contractor and its Subcontractors and Suppliers, including delegated design professionals, shall immediately investigate Contracting Officer's concern indicated in such advisory and remedy as necessary and required.
d. Neither Contracting Officer nor Owner, nor their respective consultants and subcontractors, is obligated to review any Submittal for delegated design Work beyond the limited review required by the Contract Documents. No such advisory, if issued, entitles Contractor, Subcontractor, or Supplier, including delegated design professionals, to rely on such advisory or to assume that any further such reviews or written or oral advisories are forthcoming.

1.7 RESPONSIBILITIES OF DELEGATED DESIGN PROFESSIONALS

A. Standard of Care:

1. Unless a higher standard of care is established by the Division 02-46 Specifications section where the associated delegated design Work is required, the delegated design services shall comply with the following standard of care:
   a. Except as provided in the paragraph immediately above this, the standard of care for all delegated design professional services and related services performed or furnished by delegated design professionals for the Project will be the care and skill ordinarily used by members of the subject profession practicing under similar circumstances at the same time and in the same locality.

B. Responsibilities of delegated design professionals employed on the Work include, but are not necessarily limited to, the following, unless specifically indicated otherwise in the associated elements of the Contract Documents where the delegated design is required:

1. Ethical Conduct and Professionalism: Comply with Laws and Regulations and applicable standards and guidelines relevant design professional organizations for ethical conduct and professional practice.
2. Comply with Laws and Regulations and relevant design standards applicable to the subject delegated design Work.
3. Performance and Design Criteria Indicated in the Contract Documents and Other Information:
   a. Review performance and design criteria, indicated in the Contract Documents, that the delegated design Work must satisfy.
   b. Prepare written requests for interpretations or clarifications of performance or design criteria.
   c. Review existing information about the Site that constitutes Technical Data (if any, applicable to the subject delegated design Work), as indicated in the Supplementary Conditions.
4. Site Information and Investigations: With Contractor, obtaining all other necessary dimensions, field information, and other information necessary for preparing delegated design Submittals.
5. Design and Other Professional Services: Personally perform and prepare, or actively exercise direct, personal, supervisory control over others performing or preparing:
   a. Necessary design professional evaluations of conditions, materials, and equipment.
   b. Prepare the instruments of service Submittals and calculations Submittal for the subject delegated design Work, where required by the associated Division 02-46 Specifications and other, associated Contract Documents.

c. Assist Contractor with applying for and obtaining permits and approvals (not previously obtained by Owner or those for whom Owner is responsible) necessary for the delegated design Work.

d. Review and approve or take other appropriate action on Shop Drawings (unless such Shop Drawings are sealed and signed by delegated design professional), product data, Samples, and testing plans, and other Submittals associated with the delegated design Work.

e. Prepare modifications of the delegated design instruments of service as necessary.

6. Sealing and Signing:
   a. Seal, sign, and indicate date of sealing and signing, on all of the following when such Submittals are required by the Division 02-46 Specifications where the delegated design Work is required:
      1) Instruments of service Submittals, including certification of compliance required.
      2) Calculations.
      3) Modifications to the delegated design.
      4) Other documents required to be sealed and signed by Laws or Regulations or the Contract Documents.
   b. Sealing and signing documents in accordance with Laws and Regulations and the Contract Documents, prior to submittal (through Contractor) to Contracting Officer, and for submittal to authorities having jurisdiction to obtain necessary permits and approvals.
7. Certification of Compliance by Delegated Design Professional:
   a. Schedule:
      1) Submit certification of compliance after Contracting Officer's acceptance of delegated design professional's qualifications statement.
      2) Obtain Contracting Officer's approval of certificate of compliance Submittal prior to furnishing other Submittals for delegated design Work under the same Specifications section, unless otherwise allowed by Engineer.
   b. Through Contractor, submit to Contracting Officer, delegated design professional's written certification indicating:
      1) General Information: (1) Project name and designation, (2) Contractor name and Contract designation, (3) Subcontractor or Supplier name (when applicable), (4) full name of delegated design professional's business entity under which the delegated design services were performed, (5) full name and license number of the individual sealing and signing the subject delegated design Work, (6) specific elements of delegated design Work to which the certification applies, and (7) delegated design professional's seal, signature, and date of signature.
   c. Explicit certification that the subject delegated design complies with:
      1) All applicable performance and design criteria indicated in the Contract Documents. Expressly indicate on certification of compliance the specific performance and design criteria used in the delegated design, and reaction forces of the delegated design imparted to other Work and existing construction. Reaction forces imparted from the delegated design elements to the Contracting Officer Representative's designed system shall include the following:
         a) Reaction forces imparted from the delegated design elements to the Contracting Officer Representative's designed system shall be presented as follows:
            (1) Unfactored loads per category (dead, live, wind, seismic, etc.).
            (2) Load combinations presented in Load Factor Resistance Design (LRFD) format from each element transmitting load.

b) All Laws and Regulations.
c) Applicable design standards commonly applicable to such types of construction. Expressly indicate such design standards on the certification of compliance.
d) The applicable standard of care. Expressly indicate the applicable standard of care.

8. Approvals of Other Delegated Design Submittals:
   a. Review and taking appropriate action on Submittals for delegated designs.
   b. Such reviews and approvals or other appropriate action shall be to ascertain compliance with:
      1) Delegated design professional's design intent.
      2) Delegated design professional's instruments of service and calculations.
      3) Associated requirements of the Contract Documents.
   c. Delegated design professional's review stamp or facsimile thereof, review action or disposition concerning the associated Submittal for the delegated design, date of review, and name of person performing the review shall be clearly legible on the associated Submittals (except for delegated design professional's own instruments of service Submittals, calculations, and reports of delegated design professional's visits to the Site). Prominently display delegated design professional's Submittal review stamp or facsimile thereof on: (1) each sheet of Shop Drawings, (2) each major section of product data Submittals, (3) each Sample, (4) each testing plan, and (5) each other Submittal associated with the delegated design for which such review stamp is required.
   d. Do not apply delegated design professional's Submittal review stamp and comments, if any, over other text, tables, or graphics.
   e. Where review stamp or facsimile thereof is required, submit to Contracting Officer only those Submittals for delegated design Work that bear delegated design professional's explicit approval of the Submittal.
9. Respond promptly to requests for interpretation or clarification on delegated design professional's instruments of service and other Submittals for the delegated design Work.
10. Progress and Quality of Construction of Delegated Design Work:
    a. Where appropriate for the subject delegated design Work, periodically visit the Site at appropriate intervals to observe the progress and quality of the subject delegated design Work.
    b. Where delegated design professional does not visit the Site during construction, keep informed of the progress and quality of the subject delegated design Work via discussions with Contractor, Subcontractor, and Suppliers, via photographic documentation, and other means acceptable to delegated design professional.
    c. Advise Contractor in writing when the subject delegated design Work is not in accordance with the delegated design professional's instruments of service (approved by Contracting Officer) and related Submittals approved by delegated design professional.
    d. Furnish to entity that retained delegated design professional copy of delegated design professional's written report of each visit to the Site.
11. Modifications to Design:
    a. Design appropriate modifications to the delegated design Work, including preparing new or revised certifications, reports, design drawings, sketches, design specifications, and calculations, as appropriate.
    b. Such instruments of service and calculations shall be submitted to Contracting Officer through Contractor to same extent original instruments of service Submittals and calculations, if any, where required by the Contract Documents for the subject delegated design Work.

12. Other services, as mutually agreed upon by delegated design professional and its client, or as required elsewhere in the Contract Documents.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION - (NOT USED)

END OF SECTION 01 35 73

This page intentionally left blank.

SECTION 01 35 91 - HISTORIC PRESERVATION TREATMENT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes special procedures for historic treatment on Project including, but not limited to:

1. Definitions
2. Submittals
3. Quality Assurance
4. Storage and protection of existing historic materials
5. Project site conditions
6. Historic Preservation Treatment Plan
7. Protection, General
8. Protection during application of chemicals
9. Protection during use of heat-generating equipment
10. Historic preservation treatment procedures

1.2 DEFINITIONS

A. "Rehabilitation" - To make possible a compatible use for property through repair, alterations, and additions while preserving those portions or features that convey its historical, cultural, or architectural values.

B. "Restoration" - To accurately return form, features, and character of a property to its appearance at a particular period of time by means of removal of features from other periods in its history and repair and reconstruction of missing and deteriorated features from the restoration period.

C. "Reconstruction" - To reproduce in exact form and detail, a building, structure, or artifact as it appeared at a specific period in time. Reconstructed elements do not possess historic integrity in their own right since they are-not original fabric.

D. "Protect and Maintain" - To remove deteriorating corrosion, reapply protective coatings, and install protective measures such as temporary guards; to provide the least degree of intervention.

E. "Repair" - To stabilize, consolidate, or conserve; to retain existing materials and features while employing as little new material as possible. Repair includes patching, piecing-in, splicing, consolidating, or otherwise reinforcing or upgrading materials. Within restoration, repair also includes limited replacement in kind, rehabilitation, and reconstruction, with compatible substitute materials for deteriorated or missing parts of features when there are surviving prototypes.

F. "Replace" - To duplicate in its entirety, a historic element or feature by matching its historic pattern, detail and appearance. Replacement is justified when original or historic elements are damaged beyond repair or are missing. Replacement conditions and methods include:

1. Replacement with Original or Historic Fabric: Includes fabric salvaged from other locations or projects having identical architectural qualities. Duplication of appearance using identical material possessing historical significance.
2. Replacement with New Materials: Includes replacement with new material of like kind (custom fabricated of manufactured). Duplication of appearance using like material.

3. Replacement with Substitute Materials: Includes replacement with a compatible substitute that is frequently contemporary and unlike the historic fabric. Duplication of appearance using modern (non-traditional) material Use of substitute materials is not approved unless matching materials are not available.

G. "Remove" - To demolish or detach items from existing construction and legally dispose of them off-site unless indicated to be removed and salvaged or removed and reinstalled.

H. "Remove and Reinstall" - To detach items from existing construction, repair and prepare for reuse, and reinstall where indicated.

I. "Existing to Remain" or "Retain" - Existing items of construction not to be removed and not otherwise indicated to be removed and salvaged or removed and reinstalled.

1.3 SUBMITTALS

A. Historic Preservation Treatment Plan: Park to prepare a written Historic Preservation Treatment Plan (HPTP).
   1. Contractor shall abide by the requirements developed and provided in the plan.

B. Alternative Methods and Materials: If alternative methods and materials to those indicated are proposed for any phase of work, provide a written description including evidence of successful use on other, comparable projects, and program of testing to demonstrate effectiveness for use on this Project.

C. Photographs or Videotape: In accordance with Section 01 32 33 "Photographic Documentation," show existing conditions of adjoining construction and site improvements including finish surfaces that might be misconstrued as damage caused by historic treatment operations. Submit before work begins.

1.4 QUALITY ASSURANCE

A. Historic Preservation Treatment Specialist Qualifications: Experienced firm with required certifications and training able to demonstrate through past performance they are qualified to perform this work.

1.5 STORAGE AND PROTECTION OF HISTORIC MATERIALS

A. Removed and Salvaged Historic Materials:

   1. Clean salvaged historic items.
   2. Pack or crate items after cleaning. Identify contents of containers.
   3. Store items in secure area until delivery to Contracting Officer or designated representative.
   4. Transport items to storage area indicated on Drawings.
   5. Protect items from damage during transport and storage.
   6. Do not dispose of items removed from existing construction without prior written consent of Contracting Officer.

B. Removed and Reinstalled Historic Materials:

   1. Clean and repair historic items to functional condition adequate for intended reuse.
   2. Pack or crate items after cleaning and repairing. Identify contents of containers.

3. Protect items from damage during transport and storage.
4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use as designed.

C. Existing Historic Materials to Remain: Protect construction indicated to remain against damage and soiling during historic treatment. When permitted by Contracting Officer, items may be removed to suitable, protected storage location during historic treatment and reinstalled in their original locations after historic treatment operations are complete.

D. Storage and Protection: When removed from existing location, store historic materials within weather-tight enclosure protected from wetting by rain, snow, or ground water, and temperature variations. Secure stored materials to protect from theft.

1. Identify removed items with an inconspicuous mark indicating original location.
2. Develop key plan when many similar items are scheduled for removal and reinstallation.

1.6 PROJECT-SITE CONDITIONS

A. Exterior Cleaning and Repairing:

1. Proceed with work only when forecasted weather conditions are favorable.
   a. Wet Weather: Do not attempt repairs during rainy or foggy weather. Do not apply primer, paint, putty, or epoxy when relative humidity is above 80 percent. Do not remove exterior elements of structures when rain is forecast or in progress.
   b. Do not perform exterior wet work when air temperature is below 40 degrees Fahrenheit.
   c. Do not begin cleaning, patching, or repairing given likelihood of frost or freezing.
   d. Do not begin cleaning when either air or surface temperature is below 45 degrees Fahrenheit unless approved means are provided for maintaining 45 degrees Fahrenheit temperature of air and materials during, and for 48 hours subsequent to, cleaning.

B. The National Park Service and the concessionaire will occupy portions of building immediately adjacent to historic treatment area. Conduct historic treatment so National Park Service operations will not be disrupted. Provide hours' notice to Contracting Officer of activities that will affect National Park Service operations.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 HISTORIC PRESERVATION TREATMENT PLAN

A. Park shall prepare a written plan for preservation work covering preservation components of project. Plan shall verify construction strategy and intent is compatible with Department of the Interior’s (DOI) standards for Treatment of Historic Properties, guidelines for Treatment of Cultural Landscapes, and National Park Service management policies for cultural resources. Plan shall satisfy both project scope and resource protection requirements. Plan will include the following:

1. Organized list of preservation components of project, systems, and tasks
2. Staging and sequence of work
3. Disassembly and reassembly techniques and steps
4. Equipment and tools required
5. Supplies and materials with manufacturer or supplier identified including specific clean up/storage procedure including frequency and documentation of such.
6. Skilled trades and crafts required
7. Anticipated testing and analysis of fabric
8. Additional investigations for extents or magnitude of treatments needed
9. Protective measures
10. Seasonal limitations on work
11. Alternative means if primary treatment method is unfeasible
12. Work conducted off-site (Approval from Contracting Officer required prior to taking resources off-site).

3.2 PROTECTION, GENERAL

A. Comply with manufacturer's written instructions for precautions and effects of products and procedures on adjacent building materials, components, and vegetation.

B. Ensure supervisory personnel are present when work begins and during progress.

C. Temporary Protection of Historic Materials during Construction:

1. Protect existing materials during installation of temporary protections and construction. Do not deface or remove existing materials.
2. Attachments of temporary protection to existing construction shall be approved by Contracting Officer prior to installation.

D. Protect landscape work adjacent to or within work areas as follows:

1. Provide barriers to protect tree trunks.
2. Bind spreading shrubs.
3. Coverings shall allow plants to breathe. Remove coverings at end of day. Do not cover plant material with waterproof membrane more than 8 hours at a time.
4. Set scaffolding and ladder legs away from plants.

E. Existing Drains: Prior to start of work or cleaning operations, test drains and other water removal systems to ensure drains and systems function properly. Notify Contracting Officer immediately of stopped or blocked drains or systems. Do not begin Work of this Section until drains are in working order.

1. Provide method to prevent solids including stone or mortar residue from entering drains or drain lines. Clean out drains and drain lines blocked or filled because of work performed under this Contract.
2. Protect storm drains from pollutants. Block drains or filter out sediments, allowing only clean water to pass.

3.3 PROTECTION DURING APPLICATION OF CHEMICALS

A. Protect persons, motor vehicles, surrounding surfaces of building being restored, building site, plants, and surrounding buildings from harm or damage resulting from applications of chemical cleaners and paint removers.

B. Comply with requirements in Section 01 50 00 "Temporary Facilities and Controls."

C. Cover adjacent surfaces with materials proven to resist chemical cleaners selected for Project unless chemicals being used will not damage adjacent surfaces. Use covering materials containing only waterproof, UV (ultraviolet)-resistant adhesives. Apply masking agents to comply with manufacturer's written instructions. Do not apply liquid masking agent to painted or porous surfaces. Promptly remove masking to prevent adhesive staining on completion.

D. Do not clean surfaces during winds of sufficient force to spread cleaning solutions to unprotected surfaces.

E. Neutralize and collect alkaline and acid wastes and dispose of outside park boundaries.

F. Dispose of runoff from chemical operations by legal means and in a manner preventing soil erosion, undermining of paving and foundations, damage to landscaping, and water penetration into building interiors.

G. Document daily clean-up, proper storage, and disposition of any flammable/combustible materials with daily reports.

3.4 PROTECTION DURING USE OF HEAT-GENERATING EQUIPMENT

A. Comply with following procedures while performing work with heat-generating equipment, including welding, cutting, soldering, brazing, paint removal with heat, and other operations where open flames or implements utilizing heat are used:

1. Obtain Contracting Officer’s approval for operations involving use of open-flame or welding equipment.
    a. Notification shall be given for each occurrence and location of work with heat-generating equipment.
    b. Obtain appropriate permit from the park as required.
2. As far as practical, use heat-generating equipment in shop areas or outside building.
3. Before work with heat-generating equipment commences, furnish fire watch (or watches) for location(s) where work is to be performed.
4. Do not perform work with heat-generating equipment in or near rooms or in areas where flammable liquids or explosive vapors are present or thought to be present. Use combustible gas indicator test to ensure area is safe.
5. Remove and keep area free of combustibles, including, rubbish, paper, waste, etc., within area of operations.
    a. If combustible material cannot be removed, provide fireproof blankets to cover such materials.
6. Where possible, furnish and use baffles of metal or gypsum board to prevent spraying of sparks or hot slag into surrounding combustible material.
7. Prevent extension of sparks and particles of hot metal through open windows, doors, holes, and cracks in floors, walls, ceilings, roofs, and other openings.
8. Inspect each location of day's work not sooner than 30 minutes after completion of operations to detect hidden or smoldering fires; ensure proper housekeeping is maintained.

B. Where sprinkler protection exists and is functional, maintain without interruption while operations are performed. If operations are performed near automatic sprinkler heads, shield individual heads temporarily with guards.

3.5 HISTORIC PRESERVATION TREATMENT PROCEDURES

A. The principal aim of preservation work is to halt the process of deterioration and stabilize the item's condition to sustain the integrity of the historic element, feature or structure being preserved. Cyclic maintenance is often required as well as repair work. Repair is required where specifically indicated. The following procedures shall be followed:

1. Retain as much existing material as possible; repair and consolidate rather than replace.
2. Use additional material or structure to reinforce, strengthen, prop, tie, and support existing material or structure.
3. Use reversible processes wherever possible.
4. Use traditional replacement materials and techniques if possible. New work shall be distinguishable from old work and original materials and techniques.
5. Record repair work during construction with periodic construction photos and daily inspection reporting. Photo documentation is specified in Section 01 32 33 "Photographic Documentation."

B. Prohibit smoking by personnel performing work on or near historic structures.

C. Notify Contracting Officer of visible changes in integrity of material or components due to environmental causes including biological attack, UV degradation, freezing, or thawing, or due to structural defects including cracks, movement, or distortion.

1. Do not proceed with work in question until directed by Contracting Officer.

D. Where Work requires existing features to be removed, cleaned, and reinstalled, perform operations without damage to material itself, to adjacent materials, or to substrate.

E. Identify new or replacement materials and features with inconspicuous, permanent marks to distinguish from original materials. Record legend of identification marks and locations of these marks on Record Drawings.

F. When cleaning, match samples of existing materials that have been cleaned and identified for acceptable cleaning levels. Avoid over-cleaning to prevent damage. Use gentlest methods available. Initiate cleaning using hand cleaning methods before introducing power cleaning methods and equipment.

END OF SECTION 01 35 91

SECTION 01 40 00 - QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for quality assurance and quality control.

B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements. Quality of work shall be responsibility of the Contractor.

1. Specified tests, inspections, and related actions do not limit Contractor's other quality assurance and control procedures that facilitate compliance with Contract Document requirements.

C. See Divisions 2 through 46 Sections for specific test and inspection requirements.

1.2 DEFINITIONS

A. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

B. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate actual products incorporated into the work and completed construction comply with requirements.

C. Preconstruction Testing: Tests and inspections performed specifically for project before products and materials are incorporated into work to verify performance or compliance with specified criteria.

D. Product Testing: Tests and inspections performed by a Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or a testing agency qualified to conduct product testing, to establish product performance and compliance with industry standards.

E. Source Quality Control Testing: Tests and inspections performed at the source, i.e., plant, mill, factory, or shop.

F. Field Quality Control Testing: Tests and inspections performed on-site for installation of work and for completed work.

G. Testing Agency or Laboratory: Entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

H. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.

1. Using a term such as "carpentry" does not imply certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of corresponding generic name.

1.3 CONFLICTING REQUIREMENTS

A. Reference Standards: If compliance with two or more standards is specified and standards establish different or conflicting requirements for minimum quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer before proceeding.

B. Minimum Quality Levels: Quality level shown or specified shall be minimum provided or performed. Actual installation may comply exactly with minimum quality specified, or it may exceed minimum within reasonable limits. To comply with requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Contracting Officer before proceeding.

1.4 SUBMITTALS

A. Quality Control Plan:

1. After contract award and before Pre-Construction conference, submit a written Contractor Quality Control (CQC) plan.
2. If plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
3. Government reserves the right to require changes in plan during contract period as necessary to obtain the quality specified.
4. No change in the approved plan may be made without written concurrence by Contracting Officer.

B. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in form of a recent report on inspection of testing agency by a recognized authority.

C. Contractor Quality Control (CQC) Daily Reports: Submit showing inspections and tests on first workday following date covered by report. Quality Control Supervisor shall utilize DSC Forms.

1. Review Construction Management Representative (CMR) Daily report if applicable and reconcile any differences prior to posting.

D. Test Reports

1. Test reports shall be completed by person performing test.
2. Submit Daily Test Information Sheets with Quality Control Daily Reports.
3. Submit failing test results and proposed remedial actions within four hours of noted deficiency.
4. Submit three copies of complete test results no later than one calendar day after test was performed.

E. Accessibility Inspection Report:

1. Fill out applicable sections of the Accessibility Inspection Report and attach to Contractor Quality Control Daily Report.

2. Utilize attached Accessibility Inspection form to document compliance with Architectural Barriers Act Accessibility Standards (ABAAS).
3. Inspect at various stages of construction as needed to ensure finished product meets standards.
4. Submit report no later than one calendar day after inspection was performed.

F. Off-Site Inspection Reports: Submit prior to shipment.

G. If Contractor Quality Control plan and Quality Control Daily Reports are not submitted as specified, Contracting Officer may retain payments until such time plan(s) is/are accepted and implemented, or may retain payments for work completed on days with no Quality Control Daily Reports.

H. Permits, Licenses, and Certificates: For National Park Service (NPS) records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of work.

1.5 QUALITY ASSURANCE

A. General: Qualifications paragraphs in this Article establish minimum qualification levels required; individual Specification Sections specify additional requirements.

B. Contractors Quality Control Staff:

1. Contractor's Quality Control Supervisor shall be assigned no other duties.
2. Contractor's designated Quality Control Supervisor shall be on the project site whenever contract work is in progress.
3. Contractor's job supervisory staff may be used to assist Quality Control Supervisor supplemented, as necessary, by additional certified testing technicians.

C. Installer Qualifications: Firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent indicated for Project, whose work has resulted in construction with a record of successful in-service performance.

D. Manufacturer Qualifications: Firm experienced in manufacturing products or systems similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

E. Fabricator Qualifications: Firm experienced in producing products similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

F. Professional Engineer Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and is experienced in providing engineering services of kind indicated (including Structural Tests and Special Inspections (STSI)). Engineering services are defined as those performed for installations of system, assembly, or products similar to those indicated for Project in material, design, and extent.

G. Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for activities indicated.

1. Requirement for specialists shall not supersede building codes and regulations governing Work.

H. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with experience and capability to conduct testing and inspecting indicated, according to ASTM E 329; and with additional qualifications specified in individual Sections; and where required by Contract, is acceptable to Contracting Officer.

1. Nationally Recognized Testing Laboratory (NRTL): Nationally recognized testing laboratory according to 29 CFR 1910.7 (Code of Federal Regulations).
2. National Voluntary Laboratory Accreditation Program (NVLAP): Testing agency accredited according to National Institute of Standards and Technology's (NIST) National Voluntary Laboratory Accreditation Program.
3. Measuring devices, laboratory equipment, and instruments shall be calibrated at established intervals against certified standards in accordance with NIST requirements. Measuring and testing devices shall be made available for use by Government for verification tests.

I. Factory-Authorized Service Representative Qualifications: Authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products similar in material, design, and extent to those indicated for Project.

1.6 QUALITY CONTROL

A. Contractor is responsible for testing and inspections, including Structural Tests and Special Inspections (STSI), as identified in attached STSI to inspect and test work as needed to ensure quality of materials, workmanship, construction, finish, and functional performance are in compliance with applicable specifications, drawings, and those required by the Building Code.

1. Engage qualified testing agency to perform quality-control services.
2. Submit appropriate report for each quality-control service.
3. Testing and inspecting requested by Contractor and not required by Contract Documents are Contractor's responsibility.
4. Contracting Officer may designate test locations.

B. Manufacturer's Field Services: Where indicated, engage factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.

C. Re-testing/Re-inspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and re-inspecting, for construction of replaced work that failed to comply with Contract Documents.

D. Testing Agency Responsibilities: Cooperate with NPS and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify Contracting Officer and Contractor promptly of irregularities or deficiencies observed in work during performance of services.
2. Determine location from which test samples will be taken and in which in-situ tests are conducted.
3. Conduct and interpret tests and inspections, State in each report whether tested and inspected work complies with or deviates from requirements.

4. Submit 3 copies of certified written report of each test, inspection, and similar quality-control service through Contractor.
5. Do not release, revoke, alter, or increase Contract Document requirements or approve or accept any portion of Work.

E. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide:

1. Access to Work.
2. Incidental labor and facilities necessary to facilitate tests and inspections.
3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
4. Facilities for storage and field curing of test samples.
5. Delivery of samples to testing agencies.
6. Preliminary design mix proposed for material mixes that require control by testing agency.
7. Security and protection for samples and testing and inspecting equipment at Project site.

F. Coordination: Coordinate sequence of activities to accommodate required quality assurance and control services with minimum delay and to avoid removing and replacing construction to accommodate testing and inspecting.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS

2.1 QUALITY CONTROL PLAN

A. Quality Control Plan shall include:

1. List of personnel responsible for quality control and assigned duties. Include each person's qualifications. Include alternate(s) and qualifications.
2. Copy of letter of direction to Contractor's Quality Control Supervisor(s) outlining assigned duties and authorities designated by principal or owner.
3. Names, qualifications / accreditations, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms from laboratories.
4. Methods of performing, documenting, and enforcing quality control of work including Contractor report forms and acknowledgment of NPS forms.
5. Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.
6. Specific discussion regarding mockups, off-site visits, receiving inspections, manufacturers representation, startup requirements, and other aspects of performance specific to Project.
7. Provisions for substantial completion(s) and final inspection(s) per Contract.

PART 3 - EXECUTION

3.1 OFF-SITE CONTROL

A. Items fabricated or assembled off-site shall be inspected for quality control at place of fabrication.

3.2 ON-SITE CONTROL

A. Notification:

1. Notify Contracting Officer at least 48 hours in advance of preparatory phase meeting.
2. Notify Contracting Officer at least 24 hours in advance of initial and follow-up phases.

B. Preparatory Phase: Perform before beginning each feature of work.

1. Review control submittal requirements with personnel directly responsible for quality assurance and quantity control of the work. As a minimum, Contractor's Quality Control Supervisor and foreman responsible for feature of work shall be in attendance.
2. Review applicable specifications sections and drawings related to feature of work.
3. Ensure copies of referenced standards related to sampling, testing, and execution for feature of work are available on site.
4. Ensure provisions have been made for field control testing.
5. Examine work area to ensure preliminary work has been completed.
6. Verify field dimensions and advise Contracting Officer of discrepancies with contract documents.
7. Ensure necessary equipment and materials are at project site and they comply with approved shop drawings and submittals.
8. Document preparatory phase activities and discussions on Contractor's Quality Control Daily Report.

C. Initial Phase:

1. As soon as work begins, inspect and test representative portion of particular feature of work for quality of workmanship.
2. Review control testing procedures to ensure compliance with contract requirements.
3. Document initial phase activities and discussions on Contractor's Quality Control Daily Report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.

D. Follow-Up Phase: Inspect and test as work progresses to ensure compliance with contract requirements until completion of work.

E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases may be required on same feature of work for following reasons:

1. Quality of on-going work is unacceptable.
2. Changes in quality control staff, on-site production supervision, or work crew.
3. Work on particular feature of work is resumed after substantial period of inactivity.

3.3 DOCUMENTATION

A. Maintain Quality Control Daily Reports, Daily Test Report Information Sheets, and Accessibility Inspection Reports of quality control activities and tests. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > Construction Forms section.)

B. Quality Control Daily Reports shall not be substituted for other written reports required under clauses of contract, such as Disputes, Differing Site Conditions, or Changes.

C. Quality Control Daily Reports shall accurately portray all work and materials. The materials shall match the materials as approved from the Construction Documents (CD) requirements. Combustible/flammable materials shall be documented as well.

3.4 ENFORCEMENT

A. Contractor shall stop work on any item or feature pending satisfactory correction of deficiency noted by quality control staff or Contracting Officer.

3.5 REPAIR AND PROTECTION

A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.

1. Comply with Contract Document requirements for Section 01 73 29 "Cutting and Patching."

B. Protect construction exposed by or for quality-control service activities.

C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 01 40 00

This page intentionally left blank.

STATEMENT OF STRUCTURAL TESTS AND SPECIAL INSPECTIONS

National Park Service (NPS) - Denver Service Center (DSC) | 6-29-21

Park: GRCA North Rim Utilities

Project Management Information System (PMIS) Number: 318719

Project Name: NPS GRCA North Rim Utilities

Structural Engineering Firm: HDR Engineering, Inc

This Statement of Structural Tests and Special Inspections is being submitted as required by Chapter 17 of the **2024 International Building Code** (IBC-24). It includes the following:

1. Seismic requirements
2. Wind requirements
3. Qualification Requirements for Inspectors and Testing Technicians
4. Listing of Required Structural Tests and Special Inspections

The CMR's third party testing contractor will provide copies of all special inspection reports and associated documentation to the Contracting Officer (CO). The Construction Contractor will be required to correct all deficiencies discovered in the Special Inspection and Structural Testing program.

Prepared By: Juan Coronado, PE SE
(Type or print name)

Signature: _____ Date: 05/01/2026



Stamped by Professional Engineer (PE) or Structural Engineer (SE)

PART 1 - SEISMIC AND WIND REQUIREMENTS

1.1 SEISMIC REQUIREMENTS, IBC-24 SECTION 1704.3.2

Description of seismic-force-resisting systems and designated seismic systems subject to special inspections:

North Rim Water Treatment Building – Bar joist truss roof with flexible metal deck diaphragm connected to intermediate reinforced masonry unit bearing walls. The structure is a single story structure founded upon shallow footing foundation and a mat foundation at the interior.

1.2 WIND REQUIREMENTS, IBC-24 SECTION 1704.3.3

Description of wind-force-resisting systems and designated wind systems subject to special inspections:

North Rim Water Treatment Building – Bar joist truss roof with flexible metal deck diaphragm connected to intermediate reinforced masonry unit bearing walls. The structure is a single story structure founded upon shallow footing foundation.

Periodic special inspection is required for fastening of the following systems and components at the Water Treatment Plant per IBC 1705.12.3:

1. Roof covering, roof deck, and roof framing connections for the premanufactured structures.
2. Exterior wall covering and wall connections to roof and floor diaphragms and framing for the structures.

PART 2 - INSTRUCTIONS:

1. Under Listing of Required Structural Tests and Special Inspections:
    a. **Required Qualifications** - Provide qualifications for the special inspector, using the Qualification Requirements for Inspector and Testing Technicians list below, for all required Structural Tests and Special Inspections.
    b. **Continuous** - If marked with an **X**, continuous special inspection shall be as defined in Chapter 2, IBC-24. For steel construction, CONTINUOUS is interchangeable with PERFORM as defined in AISC 360-22, Chapter N5.4.
    c. **Periodic** - If marked with an **X**, provide the minimum number of tests, i.e. 20% of all field welds, or the amount of work to be inspected (e.g. 10% of all wall surfaces). For steel construction, PERIODIC is interchangeable with OBSERVE as defined in ASIC 360-22, Chapter N5.4.
2. Attach completed Statement of Structural Tests and Special Inspections to the end of NPS DSC Division 1 Specifications **Section 01 40 00 Quality Requirements**.

PART 3 - QUALIFICATION REQUIREMENTS FOR INSPECTORS AND TESTING TECHNICIANS

PE/SE	Structural Engineer – licensed PE or SE specializing in the design of buildings and structures
PE/GE	Geotechnical Engineer – licensed PE specializing in soil mechanics and foundations
EIT	Engineer-In-Training – graduate engineer who has passed the Fundamentals of engineering examination

3.1 AMERICAN CONCRETE INSTITUTE (ACI) CERTIFICATION

ACI-CCSI	Concrete Construction Special Inspector
ACI-LTT	Concrete Laboratory Testing Technician Level 1 or 2

ACI-STT Concrete Strength Testing Technician
ACI-FTT Concrete Field Testing Technician – Grade I

3.2 AMERICAN SOCIETY OF NON-DESTRUCTIVE TESTING (ASNT) CERTIFICATION

Non-Destructive Testing Technician – Level II or III

3.3 AMERICAN WELDING SOCIETY (AWS) CERTIFICATION

AWS-CWI Certified Welding Inspector

3.4 EXTERIOR DESIGN INSTITUTE (EDI) CERTIFICATION

EDI-EIFS Certified EIFS inspector

3.5 INTERNATIONAL CODE COUNCIL (ICC) CERTIFICATION

ICC-PCSI Prestressed Concrete Special Inspector
ICC-RCSI Reinforced Concrete Special Inspector
ICC-SSI Soils Special Inspector
ICC-SFSI Spray-applied Fireproofing Special Inspector
ICC-SMSI Structural Masonry Special Inspector
ICC-SSBSI Structural Steel and Bolting Special Inspector
ICC-SWSI Structural Welding Special Inspector

3.6 NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET) CERTIFICATION

NICET-CT Concrete Technician – Levels I, II, III and IV
NICET-GET Geotechnical Engineering Technician - Levels I, II, III and IV
NICET-ST Soils Technician - Levels I, II, III and IV

3.7 OTHER

_____ _____
_____ _____
_____ _____
_____ _____
_____ _____
_____ _____
_____ _____
_____ _____
_____ _____

PART 4 - LISTING OF REQUIRED STRUCTURAL TESTS AND SPECIAL INSPECTIONS

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Steel Construction (IBC-24 Section 1705.2; American Institute of Steel Construction: AISC 360-22 Chapter N, AISC 341-22 Chapter J)				
	Prior to Welding (AISC 360-22 Table N5.4-1)				
X	1. Welder qualification records and continuity records	**Welding Personnel Performance Qualifications (WPQR)**	X		
X	2. Welding procedure specifications (WPS) available	**ICC-SSBSI; AWS-CWI**	X		
X	3. Manufacturer certifications for welding consumables available	**ICC-SSBSI; AWS-CWI**	X		
X	4. Material identification (type/grade)	**ICC-SSBSI; AWS-CWI**		X	
X	5. Welder identification system	**ICC-SSBSI; AWS-CWI**		X	
X	6. Fit-up of groove welds (including joint geometry)	**ICC-SSBSI; AWS-CWI**			
	a. Joint preparation			X	
	b. Dimensions (alignment, root opening, root face, bevel)			X	
	c. Cleanliness (condition of steel surfaces)			X	
	d. Tacking (tack weld quality and location)			X	
	e. Backing type and fit (if applicable)			X	
X	7. Fit-up of CJP (complete joist penetration) groove welds of HSS (hollow structural sections) T-, Y- and K-joints without backing (including joint geometry)	**ICC-SSBSI; AWS-CWI**			

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	a. Joint preparation		X		
	b. Dimensions (alignment, root opening, root face, bevel)		X		
	c. Cleanliness (condition of steel surfaces)		X		
	d. Tacking (tack weld quality and location)		X		
	8. Configuration and finish of access holes			X	
X	9. Fit-up of fillet welds	**ICC-SSBSI; AWS-CWI**			
	a. Dimensions (alignment, gaps at root)			X	
	b. Cleanliness (condition of steel surfaces)			X	
	c. Tacking (tack weld quality and location)			X	
X	10. Check welding equipment	**ICC-SSBSI; AWS-CWI**		X	
	During Welding (AISC 360-22 Table N5.4-2)				
X	1. Control and handling of welding consumables	**ICC-SSBSI; AWS-CWI**			
	a. Packaging			X	
	b. Exposure control			X	
X	2. No welding over cracked tack welds	**ICC-SSBSI; AWS-CWI**		X	
X	3. Environmental conditions	**ICC-SSBSI; AWS-CWI**			
	a. Wind speed within limits			X	
	b. Precipitation and temperature			X	
X	4. WPS followed	**ICC-SSBSI; AWS-CWI**			

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	a. Settings on welding equipment			X	
	b. Travel speed			X	
	c. Selected welding materials			X	
	d. Shielding gas type/flow rate			X	
	e. Preheat applied			X	
	f. Interpass temperature maintained (minimum/maximum)			X	
	g. Proper position (F, V, H, OH)			X	
	h. Intermix of filler metals avoided unless approved (reference: AISC 341-10 Table J6.2)			X	
X	5. Use of qualified welders	**Welding Personnel Performance Qualifications (WPQR)**	X		
X	6. Welding techniques	**ICC-SSBSI; AWS-CWI**			
	a. Interpass and final cleaning			X	
	b. Each pass within profile limitations			X	
	c. Each pass meets quality requirements			X	
X	7. Placement and installation of headed stud anchors	**ICC-SSBSI; AWS-CWI**	X		
	After Welding (AISC 360-22 Table N5.4-3)				
X	1. Welds cleaned	**ICC-SSBSI; AWS-CWI**		X	
X	2. Size, length and location of welds	**ICC-SSBSI; AWS-CWI**	X		
X	3. Welds meet visual acceptance criteria	**ICC-SSBSI; AWS-CWI**			

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	a. Crack prohibition		X		
	b. Weld/base-metal fusion		X		
	c. Crater cross section		X		
	d. Weld profiles		X		
	e. Weld size		X		
	f. Undercut		X		
	g. Porosity		X		
X	4. Arc strikes	**ICC-SSBSI; AWS-CWI**	X		
X	5. k-area	**ICC-SSBSI; AWS-CWI**	X		
X	6. Weld access holes in rolled heavy shapes and built-up heavy shapes	**ICC-SSBSI; AWS-CWI**	X		
X	7. Backing removed and weld tabs removed (if required)	**ICC-SSBSI; AWS-CWI**	X		
X	8. Repair activities	**ICC-SSBSI; AWS-CWI**	X		
X	9. Document acceptance or rejection of welded joint or member	**ICC-SSBSI; AWS-CWI**	X		
X	10. No prohibited welds have been added without the approval of the EOR (Engineer of Record)	**ICC-SSBSI; AWS-CWI**		X	
X	11. Placement of reinforcing or contouring fillet welds (if required) (reference: AISC 341-16)	**ICC-SSBSI; AWS-CWI**	X		
	Nondestructive Testing (AISC 360-22 Section N5 and AISC 341-22 Section J6)				
	1. Risk Category II Structures - Perform Ultrasonic Testing on 10% of CJP groove welds in butt, T- and corner joints subject to transversely			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	applied tension loading, in materials 5/16 inch thick or greater.				
X	2. Risk Category III or IV Structures - Perform Ultrasonic Testing on all CJP groove welds subject to transversely applied tension loading in butt, T- and corner joints, in materials 5/16 inch thick or greater.	**ICC-SSBSI; AWS-CWI**	X		
X	3. Access Holes – Perform Magnetic Particle Testing or Liquid Penetrant Testing when the flange thickness exceeds 2 inches for rolled shapes, or when the web thickness exceeds 2 inches for built-up shapes.	**ICC-SSBSI; AWS-CWI**	X		
	4. Welded Joints Subject to Fatigue		X		
	Nondestructive Testing (AISC 341-22 Section J6)				
	Column Splice and Column to Base Plate PJP Groove Weld (AISC 341-16 Section J6 2b)				
	1. Perform Ultrasonic Testing on 100% of PJP (partial-joint-penetration) groove welds in column splices and column to base plate welds		X		
	2. CJP Groove weld		X		
	3. Lamellar tearing		X		
	4. Beam cope and access hole		X		
	5. Reduced beam section repair		X		
	6. Weld tab removal		X		
	Prior to Bolting (AISC 360-22 Table N5.6-1)				
	These inspections are not required for snug-tight joints.				
	1. Manufacturer's certifications available for fastener materials		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	2. Fasteners marked in accordance with ASTM requirements			X	
	3. Correct fasteners selected for the joint detail (grade, type, bolt length if threads are to be excluded from shear plane)			X	
	4. Correct bolting procedure selected for joint detail			X	
	5. Connecting elements, including the appropriate faying surface condition and hole preparation, if specified, meet applicable requirements			X	
	6. Pre-installation verification testing by installation personnel observed and documented for fastener assemblies and methods used			X	
	7. Protected storage provided for bolts, nuts, washers and other fastener components			X	
	During Bolting (AISC 360-22 Table N5.6-2)				
	These inspections are not required for snug-tight joints.				
	These inspections are not required for pretensioned joints and slip-critical joints, when the installer is using the turn-of-nut method with matchmarking techniques, the direct-tension-indicator method, or the twist-off-type tension control bolt method.				
	1. Fastener assemblies, of suitable condition, placed in all holes and washers (if required) are positioned as required			X	
	2. Joint brought to the snug-tight condition prior to the pretensioning operation			X	
	3. Fastener component not turned by the wrench prevented from rotating			X	
	4. Fasteners are pretensioned in accordance with the RCSC (Resource Council on Structural			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Connections) Specification, progressing systematically from the most rigid point toward the free edges				
	After Bolting (AISC 360-22 Table N5.6-2)				
	Document acceptance or rejection of bolted connections.		X		
	Other Inspection Tasks (AISC 360-22 Section N5.7)				
	1. Verify compliance of fabricated steel with the details shown on the approved shop drawings.			X	
	2. Verify compliance of the erected steel frame with the details shown on the approved erection drawings, including braces, stiffeners, member locations and joint details.			X	
	3. Anchor rods and other embedments support structural steel				
	a. Verify the diameter, grade, type and length of the anchor rod or embedded item.			X	
	b. Verify the extent or depth of embedment into the concrete.			X	
	4. RBS (reduce beam section) requirements, if applicable (AISC 341-16)				
	a. Contour and finish			X	
	b. Dimensional tolerances			X	
	5. Protected zone - no holes and unapproved attachments made by fabricator or erector, as applicable. (AISC 341-16)			X	
	6. H-piles - Protected zone - no holes and unapproved attachments made by the responsible contractor, as applicable. (AISC 341-16)			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
X	**Cold-formed Steel Deck** (IBC-24 1705.2.2)				
X	1. Special inspections in accordance with QA/QC-2017 Standard for Quality control (QC) and Quality assurance (QA) for Installation of Steel Deck	**ICC-SSBSI; AWS-CWI;**	X		**After installation of decking to structure.**
X	**Open-Web steel Joists, Trusses, and Joist Girders** (IBC-24 Table 1705.2.3)				
X	1. Installation of open-web steel joists, trusses, and joist girders.	**ICC-SSBSI**			
	a. End connections - welding or bolted			X	**After installation of bar joist trusses**
	b. Bridging - horizontal or diagonal				
	1. Standard bridging			X	**After installation of bar joist trusses**
	2. Bridging that differs from the SJI (Steel Joist Institute) specifications listed in Section 2207.1.			X	**After installation of bar joist trusses**
	Inspection of Composite Structures Prior to Concrete Placement (AISC 341-22 Table J9.1)				
	1. Material identification of reinforcing steel (Type/Grade)			X	
	2. Determination of carbon equivalent for reinforcing steel other than ASTM A706			X	
	3. Proper reinforcing steel size, spacing and orientation			X	
	4. Reinforcing steel has not been rebent in the field			X	
	5. Reinforcing steel has been tied and supported as required			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	6. Required reinforcing steel clearances have been provided			X	
	7. Composite member has required size			X	
	Inspection of Composite Structures During Concrete Placement (AISC 341-22 Table J9.2)				
	1. Concrete: Material identification (mix design, compressive strength, maximum large aggregate size, maximum slump)			X	
	2. Limits on water added at the truck or pump			X	
	3. Proper placement techniques to limit segregation			X	
	Inspection of Composite Structures During Concrete Placement (AISC 341-22 Table J9.2)				
	1. Achievement of minimum specified concrete compressive			X	
	Cold-formed Steel Trusses Spanning 60-feet or Greater (IBC-24 Section 1705.2.4)				
	1. Verify temporary installation restraint/bracing installed in accordance with the approved shop drawings.			X	
	2. Verify permanent individual truss member restraint/bracing installed in accordance with the approved shop drawings.			X	
	Concrete Construction (IBC-24 Section 1705.3)				
X	1. Inspect reinforcing steel, including prestressing tendons, and placement.	**PE/SE; EIT; ACI-CCSI/ACI-FTT; NICET-CT**		X	**Prior to every concrete placement.**
	2. Inspection of reinforcing steel welding				
	a. Verify weldability of reinforcing bars other than ASTM A706.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	b. Inspect single pass fillet welds, maximum 5/16 inch.			X	
	c. Inspect all other welds.			X	
X	3. Inspection of anchors cast in concrete.	**PE/SE; EIT; ACI-CCSI/ACI-FTT; NICET-CT**		X	**Prior to every concrete placement.**
X	4. Inspection of anchors post-installed in hardened concrete members.	**PE/SE; EIT; ACI-CCSI/ACI-FTT; NICET-CT**			
	a. Adhesive anchors installed in horizontally or upwardly inclined orientations to resist sustained tension load		X		
X	b. Mechanical anchors and adhesive anchors not defined in 4 a			X	**During installation of at least 40% of installed anchors.**
X	5. Verify use of approved design mix.	**PE/SE; EIT; ACI-CCSI/ACI-FTT; NICET-CT**		X	**Prior to concrete placement.**
X	6. Prior to placement fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete.	**ACI-CCSI/ACI-FTT; NICET-CT; ICC-RCS**	X		
X	7. Inspect concrete and shotcrete placement for proper application techniques.	**PE/SE; EIT; ACI-CCSI/ACI-FTT; NICET-CT**	X		
X	8. Inspect for maintenance of specified curing temperature and techniques.	**PE/SE; EIT; ACI-CCSI/ACI-FTT; NICET-CT**		X	**After each concrete placement.**
	9. Inspection of prestressed concrete:				
	a. Application of prestressing forces		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	b. Grouting of bonded prestressing tendons in the seismic-force-resisting system.		X		
X	10. Erection of precast structural members			X	**After hollow core erection.**
X	11. For precast concrete diaphragm connections or reinforcement at joints classified as moderate or high deformability elements (MDE or HDE) in structures assigned to Seismic Design Category C, D, E, or F, inspect such connections and reinforcement in the field for:				
	a. Installation of the embedded parts		X		**After hollow core erection.**
	b. Completion of the continuity of reinforcement across joints		X		**After hollow core erection.**
	c. Completion of connections in the field		X		**After hollow core erection.**
	12. Inspect installation tolerances of precast concrete diaphragm connections for compliance with ACI 550.5 (American Concrete Institute).			X	
	13. Verification of in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.			X	
X	14. Inspection formwork for shape, location and dimensions of the concrete member being formed.			X	**Prior to each concrete placement.**
	Masonry Construction (IBC-24 Section 1705.4)				
X	1. Inspect masonry construction in accordance with IBC-24 Section 1705.4 and TMS 602-22 (The				

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Masonry Society)/ACI 530.1-13 Article 1.6.				
	Level 1 Quality Assurance				
	Tests: None.				
	Inspection: Verify compliance with the approved submittal and project specifications.			X	
	Level 2 Quality Assurance				
	Tests:				
	1. Verify slump flow and Visual Stability Index (VSI) as delivered to the project site in accordance with TMS 602-22 Specification Article 1.5B.1.b.3 for self-consolidating grout.		X		
	2. Verify f'm and f'aac in accordance with TMS 602-22 Specification Article 1.4B prior to construction, except where specifically exempted.		X		
	Inspection:				
	1. Verify compliance with the approved submittals and project specifications.			X	
	2. At the start of masonry construction, verify:				
	a. Proportions of site-prepared mortar.			X	
	b. Construction of mortar joints.			X	
	c. Grade and size of prestressing tendons and anchorages.			X	
	d. Location of reinforcement, connectors, prestressing tendons and anchorages.			X	
	e. Prestressing technique.			X	
	f. Properties of thin-bed mortar for AAC (autoclaved aerated		X	X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	concrete) masonry. (Continuous inspection is required for the first 5000 square feet of AAC masonry. Periodic inspection is required after the first 5000 square feet of AAC masonry.)				
	3. Prior to grouting, verify:				
	a. Grout space is clean.			X	
	b. Grade, type and size of reinforcement and anchor bolts, and prestressing tendons and anchorages.			X	
	c. Placement of reinforcing and connectors, and prestressing tendons and anchorages.			X	
	d. Proportions of site-prepared grout and prestressing grout for bonded tendons.			X	
	e. Construction of mortar joints.			X	
	4. During masonry construction, verify:				
	a. Size and location of structural members.			X	
	b. Type, size and location of anchors, including other details of anchorage of masonry to structural members, frames, or other construction.			X	
	c. Welding of reinforcement.		X		
	d. Preparation, construction and protection of masonry during cold weather (temperature below 40 degrees Fahrenheit) or hot weather (temperature above 90 degrees Fahrenheit).			X	
	e. Application and measurement of prestressing force.		X		
	f. Placement of grout and prestressing grout for bonded tendons is in compliance.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	g. Placement of AAC masonry units and construction of thin-bed mortar joints. (Continuous inspection is required for the first 5000 square feet of AAC masonry. Periodic inspection is required after the first 5000 square feet of AAC masonry.)		X	X	
	5. Observe preparation of grout specimens, mortar specimens and/or prisms.			X	
X	**Level 3 Quality Assurance**	**ICC-SMSI for all Masonry work;**			
	Tests:				
X	1. Verify f'm and f'aac in accordance with TMS 602-22/ACI 530.1-13/ASCE 6-13 Specification Article 1.4B prior to construction, and for every 5000 square feet during construction.		X		**Prior to construction then as requested at every 5000sf of placement.**
X	2. Verify proportions of materials in premixed or pre-blended mortar, prestressing grout, and grout other than self-consolidating grout as delivered to the project site.		X		**During regular construction operation.**
X	3. Verify slump flow and Visual Stability Index (VSI) as delivered to the project site in accordance with TMS 602-22/ACI 530.1-13/ASCE 6-13 Specification Article 1.5B.1.b.3 for self-consolidating grout.		X		**During regular construction operation.**
	Inspection:				
X	1. Verify compliance with the approved submittals and project specifications.			X	**Prior to batches of mortar and grout placement**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
X	2. Verify:				
	a. Proportions of site-prepared mortar, grout and prestressing grout for bonded tendons.			X	
X	b. Grade, type and size of reinforcement and anchor bolts, and prestressing tendons and anchorages.			X	**At least once per day during installation**
X	c. Placement of masonry units and construction of mortar joints.			X	**At least once per day during installation**
X	d. Placement of reinforcement, connectors and prestressing tendons and anchorages.		X		**During regular construction operation.**
X	e. Grout space prior to grouting.		X		**During regular construction operation.**
X	f. Placement of grout and prestressing grout for bonded tendons.		X		**During regular construction operation.**
X	g. Size and location of structural elements.			X	
X	h. Type, size and location of anchors, including other details of anchorage of masonry to structural members, frames or other construction.		X		**During regular construction operation.**
	i. Welding of reinforcement.		X		
X	j. Preparation, construction and protection of masonry during cold weather (temperature below 40 degrees Fahrenheit) or			X	**During regular construction operation.**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	hot weather (temperature above 90 degrees Fahrenheit).				
	k. Application and measurement of prestressing force.		X		
	l. Placement of AAC masonry units and construction of thin-bed mortar joints.		X		
	m. Properties of thin-bed mortar for AAC masonry.		X		
X	3. Observe preparation of grout specimens, mortar specimens and/or prisms.		X		
	Wood Construction (IBC-24 Section 1705.5)				
	1. Inspect prefabricated wood structural elements in accordance with Section 1704.2.5.			X	
	2. High load diaphragms:				
	a. Verify sheathing grade and thickness.			X	
	b. Verify nominal size of framing members at adjoining panel edges.			X	
	c. Verify nail or staple diameter and length.			X	
	d. Verify number of fastener lines.			X	
	e. Verify spacing between fasteners in each line and at panel edges.			X	
	3. Shearwalls:				
	a. Verify sheathing grade and thickness.			X	
	b. Verify nominal size of framing members at adjoining panel edges.			X	
	c. Verify nail or staple diameter and length.			X	
	d. Verify number of fastener lines.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	e. Verify spacing between fasteners in each line and at panel edges.			X	
	f. Location and size of holdowns.			X	
	4. Verify nailing, bolting, anchoring and fastening of:				
	a. Drag struts and collectors.			X	
	b. Braces.			X	
	c. Hold-downs.			X	
	5. Metal-plate-connected wood trusses spanning 60 feet or greater:				
	a. Verify temporary installation restraint/bracing installed in accordance with the approved shop drawings.			X	
	b. Verify permanent individual truss member restraint/bracing installed in accordance with the approved shop drawings.			X	
	Soils (IBC-24 Section 1705.6)				
X	1. Verify materials below shallow foundations are adequate to achieve the required bearing capacity.	**PE/GE; EIT; NICET-GET; NICET-ST; ICC-SSI**		X	**Prior to placement of foundation rebar and after installation of underground utility.**
X	2. Verify excavations are extended to proper depth and have reached proper material.	**PE/GE; EIT; NICET-GET; NICET-ST; ICC-SSI**		X	**After excavation is complete.**
X	3. Perform classification and testing of compacted fill materials.	**PE/GE; EIT; NICET-GET; NICET-ST; ICC-SSI**		X	**Prior to placement of foundation rebar and**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
					after installation of underground utility or process piping.
X	4. During fill placement, verify use of proper materials and procedure in accordance with the provisions of the approved geotechnical report. Verify densities and lift thicknesses during placement and compaction of compacted fill.	**PE/GE; EIT; NICET-GET; NICET-ST; ICC-SSI**	X		
X	5. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly.	**PE/GE; EIT; NICET-GET; NICET-ST; ICC-SSI**		X	**Prior to placement of foundation rebar and after installation of underground utility or process piping.**
	Driven Deep Foundation Elements (IBC-24 Section 1705.7)				
	1. Verify element materials, sizes and lengths.		X		
	2. Determine capacities of test elements and conduct additional load tests when required. Refer to project specifications.		X		
	3. Inspect driving operations and maintain complete and accurate records for each element.		X		
	4. Verify element locations and plumbness.		X		
	a. Verify type and size of hammer.		X		
	b. Record number of blows per foot of penetration.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	c. Determine required penetration to achieve specified capacity.		X		
	d. Record pile tip and butt elevations.		X		
	e. Document any damage to any foundation element.		X		
	5. For steel piling, perform additional inspection in accordance with Section 1705.2 and AISC 341-16, Table J10.1.				
	6. For concrete elements and concrete-filled elements, perform additional inspections in accordance with Section 1705.3.				
	7. For specialty elements, perform additional inspections as required in the project specifications.				
	Cast-in-Place Deep Foundation (IBC-24 Section 1705.8)				
	1. Inspect drilling operations and maintain complete and accurate records for each element.		X		
	2. Verify element locations and plumbness.		X		
	a. Verify element diameter.		X		
	b. Verify bell diameter (if applicable).		X		
	c. Verify element lengths.		X		
	d. Verify embedment depth into bedrock (if applicable).		X		
	e. Verify adequate end-bearing strata capacity.		X		
	f. Record concrete or grout volumes.		X		
	3. For concrete elements, perform additional inspections in accordance with Section 1705.3.				

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Helical Piles (IBC-24 Section 1705.10)				
	1. Verify pile locations.		X		
	a. Verify installation equipment used.		X		
	b. Verify pile dimensions.		X		
	c. Verify tip elevations.		X		
	d. Verify final depth.		X		
	e. Verify final installation torque.		X		
	f. Other data as required by the project specifications.		X		
	Fabricated Items (IBC-24 Section 1705.11)				
X	1. Inspect fabricated items in accordance with Section 1704.2.5.	**PE/SE; EIT; ICC-SSBSI;**	X		**For each prefabricated component to be installed on the sites.**
	Wind Resistance (IBC-24 Section 1705.12)				
X	1. Provide inspections when required by Section 1705.12.	**PE/SE; EIT; ICC-SSBSI;**	X		**Specifically for each prefabricated component to be installed on the sites.**
	Seismic Resistance (IBC-24 Section 1705.13)				
X	1. Provide inspections when required by Section 1705.13.	**ICC-SSBSI;**		X	**Steel collectors, and**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
					veneer connections.
	Testing and Qualification for Seismic Resistance (IBC-24 Section 1705.14)				
X	1. Test and qualify seismic resistance in accordance with IBC-24 Section 1705.14 and the project specifications.	**ICC-SSBSI;**		X	**For non-structural components**
	Sprayed Fire-Resistant Materials (IBC-24 Section 1705.15)				
	1. Inspect sprayed fire-resistant materials in accordance with IBC-24 Section 1705.15 and the project specifications.				
	Mastic and Intumescent Fire-Resistant Coatings (IBC-24 Section 1705.16)				
	1. Perform inspections in accordance with AWCI 12-B (Association of the Wall and Ceiling Industry) and IBC-24 Section 1705.16.			X	
	Exterior Insulation and Finish Systems (EIFS) (IBC-24 Section 1705.17)				
	1. Perform inspections in accordance with project specifications and IBC-24 Section 1705.17.			X	
	Fire-Resistant Penetrations and Joints (IBC-24 Section 1705.18)				
	1. Perform inspections in accordance with project specifications and IBC-24 Section 1705.18.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Smoke Control (IBC-24 Section 1705.19)				
	1. Perform testing in accordance with project specifications and IBC-24 Section 1705.19.				
	Sealing of Mass Timber (IBC-24 Section 1705.20)				
	1. Perform testing in accordance with project specifications and IBC-24 Section 1705.20.				

This page intentionally left blank.

SECTION 01 42 00 – REFERENCE STANDARDS

PART 1 - GENERAL

1.1 ENVIRONMENTAL DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114 and as specified herein.

B. Biobased Materials: As defined in the Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) that is composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.

1. Biobased content: Amount of biobased carbon in the material or product as a percentage of weight (mass) of total organic carbon in the material or product.

C. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.

D. Deconstruction: Disassembly of buildings for purpose of recovering materials.

E. DFE (Design for the Environment): A technique that includes elements of resource conservation and pollution prevention as applied in various product sectors. A technique that incorporates approaches which are part of product (or assembly) concept, need and design. Considerations involve material selection, material and energy efficiency, reuse, maintainability and design for disassembly and recyclability. Refer to International Organization for Standardization (ISO) Guide 64 for additional clarification.

F. Environmentally preferable products: Products and services that have a lesser or reduced effect on the environment in comparison to conventional products and services. Refer to EPA’s Final Guidance on Environmentally Preferable Purchasing Program.

G. Non-Renewable Resource: A resource that exists in a fixed amount that cannot be replenished on a human time scale. Non-renewable resources have potential for renewal only by geological, physical, and chemical processes taking place over of millions of years. Examples include iron ore, coal, and oil.

H. Perpetual Resource: A resource that is virtually inexhaustible on a human time scale. Examples include solar energy, tidal energy, and wind energy.

I. Recycled Content Materials: Products that contain pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent Federal Trade Commission (FTC) Guide for Use of Environmental Marketing Claims.

J. Renewable Resource: A resource that is grown, naturally replenished, or cleansed, at a rate which exceeds depletion of the usable supply of that resource. A renewable resource can be exhausted if improperly managed. However, a renewable resource can last indefinitely with proper stewardship. Examples include trees in forests, grasses in grasslands, and fertile soil.

1.2 QUALITY ASSURANCE

A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

B. Publication Dates: Comply with standards in effect as of date of Contract Documents, unless otherwise indicated.

C. Conflicting Requirements: Where compliance with two or more standards is specified, and standards may establish different or conflicting requirements for minimum quantities or quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer (CO) for decision before proceeding.

1.3 INDUSTRY STANDARDS

A. Applicability of Standards: Unless Contract Documents include more stringent requirements, applicable construction industry standards have same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

B. Publication Dates: Comply with standards in effect as of date of Contract Documents unless otherwise indicated.

1.4 ABBREVIATIONS AND ACRONYMS

A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities found in Section 01 42 00 Sources for Reference Publications, Unified Facilities Guide Specifications (UFGS) (accessible via Masters website > Downloads section > click on UFGS Master (WBDG Website). Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

B. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

IAPMO	International Association of Plumbing and Mechanical Officials www.iapmo.org	(909) 472-4100
ICC	International Code Council www.iccsafe.org	(888) 422-7233
ICC-ES	ICC Evaluation Service, Inc. icc-es.org	(800) 423-6587 (562) 699-0543

C. Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABA & ABAAS United States Access Board	Architectural Barriers Act (ABA) Architectural Barriers Act Accessibility Standards (ABAAS) www.access-board.gov
CoE	Army Corps of Engineers www.usace.army.mil
CPSC	Consumer Product Safety Commission www.cpsc.gov
DOC	Department of Commerce www.commerce.gov
DOD	Department of Defense www.defense.gov
DOJ	Department of Justice www.justice.gov
DOE	Department of Energy www.energy.gov
EPA	Environmental Protection Agency www.epa.gov
FAA	Federal Aviation Administration www.faa.gov
FCC	Federal Communications Commission www.fcc.gov
FDA	Food and Drug Administration www.fda.gov
GSA	General Services Administration www.gsa.gov
HUD	Department of Housing and Urban Development www.hud.gov
LBL	Lawrence Berkeley National Laboratory www.lbl.gov
NCHRP	National Cooperative Highway Research Program (See TRB (Transportation Resource Board))
NIST	National Institute of Standards and Technology www.nist.gov
OSHA	Occupational Safety & Health Administration www.osha.gov
PHS	U.S. Department of Health and Human Services www.hhs.gov

RUS	Rural Utilities Service (See USDA (Department of Agriculture))
SD	State Department www.state.gov
TRB	Transportation Research Board www.nationalacademies.org/trb/transportation-research-board
USDA	Department of Agriculture www.usda.gov
USP	U.S. Pharmacopeia www.usp.org
USPS	Postal Service www.usps.com

D. Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABAAS	Architectural Barriers Act Accessibility Standards www.access-board.gov
CFR	Code of Federal Regulations Available from Government Printing Office www.govinfo.gov/app/collection/cfr
DOD	Department of Defense Military Specifications and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
DSCC	Defense Supply Center Columbus (See FS (Federal Specification))
FED-STD	Federal Standard (See FS (Federal Specification))
FS	Federal Specification Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/ Available from General Services Administration www.gsa.gov Available from National Institute of Building Sciences www.nibs.org
FTMS	Federal Test Method Standard (See FS (Federal Specification))
MIL	(See MILSPEC (Military Specification and Standards))

MIL-STD	(See MILSPEC (Military Specification and Standards))
MILSPEC	Military Specification and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
UFAS	Uniform Federal Accessibility Standards Available from Access Board www.access-board.gov/guidelines-and-standards/buildings-and-sites/about-the-aba-standards/ufas *(UFAS is **only** for housing projects per Fair Housing Act. See also the Fair Housing Act Design Manual, www.huduser.gov/portal/publications/destech/fairhousing)*

1.5 ENVIRONMENTAL REFERENCE STANDARDS

A. American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE):
- **ASHRAE 52.2,** *Method of Testing General Ventilation Air Cleaning Devices for Removal Efficiency by Particle Size*
- **ASHRAE 55,** *Thermal Environmental Conditions for Human Occupancy*
- **ASHRAE 62.1,** *Ventilation for Acceptable Indoor Air Quality*
- **ASHRAE 62.2,** *Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings*
- **ASHRAE/IESNA 90.1,** *Energy Standard for Buildings, Except Low-Rise Residential Buildings*
- **ASHRAE 90.2,** *Energy Efficient Design of Low-Rise Residential Buildings*

B. American Association of State Highway and Transportation Officials (AASHTO):
- M288 Geotextile Specification for Highway Applications
- **MP009-06 Standard Specification for Compost for Erosion/Sediment Control (Filter Berms and Filter Socks)**
- **MP010-03 Standard Specification for Compost for Erosion/Sediment Control (Compost Blankets)**

C. American Society for Testing and Materials International (ASTM):
- **A580/A580M Standard Specification for Stainless Steel Wire**
- **A653/A653M Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process**
- **B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube**
- **C1240 Standard Specification for Silica Fume Used in Cementitious Mixtures**
- **C128 Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate**
- **C131 Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine**
- **C1319 Standard Specification for Concrete Grid Paving Units**
- **C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings**
- **C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates**

- **C1371 Standard Test Method for Determination of Emittance of Materials Near Room Temperature Using Portable Emissometers**
- **C1386 Standard Specification for Precast Autoclaved AERATED Concrete (PAAC) Wall Construction Units**
- **C1483 Standard Specification for Exterior Solar Radiation Control Coatings on Buildings**
- **C1549 Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer**
- **C1601 Standard Test Method for Field Determination of Water Penetration of Masonry Wall Surfaces**
- **C289 Standard Test Method for Potential Alkali-Silica Reactivity of Aggregates (Chemical Method)**
- **C311 Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use as a Mineral Admixture in Portland-Cement Concrete**
- **C33 Standard Specification for Concrete Aggregates**
- **C593 Standard Specification for Fly Ash and Other Pozzolans for Use With Lime**
- **C595 Standard Specification for Blended Hydraulic Cements**
- **C618 Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete**
- **C67 Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile**
- **C739 Standard Specification for Cellulosic Fiber (Wood-Base) Loose-Fill Thermal Insulation**
- **C936 Standard Specification for Interlocking Concrete Paver Units**
- **C989 Standard Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars**
- **D1435 Standard Practice for Outdoor Weathering of Plastics**
- **D1557 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 foot pound force per cubic foot (ft-lbf/ft3) (2,700 kilonewton meter per cubic meter (kN-m/m3))**
- **D1972 Standard Practice for Generic Marking of Plastic Products**
- **D198 Standard Test Methods of Static Tests of Lumber in Structural Sizes**
- **D2103 Standard Specification for Polyethylene Film and Sheeting**
- **D217 Standard Test Methods for Cone Penetration of Lubricating Grease**
- **D2369 Standard Test Method for Volatile Content of Coatings**
- **D3273 Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber**
- **D3786 Standard Test Method for Hydraulic Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method**
- **D3792 Standard Test Method for Water Content of Coatings by Direct Injection Into a Gas Chromatograph**
- **D3864 Standard Guide for Continual On-Line Monitoring Systems for Water Analysis**
- **D3960 Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings**
- **D4017 Standard Test Method for Water in Paints and Paint Materials by Karl Fischer Method**

- **D4263 Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method**
- **D4444 Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters**
- **D4491 Standard Test Methods for Water Permeability of Geotextiles by Permittivity**
- **D4552 Standard Practice for Classifying Hot-Mix Recycling Agents**
- **D4632 Standard Test Method for Grab Breaking Load and Elongation of Geotextiles**
- **D4716 Test Method for Determining the (In-plane) Flow Rate per Unit Width and Hydraulic Transmissivity of a Geosynthetic Using a Constant Head**
- **D4833 Standard Test Method for Index Puncture Resistance of Geotextiles, Geomembranes, and Related Product**
- **D4840 Standard Guide for Sampling Chain-of-Custody Procedures**
- **D4887 Standard Test Method for Preparation of Viscosity Blends for Hot Recycled Bituminous Materials**
- **D5106 Standard Specification for Steel Slag Aggregates for Bituminous Paving Mixtures**
- **D5116 Standard Guide for Small-Scale Environmental Chamber Determinations of Organic Emissions from Indoor Materials/Products**
- **D5199 Standard Test Method for Measuring the Nominal Thickness of Geosynthetics**
- **D5261 Standard Test Method for Measuring Mass per Unit Area of Geotextiles**
- **D5268 Standard Specification for Topsoil Used for Landscaping Purposes**
- **D5359 Standard Specification for Glass Cullet Recovered from Waste for Use in Manufacture of Glass Fiber**
- **D5505 Standard Practice for Classifying Emulsified Recycling Agents**
- **D5509 Standard Practice for Exposing Plastics to a Simulated Compost Environment**
- **D5512 Standard Practice for Exposing Plastics to a Simulated Compost Environment Using an Externally Heated Reactor**
- **D5539 Standard Specification for Seed Starter Mix**
- **D5957 Standard Guide for Flood Testing Horizontal Waterproofing Installations**
- **D5603 Standard Classification for Rubber Compounding Materials—Recycled Vulcanizate Particulate Rubber**
- **D5663 Standard Guide for Validating Recycled Content in Packaging Paper and Paperboard**
- **D5759 Standard Guide for Characterization of Coal Fly Ash and Clean Coal Combustion Fly Ash for Potential Uses**
- **D5792 Standard Practice for Generation of Environmental Data Related to Waste Management Activities: Development of Data Quality Objectives**
- **D5834 Standard Guide for Source Reduction Reuse, Recycling, and Disposal of Solid and Corrugated Fiberboard (Cardboard)**
- **D5851 Standard Guide for Planning and Implementing a Water Monitoring Program**

- **D5852 Standard Test Method for Erodibility Determination of Soil in the Field or in the Laboratory by the Jet Index Method**
- **D6002 Standard Guide for Assessing the Compostability of Environmentally Degradable Plastics**
- **D6006 Standard Guide for Assessing Biodegradability of Hydraulic Fluid**
- **D6007 Standard Test Method for Determining Formaldehyde Concentration in Air from Wood Products Using a Small Scale Chamber**
- **D6046 Standard Classification of Hydraulic Fluids for Environmental Impact**
- **D6081 Standard Practice for Aquatic Toxicity Testing of Lubricants: Sample Preparation and Results Interpretation**
- **D6108 Standard Test Method for Compressive Properties of Plastic Lumber and Shapes**
- **D6109 Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastic Lumber**
- **D6112 Standard Test Methods for Compressive and Flexural creep and Creep-Rupture of Plastic Lumber and Shapes**
- **D6117 Standard Test Methods for Mechanical Fasteners In Plastic Lumber and Shapes**
- **D6155 Standard Specification for Nontraditional Coarse Aggregates for Bituminous Paving Mixtures**
- **D6245 Standard Guide for Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation**
- **D6261 Standard Specification for Extruded and Compression Molded Basic Shapes Made from Thermoplastic Polyester (TPES)**
- **D6262 Standard Specification for Extruded, Compression Molded, and Injection Molded Basic Shapes of Poly(aryl ether ketone) (PAEK)**
- **D6270 Standard Practice for Use of Scrap Tires in Civil Engineering Applications**
- **D6329 Standard Guide for Developing Methodology for Evaluating the Ability of Indoor Materials to Support Microbial Growth Using Static Environmental Chambers**
- **D6330 Standard Practice for Determination of Volatile Organic Compounds (Excluding Formaldehyde) Emissions from Wood-Based Panels Using Small Environmental Chambers Under Defined Test Conditions**
- **D6345 Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air**
- **D6400 Standard Specification for Compostable Plastics**
- **D6435 Standard Test Method for Shear Properties of Plastic Lumber and Plastic Lumber Shapes**
- **D6629 Standard Guide for Selection of Methods for Estimating Soil Loss by Erosion**
- **D6662 Standard Specification for Polyolefin-Based Plastic Lumber Decking Boards**
- **D6712 Standard Specification for Ultra-High-Molecular-Weight Polyethylene (UHMW-PE) Solid Plastic Shapes**
- **D6886 Standard Test Method for Speciation of the Volatile Organic Compounds (VOCs) in Low VOC Content Waterborne Air-Dry Coatings by Gas Chromatography**

- **D692 Standard Specification for Coarse Aggregate for Bituminous Paving Mixtures**
- **D696 Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30°C and 30°C With a Vitreous Silica Dilatometer**
- **D698 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft3 (600 kN-m/m3))**
- **D7186 Standard Practice for Quality Assurance Observation of Roof Construction and Repair**
- **E1021 Standard Test Methods for Measuring Spectral Response of Photovoltaic Cells**
- **E1038 Standard Test Method for Determining Resistance of Photovoltaic Modules to Hail by Impact with Propelled Ice Balls**
- **E1039 Standard Test Method for Calibration of Silicon Non-Concentrator Photovoltaic Primary Reference Cells Under Global Irradiation**
- **E1040 Standard Specification for Physical Characteristics of Nonconcentrator Terrestrial Photovoltaic Reference Cells**
- **E1105 Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform or Cyclic Static Air Pressure Difference**
- **E1171 Standard Test Method for Photovoltaic Modules in Cyclic Temperature and Humidity Environments**
- **E1333 Standard Test Method for Determining Formaldehyde Concentrations in Air and Emission Rates from Wood Products Under Defined Test Conditions Using a Large Chamber**
- **E1362 Standard Test Method for Calibration of Non-Concentrator Photovoltaic Secondary Reference Cells**
- **E1433 Standard Guide for Selection of Standards on Environmental Acoustics**
- **E1462 Standard Test Methods for Insulation Integrity and Ground Path Continuity of Photovoltaic Modules**
- **E1596 Standard Test Methods for Solar Radiation Weathering of Photovoltaic Modules**
- **E1597 Standard Test Method for Saltwater Pressure Immersion and Temperature Testing of Photovoltaic Modules for Marine Environments**
- **E1609 Standard Guide for Development and Implementation of a Pollution Prevention Program**
- **E1686 Standard Guide for Selection of Environmental Noise Measurements and Criteria**
- **E1690 Standard Test Method for Determination of Ethanol Extractives in Biomass**
- **E1721 Standard Test Method for Determination of Acid-Insoluble Residue in Biomass**
- **E1755 Standard Test Method for Ash in Biomass**
- **E1758 Standard Test Method for Determination of Carbohydrates in Biomass by High Performance Liquid Chromatography**
- **E1780 Standard Guide for Measuring Outdoor Sound Received from a Nearby Fixed Source**
- **E1799 Standard Practice for Visual Inspections of Photovoltaic Modules**

- **E1802 Standard Test Methods for Wet Insulation Integrity Testing of Photovoltaic Modules**
- **E1821 Standard Test Method for Determination of Carbohydrates in Biomass by Gas Chromatography**
- **E1827 Standard Test Methods for Determining Airtightness of Buildings Using an Orifice Blower Door**
- **E1830 Standard Test Methods for Determining Mechanical Integrity of Photovoltaic Modules**
- **E1861 Standard Guide for Use of Coal Combustion By-Products in Structural Fills**
- **E1918 Standard Test Method for Measuring Solar Reflectance of Horizontal and Low-Sloped Surfaces in the Field**
- **E1971 Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings**
- **E1980 Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces**
- **E1991 Standard Guide for Environmental Life Cycle Assessment of Building Materials/Products**
- **E2047 Standard Test Method for Wet Insulation Integrity Testing of Photovoltaic Arrays**
- **E2114 Standard Terminology for Sustainability Relative to the Performance of Buildings**
- **E2128 Standard Guide for Evaluating Water Leakage of Building Walls**
- **E2129 Standard Practice for Data Collection for Sustainability Assessment of Building Products**
- **E2397 Standard Practice for Determination of Dead Loads and Live Loads associated with Green Roof Systems**
- **E2398 Standard Test Method for Water Capture and Media Retention of Geocomposite Drain Layers for Green Roof Systems**
- **E2399 Standard Test Method for Maximum Media Density for Dead Load Analysis of Green Roof Systems**
- **E2400 Standard Guide for Selection, Installation, and Maintenance of Plants for Green Roof Systems**
- **E241 Standard Guide for Limiting Water-Induced Damage to Buildings**
- **E2432 Standard Guide for General Principles of Sustainability Relative to Buildings**
- **E408 Standard Test Methods for Total Normal Emittance of Surfaces Using Inspection-Meter Techniques**
- **E413 Standard Classification for Rating Sound Insulation**
- **E477 Standard Test Method for Measuring Acoustical and Airflow Performance of Duct Liner Materials and Prefabricated Silencers**
- **E648 Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source**
- **E683 Standard Practice for Installation and Service of Solar Space Heating Systems for One- and Two-Family Dwellings**
- **E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization**

- **E781 Standard Practice for Evaluating Absorptive Solar Receiver Materials When Exposed to Conditions Simulating Stagnation in Solar Collectors With Cover Plates**
- **E782 Standard Practice for Exposure of Cover Materials for Solar Collectors to Natural Weathering Under Conditions Simulating Operational Mode**
- **E823 Standard Practice for Nonoperational Exposure and Inspection of a Solar Collector**
- **E881 Standard Practice for Exposure of Solar Collector Cover Materials to Natural Weathering Under Conditions Simulating Stagnation Mode**
- **E90 Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements**
- **E903 Standard Test Method for Solar Absorptance, Reflectance, and Transmittance of Materials Using Integrating Spheres**
- **E948 Standard Test Method for Electrical Performance of Photovoltaic Cells Using Reference Cells Under Simulated Sunlight**
- **F1869 Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride**
- **F2034 Standard Specification for Sheet Linoleum Floor Covering**
- **F2170 Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes**

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 42 00

This page intentionally left blank.

SECTION 01 45 25 - TESTING CONCRETE STRUCTURES FOR WATERTIGHTNESS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Requirements for furnishing all labor, materials, tools, equipment, and services, for all testing of concrete structures for watertightness, in accord with provisions of the Contract Documents.
2. Completely coordinate with work of all other trades.
3. Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, appurtenances and devices incidental to or necessary for a sound, secure and complete leak test.

B. Related Sections include but are not necessarily limited to:

1. Division 00 – Procurement and Contracting Requirements.
2. Division 01 – General Requirements.
3. Section 03 35 00 – Concrete Finishing and Repair of Surface Defects

C. Payment:

1. Contractor to pay all costs required for testing, retesting, patching, repair and work required to provide access for repair as required to meet watertightness requirements specified or indicated.

1.2 QUALITY ASSURANCE

A. Reference Standards:

1. American Concrete Institute (ACI):
   a. 350.1, Specification for Tightness Testing of Environmental Engineering Concrete Containment Structures and Commentary.
2. NSF International (NSF).
3. Underwriters Laboratories, Inc. (UL).
4. United States Department of Agriculture (USDA).
5. Water Quality Association (WQA).

1.3 SUBMITTALS

A. Shop Drawings:

1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
2. Watertightness testing plan:
   a. Plan shall include:
      1) Schedule for testing.
      2) Description of testing apparatus for measuring water level in structure and evaporation pan.

a) Include Drawings (plans, sections, and details) as appropriate to fully describe apparatus.
3) Location plan showing measurement location and evaporation pan location.
4) Procedures for isolation of tank or compartments to assure a constant volume during testing.
5) Narrative describing testing procedure.
6) Calculations showing:
a) Total structure volume at water elevation for commencement of test period.
b) Maximum water leakage allowed.
c) Test period: See ACI 350.1.
7) Plan shall be in accordance with ACI 350.1, Chapters 1 and 2.

3. If structure has running water leaks or otherwise fails watertightness test, submit repair and patching plan.
a. Include with plan:
1) Location and areas of leaks.
2) Repair material and procedures proposed for repair.
3) Photographs of all visible leaks and damp areas.
a) Include distant photos and close-ups to document conditions.

B. Informational Submittals:

1. Results of watertightness testing indicating the following:
a. Level of water in structure and in evaporation pan and water temperature at commencement of test period.
b. Level of water in structure and in evaporation pan and water temperature at end of test period.
c. Net leakage in percent of total volume during test period (gross leakage minus that due to evaporation).
d. Results of retesting required due to leakage exceeding specified percentages allowed.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Water for Testing:

1. See ACI 350.1.
2. Water treatment: Potable water.
3. Coordinate delivery of water for testing with Contracting Officer (CO).

B. Any patching or repair materials that may come into contact with potable water in tanks shall be approved for drinking water per NSF, UL, USDA, or WQA.

PART 3 - EXECUTION

3.1 PREPARATION BEFORE TESTING

A. General:

1. Verify the specified 28-day concrete strength has been achieved prior to testing.
    a. All concrete slabs shall be placed and shall achieve the specified 28-day concrete strength prior to testing, unless specifically noted otherwise on Drawings.
2. Testing to be performed prior to placement of exterior backfill soil.
    a. Contractor is responsible for phasing construction to minimize the impact of and to leak testing.
3. Contractor to furnish all necessary materials (such as gaskets and flange cover plates).
4. Testing to be performed prior to application of specified waterproofing.
5. Test the following tanks prior to backfilling and construction of adjacent CMU bearing walls:
    a. Volume 06: Chlorine Contact Tank

B. Source of water for watertightness testing:

1. The source of water will be the Park's finished water storage tanks and shall occur between May and September. Coordinate use of water for filling tanks with COR and CMR.
2. The cost of initial test water will be paid by NPS.
    a. The cost of additional water due to retesting will be paid by the Contractor.
3. Contractor shall provide the means of transporting the water to the structure being tested.

C. Cleaning:

1. Thoroughly clean interior of structure to be tested of all debris and dirt and hose down surfaces of all walls and slabs.

D. Patching and Finishing:

1. Prepare concrete surfaces in accordance with ACI 350.1 and Section 03 35 00.
    a. Fill all holes, voids, and honeycombed areas per Section 03 35 00. Cracks suspected to cause leakage to be filled and sealed.
    b. Review tank for areas of potential leakage before filling.

3.2 WATERTIGHTNESS TESTING

A. Provide watertightness testing for the following structures in accordance with the specified criteria:

1. Where no elevation is given for WATER ELEVATION AT COMMENCEMENT OF TEST PERIOD in the following table, use the normal operating water elevation for the indicated structure.

STRUCTURE NAME or TYPE	WATER ELEVATION AT COMMENCEMENT OF TEST PERIOD	MAXIMUM WATER LEAKAGE ALLOWED IN TEST PERIOD (PERCENT OF TOTAL VOLUME)
Volume 06: Chlorine Contact Tank (CT Tank)	6-inches below soffit of slab / top of wall	0.05 (no visible leakage on wall)

B. Perform a watertightness test as required by CO on any additional structure when in the opinion of the CO the structure contains sufficient concrete defects that could impair the watertightness of the structure.

    1. Testing to conform to requirements of this Specification Section with allowable leakage and other criteria as established by CO.

C. Test for leakage in accordance with ACI 350.1, latest edition, Chapters 1 and 2, and this Specification Section.

    1. Isolate sections of structures that can be isolated during operation.
        a. Test each section separately.
    2. Allow Contracting Officer and their representatives to witness testing for watertightness and review accompanying results.

D. During testing period, inspect structure for areas indicating leakage.

    1. Any areas evidencing running water to be repaired and patched.
    2. Patching or repair of leaks as defined above shall be completed independent of the watertightness test.
        a. Passing watertightness test does not relieve Contractor from repairing running water leaks.

E. Record level of water surface in the structure and evaporation pan and temperature every 24 HRS until end of test period.

    1. Test periods defined per ACI 350.1.

F. If leakage is greater than that allowed, repair and patch areas suspected of causing the leakage.

    1. Re-test structure using the same procedure until leakage is equal to or less than that allowed.
    2. Provide repair plan to CO for approval prior to repair of tank.
    3. Cracks suspected to cause leakage to be filled and sealed to prevent leakage.
        a. Patching to be performed after defective concrete area is cleaned of all loose material to surface of sound concrete.
    4. Prior to patching activities, Contractor to submit patching materials and procedures for review and approval by CO. Submit materials for CO approval with the repair plan required in Item 2.

G. Dispose of water used for testing.

    1. Dispose of water used for testing to an area which will not damage new or existing construction and will not interfere with construction operations.
    2. Provide hoses, temporary connections, temporary fittings and other conduits as necessary to dispose of test water without damage to structure or terrain.
    3. Point of disposal to be approved by Contracting Officer.

END OF SECTION 01 45 25

SECTION 01 50 00 - TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 DEFINITIONS

A. Permanent Enclosure: As determined by Contracting Officer (CO), permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and openings are closed with permanent construction or substantial temporary closures.

1.3 USE CHARGES

A. General: Cost or use charges for temporary facilities shall be included in Contract Sum as required.

B. Water Service: Potable Water from existing water system is not available for use due to the pipeline break caused during the 2026 flood. Contractor shall be responsible for all water required for construction and construction camp requirements until Volumes 01, 02, 04 and 05 are substantially complete and operational after which the following water service may be available for use:

1. Up to a maximum 10,000 gallons per day from April 15 through October 15 without metering and without payment of use charges.
2. Potable Water from existing water system is available for use up to a maximum 5,000 gallons per day and 250,000 gallons total from October 16 through April 14.
3. Provide connections, certified backflow devices (with annual inspection and certification provided), meter and extensions of services as required for construction operations without any additional permit costs. Contractor shall record daily water use and submit to CO.

C. Electric Power Service: Electric power from existing system is available for use without metering and without payment of use charges. Provide connections and extensions of services as required for construction operations without any additional permit costs.
1. Electric Power Service is not available for contractor use for Volume 02: Pipelines – North Rim to Roaring Springs Pumphouse for bore drilling operations.

1.4 QUALITY ASSURANCE

A. Electric Service: Comply with National Electrical Contractors Association (NECA), National Electrical Manufacturers Association (NEMA), and Underwriter Laboratories (UL) standards and regulations for temporary electric service. Install service to comply with National Fire Protection Association (NFPA) 70.

B. Environmental Protection: Provide environmental protection as required by agency(ies) with jurisdiction and as indicated in Contract Documents. Coordinate with requirements of the following:

1. Regulatory Requirements
2. Indoor Air Quality (IAQ) Management
3. Noise and Acoustics Management
4. Environmental Management
5. Construction Waste Management

1.5 PROJECT CONDITIONS

A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before NPS acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 TEMPORARY MATERIALS

A. Temporary materials may be new or used, but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.

B. Pavement: Comply with Asphalt Paving specification.

C. Chain-Link Fencing: Minimum 2 IN, 0.148 IN thick, galvanized steel, chain-link fabric fencing; minimum 6 FT high with galvanized steel pipe posts; minimum 2-3/8 IN OD (outside diameter) line posts and 2-7/8 IN OD corner and pull posts, with 1-5/8 IN OD top rails.

D. Safety Barrier Fence: Must be able to secure work areas from large wildlife such as deer.

E. Barrier Tape: Yellow tape Imprinted with "CAUTION: CONSTRUCTION AREA,".

F. Lumber and Plywood: Comply with requirements in Division 06 Section "Rough Carpentry."

2.2 TEMPORARY FACILITIES

A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.

1. Locations at South Rim Contractor Support Facilities, near the North Rim WWTP, Ballfield, and Lindberg Hill at designated staging areas as shown on the Drawings.

B. Contracting Officers Field Office: Field office shall be a separate structure from Contractor's office.

1. Prefabricated, job built, or a mobile unit; excellent condition, structurally sound, non-flammable exterior construction, weather tight, minimum 720 square feet.
2. Operable windows and security screens, adjustable ventilation.
3. Restroom (minimum of lavatory and toilet, with exhaust fan if room is windowless).
4. Air conditioner and heater.
5. Interior partition with lockable door to divide office
6. Paneling or freshly painted walls, acoustical tile or painted ceilings, and resilient flooring.

7. Two exterior doors with dead bolts keyed from outside,
8. Minimum 20-square-foot landing and steps at each exterior door.

C. Storage and Fabrication Sheds: Temporary weather tight sheds or other covered facilities for storage of materials subject to weather damage. Number and size of structures shall be subject to Contracting Officer's approval.

D. Toilets: Sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, handicap accessible, sturdy enclosures with privacy locks. Facilities shall be tied down as needed to protect from high winds.

2.3 EQUIPMENT

A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

B. Communication Devices: Contractor shall provide means of communication from the South Rim Contractor Support Facilities to North Rim base of construction operations and to all Innercanyon construction camps and crews. Provide satellite-based phone services for the construction camps. Provide radio communication and/or satellite phone service to all construction crews. Reference 01 11 00 Summary of Work Communication Equipment Plan submission requirements.

C. Heating, Ventilation, and Air Conditioning (HVAC) Equipment: Unless Contracting Officer authorizes use of permanent HVAC system, provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.

1. Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.
2. Heating Units: Listed and labeled for type of fuel being consumed, by a testing agency acceptable to agency(ies) with jurisdiction, and marked for intended use.

D. Provide two (2) Contracting Officers Field Offices

1. Submit Field Office Plan that includes at a minimum a site plan with the proposed location each office within the limits of construction and temporary utilities connections/routing and floor plan dimensioned of the specific units to be supplied.
2. Outlets: Minimum of two, quad outlets with surge protection.
3. Refrigerator: Under counter, 3.2-cubic-foot volume with 0.8-cubic-foot freezer with dedicated power receptacle.
4. Fire Extinguisher: UL listed and FM (Fire Pump Motors) approved, minimum rating of 2-A:10-B:C, dry chemical.
5. First-Aid Kit: Provide first aid kit that meets OSHA requirement.
6. Two desks with five drawers each and two chairs with casters; two drafting tables (minimum 40 inches wide by 5 feet long) and two stools; drawing rack; two 2-drawer and one 4-drawer legal size locking filing cabinets with keys; 8 feet of 12 inch deep shelving; coat rack; two additional guest chairs; two desk lamps; two drafting table lamps; and a maximum/minimum thermometer.
7. Manufactured computer work station, capable of containing CPU (central processing unit), monitor, keyboard, printer; work station chair.
8. Additional tables necessary for copier.
9. Two 5-gallon trash cans and one 30-gallon trash can with lid.
10. Ceiling mounted general lighting fixtures.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Locate the facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

A. General: Install temporary service or connect to existing service.

1. Arrange with utility company, NPS, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services. Acquire necessary permits.

B. Storm Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.

1. Connect temporary sewers within the limits of construction to the NPS collection system as directed by NPS.

C. Water for construction.

1. See Section 01 50 00 1.3 B. Install piping and connections as required in the vicinity of the water source- must be within the limits of construction as indicated on the drawings. Facilities must be cleaned and maintained in a condition acceptable to NPS. At Substantial Completion, restore these facilities to condition existing before initial use.
2. Contractor shall furnish non-potable water from a source outside park boundary at all other times.

D. Sanitary Facilities: Provide temporary toilets, and wash facilities for use by construction personnel.

1. Place in approved locations secluded from public observation and convenient to work stations. Relocate as work progress requires.
2. Maintain and clean toilet facilities at least weekly.
3. Completely remove sanitary facilities on completion of work.

E. Heating: Provide temporary heating required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.

1. Use of permanent heating system will not be allowed without written authorization from Contracting Officer. When the permanent heating system is approved for use as temporary heating, pay costs until final acceptance. Permanent heating system shall be sufficiently complete, including controls, to permit safe operation
2. Provide and maintain adequate approved facilities, as required for safety and construction requirements, during the work. Provide ample clearance around stoves, heaters, chimney and vent connections to prevent ignition of combustible material

3. Install and maintain temporary filters when air handing equipment is used for temporary heating. Install new filters before final acceptance in addition to any extra sets of filters required. Clean coils as determined by Contracting Officer.
4. Warranties for equipment used for temporary heating shall start on date of Final Acceptance.

F. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.

G. Existing Electric Power Service: Use of existing electric power service will be permitted, as long as equipment is maintained in a condition acceptable to NPS.

1. When temporary connections are removed, restore existing utility services to original condition.

H. Temporary Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.

1. Install electric power service underground and within the limits of construction, unless otherwise indicated.

I. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.

1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
2. Lighting should be limited to what is necessary to safely perform work. Lights shall be angled down and include shielding.
3. Lights shall be turned off when work site is inactive.

J. Telephone Service: No telephone service is available on site for Contractor's use.

K. Wastewater Service: No use of the North Rim summer or winter wastewater treatment systems is allowed. Contractor to provide for this service as necessary.

3.3 SUPPORT FACILITIES INSTALLATION

A. General: Comply with the following:

1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 50 feet of building lines. Comply with NFPA 241.
2. Maintain support facilities until near Substantial Completion. Remove structures, equipment, and furnishings, and terminate services after punch list is 100 percent completed or when directed by Contracting Officer. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Contracting Officer.

B. Contracting Officers Field Office:

1. Provide Heat, lights, power, temporary water pressure and sewage holding tanks.

2. Provide office, furnishings, and utility connections no later than 7 days prior to start of field activities. Exact location will be determined by Contracting Officer.
3. Maintain equipment, furnishings, and structures. Provide equipment replacement elements as needed. Provide weekly cleaning services and trash disposal. Maintain and service water and sewer holding tanks as required.

C. Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate for construction operations. Temporary roads shall only be constructed as shown on the drawings.

1. Coordinate elevations of temporary roads and paved areas with permanent roads and paved areas.
2. Vehicle traffic is often impassable on State Route 67 (SR67) from Jacob Lake to the North Rim Lodge and on other North Rim roadways from first Monday after the first full weekend in December until May 15. The contractor shall maintain access at all times if working at the North Rim during this period. Snow removal of SR67 and North Rim roads will not be provided by NPS or the ADOT. Refer to specification 01 73 00.
3. Prepare subgrade and install subbase and base for temporary roads and paved areas according to Division 31 Section "Earth Moving."
4. Recondition base after temporary use, including removing contaminated material, regrading, proof-rolling, compacting, and testing.
5. Delay installation of final course of permanent hot-mix asphalt pavement until immediately before Substantial Completion. Repair hot-mix asphalt base-course pavement before installation of final course according to Division 32 Section "Asphalt Paving."
6. Remove temporary roads and revegetate to existing conditions at completion. 90 days before the start of revegetation actions the contractor will meet with the Contracting Officer (CO) and GRCA Vegetation Program Manager to discuss planned actions.

D. Traffic Controls: Erect and maintain barricades, lights, danger signals, and warning signs in accordance with Manual on Uniform Traffic Control Devices (MUTCD), Part IV, latest edition.

1. The Contractor shall take all necessary measures to maintain a normal flow of pedestrian and vehicular traffic to prevent accidents and to protect the work throughout the construction stages until completion of the work. The Contractor shall make the necessary arrangements to reroute traffic, provide and maintain barriers, cones, guards, barricades and construction warning and regulatory signs. The Contractor shall take effective measures necessary to protect all portions of the work during construction until completion. This includes providing and maintaining all necessary barriers, barricade lights and guards in accordance with the MUTCD.
2. Any detours, both vehicular and pedestrian, shall be included in traffic control plans submitted for review. The Contractor shall be responsible for installing, maintaining and removing traffic delineators and barricades in their proper locations, as well as traffic control signs on the approaches and throughout the work area.
3. Contractor shall provide a breakdown of all traffic control measures in the Schedule of Values. The Schedule of Values must contain all elements associated with the traffic control scope of work including but not limited to traffic control devices and labor.
4. Traffic delineators shall be used to delineate centerlines and land lines as appropriate and shall be spaced a maximum of 25 feet apart throughout the construction area. All lanes for moving traffic shall be at least 10 feet in width between delineators, with clearance of 2 feet from any vertical obstruction and 5 feet of any open excavation.
5. All barriers shall be temporary, moveable barriers. No posts or poles shall be installed in the existing roadway.
6. Protect existing site improvements to remain including curbs, pavement, and utilities.
7. Maintain a minimum of one lane of travel at all times. Provide necessary flagmen to slow and reroute traffic when and where necessary to move traffic through construction zone.

8. Maintain access for fire-fighting equipment and access to fire hydrants.
9. Illuminate barricades and obstructions at night; keep safety lights burning from sunset to sunrise.
10. Adequately barricade and post open cuts in or adjacent to thoroughfares.
11. Protect pedestrian traffic by guardrails or fences.
12. When pedestrian traffic is detoured onto a roadway, provide temporary walkways with protection as required at ends and overhead. For walkways, use lumber running parallel to direction of traffic movement and provide ramps at changes of elevation.
13. Cover pipes, hoses, and power lines crossing sidewalks and walkways with troughs using beveled edge boards.
14. Install Barrier Tape where directed by Contracting Officer. Keep a minimum of two rolls on site.

E. Dewatering Facilities and Drains: Comply with requirements of NPS. Maintain Project site, excavations, and construction free of water.

1. Dispose of rainwater in lawful manner that will not result in flooding Project or adjoining properties nor endanger permanent Work or temporary facilities.
2. Remove snow and ice as required to minimize accumulations.

F. Project Identification and Temporary Signs: Provide Project identification and other signs as indicated on Drawings. Fence, barricade, or otherwise block off the immediate work area to prevent unauthorized entry.

1. Provide temporary, directional signs for construction personnel and visitors.
2. Maintain and touchup signs so they are legible at all times.
3. Erect and maintain sufficient notification of construction activities and detour signs at road closures and along detour routes.
4. Erect and maintain sufficient notification of construction activities and detour signs at trailheads, along trails and along detour routes.

G. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of NPS.

H. Existing Stair Usage: Use of existing stairs will be permitted, as long as stairs are cleaned and maintained in a condition acceptable to Contracting Officer. At Substantial Completion, restore stairs to condition existing before initial use.

I. Temporary Use of Permanent Stairs: Cover finished, permanent stairs with protective covering of plywood or similar material so finishes will be undamaged at time of acceptance.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.

B. Cleaning of Equipment Prior to Mobilization: Contractor shall ensure prior to entering park boundaries, equipment, is free of soil, seeds, vegetative matter, or other debris that could contain or hold seeds. Ensure equipment has been pressure washed and is free of soil, seeds, vegetative matter, or other debris that could contain or holds seeds or other plant material (e.g., stems, rhizomes). The contractor will make arrangements for the Contracting Officer (CO) or GRCA Vegetation Program staff to inspect each vehicle and piece of equipment before entering the project area. Equipment shall be considered free of soil, seeds, and other debris when visual

inspection does not disclose such material. Disassembly of equipment components or specialized inspection tools are not required.

C. Cleaning of Equipment On-site: Contractor shall take care in washing tools and equipment to contain washdown, and properly clean and dispose of materials and fluids. Washdown should not be introduced to the environment.

D. Temporary Erosion and Sedimentation Control: Refer to Section 01 57 23 "Temporary Storm Water Pollution Prevention".

E. Tree and Plant Protection: Refer to Section 01 11 00 "Summary of Work".

F. Pest Control: Follow NPS requirements to minimize attraction and harboring of rodents, roaches, and other pests and perform extermination and control procedures at regular intervals so Project will be free of pests and residues at Substantial Completion. Perform control operations lawfully, using environmentally safe materials.

G. Site Enclosure Fence: Before construction operations begin, furnish and install chain link fencing to prevent people and animals from easily entering site except by entrance gates.

1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
2. Locate vehicular gates to avoid interference with traffic on public thoroughfares.
3. Locate pedestrian entrance gates as required to provide controlled personnel entry.
4. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide Contracting Officer with one set of keys.

H. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.

I. Barricades, Warning Signs, and Lights: Comply with requirements of Manual on Uniform Traffic Control Devices (MUTCD), part IV, 2003 edition for erecting structurally adequate barricades, including warning signs and lighting.

J. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is not complete, insulate temporary enclosures.

K. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241.

1. Responsible Person: Capable and qualified person shall be placed in charge of fire protection. Responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.
2. Tobacco Use, Smoking, and Vaping: Smoking within buildings or temporary storage sheds is prohibited.
3. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of NPS. Check with Park; many require "burn permits" for welding.

4. Develop and supervise overall fire-prevention and -protection program for personnel at Project Site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
5. Provide temporary standpipes and hoses for fire protection. Hang hoses with warning sign stating hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.
6. Hazard Control: Take necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids in existing buildings. Provide adequate ventilation during use of volatile or noxious substances. Ensure cleanup procedures and storage requirements are followed at close of every work session.
7. Spark Arresters: Equip gasoline or diesel powered equipment used during periods of potential fire hazards or in potential forest and grass fire locations with spark arresters approved by United States Department of Agriculture (USDA) Forest Service.
    a. Written determinations of periods and areas of potential fire hazard will be issued by Contracting Officer.
8. Buildings: Furnish a minimum of one extinguisher for each 1,500 square feet of area or major fraction thereof.
    a. Travel distance from any work station to nearest extinguisher shall not exceed 75 feet.
9. Vehicles and Equipment: Provide one extinguisher on each vehicle or piece of equipment.
10. Service and Refueling Areas: Locate areas a minimum of 50 feet from buildings. Shut down equipment before refueling.

3.5 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. Minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Maintenance: Maintain facilities in good operating condition until removal.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on 24-hour basis where required to achieve indicated results and avoid possibility of damage.

C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.

D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. NPS reserves right to take possession of Project identification signs.
2. At Substantial Completion, clean and renovate permanent facilities used during construction period.

END OF SECTION 01 50 00

This page intentionally left blank.

SECTION 01 57 19.11 – INDOOR AIR QUALITY MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for Indoor Air Quality (IAQ) management during construction operations.
   a. Control of emissions during construction.
   b. Moisture control during construction.
2. Procedures for testing baseline IAQ. Baseline IAQ requirements, specify maximum indoor pollutant concentrations for acceptance of the facility.

1.2 DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114.

B. Adequate ventilation: Ventilation, including air circulation and air changes, required to cure materials, dissipate humidity, and prevent accumulation of particulates, dust, fumes, vapors, or gases.

C. Hazardous Materials: Any material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have the potential to meet the definition of a Hazardous Waste in accordance with 40 CFR 261. Throughout this specification, hazardous material includes hazardous chemicals.

1. Hazardous materials include pesticides, biocides, and carcinogens as listed by recognized authorities, such as the Environmental Protection Agency (EPA) and the International Agency for Research on Cancer (IARC).

D. Indoor Air Quality (IAQ): Composition and characteristics of air in an enclosed space that affect occupants of that space. Indoor air quality of a space refers to relative quality of air in a building with respect to contaminants and hazards and is determined by the level of indoor air pollution and other characteristics of the air, including impact on thermal comfort such as air temperature, relative humidity and air speed.

E. Interior final finishes: Materials and products exposed to interior occupied spaces; including flooring, wall covering, finish carpentry, and ceilings.

F. Packaged dry products: Materials and products installed in dry form delivered in manufacturer's packaging; including carpets, resilient flooring, ceiling tiles, and insulation.

G. Wet products: Materials and products installed in wet form, including paints, sealants, adhesives, special coatings, and materials which require curing.

1.3 QUALITY ASSURANCE

A. Inspection and Testing Lab Qualifications: Minimum of 5 years of experience in performing types of testing specified herein.

1.4 SUBMITTALS

A. Indoor Air Quality (IAQ) Management Plan: After award and before Pre-construction conference, prepare and submit IAQ Management Plan, including:

1. Procedures for control of emissions during construction.
    a. Identify schedule for application of interior finishes: Identify each interior finish that generates odors, moisture, or vapors or is susceptible to adsorption of odors and vapors. Indicate air handling zone, sequence of application, and curing times.
    b. Identify potential sources of odor and dust.
    c. Identify construction activities likely to produce odor or dust.
    d. Identify areas of project potentially affected, especially occupied areas.
    e. Evaluate potential problems by severity and describe methods of control.
    f. Describe construction ventilation to be provided, including type and duration of ventilation, use of permanent Heating HVAC systems, types of filters and schedule for replacement of filters.
    g. Describe cleaning and dust control procedures.
    h. Describe coordination with commissioning procedures.
2. Procedures for moisture control during construction.
    a. Identify porous materials and absorptive materials.
    b. Identify schedule for inspection of stored and installed porous and absorptive materials.
3. Revise and resubmit Plan as required by Contracting Officer (CO).
    a. Approval of Contractor’s Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations.

B. Product Data:

1. Submit product data for filtration media used during construction and during operation. Include Minimum Efficiency Reporting Value (MERV).
2. Material Safety Data Sheets (MSDS): Submit MSDSs for inclusion in Operation and Maintenance Manual for:
    a. Adhesives
    b. Floor and wall patching/leveling materials
    c. Caulking and sealants
    d. Insulating materials
    e. Fireproofing and firestopping
    f. Carpet
    g. Paint
    h. Clear finish for wood surfaces
    i. Lubricants
    j. Cleaning products

C. Inspection and Test Reports:

1. Moisture control inspections
2. Moisture content testing
3. Moisture penetration testing
4. Microbial Growth testing

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 IAQ MANAGEMENT - EMISSIONS CONTROL

A. During construction operations, follow the recommendations in SMACNA IAQ Guidelines for Occupied Buildings under Construction.

B. HVAC Protection:

1. Seal return registers during construction operations.
2. Provide temporary exhaust during construction operations.
3. To greatest extent possible, isolate and/or shut down return side of HVAC system during construction. When ventilation system must be operational during construction activities, provide temporary filters at air inlets (returns) and at locations for filters prescribed in the design.
4. Contractor shall bear cost of cleaning required due to failure to protect ducts and equipment from construction dust.

C. Source Control: Provide low and zero VOC materials as specified.

D. Pathway Interruption: Isolate areas of work to prevent contamination of clean or occupied spaces. Provide pressure differentials and/or physical barriers to protect clean or occupied spaces.

E. Housekeeping: During construction, maintain project and building products and systems to prevent contamination of building spaces.

F. Temporary Ventilation: For materials/products that generally require ventilation for off gassing, provide an ACH (air changes per hour) of 1.5 or more and as follows:

1. Provide minimum 48-hour pre-ventilation of packaged dry products prior to installation. Remove from packaging and ventilate in a secure, dry, well-ventilated space free from strong contaminant sources and residues. Provide a temperature range of 60 degrees Fahrenheit minimum to 90-degree Fahrenheit maximum continuously during ventilation period. Do not ventilate within limits of Work unless otherwise approved by Contracting Officer.
2. Provide adequate ventilation during and after installation of interior wet products and interior final finishes.
3. Provide filtration media with a Minimum Efficiency Reporting Value (MERV) of 8 as determined by ASHRAE 52.2 during construction. Coordinate with work of Division 23 (15), Heating Ventilating and Air Conditioning (HVAC).

G. Scheduling: Schedule construction operations involving wet products prior to packaged dry products to the greatest extent possible.

H. Flush-Out: After construction ends, prior to occupancy and with all interior finishes installed, perform a building flush-out by supplying a total air volume of 14,000 cubic feet. of outdoor air per square feet of floor area while maintaining an internal temperature of at least 60 degrees Fahrenheit and relative humidity no higher than 60%.

1. Obtain Contracting Officers concurrence that construction is complete enough before beginning flush-out.

2. If additional construction involving materials that produce particulates or any of specified contaminants is conducted during or after flush-out, then flush-out process must be restarted.
3. Install new HVAC filtration media in locations identified to have permanent filtration in contract documents after completion of flush-out and before occupancy or further testing.

3.2 IAQ MANAGEMENT - MOISTURE CONTROL

A. Housekeeping:

1. Keep materials dry. Protect stored on-site and installed absorptive materials from moisture damage.
2. Verify installed materials and products are dry prior to sealing and weatherproofing building envelope.
3. Store interior absorptive materials only after building envelope is sealed and weatherproofed.

B. Inspections: Document and report results of inspections; state whether or not inspections indicate satisfactory conditions.

1. Examine materials for dampness as they arrive. If acceptable to Contracting Officer, dry damp materials completely prior to installation; otherwise, reject materials that arrive damp.
2. Examine materials for mold as they arrive and reject materials that arrive contaminated with mold.
3. Inspect stored and installed absorptive materials regularly for dampness and mold growth. Inspect weekly.
    a. If stored or installed absorptive materials become wet, notify Contracting Officer. Inspect for damage. If acceptable to the Contracting Officer, dry completely prior to closing in assemblies; otherwise, remove (in accordance with the Waste Management Plan) and replace with new materials.
4. Basement: Monitor basement and crawlspace humidity and dehumidify when relative humidity is greater than 70 percent for more than 2 weeks or at first sign of mold growth.
5. Site drainage: Verify final grades of site work and landscaping drain surface water and ground water away from building.
6. Weatherproofing: Inspect moisture control materials as they are being installed. Include:
    a. Air barrier: Verify air barrier is installed without punctures and/or other damage. Verify air barrier is sealed completely.
    b. Flashing: Verify correct shingling of flashing for roof, walls, windows, doors, and other penetrations.
    c. Vapor Barrier: Verify vapor barrier is installed in accordance with Contract documents.
    d. Insulation layer: Verify insulation is installed without voids.
    e. Roofing: In accordance with ASTM D7186 Standard Practice for Quality Assurance Observation of Roof Construction and Repair.
7. Plumbing: Verify satisfactory pressure test of pipes and drains is performed before closing in and insulating lines.

C. Schedule:

1. Schedule work such that absorptive materials, such as porous insulations, paper-faced gypsum board, ceiling tile, and finish flooring, are not installed until they can be protected from rain and construction-related water.

2. Weather-proof as quickly as possible. Schedule installation of moisture-control materials, including but not limited to air barriers, flashing, exterior sealants and roofing, at earliest possible time.

D. Testing for Moisture Content: Test moisture content of porous materials and absorptive materials to ensure they are dry before sealing them into an assembly. Document and report results of testing. Where tests are not satisfactory, dry materials and retest. If satisfactory results cannot be obtained with retest, remove and replace with new materials.

1. Wood: Moisture test as per ASTM D4444 - Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters; unless otherwise indicated acceptable upper limits for wood products are less than 20% at center of piece; less than 15% at surface.

E. Testing for Moisture Penetration:

1. Windows: Test as per ASTM E1105 Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform or Cyclic Static Air Pressure Difference; unless otherwise indicated, acceptable upper limits are no leakage for 15 minutes**.**
2. Horizontal Waterproofing (not roofing): Test as per ASTM D5957 Standard Guide for Flood Testing Horizontal Waterproofing Installations; acceptable upper limits are no leakage for 15 minutes**.**
3. Masonry: Test as per ASTM C1601 Standard Test Method for Field Determination of Water Penetration of Masonry Wall Surfaces; acceptable upper limits are no leakage for 15 minutes.
4. Exterior Walls:
   a. Air tightness of the enclosure test: ASTM E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization or ASTM E1827 Standard Test Methods for Determining Air tightness of Buildings Using an Orifice Blower Door; acceptable upper limits are 0.25 CFM/sf or less at 50 Pascal's.
   b. Water Leakage: Review as per ASTM E2128 Standard Guide for Evaluating Water Leakage of Building Walls.

END OF SECTION 01 57 19.11

This page intentionally left blank.

SECTION 01 57 19.12 – NOISE AND ACCOUSTICS MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for noise and acoustics management during construction operations.

1.2 DEFINITIONS

A. Ambient noise level: The total noise associated with a given environment, being usually a composite of normal or existing sounds from all sources near and far, excluding the noise source at issue.

B. Daytime: The hours from 7 A.M. to 6 P.M.

C. Nighttime: All non-daytime hours.

D. Property line: The real or imaginary line along the ground surface and its vertical extension, which separates real property owned or controlled by one person from contiguous real property owned or controlled by another person or from any public right-of-way or from any public space.

E. Receiving noise area: Any real property where people live or work and where noise is heard, excluding the project or source area.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 NOISE MANGEMENT

A. Noise Control: Perform construction operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer CO).

B. Repetitive and/or intermittent, high-level noise: Permitted only during Daytime.

1. Do not exceed the following dB(A) limitations at 50 FT:

Sound Level in dB(A)	Time Duration of Impact Noise
70	More than 12 minutes in any hour
80	More than 3 minutes in any hour

2. Maximum permissible construction equipment noise levels at 50 FT:

EARTHMOVING	dB(A)	MATERIALS HANDLING	dB(A)
Front Loaders	75	Concrete Mixers	75
Backhoes	75	Concrete Pumps	75

Dozers	75	Cranes	75
Tractors	75	Derricks Impact	75
Scrapers	80	Pile Drivers	95
Graders	75	Jack Hammers	75
Trucks	75	Rock Drills	80
Pavers, Stationary	80	Pneumatic Tools	80
Pumps	75	Saws	75
Generators	75	Vibrators	75
Compressors	75		

C. Ambient Noise:

1. Maximum noise levels (dB (decibel)) for receiving noise area at property line shall be as follows:
   a. Residential receiving area
      Daytime: 65 dB
      Nighttime: 45 dB
   b. Commercial/Industrial receiving area
      Daytime: 67 dB
      Nighttime: 65 dB
   c. In the event the existing local ambient noise level exceeds the maximum allowable receiving noise level (dB), the receiving noise level maximum for construction operations shall be adjusted as follows:
   d. Residential receiving area: Maximum 3 additional dB above the local ambient as measured at property line.
   e. Commercial/Industrial receiving area: Maximum 5 additional dB above the local ambient as measured at the property line.

3.2 FIELD QUALITY CONTROL

A. Assess potential effects of construction noise on adjacent NPS operations, concessionaire, and visitors in accordance with ASTM E1686 and as follows:

1. Ambient noise measurement: Measure at property line at a height of at least four (4) feet above the immediate surrounding surface. Average the ambient noise level over a period of at least 15 minutes.
2. Ambient noise measurement at urban sites: Conduct during morning peak traffic hour between 7 A.M. and 9 A.M. and afternoon peak traffic hour between 4 P.M. and 6 P.M. In addition, conduct a 24-hour measurement at the proposed project site to document the noise pattern throughout the day. Adjust and weight for seasonal and climatic variations.

B. Monitor noise produced from construction operations in accordance with ASTM E1780.

C. Utilize industry available noise reduction and spark arrest features for mechanical equipment to reduce construction noise in all project areas.

END OF SECTION 01 57 19.12

SECTION 01 57 23 - TEMPORARY STORM WATER POLLUTION PREVENTION

PART 1 - GENERAL

1.1 SUMMARY

A. Federal Regulations for controlling discharges of pollutants (including chemicals, erodible material, and trash) from municipal separate storm sewer systems, construction sites, and industrial activities, were brought under the National Pollution Discharge Elimination System (NPDES) permit process by amendments to the Clean Water Act (CWA), and promulgation of federal stormwater regulations issued by the United States Environmental Protection Agency (USEPA). The USEPA uses amount of ground disturbance as a measure of a project potential to generate pollution from erosion. NPDES Phase I regulates discharges from construction sites that disturb 5 acres or more. NPDES Phase II regulations expand existing General Permit requirements under Phase I to include/regulated discharges from construction sites that disturb land equal to or greater than one (1) acre and less than 5 acres, known as Small Construction Activity. Construction disturbances 1 acre and above typically require a formal NPDES permit and a formal Stormwater Pollution Prevention Plan (SWPPP) must be submitted to Agency(ies) with Jurisdiction for review and approval.

B. The work of this section consists of implementing measures to prevent discharges of pollutants, including temporary storm water pollution during construction activities, either through compliance with NPDES permit program.

1.2 DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114.

B. Environmental Pollution and Damage: The presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of the environment for aesthetic, cultural, or historical purposes.

C. National Pollution Discharge Elimination System (NPDES) Phase I: Regulates discharges from construction sites that disturb 5 acres or more.

D. NPDES Phase II: Regulations expand existing General Permit requirements under Phase I to include and regulate discharges from construction sites that disturb land equal to or greater than one (1) acre and less than 5 acres, known as Small Construction Activity.

E. Storm Water Pollution Prevention Plan (SWPPP): Developed and implemented stormwater management measures to protect surface water from pollutants during construction activities disturbing an acre or more in compliance with federal, state, and local requirements for permit approval under NPDES program.

1.3 SUBMITTALS

A. After contract award and before pre-construction conference, prepare and submit:

1. A SWPPP showing SWPPP satisfies Federal and State NPDES permit requirements.

B. Inspection Schedule: Submit schedule for inspection and monitoring of pollution prevention measures.

C. Inspection Schedule: Submit schedule for inspection and monitoring of storm water pollution prevention measures.

D. Erosion Control Products: Submit manufacturer's product information and installation recommendations for silt fence, filter fabric, erosion control blanket, straw bales, and other materials proposed for use on this project.

1.4 QUALITY ASSURANCE

A. Contractor shall prepare and submit a plan to Contracting Officer (CO) for review and concurrence.

B. Orientation Meeting: Contractor shall arrange and conduct an Erosion and Sediment Control meeting/briefing to inform parties, scheduled to be on-site during project, of measures to be implemented for proper erosion and sediment control (may be included as part of Pre-Construction Meeting).

1. Installation of silt fences, storm drain protection, and other forms of erosion and sediment control shall not begin until after this meeting has occurred.

C. Pollution Prevention and Erosion Control Manager: Contractor shall designate Pollution Prevention and Erosion Control Manager responsible for implementation, inspection, maintenance, and amendments to approved plan.

1. Pollution Prevention and Erosion Control Manager shall be familiar with temporary storm water pollution prevention procedures and Best Management Practices and ensure emergency procedures and plan are updated as needed and available for inspection.
2. When changes in approved plan are required, Pollution Prevention and Erosion Control Manager shall prepare and certify an amendment and submit to Contracting Officer for review and concurrence.

PART 2 - PRODUCTS

2.1 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Provide SWPPP which satisfies Federal and State NPDES permit requirements and includes:

1. Site description.
2. Identification and contract information for Pollution Prevention and Erosion Control Manager.
3. Expected sequencing of operations and construction schedule.
4. Weather monitoring procedure.
5. Descriptions and details Best Management Practices for pollution prevention and erosion controls, including dust control.
6. Pollution prevention and erosion control plans.
7. Controls for other potential onsite storm water pollutants.
8. Applicable specifications.
9. Maintenance and inspection procedures and forms.
10. Description of potential non-storm water discharges at site.

11. Notice of Intent (NOI) form.
12. Notice of Termination (NOT) form.
13. Contractor and Sub-contractor Certification forms.
14. Other record keeping forms and procedures.
15. Housekeeping Best Management Practices, including vehicle wash-down areas, protection of equipment storage and maintenance areas, and sweeping of roadways related to hauling activities.

PART 3 - EXECUTION

3.1 ENVIRONMENTAL PROTECTION

A. Protection of Natural Resources: Comply with applicable regulations and these specifications. Preserve natural resources within project boundaries and outside limits of work performed under this Contract in their existing condition or restore to an equivalent or improved condition as approved by Contracting Officer.

B. Construction Zone: Arrange construction activities to minimize pollution (i.e., erosion, trash, etc.) to maximum practical extent.

1. Clearing, excavation, and grading shall be limited to those areas of project site necessary for construction. Minimize area exposed and unprotected.
2. Clearly mark and delineate limits of work activities.
3. Equipment shall not be allowed to operate outside limits of work or to disturb existing vegetation.
4. Excavation and grading shall be completed during dry season to maximum extent possible.
5. Material shall be stored away from locations where water is present to greatest extent practicable.

3.2 REGULATORY REQUIREMENTS

A. Permits: Contractor shall obtain required NPDES permits resulting in no impacts to scheduled work. Contractor shall account for possibility of significant lead time in scheduling and executing work.

1. Implement requirements of NPDES permit for erosion control due to storm water runoff during construction.
2. Implement good housekeeping practices, inspections and record keeping.
3. Prior to construction, Contractor and Subcontractors shall sign certifications (included in the plan) that they understand requirements of NPDES permit.
4. Subcontractors shall comply with requirements of NPDES under supervision of Contractor.
5. Accepted plan shall comply with terms and conditions of EPA permit.

B. Notice of Intent (NOI): Contractor shall file a Notice of Intent and formal SWPPP as required to the Agency(ies) with Jurisdiction.

C. Notice of Termination (NOT): After Substantial Completion of Work, file a Notice of Termination (NOT) with the Agency(ies) with Jurisdiction.

D. Contracting Officer Notification: Contractor shall notify Contracting Officer in writing and by telephone of these events:

1. Erosion and sediment control meeting/briefing.
2. Following installation of required sediment control structures.
3. Prior to removal of or modification to sediment control structures.
4. Prior to removal of sediment control structures.

3.3 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Review and Acceptance: Contractor and Contracting Officer will jointly review draft Plan and agree to needed revisions. Contractor shall incorporate revisions, sign, and submit final Plan to Contracting Officer. Final Plan will be the document enforced on the project.

1. Accepted Plan will describe and ensure implementation of practices to be used to reduce pollutants in storm water discharges.
2. Contractor shall maintain current copy of Plan and associated records and forms at jobsite throughout duration of project.
3. Plan shall be available for public inspection and inspection and use of Contracting Officer.
4. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.

B. Implementation: Implement Plan as required throughout construction period and maintain erosion control elements in proper working order.

1. Do not perform clearing and grubbing or earthwork until Plan has been implemented.

C. SWPPP (including inspection forms) and data used to complete the NOI shall be provided to Contracting Officer after Substantial Completion of project.

3.4 SITE INSPECTIONS AND PLAN REVISIONS

A. Inspections: Contractor and Contracting Officer will perform a weekly inspection onsite.

1. Inspection shall include disturbed areas not completely stabilized, areas used for storage of materials, locations where vehicles enter or exit site, and other erosion and sediment controls included in the Plan.
2. Inspections shall be documented.
3. Inspection forms shall be retained onsite in Plan notebook throughout construction period.

B. Plan Revisions: It may be necessary to revise Plan during construction to make necessary improvements, revisions, or to respond to unforeseen conditions noted during construction or site inspections.

1. Plan shall specify mechanism whereby revisions may be proposed by Contractor or Contracting Officer.
2. Contractor and Contracting Officer will jointly review each revision to Plan before changes incorporated and implemented. Contractor will then provide revised copy of Plan to Contracting Officer.
3. Accepted modifications will be implemented within 7 calendar days following date of inspection when deficiencies or necessary corrections are first noted.

C. Negligence: Provide additional temporary erosion and pollution controls made necessary by Contractor's errors or negligence at no additional cost to Government.

3.5 HOUSEKEEPING AND SITE MANAGEMENT

A. Store materials onsite in conformance to Federal, state, local, and manufacturer's regulations and specifications. Use Best Management Practices to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.

B. Manage solid waste in conformance to Federal, state, and local regulations. Best Management Practices should be used to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.

C. Include a spill prevention and control plan with provisions placed in SWPPP.

D. Manage hazardous waste (including contaminated soil) in conformance to Federal, state, local and NPS regulations and guidelines.

3.6 EROSION CONTROL MEASURES

A. Erosion control measures shall consist of Best Management Practices for storm water discharges, including silt fencing, barrier protectors, straw bales, temporary soil retention blankets, excelsior drainage filters, sediment traps and berms.

B. Berms and excelsior drainage filters shall be used to form sediment traps and control run-on and run-off into other areas, including creeks, streams, marshes, access roads, well areas, and staging areas.

C. Erosion control measures shall be used to contain only direct precipitation in construction zone. Contained water shall be allowed to percolate into ground or drain slowly through drainage filter sediment traps.

D. Earthen sediment traps or holding ponds shall not be used unless accepted by Contracting Officer.

E. Reduce runoff velocity and direct surface runoff around and away from fuel containment, storage, and borrow areas.

F. Divert surface runoff around and away from cut and fill slopes.

G. Place drainage filters around catch basins to create sediment traps to control run-off from construction area.

H. Excess water used for dust control shall be contained within demolition areas by erosion control measures.

I. Contractor shall prevent deposition of materials onto paved areas. Contractor shall inspect paved areas for deposited materials weekly and remove materials immediately.

J. Furnish, install, maintain, and operate necessary control measures and other equipment necessary to prevent erosion as described in approved SWPPP.

K. Furnish, install, maintain, and operate necessary control measures and other equipment necessary to prevent erosion as described in approved UPPP.

L. Before work begins, sufficient equipment shall be available on site to assure operation and adequacy of erosion control system can be maintained.

3.7 MAINTENANCE OF TEMPORARY FACILITIES

A. Ensure erosion and sediment control structures remain effective throughout excavation and grading operations. Relocate structures as necessary.

B. Inspect control structures after each significant rainfall. Promptly repair breaches which occur.

C. Contractor shall remove entrapped sediment from behind excelsior drainage filter after each storm.

3.8 REPORTING

A. If a discharge occurs or if project receives written notice or order from regulatory agency, Contractor shall immediately notify Contracting Officer and shall file written report to Agency(ies) with Jurisdiction within 7 days of discharge event, notice, or order. Corrective measures shall be implemented immediately following discharge, notice, or order. The report to the Agency(ies) with Jurisdiction shall contain:

1. Date, time, location, nature of operation, and type of discharge, including cause or nature of notice or order.
2. Best Management Practices deployed before discharge event, or prior to receiving notice or order.
3. Date of deployment and type of Best Management Practices deployed after discharge event, or after receiving notice or order, including additional Best Management Practices installed or planned to reduce or prevent re-occurrence.
4. An implementation and maintenance schedule for affected Best Management Practices.

3.9 SEDIMENT DISPOSAL

A. Sediment excavated from temporary sediment control structures shall be disposed on site with general fill, or with topsoil. Sediment shall be allowed to dry out as required before reuse.

B. Contractor shall place sediment removed from traps and other structures where it will not enter a storm drain or watercourse and where it will not immediately reenter the basin.

3.10 REMOVAL OF TEMPORARY STORM WATER POLLUTION CONTROL MEASURES

A. Temporary control measures shall be removed with permission of Contracting Officer within 20 working days after final acceptance of project, and/or once grading is complete and slopes have stabilized.

END OF SECTION 01 57 23

SECTION 01 61 03 - EQUIPMENT - BASIC REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Requirements of this Specification Section apply to all equipment provided on the Project including those found in other Divisions even if not specifically referenced in individual "Equipment" Articles of those Specification Sections.

B. Related Sections include but are not necessarily limited to:

1. Section 01 81 10 – Wind and Seismic Design Criteria
2. Section 03 15 19 – Anchorage to Concrete
3. Section 03 31 30 – Concrete, Materials and Proportioning
4. Section 05 50 00 – Metal Fabrications
5. Section 07 92 00 – Joint Sealants
6. Section 09 96 00 – High Performance Industrial Coatings
7. Section 10 14 00 – Identification Devices
8. Section 26 29 23 – Variable Frequency Drives - Low Voltage
9. Section 40 05 00 – Pipe and Pipe Fittings - Basic Requirements
10. Section 40 67 00 – Control System Equipment Panels and Racks
11. Section 40 71 00 – Flow Instrumentation
12. Section 40 72 00 – Level Instrumentation
13. Section 40 73 00 – Pressure Instrumentation
14. Section 40 75 00 – Process Liquid Analytical Measurement
15. Division 46 – Water and Wastewater Equipment

1.2 QUALITY ASSURANCE

A. Referenced Standards:

1. American Bearing Manufacturers Association (ABMA).
2. American Gear Manufacturers Association (AGMA).
3. ASTM International (ASTM):
   a. E1934, Standard Guide for Examining Electrical and Mechanical Equipment with Infrared Thermography.
   b. F593, Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
4. Hydraulic Institute (HI):
   a. 9.6.4, Rotodynamic Pumps for Vibration Measurements and Allowable Values.
5. International Electrotechnical Commission (IEC).
6. Institute of Electrical and Electronics Engineers, Inc. (IEEE).
7. International Organization for Standardization (ISO):
   a. 1940, Mechanical Vibration - Balance Quality Requirements for Rotors in a Constant (Rigid) State - Part 1: Specification and Verification of Balance Tolerances.

b. 21940-11, Mechanical Vibration - Rotor Balancing - Part 11: Procedures and Tolerances for Rotors with Rigid Behavior.

8. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   b. ICS 6, Enclosures for Industrial Control and System.
   c. MG 1, Motors and Generators.
9. International Electrical Testing Association (NETA):
   a. ATS, Acceptance Testing Specification for Electrical Power Distribution Equipment and Systems.
10. National Fire Protection Association (NFPA):
    a. 70, National Electrical Code (NEC):
11. National Institute for Certification in Engineering Technologies (NICET).
12. National Institute of Standards and Technology (NIST).
13. Occupational Safety and Health Administration (OSHA):
    a. 29 CFR 1910, Occupational Safety and Health Standards, referred to herein as OSHA Standards.
14. Underwriters Laboratories, Inc. (UL).
    a. 508, Standard for Safety Industrial Control Equipment.
    b. 508A, Standard for Safety Industrial Control Panels.
    c. 698A, Standard for Industrial Control Panels Relating to Hazardous (Classified) Locations.
15. Vibration Institute.

B. All testing required by this specification section to be provided and paid for by Contractor.

C. Natural frequency analysis firm:

1. An independent firm, whose sole or principal part of its business is the calculation of and analysis of natural frequencies of rotating equipment.
2. Minimum of 10 years experience.
3. Employs a registered professional engineer who has experience in finite element analysis, rotordynamic analysis and experimental modal analysis.
   a. Minimum five years combined field testing and data analysis experience.
   b. Qualified Vibration Category III certification from the Vibration Institute.

D. Vibration Testing Program:

1. Testing firm:
   a. An independent firm performing, as the sole or principal part of its business for a minimum of 10 years, the inspection, testing, calibration, and adjusting of systems.
   b. Must have an established monitoring and testing equipment calibration program with accuracy traceable in an unbroken chain, according to NIST.
2. Field personnel:
   a. Minimum of three years field experience covering all phases of field vibration testing and data gathering.
   b. Qualified Vibration Category II certification from the Vibration Institute.
3. Analysis personnel:
   a. Minimum five years combined field testing and data analysis experience.
   b. Qualified Vibration Category III certification from the Vibration Institute.

E. Infrared Thermography Testing Program:

1. Testing firm:

a. An independent firm performing, as the sole or principal part of its business for a minimum of 10 years, the inspection, testing, calibration, and adjusting of systems.
b. Must have an established monitoring and testing equipment calibration program with accuracy traceable in an unbroken chain, according to NIST.

2. Field personnel:
a. Minimum of one year field experience covering all phases of field thermography testing and data gathering.
b. Supervisor certified by NETA or NICET.

3. Analysis personnel:
a. Minimum three years combined field testing and data analysis experience.
b. Supervisor certified by NETA or NICET.

F. Electrical Equipment and Connections Testing Program:

1. Testing firm:
a. An independent firm performing, as the sole or principal part of its business for a minimum of 10 years, the inspection, testing, calibration, and adjusting of systems.
b. Must have an established monitoring and testing equipment calibration program with accuracy traceable in an unbroken chain, according to NIST.

2. Field personnel:
a. Minimum of one year field experience covering all phases of electrical equipment inspection, testing, and calibration.
b. Relay test technician having previous experience with testing and calibration of relays of the same manufacturer and type used on project and proficient in setting and testing the types of protection elements used.
c. Supervisor certified by NETA or NICET.

3. Analysis personnel:
a. Minimum three years combined field testing and data analysis experience.
b. Supervisor certified by NETA or NICET.

G. Miscellaneous:

1. A single manufacturer of a "product" shall be selected and utilized uniformly throughout Project even if:
a. More than one manufacturer is listed for a given "product" in Specifications.
b. No manufacturer is listed.

2. Equipment, electrical assemblies, related electrical wiring, instrumentation, controls, and system components shall fully comply with specific NEC requirements related to area classification and to NEMA 250 and NEMA ICS 6 designations shown on Electrical Power Drawings and defined in the Electrical specifications.

3. Variable speed equipment applications: The driven equipment manufacturer shall have single source responsibility for coordination of the equipment and VFD system and verify their compatibility.

1.3 DEFINITIONS

A. Product: Manufactured materials and equipment.

B. Major Equipment Supports - Supports for Equipment:

1. Located on or suspended from elevated slabs with supported equipment weighing 2,000 LBS or greater, or;

2. Located on or suspended from roofs with supported equipment weighing 500 LBS or greater, or;
3. Located on slab-on-grade or earth with supported equipment weighing 5,000 LBS or more.

C. Equipment:

1. One or more assemblies capable of performing a complete function.
2. Mechanical, electrical, instrumentation or other devices requiring an electrical, pneumatic, electronic or hydraulic connection.
3. Not limited to items specifically referenced in "Equipment" articles within individual Specifications.

D. Installer or Applicator:

1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
2. Installer and applicator are synonymous.

E. Baseplate or Equipment Base Plate or Machine Base:

1. Are fabricated frames of structural shapes and plates with enough strength and sturdiness to serve as the surface to which other equipment is attached to and supported by. Baseplates can be directly mounted and grouted to concrete equipment support bases or machined and bolted to a sole plate.

F. Sole Plate:

1. A thick steel machined plate that is attached to and grouted to a concrete equipment support base.
2. Base plates are bolted to a sole plate when a sole plate is specified and/or provided.

1.4 SUBMITTALS

A. Shop Drawings:

1. General for all equipment:
   a. See Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   b. Data sheets that include manufacturer's name and complete product model number.
      1) Clearly identify all optional accessories that are included.
   c. Acknowledgement that products submitted comply with the requirements of the standards referenced.
   d. Manufacturer's delivery, storage, handling, and installation instructions.
   e. Equipment identification utilizing numbering system and name utilized in Drawings.
   f. Equipment installation details:
      1) Location of anchorage.
      2) Type, size, and materials of construction of anchorage.
      3) Anchorage setting templates.
      4) Manufacturer's installation instructions.
   g. Equipment area classification rating.
   h. Shipping and operating weight.

i. Equipment physical characteristics:
    1) Dimensions (both horizontal and vertical).
    2) Materials of construction and construction details.
j. Equipment factory primer and paint data.
k. Manufacturer's recommended spare parts list.
l. Equipment lining and coatings.
m. Equipment utility requirements include air, natural gas, electricity, and water.
n. Ladders and platforms provided with equipment:
    1) Certification that all components comply fully with OSHA requirements.
    2) Full details of construction/fabrication.
    3) Scaled plan and sections showing relationship to equipment.

2. Mechanical and process equipment:
    a. Operating characteristics:
        1) Technical information including applicable performance curves showing specified equipment capacity, rangeability, and efficiencies.
        2) Brake horsepower requirements.
        3) Copies of equipment data plates.
    b. Piping and duct connection size, type and location.
    c. Equipment bearing life certification.
    d. Equipment foundation data:
        1) Equipment center of gravity.
        2) Criteria for designing vibration, special or unbalanced forces resulting from equipment operation.

3. Electric motor:
    a. Motor manufacturer and model number.
    b. Complete motor nameplate data.
    c. Weight.
    d. NEMA design type.
    e. Enclosure type.
    f. Frame size.
    g. Winding insulation class and temperature rise.
    h. Starts per hour.
    i. Performance data:
        1) Motor speed-torque curve superimposed over driven machine speed-torque curve during start-up acceleration and at rated terminal voltage a minimum permissible or specified terminal voltage for all motors over 50 HP.
        2) Time-current plots with acceleration versus current and thermal damage curves at the operating and ambient temperatures and at rated terminal voltage and minimum permissible or specified terminal voltage for all motors over 50 HP.
        3) Guaranteed minimum efficiencies at 100 PCT, 75 PCT, and 50 PCT of full load.
        4) Guaranteed minimum power factor at 100 PCT, 75 PCT, and 50 PCT of full load.
        5) Locked rotor and full load current at rated terminal voltage and minimum permissible or specified terminal voltage.
        6) Starting, full load, and breakdown torque at rated terminal voltage and minimum permissible or specified terminal voltage.
    j. Bearing data and lubrication system.
    k. Natural frequency calculations for:
        1) Completed assembly including but not limited to the equipment base, rotating piece of equipment, and the rotating piece of equipment driver.
        2) Individual piece of rotating equipment.
        3) Equipment driver and connected gear reducer, if applicable.

l. Thermal protection system including recommended alarm and trip settings for winding and bearing RTD's and/or other temperature-sensing devices.
m. Fabrication and/or layout drawings:
1) Dimensioned outlined drawing.
2) Connection diagrams including accessories (strip heaters, thermal protection, etc.).
n. Certifications:
1) When utilized with a reduced voltage starter, certify that motor and driven equipment are compatible.
2) When utilized with a variable frequency controller, certify motor is inverter duty and the controller and motor are compatible.
a) Include minimum speed at which the motor may be operated for the driven machinery.
o. Electrical gear:
1) Unless specified in a narrow-scope Specification Section, provide the following:
a) Equipment ratings: Voltage, continuous current, kVA, watts, short circuit with stand, etc., as applicable.
2) Control panels:
a) Panel construction.
b) Point-to-point ladder diagrams.
c) Scaled panel face and subpanel layout.
d) Technical product data on panel components.
e) Panel and subpanel dimensions and weights.
f) Panel access openings.
g) Nameplate schedule.
h) Panel anchorage.
i) Short Circuit Current Rating (SCCR) nameplate marking per NFPA 70. Include any required calculations.

4. Systems schematics and data:
a. Provide system schematics where required in system specifications.
1) Acknowledge all system components being supplied as part of the system.
2) Utilize equipment, instrument and valving tag numbers defined in the Contract Documents for all components.
3) Provide technical data for each system component showing compliance with the Contract Document requirements.
4) For piping components, identify all utility connections, vents and drains which will be included as part of the system.
5. For factory painted equipment, provide paint submittals in accordance with Section 09 96 00.
6. Qualifications for:
a. Natural frequency analysis firm and personnel.
b. Vibration testing firm and personnel.
c. Infrared thermography testing firm and personnel.
d. Electrical equipment and connections testing firm and personnel.
7. Equipment Monitoring and Testing plans, in accordance with PART 3 of this Specification Section:
a. Natural frequency analysis and calculations.
b. Vibration testing.
c. Thermography testing.
d. Electrical equipment and connection testing.

B. Factory Test Reports:

1. Natural frequency bump test reports where required for rotating equipment.
    a. Minimum characteristics of impact hammer.
        1) Frequency Range 1 kHz.
        2) Range (5v output) 5,000 LBF (22,200 N).
        3) Hammer Sensitivity (7pprox.) 1 mV/lbf (0.23 mV/N).
        4) Resonant Frequency 12 kHz
2. Motor, equipment and final assembled equipment including motor.
    a. Determine natural frequency of assembled motor prior to shipping to OEM or job site.
        1) Individual motor fastened to an “infinitely rigid” mass at the same bolt circle as the final assembled equipment.
    b. Determine natural frequency of the pump.
        1) Pump fastened to an “infinitely rigid” mass at the same bolt circle as the final assembled equipment.
    c. Determine natural frequency of the pump/motor assembly.
        1) Pump/motor assembly fastened to an “infinitely rigid” mass at the same bolt circle as the final field assembled equipment.
    d. For this use, the "infinitely rigid" mass shall be at least 10 times the weight of the equipment being tested.
3. Submit natural frequency report(s) for approval prior to shipment.
4. Equipment performance tests.
    a. As listed in individual equipment specifications.

C. Contract Closeout Information:

1. Operation and Maintenance Data:
    a. See Section 01 78 23 – Operation and Maintenance Data for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

D. Informational Submittals:

1. Notification, at least one week in advance, that testing will be conducted at factory.
2. Certification from equipment manufacturer that all manufacturer-supplied control panels that interface in any way with other controls or panels have been submitted to and coordinated with the supplier/installer of those interfacing systems.
3. Submit sample Manufacturer's Field Service Report (MFSR). Report shall use manufacturer’s standard report or use the form in the Exhibits and have at least the following information:
    a. Certification that equipment has been installed properly, has been initially started up, has been calibrated and/or adjusted as required, and is ready for operation.
    b. Certification for major equipment supports that equipment foundation design loads shown on the Drawings or specified have been compared to actual loads exhibited by equipment provided for this Project and that said design loadings are equal to or greater than the loads produced by the equipment provided.
    c. Motor test reports.
    d. Field noise testing reports if such testing is specified.
    e. Preliminary field quality control testing format to be used as a basis for final field quality control reporting.
    f. Provide three bound final written reports documenting natural frequency testing, vibration monitoring and testing for specified equipment.
        1) Include the acceptance criteria of all equipment tested.
        2) Provide individual tabbed sections for information associated with each piece of tested equipment.

g. Certification prior to Project closeout that electrical panel drawings for manufacturer-supplied control panels truly represent panel wiring including any field-made modifications.
h. Testing and monitoring reports in accordance with PART 3 of this Specification Section.
i. Certification that driven equipment and VFD are compatible.

4. Submit completed Manufacturer's Field Service Report (MFSR) for each piece of equipment supplied.

PART 2 - PRODUCTS

A. Product requirement are to conform with the following salient characteristics.

2.2 MANUFACTURED UNITS

A. Electric Motors:

1. Where used in conjunction with adjustable speed AC or DC drives, provide motors that are fully compatible with the speed controllers.
2. Design for frequent starting duty equivalent to duty service required by driven equipment.
3. Design for full voltage starting.
4. Design bearing life based upon actual operating load conditions imposed by driven equipment.
5. Size for altitude of Project.
6. Furnish with stainless steel nameplates which include all data required by NEC Article 430.
7. Use of manufacturer's standard motor will be permitted on integrally constructed motor driven equipment specified by model number in which a redesign of the complete unit would be required in order to provide a motor with features specified.
8. AC electric motors less than 1/3 HP:
   a. Single phase, 60 Hz, designed for the supply voltage shown on the Drawings.
   b. Permanently lubricated sealed bearings conforming to ABMA standards.
   c. Built-in manual reset thermal protector or integrally mounted manual motor starter with thermal overload element with stainless steel enclosure.
9. AC electric motors 1/3 to 1 HP:
   a. Single or 3 PH, 60 Hz, designed for the supply voltage shown on the Drawings.
   b. Permanently lubricated sealed bearings conforming to ABMA standards.
      1) For single phase motors, provide built-in manual reset thermal protector or integrally mounted manual motor starter with thermal overload element.
10. AC electric motors 1-1/2 to 10 HP:
   a. Single or 3 PH, 60 Hz, designed for the supply voltage shown on the Drawings.
   b. Permanently lubricated sealed bearings conforming to ABMA standards.
   c. For vertical motors provide 15 year, average-life thrust bearings conforming to ABMA standards.
11. AC electric motors greater than 10 HP:
   a. Single or 3 PH, 60 Hz, designed for the supply voltage shown on the Drawings.
   b. Oil or grease lubricated antifriction bearings conforming to ABMA standards.
      1) Design bearing life for 90 PCT survival rating at 50,000 HRS of operation for motors up to and including 100 HP.

2) For motors greater than 100 HP, design bearing life for 90 PCT survival rating at 100,000 HRS of operation.

c. For vertical motors provide 15 year, average-life thrust bearings conforming to ABMA standards.

d. Thermal protection:

1) For all motors controlled from a variable frequency drive and for all other motors 100 HP and above, provide integral thermal detectors with normally closed contacts that will open on overtemperature.

a) Two thermal sensing devices per phase in each phase hot-spot location.

2) For Raw Water Booster Pump motors at the Roaring Springs Pumphouse provide resistance type temperature detectors (RTD).

a) 100-ohm platinum type

b) Eight (8) RTDs total per motor:

(1) Two (2) for each motor winding

(2) One (1) for drive-end bearing and one (1) for non-drive-end bearing.

e. Vibration Detector / Accelerometer:

1) For Raw Water Booster Pump motors at the Roaring Springs Pumphouse only, provide three (3) accelerometers, mounted on a bracket 90DEG apart to monitor vibration along the X Y and Z motor axes.

a) Metrix model 162VTS-200-256-00.

b) 1.60 in/sec, peak.

c) 3/8-24UNF x 3/8" mounting.

d) Non-hazardous locations.

e) 4-20mA output signals, 2-pin terminal block, loop powered.

f) Motor manufacturer shall fabricate/install the mounting bracket.

12. Severe duty motor to have the following minimum features:

a. All cast iron construction.

b. Gasketed conduit box.

c. Epoxy finish for corrosion protection.

d. Hydroscopic varnish on windings for corrosion protection.

e. Drain plug and breather.

B. NEMA Design Squirrel Cage Induction Motors:

1. Provide motors designed and applied in compliance with NEMA and IEEE for the specific duty imposed by the driven equipment.

2. Motors to meet NEMA MG 1 (NEMA Premium) efficiencies.

3. Do not provide motors having a locked rotor kVA per HP exceeding the NEMA standard for the assigned NEMA code letter.

4. For use on variable frequency type adjustable speed drives, provide:

a. Induction motors that are in compliance with NEMA MG 1, Part 31.

b. Nameplate identification meeting NEMA MG 1 Part 31 requirements.

c. Insulated drive end bearing on all motors.

d. Insulated non-drive end bearings, at a minimum, on all motors with horizontal shaft 10 HP and larger.

e. An insulated bearing carrier on the non-drive end for vertical shaft motors 100 HP and larger.

f. Shaft grounding ring on all motors:

1) Factory installed, maintenance free, circumferential, bearing protection ring with conductive microfiber shaft contacting material.

2) Electro Static Technology AEGIS SGR Bearing Protection Ring or approved equal.

g. Have the following minimum turndown ratio without the use of additional cooling, such as a blower, to provide continuous supply of cooling air over the motor.
   1) Variable torque: 10:1.
   2) Constant torque: 6:1.

5. Design motor insulation in accordance with NEMA standards for Class F insulation with Class B temperature rise above a 40 DEGC ambient.
6. Design motors for continuous duty.
7. Size motors having a 1.0 service factor so that nameplate HP is a minimum of 15 PCT greater than the maximum HP requirements of the driven equipment over its entire operating range.
   a. As an alternative, furnish motors with a 1.15 service factor and size so that nameplate HP is at least equal to the maximum HP requirements of the driven equipment over its entire operating range.
8. Motor enclosure and winding insulation application:
   a. The following shall apply unless modified by specific Specification Sections:

MOTOR LOCATION	MOTOR ENCLOSURE / WINDING INSULATION
Unclassified Indoor Areas	TEFC, Standard Insulation
Wet Indoor Areas	TEFC, Encapsulated Windings
Wet Outdoor Areas	TEFC Encapsulated Windings, WP-II (for vertical motors)
Corrosive Areas	TEFC, Severe/ Chemical Duty
Class I, Division 1 Areas	Explosion Proof, Approved for Class I Division 1 Locations
Class I, Division 2 Areas	Explosion Proof, Approved for Division 1 Locations or TEFC with maximum external frame temperature compatible with the gas or dust in the area, Extra Dip and Bake for moisture.

NOTE: Provide TENV motors in the smaller horsepower ratings where TEFC is not available.

9. Provide oversize conduit box complete with clamp type grounding terminals inside the conduit box.
10. Balance motors to ISO G2.5 level.
    a. Submit prior to shipping to OEM or job site.

2.3 COMPONENTS

A. Gear Drives and Drive Components:

1. Size drive equipment capable of supporting full load including losses in speed reducers and power transmission.
2. Provide nominal input horsepower rating of each gear or speed reducer at least equal to nameplate horsepower of drive motor.
3. Design drive units for 24 HR continuous service, constructed so oil leakage around shafts is precluded.
4. Utilize gears, gear lubrication systems, gear drives, speed reducers, speed increasers and flexible couplings meeting applicable standards of AGMA.
5. Gear reducers:
   a. Provide gear reducer totally enclosed and oil lubricated.
   b. Utilize antifriction bearings throughout.
   c. Provide worm gear reducers having a service factor of at least 1.20.

d. Furnish other helical, spiral bevel, and combination bevel-helical gear reducers with a service factor of at least 1.50.

2.4 ACCESSORIES

A. Guards:

1. Provide each piece of equipment having exposed moving parts with full length, easily removable guards, meeting OSHA requirements.
2. Interior applications:
   a. Construct from expanded galvanized steel rolled to conform to shaft or coupling surface.
   b. Utilize non-flattened type 16 GA galvanized steel with nominal 1/2 IN spacing.
   c. Connect to equipment frame with hot-dip galvanized bolts and wing nuts.
3. Exterior applications:
   a. Construct from 16 GA stainless steel or aluminum.
   b. Construct to preclude entrance of rain, snow, or moisture.
   c. Roll to conform to shaft or coupling surface.
   d. Connect to equipment frame with stainless steel bolts and wing nuts.

B. Anchorage:

1. Cast-in-place anchorage:
   a. Provide ASTM F593, Type 316 stainless steel anchorage for all equipment.
   b. Configuration and number of anchor bolts shall be per manufacturer's recommendations.
   c. Provide two nuts for each bolt.
2. Drilled anchorage:
   a. Adhesive anchors per Section 03 15 19.
   b. Epoxy grout per Section 03 31 30.
   c. Threaded rods same as cast-in-place.

C. Data Plate:

1. Attach a stainless steel data plate to each piece of rotary or reciprocating equipment.
2. Permanently stamp information on data plate including manufacturer's name, equipment operating parameters, serial number and speed.

D. Gages:

1. Provide gages in accordance with Section 40 73 00.
2. Provide at the following locations:
   a. Inlet and outlet of all reciprocating, centrifugal and positive displacement mechanical and process equipment.
   b. At locations identified on Drawings.
3. Utilize tapping sleeves for mounting per Section 40 05 00.

E. Lifting Eye Bolts or Lugs:

1. Provide on all equipment 50 LBS or greater.
2. Provide on other equipment or products as specified in the narrow-scope Specification Sections.

F. Platforms and Ladders:

1. Design and fabricate in accordance with OSHA Standards.
2. Fabricate components from galvanized steel unless otherwise indicated on Drawings.
3. Provide platform surface: Non-skid grating, unless specified in narrow-scope Specification Sections.

2.5 FABRICATION

A. Design, fabricate, and assemble equipment in accordance with modern engineering and shop practices.

B. Manufacture individual parts to standard sizes and gages so that repair parts, furnished at any time, can be installed in field.

C. Furnish like parts of duplicate units to be interchangeable.

D. Ensure that equipment has not been in service at any time prior to delivery, except as required by tests.

E. Furnish equipment which requires periodic internal inspection or adjustment with access panels which will not require disassembly of guards, dismantling of piping or equipment or similar major efforts.

1. Quick opening but sound, securable access ports or windows shall be provided for inspection of chains, belts, or similar items.

F. Provide common, lipped base plate mounting for equipment and equipment motor where said mounting is a manufacturer's standard option.

1. Provide drain connection for 3/4 IN PVC tubing.

G. Machine the mounting feet of rotating equipment.

H. Fabricate equipment which will be subject to Corrosive Environment in such a way as to avoid back to back placement of surfaces that cannot be properly prepared and painted.

1. When such back to back fabrication cannot be avoided, provide continuous welds to seal such surfaces from contact with corrosive environment.
2. Where continuous welds are not practical, after painting seal the back to back surfaces from the environment in accordance with Section 07 92 00.

I. Natural Frequency/Critical Speed:

1. All rotating parts accurately machined and in as near perfect rotational balance as practicable.
2. Excessive vibration is sufficient cause for equipment rejection.
3. Ratio of all rotative speeds to natural frequency/critical speed of a unit or components: Greater than 1.2.

J. Control Panels Engineered and Provided with the Equipment by the Manufacturer:

1. Manufacturer’s standard design for components and control logic unless specific requirements are specified in the specific equipment Specification Section.

2. NEMA or IEC rated components are acceptable, whichever is used in the manufacturer's standard engineered design, unless specific requirements are required in the specific equipment Specification Section.
3. Affix entire assembly with a UL 508A or UL 698A label "Listed Enclosed Industrial Control Panel" prior to delivery.
    a. Control panels without an affixed UL 508A or UL 698A label shall be rejected.
4. Provide equipment or control panels with Short Circuit Current Rating (SCCR) labeling as required by NFPA 70 and other applicable codes.
    a. Determine the SCCR rating by one of the following methods:
        1) Method 1: SCCR rating meets or exceeds the available fault current of the source equipment when indicated on the Drawings.
        2) Method 2: SCCR rating meets or exceeds the source equipment's Amp Interrupting Current (AIC) rating as indicated on the Drawings.
        3) Method 3: SCCR rating meets or exceeds the calculated available short circuit current at the control panel.
    b. The source equipment is the switchboard, panelboard, motor control center or similar equipment where the control panel circuit originates.
    c. For Method 3, provide calculations justifying the SCCR rating. Utilize source equipment available fault current or AIC rating as indicated on the Drawings.

K. Equipment Base:

1. Adequate grout and vent openings to allow grout to flow under entire base.

2.6 SHOP OR FACTORY PAINT FINISHES

A. Electrical Equipment:

1. Provide factory-applied paint coating system(s) for all electrical equipment components except those specified in Section 09 96 00 to receive field painting.

B. Field paint other equipment in accordance with Section 09 96 00.

1. See Section 09 96 00 for factory applied primer/field paint compatibility requirements.

2.7 SOURCE QUALITY CONTROL

A. Motor Tests:

1. Test motors in accordance with NEMA and IEEE standards.
2. Provide routine test for all motors.
3. The Contracting Officer (CO) reserves the right to select and have tested, either routine or complete, any motor included in the project.
    a. The CO will pay all costs, including shipping and handling, for all motors successfully passing the tests.
    b. Contractor shall pay all costs, including shipping and handling, for all motors failing the tests.
    c. If two successive motors of the same manufacturer fail testing, the CO has the right to reject all motors from that manufacturer.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install equipment as shown on Drawings and in accordance with manufacturer's directions.

B. Utilize templates for anchorage placement for slab-mounted equipment.

C. For equipment having drainage requirements such as seal water, provide 3/4 IN PVC or clear plastic tubing from equipment base to nearest floor or equipment drain.

1. Route clear of major traffic areas and as approved by Contracting Officer.

D. Do not construct foundations until major equipment supports are approved.

E. Extend all non-accessible grease fittings using stainless steel tubing to a location which allows easy access of fittings from closest operating floor level.

F. Equipment Base:

1. Construct level in both directions.
2. Take particular care at anchor bolt locations so these areas are flat and level.

G. Machine Bases / Sole Plates:

1. Grease or tape anchorages and jack screws to inhibit grout from adhering to bolts and other anchors.
   a. Jack screws number and size by equipment manufacturer.
      1) Jack Screw
         a) 304 Stainless Steel minimum.
         b) 0.5 inches diameter minimum.
      2) Jack Screw Pad
         a) 2 inch diameter minimum
         b) Anchored in place with a structural epoxy adhesive.
2. Install machine base of rotating equipment on equipment base.
3. Level in both directions using jack screws, with a machinist level, according to machined surfaces on base.
   a. Base shall be level within vertical tolerance of the lesser of (a) 0.005 inch per foot with no more than 0.0005 inches difference between any two points, or (b) equipment manufacturer’s written instructions.
4. Level machine base on equipment base and align couplings between driver and driven equipment.

H. Rotating Equipment Couplings:

1. Align in the annular and parallel positions.
   a. For equipment rotating at 1200 RPM or less, align both annular and parallel within 0.001 IN tolerance for couplings 4 IN size and smaller.
   b. Couplings larger than 4 IN size: Increase tolerance 0.0005 IN per inches of coupling diameter, i.e., allow 6 IN coupling 0.002 IN tolerance, and allow a 10 IN coupling 0.004 IN tolerance.
   c. For equipment rotating at speeds greater than 1200 RPM allow both annular and parallel positions within a tolerance rate of 0.00025 IN per inch coupling diameter.

2. If equipment is delivered as a mounted unit from factory, verify factory alignment on site after installation and realigned if necessary.
3. Check surfaces for runout before attempting to trim or align units.

I. Grouting:

1. Level onto equipment base with jack screws in accordance with the Contract Documents, provide a dam or formwork around base to contain grout between equipment base and equipment support pad.
2. Preparation:
   a. Extend dam or formwork to cover leveling shims and blocks.
   b. Anchor sleeves:
      1) Required for equipment (Pumps, Mixers, Blowers) greater than 50 HP
      2) If anchor sleeves were used, fill voids in anchor sleeves with foam or room temperature vulcanizing (RTV) silicone to keep grout from filling sleeves.
   c. Do not use nuts below the machine base to level the unit.
   d. Saturate top of roughened concrete surface with water before grouting.
3. Grout Installation:
   a. Install grout until entire space under machine base is completely filled to underside of base.
      1) Voids are unacceptable.
   b. Puddle grout by working a stiff wire through the grout and vent holes, to ensure grout is installed properly and to release air entrained in grout or base cavity.
4. After Grout Installation:
   a. When grout is sufficiently hardened, remove dam or formwork and finish exposed grout surface to fine, smooth surface.
   b. Completely cover exposed grout surfaces with wet burlap and keep covering sufficiently wet to prevent too-rapid evaporation of water from grout.
   c. Check for voids by tapping along the top deck of the mounting plate. A solid thud indicates grout-filled areas while a drum-like hollow sound indicates a void requiring filling.
      1) Void areas are to be filled by drilling 1/8 INCH NPT holes in opposite corners of each void area. Grout to be pumped into one void with a grout gun until grout emerges from the other vent hole.
   d. When grout is fully hardened (after not less than seven days), remove jack screws, and tighten nuts on anchor bolts and similar anchors to required torque.
   e. Inspect and verify levelness of machine base and, if not in accordance with requirements, remedy by removing base and reinstalling in accordance with the Contract Documents.
   f. Inspect driver-driven equipment for proper alignment. When not in accordance with requirements, remedy so that the Work is not defective.

3.2 INSTALLATION CHECKS

A. For all equipment specifically required in detailed specifications, secure services of experienced, competent, and authorized representative(s) of equipment manufacturer to visit site of work and inspect, check, adjust and approve equipment installation.

1. In each case, representative(s) shall be present during placement and start-up of equipment and as often as necessary to resolve any operational issues which may arise.

B. Secure from equipment manufacturer's representative(s) a written report and deliver to CO immediately prior to leaving jobsite . Report shall certify that equipment:

1. Has been properly installed and lubricated.
2. Is in accurate alignment.
3. Is free from any undue stress imposed by connecting piping or anchor bolts.
4. Has been operated under full load conditions and that it operated satisfactorily.

C. No separate payment shall be made for installation checks.

1. All or any time expended during installation check does not qualify as Operation and Maintenance training or instruction time when specified.

3.3 IDENTIFICATION OF EQUIPMENT AND HAZARD WARNING SIGNS

A. Identify equipment and install hazard warning signs in accordance with Section 10 14 00.

3.4 FIELD PAINTING AND PROTECTIVE COATINGS

A. For required field painting and protective coatings, comply with Section 09 96 00.

3.5 WIRING CONNECTIONS AND TERMINATION

A. Clean wires before installing lugs and connectors.

B. Coat connection with oxidation eliminating compound for aluminum wire.

C. Terminate motor circuit conductors with copper lugs bolted to motor leads.

D. Tape stripped ends of conductors and associated connectors with electrical tape.

1. Wrapping thickness shall be 150 PCT of the conductor insulation thickness.

E. Connections to carry full ampacity of conductors without temperature rise.

F. Terminate spare conductors with electrical tape.

3.6 FIELD QUALITY CONTROL

A. General:

1. Furnish equipment manufacturer’s field quality control services and testing as specified in the individual equipment Specification Sections.
2. Execute pre-demonstration requirements in accordance with Section 01 75 00 and 01 75 10.
3. Perform and report on all tests required by the equipment manufacturer’s Operation and Maintenance Manual.
4. Provide testing of electrical equipment and connections in accordance with the Electrical specifications.

5. Equip testing and analysis personnel with all appropriate project related reference material required to perform tests, analyze results, and provide documentation including, but not limited to:
    a. Contract Drawings and Specifications.
    b. Related construction change documentation.
    c. Approved Shop Drawings.
    d. Approved Operation and Maintenance Manuals.
    e. Other pertinent information as required.

B. Equipment Monitoring and Testing Plans:

1. Approved in accordance with Shop Drawing submittal schedule.
2. Included as a minimum:
    a. Qualifications of firm, field personnel, and analysis personnel doing the Work.
    b. List and description of testing and analysis equipment to be utilized.
    c. List of all equipment to be testing, including:
        1) Name and tag numbers identified in the Contract Documents.
        2) Manufacturer's serial numbers.
        3) Other pertinent manufacturer identification,

C. Instruments Used in Equipment and Connections Quality Control Testing:

1. Minimum calibration frequency:
    a. Field analog instruments: Not more than 6 months.
    b. Field digital instruments: Not more than 12 months.
    c. Laboratory instruments: Not more than 12 months.
    d. If instrument manufacturer's calibration requirements are more stringent, those requirements shall govern.
2. Carry current calibration status and labels on all testing instruments.
3. See individual testing programs for additional instrumentation compliance requirements.

D. Testing and Monitoring Program Documentation:

1. Provide reports with tabbed sections for each piece of equipment tested.
2. Include all testing results associated with each piece of equipment under that equipment's tabbed section.
    a. Include legible copies of all forms used to record field test information.
3. Prior to start of testing, submit one copy of preliminary report format for Contracting Officer review and comment.
    a. Include data gathering and sample test report forms that will be utilized.
4. In the final report, include as a minimum, the following information for all equipment tested:
    a. Equipment identification, including:
        1) Name and tag numbers identified in the Contract Documents.
        2) Manufacturer's serial numbers.
        3) Other pertinent manufacturer identification,
    b. Date and time of each test.
    c. Ambient conditions including temperature, humidity, and precipitation.
    d. Visual inspection report.
    e. Description of test and referenced standards, if any, followed while conducting tests.
    f. Results of initial and all retesting.
    g. Acceptance criteria.

h. "As found" and "as left" conditions.
i. Corrective action, if required, taken to meet acceptance.
j. Verification of corrective action signed by the Contractor, equipment supplier, and CO or CO's authorized representative.
k. Instrument calibration dates of all instruments used in testing.

5. Provide three (3) bound final reports prior to Project final completion.

E. Electrical Equipment and Connections Testing Program:

1. Perform testing on Electrical equipment and connections in accordance with the Electrical specification requirements.
2. Testing of motors:
   a. After installation and prior to energizing the motor, perform inspections and tests per NETA ATS 7.15 for all motors 50 HP or above.
   b. Ensure motor has been lubricated.
   c. Bump motor to check for correct rotation.
3. Repair or replace equipment shown to be out of range of the acceptable tolerance until the equipment meets or exceeds acceptability standards.

F. Other Testing:

1. Perform tests and inspections not specifically listed but required to assure equipment is safe to energize and operate.
2. Subbase that supports the equipment base and that is made in the form of a cast iron or steel structure that has supporting beams, legs, and cross members that are cast, welded, or bolted shall be tested for a natural frequency of vibration after equipment is mounted.
   a. The ratio of the natural frequency of the structure to the frequency of the disturbing force shall not be between 0.5 and 1.5.

G. Infrared Thermography Testing Program:

1. Perform infrared thermography testing for equipment specified in other Divisions during the Equipment Demonstration Period.
   a. Perform on all rotating and reciprocating equipment having drivers 25 HP or greater.
   b. Perform on electrical equipment and connections: See Section 26 08 13.
2. Additional requirements for infrared thermography monitoring and testing equipment:
   a. Temperature range: -10 to 350 DEGC.
   b. Accuracy: ±2 PCT or 2 DEGC, whichever is greater.
   c. Repeatability: ±1 PCT or 1 DEGC, whichever is greater.
   d. Temperature indication resolution: 0.1 DEGC.
   e. Minimum focus distance: 0.3 meters.
   f. Output in color palettes: JPEG, BMP, or other digital format compatible with Windows.
3. Perform inspection per ASTM E1934.
   a. Operate VFD driven equipment at 100 PCT speed during thermographic inspection.
4. Acceptability of electrical connections and components based on temperature comparison between components and ambient air temperatures not greater than 10 DEGC per ASTM E1934.
5. Acceptability of motors and equipment bearings based on temperature rise not greater than 5 DEGC above the equipment and/or bearing manufacturers published criteria.
6. Repair or replace equipment shown to be out of range of the acceptable tolerance until the equipment meets or exceeds acceptability standards.

H. Equipment Vibration Monitoring and Testing Program:

1. None required.

3.7 DEMONSTRATION

A. Demonstrate equipment in accordance with Section 01 75 00 and 01 75 10.

3.8 ABBREVIATION TABLE

A. As indicated in the Drawings.

END OF SECTION 01 61 03

EXHIBIT A

MANUFACTURER FIELD SERVICE REPORT

This field service report is generic in nature. An electronic copy of this form will be furnished upon request from the CO. This report is to reflect that all requirements of the Operations and Maintenance Manual and the individual equipment specification requirements have been performed for the installation and operation and also to provide a baseline for amperage draw for each phase, vibration readings, rotation, alignment and all other applicable tests required to insure that the equipment has been installed properly. A MFSR will be required for each individual piece of equipment requiring a MFSR.

Definitions of Reports:

Initial service report: Required for construction preparations. Equipment delivered to site is in good condition and conforms to specification requirements. Anchor bolts, hardware and ancillary items (piping, flanges, conduits, fuel/power supply) are compatible with equipment.

Interim service report: Required for equipment installation onto base or foundation. Piping connections, electrical and control connections or structural attachment are complete. For equipment stored on site over four weeks, interim service report will document that manufacturer's long-term storage procedures have been incorporated and equipment has not been damaged, nor coatings deteriorated.

Final service report is to be completed when equipment can be started, electrical amperage and voltage draw measured, cold and hot alignments performed, vibration testing and monitoring performed and the equipment is found to be in compliance with Manufacturer's operating parameters and the requirements of the individual equipment specifications.

PROJECT: _____

Report Status:

Initial Service Report completed and submitted on _____

Interim Service Report completed and submitted on _____

Final Service Report completed and submitted on _____

Commencement of Warranty _____

I Description

A. Equipment Name and Identification: _____

_____

B. Serial Number: _____

C. Specification Section Number: _____

D. Manufacturer: _____

E. Representative: _____

F. Type of Service: Initial _____ Interim _____ Final _____

II General Review

A. The above referenced equipment/material/supplies have been inspected, checked, and adjusted. Yes _____ No _____

Summary: _____

_____

B. The above referenced equipment/material/supplies were placed upon properly prepared or suitable substrate. N/A _____ Yes _____ No _____

Summary: _____

_____

C. The above referenced equipment/material/supplies are free from any undue stress imposed by any connected piping, anchor bolts or any other load. N/A _____ Yes _____ No _____

Summary: _____

_____

D. The above referenced equipment/material/supplies have operated under design conditions. N/A _____ Yes _____ No _____

Summary: _____

_____

E. The above referenced equipment/material/supplies have been installed in accordance with the manufacturer's recommendations and the Procurement Documents, require no corrective work, and are hereby approved. Yes _____ No _____

Summary: _____

_____

F. The above referenced equipment/material/supplies are acceptable to the manufacturer as installed providing the following corrective action(s) are performed:

1. _____
2. _____
3. _____
4. _____
5. _____

III Inspection Checklist

Item	Acceptable (Yes/No)	Readings/Comments
Bearings (1)		
Belts (tension reading)		
Lubrication Levels		
Vibration (1) (2) (MILS/SEC)		
Infrared Thermography (1) (2)		
Starting AMPS		
Full Load AMPS		
Volts		
Rotation		
Jacket Temperature (DEGF)		
Seal Water Flow Rate (GPH or GPM)		
Seal Water Pressure (PSI)		
O-rings/Packing		
Alignment (1)		
Anchor Bolts		
Grout		

Item	Acceptable (Yes/No)	Readings/Comments
Substrate Approval		
Sound level (4 FT from unit) (1) (dB)		
Other		

(1) Inspection or testing reports must be attached.
(2) Provide vibration testing and monitoring procedures for Contracting Officer's review and approval prior to testing.

IV O&M Manuals

A. The O&M manual as presented contains all information required for proper operation, maintenance, and instruction of this system. N/A _____ Yes _____ No _____

Summary: _____

_____

V Preventive Maintenance

A. The preventive maintenance summary outlined in the O&M manual is acceptable for operation of the system throughout the warranty period. N/A _____ Yes _____ No _____

Summary: _____

_____

VI Operator Training/Classroom Instruction

A. Training and instruction have been performed in accordance with the requirements of the Procurement Documents. N/A _____ Yes _____ No _____

B. Final Training/Classroom Instruction Completed on: _____

Summary: _____

_____

VII Remarks

_____

_____

_____

VIII Certification

I hereby certify, that I, _____, am a duly authorized representative of the manufacturer, that I am empowered by the manufacturer to inspect, approve, and operate his equipment, and that I am authorized to make recommendations required to assure that the equipment furnished by the manufacturer is complete and operational, except as modified herein. I also certify that all information contained herein is true and accurate.

By: _____
(Authorized Representative)

For: _____

Date: _____

IX Acknowledgments

By: _____

For: _____
(Contractor)

Date: _____

By: _____

For: _____
(Contracting Officer)

Date: _____

SECTION 01 65 50 – MATERIAL TRANSPORT - HELICOPTERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Transport of materials by helicopter to the Innercanyon.
   a. Scheduling and aviation plan development
   b. Load preparation and rigging responsibilities
   c. Flight execution
   d. Flight regulations and restrictions
   e. Personnel transports.
2. See Section 01 67 00 – Product Requirements for transport, storage and handling of materials that are not going to be transported by helicopter.
   a. Section 01 67 00 also covers transport, storage and handling of materials that are scheduled to be transported by helicopter until they are delivered to the helibase.

B. Related Specification Sections include but are not necessarily limited to:

1. Division 00 – Procurement and Contracting Requirements.
2. Division 01 – General Requirements.

C. Payment:

1. The Contractor shall procure and pay for use of helicopters for material and equipment transport within the Canyon. Contractor shall generate an Aviation Plan that shall be approved by the Contracting Officer prior to the start of helicopter operations.

1.2 QUALITY ASSURANCE

A. Certifications:

1. Pilot certifications
   a. All helicopter pilots must have an active FAA Commercial Pilot License.
      1) Rotorcraft aircraft category rating
      2) Appropriate aircraft category, class, type, and instrument rating on the person's pilot certificate for any helicopter being flown
   b. Current medical certificate on file
   c. All pilots must complete the course "GCNP-SFRA – Grand Canyon NP Special Flight Rules Area (Certified Pilots) and get a SFRA waiver through the FAA prior to conducting flights into the Grand Canyon. See www.iat.gov/training for more details.
      1) The Contractor is responsible for obtaining all certifications and courses required and must confirm with FAA that they meet all requirements that may in addition to what is stated here.

2) Other Contractor personnel as needed and including the Contractor's helibase representative and radio operators should take the general public version of this course.

2. Rigger certifications
   a. Helicopter rigging and loads must be inspected by a rigger with training (A-219 or equivalent) in Helicopter Cargo Rigging prior to each flight.
   b. Contractor must utilize Longline Course A-219 training available from the Helibase.
3. Air Base Radio Operator (ABRO) certifications

B. Insurance and Responsibilities:

1. The Contractor shall procure insurance for helicopter operations for this project.
2. The Contractor's helicopter services shall exclude the transport of non-Contractor personnel and equipment, including the CO, COR, CMR and other NPS staff.

C. Coordination with Helibase:

1. Provide a dedicated representative at the South Rim Contractor Support Facilities to coordinate helicopter-based transportation with the NPS South Rim Helibase Manager when helicopter operations are occurring. This person will be the primary point of contact between the helibase and Contractor personnel and shall have authority to make decisions on behalf of the Contractor.
   a. The Representative shall have cell phone service and e-mail access.
   b. The Representative shall conduct daily briefings with the helibase that cover the planned operations for the day and any coordination needs.
2. Provide an Air Base Radio Operator (ABRO) to be co-located with the Government's ABRO at the South Rim helibase during Contractor aviation operations. The two ABRO's will work together to maintain safe aviation operations.
   a. ABRO training will be provide by NPS helibase personnel.
   b. It is the responsibility of the Contractor to obtain ABRO training from helibase personnel in accordance with all US government requirements.
3. South Rim Helibase Manager and coordination. A single point of contact will be established between the NPS South Rim Helibase Manager, or their designee depending on daily NPS work activities, and the Contractor's flight operations coordinator. The South Rim Helibase Manager will be identified by the Government at the time of construction.

D. Reference Standards

1. NWCG Standards for Helicopter Operations (NSHO), National Wildfire Coordinating Group
2. Flight Operations Standards and Procedures 351DM1, Department of Interior
3. Reference Manual 60 Aviation Management, Internal Aviation Management Plan (IAMP), National Park Service

1.3 SUBMITTALS

A. Aviation Plan: Submit a Contractor developed Aviation Plan to the Contracting Officer for acceptance within 30 calendar days following Notice to Proceed.

1. See Paragraph 3.1 B of this specification section for plan requirements.

2. Sequencing, scheduling, and load capacity of helicopter flights will be weather dependent and subject to other Government instituted delays (See Paragraph 3.3 C. below). Project schedule and Aviation Plan shall provide allowance for weather or schedule delay.

B. Certifications: submit certifications for pilots, riggers and aircraft demonstrating compliance with this Specification.

1. Submit documentation of A-219 completion following training performed by NPS helibase personnel.

PART 2 - PRODUCTS

2.1 HELICOPTER TYPES

A. It is the Contractor's responsibility to define payload capacities of the helicopter classes based on actual aircraft used for the Work.

B. Contractor furnished helicopters shall have the ability to communicate on VHF-AM with two radios monitoring two frequencies and VHF-FM with a minimum of one radio monitoring two frequencies ensuring communication with helibase, dispatch and ground personnel.

C. Contractor furnished helicopters shall use the satellite-based aircraft tracking system Automated Flight Following (AFF) through the website www.aff.gov. System compatibility and a subscription to AFF is required. Prior to helicopter operations in the Park, ensure AFF is operational.

D. Quiet Technology light duty helicopters are required for project operations.

1. The NPS Aviation Reference Manual 60 defines Quiet Technology as aircraft that are quieter on a per flight basis to technological improvements.

2.2 RIGGING

A. All helicopter rigging must be inspected and approved by an approved rigger with training in Helicopter Cargo Rigging prior to each flight.

B. Materials:

1. All rigging materials must be designed for the purpose they are being used for and have a sufficient load rating for the application.
2. All rigging materials must be maintained and in good condition for the intended use.

C. Weather and other flight conditions must be taken into consideration when determining helicopter weight capacities for flights.

PART 3 - EXECUTION

3.1 SCHEDULING

A. Sequencing of the project shall focus heavily on transportation methods, scheduling, and sequencing of all equipment and materials to be delivered and installed, and any existing equipment, materials or debris to be removed.

3.2 Project helicopters will generally fly between 8:00 (power on) am and 5:00 pm (shut down) each day with generally no flights between 5 pm and 8:00 am. In no case will flight occur before 1/2 hour prior to sunrise and 1/2 hour after sunset: Instrument Flight Rules (IFR) are prohibited. Deviations from specified flight times must be approved by the South Rim Helibase Manager and Contracting Officer.

A. Contracted commercial aviation operations in support of construction activities will operate on a separate communications system from the owner communication system that has ability to communicate with the owners system. Communications equipment from prior commercial contracted aviation operations is installed on the south rim and should be inspected for functionality and serviceability for reuse to support aviation communications for the scope of this project. Additional infrastructure may be needed to communications across the whole project area. Final type, location and access to install the equipment will require approval by owner Helibase and any compliance related requirements through the Contracting Officer.

B. Contractor use of south rim helibase (inside secure fenced helibase) will only allow for aviation operations to include load drop off and pick up of materials/equipment. Building loads and staging loads more than 24 hours within the helibase fence line is prohibited. Hanger utilities provided, contractor must provide hanger for helicopter operations, reference Volume 00: General drawing sheet 000 G107 for location of provided infrastructure within the helibase fenced area.

C. Aviation Plan Development:

1. Contractor shall develop an Aviation Plan that clearly itemizes each planned flight and identifies sling quantity, lift capacity (aircraft) required, any HAZMAT items, and a planned flight schedule. Items generally to include:
   a. Removal of demolished materials.
   b. Deliveries of all building materials, piping, pumps, treatment equipment, tanks, electrical gear, construction equipment, and crew supplies.
   c. Removal of unused materials, construction equipment and trash.
   d. How pilot and aircraft certification will be ensured.
   e. How load calculations (weight and balance) will be ensured and documented.
   f. How manifests will be completed and documented.
   g. How records of numbers of flights and total flight time will be documented.
   h. What actions will be taken in the event of a mechanical failure, dropped load, unscheduled maintenance, aircraft mishap, etc.
   i. Flight notification procedures.
   j. How flight following will be conducted.
   k. Coordination with the NPS Helibase personnel to include daily briefings, frequency monitoring, flight paths, flight delays due to NPS operations, conflict resolution, etc.
   l. PPE for ground and flight operations.

m. Flight routes and traffic (other aircraft, avoidance of helicopter tour routes, vehicles on roads, pedestrians, etc.) avoidance.
n. Record keeping.
o. How flights will be minimized.
p. Coordination for sensitive area avoidance.
q. Emergency protocols (e.g., mishap response plans)
r. Training for riggers.
s. Crash Rescue response/procedures.
t. Medical plan (in the event of contractor injury).
u. Options for ' Safety stand down', i.e., allowance for stopping for flights to re-evaluate or reset the operations until such time where mitigations for flight operations to proceed as approached by the Park Aviation Manager or Helibase Manager.

2. The Aviation Plan shall reference and conform to the requirements, including maximum flights per day, flight paths, wildlife avoidance, etc.
3. All sequencing, scheduling, and load capacity of helicopter flights will be weather dependent. Project schedule and Aviation Plan shall provide allowance for weather or schedule delay.
   a. The Contractor is responsible for determining if weather conditions are suitable for operations, and have responsibility for flight safety due to weather conditions
4. Sequencing shall include the most efficient use of helicopter transport as possible; the Contractor shall aim to always have a backhaul load combined with scheduled deliveries, so that helicopters rarely fly with an empty hook.
5. All helicopter-based transportation shall be coordinated with the NPS South Rim Helibase Manager.

3.3 LOAD PREPARATION AND RIGGING

A. Longline Course A-219 Training

1. The Contractor shall provide trained workers to perform rigging of all loads for Contractor procured aircraft. This includes rigging of loads at the South Rim Contractor Support Facilities, North Rim Ballfield, and at Innercanyon staging and construction locations.
   a. Contractor is responsible for all trained workers have up to date A-219 training and ensure workers are following the latest rigging methods.
2. NPS/Helibase will provide rigging training at the Helibase for Contractor personnel to attend. The Contractor is responsible to schedule training with helibase personnel in advance of training needs and within helibase staff schedules
   a. If utilized, Contractor shall coordinate and pay all non-trainer costs associated with the training.

B. Load Planning

1. Contractor shall provide certified scales as required by the project’s flight needs for load planning/rigging use at the South Rim Contractor Support Facilities.
2. Contractor shall use an accurate scale for off-site use in load planning and manifest development as required.
3. The Contractor shall plan and arrange all materials into loads that do not exceed the aircraft load capabilities. The Contractor shall utilize Helibase Management load planning forms as defined with National Wildfire Coordinating Group (NWCG) Standards for Helicopter Operations.

4. Weather and other flight conditions must be taken into consideration when determining helicopter weight capacities for flights.
5. Use of existing landing zones must be coordinated with the NPS Helibase Manager.

C. Staging

1. All materials and equipment scheduled for delivery to the Innercanyon shall be delivered to the following locations:
   a. Designated helicopter transport staging area located at the South Rim Contractor Facilities site outside the helibase deck.
   b. Material for sling load operations within the helibase deck (secure fenced perimeter) is only allowed for same day flight operations. Staging of material and equipment must be outside the helibase (secure fence line).
2. Equipment and material weights shall be based on modular componentry to maintain the lightest load available and reduce the need for heavy lift aircraft.

D. Rigging

1. Contractor shall provide all rigging materials and perform all loading and rigging.
2. Inspect rigging materials for rips, tears, or cracks prior to use. Replace defective items.
3. A load calculation (or weight and balance) must be completed and current for the day for each helicopter to be used, that identifies the 'allowable payload' for that helicopter.
4. A manifest shall be prepared for each load that details the items in the load and the weight of each item, and the total weight of the load.
5. Every load must be within the allowable identified in the load calculation (or weight and balance) for the helicopter transporting it.
6. Rigging must ensure that loads (or partial loads) will not be inadvertently lost.
7. Personnel must be available to receive each load and be able to manually unhook the load if necessary.

3.4 FLIGHT EXECUTION

A. Installation

1. All proposed materials, controls, equipment, and parts required to complete the project installation.

B. Delay in Aircraft Availability

1. Weather, national emergencies, local emergencies/search and rescue (SAR) operations, or medical emergencies to Park visitors or employees, or Park Helibase supervisory personnel as determined by the NPS Helibase Manager on duty, could delay the use of the Contractor’s helicopters. Government may direct a temporary delay in Contractor flight operations for any of these reasons. Delays should be estimated by the Contractor and built into the project schedule.
2. SAR Incidents will require Contractor coordination of flight operations and do not require full-day delay in aircraft availability.
3. Weather delays can include partial day delays in aircraft flight as well as full day delays. The following table identifies planned full day delays in aircraft availability:

Month	Number of No Flight Days
January	10

February	4
March	4
April	4
May	1
June	5
July	2
August	1
September	1
October	5
November	6
December	14

3.5 FLIGHT REGULATIONS AND RESTRICTIONS

A. Project Essentials Only

1. Only essential personnel, materials and equipment required to complete the work of this contract shall be transported to the work site by helicopter.
2. All materials and equipment shall be transported in sling-loads suspended under the aircraft. Small items not conducive to transporting in a sling shall be carried inside the aircraft.
3. Equipment shall be limited to the smallest and lightest equipment that will safely do the job.
4. Project helicopters are not to be used for shuttling equipment and material deliveries from the South Rim helibase to the North Rim. Equipment and materials deliveries intended for North Rim construction shall be delivered by ground transportation.
5. The aviation corridor adjacent to the North Rim WWTP will be cordoned off with construction fencing to include warning signs on both sides of the access road into the WWTP area. No material or equipment can be stored within the aviation corridor. Limited construction activities may occur within the aviation corridor as needed to complete the work if no other reasonable means of construction are available; this work will be requested 48 hours in advance to CO/COR and on-site CMR for coordination with NPS aviation operations. All personnel will receive aviation safety brief upon first arriving to the site to include the restricted activities of the aviation corridor and general safety protocols of the aviation zone.

B. Returning Construction Equipment and Materials from the Project Site

1. Construction equipment shall not be removed from the project site until its use is completed. Construction equipment shall be hauled into and out of the Innercanyon only once. Deviations from these requirements must be approved by the Contracting Officer with 7-day advance notification.

3.6 PERSONNEL TRANSPORT

A. Helicopter transport of Contractor’s personnel to worksites, staging areas, and Contractor Construction Camps is allowable.

1. Include these flights in the Aviation Plan.

B. Personnel transport to utilize existing landing zones or project specific landing zones approved by the NPS Helibase Flight Operation Managers.

1. Use of existing landing zones must be coordinated with Helibase Manager prior to use.

END OF SECTION 01 65 50

SECTION 01 67 00 - PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and environmental requirements.

1.2 DEFINITIONS

A. Products: Items purchased for incorporating into Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.

1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature, current as of date of Contract Documents.
2. New Products: Items that have not previously been incorporated into another project or facility. Products salvaged or recycled from other projects are not considered new products.
3. Comparable Product: An “Or Equal” product demonstrated and approved through submittal process, or where indicated as a product substitution, to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product (i.e. salient characteristics).
4. Basis of Design Product: A Basis of Design product identifies a Brand Name and/or manufacturer of a product of which specific salient characteristics are key aspects of the design. Refer to Section 2.1 Product Selection Procedures for potential design and cost implications associated with a product substitution.

B. Definitions pertaining to sustainable development: As defined in ASTM E2114.

C. Biobased Materials: As defined in Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.

1. Biobased content: Amount of biobased carbon in material or product as a percentage of weight (mass) of total organic carbon in material or product.

D. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.

E. Environmentally preferable products: Products and services with lesser or reduced effect on the environment in comparison to conventional products and services. Refer to Environmental Protection Agency’s (EPA) Final Guidance on Environmentally Preferable Purchasing for more information.

F. Stewardship: Responsible use and management of resources in support of sustainability.

G. Sustainability: Maintenance of ecosystem components and functions for future generations.

1. Recycled Content Materials: Products containing pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent with International Organization for Standardization (ISO) 140001 Standard for the Use of Environmental Marketing Claims.
2. Rapidly Renewable Material: Material made from plants typically harvested within a ten-year cycle.
3. Regional Materials: Materials manufactured and extracted, harvested, or recovered within a radius of 500 miles from Project location.

1.3 SUBMITTALS

A. Record Submittals as specified in – Sustainable Design Close-Out Documentation, submit:

1. Affirmative Procurement Reporting Form. Submit on form in Appendix A of this Section, or similar form as approved by Contracting Officer (CO).
2. Submit environmental data in accordance with Table 1 of ASTM E2129 for these products:
    a. Masonry
    b. Finish Carpentry
    c. Plastic Fabrications
    d. Building Insulation
    e. Roofing
    f. Joint Sealers
    g. Wood & Plastic Doors
    h. Windows
    i. Skylights
    j. Glazed Curtain Wall
    k. Gypsum Board
    l. Tile
    m. Acoustical Ceilings
    n. Resilient Flooring
    o. Carpet
    p. Wall Coverings
    q. Paints & Coatings
    r. Toilet Compartments
    s. Loading Dock Equipment
    t. Office Equipment
    u. Furnishings & Accessories
    v. Renewable Energy Equipment
    w. Elevators
    x. Plumbing fixtures and equipment.
    y. HVAC equipment
    z. Lighting equipment
3. Material Safety Data Sheets (MSDS): For each product required by OSHA to have a MSDS, submit an MSDS. MSDS shall be prepared within the previous five years. Include information for MSDS Sections 1 to 16 in accordance with ANSI Z400.1 and as follows:
    a. Section 1: Chemical Product and Company Identification
    b. Section 2: Composition/Information on Ingredients
    c. Section 3: Hazards Identification
    d. Section 4: First Aid Measures
    e. Section 5: Fire Fighting Measures
    f. Section 6: Accidental Release Measures
    g. Section 7: Handling and Storage
    h. Section 8: Exposure Controls/Person Protection
    i. Section 9: Physical and Chemical Properties
    j. Section 10: Stability and Reactivity Data

k. Section 11: Toxicological Information. Include data used to determine the hazards cited in Section 3. Identify acute data, carcinogenicity, reproductive effects, and target organ effects.

l. Section 12: Ecological Information. Include data regarding environmental impacts during raw materials acquisition, manufacture, and use. Include data regarding environmental impacts in event of accidental release.

m. Section 13: Disposal Considerations. Include data regarding proper disposal of the chemical. Include information regarding recycling and reuse. Indicate whether or not product is considered to be "hazardous waste" according to United States EPA Hazardous Waste Regulations 40 CFR 261 (Code of Federal Regulations).

n. Section 14: Transportation Information. Identify hazard class for shipping.

o. Section 15: Regulatory Information. Identify federal, state, and local regulations applicable to the material.

p. Section 16: Other Information. Include additional information relative to recycled content, biobased content, and other information regarding environmental and health impacts. Identify the date MSDS was prepared.

4. Chain of Custody: Submit chain-of-custody documentation for sustainable forestry for these products:
   a. Rough Carpentry
   b. Finish Carpentry
   c. Wood Doors
   d. Windows
   e. Wood Flooring
   f. Furnishings & Accessories

1.4 QUALITY ASSURANCE

A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle products using means and methods that prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.

B. Delivery and Handling:

1. Schedule delivery to minimize long-term storage at Project site and prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Project site in undamaged condition; in manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to ensure compliance with Contract Documents. Ensure products are undamaged and properly protected.
5. Obtain materials in biodegradable or recyclable/reusable packaging which uses minimum amount of packaging possible.

C. Storage:

1. Allow for inspection and measurement of quantity or counting of units.
2. Store materials in manner to not endanger Project structure.

3. Store products subject to damage by the elements, under cover in weather tight enclosure above ground, with ventilation adequate to prevent condensation.
4. Store cementitious products and materials on elevated platforms.
5. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
7. Protect stored products from damage and liquids from freezing.
8. Store loose granular materials in well-drained area on solid surfaces to prevent mixing with foreign matter.

1.6 PACKAGING

A. Where Contractor has option to provide one of listed products or equal, preference shall be given to products with minimal packaging and easily recyclable packaging as defined in ASTM D5834.

B. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.

C. Provide minimum 45 percent post-consumer recycled content and minimum 100 percent recovered fiber content of industrial paperboard in accordance with EPA’s Comprehensive Procurement Guidelines and ASTM D5663.

D. Provide minimum 10 percent post-consumer recycled content and minimum 10 percent recovered fiber content of carrier board in accordance with EPA’s Comprehensive Procurement Guidelines and ASTM D5663.

E. Provide minimum 5 percent post-consumer recycled content and minimum 5 percent recovered fiber content of brown papers (e.g., wrapping papers and bags) in accordance with EPA’s Comprehensive Procurement Guidelines and ASTM D5663.

1.7 ENVIRONMENTALLY PREFERABLE PRODUCTS

A. Provide environmentally preferable products to greatest extent possible.

1. To greatest extent possible, provide products and materials with a lesser or reduced effect on the environment considering raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and/or disposal of the product.
2. Eliminate use of ozone depleting compounds during and after construction where alternative environmentally preferable products are available, consistent with either Montreal Protocol and Title VI or Clean Air Act Amendments of 1990, or equivalent overall air quality benefits that take into account life cycle impacts.
3. Use products meeting or exceeding EPA’s recycled content recommendations for EPA-designated products. Use materials with recycled content such that the sum of post-consumer recycled content plus one-half of the pre-consumer content constitutes at least 10% (based on cost) of total value of the materials in project.

1.8 PRODUCT WARRANTIES

A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.

1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for product specifically endorsed by manufacturer to Owner.
2. Special Warranty: Written warranty required by or incorporated into Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for Owner.

B. Special Warranties: Prepare written document containing appropriate terms and identification, ready for execution. Submit draft for approval before final execution.

1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
2. Specified Form: When specified forms are included with Specifications, prepare written document using appropriate form properly executed.
3. Refer to Divisions 2 through 49 Sections for specific content requirements and particular requirements for submitting special warranties.

C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

A. General Product Requirements: Provide products to comply with Contract Documents, undamaged and, unless otherwise indicated, new at time of installation.

1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types produced and used successfully in similar situations on other projects.
3. Government reserves right to limit selection to products with warranties not in conflict with requirements of Contract Documents.
4. Where products are accompanied by term "as selected," Contracting Officer will make selection.
5. Where products are accompanied by term "match sample," sample to be matched is Governments.
6. Descriptive, performance, and reference standard requirements in Specifications establish "salient characteristics" of products.

B. Product Selection Procedures:

1. Product: Where Specifications name single product and manufacturer, provide named product that complies with requirements or equal.
2. Manufacturer/Source: Where Specifications name single manufacturer or source, provide product by named manufacturer or source that complies with requirements or equal.
3. Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed that complies with requirements or equal.
4. Manufacturers: Where Specifications include list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements or equal.
5. Available Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with provisions in Part 2 "Comparable Products" Article for consideration of an unnamed product.
6. Available Manufacturers: Where Specifications include list of manufacturers, provide a product by one of the manufacturers listed, or an unnamed manufacturer, that complies with requirements. Comply with provisions in Part 2 "Comparable Products" Article for consideration of an unnamed product.

7. Product Options: Where Specifications indicate sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide specified product, system, or equal.
8. Basis-of-Design Product: Where Specifications name a product as the Basis of Design, provide specified product or approved equal. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics based on the product named. While other manufacturers may exist, other products meeting the specified salient characteristics may have differences that create the need for design changes that could impact the functionality, maintenance, and/or life cycle. Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If approved, the Contractor is responsible for any design changes or additional construction costs associated with the selection of a product that is not the Basis of Design.
9. Visual Matching Specification: Where Specifications require matching an established Sample, select product that complies with requirements and matches Architect's sample. Contracting Officers decision will be final on whether a proposed product matches.
    a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.
10. Visual Selection Specification: Where Specifications include phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.
    a. Standard Range: Where Specifications include phrase "standard range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
    b. Full Range: Where Specifications include phrase "full range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 COMPARABLE PRODUCTS

A. Conditions: Contracting Officer will consider Contractor's request for comparable product when the following conditions are satisfied. If following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Evidence proposed product does not require revisions to Contract Documents, that it is consistent with Contract Documents and will produce indicated results and is compatible with other portions of Work.
2. Detailed comparison of significant qualities of proposed product with those named in Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
3. Evidence proposed product provides specified warranty.
4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
5. Samples, if requested.

PART 3 - EXECUTION

3.1 PROTECTION AFTER INSTALLATION

A. Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

END OF SECTION 01 67 00

AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

Project Name: _____ Project Number: _____

Contractor Name: _____ License Number: _____

Contractor Address: _____

Product	Total $ value provided	Total $ value with recycled content Pre-consumer	Total $ value with recycled content Post-consumer	Total $ value with biobased content	Exempted indicate 1,2,3,4	Comments
Hydraulic Mulch (paper based)						
Hydraulic Mulch (wood based)						
Compost						
Parking Stops (Concrete w/ fly ash, slag cement or low cement content)						
Parking Stops (Plastic/Rubber)						
Patio Blocks/Rubber						
Patio Blocks/Plastic						
Playground Surfaces						
Concrete with fly ash						
Concrete with slag cement						
Concrete with low cement content						
Plastic lumber						
Building Insulation						
Rock Wool						
Fiber glass						
Cellulose						
Perlite Comp Board						
Plastic Rigid Foam						
Glass Fiber Reinforced Foam						
Phenolic Rigid Foam						
Ceramic tile						
Resilient flooring						
Floor Tiles/Rubber						
Floor Tiles/Plastic						
Running Tracks						
Carpet (PET)						
Paint						

Reprocessed Latex Paint White & Light Colors						
Reprocessed Latex Dark Colors						
Consolidated Latex Paint						
Toilet/Shower partitions (plastic or steel)						
Other						

CERTIFICATION

I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content.

The following exemptions may apply to the non-procurement of recycled/recovered content materials:

1. The product does not meet appropriate performance standards.
2. The product is not available within a reasonable time frame.
3. The product is not available competitively (from two or more sources).
4. The product is only available at an unreasonable price (compared with a comparable non-recycled content product.)

Signature: _____ Date: _____

END OF
AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

This page intentionally left blank.

SECTION 01 71 21 – SPECIALTY ENGINEERING REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish engineering design, drawings, and calculations for Specialty Engineering Requirements, as indicated, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.2 DEFINITIONS

A. Structural Engineer of Record (SER):

1. Structural engineer legally eligible to seal structural Contract Documents for project.
2. Seal acknowledges SER performed or supervised analysis, design, and document preparation for building structure and has knowledge of requirements for structural system.
3. The SER is responsible for the design of the primary structural system.

B. Specialty Structural Engineer (SSE):

1. Registered Engineer other than Structural Engineer of Record (SER), licensed to practice structural engineering in state in which project is located.
2. Undertakes engineering calculations, design and drawing preparation of components, systems, or installation methods and equipment for specific temporary portions of Work or special items of permanent Work required to be furnished by Contractor.
3. Provide designs and details for items of permanent Work declared to be minor or non-structural.
4. Employee or officer of Contractor or fabricator, employee or officer of an entity providing components to a fabricator, or an independent consultant.
5. The Contractor is responsible for any structural engineering tasks that are required for construction related services.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 SYSTEM DESIGN

A. Contract Documents show conceptually detailed components describing aesthetic intent and provides a performance-type prescription for the design, fabrication, and installation.

B. Contractor is responsible for the engineering and design of components and materials as well as fabrication and installation.

C. Develop conditions not shown in Contract Documents to same level of aesthetics in compliance with performance and aesthetic criteria specified and indicated for detailed areas.

D. Provide engineering design with drawings and calculations sealed by registered Engineer, licensed to practice structural engineering where the project is located.

E. Comply with requirements of Contract Documents, codes, regulations, standards, and guidelines including:

1. Nationally published amendments.
2. Local Amendments.
3. Structural criteria provided.
4. Additional requirements indicated in specification sections.

F. Reference Standards:

1. Refer to technical specification sections for listed standards.

G. Minor deviations in dimensions and profiles may be considered provided design concept is unchanged or intended performance is not compromised as judged by the Architect.

H. Where SSE exercises professional judgment and takes exception to specified criteria or reference standards, disclose exception in writing.

3.2 DOCUMENTATION

A. Include the following items common to project:

1. Project Identification:
   a. Project name
   b. Project location
   c. Identifying project numbers
   d. North arrow
   e. Scale
2. Governing Codes:
   a. Building code and edition
   b. Referenced codes and standards
   c. Design method used for the design
3. Service Loads.
4. Strength loads or factors.
5. Design Load:
   a. Dead loads
   b. Live loads
   c. Snow loads
   d. Wind loads
   e. Seismic loads
6. Material Properties:
   a. Design properties
   b. ASTM designations
7. Computer Submittals:
   a. Documentation of computer programs including the program name and version should be included with any submittal of computer calculations. In the case of custom software or spreadsheet developed in house it may be necessary to provide hand calculation of representative elements to verify the use of the program.

B. Include maximum design loads at connection points to primary structure.

1. Indicate values consistent with method used for design including service loads or strength loads with factors.
2. Design system to apply loads to the structure through the centerlines of the supporting element.
3. Assume building supports are free to rotate. Torsional or flexural fixed supports shall not be used unless approved by the SER.
4. When fixed or eccentric supports are used, provide additional framing as deemed necessary by the SER at no additional cost.

C. Include member sizes, required reinforcing, connection details and material specifications.

D. Include statements where the SSE has exercised professional judgment and takes exception to the specified criteria or referenced standard. Final authority and responsibility for decisions concerning structural design criteria shall belong to the SER. When exceptions are stated as qualifications to the contractor's proposal, the SER shall be notified and respond prior to award.

E. SSE shall review and approve the shop drawings and special erection drawings prepared by a fabricator or supplier and attest to that review with a signed shop drawing stamp, or other means, prior to submittal of the drawings to the SER. When standardized erection drawings are used, there is no need to provide a shop drawing approval stamp.

END OF SECTION 01 71 21

This page intentionally left blank.

SECTION 01 73 00 - WINTER OPERATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. The North Rim receives an average of 142 IN of snow per year, with a record snowfall of 272 IN in 1978. It is common to have several feet of snow on the ground throughout the winter for average snow fall years.

B. Winter Closure Period

1. ADOT and NPS will close State Route 67 (SR67) to the public no later than the first Monday following the first weekend in December and will reopen SR67 no later than May 15 of each year, known as the "Winter Closure Period."
2. Access to the North Rim is permitted for administrative use only during the Winter Closure Period. Public access is not permitted.
3. ADOT and NPS does not maintain paved road access during the winter closure period.

C. Snow and ice removal to facilitate the work during the Winter Closure Period is the Contractor's responsibility.

1. Contractor shall meet all snow removal requirements required by NPS for access to and within limits of construction to support safe construction activities.
2. Snow and ice removal is only a requirement of the work if the contractor elects to conduct work during times of snow and ice accumulation.
3. Snow removal for NPS operations is the responsibility of NPS at its discretion.
   a. NPS and NPS partners may conduct limited snow and ice removal activities including around the North Rim Administrative Area.

D. Applicable locations for contractor snow and ice removal:

1. SR67 between Jacob Lake (Mile Post 579.4) and the boundary of the North Rim of Grand Canyon National Park (Mile Post 610.26)
2. SR67 from the Park entrance gate (Mile Post 610.26) to the North Rim Lodge (approximately 13 miles), including the North Rim Lodge parking area.
3. Lindberg Hill and corresponding access road
4. Admin Loop and Helibase Rd from SR67 to the North Rim Helibase
5. Norton Court from SR67 to the Ballfield access road
6. Areas within the limits of construction as designated on the drawings

1.2 REFERENCES

A. An intergovernmental agreement (IGA 24-0009595-I/P0012024001153) is in effect for snow removal and boundary management between ADOT and NPS.

1. The agreement allows the contractor to maintain access along SR67 to the NPS North Rim property boundary utilizing an encroachment permit.
2. Contractor shall comply with this agreement as the minimum requirements.

1.3 SUBMITTALS

A. Winter Work Plan: Submit a project winter work plan for the project duration within 30 days of notice to proceed and an updated annual winter work plan by June 1 of each year or six (6) months prior to the Winter Closure Period.

1. Annual winter work shall include at a minimum a work plan by project area, schedule, staffing plan, snow removal equipment and operators. The plan shall be customized to the work package and associated geographic areas.
2. Annual winter work plan shall be approved by Contracting Officer prior to Contractor obtaining a ADOT encroachment permit.

1.4 PERMITS AND COORDINATION

A. Contractor shall obtain an encroachment permit from ADOT, in accordance with established procedures form the State's Northcentral District Permit Office, on a yearly basis for snow and ice removal activities.

1. More details located: **Encroachment Permits | Department of Transportation**

B. Corridor review

1. Perform a video recorded corridor review with NPS no later than October 31st prior to the winter closure.
2. Perform a video recorded corridor review with NPS no later than end of the first week in May.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 ACCESS

A. Two entrance gates at Jacob Lake and North Rim Entrance Station shall remain closed and locked during the Winter Closure Period with a chained series of locks allowing limited access to ADOT, NPS, NPS Partners, and the Contractor.

3.2 SNOW AND ICE REMOVAL

A. Only vehicles with rubber tires will be allowed or permitted within the roadway prism, which is defined as the edge of asphalt pavement to edge of asphalt pavement.

B. Snow and ice removal shall be provided for two lane vehicular access.

C. Snow and ice removal within the limits of construction shall be provided for access to active work areas.

D. Snow and ice removal to active work areas shall occur within 8 hours of 6 IN of accumulation.

E. Snow packing shall be provided around the North Rim helibase including vehicle access between the helipad and Helibase Road.

3.3 INSPECTIONS AND REPAIRS

A. The Contractor is responsible to repair damages to SR67 caused by Contractor's Snow and Ice Removal operations.

1. Examples of damages include roadbed surfacing, culverts, debris removal, vegetation control, pothole patches, sealing cracks and joints, signs, spot repairs, etc. The GCNP Deputy Superintendent or the North Rim Manager and the ADOT District Administrator are delegated the authority to determine which repairs are temporary and nominal in nature.
2. Provide repairs to ADOT standards no later than July 1st annually for damage determined to be caused by the Contractor during the Winter Closure Period.
3. Comply with the latest edition Manual on Uniform Traffic Control Devices (MUTCD) published by the Federal Highway Administration (FHWA) and adopted by ADOT when performing any work on SR67.
4. Traffic Control plans shall be processed through the State's Northcentral District Permits office.

END OF SECTION 01 73 00

This page intentionally left blank.

SECTION 01 73 20 - OPENINGS AND PENETRATIONS IN CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Methods of installing and sealing openings and penetrations in construction.

B. Related Specification Sections include but are not necessarily limited to:

1. Division 00 – Procurement and Contracting Requirements.
2. Division 01 – General Requirements.
3. Section 05 50 00 – Miscellaneous Metals.
4. Section 06 82 00 – Fiberglass Reinforced Plastic Fabrications.
5. Section 07 62 00 – Flashing and Sheet Metal.
6. Section 07 84 00 – Firestopping.
7. Section 07 92 00 – Joint Sealants.
8. Section 09 96 00 – High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:

1. ASTM International (ASTM):
   a. A36, Standard Specification for Carbon Structural Steel.
   b. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
   c. A269, Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
   d. A312, Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
   e. A351, Standard Specification for Castings, Austenitic, for Pressure-Containing Parts.
   f. A554, Standard Specification for Welded Stainless Steel Mechanical Tubing.
   g. A653, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
   h. A666, Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar.
   i. A995, Castings, Austenitic-Ferritic (Duplex) Stainless Steel, for Pressure-Containing Parts.
2. National Fire Protection Association (NFPA):
   a. 70, National Electrical Code (NEC):
      1) Article 501, Class 1 Locations.
   b. 90A, Standard for Installation of Air Conditioning and Ventilating Systems.
   c. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA).

1.3 DEFINITIONS

A. Corrosive Areas: For the purpose of this specification section, the following areas are defined as corrosive:

1. None.

B. Hazardous Areas: Areas shown in the Contract Documents as having Class I or Class II area classifications.

C. Washdown Areas: Areas having floor drains or hose bibbs.

1.4 SUBMITTALS

A. Shop Drawings:

1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
2. For each structure provide dimensioned or scaled (minimum 1/8 IN = 1 FT) plan view drawings containing the following information:
   a. Vertical and horizontal location of all required openings and penetrations.
   b. Size of all openings and penetrations.
   c. Opening type.
   d. Seal type.
3. Manufacturer's installation instructions for standard manufactured products.

1.5 SITE CONDITIONS

A. Reference the project Geotechnical Report for groundwater levels.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Pipe Sleeves:

1. Steel, Hot-dipped galvanized after fabrication.
2. Penetrations 24 IN DIA or less: ASTM A53, Schedule 40.
3. Penetrations larger than 24 IN DIA: ASTM A36, Minimum 1/4 IN thickness.
4. Provide weep ring where wall thickness permits.
5. Where wall thickness does not permit a weep ring, use SikaSwell product, or equal.

B. Backing Rod and Sealant: See Specification Section 07 92 00.

C. Modular Mechanical Seals:

1. Acceptable manufacturers:
   a. Link-Seal or equal.
2. 316 stainless steel bolts, nuts and washers.

D. Integrally Cast Wall Pipes:

1. Match material of buried piping.
2. Grade equal to connecting piping system.
3. Provide weep ring where wall thickness permits.
4. Where wall thickness does not permit a weep ring, use SikaSwell product.

E. Firestopping Material: See Specification Section 07 84 00.

F. Sheet Metal Sleeves:

1. Galvanized steel: ASTM A653, G90.
2. Minimum 12 GA.

G. Commercial Wall Castings:

1. Ductile iron, ASTM A536.
2. Grade equal to connecting piping system.

PART 3 - EXECUTION

3.1 FABRICATION

A. Fabricate pipe sleeves in accordance with Specification Section 05 50 00.

B. Fabricate sheet metal sleeves in accordance with Specification Section 07 62 00.

C. Provide waterstop plate/anchor flange for piping, ducts, castings and sleeves cast-in-place in concrete.

1. For fabricated units, weld plate to sleeve, pipe, or ductwork.
2. For commercial castings, cast water stop/anchor with wall pipe.
3. Plate is to be same thickness as sleeve, pipe, casting or ductwork.
4. For fabricated units, diameter of plate or flange to be 4 IN larger than outside diameter of sleeve, pipe or ductwork.
5. For commercial castings, waterstop/anchor size to be manufacturer standard.
6. Provide continuous around entire circumference of sleeve, pipe, or ductwork.

D. Factory or shop-coat painted components in accordance with Specification Section 09 96 00.

3.2 INSTALLATION AND APPLICATION

A. Firestopping materials used in fire-resistance rated construction shall be in full compliance with Specification Section 07 84 00.

B. Seal openings and penetrations in non-fire-resistance-rated construction in accordance with Specification Section 07 92 00.

C. Obtain prior approval from Contracting Officer (CO) when any opening larger than 100 SQIN must be made in existing or newly completed construction.

D. Perform HVAC penetrations in accordance with NFPA 90A.

E. Perform electrical penetrations in accordance with NFPA 70, Article 501.

F. When mechanical or electrical work cannot be installed as structure is being erected, provide and arrange for building-in of boxes, sleeves, insets, fixtures or devices necessary to permit installation later.

1. Lay out chases, holes or other openings which must be provided in masonry, concrete or other work.

G. Where pipes, conduits or ducts pass through floors in washdown areas, install sleeves with top 3 IN above finish floors.

1. In non-washdown areas, install sleeves with ends flush with finished surfaces.

H. Size sleeves, blockouts and cutouts which will receive sealant seal such that free area to receive sealant is minimized and seal integrity may be obtained.

I. For insulated piping and ducts, size sleeves, blockouts and cutouts large enough to accommodate full thickness of insulation.

J. Where pipes, conduits or ducts pass through grating, provide banding at the entire perimeter of the opening.

1. Metal grating: See Specification Section 05 50 00.
2. FRP grating: See Specification Section 06 82 00.

K. Where pipes, conduits or ducts are removed where passing through grating:

1. Metal grating:
   a. Provide banding at perimeter and cover opening with 1/4 IN plate of the same material of the grating.
   b. See Specification Section 05 50 00.
2. FRP grating:
   a. Provide full depth cover meeting same loading requirement as existing material or replace grating section.
   b. See Specification Section 06 82 00.

L. Do not cut into or core drill any beams, joists, or columns.

M. Do not install sleeves in beams, joists, or columns.

N. Do not install recesses in beams, joists, columns, or slabs.

O. Field Cutting and Coring:

1. Saw or core drill with non-impact type equipment.
2. Mark opening and drill small 3/4 IN or less holes through structure following opening outline.
3. Sawcut opening outline on both surfaces.
   a. Knock out within sawcuts using impact type equipment.
   b. Do not chip or spall face of surface to remain intact.
   c. Do not allow any overcut with saw kerf.

P. Where alterations are necessary or where new and old work join, restore adjacent surfaces to their condition existing prior to start of work.

Q. Where area is blocked out to receive sheet metal sleeve at later date:

1. If blockout size is sufficient to allow placement, utilize dowels for interface of initially placed concrete and sleeve encasement concrete which is placed later.
    a. Size blockout based on sleeve size required plus 4 to 6 IN each side of sleeve for concrete encasement.
    b. Provide #4 dowels at 12 IN spacing along each side of blockout with minimum of two dowels required per side.
2. If blockout size is not sufficient to allow placement of dowels, provide keyway along all sides of blockout.
    a. Size blockout based on sleeve size required plus 2 to 4 IN each side of sleeve for concrete encasement.

R. For interior wall applications where backer rod and sealant are specified, provide backer rod and sealant at each side of wall.

S. Refer to Drawings for location of fire-rated walls, floors, and ceilings.

1. Utilize firestopping materials and procedures specified in Specification Section 07 84 00 IN conjunction with scheduled opening type to produce the required fire rating.

T. Use full depth expanding foam sealant for seal applications where single or multiple pipes, conduits, etc., pass through a single sleeve.

U. Do not make duct or conduit penetrations below high water levels when entering or leaving tankage, wet wells, or other water holding structures.

V. Modular Mechanical Seals:

1. Utilize one seal for concrete thickness less than 8 IN and two seals for concrete, 8 IN thick or greater.
2. Utilize two seals for piping 8 IN diameter and larger if concrete thickness permits.
3. Install seals such that bolt heads are located on the most accessible side of the penetration.

W. Backer Rod and Sealant:

1. Install in accordance with Specification Section 07 92 00.
2. Provide backer rod and sealant for modular mechanical seal applications.
    a. Apply on top side of slab penetrations and on interior, dry side wall penetrations.

3.3 SCHEDULES

A. General Schedule of Penetrations through Floors, Roofs, Foundation Base Slabs, Foundation Walls, Foundation Footings, Partitions and Walls for Ductwork, Piping, and Conduit:

1. Provide the following opening and penetration types:
    a. Type A - Block out 2 IN larger than outside dimensions of duct, pipe, or conduits.
    b. Type B - Saw cut or line-drill opening. Place new concrete with integrally cast sheet metal or pipe sleeve.
    c. Type C - Fabricated sheet metal sleeve or pipe sleeve cast-in-place. Provide pipe sleeve with water ring for wet and/or washdown areas.
    d. Type D - Commercial type casting or fabrication.

e. Type E - Saw cut or line-drill opening. Place new concrete with integrally cast pipe, duct or conduit spools.
f. Type F - Integrally cast pipe, duct or conduit.
g. Type G - Saw cut or line-drill and remove area 1 IN larger than outside dimensions of duct, pipe or conduit.
h. Type H - Core drill.
i. Type I - Block out area. At later date, place new concrete with integrally cast sheet metal or pipe sleeve.
j. Type J - Grating Banding for any field cut openings.

2. Provide seals of material and method described as follows.
   a. Category 1 - Modular Mechanical Seal.
   b. Category 2 - Roof curb and flashing according to SMACNA specifications unless otherwise noted on Drawings. Refer to Specification Section 07 62 00 and roofing Specification Sections for additional requirements.
   c. Category 3 - 12 GA sheet metal drip sleeve set in bed of silicon sealant with backing rod and sealant used in sleeve annulus.
   d. Category 4 - Backer rod and sealant.
   e. Category 5 - Full depth compressible sealant with escutcheons on both sides of opening.
   f. Category 6 - Full depth compressible sealant and flanges on both sides of opening. Flanges constructed of same material as duct, fastened to duct and minimum 1/2 IN larger than opening.
   g. Category 7 - Full depth compressible sealant and finish sealant or full depth expanding foam sealant depending on application.
   h. Category 8 - Banding for all grating openings and banding and cover plate of similar materials for abandoned openings.
   i. Category 9 – Preformed strip type seal. See Section 03 31 31 for seal requirements.
3. Furnish openings and sealing materials through new floors, roofs, grating, partitions and walls in accordance with Schedule A, Openings and Penetrations for New Construction.
4. Furnish openings and sealing materials through existing floors, grating, roofs, partitions and walls in accordance with Schedule B, Openings and Penetrations for Existing Construction.

SCHEDULE A. OPENINGS AND PENETRATIONS SCHEDULE FOR NEW CONSTRUCTION

APPLICATIONS	DUCTS		PIPING		CONDUIT	
	OPENING TYPE	SEAL CATEGORY	OPENING TYPE	SEAL CATEGORY	OPENING TYPE	SEAL CATEGORY
Through floors on grade above water table	C F I	4 Not Req 4	C F I (1)	7 9 7	C F I (1)	4 Not Req 7
Through slab on grade below water table	F	Not Req	F	Not Req	F	Not Req
Through floors in washdown areas	C I	4 4	C H (2) I (1)	4 3 4	F H (2) I (1)	Not Req 3 7
Through exterior wall below grade above water table	C F I	7 Not Req 7	C D F I (1)	1 Not Req Not Req 1	F I (1)	Not Req 7
Through wall from tankage or wet well (above high water level) to dry well or dry area	C F I	7 Not Req 7	C D F H (2)	1 Not Req Not Req 1	C F H (2) I (1)	7 Not Req 7 7
Through wall from tankage or wet well (below high water level) to dry well or dry area	F	Not Req	F	Not Req	F	Not Req
Through exterior wall above grade	A B C	6 6 6	A B D H (2)	5 5 Not Req 5	C H (2)	5 4
Roof penetrations	A	2	A	2	A	2
Through interior walls and slabs not covered by the above applications	A C	4 4	A C	4 4	A C F	4 4 Not Req
Grating openings and penetrations	J	8	J	8	J	8

SCHEDULE B. OPENINGS AND PENETRATIONS SCHEDULE FOR EXISTING CONSTRUCTION

APPLICATIONS	DUCTS		PIPING		CONDUIT	
	OPENING TYPE	SEAL CATEGORY	OPENING TYPE	SEAL CATEGORY	OPENING TYPE	SEAL CATEGORY
Through floors on grade above water table	B	7	B	7	B	7
Through slab on grade below water table	E	Not Req	E	Not Req	E	Not Req
Through floors in washdown areas	G	3	G H (2)	3 3	G H (2)	3 3
Through exterior wall below grade above water table	B	7	B (1) B (3) H (2)	7 1 7	B (1) (3) H (2)	7 7
Through wall from tankage or wet well (above high water level) to dry well or dry area	B E	7 Not Req	B E H (2)	1 Not Req 1	B (1) (3) E H (2)	7 Not Req 7
Through wall from tankage or wet well (below high water level) to dry well or dry area	E	Not Req	E	Not Req	E	Not Req
Through exterior wall above grade	G	6	G (1) (3) H (2)	5 5	G (1) (3) H (2)	5 7
Roof penetrations	G	2	G (1) (3) H (2)	2	G	2
Through interior walls and slabs not covered by the above applications	G	4	G (1) (3) H (2)	4 4	G (1) (3) H (2)	4 4
Grating openings and penetrations	J	8	J	8	J	8

(1) Multiple piping 3 IN and smaller or multiple conduits.
(2) Single pipe 3 IN and smaller or single conduit.
(3) Single pipe or conduit larger than 3 IN.

END OF SECTION 01 73 20

SECTION 01 73 29 - CUTTING AND PATCHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes procedural requirements for cutting and patching on buildings that do not contain Historic Fabric.

1.2 SUBMITTALS

A. Cutting and Patching Plan: Submit Plan describing procedures at least 10 days before cutting and patching will be performed, requesting approval to proceed. Include:

1. Extent: Describe cutting and patching, show how performed, and indicate why they cannot be avoided.
2. Changes to In-Place Construction: Describe anticipated results. Include changes to structural elements and operating components as well as changes in building's appearance and other significant visual elements.
3. Products: List products to be used and firms or entities that will perform Work.
4. Dates: Indicate when cutting and patching will be performed.
5. Utility Services and Mechanical/Electrical Systems: List services/systems that cutting and patching procedures will disturb or affect. List services/systems that will be relocated and those that will be temporarily out of service. Indicate how long services/systems will be disrupted.
6. Structural Elements: Where cutting and patching involve adding reinforcement to structural elements, submit details and engineering calculations showing integration of reinforcement with original structure. Do not cut and patch structural elements in a manner that could change their load carrying capacity or increase deflection.
7. Contracting Officer’s (CO) Approval: Obtain approval of cutting and patching plan before cutting and patching. Approval does not waive right to later require removal and replacement of unsatisfactory work.

1.3 QUALITY ASSURANCE

A. Structural Elements: Do not cut and patch structural elements in a manner that could change their load-carrying capacity or load-deflection ratio.

B. Operational Elements: Do not cut and patch operating elements and related components in a manner that results in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety.

C. Miscellaneous Elements: Do not cut and patch miscellaneous elements or related components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety.

D. Visual Requirements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch construction exposed on the exterior or in occupied spaces in a manner that would, in Contracting Officer’s opinion, reduce the building's aesthetic

qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

PART 2 - PRODUCTS

2.1 MATERIALS

A. General: Comply with requirements specified in other Sections.

B. In-Place Materials: Use materials identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.

1. If identical materials are unavailable or cannot be used, use materials that will match the visual and functional performance of in-place materials when installed.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.

1. Compatibility: Before patching, verify compatibility with and suitability of substrates, including compatibility with in-place finishes or primers.
2. Proceed with installation only after unsafe or unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Temporary Support: Provide temporary support of Work to be cut.

B. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.

C. Adjoining Areas: Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

D. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to prevent interruption to occupied areas.

3.3 PERFORMANCE

A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at earliest feasible time. Complete without delay.

1. Cut in-place construction to provide for installation of other components or performance of other construction subsequently. Patch as required to restore surfaces to original condition.

B. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer and original Installer's written recommendations.

1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots as small as possible, neatly to size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
2. Finished Surfaces: Cut or drill from exposed or finished side into concealed surfaces.
3. Concrete: Cut using an abrasive saw or a diamond-core drill.
4. Excavating and Backfilling: Comply with requirements in applicable Division 31 Sections where required by cutting and patching operations.
5. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
6. Proceed with patching after construction operations cutting are complete.

C. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams as invisible as possible. Provide materials and comply with installation requirements specified in other Sections.

1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate integrity of installation.
2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.
3. Floors and Walls: Where walls or partitions that are removed extend one finished area into another. Patch and repair floor and wall surfaces in new space. Provide an even surface of uniform finish, color, texture, and appearance. Remove in-place floor and wall coverings and replace with new materials, if necessary~~,~~ to achieve uniform color and appearance.
4. Ceilings: Patch, repair, or re-hang in-place ceilings as necessary to provide an even-plane surface of uniform appearance.
5. Exterior Building Enclosure: Patch components in a manner that restores enclosure to a weather tight condition.

D. Cleaning: Clean areas and spaces where cutting and patching are performed. Remove paint, mortar, oils, putty, and similar materials.

END OF SECTION 01 73 29

This page intentionally left blank.

SECTION 01 73 40 - EXECUTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes general procedural requirements governing execution of Work including:

1. Coordination with utility service providers
2. Construction layout
3. Field engineering and surveying
4. General installation of products
5. Progress cleaning
6. Starting and adjusting
7. Protection of installed construction
8. Correction of the Work

1.2 SUBMITTALS

A. Certificates: Submit certificate signed by land surveyor certifying location and elevation of improvements comply with requirements.

B. Landfill Receipts: Submit copy of receipts issued by landfill facility, licensed to accept hazardous materials, for hazardous waste disposal.

C. Certified Surveys: Submit two copies signed by land surveyor.

D. Quantity Surveys: Submit 2 copies showing quantities of work performed and actual construction completed in place.

E. Trail Access Management Plan: Submit and maintain a narrative plan with supporting drawings that identify locations of temporary trail closures, trail detours as referenced in Section 01 73 50.

F. Contractor Construction Camp Development Plan: Identify camps, as referenced in Section 01 73 60, to be used within 30 days of the Notice to Proceed.

G. Utility Management Plan: Submit and maintain a narrative plan with supporting drawings for coordination with the NPS in maintaining water utility services, wastewater utility services, and power utility services.

1.3 QUALITY ASSURANCE

A. Land Surveyor Qualifications: Professional legally qualified to practice in jurisdiction where Project is located and-is experienced in providing land-surveying services of kind indicated.

B. Helicopter Pilot Qualifications: See Section 01 65 50.

C. Construction Camp Outfitter Qualifications: See Section 01 73 60.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

A. Existing Conditions: Existence and location of site improvements and other construction indicated as existing are not guaranteed.

1. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Work.
2. Before construction, verify the location and points of connection of utility services.

B. Existing Utilities: Existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify existence and location of underground utilities and other construction affecting Work.

1. Before construction, verify location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; and underground electrical services.
2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.
3. Contractor shall hire a third party subsurface utility engineering (SUE) entity to coordinate utility locates of existing utilities and perform potholing of all proposed crossing of existing utilities.
    a. Call Arizona 811 and obtain utility locates from owning utilities.
    b. Locate NPS owned utilities on behalf of NPS.
    c. Submit a potholing plan that includes a map of the proposed locations of potholing at each utility crossing to Contracting Officer.
    d. Immediately backfill all utilities with cinders to the appropriate compaction.
    e. Submit a potholing report that includes pothole locations by project coordinates, depths below grade, and photos of exposed utilities.
    f. Contractor shall not proceed with new utility excavation until potholing report is approved by the Contracting Officer.

C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.

1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
2. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
3. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
4. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 COORDINATION WITH UTILITY SERVICE PROVIDERS

A. Coordination with Utility Service Providers: Contact following Utility Service providers, sufficiently in advance to avoid delaying the work, to coordinate Contractor’s portion of Work,

testing requirements, inspections, etc. Reference 01 50 00-1 section 1.3 for authorized and restricted use of water and power

1. Electrical:
   a. Entrance Contact: Contact Garkane Energy to coordinate electrical service requirements.
   b. Service Contact: Contact NPS Contracting Officer to coordinate electrical requirements inside the Park boundary.
2. Water Service Contact: Contact NPS Contracting Officer to coordinate Water service requirements.
3. Wastewater Service Contact: Contractor is responsible for wastewater service. No discharge to the existing system is allowed.

3.3 PREPARATION

A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of Work are indicated fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying Work.

B. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.

C. Review of Contract Documents and Field Conditions: Immediately on discovery of need for clarification of the Contract Documents caused by differing field conditions outside control of Contractor, submit request for information to Contracting Officer in accordance with Section 01 31 00 "Project Management and Coordination."

3.4 CONSTRUCTION LAYOUT

A. Verification: Verify layout information shown on Drawings, in relation to the existing benchmarks before proceeding to lay out Work. Notify Contracting Officer promptly if discrepancies are discovered.

B. General: Engage a land surveyor to lay out Work using accepted surveying practices.

1. Establish benchmarks and control points to set lines of construction and elsewhere as needed to locate each element of Project.
2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
3. Inform installers of lines and levels to which they must comply.
4. Check location, level and plumb, of every major element as Work progresses.
5. Notify Contracting Officer when deviations from required lines and levels exceed allowable tolerances.

C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.

D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.

E. Record Log: Maintain log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make log available for review by National Park Service (NPS).

3.5 FIELD ENGINEERING

A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning Work. Preserve and protect permanent benchmarks and control points during construction operations. Controls destroyed by Contractor will be replaced by Contractor at their expense.

1. Existing Monuments: All benchmarks, land corners, and triangulation points, established by other surveys, existing within construction area shall be preserved. If existing monuments interfere with Work, secure written permission before removing them.

B. Benchmarks: Establish and maintain a minimum of two permanent benchmarks on Project site, referenced to data established by survey control points. Comply with NPS requirements for type and size of benchmark.

1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.

C. Utility As-Constructed Surveys: Survey and record final locations of all new utilities and ancillary systems including, but not limited to:

1. Raw and finished waterlines, valve locations, and drain valve discharge points.
2. WTP yard piping, valve boxes, etc.
3. Power/Electrical lines, pull boxes, switching cabinets, transformers, etc.
4. Fiber conduits and pullboxes.

3.6 INSTALLATION

A. General: Locate Work and components of Work accurately in correct alignment and elevation, as indicated.

1. Make vertical work plumb and make horizontal work level.
2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.

B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.

C. Install products at the time and under conditions for best possible results. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.

E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.

F. Templates: Obtain and distribute to parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm adequate provisions are made for locating and installing products to comply with indicated requirements.

G. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of Work.

1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Contracting Officer.
2. Allow for building movement, thermal expansion, and contraction.
3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors to be embedded in concrete or masonry. Deliver to Project site in time for installation.

H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for best visual effect. Fit exposed connections together to form hairline joints.

I. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.

J. Quantity surveys: Shall be conducted, and data derived from these surveys shall be used in computing quantities of work performed and actual construction completed and in place.

1. Contractor shall conduct original and final surveys and surveys for any periods for which progress payments are requested. These surveys shall be conducted under direction of a representative of the Contracting Officer, unless Contracting Officer waives requirement in a specific instance. Government shall make such computations as are necessary to determine quantities of work performed or finally in place. Contractor shall make computations based on surveys for any periods for which progress payments are requested.
2. Promptly upon completing a survey, Contractor shall furnish originals of field notes and other records relating to survey or layout of Work to Contracting Officer. Contractor shall retain copies of all such material furnished to Contracting Officer.

3.7 PROGRESS CLEANING

A. General: Clean Project site, work areas, and common areas daily. Coordinate progress cleaning for joint-use areas where more than one Installer has worked. Enforce requirements strictly. Dispose of materials lawfully.

1. Comply with requirements in National Fire Protection Association (NFPA) 241 for removal of combustible waste materials and debris at the end of every work session.
2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 degrees Fahrenheit.
3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.

B. Site: Maintain Project site free of waste materials and debris.

C. Work Areas: Clean areas where work is in progress to level of cleanliness necessary for proper execution of Work.

1. Remove liquid spills promptly.
2. Where dust would impair proper execution of Work, broom-clean or vacuum entire work area, as appropriate.

3. Contractor shall provide progress cleaning that minimizes sources of food, water, and harborage available to pests.

D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials not hazardous to health or property and will not damage exposed surfaces.

1. Utilize non-toxic cleaning materials and methods.
   a. Comply with Green Seal Standard (GS) 37 for general purpose cleaning and bathroom cleaning.
   b. Use natural cleaning materials where feasible. Natural cleaning materials include:
      1) Abrasive cleaners: substitute 1/2 lemon dipped in borax.
      2) Ammonia: substitute vinegar, salt and water mixture, or baking soda and water.
      3) Disinfectants: substitute 1/2 cup borax in gallon water.
      4) Drain cleaners: substitute 1/4 cup baking soda and 1/4 cup vinegar in boiling water.
      5) Upholstery cleaners: substitute dry cornstarch.

E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect from damage and deterioration at time of Substantial Completion.

G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways.

H. Clean and protect construction in progress and adjoining materials already in place during handling and installation. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.

I. Clean and provide maintenance on completed construction as frequently as necessary through remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.

J. Limiting Exposures: Supervise construction operations so that no part of construction completed or in progress, is subject to harmful, dangerous, damaging, or deleterious exposure during construction period.

K. Final Cleaning: At completion of Work, remove remaining waste materials, rubbish, tools, equipment, machinery and surplus materials. Clean exposed surfaces and leave Project clean and ready for occupancy.

1. Provide final cleaning in accordance with ASTM E1971.

3.8 STARTING AND ADJUSTING

A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

B. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.

C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

D. Manufacturer's Field Service: If factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in Section 01 40 00 "Quality Requirements."

3.9 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

B. Comply with manufacturer's written instructions for temperature and relative humidity.

3.10 CORRECTION OF THE WORK

A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in Section 01 73 29 "Cutting and Patching."

1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

B. Restore permanent facilities used during construction to specified condition.

C. Remove and replace damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.

D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.

E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 01 73 40

This page intentionally left blank.

SECTION 01 73 50 – TRAIL ACCESS MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes requirements for managing public access and restoration of trails impacted by construction activities.

B. Existing trails will be impacted by construction activities. The goal of Trail Access Management is to protect the safety of the public, NPS staff and Contractor personnel during construction, which will require coordination with the NPS Backcountry Rangers, Mule Concessioner, and Contracting Officer to plan and communicate, implement procedures, and monitor effectiveness.

1.2 DEFINITIONS

A. Escorted Access: closing a portion of a trail for a specified duration for construction activities then allowing public access with a Contractor provided escort.

1. The Contractor provided escort shall be a member of the construction team that is knowledgeable with construction equipment, means/methods, and associated safety requirements and have the ability to effectively interface with the public.
2. Once the public is through the area, the trail is closed again for construction to resume. This can occur multiple times a day.

B. Trail Closure: closing a trail to the public for a defined time duration.

C. Trail Detour: an alternative trail to bypass a section of an existing trail impacted by construction activities. Trails detours can use the existing trail network system to by-pass a work area and/or create a new temporary trail within the limits of construction.

1.3 SUBMITTALS

A. Trail Access Management Plan (TAMP): Submit and maintain a narrative plan with supporting drawings that identify locations of temporary trail closures, trail detours, signage, and escorted access that is consistent with the requirements of this specification and the Drawings for each Volume of work, the plan shall be submitted in multiple phases.

1. Phase I TAMP shall be submitted within 30 days after the NTP that identifies sequencing of work and trail management referenced to the project baseline schedule. The Phase I TAMP shall identify the impacted trails and proposed locations for closures, bypasses, escort locations, traffic control. The objective of this plan is to provide the owner with the planning and communication document for communication to the public and its business partners early in the project. It is the basis for establishing and updating reservations systems that generally need a lead time of six (6) to twelve (12) months to update and manage. Phase I plan is to be submitted 30 days following NTP.
2. Phase II TAMP shall be submitted within six (6) months of the anticipated activity and include detailed site specific plans that detail out exact locations for closures to include the signage and barrier locations, logistics staging of construction camps,

materials/equipment, access routes between landing and drop zones to work and staging areas. Geographic grouping of project scope areas should be based on project schedule. Objective of the Phase II site specific plans will allow NPS and NPS partners to plan around.

a. Narrative of the construction equipment and methods, schedule of construction activities and associated trail/visitor impacts, and impact mitigation measures. Provide details or narrative on methods to secure the work site during off-hours.
b. Drawings showing limits of active construction, overall limits of construction, trail signage, temporary trail reroutes/detours, and locations of trail escorts. Include signage at major trailheads notifying visitors of construction activities.
c. Drawings are to be updated with changes in construction schedule or other factors that would require modification of the trail access management approach.
d. Incorporate comments on the Trail Access Management Plan from Contracting Officer.

3. Submit the Phase I plan within 30 days following Notice to Proceed..
4. Trail Access Management Plan Phase II will be updated with lessons learned throughout project duration .
5. Contractor shall provide NPS twelve (12) months in advance notice of the request for trail and campground closures and detours. If approved, NPS will provide public notification of the trail closure or detour. Contractor shall install closure and detour signage and limit the duration to the extent possible. The Contractor is notified that visitor use of the trail may continue despite closure and contractor shall provide adequate trail monitors and provisions for the safe passage of visitors.

B. Identify planned work sites, locations and trail management activities in each 3-week look ahead work schedule (see Section 01 31 00 Project Management and Coordination).

C. Signs: contractor is responsible for building, installing and relocating as needed sign structures that allow for posting of closure and other information notifications that the owner requests to be posted through the Contracting Officer. Contractor (CO) is responsible for requesting the sign format, the owner through the CO will provide the approved format and language for the contractor to print, waterproof and post to the signage structure.

1.4 PROJECT CONDITIONS

A. Not Used

PART 2 - PRODUCTS

2.1 SIGNAGE

A. Signs for notifying the public of planned detours, closures, and escorted access:

1. Legible, visible, and acceptable to the Contracting Officer.
2. See Specification Section 10 14 23 – Signage.

PART 3 - EXECUTION

3.1 GENERAL

A. Contractor shall at all times conduct operations to ensure the least inconvenience to the public.

B. Provide signage, flaggers, trail detours, and trail escorts in accordance with the approved Trail Access Management Plan.

C. Trail closures will be permitted at the following locations as shown on the Drawings:

1. North Kaibab Trail
   a. General – The North Kaibab Trail is within the limits of construction but will remain open to the public during construction unless otherwise indicated within the requirements for each volume. Contractor shall provide adequate access with traffic control measures including temporary signs and protection of trail users shall at all times.
   b. Volume 02 – Closure of the North Kaibab Trail will be required when the directional drilling is approaching the bore exit due to potential rockfall. The North Kaibab Trail shall be closed between Red Wall Bridge and Manzanita for an agreed upon duration of approximately 2 weeks (or less as feasible) for each bore exit as the bore approaches and daylights at the bore exit at Roaring Springs Pumphouse. Traffic control measures including temporary signs and trail monitors at each end of closure shall be maintained by the Contractor at all times.
   c. Volume 03 – Localized closures of portions of the Supai Tunnel rest area will be required for construction of the water spigot, bottle filling stations, valve vault, and associated piping for an agreed upon duration of approximately 4 weeks (or less as feasible). Temporary signs shall be provided. Full closure of the North Kaibab Trail above Supai Tunnel will be required for pipe installation between the North Rim WTP and Supai Tunnel for an agreed upon duration of approximately 4 weeks (or less as feasible). Trail shall be reopened as soon as construction is completed. Closure of the Supai Tunnel rest area and North Kaibab Trail is preferred during the period of October 16-May14 or the months of July and August as these are less trafficked by visitors.
   d. Volume 10 – Closure of the North Kaibab trail will be required during installation of the overhead powerlines above the trail between Roaring Springs Pumphouse and Lower Bench. The North Kaibab Trail shall be closed between North Rim and Manzanita for an agreed upon duration of approximately 8 weeks (or less as feasible) to conduct the work. Traffic control measures including temporary signs and trail monitors at each end of closure shall be maintained by the Contractor at all times. The Volume 10 work must occur when the Roaring Springs Pumphouse is offline, therefore, closure of the North Kaibab Trail must occur during the period of November 15 to May 15.
   e. Volume 11 - Construction within the North Kaibab Trail from the Roaring Springs Pumphouse to Manzanita will require closure of the trail for an agreed upon duration of approximately 4 weeks (or less as feasible). Trail shall be reopened as soon as construction is completed. Trail closure signs will be posted at the North Rim and at Cottonwood Campground.
   f. Volume 12 - The Contractor shall maintain open trail access through the Cottonwood Campground developed area to provide connection to the North Kaibab Trail. The Contractor shall provide trail signage on each side of the work area to notify public of work activities. The Contractor may provide a

temporary trail detour in the localized area of the work, if needed, with signage on each side of the trail closure and temporary construction fencing to identify trail detour.

2. Bridle Path
   a. General – The Bridle Path is within the limits of construction but will remain open to the public during construction unless otherwise indicated. Contractor shall provide adequate access with traffic control measures including temporary signs and protection of trail users shall at all times.
   b. Volume 01 – Contractor to develop and submit a closure and detour plan for the Bridle Path in the immediate vicinity of the work if a safety concern for visitors is presented from the work such as open trenches, working around construction equipment, and narrow trail corridors. Public will be temporarily bypassed around the construction on the side of the trail. Additionally, signage will be posted at Admin Loop Road, the Grand Canyon Lodge, Transept Trail, and the Nature Trail Junction noting that there is a rolling closure of the trail for construction. Signage will note that access is uninterrupted along the Transept Trail.
3. Transept Trail
   a. General – The Transept Trail is within the limits of construction but will remain open to the public during construction unless otherwise indicated. Contractor shall provide adequate access with traffic control measures including temporary signs and protection of trail users shall at all times.
4. Roaring Springs Day Use Trail
   a. The Roaring Springs Day Use trail will remain closed to visitors for the duration of construction. Contractor shall provide temporary signs of closure at the North Kaibab Trail branch.
5. Roaring Springs Pumphouse Access Trail
   a. The Roaring Springs Pumphouse Access Trail is an unmarked and unmaintained trail often used by NPS personnel to access Manzanita via the North Kaibab Trail. The trail will remain closed to visitors for the duration of construction. The trail is not used by visitors, however, construction activity may invite use. Contractor shall provide temporary signs of closure at the North Kaibab Trail branch.

3.2 TRAIL DETOURS

A. Restore terrain and vegetation along temporary detours to the original condition to the satisfaction of the CO once their use is no longer needed and prior to Acceptance of Work.

3.3 TRAIL INSTALLATION, MAINTENANCE, AND RESTORATION

A. Trail installation, maintenance, restoration shall be in accordance with NPS standard “Trail Management Standards for Grand Canyon National Park’s Class IV Stock Trails”.

B. Trail Installation

1. There is not an existing access trail between the Roaring Springs Day Use trail and the Roaring Springs Pumphouse. If required, Contractor can install a temporary trail suitable for construction from the Roaring Springs Day Use to the Roaring Springs Pumphouse within the limits of construction as identified in Volume 0. Contractor shall maintain trail for construction personnel access purposes, however, the trail shall be removed and the alignment restored to previous conditions once construction in this area is complete. Contractor is responsible for coordinating field route pre-work

meeting with the owner through the Contracting Officer (CO) for final route approval ensure no sensitive resources are impacted and or mitigated through the routing..

2. The existing Roaring Springs Pumphouse Access Trail from the Roaring Springs Pumphouse to the North Kaibab Trail is not suitable for safe construction access. Contractor shall improve trail and widen no more than a maximum of 10 FT (no more than 5 FT each side of centerline) for construction personnel access purposes and the trail shall remain in suitable condition after construction. Subject to pre-work field walk with owner for final approval of proposed work through final approval by the CO.

C. Trail Maintenance and Restoration

1. The North Kaibab Trail in areas impacted by construction activities and personnel access shall be rehabilitated to pre-project conditions or better at the completion of the work and prior to substantial completion.
2. The Bridle Path in areas impacted by construction activities and personnel access shall be rehabilitated to pre-project conditions or better at the completion of the work and prior to substantial completion.
3. The Transept Trail in areas impacted by construction activities and personnel access shall be rehabilitated to pre-project conditions or better at the completion of the work and prior to substantial completion.
4. Circulation trails at Cottonwood Campground and Manzanita shall be rehabilitated to pre-project conditions or better at the completion of work where such features have been disturbed by construction activities.
5. All temporary trail detours and reroute locations shall be restored to pre-project conditions or better at the completion of work and prior to substantial completion.
6. The Roaring Springs Day Use trail will be used by construction personnel for construction camps and staging. Contractor shall maintain trail for construction access purposes including clearing of rockfalls. Contractor shall install a temporary footbridge over the creek near the Day Use area to prevent damage and erosion. Additionally, Contractor shall install erosion control silt fencing/waddles along the creek to prevent sedimentation from entering. Footbridge shall be removed and area restored to pre-construction conditions after work in the area is complete and/or proposed condition agreed upon by owner through the CO.

END OF SECTION 01 73 50

This page intentionally left blank.

SECTION 01 73 60 – CONSTRUCTION CAMPS AND STAFF HOUSING

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes requirements for construction camps and staff housing for Contractor personnel located within the Grand Canyon National Park at the North Rim and innercanyon.

1.2 DEFINITIONS

A. Construction Camp: Temporary facilities to house Contractor personnel during construction.

1. The intended use may include camping units, trailer complex for living purposes, food prep stations, common areas, utilities and trash removal provisions, personnel vehicle parking, and possible construction administration trailers.
2. Construction camps are not intended for equipment and material staging and should not consist of construction equipment and tool storage, material stockpiling, construction equipment parking, and similar.

B. Professional Outfitter: Company hired by Contractor to operate each Construction Camp.

1.3 QUALITY ASSURANCE

A. Professional Outfitter: Demonstrated experience in providing similar operations within the Grand Canyon or other remote work sites associated with Construction, research, or similar activities.

1.4 PROJECT CONDITIONS

A. Establishment of construction camps will be permitted at:

1. North Rim Construction Camps
   a. Lindberg Hill approximately 7 miles north of the North Rim administrative area off of SR67 as shown on drawings.
   b. Norton Court camp is located within 0.5 miles of the ballfield.
2. Innercanyon Construction Camps
   a. Cottonwood Campground is a small campground 6.8 miles below the North Rim and approximately 2 miles south of the RSPH along the North Kaibab Trail. It is accessible by foot and helicopter. This location shall be used for all innercanyon work.
3. South Rim Contractor Support Facilities is a flat area adjacent to the South Rim Helibase. This site can be used for helicopter staging and flights. Access is off of Highway 64 approximately 5 miles from the entrance gate.

1.5 SUBMITTALS

A. North Rim Contractor Construction Camp Development Plan

1. Aramark North Rim Hospitality LLC operated staffing dorms at the North Rim for the National Park Service. The Contractor can reach out to Aramark North Rim Hospitality LLC to determine if housing can be used by the contractor. Any agreement reached is between the Contractor and Aramark North Rim Hospitality LLC.
    a. Location of waste storage and disposal services, if proposed.
    b. Method and destination of waste disposal

B. Innercanyon Contractor Construction Camp Development Plan

1. Innercanyon infrastructure has limited wastewater capacity, all solid human waste should be self contained and removed by contractor on as-needed basis. Power is limited and or not available at most innercanyon locations with exception of the RSPS and Manzanita. Potable water availability may be limited or not available. Contractor should plan on providing all water and wastewater facilities.
2. Submit a plan and drawing of the proposed Contractor's Site facilities detailing, developed by the Professional Outfitter:
    a. Location and sizes of tents or temporary structures.
    b. Any adjoining land uses and structures including residences, restrooms, campsites, trails, etc.
    c. Location of sensitive environmental areas such as watercourses, protected vegetation.
    d. Access plan to temporary structures from existing trails.
    e. Location of waste storage areas and toilets.
    f. Location of material and equipment storage areas.
    g. Fuel storage locations containment details.
    h. Location of water service connections if proposed.
    i. Location of electrical service connection if proposed.
    j. Location of electrical generator if proposed.
    k. A waste management plan to address all waste generated at the camps including but not limited to vegetation, sanitary waste, and domestic waste, detailing but not limited to:
        1) Description of waste likely to be generated from the site of the proposed facilities.
        2) Methods to avoid or minimize the volumes of waste generated.
        3) Methods of waste treatment and disposal.
        4) Method and frequency of waste removal,
        5) Destination of waste for ultimate disposal.

C. South Rim Contractor Support Facilities Development Plan

1. Submit a plan and drawing of the proposed Contractor's Site facilities within the limits of construction detailing:
    a. Helicopter staging
    b. Location and sizes of self-contained camping units
    c. Any adjoining land uses and structures including residences, restrooms, campsites, trails, etc.
    d. Vehicle access plan
    e. Overflow parking areas
    f. Location of material storage areas
    g. Location of water service connections, if proposed
    h. Location of restroom services, if proposed.
    i. Location of electrical service connections, if proposed
    j. Location of electrical generator, if proposed
    k. Location of waste storage and disposal services, if proposed.
    l. Method and destination of waste disposal

PART 2 - PRODUCTS

2.1 CONTRACTOR CONSTRUCTION CAMPS

A. Locations: Camps shall be provided only at the locations identified in Paragraph 1.5 above and as shown on the drawings.

B. Lindberg Hill

1. Facility has been used as a camp for fire personnel in the event of wildland fires near the Grand Canyon National Park. Contractor shall defer site use to wildland fire emergency personnel and equipment in the area located on the Drawings. In the event of an extreme fire, contractor may be required to stop all construction work and vacate premises for safety concerns and to allow for fire operations.
2. Site contains 9 sites with electric hook ups that can be used by the contractor.
3. A maximum of 40 FT RV or trailer will be allowed at designated electric hook up sites to not impede access for other vehicles and equipment.
4. RV and trailer length is unrestricted in other designated construction camp areas for this site.
5. Standby generators will be allowed at a maximum 70 dBA along SR67. Noise attenuation for generators is required.
6. No water or sewer hook ups are available.
7. All trash and equipment must be contained and secured at all times due to wildlife concerns.
8. Contractor shall abide by the same rules and regulations posted at the North Rim campground, unless otherwise approved.
9. Due to concern of wildland fires, smoking is allowed only in a single designated area of no larger than 500 sf. This area shall be clear of exposed vegetation and have adequate means for disposal of trash and flammable materials.

C. Norton Court

1. Facility has been used as a park housing prior to construction project.
2. Site contains 19 sites with electric, water, sewer hook ups- 16 of the sites can be used by the Contractor and 3 are reserved for the CMR.
3. Trailer size will not exceed developed pad site and/or impede road traffic.
4. All trash and equipment must be contained and secured at all times due to wildlife concerns.
5. Contractor shall abide by the same rules and regulations posted at the North Rim campground, unless otherwise approved.
6. Due to concern of wildland fires, smoking is allowed only in a single designated area of no larger than 500 sf. This area shall be clear of exposed vegetation and have adequate means for disposal of trash and flammable materials.

D. Cottonwood Campground (CCG)

1. Contractor camp shall be operated and staffed at all times by a professional outfitter hired by the general contractor.
2. Contractor will be allowed use of all designated sites at Cottonwood campground for all innercanyon work. Each site shall contain no more than two 4-person tents and 4 persons per site.
3. Cottonwood campground camping will be closed to visitors during contractor use.
4. North Kaibab Trail will remain an open thoroughfare for North Kaibab Trail hiking. Contractor shall not impede access.
5. Contractor shall be respectful of appearance and impact on users keeping noise to a minimum and campground maintained.

6. Potable water is likely not available. Contractor shall plan on providing all water required for camping at this site.
7. Contractor shall provide the following at the Construction Camp:
    a. Camping units: Portable, pre-fabricated fabric structures and wall tents.
    b. Self-contained food preparation and storage facilities that are protected from the environment, weather, and wildlife. Large mess hall tents will not be allowed.
    c. Potable water shall be made available at all times by the Contractor. Cottonwood campground.
    d. Existing pit toilets are available for contractor use and contractor shall maintain facilities.
    e. Additional sanitary facilities including toilets and washrooms shall be provided if required by the professional outfitter.
    f. Generators are allowed at a maximum of 60 dBA as measured at North Kaibab Trail. Noise attenuation for generators is required. No use is allowed between the hours of 8 pm and 6 am.
    g. Provide material and equipment storage as required.
8. All trash and equipment must be contained and secured at all times due to wildlife concerns.
9. All waste including trash, food, and toilets must be removed from the site at least weekly.
10. Contractor shall abide by the same rules and regulations posted for visitor use of this campground, unless otherwise approved.
11. Due to concern of wildland fires, no smoking is allowed at this site.

E. South Rim Contractor Support Facilities

1. Facility has been used as a camp for South Rim improvements project. Site is located next to the South Rim Helibase and the Facilities Maintenance Division (FMD) for Grand Canyon National Park.
2. Contractor Support Facilities and contractor helicopter operations will be on or adjacent to an active helibase site used by NPS for operations including transporting personnel and materials, emergencies, and search and rescue missions. Contactor shall yield all work to the needs of the helibase operations at any time. Helibase operations will have final authority to stop construction work including during helicopter operations and emergencies, if required.
3. Site has electric hook ups that can be used by the contractor.
4. RV and trailer length is unrestricted for this site.
5. Standby generators will be allowed at a maximum 70 dBA along SR67. Noise attenuation for generators is required.
6. Water and sewer hook ups are available.
7. All trash and equipment must be contained and secured at all times due to wildlife concerns.
8. The contractor shall access the site through the Contractor Access Road off of State Route 64 (SR64). All deliveries and access will utilize this roadway. Access via Center Road and Shuttle Bus Road is not permitted.
    a. The Contractor Access Road crosses the Arizona Trail where pedestrian use including hiking and biking is common throughout the year. Contractor shall adhere to all vehicular signage and maintain safe operations at all times. Contractor shall provide temporary signs and protection of trail users is required.
    b. A vehicle gate is installed at the entrance. The contractor shall maintain a locking system for the gate and provide key access to the CMR and helibase staff. The gate shall be locked when the contractor facilities area is not in use, and shall remain open at all other times unless otherwise approved by the CMR.

9. Due to concern of wildland fires, smoking is allowed only in a single designated area of no larger than 500 sf. This area shall be clear of exposed vegetation and have adequate means for disposal of trash and flammable materials.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Camp locations are limited to the areas described in this specification and shown on the Drawings.

B. Locate camps to limit site disturbance and as directed by the Contracting Officer.

C. Provide each camp ready for construction staff use prior to commencing construction activities in the area serviced by the camp. Do not remove until facilities are no longer needed. Do not mobilize a camp multiple times in the same location.

3.2 OPERATION, TERMINATION, AND REMOVAL

A. Keep camps neat and clean

B. Maintain and clean toilet facilities at least weekly.

C. Dispose of other camp waste at least weekly.

D. NPS may conduct food service inspections by a Public Health Consultant (PHC) at food preparation areas at construction camps. Contractor shall allow access and comply with directives issued by the PHC.

E. Utilize helicopter return/backhaul flights for flying out construction and human waste on as needed basis to reduce overall number of helicopter flights required.

F. Temporary electric and water service termination and removal:

1. Remove each temporary facility when need for its service has ended, or no later than Substantial Completion.
2. Complete permanent construction that may have been delayed because of interference with temporary facility.
3. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

G. Return site to pre-project condition at completion of work. It is the contractor's responsibility to document pre-project conditions.

H. Materials and facilities that constitute temporary facilities are property of Contractor.

END OF SECTION 01 73 60

This page intentionally left blank.

SECTION 01 74 15 - DISINFECTION OF FACILITIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Requirements for cleaning and disinfection of:
   a. Interior piping and associated equipment and outdoor tankage for the finished water back feed system at the Roaring Springs Pumphouse (Volume 5).
   b. Interior piping and associated equipment and indoor/outdoor tankage at the North Rim WTP (Volume 6).
2. This Section covers the protection, cleaning, flushing, and disinfection, treatment facilities and chemical feed systems that are any part of the water treatment process, including raw water through finished water.
3. See Related Sections for cleaning and disinfection of buried pipelines including:
   a. North Rim Raw Water Supply Pipeline (Volumes 01, 02, 05, 06)
   b. Innercanyon Potable Water Back Feed (Volumes 01, 02, 05, 06)
   c. Supai Tunnel Pipeline (Volumes 03, 06)
   d. Roaring Springs Cave Pipeline (Volume 04)
   e. North Rim Water Treatment Plant Site local water distribution improvements (Volume 06)

B. Related Sections include but are not necessarily limited to:

1. Division 00 – Bidding Requirements, Contract Forms, and Conditions of the Contract.
2. Division 01 – General Requirements.
3. Section 01 61 03 – Equipment: Basic Requirements.
4. Section 33 11 13 – Water Main Construction
5. Section 40 05 00 – Pipe and Pipe Fittings – Basic Requirements.

1.2 QUALITY ASSURANCE

A. Referenced Standards:

1. American National Standards Institute (ANSI)/American Water Works Association (AWWA):
   a. ANSI/AWWA B300 – Hypochlorites.
   b. ANSI/AWWA B301 – Liquid Chlorine.
   c. ANSI/AWWA C651 – Disinfection Water Mains.
   d. ANSI/AWWA C653 – Disinfection of Water Treatment Plants.
2. American National Standards Institute (ANSI)/National Sanitation Foundation (NSF):
   a. ANSI/NSF 60 - Drinking water treatment chemicals-Health effects.
   b. ANSI/NSF 61 - Drinking water system components-Health effects.
3. American Public Health Association (APHA)/American Water Works Association (AWWA)/Water Environment Federation (WEF):
   a. APHA/AWWA/WEF - Standard Methods for the Examination of Water and Wastewater.

B. Qualifications:

1. Provide qualified person to supervise use of liquid chlorine as defined in Part 2.

1.3 SUBMITTALS

A. See Section 01 33 23.

B. Shop Drawings:

1. Product technical data including:
   a. Acknowledgement that products submitted meet requirements of standards referenced.
   b. Product data for disinfectants to be used.

C. Miscellaneous Submittals

1. Qualifications of supervising personnel for use of liquid chlorine.
2. Disinfection Plan:
   a. Accompany Contractor's Facility Start-up Plan as required in Section 01 75 00.
   b. Include the following:
      1) Schedule for activities.
      2) Procedure and plan for cleaning and flushing system.
      3) Procedure and plan for disinfection and verification testing.
      4) Proposed locations where samples are to be taken.
      5) Proposed sampling intervals.
      6) Schedule of samples to be collected by Contractor.
      7) Type of disinfecting solution and method of preparation.
      8) Method of neutralizing and disposal of highly chlorinated water.
   c. Certified bacteriological verification test results.

1.4 SEQUENCING AND SCHEDULING

A. See Section 01 75 00 "System Start-up" for requirements regarding sequencing of disinfection work with facility start-up.

B. Commence disinfection after completion of the following:

1. Completion and acceptance of internal coatings systems.
2. Hydrostatic and pneumatic testing, pressure testing, functional and performance testing and acceptance of pipelines, pumping systems, structures, and equipment.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Water for Disinfection:

1. Potable water furnished by Government, when available by geographic location.

B. Equipment:

1. Furnish chemicals and equipment, such as pumps and hoses, to accomplish disinfection.
2. Provide protection as required by AWWA for cross-connection to previously disinfected sources.

C. Disinfectants:

1. Liquid Chlorine.
   a. Conforming to requirements of ANSI/AWWA B301.
   b. Certified for potable water application per ANSI/NSF 60 or ANSI/NSF 61 as applicable.
   c. May be used only if following conditions are met:
      1) Used in combination with appropriate gas-flow chlorination equipment to provide controlled high-concentration solution feed to the water to the water to be chlorinated.
      2) Used under direct supervision of a person familiar and experienced physiological, chemical, physical properties of liquid chlorine and who is trained and equipped to handle any emergency that may arise:
         a) Contracting Officer (CO) must approve qualifications of supervising person designated by Contractor.
      3) When appropriate safety practices are observed to protect working personnel and the public.
2. Sodium Hypochlorite
   a. Conforming to requirements of ANSI/AWWA B300.
   b. Certified for potable water application per ANSI/NSF 60 or ANSI/NSF 61 as applicable.
3. Calcium Hypochlorite:
   a. Conforming to requirements of ANSI/AWWA B300.
   b. Certified for potable water application per ANSI/NSF 60 or ANSI/NSF 61 as applicable.
   c. Sequestered calcium hypochlorite intended for swimming pool disinfection may not be used.

PART 3 - EXECUTION

3.1 GENERAL

A. All facilities covered by this section shall be protected, cleaned, and flushed in accordance with the requirements herein. The specific facilities to be disinfected are also listed herein.

B. Protection During Construction Period:

1. Observe Preventive and Corrective Measures During Construction as defined in ANSI/AWWA C651.
2. Keep pipe clean and dry during storage and installation.
3. Protect pipe during wet-trench installation and provide protection from flooding or storm events.
4. In the event of contamination, clean and swab pipe in accordance with ANSI/AWWA C651.

C. Disinfection procedures shall conform to ANSI/AWWA, EPA Guidance Manual, and this Specification:

1. Disinfect surfaces of materials that will contact finished water, both during and following construction, using one of the methods specified in this section.
2. Take care to avoid recontamination following disinfection.
3. Allow freshwater and disinfectant solution to flow into pipe or vessel at a measured rate so that chlorinated water is mixed and at a consistent concentration meeting or exceeding the required solution strength.
4. Do not place concentrated commercial disinfectant in pipeline or other facilities to be disinfected before it is filled with water.

D. Facilities to be Cleaned, Flushed, and Disinfected:

1. Roaring Springs Pumphouse (Volume 05):
   a. Interior process piping and tankage associated with finished water back feed system.
2. North Rim WTP (Volume 06):
   a. Interior process piping and tankage located downstream of the UF membrane skids.
   b. Chemical feed systems and piping, except “HYPO”.
      1) Non-potable water supplying water to chemical feed systems.
   c. Existing Finished Water Storage Tanks.

3.2 PREPARATION

A. Cleaning and Flushing for all Facilities:

1. Thoroughly clean and flush piping systems including supply, source and any appurtenant devices before performing disinfection.
2. Cleaning agents used shall not contain hazardous substances or deleterious compounds that would cause a violation of water quality standards or cause health effects is subsequently introduced into the water supply during any disinfection or filling operations.
3. Clean piping in accordance with requirements of Section 40 05 00.

B. Cleaning and Flushing of Piping and In-line Equipment.

1. Flush all foreign matter from pipe in accordance with ANSI/AWWA C651.
2. Provide hoses, temporary connections, ditches, and other conduits as necessary to dispose of flushing water without damage to adjacent structures or terrain.
3. Use water suitable for disinfection.
4. Flush service connections and hydrants. Flush distribution lines prior to flushing hydrants and service connections.
5. Operate valves during flushing process at least twice during each flush.

C. Cleaning of Tanks:

1. Remove all materials not part of the operating facilities including temporary works, tools, and debris.
2. Remove all water, dirt, paint chips, sediment, or foreign material by rinsing, vacuuming, or other removal techniques.
3. Thoroughly clean walls, floors, and attached structures with high-pressure water jet and by sweeping, scrubbing, or other similar means.
4. Cleaning shall:
   a. Remove all deposits of foreign nature.
   b. Remove biological growths.

c. Clean the slopes, walls, tops, and bottom.
d. Avoid damage to the structure.
e. Remove and avoid pollution or oil deposits by workers and equipment.

3.3 DISINFECTION

A. Piping and In-Line Equipment:

1. Applies to piping and inline equipment such as pumps and valves that are not covered under other disinfection provisions.
2. Disinfect in accordance with ANSI/AWWA C651.
3. Utilize any of the three disinfection procedures.
   a. Tablet Method.
   b. Continuous Feed Method.
   c. Slug Method.
4. Provide signage and tagging at all outlets from the piping being disinfected to prevent discharge of highly chlorinated water.
5. After applicable retention period, flush piping at a velocity of not less than 2.5 feet per second:
   a. Flush water shall be potable water.

B. Water Treatment Plant Facilities:

1. Disinfect in accordance with ANSI/AWWA C653.
2. Utilize one of the two disinfection procedures outlined in ANSI/AWWA C653:
   a. Disinfection Procedure.
   b. Alternate Disinfection Procedure.
3. Provide signage and tagging at all outlets from areas being disinfected to prevent discharge of highly chlorinated water.
4. After applicable retention period and testing of residual, flush tank or reservoir with potable water with a minimum free chlorine residual of 0.5 milligram per liter to remove heavily chlorinated water.

3.4 DISPOSAL OF FLUSHING AND DISINFECTION WATER

A. All water used for cleaning, flushing and disinfecting of tanks, piping and similar shall be disposed of in accordance with all federal, state and local codes and regulations. Cleaning and flushing water shall not be discharged into any portions of the water treatment system without prior approval and coordination with Contracting Officer.

B. The disposal location for disinfection water is anticipated to be:

1. Roaring Springs Pumphouse (Volume 05): Surface discharge.
2. North Rim WTP (Volume 06): Sanitary sewer.

C. Heavily chlorinated water must be dechlorinated in accordance with ANSI/AWWA C651, C652, and C653 prior to release:

1. See appendix of ANSI/AWWA standards for additional information.

3.5 VERIFICATION TESTING

A. Upon completion of flushing, provide verification in the form of bacteriological sampling meeting the requirements of applicable ANSI/AWWA standard.

B. Collection of Samples:

1. The Contractor shall collect samples and send to independent laboratory for analysis.
2. Coordinate activities to allow samples to be taken in accordance with this Section.
3. Provide valves at sampling points.
4. Provide access to sampling points.

C. Testing Equipment:

1. Clean containers, equipment, and connections used in sampling to make sure they are free of contamination.
2. The Contractor to obtain sampling bottles with instructions for handling from laboratory.

D. Chlorine Sampling and Analysis:

1. Collect samples in accordance with applicable ANSI/AWWA standard.
2. Samples of Disinfecting Solution:
   a. One sample per batch of disinfecting solution mixed and injected into pipe or vessel.
   b. If mixed solution not used, sample structure or pipe being disinfected during or immediately after filling.
3. Free Chlorine Residual Samples:
   a. As required to establish concentrations at the beginning and end of retention period.
4. Sampling Locations and Intervals:
   a. Sampling points shall be representative and accepted by CO.
5. Laboratory analysis to be performed by independent laboratory. Samples will be analyzed for disinfectant residual concentration.
6. If chlorine concentration testing results in disinfection concentrations not meeting the required standard, disinfecting procedures and verification testing shall be repeated until specified limits are met.

E. Bacteriological Sampling and Analysis:

1. Collect samples in accordance with applicable ANSI/AWWA standard.
2. Sampling Locations and Intervals:
   a. In accordance with applicable ANSI/AWWA Standard.
   b. Sampling points shall be representative and accepted by CO.
   c. If ANSI/AWWA Standard is not applicable or does not fully describe sampling procedure, utilize the following minimum requirements:
      1) A minimum of two (2) samples on two (2) consecutive days from each separable structure and every 1,000 feet of pipeline.
3. Laboratory analysis to be performed by independent laboratory certified by the Arizona Department of Environmental Quality. Samples will be analyzed for disinfectant residual and coliform concentrations using methods as described in the latest edition of Standard Methods for Examination of Water and Wastewater.

4. If verification testing results in bacterially positive samples or disinfection concentrations not meeting the required standard, disinfecting procedures and verification testing shall be repeated until specified limits are met.

F. Documentation:

1. Contractor shall communicate with regulatory authority (ADEQ) for approval of sampling as part of approval to put the tested system/component into service.
2. Contractor shall communicate and coordinate with owner utilities operator on ADEQ approvals.
3. Contractor shall conduct final communication to the Contracting Officer with documentation of ADEQ approvals.

G. Cost of Testing:

1. Contractor shall pay all testing expense associated with verification testing.

END OF SECTION 01 74 15

This page intentionally left blank.

SECTION 01 74 19 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for:

1. Salvaging nonhazardous demolition and construction waste.
2. Recycling nonhazardous demolition and construction waste.
3. Disposing of nonhazardous demolition and construction waste.

1.2 DEFINITIONS

A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.

B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.

C. Solid Waste: Garbage, debris, sludge, or other discharged material (except hazardous waste) including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations.

D. Debris: Non-hazardous solid waste generated during construction, demolition, or renovation of a structure which exceeds 2.5 inch (60 millimeter) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if mixture is comprised primarily of debris by volume, based on visual inspection.

E. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.

F. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of environment for aesthetic, cultural, or historical purposes.

G. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

H. Hazardous Materials: Material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.

I. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.

J. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.

K. Single-use plastic products: Plastic items intended to be disposed of immediately after use, including plastic and polystyrene food and beverage containers, bottles, straws, cups, cutlery, and disposable plastic bags.

1.3 PERFORMANCE REQUIREMENTS

A. General: Project shall minimize creation of construction, deconstruction, and demolition waste to protect and restore natural habitat and resources. Minimize factors contributing to waste such as over packaging, improper storage, ordering error, poor planning, breakage, mishandling, single-use plastic products, and contamination. A Waste Management Plan shall be developed to ensure that existing site and building materials are reused, salvaged, or recycled. Minimize waste disposal in landfills.

B. Material Transport: Due to the project location remoteness and required use of aircraft for material transport, the Contractor shall minimize the transport of unnecessary product packaging and other non-project materials to the site.

C. Salvage /Recycle Requirements: Develop waste management plan resulting in end-of-Project rates for salvage/recycling of 50 percent by weight of total waste generated by the Work. The following waste categories, at a minimum, shall be diverted from a landfill:

1. Land clearing debris (chipped debris can be used on site for mulch or erosion control)
2. Clean dimensional wood, palettes
3. Plywood, OSB (oriented strand board), and particle board
4. Concrete (can be ground and used for fill on site)
5. Asphaltic concrete (can be ground and used for fill on site)
6. Cardboard, paper, packaging, newsprint
7. Metals (from banding, stud trim, piping, rebar, roofing, other trim, steel, iron, galvanized sheet steel, stainless steel, aluminum, copper, zinc, lead, brass, and bronze)
8. Gypsum drywall - unpainted
9. Non-hazardous paint and paint cans
10. Beverage containers: Aluminum, glass, and plastic containers
11. Insulation
12. Ceiling grid and tiles
13. Ductwork
14. Wiring
15. Other mixed construction and demolition waste as appropriate.

D. If waste materials encountered during deconstruction/demolition or construction phase are found to contain lead, asbestos, polychlorinated biphenyls (PCBs), (such as fluorescent lamp ballasts), or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.

E. Existing items and material to be removed during deconstruction/demolition phase shall be reused in construction phase of the Project. Items that cannot be reused shall be recycled. Items considered for reuse must be in refurbishable condition and must meet quality standards set forth in these specifications. Contractor shall ensure quality of the item(s) in question will meet or exceed accepted industry or trade standards for first quality commercial grade application. During construction, deconstruction, or demolition Contracting Officer (CO) may designate other objects or materials for reuse.

1.4 SUBMITTALS

A. Waste Management Plan: After award of contract and prior to scheduled Pre-Construction Conference, Contractor shall submit a draft Waste Management Plan to Contracting Officer for approval. Submit 3 copies of plan. Revise and resubmit Plan as required by Contracting Officer. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.

B. Progress Documentation: Supplemental to Waste Management Plan, document solid waste disposal, diversion, and cost/revenue analysis and submit completed worksheet on a monthly basis. See Project Waste Management Plan Worksheet Sample, attached to the end of the Division 1 Specifications, and report totals to date for column headings.

C. Waste Reduction Calculations: Before request for Substantial Completion, submit three copies of calculated end-of-Project rates for salvage, recycling, and disposal as a percentage of total waste generated by the Work.

D. Records of Donations: Indicate receipt and acceptance of salvageable waste donated to individuals and organizations. Indicate whether organization is tax exempt.

E. Records of Sales: Indicate receipt and acceptance of salvageable waste sold to individuals and organizations. Indicate whether organization is tax exempt.

F. Recycling and Processing Facility Records: Indicate receipt and acceptance of recyclable waste by recycling and processing facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

G. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

H. Qualification Data: For Waste Management Coordinator and refrigerant recovery technician.

I. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating refrigerant that was present was recovered and recovery was performed according to Environmental Protection Agency (EPA) regulations. Include name and address of technician and date refrigerant was recovered.

J. Progress payment requirements:

1. With each Application for payment, submit an updated Project Waste Management Plan worksheet for solid waste disposal and diversion.
2. With each Application for Payment, submit manifests, weight tickets, receipts, and invoices specifically identifying the Project and waste material.

K. Closeout Submittals

1. With Closeout Submittals, submit a summary of a Project Waste Management Plan worksheet for solid waste disposal and diversion.

1.5 QUALITY ASSURANCE

A. Waste Management Coordinator Qualifications: Experienced firm, with record of successful waste management coordination of projects with similar requirements.

B. Refrigerant Recovery Technician Qualifications: Certified by EPA-approved certification program.

C. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.

D. Waste Management Meeting: Conduct separate meeting or cover in Pre-Construction Conference and comply with requirements in Section 01 31 00 "Project Management and Coordination." Review methods and procedures related to waste management including:

1. Review and discuss waste management plan including responsibilities of Waste Management Coordinator.
2. Review requirements for documenting quantities of each type of waste and its disposition.
3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
5. Review procedures for reduction of single-use plastic products on site.
6. Review waste management requirements for each trade.

PART 2 - PRODUCTS

2.1 WASTE MANAGEMENT PLAN

A. General: Develop plan consisting of waste identification and waste reduction work plan. Include separate sections in plan for demolition and construction waste. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.

B. Waste Identification: Indicate anticipated types and quantities of demolition, site-clearing and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.

C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.

1. Salvaged Materials for Reuse: For materials salvaged and reused in Project, describe methods for preparing salvaged materials before incorporation into the Work.
2. Salvaged Materials for Sale: For materials sold to individuals and organizations, include list of names, addresses, and telephone numbers.
3. Reference 01 11 00 Summary of Work for tree removal requirements.
4. Helicopter use for construction debris and human waste should to the best of the contractors ability make use of back flights for haul out.
5. Salvaged Materials for Donation: For materials donated to individuals and organizations, include list of names, addresses, and telephone numbers.
6. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
7. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
8. Handling and Transportation Procedures: Include method used for separating recyclable waste including sizes of containers, container labeling, and designated location on Project site where materials separation will be located.

9. Reduction of single-use plastic products: Include methods for water distribution to include bottle refilling stations if available for personnel.

D. Cost/Revenue Analysis: Indicate total cost of waste disposal as if there was no waste management plan and net additional cost or net savings resulting from implementing waste management plan. Include:

1. Landfill tip fees per ton.
2. If diverted, tip fee savings from landfill diversion.
3. Costs of recycling, salvage, or reuse.
4. Revenue from recycling, salvage, or reuse.
5. Total cost or savings from diversion. (Calculate by using tip fee savings and subtracting costs of recycling or adding revenue from recycling.)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

A. General: Implement waste management plan as approved by Contracting Officer. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during entire duration of Contract.

B. Waste Management Coordinator: Engage waste management coordinator responsible for implementing, monitoring, and reporting status of waste management work plan. Coordinator shall be present at Project site full time for duration of Project.

C. Contractor shall establish contacts with local recycling and reuse companies to set up lines of responsibility. Contractor shall be responsible for coordination in terms of identifying materials, pickup schedules, and standard quality for recycled materials.

D. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.

1. Distribute waste management plan to everyone concerned within three days of submittal return.
2. Distribute waste management plan to entities when they begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.

E. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

F. Separation facilities:

1. Contractor shall designate and Contracting Officer shall approve specific area or areas to facilitate separation of materials for potential reuse, salvage, recycling, and return.
2. Place waste and recycling bins near each other, and close to point of waste generation but out of traffic pattern.
3. Keep recycling and waste bin areas neat, clean, and clearly marked in order to avoid co-mingling of materials.
4. Protect bins during non-working hours from off-site contamination.
5. Check garbage dumpsters periodically for recyclables being thrown away and undocumented materials that could be recycled.

G. Materials handling procedures: Material to be recycled shall be protected from contamination and shall be handled, stored, and transported in a manner that meets requirements set by designated facilities for acceptance. Establish defined area for operations of each trade, especially woodcutting so off-cuts are kept in one area and can be sorted by dimension for future reuse.

3.2 SALVAGING DEMOLITION WASTE

A. Salvaged Items for Reuse in the Work:

1. Clean salvaged items.
2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in secure area until installation.
4. Protect items from damage during transport and storage.
5. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.

B. Salvaged Items for Sale and Donation: Not permitted on Project site.

C. Salvaged Items for Governments Use:

1. Clean salvaged items.
2. Pack or crate items after cleaning. Identify contents of containers.
3. Allow for inspection if necessary.
4. Store items in secure area until delivery to Government.
5. Transport items to storage area designated by Government.
6. Protect items from damage during transport and storage.

3.3 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

A. General: Recycle paper and beverage containers used by on-site workers.

B. Recycling Incentives: Revenues, savings, rebates, tax credits, and other incentives received for recycling waste materials shall accrue to Contractor.

C. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to maximum extent practical.

1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.
   a. Inspect containers and bins for contamination and remove contaminated materials if found.
2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
4. Store components off ground and protect from weather.
5. Remove recyclable waste off Government property and transport to recycling receiver or processor.

3.4 RECYCLING DEMOLITION WASTE

A. Asphaltic Concrete Paving: Break up and transport paving to asphalt-recycling facility.

B. Concrete: Remove reinforcement and other metals from concrete and sort with other metals.

1. Pulverize concrete to maximum 4 inch size.

C. Masonry: Remove metal reinforcement, anchors, and ties from masonry and sort with other metals.

1. Clean and stack undamaged, whole masonry units on wood pallets.

D. Wood Materials: Sort and stack members according to size, type, and length. Separate lumber, engineered wood products, panel products, and treated wood materials.

E. Metals: Separate metals by type.

1. Structural Steel: Stack members according to size, type of member, and length.
2. Remove and dispose of bolts, nuts, washers, and other rough hardware.

F. Asphalt Shingle Roofing: Separate organic and glass-fiber asphalt shingles and felts. Remove and dispose of nails, staples, and accessories.

G. Gypsum Board: Stack large clean pieces on wood pallets and store in a dry location. Remove edge trim and sort with other metals. Remove and dispose of fasteners.

H. Acoustical Ceiling Panels and Tile: Stack large clean pieces on wood pallets and store in dry location.

1. Separate suspension system, trim, and other metals from panels and tile and sort with other metals.

I. Carpet: Roll large pieces tightly after removing debris, trash, adhesive, and tack strips.

1. Store clean, dry carpet in closed container or trailer provided by Carpet Reclamation Agency or carpet recycler.

J. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.

K. Plumbing Fixtures: Separate by type and size.

L. Piping: Reduce piping to straight lengths and store by type and size. Separate supports, hangers, valves, sprinklers, and other components by type and size.

M. Lighting Fixtures: Separate lamps by type and protect from breakage.

N. Electrical Devices: Separate switches, receptacles, switchgear, transformers, meters, panelboards, circuit breakers, and other devices by type.

O. Conduit: Reduce conduit to straight lengths and store by type and size.

P. Electronic Products: Ensure non-usable electronic products are reused, donated, sold, or recycled using environmentally sound management practices at end of life.

3.5 RECYCLING CONSTRUCTION WASTE

A. Packaging:

1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in dry location.
2. Polystyrene Packaging: Separate and bag materials.
3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.

B. Site-Clearing Wastes: Chip brush, branches, and trees on-site.

C. Wood Materials:

1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
2. Clean Sawdust: Bag sawdust not containing painted or treated wood.

D. Gypsum Board: Stack large clean pieces on wood pallets and store in dry location.

1. Clean Gypsum Board: Grind scraps of clean gypsum board using small mobile chipper or hammer mill. Screen out paper after grinding.

3.6 DISPOSAL OF WASTE

A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose in landfill or incinerator acceptable to authorities having jurisdiction.

1. Except as otherwise specified, do not allow waste materials to accumulate on-site.
2. Remove and transport debris in manner preventing spillage on adjacent surfaces and areas.

B. Burning: Do not burn waste materials.

C. Disposal: Transport waste materials off Government property and legally dispose of them.

END OF SECTION 01 74 19

SECTION 01 75 00 - CHECKOUT AND START-UP PROCEDURES FOR WATER TREATMENT FACILITIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Administrative and procedural requirements for checkout and startup of equipment, systems, and facilities for the Volume 6 North Rim Water Treatment. The work described in this section may not proceed until the requirements of Section 01 91 14 – Total Building Commissioning have been approved by the Contracting Officer (CO).

B. Scope:

1. Contractor shall initially check out, start up, and place water-treatment related equipment and systems installed under the Contract into successful operation without discharging to the distribution system, in accordance with the material and equipment manufacturers' written instructions, Suppliers' recommendations at the Site, and the Contract Documents.
2. Provide the following:
   a. All labor, tools, materials, and equipment required to complete equipment and system checkout and startup.
   b. Chemicals, lubricants, and other required operating fluids necessary for checkout, startup, and initial operation of the Work.
   c. Filters and other temporary or consumable items necessary for checkout, startup, and initial operation of the Work.
   d. Fuel, electricity, water, and other temporary utilities and temporary facilities necessary for checkout and startup of equipment and systems, unless otherwise specified.
   e. Manufacturer's field services.
3. The General Conditions, as may be modified by the Supplementary Conditions, and Section 01 77 00 - Closeout Requirements, address requirements for documenting Substantial Completion.
   a. Provide on-site technical representative from supplier of the UF membrane system for five days a week, eight hour days Monday thru Friday for the first six (6) months, starting from the AOC approval to adjust and fine tune the WTP water processing system/equipment, working with the Government's operators to optimize the WTP and maintain manufacturer's warranty requirements.

C. Related Sections include but are not necessarily limited to:

1. Section 01 61 03 – Equipment - Basic Requirements.
2. Section 01 74 15 – Disinfection of Facilities.
3. Section 01 77 00 – Closeout Requirements.
4. Section 01 78 23 – Operation and Maintenance Data
5. Section 01 79 00 – Demonstration and Training.
6. Section 40 61 13 – Process Control Systems - General Requirements.
7. Section 46 61 33 – Ultrafiltration Equipment.

D. See Section 01 91 14 – Total Building Commissioning (for commissioning requirements of building systems such as electrical, HVAC, and fire supply).

1.2 ADMINISTRATIVE REQUIREMENTS

A. Coordination:

1. Do not start up equipment or system(s) for continuous operation until all components of that equipment item or system, including instrumentation and controls, have been tested to the extent practicable and proven to be operable as intended by the Contract Documents.
2. Contractor shall be responsible for providing sufficient personnel for startup activities. Government may provide operations staff to observe startup activities and participate in process decisions, but they should not be relied upon to actively participate.
3. Suppliers of specific equipment or systems shall be present during checkout, startup, and initial operation of, unless otherwise acceptable to the CO or otherwise required by the Contract Documents.
4. Do not start up equipment and system(s), without submitting acceptable preliminary operations and maintenance manuals for equipment in accordance with the Contract Documents.
5. For startup of heating equipment, air conditioning equipment, and other equipment and systems that provide temperature control see Section 01 91 14 – Total Building Commissioning.

B. Checkout and Startup Planning Meeting:

1. Contractor, with appropriate Subcontractors and Suppliers, shall attend and participate in a meeting with CO to discuss planning, scheduling, and coordination of checkout and startup activities.
2. Upon mutual concurrence of CO and Contractor, meeting may be concurrent with the Training Scheduling Conference required in Section 01 79 00 – Demonstration and Training.
3. Meeting shall be held not less than 45 days prior to first scheduled training session for the equipment and system(s) to be checked out and started-up.
4. Attend meeting prepared to knowledgably and effectively discuss:
   a. Status of the Work and schedule-to-complete for requirements prerequisite to checkout and startup.
   b. Schedule for and status of training required for each equipment item and system.
   c. Schedule for checkout, startup, and field quality control activities for the subject Work.
   d. Status and quantities of required consumables, lubricants, and utility services necessary for checkout and startup.
5. Meeting will be chaired by the Contracting Officer Representative (COR) and/or CMR. The COR/CMR will prepare and distribute a record of topics discussed and decisions made during the meeting.
   a. If meeting is concurrent with the Training Scheduling Conference required under Section 01 79 00 - Demonstration and Training, Contractor shall chair and prepare minutes of the training scheduling planning portion of the meeting and furnish its draft minutes to COR/CMR to incorporate into the overall minutes.
6. Comply with decisions made at the meeting and the Contract Documents.

C. Sequencing:

1. Comply with Section 01 11 00 - Summary of Work, regarding phasing of the Work and allowable shutdowns.

D. Scheduling:

1. Progress Schedule:
   a. Clearly indicate in the Progress Schedule planned and actual dates for checkout, startup, and field quality control activities, including all demonstration testing activities addressed in this Specifications section and elsewhere in the Contract Documents. Separately indicate checkout, startup, and field quality control activities for each equipment item and system.
   b. Perform startup and field quality control activities on the associated, scheduled dates, unless otherwise acceptable to the CO.
2. Restrictions for Scheduling:
   a. Checkout of materials, equipment, and systems by Contractor that do not involve or require Government personnel may be performed at any time during normal working hours. Where required by the Contract Documents or requested by CO, perform checkout in the presence of CO or their designated representative.
   b. Startup, including initial operation of materials, equipment, and systems, shall not be initiated on: Monday, Friday, Saturday, Sunday, Government holidays, the day immediately prior to a holiday, or the day immediately following a holiday, unless otherwise acceptable to the CO.
   c. Unless otherwise indicated in the Contract Documents or acceptable to the CO, perform all startup during normal working hours of the day shift.
   d. To the extent practicable, where extended-duration startup or field quality control activities are required by the Contract, avoid having such activities extend into evening, night, weekend, or holiday hours.
   e. The CO reserves the right to require a minimum seven days' notice of rescheduled startup when Contractor cannot perform the associated activities as scheduled.
3. Operation and Maintenance Data:
   a. Comply with Section 01 78 23 - Operation and Maintenance Data.
   b. A preliminary copy of all operation and maintenance manuals shall be received by the CO prior to the start of the Demonstration period.
4. Training:
   a. Comply with Section 01 79 00 - Demonstration and Training.
   b. Throughout the Demonstration Period, provide knowledgeable personnel to answer Government's questions, provide final field instruction on select systems (where appropriate) and to respond to problems or failures of the Work.
5. Spare Parts, Tools, and Extra Materials.
   a. Comply with Technical Specifications (Divisions 2 - 46) for furnishing spare parts, tools, and extra materials to Government.
   b. Deliver to the CO all required spare parts, tools, and extra materials prior to commencing the demonstration period, unless earlier delivery is required elsewhere in the Contract Documents.
      1) Provide documentation of CO's receipt of spare parts, tools, and extra materials.

E. Demonstration Plan:

1. Prior to preparing the plan, Contractor shall coordinate a meeting with CO to discuss system demands and limitations for the Demonstration Period with Park staff.
2. Following the meeting, Contractor shall prepare a Draft Demonstration Plan that includes the following information at a minimum:

a. Schedule break-down by day describing mode of operation (discharge to sewer, discharge to surface, or recycle) and associated WTP production rates.
b. Details for temporary bypass piping including size, material, and layout.
c. Description of any required Discharge Permits.
d. Details for discharge flow metering and dechlorination system if surface discharge option is selected.
e. Plan to monitor, collect, and submit required water quality data during Demonstration Period.
f. Data collection and reporting log for sampling, analytical data, and data to be obtained by manually recording data from field or panel indicators for each WTP.
g. Proposed certified lab for sampling analysis. (Note that Legend Analytical in Phoenix, AZ is the Park's preferred certified lab.)
h. List of Contractor's staff and contact information during the Demonstration Period.

3. Submit Draft Demonstration Plan to CO for review. Allow for a two week review period.
4. Address Government comments and submit Final Demonstration Plan.

1.3 QUALITY ASSURANCE

A. Regulatory Requirements:

1. Activities described in this Specification shall be conducted without sending water to the distribution system. At the conclusion of Demonstration, Contractor shall submit and obtain an Interim Approval of Construction (AOC) or Approval to Commence Operations from ADEQ. Treated water shall not be discharged to the distribution system until the AOC is obtained. The AOC shall contain the following:
   a. Stamped as-built drawings for the WTP.
   b. Water quality sampling results as defined in this Specification.
   c. O&M Manual for the WTP operating as a whole.
      1) Prepared by the Government and provided to the Contractor.
      2) O&M Manuals for specific equipment will be included as appendices to WTP O&M Manual.
2. If the Contractor elects to use a surface discharge during Startup and Demonstration, then Contractor shall obtain any necessary Discharge Permits from ADEQ.

1.4 DEFINITIONS

A. The following defined terms are used in this Specifications Section:

1. Instrumentation Supplier: Entity retained by Contractor, Subcontractor, or Supplier to furnish instrumentation or controls that will be part of the completed Work, including manufacturers, manufacturer representatives, wholesalers, retailers, and others, including entities retained to perform hardware systems integration Work.
2. Project Classified System (PCS): An established, distinct part of the Project, consisting of an arrangement of items, such as equipment, structures, components, piping, cabling, materials, and incidentals, so related or connected to form an identifiable, unified, functional, operational, safe, and independent system. PCSs are specifically indicated in this Specifications section.
3. Pre-Demonstration Period: The period of time, of unspecified duration after initial construction and installation activities during which Contractor, with assistance from manufacturer's representatives, performs in the following sequence:

a. Finishing type construction work to ensure each PCS has reached a state of substantial completion.
b. Equipment start-up.
c. Personnel training.

4. Demonstration Period: A period of time, of specified duration, following the Pre-Demonstration Period, during which the Contractor initiates process flow through the Project Classified Systems as indicated below and starts up and operates the PCSs, without exceeding specified downtime limitations, to prove the functional integrity of the mechanical and electrical equipment and components and the control interfaces of the respective equipment and components comprising the PCSs as evidence of Substantial Completion.
   a. As indicated below, multiple PCSs are grouped together to form a Demonstration Period phase.
5. Substantial Completion: See Government website.
6. Beneficial Occupancy (aka Beneficial Use): See Government website.
7. Government website: www.nps.gov/dscw/definitionsdc_**α**.htm
   a. Insert the first letter of the item to be defined for "α".

1.5 SUBMITTALS

A. Action Submittals: Submit the following for each WTP:

1. Data collection and reporting log for each required Demonstration Period.
2. Draft and Final Demonstration Plan.

B. Informational Submittals: Submit the following:

1. Progress Schedules indicating dates for checkout, startup, and field quality control activates.
2. Completed checkout and startup log required in Paragraph 3.2.C of this Specifications section.
3. Manufacturer's installation check letters (also known as Manufacturer's Field Services Report or Certificate of Proper Installation) required in Paragraph 3.2.C of this Specifications section.
4. Instrumentation Supplier's Instrumentation Installation Certificate.
5. Letter verifying completion of all Pre-Demonstration startup activities.
6. Report of data collected during each required Demonstration Period.
7. Qualifications Statements:
   a. Qualifications, including resume' and copy of license, of Contractor-retained licensed operator.

1.6 PROJECT CLASSIFIED SYSTEMS

A. Project Classified Systems (PCS's) at the North Rim Water Treatment are established as follows:

1. PCS #1 – Raw Water:
   a. Raw water tanks: TK-62011 and TK-62012.
   b. Raw water flow meter and transmitter: FE/FIT-62071.
   c. Raw water flow control valve: FCV-62033.
   d. Raw water pump: PMP-62011
   e. Raw water pump flow control valve: FCV-62031.
   f. Raw water sample pump: PMP-62012.
   g. All associated water quality and liquid measurement instrumentation.

2. PCS #2 – Reaction Tanks:
    a. Reaction tanks: TK-63011 and TK-63012.
    b. Rapid Mixers: MX-63012 and MX-63013.
    c. Reaction tank waste valves: FV-63031 and FV-63032.
    d. All associated water quality and liquid measurement instrumentation.
3. PCS #3 – Ultrafiltration (UF) Membrane System (reference Veolia P&IDs)
    a. UF skids: Refer to Veolia P&IDs.
    b. Master control panel: LCP-94001
    c. Compressors: Refer to Veolia P&IDs.
    d. CIP tank: Refer to Veolia P&IDs.
    e. Caustic soda feed pump: Refer to Veolia P&IDs.
    f. Sodium hypochorite feed pump: Refer to Veolia P&IDs.
    g. Citric acid feed pump: Refer to Veolia P&IDs.
    h. Hydrochloric acid metering pump: Refer to Veolia P&IDs.
    i. Sodium bisulfite metering pump: Refer to Veolia P&IDs.
    j. UF influent sample pump: PMP-64011.
    k. Combined filtrate flow meter: FE/FIT-64071
    l. All associated water quality and liquid measurement instrumentation.
4. PCS #4 – Disinfection and Chlorine Contact Tank:
    a. Chlorine Contact (CT) Tank: TK-66011
    b. CT Tank influent sample pump: PMP-68117.
    c. All associated water quality instrumentation.
5. PCS #5 – Finished Water System:
    a. Finished water tank pumps: PMP-68111 and PMP-68112.
    b. Finished water tanks influent flow meter: FE/FIT-68173
    c. Finished water tanks: TK-68411 and TK-68412.
    d. Distribution water sample pump: PMP-66011.
    e. High demand pumps: PMP-68211, PMP-68212, and PMP-68213.
    f. Upper Zone booster pumps: PMP-68311, PMP-68312, and PMP-68313.
    g. Hydroneumatic tanks: TK-68133 and TK-68114:
    h. Upper Zone distribution flow meter: FE/FIT-68171.
    i. Lower Zone distribution flow meter: FE/FIT-68172.
    j. Combined backfeed flow meter: FE/FIT-68174.
    k. Backfeed to Supai Tunnel flow meter: FE/FIT-68175.
    l. All associated water quality and liquid measurement instrumentation.
6. PCS #6 – Coagulant feed system
    a. Metering pump: PMP-67111.
    b. All associated liquid measurement instrumentation.
7. PCS #7 – Caustic soda feed system
    a. Metering pump: PMP-67311.
    b. All associated liquid measurement instrumentation.
8. PCS #8 – Sodium hypochlorite feed system
    a. Metering pumps: PMP-67211, PMP-67212, PMP-67213, and PMP-67214.
    b. All associated liquid measurement instrumentation.
9. PCS #6 – Process waste tank system
    a. Process waste tank: TK-69111.
    b. All associated liquid measurement instrumentation.
10. PCS #7 - Plant SCADA System.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION

3.1 CHECKOUT AND STARTUP – GENERAL

A. Facility Startup Divided into Two Periods:

1. Pre-Demonstration Period.
2. Demonstration Period, including.

B. Provide all consumable items required for checkout and startup including, but not limited to: fuel, oil, grease, and chemicals.

1. Coordinate with Contracting Officer for delivery of chemicals to site.

C. Provide all temporary bypass piping, bypass valving, temporary pumping, plugs and caps, required to perform checkout and startup.

3.2 PRE-DEMONSTRATION PERIOD

A. Prior to the Pre-Demonstration Period:

1. Complete the Work to the point where it is ready for checkout and startup, including operation available in all manual, automatic, and other modes.
2. Obtain the CO's approval or acceptance (as applicable) of Submittals required prior to checkout and startup, including all Shop Drawings, Samples, source quality control (shop testing) submittals, preliminary operation and maintenance manuals, and other Submittals required by the Contract Documents, other than Submittals that cannot be furnished until after startup.

B. Checkout.

1. Comply with Section 01 61 03 - Equipment - Basic Requirements, including provisions concerning installation checks.

C. Startup:

1. Comply with requirements for startup of materials, equipment, and systems indicated in the associated Specification sections and elsewhere in the Contract Documents. Specific items of note include:
   a. The Functional Test and Initial Performance Test of UF membrane equipment as described in Section 46 61 33.
   b. Pipe pressure testing as described in Section 40 05 00.
2. Prepare the Work so it will operate properly and safely and be ready to demonstrate functional integrity during the Demonstration Period.
3. Perform startup to extent possible without introducing process flow.
4. Test tanks, pumping and similar equipment requiring a fluid, using clean water supplied by Government at Contractor's expense.
5. Test chemical metering pumps using intended product and calibrate pumps.
6. Dispose of water used for Equipment Start-up to sanitary sewer.
7. Introduce process flow to complete startup for all PCS systems except PCS #7.

8. Procedures include but are not necessarily limited to the following:
    a. Test or check and correct deficiencies of:
        1) Power, control, and monitoring circuits for continuity prior to connection to power source.
        2) Voltage of all circuits.
        3) Phase sequence.
        4) Cleanliness of connecting piping systems.
        5) Alignment of connected machinery.
        6) Vacuum and pressure of all closed systems.
        7) Lubrication.
        8) Valve orientation and position status for manual operating mode.
        9) Valve actuation for automatic valves.
        10) Tankage for integrity using clean water.
        11) Pumping equipment using clean water.
        12) Instrumentation and control signal generation, transmission, reception, and response.
            a) Comply with Section 40 61 13 - Process Control System General Requirements.
        13) Tagging and identification systems.
        14) Proper connections, alignment, calibration and adjustment.
    b. Calibrate safety equipment.
    c. Calibrate chemical metering pumps.
    d. Manually rotate or move moving parts to assure freedom of movement.
    e. “Bump-start” electric motors to verify proper rotation.
    f. Perform other tests, checks, and activities required to make the Work ready for Demonstration Period.
    g. Checkout and Startup Log:
        1) Prepare a log showing each equipment item and system requiring checkout and startup. Indicate in the log activities to be accomplished during checkout and startup.
        2) Provide a place for Contractor to record date and person performing required checkout and startup. Indicate associated date(s), personnel, and employer of each.
        3) Submit completed checkout and startup log to the COR/CMR and obtain the CO’s acceptance.
        4) Provide separate logs for each WTP.
9. Provide chemicals for Start-up and Demonstration including:
    a. Sodium hypochlorite: one 55 gallon drum and one 300 gallon tote.
    b. Caustic soda: two 55 gallon drums.
    c. Coagulant: one 55 gallon drum.
    d. Hydrochloric acid: one 55 gallon drum.
    e. Sodium bisulfite: one 55 gallon drum.
    f. Citric acid: one 55 gallon drum.
10. Obtain Suppliers’ certifications of the installed and operational Work, without restrictions, and submit to the CO:
    a. Manufacturer's installation check letters (sometimes referred to as Manufacturer’s Field Services Report or COPI).
    b. Instrumentation Supplier's Instrumentation Installation Certificate.
11. Submit letter verifying completion of all Pre-Demonstration startup activities including receipt of all specified items from Suppliers as final item prior to initiation of Demonstration Period.
    a. Submit separately for each WTP.
12. Complete training of operations and maintenance personnel.

3.3 DEMONSTRATION PERIOD

A. General:

1. Demonstrate the operation and performance of mechanical, electrical, instrumentation, and control interfaces of the Work undergoing the Demonstration Period, in accordance with the Contract Documents and without discharging water to the distribution system.
2. Duration of Demonstration: 120 consecutive hours (5 days) in full automatic mode.
   a. If the aggregate time used for repair, alteration, or unscheduled adjustments to any part of the Work that renders the affected Work inoperative or operation outside of recommended ranges exceeds the criteria for acceptance as defined below, the demonstration of operation and performance will be deemed unacceptable and Contractor shall provide appropriate adjustments and remedies and re-perform the Demonstration Test, at no additional cost to the Government, until acceptable results are obtained.
   b. Re-performance of the Demonstration Period shall comply with the same requirements as the original Demonstration Period.
3. Perform the demonstration of operation and performance of the Work under minimum and maximum net finished water production rates as defined by the Contract Drawings.
4. Contractor shall develop a Demonstration Plan using the potential modes of operation available to Contractor and outlined below, which allow for flow through the WTP without discharge to the distribution system.

B. Potential Modes of Operation:

1. North Rim WTP:
   a. Discharge to Sewer: Pending the time of year and associated sanitary sewer flow rates, the Contractor may be able to use temporary bypass piping to route flow from the discharge of the Finished Water Tank Pumps (downstream of the CT Tank) to the sanitary sewer system.
      1) Provide flow metering and dechlorination system.
      2) Coordinate with the CO on discharge flow limitations.
   b. Recycle: The Contractor has the option to recycle water (i.e. run water in a “loop”) using temporary bypass piping connected from the discharge of the Finished Water Tank Pumps to the raw water piping ahead of the Reaction Tanks (within the WTP).
      1) In Recycle mode, treated water is sent back through the treatment process. Therefore, it will be necessary for the Contractor to adjust and/or turn off specific chemical feed systems as the duration of Recycle increases.

C. Government’s Right to Simulate Scenarios:

1. Government reserves the right to simulate operational variables, equipment failures, routine maintenance scenarios, and similar actions and events during the Demonstration Period to verify the operation and performance of the Work in automatic, manual, and other types of operating modes, backup systems, and alternate operating modes.

D. Demonstration Period, Evaluation, and Acceptance:

1. Responsibilities for Sampling and Data Collection:
   a. Collect samples, preserve properly and transport to acceptable lab for analysis.

1) Complete one round of sampling prior to completion of Demonstration Periods.
2) Sample collection shall include:
a) Raw Water: Turbidity, pH, temperature, alkalinity, TOC, hardness, and TDS.
b) Entry-Point-to-Distribution System (EPDS): Turbidity, pH, temperature, alkalinity, TOC, hardness, TDS, and Bac-T.
(1) Located downstream of CT Tank
c) Online instruments measurements at EPDS:
(1) Turbidity.
(2) Chlorine residual.

2. Responsibilities for Data Reporting:
a. Submit data collected to the COR/CMR for evaluation of acceptability of results.
3. Data Evaluation:
a. The CO will evaluate the data collected during the Demonstration Period and other information obtained during the Demonstration Period for compliance with the Contract Documents.
b. The CO will advise Contractor in writing of whether the data and information obtained indicate that the Demonstration Period was successfully completed.
4. Criteria for Acceptance:
a. Downtime is less than 5% (6 hours), cumulative, in fully automatic mode.
b. Treatment facilities can treat maximum net finished water production rate without overflowing and minimum net finished water production rate without failure.
1) See Contract Drawings for max/min net finished water production rates.
c. Treatment facilities can produce water that meets all State and Environmental Protection Agency (EPA) Drinking Water Regulations.
d. At the satisfactory conclusion of the Demonstration Period, Government will take beneficial use of the WTP.

END OF SECTION 01 75 00

SECTION 01 77 00 - CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for contract closeout, including:

1. Project Record Drawings
2. Closeout Submittals
3. Substantial Completion and Final Inspection
4. Permit Closure and Transfer
5. Final Acceptance of the Work
6. Warranties

1.2 PROJECT RECORD DRAWINGS

A. Maintain one complete full-size set of contract drawings and one full-size set of vendor-supplied drawings. Clearly mark changes, deletions, and additions using National Park Service (NPS) drafting standards to show actual construction conditions. Show additions in red, deletions in green and special instructions in blue.

B. Keep record drawings current. Make record drawings available to Contracting Officer (CO) for inspection at the time of monthly progress payment requests. If project record drawings are not current, Contracting Officer may retain an appropriate amount of progress payment.

C. Submit complete record drawings on completion of total project. Include shop drawings, sketches, and additional drawings to be included in final set, with clear instructions showing the location of these drawings.

1. Record drawings shall be completed by Drawing Volume as provided in the Contract Drawings.

1.3 CLOSEOUT SUBMITTALS

A. A list of closeout requirements has been attached at the end of the Division 1 Specifications for your convenience. The intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list. Submit the following before requesting final inspection:

1. Specific warranties, guarantees, workmanship bonds, final certifications, and similar documents.
2. NPS required forms for occupancy, Fire Sprinkler/Alarm acceptance, and other similar forms or certificates.
3. Project Record Documents, operation and maintenance manuals, final completion construction digital images recorded on USBR Flash Drive with index and descriptions, and similar final record information.
4. Environmental Record Documents:

a. Final Summary of Solid Waste Disposal and Diversion: As specified in Section Insert Specification section 01 74 19 Construction Waste Management and Disposal.

5. Posted Operating Instructions: As specified in individual sections. Furnish operating instructions attached to or posted adjacent to equipment. Include wiring diagrams, control diagrams, control sequence, start-up, adjustment, operation, lubrication, shut-down, safety precautions, procedures in the event of equipment failure, and other items of instruction recommended by manufacturer.
6. Deliver tools, spare parts, extra materials, and similar items to location designated by Contracting Officer. Label with manufacturer's name and model number where applicable.
   a. Special Tools: One set of special tools required to operate, adjust, dismantle, or repair equipment. Special tools are those not normally found in possession of mechanics or maintenance personnel.
7. Keys and Keying Schedule: Submit keys including duplicates. Wire keys for each lock securely together. Tag and plainly mark with lock number, equipment identification, or panel or switch number, and indicate location, building, and room name or number.
8. Make final changeover of permanent locks and deliver keys to Contracting Officer. Advise Park personnel of changeover in security provisions.
9. Approved pre-functional checklists and functional performance testing reports from commissioning documentation.
   a. Equipment start-up requires coordination with commissioning process. Refer to Section 01 75 00 and 01 91 14. Equipment shall not be "temporarily" started for commissioning.
10. Test and balance report.
11. Terminate and remove temporary facilities, mockups, construction tools, and similar elements from Project site, complete final cleaning requirements, including touchup painting.
12. Touch up and repair and restore marred exposed finishes to eliminate visual defects.
13. Instruct NPS personnel in operation, adjustment, and maintenance of products, equipment, and systems.

1.4 FINAL INSPECTION, SUBSTANTIAL COMPLETION AND ACCEPTANCE PROCEDURES

A. Request final inspection in writing when project or designated portion of project is substantially complete. Contracting Officer will proceed with inspection within 10 days of receipt of written request or will advise Contractor of items that prevent project from being substantially complete.

B. If work is determined substantially complete, following final inspection, Contracting Officer will prepare Punch List and issue a Letter of Substantial Completion.

C. If work is not determined substantially complete following final inspection, Contracting Officer will notify Contractor in writing. Contractor shall request new final inspection after completing work. Re-inspection costs may be charged against Contractor in accordance with Inspection of Construction contract clause.

D. Contractor shall complete Punch List within 30 calendar days, documented weather permitting.

1. Prior to requesting final inspection:
   a. Complete commissioning requirements of Section 01 91 14, unless approved in writing by Contracting Officer.

E. If Contractor completes items of work on Punch List and contractually required items, Contracting Officer will issue Letter of final acceptance of work.

F. If Contractor fails to complete work within the time frame, Contracting Officer may correct work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with Inspection of Construction contract clause.

1.5 PERMIT CLOSURE AND TRANSFER

A. When work covered by the permits is complete, create list of tasks required to close or transfer permits to Park. Submit to Contracting Officer for approval.

B. After substantial completion and Punch List completion, permits shall be closed and documented by Agency(ies) with Jurisdiction for the permit.

C. If responsibility for permits is to be transferred to Park, Park shall be informed of permit provisions completed and responsibilities transferring to Park staff.

1.6 WARRANTIES

A. Contractor warranty on materials and workmanship:

1. Period of one year after substantial completion.
2. Provide one contractor warranty inspection during the warranty period and address any warranty issues.
3. Provide six (6) months of on-site technical shake down support for North Rim WTP starting ADEQ approval to operate to optimize the system configuration and operations of the equipment to maintain the warranty requirements of the manufacture, reference 01 75 00 (M-F 40hrs).

B. Submittal Time: Submit written warranties on request of Contracting Officer for designated portions of Work where commencement of warranties other than date of Substantial Completion is indicated.

C. Organize warranty documents into an orderly sequence based on table of contents of Project Manual.

1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2 by 11 inch paper.
2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify product or installation. Provide typed description of product or installation, including name of product and name, address, and telephone number of Installer.
3. Identify each binder on front and spine with typed or printed title "WARRANTIES," Project name, and name of Contractor.
4. Warranty Electronic File: Scan warranties and bonds and assemble complete warranty and bond submittal package into a single indexed electronic PDF (portable document format) file with links enabling navigation to each item. Provide bookmarked table of contents at beginning of document.

D. Provide additional copies of each warranty in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 MATERIALS

A. See Division 1 Specification Section "Execution" for information on cleaning agents.

PART 3 - EXECUTION

3.1 FINAL CLEANING

A. General: Conduct final cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:
   a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
   b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
   c. Rake grounds that are neither planted nor paved to smooth, even-textured surface.
   d. Remove tools, construction equipment, machinery, and surplus material from Project site.
   e. Remove snow and ice to provide safe access to building.
   f. Clean exposed exterior and interior hard-surfaced finishes to dirt-free condition, free of stains, films, and foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to original condition.
   g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
   h. Sweep concrete floors broom clean in unoccupied spaces.
   i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo, soil or stains remain.
   j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and vision-obscuring materials. Replace chipped or broken glass and transparent materials. Polish mirrors and glass.
   k. Remove labels that are not permanent.
   l. Touch up, repair, and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.
      1) Do not paint over "UL" (Underwriters Laboratories) and similar labels, including mechanical and electrical nameplates.
   m. Wipe surfaces of mechanical and electrical equipment and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
   n. Replace parts subject to unusual operating conditions.
   o. Clean plumbing fixtures to sanitary condition, free of stains, including stains resulting from water exposure.

p. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.

q. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out and noticeably dimmed bulbs, and defective or noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.

r. Leave Project clean and ready for occupancy.

C. Pest Control: Engage experienced, licensed exterminator to make a final inspection and rid project of rodents, insects, and other pests. Provide Government with report.

D. Waste Disposal: Comply with requirements of Section 01 74 19 "Construction Waste Management and Disposal."

END OF SECTION 01 77 00

This page intentionally left blank.

SECTION 01 78 23 - OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including:

1. Manuals, general
2. Emergency manuals
3. Operation manuals for systems, subsystems, and equipment
4. Maintenance manuals for care and maintenance of systems and equipment

B. See Divisions 2 through 49 Sections for additional operation and maintenance manual requirements for Work in those Sections.

1.2 SUBMITTALS

A. Manual: Submit two copies of each manual in draft form or one electronic copy at least 15 days before final inspection. Contracting Officer (CO) will return copy or edit version with comments within 15 days of receipt.

B. Format: Submit operations and maintenance manuals in following format:

1. PDF (portable document format) electronic file. Assemble each manual into composite electronically indexed file. Submit on digital media acceptable to Contracting Officer.
   a. Name each indexed document file in composite electronic index with applicable item name. Include a complete electronically linked operation and maintenance directory.
   b. Enable inserted reviewer comments on draft submittals.
2. Hard copy manual: In accordance with Part 2 of this Section.
3. Correct or modify each manual to comply with Contracting Officers comments. Submit 4 copies of each corrected manual within 15 days of receipt of Contracting Officers comments.

1.3 QUALITY ASSURANCE

A. Coordinate with Section 01 91 14 "Total Building Commissioning." Commissioning Agent shall review Operation and Maintenance Manuals for commissioned systems.

PART 2 - PRODUCTS

2.1 MANUALS, GENERAL

A. Organization: Unless otherwise indicated, organize manual into separate sections for each system and subsystem, and separate sections for each piece of equipment not part of a system. Manual shall contain title page, table of contents, and manual contents.

B. Title Page: Enclose title page in transparent plastic sleeve. Include:

1. Project Title
2. Location
3. Park
4. Contract Number
5. Prime Contractors Name and Address
6. Date of Substantial Completion
7. Binder Volume Number

C. Table of Contents: List each product included in manual, identified by product name, indexed to content of the volume, and cross-referenced to Specification Section number in Project Manual.

D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. Assemble instructions for subsystems, equipment, and components of one system into a single binder if needed.

1. Binders: White, commercial quality, hard back, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to hold 8-1/2 by 11 inch paper; with clear plastic window sleeve on front and spine to hold label describing contents and pockets inside covers to hold folded oversize sheets.
   a. Cover Sheet: Identify binders on front and spine, with project title, location, park, contract number, prime contractor's name and address, date of substantial completion, and binder volume number. Insert cover sheet into clear plastic view pocket on front of binder. Insert sheet into clear plastic view pocket on spine with title "OPERATION AND MAINTENANCE MANUAL," Project title or name, and subject matter of contents. Indicate volume number for multiple-volume sets.
2. Data: Fill binders to no more than 75 percent of capacity. Punch holes shall not obscure any data. When contents of a single tabbed section cover more than one item, provide colored paper sheets to separate the data for each item.
   a. Manufacturers' Data: Provide originals for color or copyrighted data. Black and white data may be originals or clean, good quality reproductions. No copies produced by facsimile transmission and sheets with stamps, such as submittal approval stamps. Include only sheets that apply to items installed; cross out inapplicable data.
   b. Vendor Furnished As-Built Drawings: Maximum 24 inch by 36 inch sheets with minimum character or lettering size of 1/8 inch. Reduced-size reproductions may be provided instead of full-size drawings if reproductions are clear and legible. If reduced-size drawings are used, identify as "REDUCED SIZE" and provide graphic scales, if applicable.
   c. Custom Data: Data supplemented by drawings and schematics necessary to describe systems adequately.
   d. Equipment Data Sheet: Data, using form at end of this section.
   e. Schedules: Schedules reflecting final, as-installed conditions.
   f. Poorly reproduced or illegible data will be rejected.

3. Dividers: Divider sheets with Mylar reinforced edges and pre-printed numbered tabs aligned with numbers and title lines on index sheet. Include typed list of products and major components of equipment included in section on each divider, cross-referenced to Specification Section number and title of Project Manual.
4. Protective Plastic Sleeves: Transparent plastic sleeves designed to enclose diagnostic software diskettes for computerized electronic equipment.
5. Drawings: Attach reinforced, punched binder tabs on drawings and bind with text.
    a. If oversize drawings are necessary, fold drawings to same size as text pages and use as foldouts.
    b. If drawings are too large to be used as foldouts, fold and place drawings in labeled envelopes and bind envelopes in rear of manual. At appropriate locations in manual, insert typewritten pages indicating drawing titles, descriptions of contents, and drawing locations.

2.2 EMERGENCY MANUALS

A. Content: Organize manual into separate section for type of emergency, emergency instructions, and emergency procedures.

B. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component for fire, water leak, power failure, water outage, and equipment failure.

C. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of National Park Service (NPS) operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.

D. Emergency Procedures: Include instructions on stopping, shutdown instructions for each type of emergency, operating instructions for conditions outside normal operating limits, and required sequences for electric or electronic systems.

2.3 OPERATION AND MAINTENANCE MANUALS

A. Operation Requirements

1. Content: In addition to requirements in Section, include operation data required in individual Specification Sections and equipment descriptions, operating standards, operating procedures, operating logs, wiring and control diagrams, and license requirements.
2. Descriptions: Include:
    a. Product name and model number
    b. Manufacturer's name
    c. Equipment identification with serial number of each component
    d. Equipment function
    e. Operating characteristics
    f. Limiting conditions
    g. Performance curves
    h. Engineering data and tests
    i. Complete nomenclature and number of replacement parts
3. Operating Procedures: Include start-up, break-in, and control procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; and required sequences for electric or electronic systems.

4. Systems and Equipment Controls: Describe sequence of operation, and diagram controls as installed.
5. Piped Systems: Diagram piping as installed and identify color-coding where required for identification.

B. Maintenance Requirements for Systems and Equipment

1. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, and equipment data sheets as described below.
2. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
3. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including maintenance instructions, drawings and diagrams for maintenance, nomenclature of parts and components, and recommended spare parts for each component part or piece of equipment:
4. Maintenance Procedures: Test and inspection instructions, troubleshooting guide, disassembly instructions, and adjusting instructions, detailing essential maintenance and environmental procedures.
5. Maintenance and Service Schedules: Service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
6. Spare Parts List and Source Information: Lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
7. Warranties and Bonds: Copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.4 PRODUCT MAINTENANCE MANUAL

A. Content: Organize manual into separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.

B. Source Information: List each product included in manual identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.

C. Product Information: Include:

1. Product name and model number
2. Manufacturer's name
3. Color, pattern, and texture
4. Material and chemical composition
5. Reordering information for specially manufactured products

D. Environmental Requirements

1. Identify environmentally preferable products incorporated into Project. Include: product model; manufacturer's name, address, phone, and website; and local technical representative.
    a. Verify plastic products to be incorporated into Project are labeled in accordance with ASTM D1972. Where products are not labeled, provide product data indicating polymeric information in Operation and Maintenance Manual.
        1) Type 1: Polyethylene Terephthalate (PET, PETE)
        2) Type 2: High Density Polyethylene (HDPE)
        3) Type 3: Vinyl (Polyvinyl Chloride or PVC)
        4) Type 4: Low Density Polyethylene (LDPE)
        5) Type 5: Polypropylene (PP)
        6) Type 6: Polystyrene (PS)
        7) Type 7: Other. Use of this code indicates that package in question is made with a resin other than the six listed above or is made of more than one resin listed above and used in a multi-layer combination.
    b. Describe maintenance procedures associated with environmentally preferable materials and systems. Provide cleaning recommendations in accordance with ASTM E1971 and approved Integrated Pest Management (IPM) plan.
        1) Include potential environmental impacts of recommended maintenance procedures and materials.
        2) Include potential indoor air quality impacts of recommended maintenance procedures and materials.
        3) Where proposed maintenance procedures incorporate composting of plastics, assess potential effect of each type of plastic to be included in composting process in accordance with ASTM D5509 or ASTM D6002
    c. Material Safety Data Sheets (MSDS): Include MSDSs as specified.
2. Develop environmental management programs for facility as follows:
    a. Waste management program: Develop in accordance with ASTM E1609. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.
    b. Indoor Air Quality (IAQ) management program: Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of VOCs (volatile organic compounds) in indoor air in accordance with ASTM D6345.
    c. Water management program: Develop water monitoring program for surface and ground water on project site in accordance with ASTM D5851 and consistent with water management program utilized during construction operations.

E. Maintenance Procedures: Include manufacturer's written recommendations and inspection procedures, types of cleaning agents, methods of cleaning, schedule for cleaning and maintenance, and repair instructions.

F. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.

G. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.5 SYSTEMS AND EQUIPMENT MAINTENANCE MANUALS

A. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, as described below.

B. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.

C. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including the following information for each component part or piece of equipment:

1. Standard maintenance instructions and bulletins.
2. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
3. Identification and nomenclature of parts and components.
4. List of items recommended to be stocked as spare parts.

D. Maintenance Procedures: Include the following and items detailing essential maintenance procedures:

1. Test and inspection instructions
2. Troubleshooting guide
3. Precautions against improper maintenance
4. Disassembly; component removal, repair, and replacement; and reassembly instructions
5. Aligning, adjusting, and checking instructions
6. Demonstration and training video recording

E. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.

1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.

F. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.

G. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.

H. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

1. Include procedures and required notifications for warranty claims.

PART 3 - EXECUTION

3.1 GENERAL

A. At start of project, begin accumulating operation and maintenance data and initiate index. Install and index data in binders within 30 days after delivery of items. As custom written data and test results are produced, add to operation and maintenance data file.

B. List of Operation and Maintenance requirements has been attached at end of the Division 1 Specifications for your convenience. Intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list.

C. Keep operation and maintenance data current. Make operation and maintenance binders available to Contracting Officer for inspection at time of monthly progress payment requests. If operation and maintenance binders are not current, Contracting Officer may retain an appropriate amount of the progress payment.

3.2 MANUAL PREPARATION

A. Manual Types

1. Emergency Manual: Assemble complete set of emergency information indicating procedures for use by emergency personnel and by NPS operating personnel for types of emergencies indicated.
2. Product Maintenance Manual: Assemble complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into Work.
3. Operation and Maintenance Manuals: Assemble complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.

B. Manual Contents: Including:

1. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark sheet to identify product or component incorporated into Work. If data include more than one item in a tabular format, identify each item using appropriate references from Contract Documents. Identify data applicable to Work and delete references to information not applicable.
2. Equipment Data Sheets: For each item of equipment included in operation and maintenance data, provide Equipment Data Sheet using form at the end of this section. For equipment consisting of a driven machine and a driver (for example, a pump and a motor), equipment data shall cover both the driven machine and the driver. For similar type equipment (for example, multiple exhaust fans of the same model and type), provide a single equipment data sheet with an attached schedule listing individual equipment items.
3. Vendor Furnished As-Built Drawings: Provide for each electrical and each mechanical control system.
    a. For each control system, provide control circuit schematic drawings. Identify each wire and terminal block number. Show terminal numbers on control devices. Show control wires and devices remote from control panel.
    b. For each control panel, provide general arrangement drawing showing location of each control component and terminal block on the panel front and interior. Include materials list of panel-mounted control components as well as field-installed control components remote from the panel, identifying components, manufacturer, model number, and initial set points or sensing ranges of devices where applicable.
    c. For packaged equipment systems, provide general arrangement drawings showing interrelationships of the various items of equipment and components.
    d. In addition to control wiring schematic, provide power wiring schematic drawing showing power flow to each motor. Identify each power conductor. Show over-current protection and motor starting devices.

C. Comply with Section 01 77 00 "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

END OF SECTION 01 78 23

EQUIPMENT DATA SHEET	
Equipment Item: _____ Designation: _____ Function: _____ Location: _____ Project: _____ Model Number: _____ Serial Number: _____	
Manufacturer Address and Phone:	Supplier Address and Phone:
Preventive Maintenance Tasks:	
Nameplate Data:	
Spare Parts Furnished and Other Information:	

This page intentionally left blank.

Closeout and Operation & Maintenance (O&M) Requirements

GRCA 318719

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

March 2026

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
Project Record Drawings	01 77 00	Final Project Record Drawings			
		Volume 1			
		Volume 2			
		Volume 3			
		Volume 4			
		Volume 5			
		Volume 6			
		Volume 8			
		Volume 9			
		Volume 10			
		Volume 11			
		Volume 12			
	01 91 14	Final Project Record Drawings			
		Volume 1			
		Volume 2			
		Volume 3			
		Volume 4			
		Volume 5			
		Volume 6			
		Volume 8			
		Volume 9			
		Volume 10			
		Volume 11			
		Volume 12			
	40 61 96	Process Control Descriptions			
		Volume 5: RSPH			
		Volume 6: NR WTP			
	40 67 00	Control System Panels: As-Built Shop Drawings for Volume 5: RSPH			
		Control System Panels: As-Built Shop Drawings for Volume 6: NR WTP			
System Demonstration and Training	01 75 00	Volume 6: NR WTP: Pre-demonstration and Demonstration requirements for Water Treatment Facilities			
	01 79 00	Volume 5: RSPH, Volume 6: NR WTP, Vol 9: PDC Generation & Fuels: Demonstration of operation of systems, subsystems, and equipment. Training in operation and maintenance of systems, subsystems, and equipment including environmental considerations. Demonstration and training video.			
	26 09 13	Provide one set of printer ink catridges and one eam of printer paper (500pages)			
	26 09 16	Provide contacts for all required control plus two spares			
	26 12 19	Provide two (2), one (1) quart cans of touch up paint for equipment			
	26 13 16	Provide touch up paint, one set of three grounding jumpers with storage bag			
	26 13 23	Provide touch up paint			
	26 24 19	Two additional normally open spare field replaceable auxiliary contacts			
	26 25 00	Furnish a minimum of 10% of total of each type and amperage of fuses for fixtures indicated to be fused.			
	26 29 23	Provide manufacturer's recommended renewable spare parts (e.g., power and control fuses)			
	26 31 00	Contractor shall propose a recommended quantity of spare modules in excess of proposed project installed DC capacity			
	26 32 14	Furnish spare parts and extra materials sufficient for one year of operation as recommended by generator system manufacturer, for each generator system furnished			
	27 10 00	Patch panels mounted in equipment racks with sufficient ports to accommodate all installed cable plus 25% spare			
	27 10 00	Provide space for the number of horizontal and backbone cables terminated on the blocks plus 25% spare			
	27 10 00	Patch panels shall be mounted in equipment racks with sufficient modular jacks to accommodate the installed cable plant plus 25% spares			
	27 10 00	Patch panels shall be mounted in equipment racks with sufficient connectors to accommodate the installed cable plant plus 25% spares			
	40 61 21	Terminal block, fuse, breaker, and other required installed spares			
	40 61 21	General arrangement and space allocation, including spare usable space in panel and on panel door			
	40 61 21	PLC equipment (chassis, power supply, power conditioners, processor, adapters, I/O and specialty modules, installed spares, spare usable space) is appropriate			
	40 62 05	Two vertical power strips shall be provided at rear of panel. Power strip capacity shall be sized with minimum 25% spare plug space for future expansion			
	40 62 05	One spare toner cartridge per laser printer provided			
	40 62 05	One spare port lock key for each lock type provided			
	40 62 05	Unless otherwise noted, provide at least two spare 10/100/1000 MBit/s port (twisted pair) at each Ethernet switch			
	40 63 43	One spare CPU for every four field CPUs or fraction thereof installed			
	40 63 43	One spare communication module by type installed in the Plant			
	40 63 43	One spare I/O card of each card type for every 10 cards or fraction thereof installed			
	40 63 43	One spare power supply for every 10 power supplies or fraction thereof installed			
	40 63 43	A set of spare fuses for all installed types			
	40 66 05	Unless otherwise noted, provide at least two spare 100/1000 Mbps port (twisted pair) at each Ethernet switch			
	40 66 05	Unless otherwise noted, provide at least two spare 10/100 MBit/s port (twisted pair) at each Ethernet switch			
	40 66 05	Unless otherwise noted, provide at least two spare 10/100/1000 MBit/s port (twisted pair) at each Ethernet switch			
	40 66 05	Unless otherwise noted, provide at least two spare 10/100/1000 MBit/s port (twisted pair) at each Ethernet switch			
	40 66 05	One spare Ethernet switch of each type utilized			
	40 66 40	One spare power supply of each type for every 10 power supplies or fraction thereof installed			
	40 66 40	One spare radio of each type installed			
	40 66 40	One spare surge arrestor			
	40 67 00	Install minimum of 20 % spare terminals			
	40 75 93	25 feet of sample collection tubing of the same size, type, and material installed			
	40 75 93	One additional sample container of the same size, material, and configuration of that provided with autosamplers			
	43 23 22	Impellers: Two			
	43 23 22	Wearing Rings: One set			
	43 23 22	Mechanical seal: One unit			
	43 23 31.29	Wearing rings: One set			
	43 23 31.29	Mechanical Seal: One cartridge			
	46 33 11	Spare parts for Chemical Feed - Liquid Systems			
	46 61 33	Spare parts for Membrane systems			
	07 92 00	Stock Material for Joint Sealants			
	09 30 13	2 percent of each different size, pattern, style and/or color of tile used including trim shapes with minimum of one full size tile of each for maintenance purposes			
	09 30 13	Enough extra grout mix of each different color and/or type to allow for full 2 percent of tile to be replaced			
	09 51 00	Provide 96 square feet of each type, color, and size of acoustic ceiling tile specified			
	09 51 00	Provide acoustic free-hanging elements (Clouds and Baffles) – Minimum 5 percent of each type installed			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
Extra Stock	09 51 00	Provide Suspension System Components – Minimum 5 percent of each type installed for acoustic ceiling tiles, clouds and baffles			
	09 65 36	Remaining portion of one partially used roll of resilient base material with a minimum of 10 linear foot of each height, color and type			
	11 24 26	Furnish Contracting Officer two, 12 OZ spray cans of touch-up paint for each equipment color provided			
	23 31 00	3 complete filter media change(s) for each type of filter			
	26 32 14	Furnish spare parts and extra materials sufficient for one year of operation as recommended by generator system manufacturer, for each generator system furnished			
Reports	01 74 19	Final Summary of solid waste disposal and diversion			
	01 75 00	Draft and Final Demonstration Plans			
	01 77 00	Approved pre-functional checklists and functional performance testing reports from commissioning documentation			
	01 77 00	Test and balance report			
	01 57 19.11	Moisture control inspections			
	01 57 19.11	Moisture content testing			
	01 57 19.11	Moisture penetration testing			
	01 57 19.11	Microbial Growth testing			
	01 61 03	Motor test reports			
	01 61 03	Field noise testing reports			
	01 61 03	Testing and monitoring reports			
	01 91 14	Commissioning reports			
	01 91 14	Test and inspection reports and certificates			
	01 91 14	Testing, adjusting, and balancing reports			
	03 01 30	Laboratory test reports			
	03 01 30	Test and Evaluation Reports			
	03 05 05	Test reports and inspection reports			
	03 31 30	Cement and SCM mill certificates for all materials to be supplied			
	03 31 30	Test results for AAR impurities of coarse aggregates within proposed mixes, in accordance with ASTM standards cited in this Section			
	03 41 33	Test reports			
	04 05 13	Test results for AAR impurities of coarse aggregates within proposed mixes, in accordance with ASTM standards cited in this Section			
	04 22 00	Provide special inspection in accordance with and including observation of masonry work for conformance to the Contract Documents			
	05 12 00	Certificates of compliance with standards specified for all major components and fasteners incorporated into work			
	05 12 00	Certified copies of mill tests			
	05 12 00	Manufacturer's load test and temperature sensitivity data for post-installed anchor bolts			
	05 12 00	Test reports for all structural steel work			
	05 14 00	Certified copies of material mill tests			
	05 14 00	Manufacturer's load test and temperature sensitivity data for expansion anchor bolt and adhesive anchor bolts			
	05 30 00	Powder driven fastener ICC-ES Evaluation Report(s)			
	05 30 00	Manufacturer's certified test and mill reports for actual materials furnished			
	07 84 00	Special Inspection Reports			
	08 11 16	Testing reports and evaluations provided by manufacturers conducted by and accredited independent testing agency certifying doors and frames comply with specified performance requirements			
	08 70 00	Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency			
	09 84 00	UL certifications that products comply with specified requirements, including laboratory reports showing compliance with specified tests and standards. Acoustic performance, products must be tested to the A and D-20, methods			
	22 05 03	Copies of pressure test results on all piping systems			
	22 05 03	Disinfection test report			
	22 05 03	Notification of time and date of piping pressure tests			
	23 05 93	Completed test reports and data forms upon completion of installation, balance and testing of HVAC systems			
	23 34 00	Test and balance reports			
	23 74 36	Test reports of the refrigerant piping leak tests for all refrigerant piping systems installed			
	26 05 13	Cable test report(s)			
	26 05 26	grounding systems continuity test and grounding resistance test.			
	26 08 13	RSPH: results of electrical acceptance tests required in this Specification (numerous tests)			
	26 08 13	NR WTP: results of electrical acceptance tests required in this Specification (numerous tests)			
	26 08 13	Electric distribution system: results of electrical acceptance tests required in this Specification (numerous tests)			
	26 09 43	Commissioning and training certification of digital lighting control system			
	26 13 16	NR WTP: manufacturer's certification of installation for metal-clad load interrupter switchgear			
	26 05 46	Pole Inspection Reports			
	26 08 13	Testing and monitoring reports			
	26 09 13	Operating instructions of the meter(s) supplied			
	26 09 13	Operating instructions of the Power Management software			
	26 09 43	Device list report: All devices in a project listed by type			
	26 09 43	Load binding report: All load controller bindings showing interaction with sensors, switches, and daylighting			
	26 09 43	BACnet points report: Per room identify Device ID and associated BACnet point mapping			
	26 09 43	Device parameter report: Per-room lists of all configured parameters for all control devices for use with O&M documentation			
	26 09 43	Scene setting reports: All project scene setting values associated with each named scene			
	26 13 16	Certified reports of all factory production tests			
	26 13 23	manufacturer's certification of installation for sectionalizing equipment			
	26 23 00	Ground fault protection system test reports signed by the projects supervising electrical foreman			
	26 28 00	Ground fault protection system test reports signed by the projects supervising electrical foreman			
	26 28 02	Short circuit study report			
	26 28 02	Protective coordination study report			
	26 28 02	Arc-flash hazard study			
	26 29 23	Certified factory test reports confirming compliance with specified requirements			
	26 29 23	Certified field service reports			
	26 32 14	generator start-up testing reports, sound attenuation report.			
	26 41 13	UL Master Label Certificate for lightning protection system			
	26 50 00	IESNA LM-79 Test Report for Solid-State Luminaire			
	26 50 00	IESNA LM-80 Test Report Solid-State Light Source			
	27 10 00	Telecommunications cabling test results for all horizontal and backbone links			
	31 22 00	Report and certification of aggregate fill and drainage fill			
	31 22 00	Test reports on borrow material			
	31 22 00	Field reports; in-place soil density and moisture tests			
	31 23 00	Report and certification of aggregate fill and drainage fill			
	31 23 00	Test reports on borrow material			
	31 23 00	Field reports; in-place soil density and moisture tests			
	31 23 33	Geotechnical reports			
	32 15 40	Certified test reports and service records to determine acceptability and application of stone material			
	32 32 16	Geotechnical investigation reports			
	32 91 13	Soil Survey Report			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
	33 05 15	Copies of source quality control tests, including compressive strength and air content, for units provided			
	33 05 24	Chemical and physical test reports showing data consistent with specified requirements for each heat of steel proposed for use			
	33 05 24	Source Quality Control Test Reports			
	33 05 24	Field Quality Control Test Reports			
	33 11 13	Bacteriological test reports on disinfection requirements			
	33 11 13	Bacteriological reports on samples from the system			
	33 31 11	Pipe Joint Testing Eqipment			
	33 52 10	Pressure test reports			
	33 56 13	Factory Test Reports			
	33 56 13	Field Test Reports			
	33 56 13	Mill test reports			
	33 71 16.23	Certified Material Test Reports			
	33 71 16.23	Certified welding reports			
	33 71 16.23	Test reports on coating thickness			
	40 05 00	Reports defining results of dielectric testing and corrective action taken			
	40 05 00	Bacteriological reports on samples taken from system			
	40 05 63	One set of any special tools or wrenches required for operation or maintenance for each type valve			
	40 05 63	One 4" Spare actuator (V-52338)			
	40 05 63	One 3" Spare actuator (V-52331)			
	40 41 13	Megger test results			
	43 21 00	Factory hydrostatic test			
	43 41 15	Certified copies of test results from radiographs and inspections of test segments			
	40 61 21	Site Acceptance Test (SAT) Reports			
	40 61 21	Completed Loop Checkout Sheets			
	40 61 21	Completed Instrument Certification Sheets			
	40 61 21	Completed Final Control Element Certification Sheets			
	40 61 21	All system startup and checkout records, test records, recommendations and information furnished by Manufacturer's representatives during testing and startup			
	40 61 21	Completed Loop Checkout Sheets			
	40 61 21	Completed Instrument Certification Sheets			
	40 61 21	Completed Final Control Element Certification Sheets			
	43 41 45	Hydrostatic Test Reports			
	43 41 45	Inspection Test Report according to ASTM C582 Standards for FRP Tanks			
Keys & Keying Schedule	08 70 00	NR WTP, RSPH: Keying schedule			
	05 52 46	Provide owner with manufacturer's standard limited warranty for materials and installation - (1-year standard)			
	06 10 00	Interior fire-retardant treated wood: Manufacturer's standard 20-year warranty			
	06 41 00	Furnish a written two-year manufacturer's warranty covering all items furnished under this section against defective materials and workmanship			
	07 14 00	Warranty for Fluid Applied Waterproofing			
	07 27 46	Five-year warranty for Fluid Applied Air and Moisture Barrier and accessories			
	07 46 10	Pre-Finished Fiber-Cement Board Lap Siding for 30 years			
	07 46 10	Application limited warranty for 2 years			
	07 61 13	Provide 20 year minimum complete system warranty, including material for air and weather tightness of entire roof assembly signed by manufacturer			
	07 61 13	Provide manufacturer's 20-year - non-prorated - warranty on panel finish against fading, chipping, cracking and peeling of the panel exterior finish and erosion of substrate metal			
	07 62 00	Warranty for Flashing			
	07 62 00	NR WTP: Warranty for Flashing			
	08 11 00	RSPH: O&M Data for Metal Doors and Hardware			
	08 11 00	O&M Data for Metal Doors and Hardware			
	08 11 16	Factory applied high performance organic coatings utilizing PVDF resins shall be provided with manufacturer's standard 20-year warranty against color fade, chalk and film integrity			
	08 14 16	Warrant doors in writing for life of installation against defects			
	08 31 00	O&M Data for Access Doors			
	08 31 00	NR WTP: O&M Data for Access Doors			
	08 31 00	RSPH: Warranty for Access Doors			
	08 31 00	Warranty for Access Doors			
	08 33 22	Manufacturer shall warrant the doors and its component parts for one (1) year against defects in material and workmanship			
	08 70 00	Unless otherwise indicated, warranty shall be one year from date of Substantial Completion			
	08 70 00	Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation			
	08 81 00	Provide manufacturer's written 10-year warranty to cover deterioration of glass, glass units and coating. Insulating glass units shall be warranted against failure of hermetic seal resulting in fogging or film formation on the interior glass surfaces			
	08 81 00	Provide a 20-year warranty to cover delamination and failure of the structural integrity of the warm edge spacers			
	08 81 00	Insulating Glass Spacers: Provide a 20-year warranty for delaminating and against failure of structural integrity			
	08 81 00	Provide insulating glass manufacturer's written warranty, agreeing to, provide and replace insulating glass units which have defective hermetic seals for a period of ten (10) years			
	09 30 13	O&M Data for Ceramic tile			
	09 51 00	Warranty for Acoustical Ceilings			
	09 51 00	RSPH: O&M Data for Acoustical Ceilings			
	09 51 00	O&M Data for Acoustical Ceilings			
	09 65 36	RSPH: Warranty for Resilient Flooring			
	09 65 36	RSPH: O&M Data for Resilient Flooring			
	09 65 36	O&M Data for Resilient Flooring			
	09 67 26	Applicator will provide a written one-year warranty from date of substantial completion covering both labor and materials			
	09 84 00	Defects in materials or factory workmanship warranty - Thirty (30) years from date of substantial completion			
	10 11 00	O&M Data for Marker Boards			
	10 14 23	O&M Data for Signage			
	10 28 13	O&M Data for Toilet accessories			
	10 44 33	O&M Data for Fire Extinguisher			
	10 51 13	Warranty for Lockers			
	10 51 13	O&M Data for Lockers			
	11 24 26	Provide Manufacturers 5 year Warranty on all curtains			
	11 24 26	Provide Manufacturers Lifetime Warranty on all Hardware			
	11 30 13	Warranty for Appliances			
	11 30 13	O&M Data for Appliances			
	11 53 00	SR: O&M Data for Laboratory Instrumentation			
	11 53 00	NR WTP: O&M Data for Laboratory Instrumentation			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
O&M Data, Warranties, and Guarantees	11 53 00	Manufacturer's standard warranty against equipment failure, including cost of shipping, repair, replacement, and legal discard of waste materials			
	11 53 43	O&M Data for Laboratory Service Fittings and Fixtures			
	12 24 13	Roller Shade Hardware: Manufacturer's standard non-depreciating 25-year limited warranty			
	12 24 13	Shadecloth: Manufacturer's standard non-depreciating 25-year limited warranty			
	12 35 53	O&M Data for Laboratory Casework and Furnishings			
	12 24 13	Manufacturer's standard two year warranty			
	13 47 13.17	One year from CP system startup and acceptance by owner. If the system is tested and not left operating, the warranty shall start when the system is left in operation after any time for the coating to “cure”			
	23 09 00	HVAC controls shop drawings			
	23 34 00	O&M manual for inline centrifugal fans			
	23 34 00	O&M manual for axial fans			
	23 31 00	Ductwork shop drawings			
	23 74 36	The completed refrigerant piping system shall be guaranteed to be sufficiently free from leaks so that the loss of refrigerant for 18 months from the date of final payment shall not exceed 5 percent			
	23 80 00	O&M manual for air handling units			
	23 80 00	O&M manual for condensing units			
	23 81 29	For compressors only, under normal use and maintenance for comfort cooling and conditioning applications, the products will be free from defects in material or workmanship for a six-year period			
	26 09 43	Manufacturer's warranty agreeing to repair or replace components of lighting controls system that fail in materials or workmanship within specified warranty period following substantial completion - Warranty Period: One year			
	26 09 43	Software and firmware updates and onsite software and firmware support: One (1) year			
	26 12 19	O&M manual for distribution transformers			
	26 13 16	O&M manual for metal-enclosed load interrupter switchgear			
	26 13 23	O&M Manual for sectionalizing equipment			
	26 22 13	O&M manual for dry-type transformers			
	26 33 00	O&M manual for Battery and Battery Charging Systems			
	26 24 13	O&M manual for switchboards			
	26 24 16	O&M manual for panelboards			
	26 24 19	O&M manual for motor control centers and other motor control equipment under this specification			
	26 26 13	O&M manual for packaged power supply			
	26 28 00	O&M information for overcurrent protective devices			
	26 29 23	O&M manuals for Variable Frequency Drives, recommended spare parts lists			
	26 32 14	O&M manuals for engine-generator			
	26 35 26	RSPH: O&M manuals for harmonic correction equipment			
	26 35 26	NR WTP: O&M manuals for harmonic correction equipment			
	26 35 26	The power correction system manufacturer shall warrant products against defects in material and workmanship for a period of two years from date of shipment			
	26 41 13	O&M manual for lightning protection system			
	26 43 13	Minimum of a five year Warranty from date of shipment against failure when installed in compliance with applicable national/local electrical codes and the manufacturer's installation, operation and maintenance instructions			
	26 50 00	Minimum of a five year Warranty from date of manufacture against failure for solid-state luminaire including LED arrays, LED drivers and integral control devices. The solid-state product is considered defective if more than 15% of the individual light emitting diodes fail to illuminate			
	33 11 13	O&M Manual for inline valves			
	33 16 33	All work shall be guaranteed for a minimum of two years after Substantial Completion. Guarantee that the performance of the equipment will meet the specified requirements, and that a safe operating system will result			
	33 31 11	O&M Data for pipeline and manhole construction			
	33 56 13	Tank will not fail for a period of 30 years from the date of original purchase due to external corrosion			
	33 56 13	Tank will not fail for a period of 30 years from the date of original purchase due to internal corrosion provided the tank is used solely with specified fuel			
	33 56 13	Will not fail for a period of 30 years from the date of original purchase due to structural failure (defined as breaking or collapse) provided the installation is performed and validated by a qualified installation contractor and the tank is used as stated above			
	33 56 16	2-year warranty which covers all parts and all labor for guarantee of parts and workmanship for generator fuel transfer pumps and polishing system			
	40 05 51	O&M Manual for valves (basic requirements)			
	40 05 52	O&M Manual for miscellaneous valves			
	40 05 61	O&M Manual for gate valves			
	40 05 63	O&M Manual for ball valves			
	40 05 64	O&M Manual for butterfly valves			
	40 05 66	O&M Manual for check valves			
	40 61 43	O&M Manual for surge protection devices			
	40 61 43	5 year limited warranty for surge protection devices			
	40 61 93	Post-commissioning final I/O List			
	40 62 05	O&M Manual for Computer Hardware and Ancillaries			
	40 63 43	O&M Manual for Programmable Logic Controllers and associated hardware			
	40 66 05	O&M Manual for Network Equipment			
	40 66 40	O&M Manual for radio equipment			
	40 67 00	O&M Manual for Control Panels			
	40 67 63	O&M Manual for UPS			
	40 71 00	O&M Manual for flow instruments			
	40 72 00	O&M Manual for level instruments			
	40 73 00	O&M Manual for pressure instruments			
	40 73 00	O&M Manual for pressure instruments			
	40 74 00	O&M Manual for temperature instruments			
	40 75 00	O&M Manual for process liquid analytical instruments			
	43 41 16	Tanks shall be warranted for a minimum of three (3) years against any defects in material or workmanship. A written warranty shall be provided. Manufacturer shall replace tank in whole, or any individual component, found to be defective during the warranty period			
	43 41 45	Provide with a manufacturer’s warranty certificate for defects in workmanship and materials for of period of five (5) years from the date of delivery to the project site			
	43 41 15	Tank supplier shall guarantee that material, equipment, and components in its scope of supply are free of defect and that workmanship will be unenforcible, suitable for intended usage and environment. The guarantee shall be in effect for a period of 12 months after acceptance of the installation			
	46 61 33	Twenty-four (24) months for all components except where otherwise specified			
	46 61 33	A ten (10) year warranty on membrane modules with the first thirty-six (36) months offered as full replacement warranty and the remaining eighty-four (84) months as a prorated warranty			

SECTION 01 79 00 - DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for instructing National Park Service (NPS) personnel, including:

1. Demonstration of operation of systems, subsystems, and equipment.
2. Training in operation and maintenance of systems, subsystems, and equipment, including environmental considerations.
3. Demonstration and training video.

B. See Divisions 2 through 49 Sections for specific requirements for demonstration and training for products in those Sections.

1.2 GENERAL REQUIRMENTS

A. List of System Demonstration and Training requirements has been attached at end of Division 1 Specifications for your convenience. Intent is to provide overall summary of requirements and not a comprehensive list. Terms and conditions of contract still require satisfaction of requirements of individual specification sections regardless of what is shown on list.

1.3 SUBMITTALS

A. Instruction Program: Submit two copies of outline of instructional program for demonstration and training, including schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.

1. For each training session, Contractor shall submit for approval a proposed outline of subjects to be covered. Training shall not be conducted until outline is approved.

B. Demonstration and Training Video: Submit digital copies of each (.h264 or .mov) for training sessions within seven days of end of each training module.

1. Label each with date of demonstration or training, instructor's name, and provide an index of contents. Index shall list start and end time of each subject covered during training session. Sequence of training subjects shall follow sequence listed in approved training outline or as actually conducted

1.4 QUALITY ASSURANCE

A. Facilitator Qualifications: Firm or individual experienced in training or educating maintenance personnel in training program similar in content and extent for this Project, and whose work has resulted in training or education with a record of successful learning performance.

B. Instructor Qualifications: Factory-authorized service representative, complying with requirements in Section 01 40 00 "Quality Requirements," experienced in operation and maintenance procedures and training.

C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by Contracting Officer (CO).

PART 2 - PRODUCTS

2.1 INSTRUCTION PROGRAM

A. Program Structure: Develop instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, and:

1. Equipment systems and controls.
2. Environmental Topics
    a. Overview of environmental issues related to building industry.
    b. Overview of environmental issues related to Project.
    c. Review of site-specific procedures and management plans implemented during construction:
        1) Regulatory Requirements
        2) Indoor Air Quality (IAQ) Management
        3) Noise and Acoustics Management
        4) Environmental Management
        5) Construction Waste Management
    d. Review of site-specific procedures and management plans to be implemented during operation and maintenance.
        1) Include review of environmentally related aspects of Operations and Maintenance Manual.
        2) Integrated Pest Management (IPM)

B. Training Modules: Develop a learning objective and teaching outline for each module. Include description of specific skills and knowledge that participant is expected to master. For each module, include instruction for:

1. Basis of System Design, Operational Requirements, and Criteria: Include system and equipment descriptions, operating standards, regulatory requirements, equipment function, operating characteristics, limiting conditions, and performance curves.
2. Documentation: Review emergency, operations, and maintenance manuals; Project Record Documents; identification systems; warranties and bonds; and maintenance service agreements.
3. Emergencies: Instructions on stopping; shutdown instructions; operating instructions for conditions outside normal operating limits; instructions on meaning of warnings, trouble indications, and error messages; and required sequences for electric or electronic systems.
4. Operations: Startup, break-in, control, and safety procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; operating procedures for emergencies and equipment failure; and required sequences for electric or electronic systems.
5. Adjustments: Alignments and checking, noise, vibration, economy, and efficiency adjustments.
6. Troubleshooting: Diagnostic instructions and test and inspection procedures.

7. Maintenance: Inspection procedures, types of cleaning agents, methods of cleaning, procedures for preventive and routine maintenance, and instruction on use of special tools.
8. Repairs: Diagnosis, repair, and disassembly instructions; instructions for identifying parts; and review of spare parts needed for operation and maintenance.

PART 3 - EXECUTION

3.1 INSTRUCTION

A. Facilitator: Engage qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Contracting Officer for number of participants, instruction times, and location.

B. Engage qualified instructors to instruct NPS personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.

C. Scheduling: Provide instruction at mutually agreed on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.

1. Schedule training with NPS through Contracting Officer with at least seven days' advance notice.
2. Conduct training sessions after equipment or system has been accepted and turned over to Government. Coordinate with commissioning requirements.
3. Coordinate with Integrated pest management requirements. Refer to specifications section and approved IPM plan.
4. Individual sections specify duration of training required. If no duration is listed, provide training of sufficient duration to adequately cover subjects.

3.2 DEMONSTRATION AND TRAINING VIDEO

A. General: Engage qualified commercial photographer to record demonstration and training video. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice.

1. At beginning of training module, record each chart containing learning objective and lesson outline.

B. Video Format: Digital (.h264 or .mov).

C. Video Recording: Record sessions with high resolution equipment (min of 1080p). Instructor's voice shall be clearly audible and understandable on digital file. Utilize supplemental microphone worn by instructor.

D. Narration: Describe scenes on video by audio narration by microphone while video is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

1. Digital Files with poor video or audio quality will be rejected and training recorded again.
2. The attached table outlines the operator training requirements for this project.

END OF SECTION 01 79 00

This page intentionally left blank.

Equipment	Section	No. of Training Days Req'd[(2)]	No. of Training Sessions Per Day Req'd	Duration of Each Training Session (hr.)	Training Session Conducted in Classroom and/or Field	Video Taping Req'd During Training (Yes/No)	Other Special Training Requirements
Cathodic Protection of Steel Water Tanks – Internal Galvanic	13 47 13.17	1	1	2	Field Only	Yes	
Packaged Domestic Water Pressure Booster Systems	22 11 26	1	1	1	Field Only	Yes	
Plumbing Fixtures and Equipment	22 20 00	1	1	2	Field Only	Yes	
Instrumentation and Control for HVAC Systems	23 09 00	1	1	4	Field Only	Yes	
HVAC: Fans	23 34 00	1	1	4	Field Only	Yes	
HVAC: Equipment	23 80 00	1	1	4	Field Only	Yes	
Variable Refrigerant Flow Heat Pumps	23 81 29	1	1	4	Field Only	Yes	
Electrical Metering Devices	26 09 13	1	1	8	Field Only	Yes	
Low Voltage Lighting Control System	26 09 43	1	1	2	Field Only	Yes	
Medium Voltage Metal Enclosed Load Interrupter Switchgear	26 13 16	1	1	4	Field Only	Yes	
Sectionalizing Equipment	26 13 23	1	1	2	Field Only	Yes	
Generator Paralleling Switchgear	26 23 00	1	1	4	Field Only	Yes	
Motor Control Equipment	26 24 19	1	2	2-4	Both	Yes	
Variable Frequency Drives: Low Voltage	26 29 23	1	2	4	Both	Yes	
Battery and Battery Charging Systems	26 33 00	1	1	4	Field Only	Yes	

Equipment	Section	No. of Training Days Req'd[(2)]	No. of Training Sessions Per Day Req'd	Duration of Each Training Session (hr.)	Training Session Conducted in Classroom and/or Field	Video Taping Req'd During Training (Yes/No)	Other Special Training Requirements
Engine Generator: Diesel and Automatic Transfer Switch	26 32 14, 26 36 00	1	2	4	Both	Yes	
Harmonic Correction Equipment	26 35 26	1	1	2	Both	Yes	
Fire Alarm Systems	28 46 00	2	1	2-4	Field Only	Yes	
Interior and Exterior Lighting	26 50 00	1	1	2	Field Only	Yes	
Service Piping – Fuel Systems	33 52 10	1	1	4	Field Only	Yes	
Aboveground Fuel Storage Tanks	33 56 13	1	1	4	Field Only	Yes	
Fuel Day Tank Systems	33 56 16	1	1	4	Field Only	Yes	
Actuators for Process Valves and Gates	40 05 57	1	1	4	Field Only	Yes	
Primary Meters and Transmitters	40 71 00, 40 72 00, 40 73 00, 40 74 00	2	1	2	Field Only	Yes	
Process Liquid Analytical Measurement	40 75 00	2	1	2	Both	Yes	
Hoists, Trolleys, and Monorails	41 22 23	1	1	4	Field Only	Yes	
Pumping Equipment – Horizontal Multi-Stage Centrifugal Pumps	43 23 22	2	1	2	Both	Yes	
Pumping Equipment – Vertical Inline Centrifugal Pumps	43 23 31.29	1	1	2	Field Only	Yes	

Equipment	Section	No. of Training Days Req'd[(2)]	No. of Training Sessions Per Day Req'd	Duration of Each Training Session (hr.)	Training Session Conducted in Classroom and/or Field	Video Taping Req'd During Training (Yes/No)	Other Special Training Requirements
Fiberglass Reinforced Plastic Tanks	43 41 45	1	1	2	Field Only	Yes	
Chemical Feed – Liquid Systems	46 33 11	1	1	8	Both	Yes	
Vertical Mixers	46 41 11	1	1	8	Both	Yes	
Ultrafiltration Equipment	46 61 33	4	1	4	Both	Yes	See specification.

Notes:

(1) Training agenda must be submitted to the Contracting Officer minimum 1 week prior to the scheduled training.
(2) Number of training days is per water treatment plant if the equipment/system is installed at both plants.

This page intentionally left blank.

SECTION 01 81 10 – WIND AND SEISMIC DESIGN CRITERIA

PART 1 - GENERAL

1.1 SUMMARY

A. This Section is intended to be used for all aspects of this project.

1. When there are conflicts between this Section and other wind, seismic, and snow design criteria given in the Contract Documents, the more stringent loading shall control unless clarified in writing during the Bid phase.
2. Obtain clarification of all conflicts in writing prior to construction.

B. Section Includes:

1. The wind, seismic, and snow design criteria for this project including all items directly specified in the Contract Documents as well as all items that are specified to be designed by the Contractor and submitted for approval. Items include but are not necessarily limited to the following:
   a. Anchorage of mechanical and electrical equipment.
   b. Anchorage of pipe support structures.
   c. Design and anchorage of tanks and vessels fabricated off site and shipped to Project site.
   d. Self-contained enclosures
   e. Packaged equipment systems.
   f. Other structures or items as specified or indicated in the Contract Documents.

C. Related Specification Sections include but are not necessarily limited to:

1. Division 00 – Procurement and Contracting Requirements.
2. Division 01 – General Requirements.
3. Section 03 15 19 – Anchorage to Concrete.
4. Section 05 21 00 – Steel Joist
5. Section 05 30 00 – Metal Decking

1.2 QUALITY ASSURANCE

A. Referenced Standards:

1. American Society of Civil Engineers (ASCE):
   a. 7-22, Minimum Design Loads and Associated Criteria for Buildings and Other Structures.
2. International Code Council (ICC)
   a. International Building Code and associated standards, 2024 Edition including all amendments, referred to herein as Building Code.
3. When referenced standards conflict the most stringent shall apply unless specifically indicated otherwise in the Contract Documents or unless approved otherwise in writing by the Contracting Officer.

B. Qualifications:

1. Engineer for Contractor designed items: Professional Engineer licensed in the State of Arizona.

1.3 GENERAL DESIGN CRITERIA

A. This paragraph is applicable to both wind, seismic, and snow design criteria.

1. Risk Category: IV.
2. Design in accordance with the Building Code load combinations for service level or factored level at Contractor's option.
   a. Mechanical and electrical equipment loads will bear considered dead loads.

1.4 SEISMIC DESIGN CRITERIA

A. North Rim WTP (06 Series Drawings) Seismic Design Load Criteria:

1. Design spectral acceleration at short period: S_{DS} = 0.300.
2. Design spectral acceleration at 1-second period: S_{D1} = 0.110.
3. Importance Factor: Ie = 1.5.
4. Seismic Design Category: C.
5. Component or system amplification factor, (a_P) and Component response modification factor, (R_P): In accordance with ASCE 7-22, Tables 13.5-1 and 13.6-1.
6. Component importance factor: I_P = 1.50.

B. Roaring Springs Pumphouse (05 Series Drawings) Seismic Design Load Criteria:

1. Design spectral acceleration at short period: S_{DS} = 0.310.
2. Design spectral acceleration at 1-second period: S_{D1} = 0.110.
3. Importance Factor: Ie = 1.5.
4. Seismic Design Category: C.
5. Component or system amplification factor, (a_P) and Component response modification factor, (R_P): In accordance with ASCE 7-22, Tables 13.5-1 and 13.6-1.
6. Component importance factor: I_P = 1.50.

C. Seismic forces must be resisted by direct load transfer through fasteners to all seismic resisting elements.

1. Do not use connections that use friction to transfer seismic forces.

1.5 WIND DESIGN CRITERIA

A. Wind design load criteria:

1. Occupancy category I is not listed as an option below since it is only applicable to agricultural facilities, certain temporary facilities, and minor storage facilities and should not be applicable to any of the typical Projects designed and constructed under these Specifications.
2. Applies to North Rim WTP (06 Series Drawings).
3. Basic wind speed (V_{ULT}): 111 MPH.
4. Exposure category: C.
5. Topographic factor: K_{zt} = 1.0.
6. Wind importance factor: I_w = Not Applicable.

7. Building Description for wind design is Enclosed, unless specifically noted otherwise on Drawings.

B. Wind forces must be resisted by direct load transfer through fasteners to wind resisting elements.

1. Do not use connections that use friction to transfer wind forces.

1.6 SNOW DESIGN CRITERIA:

A. Snow Design Criteria:

1. Applies to North Rim WTP (06 Series Drawings):
2. Design ground snow load: pg = 256 psf.
3. Design exposure factor: Ce = 0.9
4. Design slope factor: Cs = 1.0
5. Design thermal factor: Ct = 1.0
6. Winter Wind Parameter: $W_2 = 0.35$
7. Design considerations for partial loading, unbalanced snow, snow drift in the vicinity of adjacent structures or projections, sliding snow, and other applicable factors shall be considered at locations required by Laws and Regulations, including applicable building codes.
8. In no event shall snow design load criteria or minimum snow load be less than required by the National Park Service.

1.7 SUBMITTALS

A. Informational Submittals:

1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
2. Structural Calculations:
   a. Submit calculations for each Contractor designed item under the Specification Section number for that item.
   b. Indicate compliance with specific referenced documents of the Building Code.
   c. Provide basis of design and lateral analysis as required to derive all loads and to show system stability including compatibility of deflections and compatibility with allowable soil parameters as applicable.
   d. Indicate design load to each connection.
   e. Provide a complete lateral load resisting system that transfers all wind and seismic loads through a load path to ground.
   f. Sealed by a professional engineer licensed in the State of Arizona.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION - (NOT USED)

END OF SECTION 01 81 10

This page intentionally left blank.

SECTION 01 81 13 - SUSTAINABLE DESIGN REQUIREMENTS FOR NON-LEED™ PROJECTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes general requirements and procedures for compliance with Federal Sustainability requirements. This project is not seeking Leadership in Energy & Environmental Design (LEED™) certification but shall comply with applicable Federal Sustainability requirements. Requirements include laws (Executive Orders (EO) and regulations), management policies, building codes and standards, Federal directives, and National Park Service (NPS) guidelines.

B. Many Federal requirements can be achieved only through intelligent and integrated design of the project and are beyond control of the Contractor. Certain requirements relate to the products and procedures used for construction, therefore, full cooperation of the Contractor and Subcontractors is essential to successful compliance with Federal requirements.

C. Contractors shall familiarize themselves with relevant requirements and provide necessary information and instruction to subcontractors and installers.

1. Some requirements involve quantifying percentages by weight; these require careful recordkeeping and reporting by Contractor.
2. See Denver Service Center (DSC) Workflows Sustainability Standards for a list of Federal Sustainability requirements. Applicable Federal Sustainability requirements are also summarized on the project's NPS Project Sustainability Checklist. Contractor is responsible for ensuring the elements in the NPS Project Sustainability Checklist, identified by the Architect/Engineer (A/E) team, are incorporated into the construction of the project.

D. Related Sections:

1. See Divisions 1 through 49 Sections for sustainability requirements specific to work of each of these Sections.

1.2 DEFINITIONS

A. Chain-of-Custody Certificates: Certificates signed by manufacturers certifying wood used to make products was obtained from forests certified by a Forest Stewardship Council (FSC)-accredited certification body to comply with FSC Standard STD-01-001, "FSC Principles and Criteria for Forest Stewardship." Certificates shall include evidence manufacturer is certified for chain of custody by an FSC-accredited certification body.

B. LEED™: Leadership in Energy & Environmental Design. Sustainability rating system developed by United States Green Building Council (USGBC).

C. Rapidly Renewable Materials: Materials made from plants typically harvested within 10-year or shorter cycle. Rapidly renewable materials include products made from bamboo, cotton, flax, jute, straw, sunflower seed hulls, vegetable oils, or wool.

D. Recycled Content: Recycled content value of a material assembly shall be determined by weight.

1. "Post-consumer" material: Waste material generated by households or by commercial, industrial, and institutional facilities in their role as end users of product, which can no longer be used for intended purpose.
2. "Pre-consumer" material: Material diverted from waste stream during manufacturing process. Reutilization of materials such as rework, regrind, or scrap generated in a process and capable of being reclaimed within the same process that generated it is excluded.

E. Biopreferred Products: Commercial or industrial products (other than food or feed) composed in whole, or in significant part, of biological products, renewable agricultural materials (including plant, animal, and marine materials), or forestry materials and includes biobased intermediate ingredients or feedstocks.

1.3 FEDERAL SUSTAINABILITY DOCUMENTATION SUBMITTALS

A. Most of Federal sustainability documentation submittals are aggregations of submittals already required in relevant technical specifications. They are mentioned here to ensure they are collected and organized together to efficiently document compliance with sustainability requirements.

B. Provide preliminary submittals to NPS indicating how the following Federal requirements will be met:

1. Recycled Content: List of specified/proposed materials with recycled content. Indicate cost, post-consumer recycled content, and pre-consumer recycled content for each product having recycled content.
2. Certified Wood: Product data and/or chain-of-custody certificates for products containing certified wood. Include statement indicating cost for each certified wood product.
3. Construction Indoor Air Quality (IAQ) Management Plan – During Construction:
    a. Construction indoor-air-quality management plan.
    b. Product data for temporary filtration media.
    c. Product data for filtration media used during occupancy.
    d. Construction Documentation: Six photographs at three different times during construction period, along with brief description of the Sheet Metal and Air Conditioning Contractors' National Association (SMACNA) approach employed, documenting implementation of indoor-air-quality management measures, such as protection of ducts and on-site stored or installed absorptive materials.
4. Construction IAQ Management Plan – Before Occupancy:
    a. Signed statement describing building air flush-out procedures including dates when flush-out was started and completed and statement filtration media was replaced after flush-out.
    b. Product data for filtration media used during flush-out and during occupancy.
5. Low Emitting Materials - Adhesives and Sealants: Product data for adhesives and sealants used inside weatherproofing system indicating Volatile Organic Compound (VOC) content of each product used. Indicate VOC content in g/L (grams per liter) calculated according to 40 CFR 59 Subpart D (Code of Federal Regulations).
6. Low Emitting Materials - Paints and Coatings: Product data for paints and coatings used inside weatherproofing system indicating chemical composition and VOC content of each product used. Indicate VOC content in g/L calculated according to 40 CFR 59 Subpart D.
7. Low Emitting Materials - Flooring: Product data for products containing composite wood or agrifiber products or wood glues indicating they do not contain urea-formaldehyde resin.
8. Biopreferred Products: Provide list of bio-based products used on project.

PART 2 - PRODUCTS

2.1 RECYCLED CONTENT OF MATERIALS

A. Recycled Content: Provide building materials with recycled content such that post-consumer recycled content plus one-half of pre-consumer recycled content constitutes a minimum of 10 percent of cost of materials used for Project.

1. Determine cost of post-consumer recycled content by dividing weight of post-consumer recycled content in item by total weight of item and multiplying by cost of item.
2. Determine cost of pre-consumer recycled content of an item by dividing weight of pre-consumer recycled content in item by total weight of item and multiplying by cost of item.
3. Do not include mechanical and electrical components, and specialty items such as elevators and equipment in calculation.

2.2 BIOPREFERRED PRODUCTS

A. Use bio-based products found on United States Department of Agriculture (USDA) Biopreferred Products list where applicable on project.

2.3 LOW-EMITTING MATERIALS

A. For applications inside the weatherproofing system, use adhesives and sealants that comply with VOC content limits in Specification Divisions 2 through 49.

B. For field applications inside the weatherproofing system, use paints and coatings that comply with VOC content limits in Specification Divisions 2 through 49.

C. Do not use composite wood or agrifiber products or adhesives that contain urea-formaldehyde resin.

PART 3 - EXECUTION

3.1 REFRIGERANT REMOVAL

A. Remove Chlorofluorocarbons **(**CFC)-based refrigerants from existing Heating, Ventilation, Air Conditioning, and Refrigeration (HVAC&R) equipment indicated to remain and replace with refrigerants that are not CFC based. Replace or adjust existing equipment to accommodate new refrigerant as described in Division 23 Sections.

B. Remove clean-agent fire-extinguishing agents that contain Hydrochlorofluorocarbons (HCFCs) or halons and replace with agent that does not contain HCFCs or halons. See Division 21 Section "Clean-Agent Fire Extinguishing Systems" for additional requirements.

3.2 MEASUREMENT AND VERIFICATION

A. Coordinate with Divisions 2 through 49 for project requirements regarding installation of building level metering equipment to measure energy, water, and electric usage.

3.3 INDOOR-AIR-QUALITY MANAGEMENT

A. Coordinate with Section 01 57 19.11 "Indoor Air Quality Management" for managing indoor air quality during construction and prior to occupancy.

END OF SECTION 01 81 13

SECTION 01 81 33 CYBER SECURITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. The supplier and all other parties providing parts and services under this contract are subject to these conditions. This includes hardware and software, licensing, intellectual equipment, patent restrictions and use, copyright, and similar equipment, systems, and appurtenances covered under this Section.
2. Requirements for vendors, suppliers and / or Contractor-furnished equipment with microprocessor-based equipment including:
    a. Providing full and unrestricted use by the Government for equipment supplied under this contract including access, review, updates, and modification of software code, firmware and configurations of PLCs, PACs, RTUs, HMIs, OITs or any other microprocessor-based equipment.
    b. Allowing the Government to transfer to designated agents the right to perform this work on their behalf.
    c. Furnishing software source code for the system supplied. Coordinate with the COR to remove default usernames and passwords, to remove contractor passwords at time of system acceptance, and turn-over of all passwords utilized in the system.
3. Requirements for vendors, suppliers and / or Contractor-furnished items for furnishing program code, passwords, keys, and configuration files upon project milestones and / or as requested by the CO after project completion.

B. Scope:

1. The supplier agrees to assist in cybersecurity basic hygiene practices as dictated by best practices included in this section, government regulation at time of bid, and as provided by the Government's cyber security team. At a minimum, the supplier will coordinate with the COR and assign IP addresses and subnet masking, utilized of managed switches with unused ports disabled, provide all equipment with latest firmware at time of field installation.
2. To this end the following are expected standard industry practices that vendors shall abide by:
    a. All equipment supplied will be current and not be obsolete or subject to a manufacturer's end-of-life notice of less than 2 years after startup.
    b. Provide any notices of manufacturer's end-of-life notices for all equipment at the time of submittal for approval of equipment. This shall include searching equipment websites during the submittal process to ensure the products are not nearing end of life.
3. The Government shall be granted full use of equipment and services provided by equipment and its associated automation under this Section and shall include:
    a. Full access and use of supplied software within any equipment supplied including PLCs, OIT, networking and communications equipment, gateways, and HMI code (including fully documented source code), internal function blocks or code shall be ceded to the Government for their exclusive use at the location(s) of the equipment. These rights shall transfer to future locations.

b. Right to modify, enhance or make changes to the equipment as deemed necessary or desired by the Government, including hardware, software, networking settings, passwords or other information or settings that the Government may change for maintenance, upgrade or cybersecurity reasons.
c. Full release of claims to Intellectual Equipment, Trade Secrets, or Sensitive Information as supplied with this equipment as may be claimed by the supplier or any of their subcontractors, subconsultants and the like as regards equipment or services provided herein to the Government.
d. No provisions for Non-Disclosure Agreements (NDAs) or other restrictions to restrict or prevent the Government from obtaining or using information, licenses, electronic data, software are allowable.

4. The supplier shall furnish software source code for the equipment supplied. The code or any part of the system may not be hidden or protected by encryption, passwords or other means without providing these passwords or access to the Government's designated personnel.

a. Government will coordinate with manufacturer regarding any changes to parameters that may void manufacturer's warranty.

5. The supplier shall take measures to limit the distribution of information on Government's SCADA network to the minimum personnel required. The supplier shall not publish to public facing media this information at any time.

C. Related Requirements:

1. Section 01 33 23 – Submittal Procedures.
2. Section 01 67 00 – Product Requirements
3. Section 01 75 00 – Checkout and Startup Procedures.
4. Section 01 77 00 – Closeout Procedures

1.2 SUBMITTALS

A. Submit the following:

1. Comprehensive asset inventory of all networked components:
   a. Provide in Excel spreadsheet format.
   b. Coordinate with the CO to determine the preferred method of delivery to assure security of information contained in asset inventory.
   c. Include:
      1) Device ID
      2) Manufacturer
      3) Model Number
      4) Serial Number
      5) IP Address
      6) Device Use description
      7) Firmware
2. Network Diagrams:
   a. Provide in both AutoCAD and PDF formats.
   b. Coordinate with the Government to determine the preferred method of delivery to assure security of information contained in Network Diagrams.
   c. Logical Network Diagram(s):
      1) Depict information flow through network(s), and include:
         a) Major network devices, subnets, and VLANs.
         b) Include all wireless communication devices.
         c) Include the following information for each networked device:
            (1) Device ID

(2) Device description
(3) Manufacturer/model number
(4) IP address
(5) Ports and Protocols

d. Physical Network Diagram(s):
1) Show all network components, ports, protocols, connections and cables.
a) Include all wireless communication devices.

3. Software source code for all PLCs, HMIs, OITs and other devices.
4. Configuration files, passwords, encryption information and other sensitive information for the full project scope.

B. Timing: Submittals will be required at the following project stages:

1. Network Diagrams and Asset Inventory – with control panel submittals
2. Draft Software and Configuration Files, and As-Manufactured Network Diagrams – Available at Factory Acceptance Testing
3. All documents available on-site upon equipment delivery to the project site
4. All documents contained within Operation & Maintenance manuals
5. Final As built of all documents and electronic files at project closeout

1.3 COPIES OF PROGRAM CODE AND CONFIGURATION FILES

A. Copies of Program Code and Configuration Files – General:

1. Submit as Electronic Documents only. Paper Submittals are not required for program code and configuration files.
2. Files to be securely transferred with limited distribution as requested by the CO to protect sensitive information. Provide proof of file transfer for submittal record purposes.
3. In accordance with the Contract Documents, following Substantial Completion, the Government shall have rights to: (a) modify program code and configuration files, (b) update software and firmware, (c) revise system security settings, such as passwords, IP addresses, and other security settings, and (d) implement related modifications, without restriction or interference from Contractor, Subcontractor, Supplier, and others.
4. The Government agrees to use program code and configuration files only with Government facilities and may be transferred to the Government's assignees.
5. The Government will not be subject to any Supplier-requested non-disclosure agreement that is not part of the Contract Documents.
6. The CMR agrees to not distribute program code and configuration files obtained under the Project, except in exchanging such files with the Government. The CMR will not be party to any Supplier-requested non-disclosure agreement.

B. Configuration Files:

1. Submit copies of system configuration prepared for the Project, such as setpoints for programmable controllers, facility SCADA display configurations, and similar configuration files.
2. Files to be securely transferred with limited distribution as requested by the Government or supplier to protect sensitive information. Provide proof of file transfer for submittal record purposes.
3. Submit as separate files configuration files for each separate control and monitoring device for which configuration files are furnished. Clearly distinguish the device(s) associated with each file.

4. Contractor (including Subcontractors and Suppliers) is not responsible for configurations and control setpoints subsequently changed by the Government, facility manager, or others for whom either is responsible, not in accordance with Supplier's written recommendations and operation and maintenance instructions.

C. Program Code:

1. Submit copies of program code for programmable logic controllers (PLC), human-machine interfaces (HMI), operator interface terminals (OIT), and other programmable controllers, subject to the following:
   a. Two alternative versions of the paragraph below are provided; retain one and delete or do not use the other. The second alternative version is in optional text. Retain the first alternative version, below, when the Owner or facility manager requires copies of ***all*** program code. The second alternative version, in optional text, is likely more-desirable to equipment manufacturers; when the second alternative version is used, retain the optional text and delete the first version.
   b. Submit for all PLCs, HMI, OITs, and other programmable controllers furnished as part of the Work, and where the Government's existing devices were modified as part of the Work, regardless of whether such program code is manufacturer's standard, or developed specifically for the Project, or a combination of manufacturer's standard program code and Project-specific program code. Contractor and associated Subcontractors and Suppliers are not responsible for program code modifications made by the Government that result in improper operation of materials, equipment, or systems or that invalidate applicable warranties and manufacturer's recommended operating instructions.
   c. Third-party, licensed, commercially available software (such as, but not limited to, Microsoft operating system software sold at retail, and commercial SCADA system software platforms, PLC programming software) is excluded from requirements of this Article. Furnish copies of commercially available, licensed, third-party software, where required, in accordance with the Contract Documents.
2. Submit complete logic listings in all formats (ladder logic, function block diagram, sequential function chart, instruction list, structured text).
3. Format Requirements:
   a. Retain one of the paragraphs below, in coordination with the requirements of the paragraph immediately above this user note. Delete inapplicable versions. Edit to suit the project.
   b. For ladder diagram logic, include complete cross-referencing of all logic elements. Annotate all elements with clearly understandable tags or descriptive labels.
   c. For function block diagram, label each function block with understandable tags or descriptive labels. Describe purpose and action of each function block.
   d. For sequential function chart, include extensive comments for each step to describe program step function.
   e. For instruction list and structured text, include extensive comments for each program line to describe program line function.
4. Submit complete programmable logic controller listing of all input/output address assignments, tag assignments, and pre-set constant values, with functional point descriptions.
5. Submit complete manufacturer's program code manuals.

1.4 REMOTE ACCESS

A. Suppliers will not be allowed open access to the furnished equipment after installation on the site.

B. Remote access appliances for interaction with programmable devices will not be accepted during construction or operation.

PART 2 - EQUIPMENT - (NOT USED)

PART 3 - EXECUTION

3.1 COORDINATION WITH OTHER INTEGRATORS

A. Supplier shall provide full access to the PLC and HMI code for the Government's Software System Integrator to develop and configure the main Human Machine Interface (HMI), as requested. This code will be transmitted in its native with all passwords supplied through secure transmission portal as allowed by the Government policies.

B. Supplier shall coordinate programming efforts to allow for data to be passed from PLC to PLC for operations and coordination between vendor and balance of plant or other PLCs. This may involve a variety of native protocols including:

1. Ethernet/IP between AB PLCs and Ethernet/IP devices including the Do-More CPU.
2. Produced and Consumed shall be used between AB PLCs.
3. Modbus RTU or Modbus TCP/IP.

C. Other protocols shall be implemented only if mutually agreeable.

3.2 ON SITE PROGRAMMING ACCESS

A. Only Government-provided programming laptops will be allowed to connect/access on site networks.

1. All programming software will be a common version. This version will be provided during construction by the CO.

END OF SECTION 01 81 33

This page intentionally left blank.

SECTION 01 91 14 – TOTAL BUILDING COMMISSIONING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. General requirements for coordinating and scheduling commissioning.
2. Commissioning meetings
3. Commissioning reports
4. Use of test equipment, instrumentation, and tools for commissioning.
5. Construction checklists, including, but not limited to, installation checks, startup, and performance tests.
6. Commissioning tests
7. Adjusting, verifying, and documenting identified systems and assemblies.

B. Work included under this section includes a complete and thorough investigation of equipment and systems indicated in Part 3 of section. In order to ensure proper installation and operation of components and systems. Contractor shall perform commissioning as described herein to accomplish the tasks, and goals of commissioning. Systems to be evaluated include but are not limited to:

1. HVAC (Heating, Ventilation, and Air Conditioning) components and equipment.
2. HVAC system: Interaction of cooling, heating, and comfort delivery systems.
3. Building Automation System (BAS): Control hardware and software, sequence of operations, and integration of factory controls with BAS.
4. Lighting Control System and interface with daylighting.
5. Building Envelope (walls, roof, windows, infiltration, etc.)
6. Life Safety Systems (Fire Alarm & Suppression)
7. Electrical and Alternative Energy Systems (propane generator, photovoltaic panels, etc.)
8. Access Control/ Security Systems
9. Lightning Protection

C. Building commissioning activities and documentation are described in the following reference material: United States Green Building Council (USGBC) Leadership in Energy and Environmental Design (LEED™) rating program, American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE) Guideline 0-2005, The Commissioning Process, and National Institute of Building Sciences (NIBS) Guidelines.

D. National Park Service (NPS) personnel, Green Consultant, and Architect/Engineer, are not responsible for construction means, methods, job safety, or management function related to commissioning on job site.

E. Related Sections:

1. 01 31 00 - Project Management and Coordination
2. 01 33 23 - Submittal Procedures
3. 01 40 00 - Quality Requirements
4. 01 57 19.11 - Indoor Air Quality (IAQ) Management
5. 01 57 19.12 - Noise and Acoustics Management
6. 01 77 00 - Closeout Procedures

7. 01 78 23 - Operation and Maintenance Data
8. 01 79 00 - Demonstration and Training
9. 01 81 13 - Sustainable Design Requirements

1.2 DEFINITIONS

A. Acceptance Criteria: Threshold of acceptable work quality or performance specified for a commissioning activity.

B. Basis-of-Design Document: Document prepared by Designer that records concepts, calculations, decisions, and product selections used to comply with Owner's Project Requirements and to suit applicable regulatory requirements, standards, and guidelines.

C. Total Building Commissioning (TBC): Quality-focused process for verifying and documenting that facility, systems and assemblies are planned, designed, installed, and tested to comply with Owner's Project Requirements. Requirements specified here are limited to construction phase commissioning activities.

D. Construction Checklist: Form used by Contractor to verify appropriate components are on site, ready for installation, correctly installed and functional.

E. Contractor’s Commissioning Representative: (CCxR) Contractor’s designated individual to coordinate, manage, and execute commissioning processes of the contracting organizations.

F. Commissioning Plan (CCxP): Plan that provides structure, schedule and coordination planning for commissioning process proposed specifically for this project. CCxP includes Personnel, activities, and a description of Infrastructure, and list of instruments and logging devices that will be used during Commissioning.

G. Deficiency: Condition in the installation or function of a component, piece of equipment or system not in compliance with Contract Documents, does not perform properly or is not complying with Basis of Design.

H. Functional Performance Test (FPT): Test of dynamic function and operation of equipment and systems using manual (direct observation) or monitoring methods. Functional testing is the dynamic testing of systems (rather than just components) under full operation (e.g., the chiller pump is tested interactively with the chiller functions to see if the pump ramps up and down to maintain the differential pressure setpoint). Systems are tested under various modes, such as during low cooling or heating loads, high loads, component failures, unoccupied, varying outside air temperatures, fire alarm, power failure, etc. The systems are run through all the control system’s sequences of operation and components are verified to be responding as the sequences state. Traditional air or water test and balancing (TAB) is not functional performance testing, in the commissioning sense of the word. TAB’s primary work is setting up the system flows and pressures as specified, while functional testing is verifying that which has already been set up. The CCxR develops the sequentially written functional test procedure forms, and oversees and documents the actual testing, which is performed by the installing contractor or vendor. The CCxR creates worksheets from these forms which include procedures required to accommodate actual equipment, means and methods used in the project. Functional Performance Tests are performed after pre-functional checklists and startup is complete.

I. Manual Test: Using hand-held instruments, control system readouts or direct observation to verify performance (contrasted to analyzing electronically monitored data taken over time to make the "observation").

J. Monitoring: Recording of parameters (flow, current, status, pressure, etc.) of equipment operation using data loggers or the trending capabilities of control systems.

K. Owner's Project Requirements: Document originated by Designer that details functional requirements of project and expectations of use and operation, including project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information. Document is updated, with input from Contracting Officer (CO) as required as project is finished.

L. Pre-functional Checklist: List of items to inspect and elementary component tests to conduct to verify proper installation of equipment. Pre-functional checklists are primarily static inspections and procedures to prepare equipment or system for initial operation (e.g., belt tension, oil levels ok, labels affixed, gages in place, sensors calibrated, etc.). However, some pre-functional checklist items entail simple testing of the function of a component, a piece of equipment or system (such as measuring voltage imbalance on a three-phase pump motor of a chiller system). Pre-functional" refers to "before" functional testing. Pre-functional checklists augment and are combined with the equipment manufacturer’s start-up checklist.

M. Seasonal Performance Tests: Functional Performance Tests deferred until system(s) will experience seasonal conditions closer to their design conditions.

N. Systems Manual: System focused composite document that includes operational manual, maintenance manual, and additional information of use to Government during Occupancy and Operation Phase.

1.3 COMMISSIONING TEAM

A. Members Appointed by Contractor(s): Individuals, each having authority to act on behalf of the entity he or she represents, explicitly organized to implement the commissioning process through coordinated action.

1. CCxR (Contractor’s Commissioning Representative): CCxR shall be approved by Contracting Officer and satisfy as many of the following requirements as possible:
    a. Certified in Commissioning by nationally accredited organization (i.e. Associated Air Balance Council (AABC), Association of Energy Engineers (AEE), Building Commissioning Association (BCA), and National Environmental Balancing Bureau (NEBB))
    b. Acted as principal Commissioning Authority where total building commissioning approach (including building envelope) was used for at least three projects of comparable size, type, and scope.
    c. Technical training in Mechanical, Electrical, and/or fire protection engineering.
    d. Past commissioning experience.
    e. Knowledge of national codes.
    f. Leadership in Energy and Environmental Design (LEED) Accredited Professional.
    g. Experience in energy-efficient design and control strategy optimization.
    h. Specific experience with specialty systems relative to particular facility type (i.e. Federal blast and progressive collapse requirements, security systems, etc.).
2. Contractor Quality Control (CQC) Supervisor
3. Other Representatives may include Project superintendents, installers, suppliers, and specialists.

B. Members Appointed by Contracting Officer:

1. Representatives of facility user and operation and maintenance personnel.

2. Architect and engineering design professionals.

1.4 CONTRACTOR'S RESPONSIBILITIES

A. Contractor shall assign representatives with expertise and authority to act on its behalf to participate in and perform commissioning process activities including:

1. Perform commissioning tests, as required by technical specifications. Evaluate performance deficiencies identified in test reports and, in collaboration with entity responsible for system and equipment installation, recommend corrective action.
2. Record and resolve commissioning issues.
3. Attend commissioning team meetings held on variable basis.
4. Integrate and coordinate commissioning process activities with overall project schedule.
5. Review Construction Checklist attached at end of specification section.
6. Complete electronic construction checklists as contract work is completed and provide to Contracting Officer on a weekly basis.
7. Complete commissioning process test procedures.
8. Provide maintenance orientation and inspection for systems, assemblies, equipment, and components based on contract requirements.
9. Provide Commissioning Plan and documentation for final commissioning documentation.
10. Provide measuring instruments and logging devices to record test data and provide data acquisition equipment to record data for complete range of testing for required test period.

1.5 COMMISSIONING DOCUMENTATION

A. Provide the following information:

1. Review of systems manual, submittals, documents, and other commissioning reports
2. Identification of installed systems, assemblies, equipment, and components including design changes that occurred during the construction phase
3. Commissioning Plan including Process activities and schedule for completing construction checklists and manufacturer's pre-start and startup checklists for systems, assemblies, equipment, and components to be verified and tested
4. Certificate of readiness certifying systems, subsystems, equipment, and associated controls are ready for testing
5. Test and inspection reports and certificates
6. Corrective action documents
7. Testing, adjusting, and balancing reports

1.6 SUBMITTALS

A. Two-week look-ahead schedules: Schedule showing the next two weeks of commissioning related construction activity to include completion dates for each element of commissioning documentation for each major system or subsystem as identified in 1.1.B.

B. Certificates of readiness.

C. Contractor’s Commissioning Representative Qualifications.

D. Commissioning Plan: Submit within 30 calendar days of authorization to proceed.

1. Update as necessary during the work to reflect progress on components and systems.

E. Pre functional checklists.

F. Owner's project requirements.

G. Functional performance test forms: Submit minimum 30 calendar days prior to testing

H. List of test instrumentation, equipment, and monitoring devices. Include:

1. Make, model, serial number, and application for each instrument, equipment, and monitoring device.
2. Brief description of intended use.
3. Calibration record showing:
    a. Calibration agency, including name and contact information
    b. Last date of calibration
    c. Range of values for which calibration is valid
    d. Certification of accuracy
    e. National Institute of Standards and Technology (NIST) traceability certification for calibration equipment.
    f. Due date of the next calibration.

I. Deficiency Report and Resolution Record: Document items of non-compliance in materials, installation or operation. Document results from start-up/pre-functional checklists, functional performance testing, and short-term diagnostic monitoring. Include details of components or systems found to be non-compliant with drawings and specifications. Identify adjustments and alterations required to correct system operation and identify who is responsible for making corrective changes.

1. Update as necessary during work to reflect progress on components and systems. Submit updated versions monthly.

J. Closeout Documentation

1. Closeout documents for commissioned equipment and systems shall be submitted prior to functional performance testing. These include:
    a. Record Documents and Drawings
    b. Start-up certificates for commissioned equipment with start-up requirements
    c. Systems Manual
    d. Include TAB, startup, and Control System check-out reports.
    e. Other documents required by Section 01 77 00.
2. Operation and Maintenance (O&M) Submittals (refer to requirements of technical specifications):
    a. Training plan: Include for each training session:
        1) Dates, start and finish times, and locations
        2) Outline of the information to be presented
        3) Names and qualifications of presenters
        4) List of texts and materials required to support training

PART 2 - PRODUCTS

2.1 TEST EQUIPMENT

A. Instrumentation shall:

1. Be of sufficient quality and accuracy to test and measure system performance within tolerances required to determine adequate performance.
2. Be calibrated on manufacturer's recommended intervals calibration tags permanently affixed to instrument being used.
3. Be maintained in good repair and operation condition throughout duration of use on this project.

B. Standard testing equipment required to perform startup and initial checkout and required functional performance testing shall be provided by Contractor for equipment being tested.

C. Required commissioning equipment (sensors, transducers, data loggers, etc.) not integral to the systems or equipment installed shall be provided by Contractors Commissioning Representative and shall not become property of the Government.

2.2 PRE-FUNCTIONAL CHECKLIST:

A. Prepare pre functional checklists for equipment and systems to be commissioned.

B. Pre-functional checklists shall be complementary to Commissioning Plan and Commissioning Schedule.

2.3 FUNCTIONAL TEST PROCEDURE FORMS: Prepare functional test procedure forms for each piece of equipment and each system to be commissioned.

2.4 FUNCTIONAL PERFORMANCE WORKSHEETS:

A. Prepare Functional Performance worksheets, consisting of test procedures and expected results of testing.

2.5 REPORT FORMAT AND ORGANIZATION

A. General Format and Organization:

1. Bind report in three-ring binders.
2. Label front cover and spine of each binder with report title, volume number, project name, Contractor's name, and date of report.
3. Record report on compact disk.
4. Electronic Data: Portable document format (PDF); a single file with outline-organized bookmarks for major and minor tabs and tab contents itemized for specific reports.

B. Commissioning Report:

1. Include table of contents and an index to each test.
2. Include major tabs for each Specification Section.
3. Include minor tabs for each test.
4. Within each minor tab, include:
    a. Test specification.
    b. Pre-startup reports.
    c. Approved test procedures.
    d. Test data forms, completed and signed.

e. Commissioning issue reports, showing resolution of issues, and documentation related to resolution of issues pertaining to a single test. Group data forms, commissioning issue reports showing resolution of issues, and documentation related to resolution of issues for each test repetition together within minor tab, in reverse chronological order (most recent on top).

PART 3 - EXECUTION

3.1 COMMISSIONING PROCESS

A. Following activities outline general commissioning tasks (requiring development, execution, etc.) and order in which they occur.

1. Commissioning Scoping Meeting
2. Finalize Owner's Project Requirements
3. Commissioning Plan
4. Prepare pre-functional checklists.
5. Prepare functional performance worksheets.
6. Perform Start-Up/Pre-Functional Checks in accordance with manufacturer's recommendations and pre-functional checklists.
7. Functional Performance Testing in accordance with functional performance worksheets
8. Deficiency Report and Resolution Record
9. Operation and Maintenance Documentation
10. Operations and Maintenance Training
11. Deferred Testing

3.2 TOTAL BUILDING COMMISSIONING (TBC) REQUIREMENTS

A. TBC during construction, acceptance, and warranty phases is intended to achieve following specific objectives:

1. Verify that systems and equipment meet Owner's Project Requirements.
2. Verify equipment is what was submitted and approved.
3. Verify and document equipment is installed and started per manufacturer's recommendations, industry accepted minimum standards, and Contract Documents.
4. Verify and document equipment and systems receive complete operational checkout by installing contractors.
5. Verify and document equipment capacity and system efficiency.
6. Verify performance of building envelope. Document testing and conformance to Contract Documents.
7. Verify completeness of operations and maintenance materials.
8. Ensure Governments operating personnel are adequately trained on operation and maintenance of building equipment.

3.3 COMMISSIONING SCOPING MEETING

A. Commissioning Scoping Meeting:

1. Schedule, coordinate, and facilitate a scoping meeting.
2. Review each building system to be commissioned, including intended operation, commissioning requirements, and completion and start-up schedules.

3. Establish scope of work, tasks, schedules, deliverables, and responsibilities for implementation of Commissioning Plan.
4. Attendance: Commissioning Team members.

3.4 COMMISSIONING PLAN

A. Commissioning Plan: Develop commissioning plan to identify how commissioning activities will be integrated into general construction and trade activities. Commissioning plan shall identify how commissioning responsibilities are distributed. Intent of plan is to evoke questions, expose issues, and resolve issues with input from entire commissioning team early in construction.

1. Identify who will be responsible for producing various procedures, reports, Contracting Officer notifications and forms.
2. Include commissioning tasks and activities in overall project schedule. Tag individual activities so they can be filtered at later date.
3. List and describe each test/acceptance procedure, including acceptance criteria.

3.5 START-UP/PRE-FUNCTIONAL CHECKLISTS

A. Start-Up/Pre-Functional Checklists: Complete pre-functional checklists prior to start up. Checklist shall help verify that systems are complete and operational, so functional performance testing can be scheduled.

1. Verify equipment installed is what was approved on Submittal.
2. Manufacturer’s start-up checklists and other technical documentation guidelines may be used as basis for pre-functional checklists.

3.6 FUNCTIONAL PERFORMANCE TESTING

A. Functional Performance Testing: Test procedures fully describe system configuration and steps required for each test.

1. Test Methods: Functional performance testing and verification may be achieved by direct manipulation of system inputs (i.e. heating or cooling sensors), manipulation of system inputs with building automation system (i.e. software override of sensor inputs), trend logs of system inputs and outputs using building automation system, or short-term monitoring of system inputs and outputs using standalone data loggers. A combination of methods may be required to completely test complete sequence of operations. CCxR shall determine which method or combination of methods is most appropriate.
2. Setup: Each test procedure shall be performed under conditions that simulate normal operating conditions as closely as possible. Where equipment requires integral safety devices to stop/prevent equipment operation unless minimum safety standards or conditions are met, functional performance test procedures shall demonstrate actual performance of safety shutoffs in real or closely simulated conditions of failure.
3. Sampling: Multiple identical pieces of non-life-safety or non-critical equipment may be functionally tested using a sampling strategy. If, after three attempts at testing the specified sample percentage, failures are still present, remaining units shall be tested at Contractors’ expense.

B. Prepare functional performance test procedure forms to accommodate actual installed equipment and systems.

C. Coordinate, execute, and record results of functional performance testing.

1. Coordinate retesting as necessary until satisfactory performance is verified.
2. Verify intended operation of individual components and system interactions under various conditions and modes of operation.

3.7 DEFICIENCY REPORT AND RESOLUTION RECORD

A. Deficiency Report and Resolution Record: Document items of non-compliance in materials, installation or operation.

B. Non-Conformance. Non-conformance and deficiencies observed shall be addressed immediately. Notify responsible parties and provide recommended actions to correct deficiencies.

1. Corrections of minor deficiencies identified may be made during tests at discretion of CCxR. In such cases the deficiency and resolution shall be documented on procedure form.
2. For identified deficiencies:
    a. If no dispute on deficiency and responsibility to correct it:
        1) CCxR documents deficiency and adjustments or alterations required to correct it. Contractor corrects deficiency and notifies CCxR that equipment is ready to be retested.
        2) CCxR reschedules test and test is repeated until satisfactory performance is achieved.
    b. If there is a dispute about a deficiency or who is responsible:
        1) Deficiency is documented CCxR on non-compliance form.
        2) Resolutions are made at lowest management level possible. Additional parties are brought into discussions as needed. Contractor shall have responsibility for resolving construction deficiencies. If a design revision is deemed necessary and approved by Contracting Officer, Architect/Engineer (A/E) shall have responsibility for providing design revision. CCxR documents resolution process.
        3) Once interpretation and resolution have been decided, appropriate party corrects deficiency and notifies CCxR that equipment is ready to be retested. CCxR reschedules test and test is repeated until satisfactory performance is achieved.
3. Cost of Retesting: Costs for retesting shall be charged to Contractor.

3.8 OPERATIONS AND MAINTENANCE TRAINING

A. Training: Develop Training Plan. Coordinate and execute training programs with .

1. Stress and enhance importance of system interactions, troubleshooting, and long-term preventive maintenance and operation programs.

3.9 DEFERRED TESTING

A. Unforeseen Deferred Tests: If test cannot be completed due to building structure, required occupancy condition, or other deficiency, functional testing may be delayed upon recommendation of CCxR and approval of Contracting Officer. These tests are conducted in same manner as the seasonal tests, as soon as possible.

B. Seasonal Testing

1. Schedule, coordinate, execute, and document additional testing for seasonal variation in operations and control strategies during appropriate season to verify performance of HVAC system and controls. Complete testing during warranty period to fully test sequences of operation.
2. Update O&M manuals and Project Record Drawings as necessary due to testing.

3.10 EQUIPMENT & SYSTEM SCHEDULE

A. Commissioned Equipment and Systems List: Following is a list of systems and equipment to be commissioned organized by system. It includes the percentage of each category that will undergo testing. The intent is to provide an overall summary of commissioned equipment and systems, and not a comprehensive list. Refer to applicable specification sections for more information.

COMMISSIONED EQUIPMENT AND SYSTEMS LIST			
SYSTEM	**EQUIPMENT**	**CC** (Commissioning Coordinator)	**FPT** (Functional Performance Test)
ELECTRICAL	BUILDING ELECTRICAL SYSTEMS; NON-PROCESS RELATED	100	100
LIGHTING	BUILDING LIGHTING SYSTEMS	100	100
BACKUP POWER	GENERATOR AND FUEL SYSTEM	100	100
HVAC	BUILDING HVAC SYSTEMS	100	100
PLUMBING	BUILDING PLUMBING SYSTEMS	100	100
FIRE PROTECTION	BUILDING FIRE PROTECTION SYSTEMS	100	100

END OF SECTION 01 91 14